

MTL-730: Cryptography

(Combined Lecture Notes)

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Chapter 1

Cryptography: Classical Cryptography

1.1 Introduction

In today's digital age the topmost priority of a human being is to communicate securely. But it is not easy to have secure communication over any channel. It has been always a topic of interest "how to communicate securely". In past, people used to communicate through handwritten letters, and letters were sent via hands. It was a tough challenge to maintain the secrecy of information. For secure communication, they were use riddles, puzzles, some special characters in place of the original message such a way that only the receiver could decode.

The practice and study of hiding information is known as Cryptography. Cryptography is a Mathematical science that comes up with a way for secure communication between two or more parties by converting the confidential information into a scrambled form so that only authorized users can remove scrambles and read information. For instance a sender Alice wants to send secret information to a receiver Bob over an insecure communication channel i.e via the internet or telephone. Then it is possible that Eavesdropper can intercept and read the information.

1.2 Security Services of Cryptography

The main goal of cryptography is to provide the following four fundamental information security services.

- **Confidentiality:** Confidentiality is the fundamental security service provided by cryptography. Confidentiality means protecting information from unauthorized users. It provides a surety to a user that his information is not to be shared with anyone. It is sometimes referred to as privacy or secrecy.
- **Data Integrity:** It deals with identifying any alteration to the data. The data may get modified by an unauthorized entity intentionally or accidentally. Data integrity cannot prevent the alteration of data, but provides a means for detecting whether data has been manipulated.
- **Authentication:** Authentication provides the identification of the originator. It confirms to the receiver that the data received has been sent only by an identified and verified sender. Apart from the originator, authentication may also provide assurance about other parameters related to data such as the date and time of creation/transmission.
- **Non-repudiation:** It ensures that an entity cannot refuse the ownership of a previous commitment or an action. Non-repudiation is a property that is most desirable in situations where there are chances of a dispute over the exchange of data.

1.3 Basic Terminology:

- **Plaintext ($\mathcal{P}$):** It is the message to be protected during transmission.
- **Ciphertext ($\mathcal{C}$):** It is the scrambled version of the plaintext produced by the encryption.
- **Encryption ($\mathcal{E}$):** It is a process of converting plain text to cipher text i.e.

$$\begin{aligned}\mathcal{E} : \mathcal{P} &\mapsto \mathcal{C} \\ m &\mapsto \mathcal{E}(m) = c\end{aligned}$$

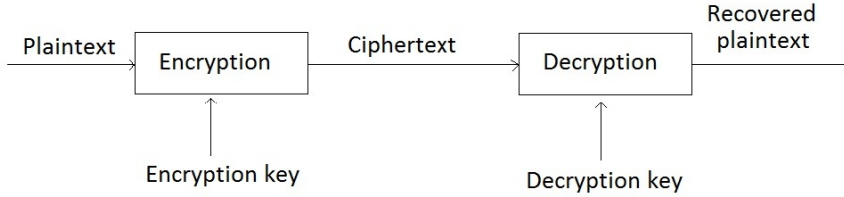


Figure 1.1: Block-diagram

- **Decryption ($\mathcal{D}$):** It is a process of reconvertng cipher text to plain text i.e.

$$\begin{aligned}\mathcal{D} : \mathcal{C} &\mapsto \mathcal{P} \\ c &\mapsto \mathcal{D}(c) = m\end{aligned}$$

- **Encryption Key:** It is a value that is known to the sender and used for encryption.
- **Decryption Key:** It is a value that is known to the receiver and used for decryption.
- **Cryptanalysis:** It is the study of deciphering ciphertext without knowing key.

Note:- The plaintext and ciphertext are written in some symbols (usually, but not always, they are written in the same symbols) consisting of a certain number n of letters. The term “letter” (or “character”) can refer not only to the familiar A-Z, but also to numerals, blanks, punctuation marks, or any other symbols that we allow ourselves to use when writing the messages.

6 CHAPTER 1. CRYPTOGRAPHY: CLASSICAL CRYPTOGRAPHY

Let Plain text space is $\mathcal{P} = \{A, B, \dots, Z\}$ and Cipher text space is $\mathcal{C} = \{A, B, \dots, Z\}$.

Then Encryption function is defined by

$$\mathcal{E} : \mathcal{P} \rightarrow \mathcal{C}$$

And Decryption function is defined by

$$\mathcal{D} : \mathcal{C} \rightarrow \mathcal{P}$$

Such that

$$\mathcal{D}\mathcal{E} = \mathcal{E}\mathcal{D} = I$$

$\mathcal{E}_0 :$	A	B	C	...	X	Y	Z	(No shift)
$\mathcal{E}_1 :$	B	C	D	...	Y	Z	A	(Shift by 1)
$\mathcal{E}_2 :$	C	D	E	...	Z	A	B	(Shift by 2)
$\mathcal{E}_3 :$	D	E	F	...	A	B	C	(Shift by 3)

$\mathcal{D}_0 :$	A	B	C	...	X	Y	Z
$\mathcal{D}_1 :$	Z	A	B	...	W	X	Y
$\mathcal{D}_2 :$	Y	Z	A	...	V	W	X
$\mathcal{D}_3 :$	X	Y	Z	...	U	V	W

Decrypt “RK!LZLVKLNQHZ”

$\mathcal{D}_1(\text{RK}) = \text{QJ}$ (make no sense i.e no English word)

$\mathcal{D}_2(\text{RK}) = \text{PI}$ (possibility of English words like PIN, PING etc)

$\mathcal{D}_2(\text{LZL}) = \text{JXJ}$

Combining two decryptions we have “PI JXJ” (make no sense i.e no English word)

$\mathcal{D}_3(\text{RK}) = \text{OH}$ (make sense)

$\mathcal{D}_3(\text{LZL}) = \text{IWI}$ (make sense i.e OH I WI)

$\mathcal{D}_3(\text{VKL}) = \text{SHI}$ (make sense i.e OH I WISH I)

$\mathcal{D}_3(\text{NQHZ}) = \text{KNEW}$ (make sense)

So Cipher text is decrypted by $\mathcal{D}_3$ and decrypted message is “OH! I WISH I KNEW”.

Numerical equivalent of Symbols (Alphabets, other text symbols):-

To perform mathematical operations on plain texts and cipher texts we need some numeric value corresponding to alphabets.

- **Monograph:** Monograph means single symbol at a time for encryption and decryption.

Numerical equivalent of monographs (Single symbol):

A	B	C	D	E	F	G	H	I	J	K	L	M	N
0	1	2	3	4	5	6	7	8	9	10	11	12	13
O	P	Q	R	S	T	U	V	W	X	Y	Z	?	!
14	15	16	17	18	19	20	21	22	23	24	25	26	27

i.e $\{A, B, \dots, Y, Z, ?, !, \dots, \hat{b}\} = \{0, 1, \dots, 24, 25, \dots, n-1\}$ for n symbols and mathematical operations are same as $\mathbb{Z}_n$ with addition and multiplication modulo n (=no of symbols).

- **Digraphs:** Digraphs means two symbols at a time for encryption and decryption. There is two ways for numerical equivalent of digraphs:

1. Numerical equivalent of symbols are as follows:

A	B	C	D	E	F	G	H	I	J	K	L	M	N
00	01	02	03	04	05	06	07	08	09	10	11	12	13
O	P	Q	R	S	T	U	V	W	X	Y	Z	?	!
14	15	16	17	18	19	20	21	22	23	24	25	26	27

Numerical equivalent of digraph XY is equal to

$XY = \text{Numerical equivalent of X} \times \text{Numerical equivalent of Y}$ (each having two digits).

e.g. NO=1314, AN=0013, NI=1308 O!=1427 etc.

2. Numerical equivalent of digraphs XY to base n can be written as

$$XY = n \times N(X) + N(Y)$$

$N(X)$ = Numerical equivalent to X

$N(Y)$ = Numerical equivalent to Y

E.g. AN = $0 \times n + 13$

NO = $13 \times n + 14$

O! = $14 \times n + 27$

In particular, let English alphabets be set of symbols ($n=26$) then we have

AN = $A \times 26 + N = 0 \times 26 + 13$

NO = $13 \times 26 + 14$

NI = $13 \times 26 + 8$

- Similarly for trigraphs (3 alphabets at a time for encryption and decryption)

$$XYZ = n^2 \times N(X) + n \times N(Y) + N(Z)$$

Remark: In general, we can label blocks of k letters in an N -letter alphabet by integers between 0 and $N^k - 1$ by regarding each such block as a k -digit integer to the base N .

Assumption(Kirchoff's law):- The attacker knows all the details(of the cryptosystem) except the keys (excepts) the cryptosystem is still secure. No cryptosystem is secure only time matters.

1.4 Classical Cipher:

Classical ciphers are ciphers that were used upto 1970s that is in the pre computer era. But since then they have fallen out of use. This is due to the fact that most of them can be solved by hand or using modern cryptographic algorithms.

There are two type of classical ciphers.

Substitution Cipher: In substitution ciphers each symbol of the plaintext message is substituted by other symbol in a systematic form. Substitution of symbols can be done mono alphabetically or poly alphabetically.

Transposition cipher: In transposition cipher symbols of the plaintext message remain same but ordering of these symbols are changed in a specific form.

1.5 Substitution Ciphers

1.5.1 Shift Cipher:

Suppose we are using an n -letter symbols(alphabets) with numerical equivalents $0, 1, \dots, n - 1$, i.e. $|\mathcal{P}| = n$. Let b be a fixed integer with $0 \leq b \leq n - 1$. In shift cipher we do encryption by shift transformation.

- **Encryption:** The encryption function is defined by

$$\begin{aligned} \mathcal{E}: \mathcal{P} &\rightarrow \mathcal{C} \\ m &\mapsto c = \mathcal{E}(m) \equiv m + b \pmod{n} \end{aligned}$$

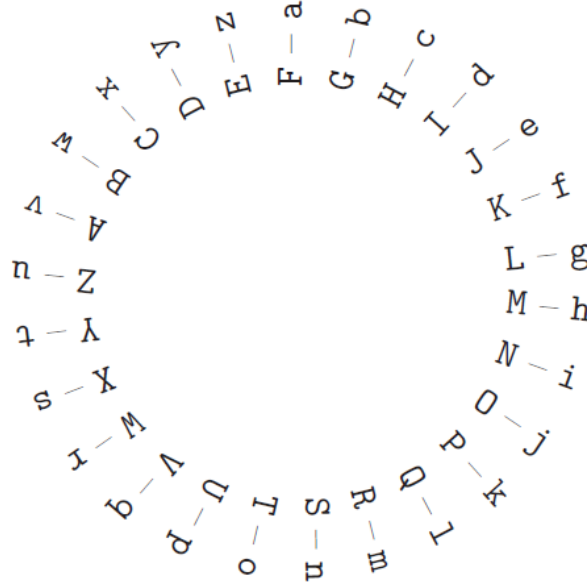


Figure 1.2: Cyclic Shift by 5 letter

where ‘ m' ’ is numerical equivalent of symbol (alphabet) to be encrypted.

- **Decryption:** The decryption function is defined by

$$\begin{aligned} \mathcal{D}: \mathcal{C} &\rightarrow \mathcal{P} \\ c &\mapsto m = \mathcal{D}(c) \equiv c - b \pmod{n} \end{aligned}$$

Note1: Here, $\mathcal{P}$ and $\mathcal{C}$ are equivalent to $\mathbb{Z}_n$.

Note2: In particular, when $b = 3$ shift cipher is known as **Caesar cipher**.

Example:

Let $\mathcal{P} = \{A=0, B=1, \dots, Y=24, Z=25\}$ be set of symbols (alphabets), then encryption function is given by

Encryption: Let $b = 2$

$$\begin{aligned} \mathcal{E}: \mathbb{Z}_{26} &\rightarrow \mathbb{Z}_{26} \\ m &\mapsto c = \mathcal{E}(m) \equiv m + 2 \pmod{26} \end{aligned}$$

To encrypt “HELLO”

$$\begin{aligned}\mathcal{E}(m_1 = H) &= \mathcal{E}(7) = 7 + 2 \equiv 9 \pmod{26} = \mathbf{J} \\ \mathcal{E}(m_2 = E) &= \mathcal{E}(4) = 4 + 2 \equiv 6 \pmod{26} = \mathbf{G} \\ \mathcal{E}(m_3 = L) &= \mathcal{E}(11) = 11 + 2 \equiv 13 \pmod{26} = \mathbf{N} \\ \mathcal{E}(m_4 = L) &= \mathcal{E}(11) = 11 + 2 \equiv 13 \pmod{26} = \mathbf{N} \\ \mathcal{E}(m_5 = O) &= \mathcal{E}(14) = 14 + 2 \equiv 16 \pmod{26} = \mathbf{Q}\end{aligned}$$

So encryption of 'HELLO' is 'JGNNQ'.

Decryption: Decryption function is given by

$$\begin{aligned}\mathcal{D}: \mathbb{Z}_{26} &\rightarrow \mathbb{Z}_{26} \\ c &\mapsto m = \mathcal{D}(c) \equiv c - 2 \pmod{26}\end{aligned}$$

To decrypt “JGNNQ”

$$\begin{aligned}\mathcal{D}(C_1 = J) &= \mathcal{D}(9) = 9 - 2 \equiv 7 \pmod{26} = \mathbf{H} \\ \mathcal{D}(C_2 = G) &= \mathcal{D}(6) = 6 - 2 \equiv 4 \pmod{26} = \mathbf{E} \\ \mathcal{D}(C_3 = N) &= \mathcal{D}(13) = 13 - 2 \equiv 11 \pmod{26} = \mathbf{L} \\ \mathcal{D}(C_4 = N) &= \mathcal{D}(13) = 13 - 2 \equiv 11 \pmod{26} = \mathbf{L} \\ \mathcal{D}(C_5 = Q) &= \mathcal{D}(16) = 16 - 2 \equiv 14 \pmod{26} = \mathbf{O}\end{aligned}$$

So decryption of 'JGNNQ' is 'HELLO'.

1.5.2 Hill Cipher:

Let $\mathcal{P}$ and $\mathcal{C}$ be a plain text space and cipher text space on $\mathbb{Z}_n$.

- **Encryption:** The encryption function is defined by

$$\begin{aligned}\mathcal{E}: \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ m &\mapsto \mathcal{E}(m) = c \equiv am \pmod{n}\end{aligned}$$

where $a \in \mathbb{Z}_n$ is a fixed integer.

- **Decryption:** The decryption function is defined by

$$\begin{aligned}\mathcal{D} : \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ c &\mapsto \mathcal{D}(c) \equiv a^{-1}c \pmod{n}\end{aligned}$$

(Provided a^{-1} exist) it exist when $(a, n) = 1$ and can be computed by Euclidean algorithm.

Example:

Let $\mathcal{P} = \{A, B, \dots, Z, \}$ and $n = 26$
Plain text- “SPARKY”

Encryption: Let $a = 3$

$$\mathcal{E}(m) = c \equiv 3m \pmod{26}$$

$$\mathcal{E}(S) = \mathcal{E}(18) \equiv 3 \cdot 18 \pmod{26} = 2 = \mathbf{C}$$

$$\mathcal{E}(P) = \mathcal{E}(15) \equiv 3 \cdot 15 \pmod{26} = 19 = \mathbf{T}$$

$$\mathcal{E}(A) = \mathcal{E}(0) \equiv 3 \cdot 0 \pmod{26} = 0 = \mathbf{A}$$

$$\mathcal{E}(R) = \mathcal{E}(17) \equiv 3 \cdot 17 \pmod{26} = 25 = \mathbf{Z}$$

$$\mathcal{E}(K) = \mathcal{E}(10) \equiv 3 \cdot 10 \pmod{26} = 4 = \mathbf{E}$$

$$\mathcal{E}(Y) = \mathcal{E}(24) \equiv 3 \cdot 24 \pmod{26} = 20 = \mathbf{U}$$

Encrypted message is “CTAZEU”

Decryption:

$$\mathcal{D}(c) \equiv 3^{-1}c \pmod{26}$$

Since $(3, 26) = 1$, hence 3^{-1} exist in *mod* 26. By Euclidean algorithm we can find inverse of 3 in *mod* 26.

i.e $3^{-1} \equiv 9 \pmod{26}$

$$\begin{aligned}\mathcal{D}(C) &= \mathcal{D}(2) \equiv 9 \cdot 2 \pmod{26} = 18 = \mathbf{S} \\ \mathcal{D}(T) &= \mathcal{D}(9) \equiv 9 \cdot 19 \pmod{26} = 15 = \mathbf{P} \\ \mathcal{D}(A) &= \mathcal{D}(0) \equiv 9 \cdot 0 \pmod{26} = 0 = \mathbf{A} \\ \mathcal{D}(Z) &= \mathcal{D}(25) \equiv 9 \cdot 25 \pmod{26} = 17 = \mathbf{R} \\ \mathcal{D}(E) &= \mathcal{D}(4) \equiv 9 \cdot 4 \pmod{26} = 10 = \mathbf{K} \\ \mathcal{D}(U) &= \mathcal{D}(20) \equiv 9 \cdot 20 \pmod{26} = 24 = \mathbf{Y}\end{aligned}$$

Decrypted message is “SPARKY”

1.5.3 Affine Cipher: (Shift as well as multiplication)

Let $\mathcal{P}$ and $\mathcal{C}$ be a plain text space and cipher text space on $\mathbb{Z}_n$.

- **Encryption:** The encryption function is defined by

$$\begin{aligned}\mathcal{E} : \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ m &\mapsto \mathcal{E}(m) = c \equiv am + b \pmod{n}\end{aligned}$$

where $a, b \in \mathbb{Z}_n$ are fixed integer.

- **Decryption:** The decryption function is defined by

$$\begin{aligned}\mathcal{D} : \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ c &\mapsto \mathcal{D}(c) \equiv a'c + b' \pmod{n}\end{aligned}$$

where $a' \equiv a^{-1} \pmod{n}$ and $b' \equiv -a^{-1}b \pmod{n}$ (Provided a^{-1} exists)

$$\begin{aligned}c - b &= am \implies a^{-1}(c - b) = m \\ &\implies a^{-1}c - a^{-1}b = m \\ &\implies a'c + b' = m = \mathcal{D}(c)\end{aligned}$$

Example:

Let $\mathcal{P} = \{A, B, \dots, Z, _\}$ and $n = 27$

Plain text- “HELP ME”.

Encryption: Let $a = 13, b = 7$

$$\mathcal{E}(m) = c \equiv 13m + 7 \pmod{27}$$

$$\mathcal{E}(H) = \mathcal{E}(7) \equiv 13 \cdot 7 + 7 \pmod{27} = 17 = \mathbf{R}$$

$$\mathcal{E}(E) = \mathcal{E}(4) \equiv 13 \cdot 4 + 7 \pmod{27} = 5 = \mathbf{F}$$

$$\mathcal{E}(L) = \mathcal{E}(11) \equiv 13 \cdot 11 + 7 \pmod{27} = 15 = \mathbf{P}$$

$$\mathcal{E}(P) = \mathcal{E}(15) \equiv 13 \cdot 15 + 7 \pmod{27} = 13 = \mathbf{N}$$

$$\mathcal{E}(V) = \mathcal{E}(26) \equiv 13 \cdot 26 + 7 \pmod{27} = 21 = \mathbf{V}$$

$$\mathcal{E}(M) = \mathcal{E}(12) \equiv 13 \cdot 12 + 7 \pmod{27} = 1 = \mathbf{B}$$

Encrypted message is “RFPNVBF”

Decryption:

$$\mathcal{D}(c) \equiv a'c + b' \pmod{27}$$

where $a' \equiv 13^{-1} \pmod{27}$ and $b' \equiv -13^{-1}7 \pmod{27}$

Since $(13, 27) = 1$, hence 13^{-1} exist in *mod* 27. By using Euclidean algorithm we can find inverse of 13 in *mod* 27.

Hence $a' \equiv 25 \pmod{27}$ and $b' \equiv -25 \cdot 7 \pmod{27} \equiv 14 \pmod{27}$

$$\mathcal{D}(R) = \mathcal{D}(17) \equiv 25 \cdot 17 + 14 \pmod{27} = 7 = \mathbf{H}$$

$$\mathcal{D}(F) = \mathcal{D}(5) \equiv 25 \cdot 5 + 14 \pmod{27} = 4 = \mathbf{E}$$

$$\mathcal{D}(P) = \mathcal{D}(15) \equiv 25 \cdot 15 + 14 \pmod{27} = 11 = \mathbf{L}$$

$$\mathcal{D}(N) = \mathcal{D}(13) \equiv 25 \cdot 13 + 14 \pmod{27} = 15 = \mathbf{p}$$

$$\mathcal{D}(V) = \mathcal{D}(21) \equiv 25 \cdot 21 + 14 \pmod{27} = 26 = \mathbf{-}$$

$$\mathcal{D}(B) = \mathcal{D}(1) \equiv 25 \cdot 1 + 14 \pmod{27} = 12 = \mathbf{M}$$

Decrypted message is “HELP ME”

Activity: Read a 500 words article in English. Count how many times each English alphabet appear in that article.

Practically, Code breakers always try their best to break cipher text with their best strategy. So encrypted text is secure only when it is secure from best known methods presently. As new improved methods are developed, the old encryption system can be less secure only, never become better . In previous lecture we have studied about monoalphabetic substitution ciphers which are not more secure. In monoalphabetic cipher, each symbol is replaced by a unique symbol every time on encryption due to this fact we can apply frequency analysis to break such ciphers. In this lecture we will discuss cryptanalysis of some ciphers.

1.6 Frequency Analysis

From the activity given in last lecture, you would have observed that the letter **q** is always followed by letter **u**. The fact is that many letters appear, such as **e**, **t** and **a**, more frequently than others letters such as **f**, **h**, and **z**. Here we have frequency analysis of English Alphabets:

By decreasing frequency			
E	13.11%	M	2.54%
T	10.47%	U	2.46%
A	8.15%	G	1.99%
O	8.00 %	Y	1.98%
N	7.10%	P	1.98%
R	6.83%	W	1.54%
I	6.35%	B	1.44%
S	6.10%	V	0.92%
H	5.26%	K	0.42%
D	3.79%	X	0.17%
L	3.39%	J	0.13%
F	2.92%	Q	0.12%
C	2.76%	Z	0.08%

Table 1.1: Frequency Analysis of English Alphabets

Example: Decrypt the following cipher text over the set of symbols $\mathcal{A} = \{A=0, B=1, \dots, Y=24, Z=25\}$, $|\mathcal{A}| = 26$ decrypted by substitution cipher.

DGMNM XNM TXUQ ODEUDO RU DGRO AWXOO
 MXAG ODEUD FUYJO GRO UXTM MZMNQYUM
 GXZM DGNMM HMU

First we observe frequency of letters in cipher text

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Count	2	0	0	7	2	0	6	1	0	1	0	0	12

Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Count	4	8	0	2	3	0	2	7	0	1	6	2	2

(Verify !)

- M is most frequent and appears 12 times in cipher text, So M corresponds to E.
- O is second most frequent and appears 8 times in cipher text, So O corresponds to T.
- D, U appears 7 times each in cipher text, So D, U corresponds to A or O, first we take D corresponds to A and U corresponds to O.
- G, X appears 6 times each in cipher text, So G, X corresponds to R or N, first we take G corresponds to R and X corresponds to N.

i.e $\mathcal{D}(M)=E$, $\mathcal{D}(O)=T$, $\mathcal{D}(D)=A$, $\mathcal{D}(U)=O$, $\mathcal{D}(X)=N$, $\mathcal{D}(G)=R$ and so on.

Cipher text	DGMNM	XNM	TXUQ	ODEUDO	RU	DGRO	AWXOO	MXAG
Plain text	ARE-E	N-E	-NO-	TA-AT	-O	AR-T	-NTT	EN-R

Obtained plain text is not meaningful.

So we will change the decryptions to nearby frequencies like $\mathcal{D}(D)=T, \mathcal{D}(X)=A$.

We have decryptions:

$\mathcal{D}(M)=E$, $\mathcal{D}(O)=S$, $\mathcal{D}(D)=T$, $\mathcal{D}(U)=N$, $\mathcal{D}(X)=A$, $\mathcal{D}(G)=H$, $\mathcal{D}(N)=R$, $\mathcal{D}(R)=I$, $\mathcal{D}(M)=U$, $\mathcal{D}(Q)=Y$, $\mathcal{D}(A)=C$, $\mathcal{D}(W)=L$ and so on.

Cipher text	DGMNM	XNM	TXUQ	ODEUDO	RU	DGRO	AWXOO	MXAG
Plain text	THERE	ARE	MANY	STUDENTS	IN	THIS	CLASS	EACH

In the same way, we can decrypt remaining cipher text.

Frequency Analysis attack: When attacker (unauthorized person) decrypt the cipher text using frequency analysis of symbols, called frequency analysis attack.

Note 1. In practice we do frequency analysis over large text for accuracy. In above example we have taken very small text (70 letters only).

Similarly frequency analysis of digraphs and trigraphs over English alphabets are:

Decreasing order of digraphs

Order	Digraph	Order	Digraph
1	TH	20	HI
2	HE	21	IS
3	IN	22	OR
4	ER	23	TI
5	AN	24	AS
6	RE	25	TE
7	ND	26	ET
8	AT	27	NG
9	ON	28	OF
10	NT	29	AL
11	HA	30	DE
12	ES	31	SE
13	ST	32	LE
14	EN	33	SA
15	ED	34	SI
16	TO	35	AR
17	IT	36	VE
18	OU	37	RA
19	EA	38	LD

Order	Trigraph	Order	Trigraph
1	THE	16	ATI
2	AND	17	HAT
3	ING	18	ATE
4	ENT	19	ALL
5	ION	20	ETH
6	HER	21	HES
7	FOR	22	VER
8	THA	23	HIS
9	NTH	24	OFT
10	INT	25	ITH
11	ERE	26	FTH
12	TIO	27	STH
13	TER	28	OTH
14	EST	29	RES
15	ERS	30	ONT

1.7 Vigenère cipher

Vigenère Cipher is a polyalphabetic cipher. It is named after Blaise de Vigenere (1523–1596), whose 1586 book *Traicte des Chiffres* describes the known ciphers of his time.

In Vigenère cipher, encryption of different symbols is done by different shift ciphers. To decide how far to shift each symbol, sender (Bob) and receiver (Alice) share on a common keyword (common knowledge).

Now, Bob wants to send a secret message to Alice. Bob will divide the plain text (message) into blocks of length l (l =length of the keyword). Then Bob will shift the symbols of each block corresponding to the keyword, i.e., Bob will shift the first symbols of the block by the first symbol of the keyword, the second symbol of the block by the second symbol of keyword, and so on, Bob will do this for each block to encrypt the whole message.

To decrypt the ciphertext, Alice will divide ciphertext into the blocks of length l , (l =length of the keyword). After that, Alice will perform a backward shift of each cipher symbol of each block corresponding to the keyword, i.e., first cipher symbols backward shift by the first symbol of keyword, second cipher symbols backward shift by the second symbol of keyword, and so on, to get the plaintext.

To understand above process of encryption and decryption, suppose keyword

$$K = k_1 k_2 k_3 \dots k_l, \quad k_i \in \mathbb{Z}_n$$

message block is therefore

$$x = x_1 x_2 \dots x_l, \quad x_i \in \mathbb{Z}_n$$

then encryption function is given by

$$\mathcal{E}(x) = y = y_1 y_2 \dots y_l \quad \text{where} \quad y_i \equiv x_i + k_i \pmod{n}$$

decryption function is given by

$$\mathcal{D}(y) = x = x_1 x_2 \dots x_l \quad \text{such that} \quad x_i \equiv y_i - k_i \pmod{n}$$

where $n = |\mathcal{A}|$ =number of symbols written in their numerical equivalent.

We can treat

$$x = (x_1, x_2, \dots, x_l) \in \underbrace{\mathbb{Z}_n \times \mathbb{Z}_n \times \dots \times \mathbb{Z}_n}_{l \text{ times}} = \mathbb{Z}_n^l$$

Now consider plaintext space $\mathcal{P} = \mathbb{Z}_n^l$ and cipher text space $\mathcal{C} = \mathbb{Z}_n^l$. Then encryption is defined as

$$\mathcal{E} : \mathcal{P} \rightarrow \mathcal{C}$$

$$\mathcal{E}(x) = y = (y_1, y_2, \dots, y_l) \text{ where each } y_i \in \mathbb{Z}_n$$

Example: Let $\mathcal{A} = \{A=0, B=1, \dots, Y=24, Z=25, ?=26\}$ i.e. $|\mathcal{A}| = 27$
Encrypt the following plain text with keyword “EGYPT”

“HELLO HOW ARE YOU?”

Solution:

Keyword: EGYPT, Keylength=5

Encrypt: HELLO HOW ARE YOU?

First divide the message into blocks of length 5 i.e.

$$b_1 = \text{HELLO} \quad b_2 = \text{HOWAR} \quad b_3 = \text{EYOU?}$$

Encryption of block $b_1 = \text{HELLO} = x = x_1 x_2 x_3 x_4 x_5 = [7 \ 4 \ 11 \ 11 \ 14]$

$$K = \text{EGYPT} = k_1 k_2 k_3 k_4 k_5 = [4 \ 6 \ 24 \ 15 \ 19]$$

$$\mathcal{E}(x) = y = y_1 y_2 y_3 y_4 y_5 \text{ where } y_i \equiv x_i + k_i \pmod{27}$$

$$y_1 = x_1 + k_1 = 7 + 4 \equiv 11 \pmod{27}$$

$$y_2 = x_2 + k_2 = 4 + 6 \equiv 10 \pmod{27}$$

$$y_3 = x_3 + k_3 = 11 + 24 = 35 \equiv 8 \pmod{27}$$

$$y_4 = x_4 + k_4 = 11 + 15 \equiv 26 \pmod{27}$$

$$y_5 = x_5 + k_5 = 14 + 19 = 33 \equiv 6 \pmod{27}$$

$$\mathcal{E}(x) = y = y_1 y_2 y_3 y_4 y_5 = [11 \ 10 \ 8 \ 26 \ 6] = \text{LKI?G}$$

So encryption of ‘HELLO’ is LKI?G

Similarly for block $b_2 = \text{HOWAR} = [7 \ 14 \ 22 \ 0 \ 17] = x = x_1 x_2 x_3 x_4 x_5$
 $\mathcal{E}(x) = y = y_1 y_2 y_3 y_4 y_5$ where $y_i \equiv x_i + k_i \pmod{27}$

$$\begin{aligned} y_1 &= x_1 + k_1 = 7 + 4 \equiv 11 \pmod{27} \\ y_2 &= x_2 + k_2 = 14 + 6 \equiv 20 \pmod{27} \\ y_3 &= x_3 + k_3 = 22 + 4 = 46 \equiv 19 \pmod{27} \\ y_4 &= x_4 + k_4 = 0 + 15 \equiv 15 \pmod{27} \\ y_5 &= x_5 + k_5 = 17 + 19 = 33 \equiv 9 \pmod{27} \end{aligned}$$

$\mathcal{E}(x) = y = y_1 y_2 y_3 y_4 y_5 = [11 \ 20 \ 19 \ 15 \ 9] = \text{LUTPJ}$
 So encryption of 'HOWAR' is LUTPJ.

Similarly do for block b_3 .

Decryption: $y = \text{LKI?G} = y_1 y_2 y_3 y_4 y_5 = [11 \ 10 \ 8 \ 26 \ 6]$
 $\mathcal{D}(y) = x = x_1 x_2 x_3 x_4 x_5$ such that $x_i \equiv y_i - k_i \pmod{27}$

$$\begin{aligned} x_1 &= y_1 - k_1 = 11 - 4 \equiv 7 \pmod{27} \\ x_2 &= y_2 - k_2 = 10 - 6 \equiv 4 \pmod{27} \\ x_3 &= y_3 - k_3 = 8 - 24 = -16 \equiv 11 \pmod{27} \\ x_4 &= y_4 - k_4 = 26 - 15 \equiv 11 \pmod{27} \\ x_5 &= y_5 - k_5 = 6 - 19 = -13 \equiv 14 \pmod{27} \end{aligned}$$

$\mathcal{D}(y) = x = [7 \ 4 \ 11 \ 11 \ 14] = \text{HELLO}$

So decryption of 'LKI?G' is 'HELLO'.

In the same way we can decrypt any Vigenère ciphers when keyword is known.

Autokey Cipher: Autokey cipher is similar to Vigenère cipher. For encryption, shift few letters with the help of keyword after that shift the remaining plaintext by plaintext itself, that is $\text{key} = \text{keyword} + \text{plaintext}$.

Example: Encryption of "DO NOT WASTE THE TIME"
 by keyword "CAT"

Plain Text: D O N O T W A S T E T H E T I M E
 Key: C A T D O N O T W A S T E T H E T
 Cipher Text: F O G R H J O L P E L A I M P Q X

1.8 Techniques of decrypting Vigenère Cipher

• When key length l is known

Suppose the key length of the chosen keyword is l . Then lines up all ciphertext in l columns. So one can consider that 1st column is shifted by the first symbol of the keyword and 2nd column is shifted by 2nd symbol of the keyword and so on. That is each column can be seen as a monoalphabetic substitution cipher. Therefore, for each column, we can use frequency analysis to guess the plaintext.

$$\begin{array}{rcl} C_1 & = & y_1^1 \quad y_2^1 \quad \dots \quad y_l^1 \\ C_2 & = & y_1^2 \quad y_2^2 \quad \dots \quad y_l^2 \\ C_3 & = & y_1^3 \quad y_2^3 \quad \dots \quad y_l^3 \\ & \vdots & \\ & \vdots & \\ C_m & = & y_1^m \quad y_2^m \quad \dots \quad y_l^m \end{array}$$

Hence we can write

$$(y_1^1, y_1^2, y_1^3, \dots, y_1^m)(y_2^1, y_2^2, y_2^3, \dots, y_2^m) \dots (y_l^1, y_l^2, y_l^3, \dots, y_l^m)$$

We now do the frequency analysis on each subset and get the plaintext.

Example: Decrypt the following ciphertext using Vigenère Cipher for keylength $l = 7$.

**XHUEHXZHRIUHTIHGEUZSWEIATUMEZXHTVYXAAHKJLN
ECSBMIXONQDPVWKEKAIEEMGJAIELIMRVAVKWAPKGIXET
GZEGHWIVPZVWIVGSSGFINVHLVETKVRXBZEUHRZGHIXPH
LTPTJLOEHALGKKVTRDURMCEWTQDSIDAIVOZZKXUG**

Step-1: At first divide ciphertext into blocks of length 7 i.e

XHUEHXZ HRIUHTI HGEUZSW EIATUME ZXHTVYX AAHKJLN
XECSBMI XONQDPV WKEKAIE EMGJAEI LIMRVAV KWAPKGI
XETGZEG HWIVPZV WIVGSSG FINVLHL VETKVRX BZEUHRZ

GHIXPHL TPTJLOE HALGKKV TRDURMC EWTQDSI DAIVZZ
KXUG.

Step-2: Now lines up all ciphertext in 7 column

X	H	U	E	H	X	Z
H	R	I	U	H	T	I
H	G	E	U	Z	S	W
E	I	A	T	U	M	E
Z	X	H	T	V	Y	X
A	A	H	K	J	L	N
X	E	C	S	B	M	I
X	O	N	Q	D	P	V
W	K	E	K	A	I	E
E	M	G	J	A	I	E
L	I	M	R	V	A	V
K	W	A	P	K	G	I
X	E	T	G	Z	E	G
H	W	L	V	P	Z	V
W	I	V	G	S	S	G
F	I	N	V	L	H	L
V	E	T	K	V	R	X
B	Z	E	U	H	R	Z
G	H	I	X	P	H	L
T	P	T	J	L	O	E
H	A	L	G	K	K	V
T	R	D	U	R	M	C
E	W	T	Q	D	S	I
D	A	I	V	O	Z	Z
K	X	U	G			

Step-3: Now each column can be written as

(XHHEZAXXWELKXHW FVBGTHTEDK)
 (HRGIXAEOKMIWEWII EZH PARWAX)
 (UIEAHHCNEGMATIVNTEITLDTIU)
 (EUUTTKSQKJRPGVGVKUXJGUQVG)
 (HHZUVJBDAAVKZPSLVHPLKRDO)
 (XTSMYLMPIIAGEZSHRRHOKMSZ)
 (ZIWEXNIVEEVIGVGLXZLEVCIZ)

Step-4: Now use frequency analysis on each column

On first column - X occurs 4 times which is the highest frequency in first column hence E may be encrypted to X i.e

$$\mathcal{E}(4) = 4 + k_1 = 23 \pmod{26}$$

$$\Rightarrow k_1 = 19 = \mathbf{T}$$

On second column- I occurs 4 times which is the highest frequency in second column hence E may be encrypted to I i.e

$$\mathcal{E}(4) = 4 + k_2 = 8 \pmod{26}$$

$$\Rightarrow k_2 = 4 = \mathbf{E}$$

On third column- T occurs 4 times which is the highest frequency in third column hence E may be encrypted to T i.e

$$\mathcal{E}(4) = 4 + k_3 = 19 \pmod{26}$$

$$\Rightarrow k_3 = 15 = \mathbf{P}$$

(When we decrypt our cipher text taking third letter of keyword as P then plaintext does not make sense (Verify))

Now take second more frequent English alphabet i.e T hence T (second more frequent in English alphabet) may be encrypted to T (most frequent symbol in third column of cipher)

$$\mathcal{E}(19) = 19 + k_3 \pmod{26}$$

$$\implies k_3 = 0 = \mathbf{A}$$

On fourth column- G occurs 4 times which is the highest frequency in fourth column hence E may be encrypted to G i.e

$$\mathcal{E}(4) = 4 + k_4 = 6 \pmod{26}$$

$$\implies k_4 = 2 = \mathbf{C}$$

On fifth column- H and V occur more frequent but if we consider encryption of E either H or V then plaintext does not make sense (Verify). Similarly if we take encryption of T is either H or V then also plain text does not make sense. Now if we consider the 3rd more frequent letter in English alphabet i.e A may be encrypted to H then

$$\mathcal{E}(0) = 0 + k_5 = 7 \pmod{6}$$

$$\implies k_5 = 7 = \mathbf{H}$$

Similarly we can do frequency analysis on 6th and 7th column. After doing this we find that our key word is **TEACHER**.

Now we know keyword hence we can decrypt the cipher text easily and we get plaintext as follows.

“Education is a process of learning through which we acquire knowledge. It enlightens, empowers, and creates a positive development. Education gives an individual the knowledge and skills to work with virtue.”

• When key length is unknown

If the key length of the keyword is not known then first we find the key length by Kasiski Method. Kasiski method utilizes the fact of repetition of words in the ciphertext. In this method, it is considered that repeated substring in plaintext is encrypted by the same substring in the keyword. That's why in the ciphertext, we get repeated substring, and the distance between the occurrence of repetition is multiple of the key lengths. Hence we can calculate key length by taking gcd of the distances between repetitions. After

finding the key length, we apply the above method of frequency analysis for decrypting the ciphertext.

Note: Not every repetition occur in the above way but the probability of not occurring the repetition as above is very small.

A long ciphertext probably has more repetition and additionally, long repeated substrings are not likely to be by chance but short repetition may appear more often.

Example: Find the keylength and keyword for following cipher text

LIBVQSTNEZLQMEDLIVMAMPAKUFAVATLJVDAYYVN
FJQLNPLJVKVTRNFLJVCMLKETALJVHUYJVSFKRF
TTWEFUXVHZNP

Solution: From the cipher text, we can see that

LJVH occurs 2 times
LJV occurs 4 times
JVH occurs 2 times
JV occurs 5 times
LJ occurs 4 times
VH occurs 3 times
LI, NF, NP occurs 2 times

Now the distance (number of symbols between two consecutive repetition) between two 4-graphs LJVH is $20 = 5 \times 2^2$

distance between trigraphs LJV is $15 = 5 \times 3$ and $10 = 5 \times 2$

So keylength = gcd of distances between repetitions

$$\gcd(15, 10, 20) = 5$$

So keylength=5

Now we know the keylength, find keyword by above discussed method.

1.9 Exercise

Note : Let $\mathcal{P}=\{a=0,b=1,...,y=24,z=25\}$ be set of symbols (alphabets).

1. Encrypt the following text using shift cipher:

- (a) This degradation of the environment and infrastructure causes air pollution, further destroys buildings, and ruins the economy of tourism in the region. The Himalayas utilize hydroelectric power, but need to further advantage of it to make it work entirely. The formation of dams not only displaced locals' homes, but also greatly affected the environment for both land-going and aquatic animals. These dams, filled with pollution because of all the waste surrounding it, flow all the way into larger bodies of water all the way to the ocean. Due to climate change in the Himalayas, there is significantly less snow and ice, but more water is running down and being polluted. (Encrypt using $b = 11$ (Ignore 'comma' , 'full stop' , 'apostrophe' , ' space '))
- (b) A problem has arisen in the Punjab area of India among government officials and farmers. Forest fires, as a result of clearing land for crop planting of the already oppressed farmers, have been made illegal in the past, only to be protested and still affecting hundreds of millions. It is especially dangerous in the covid ridden country of India, where air quality matters even more to keep people's immune systems healthy. (Encrypt using $b = 17$)

2. Decrypt the following text using shift cipher using given b:

- (a) Trarennyyl snezref ner abg abg gerngrq pbeerpgyl sbe gur jbex gurl qb va n gbhtu znexrg rferpvnyyl vs pebcf ner abg tebjvat jryy Gur fgerff bs abg xabjvat vs pebcf jvyy tebj vs gur znexrg jvyy punetr rabhtu sbe gur lrnef jbex cnlvat onpx onaxf naq zber nqqf gb guvf ahzore vapernfvat ($b = 13$)
- (b) Bpm lmvamtg xwxctibml izmia wn Lmtpq ivl Cbbiz Xzilmap ivl uivg wbpzmz bweva ivl dqttioma aczzwcvl bpm zqdmz epmzm qb mdmvbcittg ummba eqbp bpm Oivoma Zqdmz Vwb wvtg pil xmwxtm lcuxml xwttecbqvv ivl oizjiom qvbw bpm eibmz jcb aw lql qvlcabzqit jecqlqvoa Lmbmzomvb kpmuqkit eiabm bziap ivl

uwzm kiv jm nwevl qv bpm Giucvi lmaxqbm bpm owdmzvumvba
mnnwzba qv bzgqvo bw abwx qb ($b = 8$)

3. Decrypt the following text using shift cipher:

- (a) ncjaezrc lasj
- (b) aolyl hyl vul obukylk huk mvyaf adv zabkluz pu aopz jvbyzl
- (c) li hqfubswlrq lv grqh eb wkuhh vkliwv wkhq lw lv fdoohg dv fdhvdv flskhu
- (d) Vadqpa lpgbxcv xh p itgb duitc jhts xcitgrwpcvtpqan lxiw raxbpit rwpvct ph xi xh dct du iwt bdhi xbedgipci btpbjgth du vadqpa rwpvcth Vadqpa lpgbxcv gtutgh id iwt gxht xc pktgpvt vadqpa itbetgpjgth lwxrw xh axczts id hxvexuxrpei xbeprih dc wjbpc h l xasaxut pcs trdhnhitbh pgdjcs iwt ldgas
- (e) pusufmx xurq ue mgsyqzfuzs tgy m z ombmoufuge mzp pue d gbfuzs qaze axp tgy m z mofuhufuge oapqLpduhqz ekefye tmhq ebdqmp fa yadq ftmz tmxr ar ftq iadxpe uztmnufmzfe uz mynuqzf uzradymfuaz mzp oazzqofuhufk arrqduzs bdqhuagexk gzu ymsuzqp abbadfgzufuge mzp gzbdqoqpqzfqp ftdqmfe

4. Encrypt the following text using hill cipher:

- (a) WELCOME (Encrypt using $a = 5$)
- (b) ENCRYPTION (Encrypt using $a = 7$)
- (c) DECRYPTION (Encrypt using $a = 9$)

5. Decrypt the following text using hill cipher:

- (a) OPMBDUQPABXM ($a = 7$)
- (b) ZKRRWKJZSF ($a = 11$)
- (c) AIFVMNFYGA FYQN

6. Plain text "DIGITALSIGNATURE" is encoded as "ZGYGLAFUGY-NALCDQ" using hill cipher. Find the value of $a \pmod{26}$.

7. Encrypt the following text using affine cipher, encrypting key (a,b) :

- (a) affinecipher ($a = 5, b = 11$)

- (b) artificialintelligence ($a = 7, b = 6$)
 - (c) bitcoinisthefirstcryptocurrency ($a = 11, b = 5$)
8. Find decrypting key (a', b'), using the given affine cipher encrypting key (a, b). Also decrypt the corresponding text:
 - (a) qlgdopbmdolqzc ($a = 7, b = 12$)
 - (b) nbkdqwr dwabixwyxj ($a = 5, b = 9$)
 - (c) fjsigixcapaehxohoi xpsjsuwfoxaplfahisap ($a = 3, b = 2$)
 9. The text 'SUBSTITUTIONCIPHER' is encrypted as 'TDMTYVY-DYVZURVEQBO'. The encryption function maps message m to $am+b$. The same text is encrypted as 'AKTAFCKFKFCGBYCLXIV' if the encryption function maps message m to $am + 2b$. Find the values of a and b .
 10. Suppose you try to use the affine function $2x + 1$ to encrypt. Find two letters that encrypt to same cipher text letter in the 26 letter English alphabet. What happens when you try to decrypt the cipher text?
 11. Decrypt the given cipher text obtained from shift cipher using frequency analysis:
 YMJUJTUQJTKNSINFJQJHYYMJNWLTAJWSRJSYXG
 DFXDXYJRTKZSNAJWXFQFIZQYKWFSHMNXJUTUZ
 QFWQDPSTBSFXTSJUJWXTSTSJATYJ
 12. Encrypt the following text using Vigenère cipher: 'WISH YOU A MERRY CHRISTMAS AND HAPPY NEW YEAR' using the key "GOAL".
 13. The string 'FREDFEDTEDBREADANDTEDEFEDFREDBREAD' is encrypted with a Vigenere key of unknown length. The resulting cipher text is 'PVCNJCNCNFPOEBKRBDIBPIBPVCNFP OEB'. What is the length of the key and decrypt the message?
 14. A certain message is encrypted with a two letter key. The encrypted message is 'FWKWFHUSNBFKRF'. We also happen to know that the second character of the plain-text is 'I'. Break the cipher and find the plaintext.

(Hint: Since we know that the second letter of the plain-text is I, we can deduce that the second character in the key is O)

15. Following is the ciphertext, obtained from a Vigenere Cipher:
'IMFIKXUTXZBMVSQHTXQGDWRTBKWPXLBIIXQVL IBIVSAJI
QOOEMPTXXBIKABIMMDTBXQVIQOOXLUJIMFIKXUTXZBMV
SQHTXQGDWRTBKWPXLBIIXQVLETIKMUWMPQTXKWSYXU
GDTQHIMBTXZETXBQVIQBIKXUGDMP'.

Determine the key. Decrypt the message if key starts with T and ends with E.

16. Decrypt the message: 'uege mnvve pgj e ckun. le pgj zeke klikykc. hx lciw akii agj klekk zr sxgigh hl netxx. syt ak tsuej est yulrd pgkir bt rrypnvve. tz cesm nv vetiyid t mrvdxt. yi stc r tom oe xhx crxek. zyi ckun aal bvvy aggty. mnvve pgj e vxyp pimzci wtzvv ig ok. le vulpd guk hrbtb mt. ak klonmyx a irrr. hx hisuznk wofk gmevkj sf lzfrel geh pnz klef oexo mnv tom. zyi wtzvv rhyvyp. ak uvagq netxx rrd yrva apgp.'.

R.K.Sharma

Chapter 2

Matrix Ciphers

We have studied many encryption methods in previous lectures like Substitution/shift cipher, Hill cipher and affine ciphers. In this lecture we will discuss some of previous discussed encryption methods with the help of matrices with usual matrix addition and multiplication.

2.1 Hill Cipher

Let $\mathcal{A}$ be the set of symbols (like English alphabets A,B ...Z, ? , !, _ (blank space) and $\mathcal{P}$ be plain text.

Encryption: First divide the plain text into n -length message units. Choose a $n \times n$ matrix A , then encryption function $\mathcal{E}$ is defined by

$$\begin{aligned}\mathcal{E}: \mathcal{P} &\rightarrow \mathcal{A}^n \\ X &\mapsto C = \mathcal{E}(X) = AX\end{aligned}$$

where $X = [x_1, x_2, \dots, x_n]^T$ is message unit.

Decryption: Suppose $C = \mathcal{E}(X) = AX$, then decryption function $\mathcal{D}$ is defined by

$$\begin{aligned}\mathcal{D}: \mathcal{A}^n &\rightarrow \mathcal{P} \\ C &\mapsto X = \mathcal{D}(C) = A^{-1}C\end{aligned}$$

provided A^{-1} exists.

Example: Let $\mathcal{A}=\{A=0,B=1,C=2,\dots,Y=24,Z=25\}$ be set of symbols(English alphabets).

$|\mathcal{A}| = 26$

Plain text: “NO I DO NOT KNOW”

First We divide plain text into 2-length ($n=2$) message units
i.e NO ID ON OT KN OW

Encryption: $X_1=NO, X_2=ID, X_3=ON, X_4=OT, X_5=KN, X_6=OW$

Now encryption of $X_1=NO=[13 \ 14]^T = \begin{bmatrix} 13 \\ 14 \end{bmatrix}$

Let $A=\begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix}$, Fixed matrix for encryption in $\mathbb{Z}_{26} = |\mathcal{A}|$

Now encryption $\mathcal{E}(X_1) = AX_1$

$$\begin{aligned} \mathcal{E}(NO) &= \begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 13 \\ 14 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 16 \\ 21 \end{bmatrix} \pmod{26} = \begin{bmatrix} Q \\ V \end{bmatrix} \end{aligned}$$

The Encryption of 'NO' is 'QV'.

Decryption: For $C_1 = \begin{bmatrix} Q \\ V \end{bmatrix} = \begin{bmatrix} 16 \\ 21 \end{bmatrix}$

$\mathcal{D}(C_1) = A^{-1}C_1$, So first we calculate A^{-1} (if exists)

$$A = \begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix}$$

$\det(A) \equiv 16 - 21 \equiv -5 \equiv 21 \pmod{26} \not\equiv 0 \pmod{26}$

So, $A^{-1} \pmod{26}$ exists.

$$A^{-1} = \frac{1}{\det(A)} \text{cofactor}(A)$$

$$\begin{aligned} A^{-1} &= -\frac{1}{5} \begin{bmatrix} 8 & -3 \\ -7 & 2 \end{bmatrix} \pmod{26} \\ &= -21 \begin{bmatrix} 8 & -3 \\ -7 & 2 \end{bmatrix} \pmod{26} \quad (\text{since } 5^{-1} = 21 \pmod{26}) \\ &= 5 \begin{bmatrix} 8 & -3 \\ -7 & 2 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix} \pmod{26} \end{aligned}$$

Now

$$\begin{aligned} \mathcal{D}(C_1) &= \mathcal{D} \left(\begin{bmatrix} Q \\ V \end{bmatrix} \right) = \mathcal{D} \left(\begin{bmatrix} 16 \\ 21 \end{bmatrix} \right) = A^{-1}C_1 \\ &= \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix} \begin{bmatrix} 16 \\ 21 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 224 + 231 \\ 272 + 210 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 455 \\ 482 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} 13 \\ 14 \end{bmatrix} \pmod{26} \\ &= \begin{bmatrix} N \\ O \end{bmatrix} \end{aligned}$$

So decryption of 'QV' is 'NO'.

In the same way we can do encryption and decryption for $X_2=\text{ID}$, $X_3=\text{ON}$, $X_4=\text{OT}$, $X_5=\text{KN}$, $X_6=\text{OW}$.

2.2 Affine Cipher

Let $\mathcal{A}$ be the set of symbols (like English Alphabtes, ? , !, _) and $\mathcal{P}$ be plain text.

Encryption: First divide plain text into n-length message units. Let A be $n \times n$ matrix and B be $n \times 1$ vector. Encryption function $\mathcal{E}$, is defined by

$$\begin{aligned}\mathcal{E}: \mathcal{P} &\rightarrow \mathcal{A}^n \\ X &\mapsto C = \mathcal{E}(X) = AX + B\end{aligned}$$

where $X = [x_1, x_2, \dots, x_n]^T$ is message unit.

Decryption: Suppose $C = \mathcal{E}(X) = AX + B$, Decryption function $\mathcal{D}$ is defined by

$$\begin{aligned}\mathcal{D}: \mathcal{A}^n &\rightarrow \mathcal{P} \\ C &\mapsto X = \mathcal{D}(C) = A^{-1}(C - B)\end{aligned}$$

i.e $\mathcal{D}(C) = A'C + B'$ where $A' = A^{-1}$ and $B' = -A^{-1}B$, provided A^{-1} exists.

Example: Let $\mathcal{A} = \{A = 0, B = 1, C = 2, \dots, Y = 24, Z = 25\}$ be set of alphabets.

$|\mathcal{A}| = 26$

Plain text $\mathcal{P}$: NOANSWER

We first divide plain text into 2-length message unit (n=2) i.e $X_1=NO$, $X_2=AN$, $X_3=SW$, $X_4=ER$

Let $A = \begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix}$, Fixed matrix for encryption in $\mathbb{Z}_{26} = |\mathcal{A}|$, $B = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

Encryption: $X_1=NO = [13 \ 14]^T = \begin{bmatrix} 13 \\ 14 \end{bmatrix}$

$$\begin{aligned}
\mathcal{E}(X_1) &= C_1 = AX_1 + B \\
&= \begin{bmatrix} 2 & 3 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 13 \\ 14 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \end{bmatrix} \pmod{26} \\
&= \begin{bmatrix} 16 \\ 21 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \end{bmatrix} \pmod{26} \\
&= \begin{bmatrix} 20 \\ 23 \end{bmatrix} \pmod{26} \\
&= \begin{bmatrix} U \\ X \end{bmatrix}
\end{aligned}$$

So encryption of 'NO' is 'UX'.

Decryption: $C_1=UX$

$$\mathcal{D}(C_1) = A'C_1 + B' = A^{-1}C_1 - A^{-1}B = A^{-1}(C_1 - B)$$

From previous example we have

$$\begin{aligned}
A^{-1} &= \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix} \pmod{26} \\
\mathcal{D}(C_1) &= \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix} \left(\begin{bmatrix} 20 \\ 23 \end{bmatrix} - \begin{bmatrix} 4 \\ 2 \end{bmatrix} \right) \pmod{26} \\
&= \begin{bmatrix} 14 & 11 \\ 17 & 10 \end{bmatrix} \begin{bmatrix} 16 \\ 21 \end{bmatrix} \pmod{26} \\
&= \begin{bmatrix} 13 \\ 14 \end{bmatrix} \pmod{26} \\
&= \begin{bmatrix} N \\ O \end{bmatrix}
\end{aligned}$$

So decryption of 'UX' is 'NO'.

In the the same way we can do encryption and decryption for $X_2=AN$, $X_3=SW$, $X_4=ER$.

Question: Decrypt the message

$$WKNCMSLRTQ$$

obtained by digraph encryption on English alphabets (A=0 to 25=Z) with ". "(dot=26), "," (comma=27), "?" (question mark=28) (29 letters) using Hill Cipher. Message started with 'GIVE'.

Solution: Digraph encryption means $n=2$, $|\mathcal{A}| = 29$

We know in Hill cipher encryption function is given by

$$\mathcal{E}(X) = AX$$

where A is 2×2 matrix in $\mathbb{Z}_{29}$ and X is 2-length message unit.

Decryption function is given by $\mathcal{D}(C) = A^{-1}C$

So, to decrypt the given cipher we have to calculate A^{-1} in $\mathbb{Z}_{29}$ i.e we have to find matrix A in $\mathbb{Z}_{29}$.

Let matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Message started with 'GIVE' means $\mathcal{E}(GI) = WK$ and $\mathcal{E}(VE) = NC$

Let $X_1=GI$, $C_1=WK$, $X_2=VE$, $C_2=NC$

by encryption function we have $\mathcal{E}(X_1) = AX_1 = C_1$, $\mathcal{E}(X_2) = AX_2 = C_2$

i.e $AX_1 = C_1$, $AX_2 = C_2$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 22 \\ 10 \end{bmatrix} \text{ and } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 21 \\ 4 \end{bmatrix} = \begin{bmatrix} 13 \\ 2 \end{bmatrix}$$

from above, we have system of 4 linear equations in a, b, c, d

$$6a + 8b = 22$$

$$6c + 8d = 10$$

$$21a + 4b = 13$$

$$21c + 4d = 2$$

Solve the above system of equation for a, b, c, d

We have

$$\begin{aligned} a &= \frac{1}{9} = 1 \times 9^{-1} = 1 \times 13 = 13 \pmod{29}, \\ b &= \frac{8}{3} = 8 \times 3^{-1} = 8 \times 10 = 22 \pmod{29}, \\ c &= \frac{-1}{6} = -1 \times 6^{-1} = -1 \times 5 = 24 \pmod{29}, \\ d &= \frac{11}{8} = 11 \times 8^{-1} = 11 \times 11 = 5 \pmod{29} \end{aligned}$$

So we have matrix $A = \begin{bmatrix} 13 & 22 \\ 24 & 5 \end{bmatrix}$,

$$\det(A) = 13 \times 5 - 22 \times 24 = 65 - 528 = -463 = 1 \pmod{29}$$

$$\text{and } A^{-1} = \begin{bmatrix} 5 & -22 \\ -24 & 13 \end{bmatrix} \pmod{29} = \begin{bmatrix} 5 & 7 \\ 5 & 13 \end{bmatrix} \pmod{29}$$

$$C_3 = MS = \begin{bmatrix} 12 \\ 18 \end{bmatrix}$$

$$C_4 = LR = \begin{bmatrix} 11 \\ 17 \end{bmatrix}$$

$$C_5 = TQ = \begin{bmatrix} 19 \\ 16 \end{bmatrix}$$

Now

$$X_3 = A^{-1}C_3 = \begin{bmatrix} 5 & 7 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} 12 \\ 18 \end{bmatrix} = \begin{bmatrix} 12 \\ 4 \end{bmatrix} \pmod{29} = \begin{bmatrix} M \\ E \end{bmatrix}$$

$$X_4 = A^{-1}C_4 = \begin{bmatrix} 5 & 7 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} 11 \\ 17 \end{bmatrix} = \begin{bmatrix} 0 \\ 15 \end{bmatrix} \pmod{29} = \begin{bmatrix} A \\ P \end{bmatrix}$$

$$X_5 = A^{-1}C_5 = \begin{bmatrix} 5 & 7 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} 19 \\ 16 \end{bmatrix} = \begin{bmatrix} 4 \\ 13 \end{bmatrix} \pmod{29} = \begin{bmatrix} E \\ N \end{bmatrix}$$

So decrypted message is "GIVEMEAPEN" i.e. "GIVE ME A PEN"

2.3 Vigenère Cipher

Recall Vigenère Cipher: Suppose keyword

$$K = k_1 k_2 k_3 \dots k_l, \quad k_i \in \mathbb{Z}_n$$

message block is therefore

$$x = x_1 x_2 \dots x_l, \quad x_i \in \mathbb{Z}_n$$

then encryption function is given by

$$\mathcal{E}(x) = y = y_1 y_2 \dots y_l \quad \text{where} \quad y_i \equiv x_i + k_i \pmod{n}$$

decryption function is given by

$$\mathcal{D}(y) = x = x_1 x_2 \dots x_l \quad \text{such that} \quad x_i \equiv y_i - k_i \pmod{n}$$

where $n = |\mathcal{A}|$ =number of symbols written in their numerical equivalent.

We can write keyword K , message block x in matrix of order $l \times 1$ where l =keylength

$$K = \begin{bmatrix} k_1 \\ k_2 \\ \cdot \\ \cdot \\ \cdot \\ k_l \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_l \end{bmatrix}$$

then encryption function is defined by

$$\mathcal{E}: M_{l \times 1} \rightarrow M_{l \times 1}$$

$$\mathcal{E}(x) = y = \mathcal{E} \left(\begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_l \end{bmatrix} \right) = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_l \end{bmatrix} + \begin{bmatrix} k_1 \\ k_2 \\ \cdot \\ \cdot \\ \cdot \\ k_l \end{bmatrix} = \begin{bmatrix} x_1 + k_1 \\ x_2 + k_2 \\ \cdot \\ \cdot \\ \cdot \\ x_l + k_l \end{bmatrix} \pmod{n} = \begin{bmatrix} y_1 \\ y_2 \\ \cdot \\ \cdot \\ \cdot \\ y_l \end{bmatrix}$$

and decryption function is defined by

$$\mathcal{D}: M_{l \times 1} \rightarrow M_{l \times 1}$$

$$\mathcal{D}(y) = x = \mathcal{D} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_l \end{pmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_l \end{bmatrix} - \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_l \end{bmatrix} = \begin{bmatrix} y_1 - k_1 \\ y_2 - k_2 \\ \vdots \\ y_l - k_l \end{bmatrix} \pmod{n} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_l \end{bmatrix}$$

Example: Let $\mathcal{A} = \{A=0, B=1, \dots, Y=24, Z=25, ?=26\}$ i.e. $|\mathcal{A}| = 27$
 Encrypt the following plain text with keyword “EGYPT”

“HELLO HOW ARE YOU?”

Solution: We have already done this example, here we do encryption and decryption by matrices

Keyword: EGYPT, Keylength=5

Encrypt: HELLO HOW ARE YOU?

First divide the message into blocks of length 5 i.e.

$b_1 = \text{HELLO}$ $b_2 = \text{HOWAR}$ $b_3 = \text{EYOU?}$

$$\text{Encryption of block } b_1 = \begin{bmatrix} H \\ E \\ L \\ L \\ O \end{bmatrix} = x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ 11 \\ 11 \\ 14 \end{bmatrix}$$

$$K = \text{EGYPT} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \\ k_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 24 \\ 15 \\ 19 \end{bmatrix}$$

$$\mathcal{E}: M_{5 \times 1} \rightarrow M_{5 \times 1}$$

$$\mathcal{E}(x) = y = \mathcal{E} \left(\begin{bmatrix} 7 \\ 4 \\ 11 \\ 11 \\ 14 \end{bmatrix} \right) = \begin{bmatrix} 7 \\ 4 \\ 11 \\ 11 \\ 14 \end{bmatrix} + \begin{bmatrix} 4 \\ 6 \\ 24 \\ 15 \\ 19 \end{bmatrix} = \begin{bmatrix} 7+4 \\ 4+6 \\ 11+24 \\ 11+15 \\ 14+19 \end{bmatrix} \pmod{27} = \begin{bmatrix} 11 \\ 10 \\ 8 \\ 26 \\ 6 \end{bmatrix} \pmod{27} = \begin{bmatrix} L \\ K \\ I \\ ? \\ G \end{bmatrix}$$

So encryption of 'HELLO' is 'LKI?G'

$$\text{Decryption: } y = \begin{bmatrix} L \\ K \\ I \\ ? \\ G \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 11 \\ 10 \\ 8 \\ 26 \\ 6 \end{bmatrix}$$

$$\mathcal{D}: M_{5 \times 1} \rightarrow M_{5 \times 1}$$

$$\mathcal{D}(y) = x = \mathcal{D} \left(\begin{bmatrix} 11 \\ 10 \\ 8 \\ 26 \\ 6 \end{bmatrix} \right) = \begin{bmatrix} 11 \\ 10 \\ 8 \\ 26 \\ 6 \end{bmatrix} - \begin{bmatrix} 4 \\ 6 \\ 24 \\ 15 \\ 19 \end{bmatrix} = \begin{bmatrix} 11-4 \\ 10-6 \\ 8-24 \\ 26-15 \\ 6-19 \end{bmatrix} \pmod{27} = \begin{bmatrix} 7 \\ 4 \\ 11 \\ 11 \\ 14 \end{bmatrix} \pmod{27} = \begin{bmatrix} H \\ E \\ L \\ L \\ O \end{bmatrix}$$

So decryption of 'LKI?G' is 'HELLO'

In the same way do encryption and decryption for other blocks.

2.4 Exercise

Note : Let $\mathcal{S}$ be set of symbols.

1. Let $\mathcal{S} = \{A = 0, B = 1, \dots, Y = 24, Z = 25\}$ and "HILLCIPHER" be the plain text. Encrypt the plain text using the matrix Hill cipher if matrix A is given by:

(a) $\begin{bmatrix} 4 & 3 \\ 3 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 5 & 2 \\ 4 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 4 & 7 & 9 \\ 5 & 4 & 5 \\ 3 & 7 & 1 \end{bmatrix}$

2. Let $\mathcal{S} = \{A = 0, B = 1, \dots, Y = 24, Z = 25, ? = 26, \}$. Decrypt the cipher text "RENKOOYCWGOQDUM" using the matrix Hill cipher if matrix A is given by

$$A = \begin{bmatrix} 17 & 5 & 11 \\ 25 & 21 & 21 \\ 17 & 0 & 10 \end{bmatrix}$$

3. Let $\mathcal{S} = \{A = 0, B = 1, \dots, Y = 24, Z = 25, ! = 26, \}$. The decryption of the plain text "MATRIXHILLCIPHER" is "TJTKUXVULZFIBEEE" using the matrix Hill cipher if matrix A is given by

$$A = \begin{bmatrix} a & 3 & 7 \\ 0 & b & 9 \\ 0 & 0 & 1 \end{bmatrix} \text{ Find the value of } a \text{ and } b \text{ in } \textit{mod } 27.$$

4. Encrypt the message "MATRIXAFFINECIPHER" using the matrix given in question 1(c) and assuming matrix B as:

$$B = \begin{pmatrix} 12 \\ 15 \\ 11 \end{pmatrix}$$

5. Let $S = \{0, 1, 2, \dots, p-1\}$ in modulo p

Consider the matrix affine cipher:

$$e_K(m) \equiv K_1 m + K_2 \pmod{p} \text{ and } d_K(c) \equiv (K_1)^{-1}(c - K_2) \pmod{p}$$

where m , c , and K_2 are column vectors of dimension n , and K_1 is an $n \times n$ matrix.

- (a) We use the matrix affine cipher with $p = 7$ and the key

$$K_1 = \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$$

and

$$K_2 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

- i. Encrypt the message $m = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$
 - ii. What is the matrix K^{-1} used for decryption?
 - iii. Decrypt the message $c = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$
- (b) The following plaintext/ciphertext pairs were generated using a matrix affine cipher with the prime $p = 11$. Find the keys K_1 and K_2 . $m_1 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$ $c_1 = \begin{pmatrix} 1 \\ 8 \end{pmatrix}$; $m_2 = \begin{pmatrix} 8 \\ 10 \end{pmatrix}$ $c_2 = \begin{pmatrix} 8 \\ 5 \end{pmatrix}$; $m_3 = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$
 $c_3 = \begin{pmatrix} 8 \\ 7 \end{pmatrix}$
6. Let $\mathcal{S} = \{A = 0, B = 1, \dots, Y = 24, Z = 25, \text{ , } = 26, . = 27, ? = 28\}$. Decrypt the cipher text ".IVXVCXQDX,PBSASPF" using the matrix affine cipher if matrix A and B is given by

$$A = \begin{bmatrix} 9 & 3 & 5 \\ 1 & 8 & 9 \\ 1 & 0 & 6 \end{bmatrix} \text{ and } B = \begin{bmatrix} 10 \\ 4 \\ 24 \end{bmatrix}$$

7. Let $\mathcal{S} = \{A = 0, B = 1, \dots, Y = 24, Z = 25, \text{ , } = 26, . = 27, ? = 28\}$. The encryption of the plain text "QUANTUM COMPUTING" is "DATVXMUX.DKV?QXMP" using the matrix affine cipher if matrix A and B is given by

$$A = \begin{bmatrix} a & 5 & 9 \\ 0 & 2a & 3 \\ 0 & 0 & a \end{bmatrix} \text{ and } B = \begin{bmatrix} b \\ b+1 \\ b+6 \end{bmatrix}$$

Find the value of a and b in *mod* 29.

Chapter 3

Transposition Ciphers & Some More classical Ciphers

In transposition cipher, symbols of the cipher text remain the same as plain-text, but the ordering of these symbols are changed in a specific form. Transposition Ciphers are not very secure as the cipher is the rearrangement of plain text. With modern computers, one can check all possible rearrangements of cipher text and get the plain text. But before the computer era, transposition ciphers were used and considered to be a secure cipher.

3.1 Columnar Transposition Cipher

3.1.1 Simple Columnar Transposition Cipher

In a simple columnar transposition cipher, sender (Alice) and receiver (Bob) fix a number k . Alice will write the plain text in k columns to encrypt the message. Encrypted cipher text is obtained by writing column array from left to right, i.e., first, write symbols of the first column, then write the symbols of the second column and so on.

Example: To encrypt “MATHEMATICS IS A LANGUAGE”. Suppose fix number is $k = 4$ (known to both sender and receiver).

Now write the plain text into 4 columns as follows:

Column-1	Column-2	Column-3	Column-4
M	A	T	H
E	M	A	T
I	C	S	I
S	A	L	A
N	G	U	A
G	E		

The encryption of plain text is

“[Column-1][Column-2][Column-3][Column-4]”

i.e.

Cipher Text: “MEISNGAMCAGETASLUHTIAA”

For decryption, Count the numbers of symbols in cipher text say n . The receiver (Bob) knows the number of columns, which is k . Now Bob will perform the following calculations to know the number of symbols in each column.

By division algorithm $n = k \times q + r$, where q is quotient and r is remainder on dividing number of symbols in cipher text n by number of columns k .

If $r = 0$ then each column have q symbols.

If $r \neq 0$ then first r columns have $q + 1$ symbols and other columns (i.e. last $(k - r)$ columns) have q symbols. Plain text is obtained by writing symbols row-wise.

Example: Decrypt the following transposition cipher when the number of columns is 4.

Cipher Text: “MEISNGAMCAGETASLUHTIAA”

Solution:

Number of symbols in cipher text = 22

Number of columns = 4

By division algorithm $22 = 4 \times 5 + 2$, i.e $q = 5$ and $r = 2$

This implies first two columns have $q + 1 = 6$ symbols and last 2 columns have $q = 5$ symbols. So divide the cipher text into blocks of length 6, 6, 5

and 5, each block corresponds to column array.
So we have column array :

Column-1	Column-2	Column-3	Column-4
M	A	T	H
E	M	A	T
I	C	S	I
S	A	L	A
N	G	U	A
G	E		

Now plain text is “[row-1][row-2][row-3][row-4][row-5][row-6]” i.e.

“MATHEMATICSISALANGUAGE”

i.e.

“MATHEMATICS IS A LANGUAGE”

3.1.2 Keyword Columnar Transposition Cipher

In keyword columnar transposition ciphers, the sender (Alice) and receiver (Bob) choose a common keyword. To encrypt the plain text, sender will write the plain text into l columns (l = number of symbols in keyword). Cipher text is obtained by writing column array in alphabetical order of the keyword.

Example: Encrypt the following plain text using keyword columnar transposition cipher with keyword “WORLD”

“MATHEMATICS IS A LANGUAGE”

Solution: Keyword is WORLD that has 5 symbols, so write the plaintext in five columns namely

Column-W	Column-O	Column-R	Column-L	Column-D
M	A	T	H	E
M	A	T	I	C
S	I	S	A	L
A	N	G	U	A
G	E			

alphabetical order of keyword is 'DLORW', thus ciphertext is

“[Column-D][Column-L][Column-O][Column-R][Column-W]”

i.e.

“ECLAHIAUAAINETTSGMMSAG”

Decryption: Since receiver knows the keyword, suppose keyword has k symbols (i.e. $\text{keylength}=k$). Perform the following calculations to know the number of symbols in each column according to keyword.

By division algorithm $n = k \times q + r$, where q is quotient and r is remainder on dividing number of symbols in cipher text n by number of columns k .

If $r = 0$ then each column have q symbols.

If $r \neq 0$ then first r columns of keyword have $q+1$ symbols and other columns of keyword (i.e. last $(k-r)$ columns) have q symbols. Now arrange the k columns in alphabetical order of keyword. Plain text is obtained by writing symbols row-wise.

Example: Decrypt the following keyword columnar transposition cipher using keyword 'WORLD'

“ECLAHIAUAAINETTSGMMSAG”

Solution: No. of symbols in ciphertext =22

keylength is 5, so no. of columns is 5 By division algorithm $22 = 5 \times 4 + 2$, i.e $q = 4$ and $r = 2$

this implies first two columns, column-W and column-O have $q + 1 = 5$ symbols and last 3 columns column-R, column-L and column-D have $q = 4$ symbols. Now write columns in alphabetical order of keyword i.e.

Column-D Column-L Column-O Column-R Column-W

Now put first 4 cipher symbols in column-D,next 4 cipher symbols in column-L,next 5 cipher symbols in column-O, next 4 cipher symbols in column-R, next 5 cipher symbols in column-W i.e.

Column-D	Column-L	Column-O	Column-R	Column-W
E	H	A	T	M
C	I	A	T	M
L	A	I	S	S
A	U	N	G	A
		E		G

Now rearrange above array in keyword order i.e ‘WORLD’

Column-W	Column-O	Column-R	Column-L	Column-D
M	A	T	H	E
M	A	T	I	C
S	I	S	A	L
A	N	G	U	A
G	E			

Plain text is obtained by writing symbols row-wise from above array

“MATHEMATICSISALANGUAGE”

i.e

“MATHEMATICS IS A LANGUAGE”

3.2 Double Transposition

In double transposition, we apply columnar transposition twice. First apply columnar transposition on plain text. Again apply columnar transposition on obtained cipher text.

Example: Encrypt the following plain text using double transposition with fix number (number of columns) is equal to 4

“MATHEMATICS IS A LANGUAGE”

Solution: First, apply columnar transposition on given plain text we obtained cipher text (from above example done in columnar transposition)

Cipher Text: “MEISNGAMCAGETASLUHTIAA”

again apply columnar transposition on the above cipher text

Column-1	Column-2	Column-3	Column-4
M	E	I	S
N	G	A	M
C	A	G	E
T	A	S	L
U	H	T	I
A	A		

The encryption of plain text is

“[Column-1][Column-2][Column-3][Column-4]”

i.e.

Cipher Text: “MNCTUAEGAAHAIA GSTSMELI”

So encryption of “MATHEMATICS IS A LANGUAGE” is “MNCTUAEGAAHAIA GSTSMELI” by double transposition.

Decryption: To decrypt double transposition cipher apply decryption of columnar transposition twice.

Example: Decrypt the following double transposition cipher

Cipher Text: “MNCTUAEGAAHAIA GSTSMELI”

Given, number of columns=4

Solution: Number of symbols in cipher text =22

number of columns=4

By division algorithm $22 = 4 \times 5 + 2$, i.e $q = 5$ and $r = 2$

this implies first two columns have $q + 1 = 6$ symbols and last 2 columns have $q = 5$ symbols. So divide the cipher text into blocks of length 6,6,5 and 5, each block corresponds to columns of transposition cipher.

So we have columns :

Column-1	Column-2	Column-3	Column-4
M	E	I	S
N	G	A	M
C	A	G	E
T	A	S	L
U	H	T	I
A	A		

After applying columnar transposition decryption one time we get

“MEISNGAMCAGETASLUHTIAA”

Again apply columnar transposition decryption on

“MEISNGAMCAGETASLUHTIAA”

(Similar to example of columnar transposition)

Number of symbols =22

number of columns=4

By division algorithm $22 = 4 \times 5 + 2$, i.e $q = 5$ and $r = 2$

this implies first two columns have $q + 1 = 6$ symbols and last 2 columns have $q = 5$ symbols. So divide the cipher text into blocks of length 6,6,5 and 5, each block corresponds to columns of transposition cipher.

So we have columns :

Column-1	Column-2	Column-3	Column-4
M	A	T	H
E	M	A	T
I	C	S	I
S	A	L	A
N	G	U	A
G	E		

Now, plain text is obtained by writing symbols row-wise

“MATHEMATICSISALANGUAGE”

i.e.

“MATHEMATICS IS A LANGUAGE”

Similarly, We can do double transposition with two different number of columns or keywords for two steps of columnar transposition.

3.3 ADFGX Cipher

ADFGX ciphers were used by the German military in March 1918 for the first time. This cipher is the combination of substitution cipher and transposition cipher. This cipher is named ADFGX because in the ciphertext only five symbols (A, D, F, G, X) can appear. It involves two steps.

In the first step, it uses the substitution cipher to encrypt the plaintext. To apply the substitution cipher, users (Sender and Receiver) agree upon a 5×5 array of the English alphabet. In the 5×5 array, there are 25 places at which 26 symbols are placed by considering I and J at the same place. The rows and columns of this ADFGX array are each labeled with the symbols A, D, F, G, and X in order as follows.

	A	D	F	G	X
A					
D					
F					
G					
X					

Now each symbol of plaintext is mapped to a pair of corresponding row and column symbol label. Therefore we get twice no. of the symbols in the ciphertext as compare to plaintext and it contains only five symbols A, D, F, G, and X.

In the 2nd step, it performs the transposition cipher's technique i.e keyword columnar transposition cipher on the obtained ciphertext in the first

step.

Example: Consider the following 5×5 array of symbols

	A	D	F	G	X
A	T	F	I/J	K	E
D	H	N	C	S	Z
F	D	A	X	M	G
G	O	V	R	U	Y
X	L	B	Q	W	P

Take keyword 'WATER' and encrypt the following message using ADFGX cipher

“PEACOCK IS A NATIONAL BIRD OF INDIA”

Solution:

Encryption: To encrypt “PEACOCK IS A NATIONAL BIRD OF INDIA” we map each letter of plain text to a pair of corresponding row and column symbol label in 5×5 array of symbols i.e to encrypt P we write XX , For E we write AX and so on. And we get a first cipher text as follows.

Plain text	P	E	A	C	O	C	K	I	S	A	N
Cipher text	XX	AX	FD	DF	GA	DF	AG	AF	DG	FD	DD
Plain text	A	T	I	O	N	A	L	B	I	R	D
Cipher text	FD	AA	AF	GA	DD	FD	XA	XD	AF	GF	FA
Plain text	O	F	I	N	D	I	A				
Cipher text	GA	AD	AF	DD	FA	AF	FD				

Hence cipher text obtained from the first step of ADFGX cipher is

**XXAXFDDFGADFAGAFDGFDDDDFDAAAFGADDDFDXAXDAFG
FFAGAADAFFDDFAAFFD**

Now we apply key columnar transposition cipher using key word 'WATER' on the cipher text obtained from the first step to get final ciphertext.

Column array of key columnar transposition cipher:

Column-W	Column-A	Column-T	Column-E	Column-R
X	X	A	X	F
D	D	F	G	A
D	F	A	G	A
F	D	G	F	D
D	D	F	D	A
A	A	F	G	A
D	D	F	D	X
A	X	D	A	F
G	F	F	A	G
A	A	D	A	F
D	D	F	A	A
F	F	D		

Therefore, final cipher text is obtained by wrtining column arrays in alphabetical order of keyword i.e.

**XDFDDADXFADFXG GFDGD AAAFAADAAXFGFAAFAGFF
FDFDFDXDDFDADAGADF**

Decryption: To decrypt the cipher text we go in the reverse order of encryption i.e. first, we apply decryption of keyword columnar transposition cipher follwed by decryption of substitution cipher.

When we apply decrytpion of keyword columnar transposition cipher, we get

**XXAXFDDFGADFAGAFDGFDDDDFDAAAFGADDDFDXAXDAFG
FFAGAADAFFDDFAAFFD**

Now, decrypt the above text with the help of 5×5 array to get the plain text

To do so, divide the text in blocks of length 2. In each block, first symbol corresponds to row label and second symbol corresponds to column label. After dividing text in blocks of length 2 we have

**XX AX FD DF GA DF AG AF DG FD DD FD AA AF GA DD
FD XA XD AF GF FA GA AD AF DD FA AF FD**

And 5×5 array is

	A	D	F	G	X
A	T	F	I/J	K	E
D	H	N	C	S	Z
F	D	A	X	M	G
G	O	V	R	U	Y
X	L	B	Q	W	P

So,

XX corresponds to P (row-X, column-X)

AX corresponds to E (row-A, column-X)

FD corresponds to A (row-F, column-D)

and so on.

Finally, we get plain text

“PEACOCK IS A NATIONAL BIRD OF INDIA”

3.4 ADFGVX Cipher:

In June 1918, Germans built this cipher from ADFGX, by increasing the size of the array to 6×6 . They added the digit 0-9 in the plaintext. Therefore they had to increase the size of the array from 5×5 to 6×6 . This array takes the values 0-9 and the English alphabet, by considering I and J are placed at a different position in the array. Now to label this array there was a requirement of an extra letter and this additional letter was ‘V’.

Therefore the rows and columns of this ADFGVX array are each labeled with the symbols A, D, F, G, V and X in order as follows.

	A	D	F	G	V	X
A						
D						
F						
G						
V						
X						

Encryption and Decryption technique is similar to ADFGX cipher.

Example: Using ADFGVX cipher encrypt the plaintext “**EARTH IS ESTIMATED TO BE 4540 MILLION YEARS OLD**”. Keyword is “**STAR**”

	A	D	F	G	V	X
A	1	9	A	J	M	R
D	C	B	I	2	Q	Y
F	E	0	Z	3	D	V
G	7	8	O	F	G	S
V	6	H	4	X	T	W
X	P	5	N	K	L	U

Solution: We map each letter of plain text to a pair of corresponding row and column symbol label in 6×6 array of symbols i.e to encrypt E we write FA , For A we write AF and so on. And we get a first cipher text as follows.

Plaint text	E	A	R	T	H	I	S	E	S	T	I
Cipher text	FA	AF	AX	VV	VD	DF	GX	FA	GX	VV	DF
Plaint text	M	A	T	E	D	T	O	B	E	4	5
Cipher text	AV	AF	VV	FA	FV	VV	GF	DD	FA	VF	XD
Plain text	4	0	M	I	L	L	I	O	N	Y	E
Cipher text	VF	FD	AV	DF	XV	XV	DF	GF	XF	DX	FA
Plain text	A	R	S	O	L	D					
Cipher text	AF	AX	GX	GF	XV	FV					

Encrypted cipher text

**“FAAFAXVVVDDFGXFAGXVVDFAVAFVVFAFVVVGFDDDFA
VFXDVFDDAVDFXVXVDFGFXFDXFAAFAXGXGFXVVFV”**

Now we apply key columnar transposition cipher using key word ‘STAR’ on the ciphet text obtained from the first step to get final ciphertext.

Column-S	Column-T	Column-A	Column-R
F	A	A	F
A	X	V	V
V	D	D	F
G	X	F	A
G	X	V	V
D	F	A	V
A	F	V	V
F	A	F	A
V	V	G	F
D	D	F	A
V	F	X	D
V	F	F	D
A	V	D	F
X	V	X	V
D	F	G	F
X	F	D	X
F	A	A	F
A	X	G	X
G	F	X	V
F	V		

Final cipher text is :

“AVDFVAVFGFXFDXGDAGXFVFAVVVAFADDFVFXFXVF
AVGGDAFVDVVAXD XFAGFAXDXXFFAVDFFVVFFAXFV”

3.5 Permutation cipher:

Let S_n be the symmetric group on n symbols where n is cardinality of Alphabets. The plaintext and ciphertext space is $X = \{m_1, m_2, \dots, m_n\}$.

Then for plaintext $m = (m_1, m_2, \dots, m_k)$ and a permutation $\sigma \in S_n$

- **Encription:** The encryption function is defined by

$$\mathcal{E}: X \longrightarrow X$$

$$m \longmapsto (m)\sigma = (m_1, m_2, \dots, m_k)\sigma = (m_{(1)\sigma}, m_{(2)\sigma}, \dots, m_{(k)\sigma})$$

- **Decryption:** Denote σ^{-1} to be the inverse of σ in S_n . Let $m' = (m'_1, m'_2, \dots, m'_k)$ is a ciphertext, encrypted with permutation cipher by using permutation σ , then decryption function is defined by

$$\mathcal{D}: X \longrightarrow X$$

$$m' \longmapsto (m')\sigma^{-1} = (m'_1, m'_2, \dots, m'_k)\sigma^{-1} = (m'_{(1)\sigma^{-1}}, m'_{(2)\sigma^{-1}}, \dots, m'_{(k)\sigma^{-1}})$$

Example: Consider the English alphabets as plaintext and ciphertext space $X = \{A = 1, B = 2, \dots, Y = 25, Z = 26\}$.

Let $\sigma \in S_{26}$ and $\sigma = (1 \ 3)(4 \ 5)(16 \ 17 \ 18)(20 \ 21 \ 22)$

Plain text-“NEVER GIVE UP ”

	N	E	V	E	R	G	I	V	E	U	P
	14	5	22	5	18	7	9	22	5	21	16
Encryption:-	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
	14	4	20	4	16	7	9	20	4	22	17
	N	D	T	D	P	G	I	T	D	V	Q

Encrypted Message is: “NDTDP GITD VQ”

$$\sigma^{-1} = (1 \ 3)(4 \ 5)(17 \ 16 \ 18)(21 \ 20 \ 22)$$

	N	D	T	D	P	G	I	T	D	V	Q
	14	4	20	4	16	7	9	20	4	22	17
Decryption:-	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
	14	5	22	5	18	7	9	22	5	21	16
	N	E	V	E	R	G	I	V	E	U	P

Decrypted message is: “NEVER GIVE UP ”

Encryption in English alphabet by using key:

$X = \{A = 1, B = 2, \dots, X = 24, Y = 25, Z = 26\}$.

Plain text:- “LET US GO FOR PLAYING CRICKET”

Keyword: “SHARMA”

Arrange it in alphabetic order and drop the duplicates i.e AHMRS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
A	H	M	R	S	B	C	D	E	F	G	I	J	K	L	N	O	P	Q	T
1	8	13	18	19	2	3	4	5	6	7	9	10	11	12	14	15	16	17	20

21	22	23	24	25	26
U	V	W	X	Y	Z
↓	↓	↓	↓	↓	↓
U	V	W	X	Y	Z
21	22	23	24	25	26

Now $\sigma = (1)(2\ 8\ 4\ 18\ 16\ 14\ 11\ 7\ 3\ 13\ 10\ 6)(5\ 19\ 17\ 15\ 12\ 9)(20)(21)(22)(23)(24)(25)(26)$

i.e. $\sigma = (2\ 8\ 4\ 18\ 16\ 14\ 11\ 7\ 3\ 13\ 10\ 6)(5\ 19\ 17\ 15\ 12\ 9)$

Cipher text is :- **“IST UQ CL BLP NIAYEKC MPEMGST”**

For Decryption we use

$\sigma^{-1} = (1)(6\ 10\ 13\ 3\ 7\ 11\ 14\ 16\ 18\ 4\ 8\ 2)(9\ 12\ 15\ 17\ 19\ 5)(20)(21)(22)(23)(24)(25)(26)$

i.e. $\sigma^{-1} = (6\ 10\ 13\ 3\ 7\ 11\ 14\ 16\ 18\ 4\ 8\ 2)(9\ 12\ 15\ 17\ 19\ 5)$

Hence decrypted message is **“LET US GO FOR PLAYING CRICKET”**

3.6 Playfair Cipher:

The playfair cipher was the first practical digraph substitution cipher. The scheme was invented in 1854 by Charles Wheatstone. But it was named after Lord Playfair because he started the use of this cipher.

We have a grid of 5×5 cipher. There are 25 spaces but we have 26 letters, by convention we put I,J together.

A	B	C	D	E
F	G	H	I,J	K
L	M	N	O	P
Q	R	S	T	U
V	W	X	Y	Z

Encryption: There are four rules for encryption using playfair cipher. Suppose P,Q represent the plain text symbols and C,D represent cipher symbols.

1. If P,Q are in different rows and columns then they form a rectangle of which they are opposite diagonal entries, then the encryption of digraph PQ is given by CD where C is in the same column as of P and D is also in same column as of Q.
Eg: $\mathcal{E}(CT) = SD$, $\mathcal{E}(WA) = BV$
2. If P,Q are different and lie in the same row then the encryption of digraph PQ is given by CD where C is just right from P and D is just right from Q. If needed wrap it.
Eg: $\mathcal{E}(CE) = DA$, $\mathcal{E}(FK) = GF$
3. If P,Q are different and lie in the same column then the encryption of digraph PQ is given by CD where C is just below of P and D is just below Q. If needed wrap it.
Eg: $\mathcal{E}(NH) = SN$, $\mathcal{E}(AQ) = FV$
4. If P,Q are same then by convention put letter X in between them and then do encryption by using rules 1,2,3.

Decryption: Decryption is reverse process of encryption. Decryption rules for playfair cipher are :

1. If C,D are in different rows and columns then they form a rectangle of which they are opposite diagonal entries, then the decryption of digraph CD is given by PQ where P is in the same column as of C and Q is also in same column as of D.
Eg: $\mathcal{D}(NI) = HO$, $\mathcal{D}(BV) = WA$
2. If C,D are different and lie in the same row then the decryption of digraph CD is given by PQ where P is just left from C and Q is just left from D. If needed wrap it.
Eg: $\mathcal{D}(DA) = CE$, $\mathcal{D}(GF) = FK$
3. If C,D are different and lie in the same column then the decryption of digraph CD is given by PQ where P is just above of C and Q is just above D. If needed wrap it.
Eg: $\mathcal{D}(SN) = NH$, $\mathcal{D}(FV) = AQ$

Example: Key: RULE

Plaintext: INSTRUMENTS

Arrange key in alphabetical order i.e ELRU

In 5×5 grid, first we write row-wise alphabets of keyword in alphabetical order then remaining English alphabet in order.

E	L	R	U	A
B	C	D	F	G
H	I,J	K	M	N
O	P	Q	S	T
V	W	X	Y	Z

Plaintext: "INSTRUMENTS"

After split: 'IN' 'ST' 'RU' 'ME' 'NT' 'SZ'

$$\mathcal{E}(IN) = KH$$

$$\mathcal{E}(ST) = TO$$

$$\mathcal{E}(RU) = UA$$

$$\mathcal{E}(ME) = UH$$

$$\mathcal{E}(NT) = TZ$$

$$\mathcal{E}(SZ) = YT$$

Encrypted message is: "KHTOUAUHTZYT"

After split: 'KH' 'TO' 'UA' 'UH' 'TZ' 'YT'

$$\mathcal{D}(KH) = IN$$

$$\mathcal{D}(TO) = ST$$

$$\mathcal{D}(UA) = RU$$

$$\mathcal{D}(UH) = ME$$

$$\mathcal{D}(TZ) = NT$$

$$\mathcal{D}(YT) = SZ$$

Message is "INSTRUMENTS"

Note- When we perform encryption without keyword then we write 5×5 grid in English alphabets order i.e. A, B, ... Y, Z

A	B	C	D	E
F	G	H	I, J	K
L	M	N	O	P
Q	R	S	T	U
V	W	X	Y	Z

3.7 Enigma Machine Cipher

During World war II Germany, Italy and Japan were far outnumbered by the Allies. Their only conceivable path to victory was the ability to launch widespread surprise attack. So the goal of encryption technology was to automate the one-time pad using an encryption machine. Ideally, this machine would accept an input letter, apply a random shift and output the encrypted letter. However all machines follow the same principle. They begin in some initial configuration known as a state, they accept some input they do an operation with the input, and then they produce an output. The operation from initial state to final state is always predictable and repeatable. So the goal is to produce identical machines that output a scrambled sequence of shifts which took a long time to repeat.

The Enigma machine is a cipher device developed and used in the early-to mid-20th century to protect commercial, diplomatic, and military communication. The Enigma machine was invented by German engineer Arthur Scherbius at the end of World War I. It was employed extensively by Nazi Germany during World War II, in all branches of the German military. The Enigma machine was considered so secure that it was used to encipher the most top-secret messages.

The Enigma has an electromechanical rotor mechanism that scrambles the 26 letters of the alphabet. In typical use, one person enters text on the Enigma's keyboard and another person writes down which of 26 lights above the keyboard illuminated at each key press. If plain text is entered, the illuminated letters are the encoded ciphertext.

The security of the system depends on machine settings that were generally changed daily, based on secret key lists distributed in advance, and on other settings that were changed for each message. *The receiving station would have to know and use the exact settings employed by the transmitting station to successfully decrypt a message.*

Operation

While using the Enigma machine one would be given a plaintext message to encrypt. After setting up his machine, he would type the message on the Enigma keyboard. For each letter pressed, one lamp lit indicating a different letter according to a pseudo-random substitution determined by the electrical pathways inside the machine. The letter indicated by the lamp would be recorded, typically by a second operator, as the cyphertext letter. The action of pressing a key also moved one or more rotors so that the next key press used a different electrical pathway, and thus a different substitution would occur even if the same plaintext letter were entered again. For each key press there was rotation of at least the right hand rotor and less often the other two, resulting in a different substitution alphabet being used for every letter in the message. This process continued until the message was completed.

Design of Enigma machine

Electrical pathway: An electrical pathway is a route for current to travel. By manipulating this phenomenon the Enigma machine was able to scramble messages. The repeated changes of electrical path through an Enigma scrambler implement a **polyalphabetic substitution cipher** that provides Enigma's security.

Rotors: The rotors (wheels) is the main tool of an Enigma machine. Each rotor is a disc approximately 10 cm in diameter made, spring-loaded, electrical contact pins arranged in a circle on one face, with the other face housing 26 (named *A – Z*) corresponding electrical contacts in the form of circular plates. A Rotor performs only a very simple type of encryption, a simple substitution cipher.

An Enigma machine also consists of an Entry wheel, Reflector and a Plug-board. There are 10 cables placed onto the plugboard which connects letters

in pairs.

Mathematical Analysis

The Enigma transformation for each letter can be specified mathematically as a product of permutations. Assuming a three-rotor enigma, let P denote the plugboard transformation, U denote that of the reflector, and L , M , R , denote those of the left, middle and right rotors respectively. Then the encryption E can be expressed as

$$E = PRMLUL^{-1}M^{-1}R^{-1}P^{-1}.$$

After each key press, the rotors turn, changing the transformation. For example, if the right-hand rotor R is rotated n positions, the transformation becomes $\rho^n R \rho^{-n}$,

where ρ is the cyclic permutation mapping A to B , B to C , and so forth. Similarly, the middle and left-hand rotors can be represented as j and k rotations of M and L . The encryption transformation can then be described as

$$E = P (\rho^n R \rho^{-n}) (\rho^j M \rho^{-j}) (\rho^k L \rho^{-k}) U (\rho^k L^{-1} \rho^{-k}) (\rho^j M^{-1} \rho^{-j}) (\rho^n R^{-1} \rho^{-n}) P^{-1}.$$

Number of Settings in Enigma

- Choosing 3 rotors from a set of 5 rotors, 60 possibilities,
- Each of the 3 rotor settings with 26 positions, $26 \times 26 \times 26 = 17576$ possibilities,
- Plugboard with ten pairs of letters connected, making $\frac{26!}{6! \cdot 10! \cdot 2^{10}} = 150,738,274,937,250$ possibilities

That is there is a total of 158,962,555,217,826,360,000 (nearly 159 *quintillion*) different settings. So it was next to impossible to break(decipher) the Enigma code. But in early 1943, British Mathematician Alan Turing break the Enigma Code.

Enigma machine's setting and an example

- Wheel order – the choice of rotors and the order in which they are fitted.

- Ring settings – the position of each alphabet ring relative to its rotor wiring.
- Plug connections – the pairs of letters in the plugboard that are connected together.
- Starting position of the rotors – chosen by the operator, should be different for each message.

For a message to be correctly encrypted and decrypted, both sender and receiver had to configure their Enigma in the same way; rotor selection and order, ring positions, plugboard connections and starting rotor positions must be identical. Except for the starting positions, these settings were established beforehand, distributed in key lists and changed daily.

Example: Let us take the following setting to the Enigma machine:

- Wheel order – III V IV (3 rotors selected out 5)
- Ring settings – 15, 23, 26
- Plug connections – AR BH DO EQ FG IN JY KT SZ UV

then the plaintext **NUMBERPHILE** is encrypted into **YTHMYIUR-FGW**.

3.8 Exercise

Note : Let $\mathcal{S} = \{a = 0, b = 1, \dots, y = 24, z = 25\}$ be set of symbols.

1. Decrypt the cipher text "DTODLAIANINALEEIECIHIGSTD" using Column Transposition Cipher using the keyword "PLANE".
2. The plain text "SUNRISESINTHEEASTANDSETSINTHEWEST" is encrypted as "RSHSDSHSUSNEAENWNETANTTESIIETSIET" using column Transposition Cipher. Find the 4 letter keyword if it starts with T.
3. Encrypt the message "AN ECOSYSTEM HAS TWO COMPONENTS" using double column transposition with number of columns = 5.

4. Decrypt the cipher text "AOBNLDEIBYEUNWSDVBITSN" using Double Transposition if keyword is "HARD".
5. Encrypt the plain text "DO NOT WASTE TIME" using the permutation $\sigma = (1\ 3)(5\ 7)(6\ 8\ 9) \in S_{26}$. Also show the decryption process.
6. Decrypt the cipher "MGTOIDONLRGEGSTDAPOWARDOSAOT-DAIAKK" using permutation cipher if keyword is "LIFE".
7. Decrypt the cipher text "TAPQXTODHOCGQTKADFOAOWPSIRIKUCTA-WONBSIEXXKPV" using playfair cipher if keyword is "START".
8. Encrypt the plain text "TIME IS MONEY" using ADFGX cipher if keyword is "IDEA".
9. Decrypt the cipher text "AAXFAAAAFAFDXAVGFXXXXAFAFVDDF-GADGDADGVDDVADFFAFAAAGGDAGXGGGDAVAFXDDDDXGFFVAXXA" using ADFGVX if keyword is "PRIVACY" and array is given by:

	A	D	F	G	V	X
A	S	0	E	M	1	H
D	F	R	3	U	D	2
F	4	J	B	Q	P	K
G	A	5	T	L	6	X
V	V	N	Y	7	G	8
X	I	Z	C	W	9	O

Chapter 4

Basic Number Theory

4.1 Greatest Common Divisor

Greatest Common Divisor: Greatest Common Divisor (gcd) of integers a_1, a_2 (atleast one of them is not 0) is the largest positive integer that divides a_1, a_2 .

Note :

1. If more than one integer is 0, then gcd does not exist.
2. $\gcd(a_1, a_2) = \gcd(a_2, a_1)$
3. $\gcd(a, 0) = |a|$ where a is non zero integer.
4. $\gcd(a_1, a_2, \dots, a_n) = \gcd(a_1, a_2, \dots, \gcd(a_i, a_j), \dots, a_n) \quad \forall i, j \in \{1, 2, \dots, n\}$.

Relatively Prime Integers: Integers a and b are said to be relatively prime (or coprime) if the greatest common divisor of a and b is equal to 1.

Division Algorithm: If $a, b \in \mathbb{Z}$, $b > 0$, then there exist integers q and r such that $a = bq + r$, where $0 \leq r < b$.

Lemma: If $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$

Proof: If $d = \gcd(a, b) \Rightarrow d|a$ and $d|b \Rightarrow d|(a - bq)$ i.e $d|r$

Thus, d is a common divisor of b and r . Now claim is that d is greatest divisor. Let c be another divisor of b and r then $c|b$ and $c|r \Rightarrow c|bq + r \Rightarrow c|a$.

Since $c|a$ and $c|b \Rightarrow c|d$. This implies that d is the greatest common divisor of b and r i.e $\gcd(a, b) = \gcd(b, r)$.

Theorem: If $a, b \in \mathbb{Z}$, (atleast one of them is not 0) then $\gcd(a, b)$ exists.

Proof: Case 1 : $b > 0$

Applying division algorithm on a, b ($b > 0$) , $\exists q_1, r_1 \in \mathbb{Z}$ such that

$$a = bq_1 + r_1, \text{ where } 0 \leq r_1 < b$$

If $r_1 = 0$, then $a = bq_1 \Rightarrow b|a$ also $b|b \Rightarrow b|\gcd(a, b)$.
Since, $\gcd(a, b)|a$ and $\gcd(a, b)|b \Rightarrow \gcd(a, b) = b$

If $r_1 > 0$, then $0 < r_1 < b$ and thus we can apply division algorithm on b and r_1 .

Therefore, $\exists q_2, r_2 \in \mathbb{Z}$ such that

$$b = r_1q_2 + r_2, \text{ where } 0 \leq r_2 < r_1$$

If $r_2 = 0$, then $b = r_1q_2 \Rightarrow r_1|b$ also $r_1|r_1 \Rightarrow \gcd(b, r_1) = r_1$.

If $r_2 \neq 0$, we can apply division algorithm on r_1 and r_2 .
Thus $\exists q_3, r_3 \in \mathbb{Z}$ such that

$$r_1 = r_2q_3 + r_3 \text{ where } 0 \leq r_3 < r_2$$

Now proceed till we get n such that $r_n \neq 0$, $r_{n+1} = 0$ and $\gcd(r_{n-1}, r_n) = r_n$.

We get following system of equation:

$$\begin{aligned} a &= bq_1 + r_1 & \text{where } 0 \leq r_1 < b \\ b &= r_1q_2 + r_2 & \text{where } 0 \leq r_2 < r_1 \\ r_1 &= r_2q_3 + r_3 & \text{where } 0 \leq r_3 < r_2 \\ &\vdots \\ r_{n-2} &= r_{n-1}q_n + r_n & \text{where } 0 \leq r_n < r_{n-1} \\ r_{n-1} &= r_nq_{n+1} + r_{n+1}(= 0) \end{aligned}$$

Note that

$$\gcd(a, b) = \gcd(b, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_{n-1}, r_n) = r_n.$$

Case 2 : $b < 0$

Take $b' = -b$, then $b' > 0$. Apply division algorithm on a and b' and solve.
Note that $\gcd(a, b) = \gcd(a, b')$

Euclidean Algorithm provides a way to find the gcd of any two integers and thus proves the existence of greatest common divisor.

Theorem (Extended Euclidean Algorithm): If $a, b \in \mathbb{Z}$, then $\gcd(a, b)$ exists. Let $\gcd(a, b) = d$ then, $\exists$ integers x, y such that $d = ax + by$.

Proof: Let us suppose a set S of all linear combinations of a and b

$$S = \{ax + by \mid ax + by > 0; x, y \in \mathbb{Z}\}$$

One of a, b is not zero. Without loss of generality, suppose $a \neq 0$ and if $a > 0$ then $0 < a = a \cdot 1 + b \cdot 0 \in S$, if $a < 0$ then $0 < -a = a \cdot (-1) + b \cdot 0 \in S$ therefore, S is non empty set of natural numbers.

Well Ordering Principle: Any non empty subset of natural numbers has a least element.

By Well Ordering Principle, S have a least element d (say). So by definition of set S , there exists integers x and y such that $d = a \cdot x + b \cdot y$

Claim: $d = \gcd(a, b)$

Apply division algorithm on a and d , there exists integers q and r such that $a = qd + r$, where $0 \leq r < d$. We can write $r = a - qd = a - q(ax + by) = a(1 - qx) + b(-qy)$. If $r > 0$ then $r \in S$ (as r is linear combination of a and b) and $r < d$ which is a contradiction to d is the least element in S . Therefore, $r = 0$ this implies $a = qd$ i.e. $d \mid a$.

Similarly, apply division algorithm on b and d , there exists integers q_1 and r_1 such that $b = q_1d + r_1$, where $0 \leq r_1 < d$.

We can write $r_1 = b - q_1d = b - q_1(ax + by) = a(-q_1x) + b(1 - q_1y)$

If $r_1 > 0$ then $r_1 \in S$ (as r_1 is linear combination of a and b) and $r_1 < d$ which is again a contradiction to d is the least element in S . Therefore, $r_1 = 0$ this implies $b = q_1d$ i.e. $d \mid b$

So, d is common divisor of a and b .

If c is an arbitrary positive common divisor of a and b then c divides $a \cdot u + b \cdot v$ for all $u, v \in \mathbb{Z}$. In particular, c divides $ax + by = d$ this implies $c \leq d$

So d is greatest common divisor of a and b i.e. $\gcd(a, b) = d$.

Example: Let $a = 612$, $b = 2017$, then Find $\gcd(612, 2017)$ using Euclidean Algorithm.

Solution: To compute $\gcd$, we apply Division Algorithm on $(2017, 612)$ as :

$$2017 = 612 \cdot 3 + 181$$

Note that $\gcd(2017, 612) = \gcd(612, 181)$, we apply Division Algorithm on $(612, 181)$ as :

$$612 = 181 \cdot 3 + 69$$

Similarly $\gcd(612, 181) = \gcd(181, 69)$, we apply Division Algorithm on $(181, 69)$ as :

$$181 = 69 \cdot 2 + 43$$

Similarly $\gcd(181, 69) = \gcd(69, 43)$, we apply Division Algorithm on $(69, 43)$ as :

$$69 = 43 \cdot 1 + 26$$

Proceeding the same way

$$43 = 26 \cdot 1 + 17$$

$$26 = 17 \cdot 1 + 9$$

$$17 = 9 \cdot 1 + 8$$

$$9 = 8 \cdot 1 + 1$$

$$8 = 1 \cdot 8 + 0$$

Thus $\gcd(612, 2017) = 1$

Example: Find the greatest common divisor d of 218 and 66, and find integers x and y solving the equation $218x + 66y = d$.

Solution: To compute $\gcd(218, 66)$, we apply Division Algorithm on $(218, 66)$ as :

$$218 = 66 \cdot 3 + 20$$

Note that $\gcd(218, 66) = \gcd(66, 20)$, we apply Division Algorithm on $(66, 20)$ as :

$$66 = 20 \cdot 3 + 6$$

Similarly $\gcd(66, 20) = \gcd(20, 6)$, we apply Division Algorithm on $(20, 6)$ as :

$$20 = 6 \cdot 3 + 2$$

Similarly $\gcd(20, 6) = \gcd(6, 2)$, we apply Division Algorithm on $(6, 2)$ as :

$$6 = 2 \cdot 3 + 0$$

Thus $\gcd(218, 66) = d = 2$. To find to integers x, y such that $d = 218x + 66y$, we do the back substitution of the above steps as follows:

$$\begin{aligned} 2 &= 20 - (6 \cdot 3) \\ &= 20 - (66 - (20 \cdot 3)) \cdot 3 \\ &= 20 - (66 \cdot 3 - 20 \cdot 3 \cdot 3) \\ &= 20 \cdot (1 + 9) - 66 \cdot 3 \\ &= 20 \cdot 10 - 66 \cdot 3 \\ &= (218 - (66 \cdot 3)) \cdot 10 - 66 \cdot 3 \\ &= 218 \cdot 10 - 66 \cdot (3 \cdot 10 + 3) \\ &= 218 \cdot 10 + 66 \cdot (-33) \end{aligned}$$

Thus, $(10, -33)$ satisfies the equation $218x + 66y = d$.
We can find the other solutions.

Application of Division Algorithm: Various Representations of an Integer

Result : Let n be a fixed positive integer greater than 1. Then every positive integer m can be written as

$$m = a_{k-1}n^{k-1} + a_{k-2}n^{k-2} + \dots + a_1n + a_0 \quad (4.1)$$

where each $a_i \in \{0, 1, 2, \dots, n-1\}$.

Further, for a given m , the integers $a_0, a_1, \dots, a_{k-1}$ with $a_{k-1} \neq 0$ are unique and the integer $k = [\log_n m] + 1$.

Here $[x]$ for a real number x denotes the greatest integer less or equal to x .

Proof : First Observe that if an integer k exists such that m can be written as a polynomial of degree $k - 1$ in n as (1), then k must be $[\log_n m] + 1$.

$$\text{For } a_{k-1}n^{k-1} = m - \sum_{i=0}^{k-2} a_i n^i \leq m$$

Now

$$a_{k-1} \neq 0 \Rightarrow a_{k-1} \geq 1 \Rightarrow n^{k-1} \leq a_{k-1}n^{k-1} \leq m$$

$$\Rightarrow n^{k-1} \leq m$$

$$\text{Also } m = \sum_{i=0}^{k-1} a_i n^i$$

Since every $a_i \in \{0, 1, 2, \dots, n-1\} \Rightarrow a_i \leq n-1$, we get $m \leq \sum_{i=0}^{k-1} (n-1)n^i$

and for

$$\sum_{i=0}^{k-1} (n-1)n^i = \sum_{i=0}^{k-1} n^{i+1} - n^i = n^k - 1$$

Hence $n^{k-1} \leq m \leq n^k - 1$. But then $k-1 = [\log_n m] \Rightarrow k = [\log_n m] + 1$. Thus if such an integer k exists, then it is unique and is equal to $[\log_n m] + 1$.

Existence of k :

By division algorithm, $\exists$ integers q_1 and a_0 such that

$$m = q_1 n + a_0 \quad (0 \leq a_0 < n) ; \quad q_1 < m$$

If $0 \leq q_1 < n$, then $m = q_1 n + a_0 = a_1 n + a_0$, where $0 \leq a_0, a_1 \leq n-1$ is already in the desired form.

If $q_1 \geq n$, then again by division algorithm, $\exists$ integers q_2 and a_1 such that

$$q_1 = q_2 n + a_1 \quad (0 \leq a_1 < n)$$

On substituting this value of q_1 in the previous equation, we get:

$$m = q_1 n + a_0 = (q_2 n + a_1)n + a_0 = q_2 n^2 + a_1 n + a_0$$

where $0 \leq a_0, a_1 \leq n - 1$.

Again if $q_2 < n$, we get

$m = a_2 n^2 + a_1 n + a_0$ with $a_2 = q_2$ in the desired form.

If $q_2 \geq n$, we apply again division algorithm, to get integers q_3 and a_2 such that

$$q_2 = q_3 n + a_2 \quad (0 \leq a_2 \leq n - 1)$$

Substituting this value of q_2 in $m = q_2 n^2 + a_1 n + a_0$, we get

$$m = (q_3 n + a_2) n^2 + a_1 n + a_0 = q_3 n^3 + q_2 n^2 + a_1 n + a_0$$

where $0 \leq a_0, a_1, a_2 \leq n - 1$.

If $0 \leq q_3 < n$, we get m to be in the desired form, otherwise apply again the division algorithm on n and q_3 and continue to get

$$m = q_{j-1} n^{j-1} + a_{j-2} n^{j-2} + \dots + a_1 n + a_0 ; \quad 0 \leq a_0, a_1, \dots, a_{j-2} \leq n - 1$$

where $q_{j-1} = q_j n + a_{j-1}$ such that $q_{j-1} > n$.

Note that $m > q_1 > q_2 > q_3 > q_4 > \dots \geq 0$ is a strictly decreasing sequence of non-negative integers and hence to terminate after a finite number of steps. Thus for some integer $k - 1$ such that

$$m = q_{k-1} n^{k-1} + a_{k-2} n^{k-2} + \dots + a_1 n + a_0$$

where

$q_{k-2} = q_{k-1} n + a_{k-2}$, $m > q_1 > q_2 > q_3 > q_4 > \dots > q_{k-2} > q_{k-1}$ and $q_{k-1} < n$.

But then $m = a_{k-1} n^{k-1} + a_{k-2} n^{k-2} + \dots + a_1 n + a_0$ for $a_{k-1} = q_{k-1}$ is in the desired form.

Uniqueness of the expression:

Let $m = a_{k-1} n^{k-1} + a_{k-2} n^{k-2} + \dots + a_1 n + a_0$ and $m = b_{k-1} n^{k-1} + b_{k-2} n^{k-2} + \dots + b_1 n + b_0$ be two different such expressions for m .

Then some $a_i \neq b_i$.

Let j be the smallest such that $a_j \neq b_j$.

But then $m = a_{k-1} n^{k-1} + \dots + a_0$ and $m = b_{k-1} n^{k-1} + \dots + b_0$ implies

$$\Rightarrow (a_{k-1} - b_{k-1}) n^{k-1} + \dots + (a_j - b_j) n^j = 0 ; \quad 0 < j \leq k - 1$$

$$\begin{aligned} \Rightarrow (a_j - b_j) n^j &= -(a_{k-1} - b_{k-1}) n^{k-1} - (a_{k-2} - b_{k-2}) n^{k-2} \dots - (a_{j+1} - b_{j+1}) n^{j+1} \\ &= -n^{j+1} \{ (a_{k-1} - b_{k-1}) n^{k-1-\{j+1\}} + \dots + (a_{j+1} - b_{j+1}) \} \end{aligned}$$

$$= -n^{j+1}\{(a_{k-1} - b_{k-1})n^{k-j} + \dots + (a_{j+1} - b_{j+1})\}$$

$$\Rightarrow a_j - b_j = -n\{(a_{k-1} - b_{k-1})n^{k-j} + \dots + (a_{j+1} - b_{j+1})\}$$
 Since $a_j \neq b_j$ and $n > 1$, we get $n|(a_j - b_j) \Rightarrow n| |a_j - b_j|$
 Both a_j, b_j satisfies $0 \leq a_j, b_j < n$. This gives that $0 \leq |a_j - b_j| < n$, which is a contradiction.
 Hence $a_j - b_j = 0 \Rightarrow a_j = b_j$.
 Thus m has unique representation.

Remark 1:

Every positive integer m can be written as a polynomial in n for any fixed positive integer $n > 1$ as:

$$m = a_{k-1}n^{k-1} + a_{k-2}n^{k-2} + \dots + a_1n + a_0$$

where $a_{k-1} \neq 0$ and $k = [\log_n m] + 1$.

This is often represented by just noting the coefficient of power of n , that is

$$\begin{aligned}
 m &= (a_{k-1}, a_{k-2}, \dots, a_1, a_0)_n \\
 &\text{or} \\
 &= (a_{k-1} \ a_{k-2} \ \dots \ a_1 \ a_0)_n \\
 &\text{or} \\
 &= a_{k-1} \ a_{k-2} \ \dots \ a_1 \ a_0
 \end{aligned}$$

This is called the n -nary representation or n -adic representation of m (to the base n) and a_i 's are called the digits of m in the representation. If you recall the process, then the digits are obtained by successively dividing by n the quotients, starting with a_0 as the remainder of m on dividing by n .

Divisor		Digits
n	m $q_1 n + a_0$	a_0
n	q_1 $q_2 n + a_1$	$a_1 = q_1 \text{ if } q_1 < n$
n	q_2 $\vdots$	Till $q_{k-1} < n$ Put $a_{k-1} = q_{k-1}$

Remark 2:

When $n = 2$, the representation is called binary, when $n = 3$, the representation is called ternary, when $n = 10$, the representation is called decimal and when $n = 16$, the representation is called hexadecimal. Binary and hexadecimal are frequently used for machine implementation in a computer.

For hexadecimal representation of an integer, we need 16-digits, having values 0,1,2,...,15. For this 0,1,2,...,9 are used to represent 0,1,2,...,9 digits and A,B,C,D,E, and F are used to represent 10,11,12,13,14, and 15 respectively.

Thus DEED represents the integers:

$$D 16^3 + E 16^2 + E 16 + D = 13 \times 16^3 + 14 \times 16^2 + 14 \times 16 + 13 = 57069 \text{ in decimal.}$$

Remark 3:

To convert 57069 (in decimal form) to binary; observe the following:

$$m = 57069, \quad n = 2$$

$$m = 28534 \times 2 + 1 = q_1 n + a_0 ; \quad a_0 = 1$$

$$q_1 = 28534 = 14267 \times 2 + 0 = q_2 n + a_1 ; \quad a_1 = 0$$

$$q_2 = 14267 = 7133 \times 2 + 1 = q_3 n + a_2 ; \quad a_2 = 1$$

$$q_3 = 7133 = 3566 \times 2 + 1 = q_4 n + a_3 ; \quad a_3 = 1$$

$$q_4 = 3566 = 1783 \times 2 + 0 = q_5 n + a_4 ; \quad a_4 = 0$$

$$q_5 = 1783 = 891 \times 2 + 1 = q_6 n + a_5 ; \quad a_5 = 1$$

$$q_6 = 891 = 445 \times 2 + 1 = q_7 n + a_6 ; \quad a_6 = 1$$

$$q_7 = 445 = 222 \times 2 + 1 = q_8 n + a_7 ; \quad a_7 = 1$$

$$q_8 = 222 = 111 \times 2 + 0 = q_9 n + a_8 ; \quad a_8 = 0$$

$$q_9 = 111 = 55 \times 2 + 1 = q_{10} n + a_9 ; \quad a_9 = 1$$

$$q_{10} = 55 = 27 \times 2 + 1 = q_{11} n + a_{10} ; \quad a_{10} = 1$$

$$q_{11} = 27 = 13 \times 2 + 1 = q_{12} n + a_{11} ; \quad a_{11} = 1$$

$$q_{12} = 13 = 6 \times 2 + 1 = q_{13} n + a_{12} ; \quad a_{12} = 1$$

$$q_{13} = 6 = 3 \times 2 + 0 = q_{14} n + a_{13} ; \quad a_{13} = 0$$

$$q_{14} = 3 = 1 \times 2 + 1 = q_{15} n + a_{14} ; \quad a_{14} = 1$$

$$q_{15} = 1 = 0 \times 2 + 1 = q_{16} n + a_{15} ; \quad a_{15} = 1$$

Hence the binary representation of $m = 57069$ is 1101 1110 1110 1101.

Remark 4:

To convert a hexadecimal number to its binary form, it is not necessary

to convert it to its decimal form. It is rather simple. Observe that all the fifteen hexadecimal digits $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F$ can be represented in its binary form as $a_3a_2a_1a_0 = a_3.2^3 + a_2.2^2 + a_1.2 + a_0$, where $a_3, a_2, a_1, a_0 \in \{0, 1\}$.

Hence if $m = h_jh_{j-1}...h_i...h_0$ in hexadecimal representation, and if $h_i = b_{3,i}b_{2,i}b_{1,i}b_{0,i}$ is the binary representation of h_i , then

$$m = b_{3,j}b_{2,j}b_{1,j}b_{0,j} b_{3,j-1}b_{2,j-1}b_{1,j-1}b_{0,j-1} \dots b_{3,i}b_{2,i}b_{1,i}b_{0,i} \dots b_{3,0}b_{2,0}b_{1,0}b_{0,0}$$

is the binary representation of m . Prove it.

Hence the binary representation of $m = DEED$ in hexadecimal can be written by observing the binary representation of $D = 13$ and $E = 14$ as $13 = 1101$ and $14 = 1110$, we get $(DEED)_{16} = (1101\ 1110\ 1110\ 1101)_2$

Remark 5:

To convert an integer in decimal form to its hexadecimal form, simply keep on dividing by 16 and noting the remainders and quotients. For example, to convert 57069 to its hexadecimal form, observe the following:

$$m = 57069 = 3566 \times 16 + 13 = q_1n + a_0 ; a_0 = 13$$

$$q_1 = 3566 = 222 \times 16 + 14 = q_2n + a_1 ; a_1 = 14$$

$$q_2 = 222 = 13 \times 16 + 14 = q_3n + a_2 ; a_2 = 14$$

$$q_3 = 13 = 0 \times 16 + 13 = ; a_3 = 13$$

Hence the hexadecimal representation of 57069 is 13 14 14 13, that is DEED.

Remark 6:

To convert a binary number to its hexadecimal form, write the number into blocks of four bits (binary digits 0, 1) starting from the right most, by padding by 0's if need on the left for the left most block.

For example, to convert $m = (110111101101)_2$ to hexadecimal, we write

$$m = 0001\ 1011\ 1110\ 1101$$

in hexadecimal as

$$m = 1\ 11\ 14\ 13$$

. That is $m = 1\ B\ E\ D$ in hexadecimal.

Remark 7:

The binary digits 0, 1 are called bits. A string of eight bits is called a byte.

For example, 11011001 ($= 2^7 + 2^6 + 2^4 + 2^3 + 1 = 217$) is a byte, but 1011001 ($= 2^6 + 2^4 + 2^3 + 1 = 89$) is not a byte as the string has 7-digits only.

But it can be padded on the left by a 0 so that it becomes a byte, namely 01011001 without changing its value. Computers usually store information in bytes using the ASCII value of the characters in decimals by converting into bytes.

Remark 8: (Addition, subtraction, multiplication and division of integers.)

While we are used to add, subtract, multiply and divide when integers are given in their decimal form, to do these operations, when the integers are given in any other form, bring them first into the same form, say n -nary as a polynomial in n , and then do these operations among the corresponding polynomials.

For example, divide $(DEED)_{16}$ by 16. First convert $(DEED)_{16}$ to its decimal form, that is, 57069 and divide by 16 to get $57069 = 3566 \times 16 + 13$. The quotient is 3566 and the remainder is 13 in decimal. This is in hexadecimal D E E and D respectively.

4.2 Modular Arithmetic

Modular Arithmetic is a system of arithmetic for integers, where numbers "wrap around" when reaching a certain value, called the modulus.

Congruence

Let a and b be two integers and $n > 1$ be a positive integer. Then a is said to be congruent to b in modulo n if n divides $a - b$ and it is denoted as $a \equiv b \pmod{n}$.

Properties of Congruence modulo n :

1. The congruence relation is an equivalence relation.

2. $a \equiv b \pmod{n} \Leftrightarrow a + k \equiv b + k \pmod{n} \quad \forall k \in \mathbb{Z}$
3. $a \equiv b \pmod{n} \Rightarrow ka \equiv kb \pmod{n} \quad \forall k \in \mathbb{Z}$
4. $a \equiv b \pmod{n} \Rightarrow a^k \equiv b^k \pmod{n} \quad \forall k \in \mathbb{Z}^+ \cup \{0\}$
5. $a \equiv b \pmod{n} \Rightarrow p(a) \equiv p(b) \pmod{n} \quad \forall$ polynomials $p(x)$ with integer coefficients.
6. $a_1 \equiv b_1 \pmod{n}$ and $a_2 \equiv b_2 \pmod{n} \Rightarrow a_1 + a_2 \equiv b_1 + b_2 \pmod{n}$
7. $a_1 \equiv b_1 \pmod{n}$ and $a_2 \equiv b_2 \pmod{n} \Rightarrow a_1 a_2 \equiv b_1 b_2 \pmod{n}$
8. $ka \equiv kb \pmod{n}$ such that k is coprime to $n \Rightarrow a \equiv b \pmod{n}$
9. $ka \equiv kb \pmod{kn} \Rightarrow a \equiv b \pmod{n}$
10. $c \equiv d \pmod{\phi(n)}$, where ϕ is Euler's totient function $\Rightarrow a^c \equiv a^d \pmod{n}$, where a is coprime to n

Modular Multiplicative Inverse: Let a be an integer. Then inverse of a in modulo n exists if $\gcd(a, n) = 1$. Inverse of a in modulo n is denoted by a^{-1} and satisfies $a^{-1}a \equiv 1 \pmod{n}$.

Fermat's Little Theorem: Let p be a prime and a be an integer, then $a^p \equiv a \pmod{p}$.

Proof:

Case 1 : $\gcd(p, a) \neq 1$, then $p|a$ and $a \equiv 0 \pmod{p}$. Hence, Fermat's Little Theorem holds.

Case 2 : $\gcd(p, a) = 1$, then p does not divide a .

Consider the set $\{a, 2a, 3a, \dots, (p-1)a\}$.

Since p does not divide $a \Rightarrow p$ does not divide $ia \quad \forall 1 \leq i \leq (p-1)$.

Since $a^{-1} \pmod{p}$ exists, thus $ia = ja \Rightarrow i = j \quad \forall 1 \leq i, j \leq (p-1)$.

Thus the set $\{a, 2a, 3a, \dots, (p-1)a\}$ is equivalent to the set $\{1, 2, 3, \dots, (p-1)\}$ in $\pmod{p}$

Hence,

$$\begin{aligned} a(2a)(3a) \cdots ((p-1)a) &\equiv 1 \cdot 2 \cdot 3 \cdots (p-1) \pmod{p} \\ \Rightarrow a^{p-1} (p-1)! &\equiv (p-1)! \pmod{p} \end{aligned}$$

Since $\gcd(i, p) = 1 \quad \forall 1 \leq i \leq (p-1) \Rightarrow a^{p-1} \equiv 1 \pmod{p}$

Euler totient function:- Let n be a positive integer. The Euler totient function $\phi(n)$ is defined to be the number of non-negative integers b less than n which are prime to n :

$$\phi(n) = |\{0 \leq b < n | (b, n) = 1\}|$$

It is easy to see that $\phi(1) = 1$ and that $\phi(p) = p - 1$ for any prime p . We can also see that for any prime power

$$\phi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^\alpha \left(1 - \frac{1}{p}\right)$$

The Euler totient function is "multiplicative", meaning that

$$\phi(mn) = \phi(m)\phi(n) \text{ whenever } (m, n) = 1.$$

Since every n can be written as a product of prime powers, i.e $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$

$$\phi(n) = p_1^{\alpha_1} \left(1 - \frac{1}{p_1}\right) p_2^{\alpha_2} \left(1 - \frac{1}{p_2}\right) \dots p_r^{\alpha_r} \left(1 - \frac{1}{p_r}\right) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

Euler's Theorem: Let a and n be positive integer such that $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$, where $\phi(n)$ is Euler's totient function.

Proof: Consider the set $U(n) = \{x \in \mathbb{Z} | \gcd(x, n) = 1 \text{ and } 1 \leq x \leq n-1\}$. Note that $U(n)$ is a commutative group with respect to the multiplication modulo n . Order of $U(n)$ is $\phi(n)$. Also $a \in \mathbb{Z}$. Thus, by Lagrange's theorem order of a divides $\phi(n)$.

So, $\exists k$ such that $a^k \equiv 1 \pmod{n}$ and k divides $\phi(n) \Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}$.

Note : Case 2 of Fermat's Little Theorem is a particular case of Euler's Theorem

Example: Let $p = 11$

Let $a = 2$, then $\gcd(2, 11) = 1 \Rightarrow 2^{(11-1)} \equiv 1 \pmod{11}$

$$\begin{aligned}
\text{Let } a = 3, \text{ then } \gcd(3, 11) = 1 &\Rightarrow 3^{(11-1)} \equiv 1 \pmod{11} \\
\text{Let } a = 4, \text{ then } \gcd(4, 11) = 1 &\Rightarrow 4^{(11-1)} \equiv 1 \pmod{11} \\
\text{Let } a = 5, \text{ then } \gcd(5, 11) = 1 &\Rightarrow 5^{(11-1)} \equiv 1 \pmod{11} \\
\text{Let } a = 6, \text{ then } \gcd(6, 11) = 1 &\Rightarrow 6^{(11-1)} \equiv 1 \pmod{11} \\
\text{Let } a = 7, \text{ then } \gcd(7, 11) = 1 &\Rightarrow 7^{(11-1)} \equiv 1 \pmod{11} \\
\text{Let } a = 8, \text{ then } \gcd(8, 11) = 1 &\Rightarrow 8^{(11-1)} \equiv 1 \pmod{11} \\
\text{Let } a = 9, \text{ then } \gcd(9, 11) = 1 &\Rightarrow 9^{(11-1)} \equiv 1 \pmod{11} \\
\text{Let } a = 10, \text{ then } \gcd(10, 11) = 1 &\Rightarrow 10^{(11-1)} \equiv 1 \pmod{11}
\end{aligned}$$

Example: Let $n = 15$, then $\phi(15) = \phi(3)\phi(5) = 2 \cdot 4 = 8$

$$\begin{aligned}
\text{Let } a = 2, \text{ then } \gcd(2, 15) = 1 &\Rightarrow 2^8 \equiv 1 \pmod{15} \\
\text{Let } a = 3, \text{ then } \gcd(3, 15) \neq 1 &\Rightarrow 3^8 \equiv 6 \pmod{15} \\
\text{Let } a = 4, \text{ then } \gcd(4, 15) = 1 &\Rightarrow 4^8 \equiv 1 \pmod{15} \\
\text{Let } a = 5, \text{ then } \gcd(5, 15) \neq 1 &\Rightarrow 5^8 \equiv 10 \pmod{15} \\
\text{Let } a = 6, \text{ then } \gcd(6, 15) \neq 1 &\Rightarrow 6^8 \equiv 6 \pmod{15} \\
\text{Let } a = 7, \text{ then } \gcd(7, 15) = 1 &\Rightarrow 7^8 \equiv 1 \pmod{15} \\
\text{Let } a = 8, \text{ then } \gcd(8, 15) = 1 &\Rightarrow 8^8 \equiv 1 \pmod{15} \\
\text{Let } a = 9, \text{ then } \gcd(9, 15) \neq 1 &\Rightarrow 9^8 \equiv 6 \pmod{15} \\
\text{Let } a = 10, \text{ then } \gcd(10, 15) \neq 1 &\Rightarrow 10^8 \equiv 10 \pmod{15} \\
\text{Let } a = 11, \text{ then } \gcd(11, 15) = 1 &\Rightarrow 11^8 \equiv 1 \pmod{15} \\
\text{Let } a = 12, \text{ then } \gcd(12, 15) \neq 1 &\Rightarrow 12^8 \equiv 6 \pmod{15} \\
\text{Let } a = 13, \text{ then } \gcd(13, 15) = 1 &\Rightarrow 13^8 \equiv 1 \pmod{15} \\
\text{Let } a = 14, \text{ then } \gcd(14, 15) = 1 &\Rightarrow 14^8 \equiv 1 \pmod{15} \\
\text{Note that } U(15) = \{1, 2, 4, 7, 8, 11, 13, 14\}
\end{aligned}$$

Example: Reduce $50^{250} \pmod{83}$ to a number in the range $\{0, 1, \dots, 82\}$.

(Note: 83 is prime.

If you multiply out 50^{250} , here's what you get:

52714787526044456024726519219225572551424023323922008641517022
 0907898754023953317101764802222644649987502681255357847020768
 63325972445883937922417317167855799198150634765625000000000000
 00
 00
 00
 00
 00

But we can simply compute it using Fermat's Little Theorem.
 Thus, by using Fermat's Little Theorem, we have

$$50^{250} \equiv 50^{82 \cdot 3 + 4} \equiv 50^4 \equiv 2500^2 \equiv 10^2 \equiv 17 \pmod{83}$$

Example: Let $n = 1001$, then $\phi(1001) = \phi(7)\phi(11)\phi(13) = 6 \cdot 10 \cdot 12 = 720$
 Let $a \in U(1001)$, then $\gcd(a, 1001) = 1$
 $\Rightarrow a^{\phi(1001)} \equiv a^{720} \equiv 1 \pmod{1001}$

Compute 9^{2883}
 $9^{2883} \equiv 9^{720 \cdot 4 + 3} \equiv (9^{720})^4 \cdot 9^3 \equiv 1^4 \cdot 9^3 \equiv 9^3 \equiv 729 \pmod{1001}$

Compute 2^{4400}
 $2^{4400} \equiv 2^{720 \cdot 6 + 80} \equiv (2^{720})^6 \cdot 2^{80} \equiv (2^{10})^8 \equiv 1024^8 \equiv 23^8 \equiv 562^2 \equiv 529 \pmod{1001}$

Linear Diophantine equation: The simplest linear Diophantine equation takes the form $ax + by = c$, where a, b and c are given integers.

- The Diophantine equation $ax + by = c$ has a solution (where x and y are integers) if and only if c is a multiple of the greatest common divisor of a and b .
- Moreover, if (x, y) is a solution, then the other solutions have the form $(x + \frac{bt}{d}, y - \frac{at}{d})$, where t is an arbitrary integer, and $d = \gcd(a, b)$.

4.3 Linear Congruence

Theorem: The linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if d divides b , where $d = \gcd(a, n)$. If $d|b$, then it has d mutually incongruent solutions modulo n .

Corollary:

1. If $\gcd(a, n) = 1$, then the linear congruence $ax \equiv b \pmod{n}$ has a unique solution.
2. If $\gcd(a, n) = d$, then d incongruent solutions of the equation $ax \equiv b \pmod{n}$ is given by

$$x_0, x_0 + \left(\frac{n}{d}\right), x_0 + 2\left(\frac{n}{d}\right), \dots, x_0 + (d-1)\left(\frac{n}{d}\right)$$

where x_0 is the solution of $\left(\frac{a}{d}\right)x \equiv \left(\frac{b}{d}\right) \pmod{\left(\frac{n}{d}\right)}$.

Example 1: Solve $3x \equiv 15 \pmod{20}$

Solution: First we calculate $\gcd(3, 20)$ by Euclidean Algorithm

$$\begin{array}{ll} 20 = 3 \cdot 6 + 2 & q_1 = 6, \quad r_1 = 2 \\ 3 = 2 \cdot 1 + 1 & q_2 = 1, \quad r_2 = 1 \\ 2 = 1 \cdot 2 + 0 & q_3 = 2, \quad r_3 = 0 \end{array}$$

$\Rightarrow \gcd(3, 20) = 1$, hence given equation has unique solution
Therefore 3^{-1} exists in modulo 20.

apply Extended Euclidean Algorithm as follows

$$\begin{aligned} 1 &= 3 - 1(2) \\ 1 &= 3 - 1(20 - (3 \cdot 6)) \\ 1 &= 7 \cdot 3 - 20 \\ \Rightarrow 3^{-1} &\equiv 7 \pmod{20} \end{aligned}$$

Now,

$$\begin{aligned} 3x &\equiv 15 \pmod{20} \\ x &\equiv 3^{-1} \cdot 15 \pmod{20} \\ x &\equiv 5 \pmod{20} \end{aligned}$$

Example 2: Solve $20x \equiv 12 \pmod{36}$

Solution: Since $\gcd(20,36)=4$ (verify by Euclidean Algorithm)

This implies given equation has 4 solutions under modulo 36.

To calculate all the solutions of the equation $20x \equiv 12 \pmod{36}$, we will find the solution of the equation

$$\frac{20}{4}x \equiv \frac{12}{4} \pmod{\frac{36}{4}}$$

$$\implies 5x \equiv 3 \pmod{9}$$

Since $\gcd(5, 9) = 1$, hence 5^{-1} exist in modulo 9. And

$$5^{-1} \equiv 2 \pmod{9}$$

Now,

$$\begin{aligned} 5x &\equiv 3 \pmod{9} \\ \implies x &\equiv 5^{-1} \cdot 3 \pmod{9} \\ \implies x &\equiv 6 \pmod{9} \end{aligned}$$

Therefore $x_0 = 6$ and other solutions are given by

$$6 + \left(\frac{36}{4}\right), 6 + 2\left(\frac{36}{4}\right), 6 + 3\left(\frac{36}{4}\right)$$

i.e solutions of the linear congruence $20x \equiv 12 \pmod{36}$ are

$$x_0 = 6, x_1 = 15, x_2 = 24, x_3 = 33$$

Example 3: Decrypt the following ciphertext which is encrypted by Hill cipher $\mathcal{E}(m) = am \pmod{26}$ over the set of symbols $\mathcal{A} = \{A = 0, B = 1, \dots, Z = 25\}$. $|\mathcal{A}| = 26$

“KAFVMKAFYGC YC FVM HANSIASM QP FVM INYLMZCM”

where ‘a’ is a solution of following congruence equation

$$2x \equiv 6 \pmod{26}$$

Solution: To decrypt the message we will find the solutions of the given linear equation $2x \equiv 6 \pmod{26}$

Since $\gcd(2, 26) = 2$ and $2|6$, hence linear congruence has 2 (in congruent $\text{mod } 26$) solutions.

To find the solutions, first we will find the solution of the following equation

$$\begin{aligned} \frac{2}{2}x &\equiv \frac{6}{2} \pmod{\frac{26}{2}} \\ \implies x &\equiv 3 \pmod{13} \end{aligned}$$

Hence solutions of $2x \equiv 6 \pmod{26}$ are given by

$$x_0 = 3, x_1 = 3 + \left(\frac{26}{2}\right) = 16$$

So we have two values of $a=3$ and $a=16$

First we decrypt the cipher by taking $a=16$

To decrypt the Hill cipher we have to find a^{-1} under modulo 26.

Note that $\gcd(16, 26) \neq 1$, therefore inverse of 16 does not exist in modulo 26. So $a=16$ will not work for decryption.

Now take $a=3$

Since $\gcd(3, 26) = 1$, 3^{-1} exist in modulo 26 and $3^{-1} \equiv 9 \pmod{26}$

Decryption function is $\mathcal{D}(c) = a^{-1}c \equiv 9c \pmod{26}$

$$\mathcal{D}(K) = \mathcal{D}(10) = 9 \cdot 10 = 90 \equiv 12 \pmod{26} = \mathbf{M}$$

$$\mathcal{D}(A) = \mathcal{D}(0) = 9 \cdot 0 \equiv 0 \pmod{26} = \mathbf{A}$$

$$\mathcal{D}(F) = \mathcal{D}(5) = 9 \cdot 5 = 45 \equiv 19 \pmod{26} = \mathbf{T}$$

and so on.

We have plaintext

MATHEMATICS IS THE LANGUAGE OF THE UNIVERSE

4.4 System of Linear Congruence

Theorem: The system of linear congruence

$$ax + by \equiv r \pmod{n} \tag{4.2}$$

$$cx + dy \equiv s \pmod{n} \tag{4.3}$$

has unique solution modulo n whenever $\gcd(ad - bc, n) = 1$.

How to find solution of above system of congruence

Multiply by d in equation(1) and multiply by b in equation(2), we have

$$adx + bdy \equiv dr \pmod{n}$$

$$bcx + bdy \equiv bs \pmod{n}$$

Subtract the above equations, we have

$$(ad - bc)x \equiv (dr - bs) \pmod{n}$$

Suppose $a' = (ad - bc)$ and $b' = (dr - bs)$ then

$$a'x \equiv b' \pmod{n}$$

If $d|b'$ then $a'x \equiv b' \pmod{n}$ has d solutions modulo n where $\gcd(a', n) = d$. To calculate y , put the values of x (congruent $\pmod{n}$) in given linear congruences equation(1) and equation(2), we get values of y (in congruent $\pmod{n}$). So, solution of system of linear congruences is given by order pair (x, y) which satisfies each congruence.

Example: Solve the system of linear congruences

$$5x + 3y \equiv 10 \pmod{12} \quad (4.4)$$

$$2x + 7y \equiv 6 \pmod{12} \quad (4.5)$$

Solution: Note that $\gcd(5 \cdot 7 - 3 \cdot 2, 12) = \gcd(29, 12) = 1$, so solution exists. Multiply equation(3) by 7 and equation(4) by 3, we have

$$35x + 21y \equiv 70 \pmod{12}$$

$$6x + 21y \equiv 18 \pmod{12}$$

Subtract the above equations we have

$$(35 - 6)x + (21 - 21)y \equiv (70 - 18) \pmod{12}$$

$$29x \equiv 52 \pmod{12}$$

$$5x \equiv 4 \pmod{12}$$

since $\gcd(5, 12) = 1 \implies 5^{-1} \pmod{12}$ exists and $5^{-1} = 5 \pmod{12}$

So,

$$x \equiv 5^{-1} \cdot 4 \equiv 5 \cdot 4 \equiv 20 \equiv 8 \pmod{12}$$

Now put the value of x in equation(3) we have

$$5 \cdot 8 + 3y \equiv 10 \pmod{12}$$

$$40 + 3y \equiv 10 \pmod{12}$$

$$3y \equiv (10 - 40) \equiv -30 \pmod{12}$$

$$3y \equiv 6 \pmod{12}$$

Note that $\gcd(3, 12) = 3$ and $3|6$, So above linear congruence has 3 solutions
To find all solutions first we solve following linear congruence

$$\left(\frac{3y}{3}\right) \equiv \left(\frac{6}{3}\right) \pmod{\frac{12}{3}}$$

$$y \equiv 2 \pmod{4}$$

Hence all 3 solutions are $2, 2 + \left(\frac{12}{3}\right), 2 + 2 \cdot \left(\frac{12}{3}\right)$ i.e. 2, 6, 10 modulo 12.
Solutions (x, y) for equation(3) are $(8, 2), (8, 6)$ and $(8, 10)$.

Now put value of $x \equiv 8 \pmod{12}$ in equation(4), we have

$$2 \cdot 8 + 7y \equiv 6 \pmod{12}$$

$$16 + 7y \equiv 6 \pmod{12}$$

$$7y \equiv (6 - 16) \equiv -10 \pmod{12}$$

$$7y \equiv 2 \pmod{12}$$

Also, $\gcd(7, 12) = 1$ and $7^{-1} = 7 \pmod{12}$

So

$$y \equiv 7^{-1} \cdot 2 \equiv 7 \cdot 2 \equiv 14 \pmod{12}$$

$$y \equiv 2 \pmod{12}$$

Alternate way to calculate y , Multiply equation(3) by 2 and equation(4) by 5, we have

$$10x + 6y \equiv 20 \pmod{12}$$

$$10x + 35y \equiv 30 \pmod{12}$$

Subtract the above equations we have

$$\begin{aligned}
 (10 - 10)x + (35 - 6)y &\equiv (30 - 20) \pmod{12} \\
 29y &\equiv 10 \pmod{12} \\
 5y &\equiv 10 \pmod{12} \\
 y &\equiv 5^{-1} \cdot 10 \pmod{12} \quad (\because 5^{-1} = 5 \pmod{12}) \\
 y &\equiv 5 \cdot 10 \pmod{12} \\
 y &\equiv 50 \equiv 2 \pmod{12}
 \end{aligned}$$

Hence Solution for equation(4) is (8, 2).

Note that only the pair $(x, y) = (8, 2)$ is solution for both linear congruences. Therefore solution of given system of linear congruence is (8,2) i.e $x \equiv 8 \pmod{12}$, $y \equiv 2 \pmod{12}$.

Question: Decrypt the following cipher encrypted by affine cipher

NDXBHO

Let $\mathcal{A} = \{A = 0, B = 1 \dots, Y = 24, Z = 25, \text{blank space}(_) = 26\}$ be set of symbols. Suppose that frequency analysis is applied on digraphs, frequency analysis on cipher text tells that the most occurring digraphs (in decreasing order) : 'ZA', 'IA', 'IW'.

Also, suppose 'E_', 'S_', ' _T' appear maximum number of time (in the descending order) in the plain text.

Solution: Divide the cipher text into digraphs

$c_1=ND, c_2=XB, c_3=HO$

From given Frequency analysis of Plain text and cipher text, most occurring digraph in cipher text corresponds to most occurring digraph in plain text (in decreasing order) i.e.

$\mathcal{D}(ZA)=E_, \mathcal{D}(IA)=S_ \text{ and } \mathcal{D}(IW)=_T$

$|\mathcal{A}| = 27$, we are working on digraphs so $n = 27 \times 27 = 729$

Numerical equivalents of digraphs

$$ZA = 25 \cdot 27^1 + 0 \cdot 27^0 = 675 \pmod{729}$$

$$IA = 8 \cdot 27^1 + 0 \cdot 27^0 = 216 \pmod{729}$$

$$IW = 8 \cdot 27^1 + 22 \cdot 27^0 = 238 \pmod{729}$$

$$E_- = 4 \cdot 27^1 + 26 \cdot 27^0 = 134 \pmod{729}$$

$$S_- = 18 \cdot 27^1 + 26 \cdot 27^0 = 512 \pmod{729}$$

$$_T = 26 \cdot 27^1 + 19 \cdot 27^0 = 721 \pmod{729}$$

Decryption function of affine cipher is defined by

$$\mathcal{D}(c) = ca' + b' \pmod{n}$$

by given information we have following 3 equations

$$\mathcal{D}(ZA) = E_- \implies \mathcal{D}(675) = 134 \pmod{729} \text{ i.e. } 675a' + b' \equiv 134 \pmod{729} - (1)$$

$$\mathcal{D}(IA) = S_- \implies \mathcal{D}(216) = 512 \pmod{729} \text{ i.e. } 216a' + b' \equiv 512 \pmod{729} - (2)$$

$$\mathcal{D}(IW) = _T \implies \mathcal{D}(238) = 721 \pmod{729} \text{ i.e. } 238a' + b' \equiv 721 \pmod{729} - (3)$$

Subtract, equation(2) from equation(1), We have

$$(675 - 216)a' \equiv (134 - 512) \equiv -378 \pmod{729}$$

$$\text{i.e. } 459a' \equiv 351 \pmod{729}$$

We calculate $\gcd(459, 729)$:

$729 = 1 \times 459 + 270$	$q_1 = 1, r_1 = 270$
$459 = 1 \times 270 + 189$	$q_2 = 1, r_2 = 189$
$270 = 1 \times 189 + 81$	$q_3 = 1, r_3 = 81$
$189 = 2 \times 81 + 27$	$q_4 = 2, r_4 = 27$
$81 = 3 \times 27 + 0$	$q_5 = 3, r_5 = 0$

$\gcd(459, 729) = 27$, So 459^{-1} does not exist modulo 729.

So $459a' \equiv 351 \pmod{729}$ has 27 solutions modulo 729 obtained by solution of equation

$$\left(\frac{459a'}{27} \right) \equiv \frac{351}{27} \pmod{\frac{729}{27}} \text{ i.e. } 17a' \equiv 13 \pmod{27}$$

Note that by Extended Euclidean Algorithm $17^{-1} \equiv 8 \pmod{27}$

So

$$a' \equiv 17^{-1} \cdot 13 \pmod{27} \equiv 8 \cdot 13 \equiv 23 \pmod{27}$$

Hence solutions of $459a' \equiv 351 \pmod{729}$ are

$$a' \equiv 23 + 27k \pmod{729} \text{ for } k = 0, 1, 2, \dots, 26$$

In particular for $k = 13$ $a' = 374$

Now subtract equation (3) from equation(1), We have

$$(675 - 238)a' \equiv (134 - 721) \pmod{729}$$

$$437a' \equiv -587 \pmod{729}$$

$$437a' \equiv 142 \pmod{729}$$

$$a' \equiv 437^{-1} \cdot 142 \pmod{729} \text{ (if } 437^{-1} \pmod{729} \text{ exists)}$$

Apply Euclidean algorithm to calculate $\gcd(437, 729)$

$$729 = 1 \times 437 + 292 \quad q_1 = 1 \quad r_1 = 292$$

$$437 = 1 \times 292 + 145 \quad q_2 = 1 \quad r_2 = 145$$

$$292 = 2 \times 145 + 2 \quad q_3 = 2 \quad r_3 = 2$$

$$145 = 72 \times 2 + 1 \quad q_4 = 72 \quad r_4 = 1$$

$$2 = 2 \times 1 + 0 \quad q_5 = 2 \quad r_5 = 0$$

$\gcd(437, 729) = 1$, hence $437^{-1} \pmod{729}$ exists

and

$$437^{-1} = 362 \pmod{729} \text{ (by Extended Euclidean Algorithm)}$$

$437a' \equiv 142 \pmod{729}$ has unique solution.

$$a' \equiv 437^{-1} \cdot 142 \equiv 362 \cdot 142 \equiv 51404 \equiv 374 \pmod{729}$$

Now put the value of $a' = 374 \pmod{729}$ in equation(1),(2),(3)

$$b' \equiv 134 - (675 \cdot 374) \equiv 647 \pmod{729} \text{ From equation (1)}$$

$$b' \equiv 512 - (216 \cdot 374) \equiv 647 \pmod{729} \text{ From equation (2)}$$

$$b' \equiv 721 - (238 \cdot 374) \equiv 647 \pmod{729} \text{ From equation (3)}$$

So solution of system of linear equation (1),(2) and (3) is

$$a' = 374 \pmod{729} \quad b' = 647 \pmod{729}$$

We have cipher text $c_1=ND$, $c_2=XB$, $c_3=HO$, Decryption key is $(a', b')=(374, 647)$ and decryption function is $\mathcal{D}(c) = a'c + b' \equiv 374 \cdot c + 647 \pmod{729}$

$$c_1 = ND = 13 \cdot 27^1 + 3 \cdot 27^0 = 354 \pmod{729}$$

$$c_2 = XB = 23 \cdot 27^1 + 1 \cdot 27^0 = 622 \pmod{729}$$

$$c_3 = HO = 7 \cdot 27^1 + 14 \cdot 27^0 = 203 \pmod{729}$$

decryptions are

$$\mathcal{D}(c_1) = \mathcal{D}(354) = 374 \cdot 354 + 647 \equiv 365 \pmod{729} = 13 \cdot 27^1 + 14 \cdot 27^0 = NO$$

$$\mathcal{D}(c_2) = \mathcal{D}(622) = 374 \cdot 622 + 647 \equiv 724 \pmod{729} = 26 \cdot 27^1 + 22 \cdot 27^0 = _W$$

$$\mathcal{D}(c_3) = \mathcal{D}(203) = 374 \cdot 203 + 647 \equiv 24 \pmod{729} = 0 \cdot 27^1 + 24 \cdot 27^0 = AY$$

Therefore decrypted text is “NO_WAY” i.e “NO WAY”

4.5 Chinese Remainder Theorem:

Statement: Let $a_1, a_2, \dots, a_n \in \mathbb{Z}$ and $m_1, m_2, \dots, m_n \in \mathbb{Z}$ such that

$$x \equiv a_1 \pmod{m_1},$$

$$x \equiv a_2 \pmod{m_2},$$

$$\cdot$$

$$\cdot$$

$$\cdot$$

$$x \equiv a_n \pmod{m_n}$$

and $m_1, m_2, \dots, m_n$ are pairwise relatively prime, then there exist a unique solution *modulo* m where $m = m_1 m_2 \cdots m_n$.

Proof: Write $m = \prod_{i=1}^n m_i$ and $M_i = \frac{m}{m_i}$

Observe that $(M_i, m_i) = 1 \quad \forall i = 1, 2, \dots, n$.

So $y_i \equiv M_i^{-1} \pmod{m_i}$ exist $\forall i = 1, 2, \dots, n$ i.e $y_i M_i \equiv 1 \pmod{m_i} \quad \forall i =$

$1, 2, \dots, n$.

Let $x = \sum_{i=1}^n a_i y_i M_i \pmod{m}$.

We claim that x is a solution of the system of congruence, $x \equiv a_i \pmod{m_i} \forall i = 1, 2, \dots, n$.

For $x = \sum_{i=1}^n a_i y_i M_i \pmod{m}$

$$\implies x = mk + \sum_{i=1}^n a_i y_i M_i \text{ for some } k \in \mathbb{Z}$$

$$\implies x \equiv \sum_{i=1}^n a_i y_i M_i \pmod{m_i} \quad \forall i = 1, 2, \dots, n \quad (\text{Since } m_i | mk)$$

$$\implies x \equiv a_i y_i M_i \pmod{m_i} \quad \forall i = 1, 2, \dots, n \quad (\text{Since } m_i | M_j \text{ for } i \neq j)$$

$$\implies x \equiv a_i \pmod{m_i} \quad \forall i = 1, 2, \dots, n \quad (\text{Since } y_i M_i \equiv 1 \pmod{m_i})$$

Example: Solve the simultaneous congruences

$$x \equiv 3 \pmod{5},$$

$$x \equiv 1 \pmod{7},$$

$$x \equiv 6 \pmod{8}$$

Solution: Since 5, 7, 8 are pairwise relatively prime, the Chinese Remainder Theorem tells us that there is a unique solution modulo m , where $m = 5 \cdot 7 \cdot 8 = 280$.

Let $a_1 = 3, m_1 = 5, a_2 = 1, m_2 = 7, a_3 = 6, m_3 = 8$.

Now

$$M_1 = \frac{m}{m_1} = \frac{280}{5} = 56$$

$$M_2 = \frac{m}{m_2} = \frac{280}{7} = 40$$

$$M_3 = \frac{m}{m_3} = \frac{280}{8} = 35$$

$$y_1 \equiv M_1^{-1} \pmod{m_1} \equiv 56^{-1} \pmod{5} \equiv 1^{-1} \pmod{5} \equiv 1 \pmod{5}$$

$$y_2 \equiv M_2^{-1} \pmod{m_2} \equiv 40^{-1} \pmod{7} \equiv 5^{-1} \pmod{7} \equiv 3 \pmod{7}$$

$$y_3 \equiv M_3^{-1} \pmod{m_3} \equiv 35^{-1} \pmod{8} \equiv 3^{-1} \pmod{8} \equiv 3 \pmod{8}$$

The solution, which is unique modulo 280, is

$$\begin{aligned} x &\equiv a_1y_1M_1 + a_2y_2M_2 + a_3y_3M_3 \pmod{m} \\ x &\equiv 3 \cdot 1 \cdot 56 + 1 \cdot 3 \cdot 40 + 6 \cdot 3 \cdot 35 \pmod{280} \\ x &\equiv 78 \pmod{280} \end{aligned}$$

4.6 Some Applications to Factoring

Result 1. Suppose that $\gcd(b, m) = 1$ and $a, c \in \mathbb{Z}^+$. If $b^a \equiv 1 \pmod{m}$ and $b^c \equiv 1 \pmod{m}$ and if $d = \gcd(a, c)$, then $b^d \equiv 1 \pmod{m}$.

Proof: Since $\gcd(a, c) = d$, Using Euclidean Algorithm, We can write

$$d = ax + cy \text{ for some } x, y \in \mathbb{Z}.$$

Since, $x, y \in \mathbb{Z}$ and $ax + cy = d$, one can observe that one of them must be positive and other must be zero or negative.

Without loss of generality, Suppose $x > 0, y < 0$

Since,

$$b^a \equiv 1 \pmod{m} \implies b^{ax} \equiv 1 \pmod{m} \quad (4.6)$$

Similarly,

$$b^c \equiv 1 \pmod{m} \implies b^{-cy} \equiv 1 \pmod{m} \quad (4.7)$$

Divide the congruence of equation (4.6) and (4.7), we get

$$b^{ax-(c(-y))} \equiv 1 \pmod{m} \implies b^d \equiv 1 \pmod{m}$$

Result 2. Let p be a prime such that $p \nmid b^n - 1$. Then either $p \mid b^d - 1$ for some proper divisor d of n or $p \equiv 1 \pmod{n}$. Also, if $p > 2$ and n is odd, then $p \equiv 1 \pmod{2n}$ also holds.

Proof: It is given that $p \nmid b^n - 1$ that is $b^n - 1 \equiv 0 \pmod{p} \implies b^n \equiv 1 \pmod{p}$. Let $d = \gcd(n, p - 1)$.

Case 1. If $d < n$, then $d \mid n$ and $b^{p-1} \equiv 1 \pmod{p}$, Using above result, we have $b^d \equiv 1 \pmod{p}$.

Case 2. If $d = n$, then $n = \gcd(n, p - 1) \implies n \mid p - 1 \implies p \equiv 1 \pmod{n}$. Also, if $p > 2$ and n is odd, then $n \mid p - 1$ and $2 \mid p - 1$ implies $p \equiv 1 \pmod{2n}$.

Result 3. Let n be a positive integer and b be any integer, then $b^n - 1$ is

divisible by $b - 1$ with quotient $b^{n-1} + b^{n-2} + \dots + b^2 + b + 1$.
Moreover, for any positive integer m, n and any integer b ,

$$b^{mn} - 1 = (b^m - 1)(b^{m(n-1)} + b^{m(n-2)} + \dots + b^{2m} + b^m + 1)$$

Proof: Proof is left to the readers

Question: Prove that if $d = \gcd(m, n)$ and $a > 1$ is an integer, then $\gcd(a^m - 1, a^n - 1) = a^d - 1$.

Proof: Since $\gcd(m, n) = d$, then $\exists m_1, n_1 \in \mathbb{Z}$, such that $m = m_1d$ and $n = n_1d$ such that $\gcd(m_1, n_1) = 1$.

Using above result, we have

$$a^m - 1 = a^{m_1d} - 1 = (a^d - 1)f_{m_1}(a)$$

and

$$a^n - 1 = a^{n_1d} - 1 = (a^d - 1)g_{n_1}(a)$$

where $f_{m_1}(x) = x^{d(m_1-1)} + x^{d(m_1-2)} + \dots + 1$ and $g_{n_1}(x) = x^{d(n_1-1)} + x^{d(n_1-2)} + \dots + 1$.

Also note that $\gcd(f_{m_1}(x), g_{n_1}(x)) = 1$ (Prove it).

Thus $\gcd(a^m - 1, a^n - 1) = a^d - 1$.

4.7 Exercise

1. Find greatest common divisor of 20223 and 29121 using Euclidean Algorithm.
2. Find greatest common divisor d of 127 and 89 and find integers x and y solving the equation $127x + 89y = d$.
3. Consider the word 'FADE'. Represent this word in decimal form. Then convert it in binary form.
4. Compute:
 - (a) $\phi(12285)$
 - (b) $\phi(6195)$

(c) $\phi(31752)$

5. Find the value of x if $2^{1571} \equiv x \pmod{197}$.
6. Find the value of x if $5^{1825} \equiv x \pmod{119}$.
7. Prove that $x_1 \equiv x_2 \pmod{pq}$ if and only if $x_1 \equiv x_2 \pmod{p}$ and $x_1 \equiv x_2 \pmod{q}$ for distinct primes p and q .
8. Let, either m be an odd integer such that $m = p^\alpha$, or $m = 2p$ for some prime $p \neq 2$. If $x^2 \equiv 1 \pmod{m}$, then prove that $x \equiv 1 \pmod{m}$ or $x \equiv -1 \pmod{m}$.
9. Find an integer solution of Diophantine equation $17x + 21y = 1$.
10. Prove that $n^5 \equiv n \pmod{30}$.
11. Solve the following systems of simultaneous congruence:
 - (a) $2x + 3y \equiv 1 \pmod{26}$
 $7x + 8y \equiv 2 \pmod{26}$
 - (b) $2x + 3y \equiv 2 \pmod{21}$
 $7x + 8y \equiv 10 \pmod{21}$
12. If $a \equiv b \pmod{m}$ and $a \equiv b \pmod{n}$, then prove that $a \equiv b \pmod{l.c.m.(m, n)}$.
13. Solve the following systems of simultaneous congruence:
 - (a) $x \equiv 2 \pmod{7}$
 $x \equiv 7 \pmod{11}$
 $x \equiv 9 \pmod{17}$
 - (b) $x \equiv 3 \pmod{17}$
 $x \equiv 10 \pmod{19}$
 $x \equiv 13 \pmod{21}$
 - (c) $27x \equiv 25 \pmod{256}$
 $27x \equiv 72 \pmod{900}$
 $103x \equiv 612 \pmod{676}$
14. Solve the following systems of simultaneous congruence:

(a) $x \equiv 15 \pmod{20}$
 $x \equiv 7 \pmod{30}$

(b) $x + y + z = 100$
 $x + 8y + 50z = 156$

15. Prove Chinese Remainder's Theorem by induction on the number of congruences.

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Chapter 5

Modern Cryptography

So far, we have studied some classical cipher. There are many other classical ciphers we have not learned. Classical ciphers are not secure with present advanced technologies. Now a days, Classical ciphers are not in use. Cryptographers have developed modern encryption techniques that are secure with modern technology, especially Public Key Cryptosystem. In classical ciphers, encryption and decryption keys are the same and known to sender and receiver both and no one else for secure communication.

Public Key Cryptosystem:

In Public Key Cryptosystem, two keys are used, one public key and a second private key. The public key is used for encryption, and the private key is used for decryption.

Suppose Alice needs some information from Bob, then Alice will make two keys, one public and one private. The public key will be known to all, but the private key will not be known to anyone (not even to Bob) except Alice. Now anyone can send an encrypted message (encryption using Alice's public key) to Alice, and encrypted messages are public. But only Alice can decrypt the message with her private key.

In this chapter, we will study some modern cryptography, especially public-key cryptography.

5.1 Silver Pohlig - Hellman Exponentiation Cipher

Let p be a prime and e (called exponent) an integer such that $0 < e < p$ and $\gcd(e, p-1) = 1$.

Encryption: The encryption function is defined by

$$\begin{aligned}\mathcal{E}: \mathbb{Z}_p &\longrightarrow \mathbb{Z}_p \\ m &\longmapsto c = \mathcal{E}(m) \equiv m^e \pmod{p}\end{aligned}$$

Decryption: The decryption function is defined by

$$\begin{aligned}\mathcal{D}: \mathbb{Z}_p &\longrightarrow \mathbb{Z}_p \\ c &\longmapsto \mathcal{D}(c) \equiv c^d \pmod{p}\end{aligned}$$

Where $d \equiv e^{-1} \pmod{(p-1)}$, such d exists since $\gcd(e, p-1) = 1$ and can be determined by using Extended Euclidean Algorithm.

Now observe that, If $c = \mathcal{E}(m)$ then $c \equiv m^e \pmod{p}$.

Therefore $\mathcal{D}(c) \equiv c^d \equiv (m^e)^d \equiv m^{ed} \equiv m^{1+k(p-1)} \equiv m(m^{p-1})^k \pmod{p}$

for some $k \in \mathbb{Z}$, since $ed \equiv 1 \pmod{(p-1)}$.

This gives that $\mathcal{D}(c) \equiv c^d \equiv m \pmod{p}$ as $m^{p-1} \equiv 1 \pmod{p}$ by Fermat's little theorem.

Example: Discuss Encryption and Decryption of the plain text.

HELLO HOW ARE YOU?

Solution: Encryption: To encrypt this message take

$\mathcal{A} = \{ A = 0, B = 1, \dots, Z = 25, _ = 26, ? = 27, . = 28 \}$

So $p = 29 = |\mathcal{A}|$ and take $e = 3$.

Observe that $\gcd(e, p-1) = 1$ i.e $(3, 28) = 1$.

$$\begin{aligned}\mathcal{E}(H) &= \mathcal{E}(7) = 7^3 \pmod{29} \\ &= 343 \pmod{29} \\ &= 24 \pmod{29} = \mathbf{Y}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(E) &= \mathcal{E}(4) \equiv 4^3 \pmod{29} \\ &\equiv 64 \pmod{29} \\ &\equiv 6 \pmod{29} = \mathbf{G}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(L) &= \mathcal{E}(11) \equiv 11^3 \pmod{29} \\ &\equiv 1331 \pmod{29} \\ &\equiv 26 \pmod{29} = \mathbf{-}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(O) &= \mathcal{E}(14) \equiv 14^3 \pmod{29} \\ &\equiv 2744 \pmod{29} \\ &\equiv 18 \pmod{29} = \mathbf{S}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(_) &= \mathcal{E}(26) \equiv 26^3 \pmod{29} \\ &\equiv (-3)^3 \pmod{29} \\ &\equiv 2 \pmod{29} = \mathbf{C}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(W) &= \mathcal{E}(22) = 22^3 \pmod{29} \\ &\equiv (-7)^3 \pmod{29} \\ &\equiv 5 \pmod{29} = \mathbf{F}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(A) &= \mathcal{E}(0) \equiv 0^3 \pmod{29} \\ &\equiv 0 \pmod{29} = \mathbf{A}\end{aligned}$$

$$\begin{aligned}
\mathcal{E}(R) = \mathcal{E}(17) &\equiv 17^3 \pmod{29} \\
&\equiv 4913 \pmod{29} \\
&\equiv 12 \pmod{29} = \mathbf{M}
\end{aligned}$$

$$\begin{aligned}
\mathcal{E}(Y) = \mathcal{E}(24) &\equiv 24^3 \pmod{29} \\
&\equiv (-5)^3 \pmod{29} \\
&\equiv 20 \pmod{29} = \mathbf{U}
\end{aligned}$$

$$\begin{aligned}
\mathcal{E}(U) = \mathcal{E}(20) &\equiv 20^3 \pmod{29} \\
&\equiv (-9)^3 \pmod{29} \\
&\equiv 25 \pmod{29} = \mathbf{Z}
\end{aligned}$$

$$\begin{aligned}
\mathcal{E}(?) = \mathcal{E}(27) &\equiv 27^3 \pmod{29} \\
&\equiv (-2)^3 \pmod{29} \\
&\equiv 21 \pmod{29} = \mathbf{V}
\end{aligned}$$

Encrypted message is **YG__SCYSFCAMGCUSZV**

Decryption: For decryption we have to find d .
We know that $d \equiv e^{-1} \pmod{(p-1)}$. Here $e = 3, p = 29$.

$$\begin{aligned}
d &\equiv 3^{-1} \pmod{28} \\
d &\equiv 19 \pmod{28}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(Y) = \mathcal{D}(Y) &\equiv 24^{19} \pmod{29} \\
24 &\equiv (-5) \pmod{29} \\
24^2 &\equiv 25 \equiv (-4) \pmod{29} \\
24^4 &\equiv 16 \pmod{29} \\
24^6 &\equiv (-64) \equiv (-6) \pmod{29} \\
24^7 &\equiv 1 \pmod{29} \\
24^{14} &\equiv 1 \pmod{29} \\
24^{18} &\equiv 16 \pmod{29} \\
24^{19} &\equiv (-80) \equiv 7 \pmod{29} = \mathbf{H}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(G) = \mathcal{D}(G) &\equiv 6^{19} \pmod{29} \\
6^2 &\equiv 7 \pmod{29} \\
6^3 &\equiv 13 \pmod{29} \\
6^4 &\equiv 20 \pmod{29} \\
6^7 &\equiv 1 \pmod{29} \\
6^{14} &\equiv 1 \pmod{29} \\
6^{19} &\equiv 4 \pmod{29} = \mathbf{E}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(_) = \mathcal{D}(_) &\equiv 26^{19} \pmod{29} \\
26 &\equiv (-3) \pmod{29} \\
26^3 &\equiv (-27) \equiv 2 \pmod{29} \\
26^{18} &\equiv 6 \pmod{29} \\
26^{19} &\equiv 11 \pmod{29} = \mathbf{L}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(S) = \mathcal{D}(S) &\equiv 18^{19} \pmod{29} \\
18 &\equiv (-11) \pmod{29} \\
18^2 &\equiv 5 \pmod{29} \\
18^4 &\equiv (-4) \pmod{29} \\
18^7 &\equiv (-12) \pmod{29} \\
18^{14} &\equiv (-1) \pmod{29} \\
18^{18} &\equiv 4 \pmod{29} \\
18^{19} &\equiv 14 \pmod{29} = \mathbf{O}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(C) = \mathcal{D}(C) &\equiv 2^{19} \pmod{29} \\
2^7 &\equiv 12 \pmod{29} \\
2^{14} &\equiv (-1) \pmod{29} \\
2^{19} &\equiv 26 \pmod{29} = _
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(F) = \mathcal{D}(F) &\equiv 5^{19} \pmod{29} \\
5^2 &\equiv (-4) \pmod{29} \\
5^4 &\equiv 16 \pmod{29} \\
5^7 &\equiv (-1) \pmod{29} \\
5^{14} &\equiv 1 \pmod{29} \\
5^{19} &\equiv 22 \pmod{29} = \mathbf{W}
\end{aligned}$$

$$\mathcal{D}(A) = \mathcal{D}(A) = 0^{19} \pmod{29} = \mathbf{A}$$

$$\begin{aligned}
\mathcal{D}(M) = \mathcal{D}(M) &\equiv 12^{19} \pmod{29} \\
12^2 &\equiv (-1) \pmod{29} \\
12^{18} &\equiv (-1) \pmod{29} \\
12^{19} &\equiv 17 \pmod{29} = \mathbf{R}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(U) = \mathcal{D}(U) &\equiv 20^{19} \pmod{29} \\
20 &\equiv (-9) \pmod{29} \\
20^2 &\equiv (-6) \pmod{29} \\
20^4 &\equiv 7 \pmod{29} \\
20^7 &\equiv 1 \pmod{29} \\
20^{14} &\equiv 1 \pmod{29} \\
20^{19} &\equiv 24 \pmod{29} = \mathbf{Y}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(Z) &= \mathcal{D}(Z) \equiv 25^{19} \pmod{29} \\
25 &\equiv (-4) \pmod{29} \\
25^2 &\equiv 16 \pmod{29} \\
25^4 &\equiv (-5) \pmod{29} \\
25^7 &\equiv 1 \pmod{29} \\
25^{14} &\equiv 1 \pmod{29} \\
25^{19} &\equiv 20 \pmod{29} = \mathbf{U}
\end{aligned}$$

$$\begin{aligned}
\mathcal{D}(V) &= \mathcal{D}(V) \equiv 21^{19} \pmod{29} \\
21 &\equiv (-8) \pmod{29} \\
21^2 &\equiv 6 \pmod{29} \\
21^4 &\equiv 7 \pmod{29} \\
21^7 &\equiv 12 \pmod{29} \\
21^{14} &\equiv (-1) \pmod{29} \\
21^{19} &\equiv 27 \pmod{29} = ?
\end{aligned}$$

Decryption of cipher text is **HELLO HOW ARE YOU?**

Silver Pohlig - Hellman Exponentiation Cipher was developed in 1976. This is not considered to be a public-key cryptosystem because the public key and private key are the same, which is (p, e) .

From the idea of Silver Pohlig - Hellman Exponentiation, RSA cryptosystem was developed, which is the most secure public-key cryptosystem till today.

5.2 RSA Cryptosystem

RSA cryptosystem (from the last names of the inventors' Rivest, Shamir, and Adleman) was developed after one year of Silver Pohlig - Hellman Exponentiation Cipher in 1977, which is one of the most popular public-key cryptosystems, is based on the tremendous difficulty of factorization of integers.

Encryption and Decryption Techniques : Let p and q be two distinct large prime numbers and the product $n = pq$. Also let e such that $0 < e < n$ and $\gcd(e, \phi(n)) = 1$ i.e $\gcd(e, (p-1)(q-1)) = 1$.

Now (p, q) is private key and make (n, e) public key.

Encryption: The encryption function is defined by

$$\begin{aligned}\mathcal{E}: \mathbb{Z}_n &\longrightarrow \mathbb{Z}_n \\ m &\longmapsto c = \mathcal{E}(m) \equiv m^e \pmod{n}\end{aligned}$$

Decryption: The decryption function is defined by

$$\begin{aligned}\mathcal{D}: \mathbb{Z}_n &\longrightarrow \mathbb{Z}_n \\ c &\longmapsto m = \mathcal{D}(c) \equiv c^d \pmod{n}\end{aligned}$$

where $d \equiv e^{-1} \pmod{(p-1)(q-1)}$ exists, since $\gcd(e, (p-1)(q-1)) = 1$

Recall

1. **Euler totient function:-** Let n be a positive integer. The Euler totient function $\phi(n)$ is defined to be the number of non-negative integers b less than n which are prime to n :

$$\phi(n) = |\{0 \leq b < n \mid \gcd(b, n) = 1\}|$$

2. **Euler's Theorem:-** It states that if a and n is a positive integer such that $\gcd(a, n) = 1$ then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Validation of Encryption and Decryption

If $\mathcal{E}(m) = c \equiv m^e \pmod{n}$

Case-1 If $\gcd(m, n) = 1$ then By Euler's theorem

$$\begin{aligned}m^{\phi(n)} &\equiv 1 \pmod{n} \\ m^{(p-1)(q-1)} &\equiv 1 \pmod{n} \quad \text{since } n = pq\end{aligned}$$

Note that $d \equiv e^{-1} \pmod{(p-1)(q-1)} \implies ed = 1 + k(p-1)(q-1)$ for some integer k .

$$\begin{aligned}\text{So, } \mathcal{D}(c) &\equiv c^d \pmod{n} \\ &\equiv (m^e)^d \pmod{n} \\ &\equiv m^{ed} \pmod{n} \equiv m^{1+k(p-1)(q-1)} \equiv m \left(m^{(p-1)(q-1)}\right)^k \equiv m \pmod{n}\end{aligned}$$

Case-2 If $\gcd(m, n) \neq 1$ then $\gcd(m, n)$ can be either p or q . So we can assume WLOG $\gcd(m, n) = q$ then $q|m$ and

$$m^r \equiv m \equiv 0 \pmod{q} \text{ for any } r > 0$$

$$\Rightarrow m^{(p-1)(q-1)} \equiv m \pmod{q}$$

Also observe that $p \nmid m$ i.e $\gcd(m, p) = 1 \Rightarrow m^{(p-1)} \equiv 1 \pmod{p}$.

Now,

$$c^d \equiv m^{ed} \equiv m^{1+k(p-1)(q-1)} \equiv m \left(m^{(q-1)}\right)^{k(p-1)} \equiv m \pmod{q}$$

$$c^d \equiv m^{ed} \equiv m^{1+k(p-1)(q-1)} \equiv m \left(m^{(p-1)}\right)^{k(q-1)} \equiv m \pmod{p}$$

So we get,

$$c^d \equiv m \pmod{q}$$

$$c^d \equiv m \pmod{p}$$

Let $c^d = x$ then we have

$$x \equiv m \pmod{q}$$

$$x \equiv m \pmod{p}$$

Since p and q are distinct primes. By chinese remainder theorem, we have

$$x \equiv m \pmod{pq}$$

i.e.

$$c^d \equiv m \pmod{pq}$$

i.e $\mathcal{D}(c) \equiv m \pmod{n}$

Example: Plain Text:- “CRYPTOGRAPHY”

$\mathcal{A} = \{A = 0, B = 1, \dots, Z = 25, _ = 26, ? = 27, . = 28, @ = 29, ! = 30, \chi = 31, \backslash = 32\}$

Let $p = 3, q = 11$ and $e = 3$, observe that $0 < e < n = 33$, $\phi(n = 3 \cdot 11) = 20$ and $\gcd(e, \phi(n)) = 1$ i.e $\gcd(3, 20) = 1$

Encryption:

$$\begin{aligned}\mathcal{E}(C) &= \mathcal{E}(2) = 2^3 \pmod{33} \\ &= 8 \pmod{33} = \mathbf{I}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(R) &= \mathcal{E}(17) = 17^3 \pmod{33} \\ &= 29 \pmod{33} = \mathbf{@}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(Y) &= \mathcal{E}(24) = 24^3 \pmod{33} \\ &= 30 \pmod{33} = \mathbf{!}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(P) &= \mathcal{E}(15) = 15^3 \pmod{33} \\ &= 9 \pmod{33} = \mathbf{J}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(T) &= \mathcal{E}(19) = 19^3 \pmod{33} \\ &= 28 \pmod{33} = \mathbf{.}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(O) &= \mathcal{E}(14) = 14^3 \pmod{33} \\ &= 5 \pmod{33} = \mathbf{F}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(G) &= \mathcal{E}(6) = 6^3 \pmod{33} \\ &= 18 \pmod{33} = \mathbf{S}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(A) &= \mathcal{E}(0) = 0^3 \pmod{33} \\ &= 0 \pmod{33} = \mathbf{A}\end{aligned}$$

$$\begin{aligned}\mathcal{E}(H) &= \mathcal{E}(7) = 7^3 \pmod{33} \\ &= 13 \pmod{33} = \mathbf{N}\end{aligned}$$

Encrypted message is “I@!J.FS@AJN!”

Decryption:

$$\begin{aligned}\mathcal{D}(I) = \mathcal{D}(8) &\equiv 8^7 \pmod{33} \\ &\equiv 2 \pmod{33} = \mathbf{C}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(@) = \mathcal{D}(29) &\equiv 29^7 \pmod{33} \\ &\equiv 17 \pmod{33} = \mathbf{R}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(!) = \mathcal{D}(30) &\equiv 30^7 \pmod{33} \\ &\equiv 24 \pmod{33} = \mathbf{Y}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(J) = \mathcal{D}(9) &\equiv 9^7 \pmod{33} \\ &\equiv 15 \pmod{33} = \mathbf{P}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(.) = \mathcal{D}(28) &\equiv 28^7 \pmod{33} \\ &\equiv 19 \pmod{33} = \mathbf{T}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(F) = \mathcal{D}(5) &\equiv 5^7 \pmod{33} \\ &\equiv 14 \pmod{33} = \mathbf{O}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(S) = \mathcal{D}(18) &\equiv 18^7 \pmod{33} \\ &\equiv 6 \pmod{33} = \mathbf{G}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(A) = \mathcal{D}(0) &\equiv 0^7 \pmod{33} \\ &\equiv 0 \pmod{33} = \mathbf{A}\end{aligned}$$

$$\begin{aligned}\mathcal{D}(N) = \mathcal{D}(13) &\equiv 13^7 \pmod{33} \\ &\equiv 7 \pmod{33} = \mathbf{H}\end{aligned}$$

Decrypted message is “CRYPTOGRAPHY”

5.2.1 Cryptanalysis of RSA

RSA is a Public key cryptosystem.

Alice
 Public (n, e)
 Private (p, q)
 and $n = pq$

To send a message to Alice, say the message $m \in \mathbb{Z}_n$
 send her $c = m^e \pmod{n}$.

She decrypts the message $D(c) = c^d \pmod{n}$

She compute the decryption key d as

$$d = e^{-1} \pmod{(\phi(n))} = e^{-1} \pmod{(p-1)(q-1)}$$

To know d , one need to know $(p-1)(q-1) = \phi(n)$

When p and q are known then $\phi(n) = (p-1)(q-1)$ can be easily calculated.

Conversely when n and $\phi(n)$ are known then p and q can be calculated as follows

p, q are the roots of the equation

$$x^2 - (p+q)x + pq = 0$$

or that

$$x^2 - (p+q)x + n = 0$$

since $n = pq$ or

$$x^2 + (\phi(n) - n - 1)x + n = 0$$

Since $\phi(n) = pq - p - q + 1 = n - (p+q) + 1$

That is, when n and $\phi(n)$ is known then p and q are roots of the equation $x^2 + (\phi(n) - n - 1)x + n = 0$ such that $n = pq$.

5.3 Quadratic Cipher

Suppose $\mathcal{A}$ has n symbols (letters) then plain text space and cipher text space is treated as $\mathbb{Z}_n$.

Encryption: Encryption function is defined by

$$\begin{aligned}\mathcal{E}: \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ m &\mapsto c = \mathcal{E}(m) \equiv m^2 \pmod{n}\end{aligned}$$

Decryption: Decryption function is defined by

$$\begin{aligned}\mathcal{D}: \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ c &\mapsto m = \mathcal{D}(c) \equiv \sqrt{c} \pmod{n}\end{aligned}$$

provided $\sqrt{c}$ exists in $\mathbb{Z}_n$ i.e congruence equation $x^2 \equiv c \pmod{n}$ has solution in $\mathbb{Z}_n$

Example: Suppose $\mathcal{A} = \{A = 0, B = 1, \dots, Z = 25\}$, $|\mathcal{A}| = n = 26$

To encrypt : **HELLO**

Encryption :

$$\begin{aligned}\mathcal{E}(H) &= \mathcal{E}(7) = 7^2 = 49 \equiv 23 \pmod{26} = \mathbf{X} \\ \mathcal{E}(E) &= \mathcal{E}(4) = 4^2 = 16 \equiv 16 \pmod{26} = \mathbf{Q} \\ \mathcal{E}(L) &= \mathcal{E}(11) = 11^2 = 121 \equiv 17 \pmod{26} = \mathbf{R} \\ \mathcal{E}(O) &= \mathcal{E}(14) = 14^2 = 196 \equiv 14 \pmod{26} = \mathbf{O}\end{aligned}$$

So encryption of 'HELLO' is '**XQRRO**'

For decryption we have to know square root of elements of $\mathbb{Z}_n$. For that we calculate following table

a	$a^2 \pmod{26}$	a	$a^2 \pmod{26}$
0	0	13	13
1	1	14	14
2	4	15	17
3	9	16	22
4	16	17	3
5	25	18	12
6	10	19	23
7	23	20	10
8	12	21	25
9	3	22	16
10	22	23	9
11	17	24	4
12	14	25	1

Decryption function is $\mathcal{D}(c) = \sqrt{c} \pmod{26}$ i.e $\sqrt{c} = x \iff c = x^2 \pmod{26}$

To decrypt ‘**XQRRO**’

$$\begin{aligned}
 \mathcal{D}(X) &= \mathcal{D}(23) = \sqrt{23} \pmod{26} \equiv a \pmod{26} \text{ (say)} \\
 &\implies a^2 = 23 \pmod{26} \\
 &\implies a = 7 \text{ or } 19 \text{ (from above table)} \\
 \text{i.e. } \mathcal{D}(X) &= \mathbf{H \text{ or } T}
 \end{aligned}$$

$$\begin{aligned}
 \mathcal{D}(Q) &= \mathcal{D}(16) = \sqrt{16} \pmod{26} \equiv a \pmod{26} \text{ (say)} \\
 &\implies a^2 = 16 \pmod{26} \\
 &\implies a = 4 \text{ or } 22 \text{ (from above table)} \\
 \text{i.e. } \mathcal{D}(Q) &= \mathbf{E \text{ or } W}
 \end{aligned}$$

$$\begin{aligned}
 \mathcal{D}(R) &= \mathcal{D}(17) = \sqrt{17} \pmod{26} \equiv a \pmod{26} \text{ (say)} \\
 &\implies a^2 = 17 \pmod{26} \\
 &\implies a = 11 \text{ or } 15 \text{ (from above table)} \\
 \text{i.e. } \mathcal{D}(R) &= \mathbf{L \text{ or } P}
 \end{aligned}$$

$$\begin{aligned}
\mathcal{D}(O) = \mathcal{D}(14) &= \sqrt{14} \pmod{26} \equiv a \pmod{26} \text{ (say)} \\
&\implies a^2 = 14 \pmod{26} \\
&\implies a = 14 \text{ or } 12 \text{ (from above table)} \\
\text{i.e. } \mathcal{D}(O) &= \mathbf{O \text{ or } M}
\end{aligned}$$

Possible decryption of ‘**XQRRO**’ is “**HELLO, TWPPM, HWPPM, HEPPM, HELLM, ...**”

“HELLO” is meaningful, so decryption of ‘**XQRRO**’ is ‘**HELLO**’.

Note: It is possible that for some $a \in \mathbb{Z}_n$, $\sqrt{a} \pmod{n}$ does not exist.

E.g. Take $\mathbb{Z}_{26} = \{0, 1, \dots, 25\}$

$5 \in \mathbb{Z}_{26}$, from above table we can verify that there is no $a \in \mathbb{Z}_{26}$ such that $a^2 \equiv 5 \pmod{26}$. Therefore $\sqrt{5} \pmod{26}$ does not exist.

Also, It is possible $\sqrt{a} \pmod{n}$ has more than 2 values in modulo n i.e. congruence $x^2 \equiv a \pmod{n}$ may have more than 2 solutions in modulo n .

E.g. Take $\mathbb{Z}_n = \mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$

Observe that

$$\begin{aligned}
1^2 &\equiv 1 \pmod{8} \\
3^2 &\equiv 1 \pmod{8} \\
5^2 &\equiv 1 \pmod{8} \\
7^2 &\equiv 1 \pmod{8}
\end{aligned}$$

So congruence $x^2 \equiv 1 \pmod{8}$ has four solutions.

Therefore, $\sqrt{1} \pmod{8} = 1, 3, 5, 7 \pmod{8}$.

5.4 Exercise

1. Let $p = 31$ and $e = 7$. Decrypt the cipher 29 using Silver - Pohlig - Hellman Exponentiation Cipher.
2. Let $n = 899$ and $e = 47$. Decrypt the cipher 169 using RSA cryptosystem.

3. Let $n = 20291$ and $\phi(n) = 19992$. Find the prime factorization of n using RSA cryptanalysis.
4. Show that for the RSA Cryptosystem, the number of fixed plaintexts ($x \in \mathbb{Z}$) is equal to $\gcd(e-1, p-1) \times \gcd(e-1, q-1)$.

R.K.Sharma

Chapter 6

Quadratic Residue and Legendre Symbol

6.1 Quadratic Residue

An integer $a \in \mathbb{Z}$ is said to be a quadratic residue *mod* n (where n is a fixed positive integer such that $\gcd(a, n) = 1$). If the quadratic congruence $x^2 \equiv a \pmod{n}$ has a solution. That is, if there exists $x \in \mathbb{Z}$ such that $x^2 \equiv a \pmod{n}$. If no such $x \in \mathbb{Z}$ exist, a is said to be a non-quadratic residue *mod* n .

Example: 1. 0 is quadratic residue modulo 4 since $x^2 \equiv 0 \pmod{4}$ has two solutions i.e. $x = 0, 2$.

2. 3 is non-quadratic residue modulo 4 since $x^2 \equiv 3 \pmod{4}$ has no solution.

3. 0 is quadratic residue modulo 9 since $x^2 \equiv 0 \pmod{9}$ has three solutions i.e. $x = 0, 3, 6$.

Recall: Fermat's little theorem: If a is a positive integer such that $\gcd(a, p) = 1$ where p is prime then $a^{p-1} \equiv 1 \pmod{p}$.

Euler's Criteria: Let p be an odd prime. An integer $a \in \mathbb{Z}$ is a quadratic residue modulo p iff $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$.

Proof: Suppose a is quadratic residue *mod* p . So there exists $y \in \mathbb{Z}$ such that $y^2 \equiv a \pmod{p}$.

Since $\gcd(a, p) = 1 \implies p \nmid a \implies p \nmid y^2 \implies p \nmid y \implies \gcd(p, y) = 1$

Again by Fermat's little theorem

$$\begin{aligned} y^{p-1} &\equiv 1 \pmod{p} \\ (y^2)^{\frac{p-1}{2}} &\equiv 1 \pmod{p} \\ a^{\frac{p-1}{2}} &\equiv 1 \pmod{p} \end{aligned}$$

Conversely, suppose $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. We want to show that a is a quadratic residue *mod* p . That is $\gcd(a, p) = 1$ and $x^2 \equiv a \pmod{p}$ has an integer solution. Obviously $\gcd(a, p) = 1$ otherwise $p|a$

$$\implies a \equiv 0 \pmod{p} \implies a^{\frac{p-1}{2}} \equiv 0 \pmod{p}$$

which is a contradiction. Hence $\gcd(a, p) = 1$. Now we will show that $x^2 \equiv a \pmod{p}$ has an integer solution.

Recall: If h is the least positive integer such that $a^h \equiv 1 \pmod{n}$ then we say that a has exponent h *modulo* n .

If g is an integer belonging to the exponent $\phi(n)$ then g is called a primitive root *modulo* n . i.e $g^{\phi(n)} \equiv 1 \pmod{n}$ and $g^k \not\equiv 1 \pmod{n} \forall 1 \leq k < \phi(n)$.

In this case g becomes generator of $U(\mathbb{Z}_n)$, hence $g, g^2, \dots, g^{\phi(n)-1}$ are all incongruent *modulo* n .

Now let g be a primitive root *mod* p . So $g^{\phi(p)} \equiv 1 \pmod{p}$ and $g^k \not\equiv 1 \pmod{p} \forall 1 \leq k < p-1$.

Let us assume that $1 \leq a \leq p-1$.

(Since $U(\mathbb{Z}_r)$ is cyclic if r is prime).

Hence $U(\mathbb{Z}_p)$ is cyclic $\implies \langle g \mid g^{p-1} = 1 \rangle = U(\mathbb{Z}_p)$.

Since $a \in U(\mathbb{Z}_p)$

$$\implies a \equiv g^r \pmod{p} \text{ for some } r \in \mathbb{Z}$$

Now,

$$\begin{aligned} 1 &\equiv a^{\frac{p-1}{2}} \equiv (g^r)^{\frac{p-1}{2}} \pmod{p} \\ \implies g^{\frac{r \cdot (p-1)}{2}} &\equiv 1 \pmod{p} \\ \implies (p-1) \mid \frac{r}{2} \cdot (p-1) &\quad \because |U(\mathbb{Z}_p)| = p-1 \\ \implies \frac{r}{2} &\in \mathbb{Z} \end{aligned}$$

Take $y = g^s$ where $s = \frac{r}{2}$ this gives $y^2 = g^{2s} = g^r \equiv a \pmod{p}$.

Since $y \in \mathbb{Z}$ such that $y^2 \equiv a \pmod{p}$. We get a is a quadratic residue *mod* p .

Note: Recall that if $p = 2$ and $\gcd(a, p) = 1$ then $a \equiv 1 \pmod{2}$ i.e a is an odd integer. The congruence will be $x^2 \equiv 1 \pmod{2}$, and also has a solution $x \equiv 1 \pmod{2}$.

Theorem 1: Let p be a prime of the form $p = 4k + 3$ for some $k \in \{0\} \cup \mathbb{N}$. If $a \in \mathbb{Z}$ such that $p \nmid a$ i.e $\gcd(a, p) = 1$ and if $x^2 \equiv a \pmod{p}$ has a solution $x \in \mathbb{Z}$, then all the solutions are $x = \pm a^{\frac{p+1}{4}}$.

Proof: Since x is a solution of $x^2 \equiv a \pmod{p}$ and $\gcd(a, p) = 1$, we get ' a ' to be quadratic residue *mod* p .
By Euler's Criteria, therefore

$$a^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

Since $p = 4k + 3$

$$\implies \frac{p+1}{4} = \frac{4k+3+1}{4} = k+1 \in \mathbb{Z}$$

Now

$$\begin{aligned} \left(\pm a^{\frac{p+1}{4}}\right)^2 &\equiv a^{\frac{p+1}{2}} \pmod{p} \\ &\equiv a^{\frac{p+1}{2}} \cdot 1 \pmod{p} \\ &\equiv a^{\frac{p+1}{2}} \cdot a^{\frac{p-1}{2}} \pmod{p} \\ &\equiv a^p \pmod{p} \\ &\equiv a \cdot a^{p-1} \pmod{p} \\ &\equiv a \pmod{p} \quad (\text{By Fermat's little theorem}) \end{aligned}$$

Hence $x \equiv \pm a^{\frac{p+1}{4}} \pmod{p}$ are solutions.

Then two solution namely $x \equiv a^{\frac{p+1}{4}} \pmod{p}$ and $x \equiv -a^{\frac{p+1}{4}} \pmod{p}$ are incongruent *mod* p .

Since $x^2 \equiv a \pmod{p}$ is a quadratic congruence under modulo p (prime). It

can not have more than two incongruent solutions and therefore if the congruence $x^2 \equiv a \pmod{p}$ has a solution then $x \equiv \pm a^{\frac{p+1}{4}} \pmod{p}$ are all the solutions.

Theorem 2: If $f(x) = a_0 + a_1x + \cdots + a_nx^n$ ($a_n \not\equiv 0 \pmod{p}$) has a solution, then the number of mutually incongruent *modulo* p solutions is at most n , where p is a prime.

Proof: By induction on n (the degree of $f(x)$).

If $n = 0$, then $f(x) = a_0 \not\equiv 0 \pmod{p}$. But then $f(x) \equiv 0 \pmod{p}$ has no solution.

If $n = 1$ then $f(x) = a_1x + a_0$, $a_1 \not\equiv 0 \pmod{p}$. But $f(x) \equiv 0 \pmod{p}$ becomes $a_1x + a_0 \equiv 0 \pmod{p}$

$$\begin{aligned} \implies a_1x &\equiv -a_0 \pmod{p} \\ \implies x &\equiv -a_1^{-1}a_0 \pmod{p} \end{aligned}$$

Since $-a_1^{-1}$ is unique. We get $f(x) \equiv 0 \pmod{p}$ has at most 1 solution.

We assume that theorem holds for all polynomial of degree less than k (say).

Now let $f(x) = a_kx^k + a_{k-1}x^{k-1} + \cdots + a_1x + a_0$ ($a_k \not\equiv 0 \pmod{p}$) be such that $f(x) \equiv 0 \pmod{p}$ has $k+1$ incongruent *modulo* p solutions say

$$x = \omega_1, \omega_2, \dots, \omega_{k+1} \pmod{p}$$

Let

$$g(x) = f(x) - a_k(x - \omega_1)(x - \omega_2) \cdots (x - \omega_k)$$

Then $g(x)$ is a polynomial of degree less than k .

By induction hypothesis $g(x) \equiv 0 \pmod{p}$ can have at most $(k-1)$ (incongruent *modulo* p) solutions. But $g(x)$ has k incongruent *modulo* p solution $\omega_1, \omega_2, \dots, \omega_k$. So $g(x) \equiv 0 \pmod{p} \forall x \in \mathbb{Z}$.

But then for $x = \omega_{k+1}$

$$\begin{aligned} \implies g(\omega_{k+1}) &\equiv 0 \pmod{p} \\ \implies 0 &= g(\omega_{k+1}) = f(\omega_{k+1}) - a_k(\omega_{k+1} - \omega_1)(\omega_{k+1} - \omega_2) \cdots (\omega_{k+1} - \omega_k) \pmod{p} \\ \implies g(\omega_{k+1}) &\equiv -a_k(\omega_{k+1} - \omega_1)(\omega_{k+1} - \omega_2) \cdots (\omega_{k+1} - \omega_k) \pmod{p} \\ &\quad (\because f(\omega_{k+1}) \equiv 0 \pmod{p}) \end{aligned}$$

$$\implies g(\omega_{k+1}) \not\equiv 0 \pmod{p}$$

(Since $p \nmid a_k$ and $\omega_i, i = 1, 2, \dots, k+1$ are incongruent)

Hence we get a contradiction since $g(x) = 0 \forall x \in \mathbb{Z}$.

Hence $f(x) \equiv 0$ has at most k incongruent solutions. This proves the theorem for $n = k$, and this completes the induction.

Hence the theorem holds for all $n \geq 0$.

6.2 Legendre Symbol

Let p be odd prime. The Legendre Symbol $\left(\frac{a}{p}\right)$ is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a \text{ i.e. } p \text{ divides } a \\ 1 & \text{if } p \nmid a \text{ and } a \text{ is a quadratic residue modulo } p \\ -1 & \text{if } p \nmid a \text{ and } a \text{ is a non-quadratic residue modulo } p \end{cases}$$

Example: 1. $\left(\frac{2}{3}\right) = -1$ since $x^2 \equiv 2 \pmod{3}$ has no solution i.e. 2 is non-quadratic residue modulo 3 and $3 \nmid 2$

2. $\left(\frac{0}{p}\right) = 0$

3. $\left(\frac{3}{13}\right) = 1$ since $x^2 \equiv 3 \pmod{13}$ has solutions $x = 4, 9$ i.e. 3 is quadratic residue modulo 13

Note: Legendre symbol can be defined in general for any ' n ' instead of prime p .

Properties of Legendre Symbol

Let p be an odd prime and let a and b be integers that are relatively prime to p . Then the Legendre symbol has the following properties:

(a) If $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$

(b) $\left(\frac{a^2}{p}\right) = 1$

(c) $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$. (**Euler's Criteria**)

(d) $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$

Theorem : If p is an odd prime, then

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p = 4k + 1, \quad k \in \mathbb{N} \cup \{0\} \\ -1 & \text{if } p = 4k + 3, \quad k \in \mathbb{N} \cup \{0\} \end{cases}$$

and

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p = 8k + 1, 8k - 1, \quad k \in \mathbb{N} \cup \{0\} \\ -1 & \text{if } p = 8k + 3, 8k - 3, \quad k \in \mathbb{N} \cup \{0\} \end{cases}$$

Gauss Law of Quadratic Reciprocity:

Let p and q be distinct odd primes

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)}$$

equivalently

$$\left(\frac{p}{q}\right) = \begin{cases} -\left(\frac{q}{p}\right) & \text{if } p = 4k_1 + 3, q = 4k_2 + 3 \\ & k_1, k_2 \in \mathbb{N} \cup \{0\} \\ \left(\frac{q}{p}\right) & \text{if } p = 4k_1 + 1, q = 4k_2 + 1, \\ & p = 4k_1 + 1, q = 4k_2 + 3, \\ & p = 4k_1 + 3, q = 4k_2 + 1 \\ & k_1, k_2 \in \mathbb{N} \cup \{0\} \end{cases}$$

Example: Check whether the congruence $x^2 \equiv -46 \pmod{17}$ has solutions.

Solution: We will use Legendre symbol to find solution exist or not.

Note that $17 \nmid -46$

By definition of Legendre symbol

If $\left(\frac{-46}{17}\right) = 1$ then -46 is a quadratic residue modulo 17 i.e. it has solution.

If $\left(\frac{-46}{17}\right) = -1$ then -46 is a non-quadratic residue modulo 17 i.e. it has no solution.

So we have to find

$$\left(\frac{-46}{17}\right) = ?$$

By property (d) we have

$$\begin{aligned}\left(\frac{-46}{17}\right) &= \left(\frac{-1}{17}\right) \left(\frac{46}{17}\right) \\ &= (1) \left(\frac{46}{17}\right) = \left(\frac{46}{17}\right) \quad (\text{by above theorem})\end{aligned}$$

Also

$$46 \equiv 12 \pmod{17} \implies \left(\frac{46}{17}\right) = \left(\frac{12}{17}\right) \quad (\text{by property (a)})$$

$$\begin{aligned}\left(\frac{12}{17}\right) &= \left(\frac{2^2 \cdot 3}{17}\right) \\ &= \left(\frac{2^2}{17}\right) \left(\frac{3}{17}\right) = \left(\frac{3}{17}\right)\end{aligned}$$

since $\left(\frac{2^2}{17}\right) = 1$ by property (b)

Now by property (c) we have

$$\begin{aligned}\left(\frac{3}{17}\right) &\equiv 3^{\frac{17-1}{2}} \pmod{17} \equiv 3^8 \equiv (3^4)^2 \pmod{17} \\ &\equiv 81^2 \pmod{17} \equiv (-4)^2 \pmod{17} \\ &\equiv 16 \pmod{17} \\ &\equiv -1 \pmod{17}\end{aligned}$$

So

$$\left(\frac{-46}{17}\right) = -1$$

Therefore congruence $x^2 = -46 \pmod{17}$ has no solution.

Theorem 3: There are infinitely many primes of the form $4k + 1$ for $k \in \mathbb{N}$.

Proof: Let there are only finitely many primes of the form $4k + 1$ namely $p_1, p_2, \dots, p_n$.

To prove the theorem it is sufficient to show that there is a prime p of the

form $4k + 1$ and $p \neq p_i$ for $i = 1, 2, \dots, n$.

Let

$$N = (2p_1p_2 \cdots p_n)^2 + 1$$

Observe that N is odd and greater than 1. Therefore N has an odd prime divisor p (say).

If $p = p_i$ for some $i = 1, 2, \dots, n$ then $p_i \mid N$ and $p_i \mid (2p_1p_2 \cdots p_n)^2$ this implies $p_i \mid N - (2p_1p_2 \cdots p_n)^2 = 1$ which is not possible since p_i is prime.

Therefore $p \neq p_i$ for any $i = 1, 2, \dots, n$.

Since $p \mid N$ therefore

$$(2p_1p_2 \cdots p_n)^2 \equiv -1 \pmod{p} \quad (1)$$

Also $\gcd((2p_1p_2 \cdots p_n), p) = 1$, by Fermat's Little Theorem

$$(2p_1p_2 \cdots p_n)^{p-1} \equiv 1 \pmod{p}$$

$$((2p_1p_2 \cdots p_n)^2)^{\frac{p-1}{2}} \equiv 1 \pmod{p}$$

From (1), we have

$$(-1)^{\frac{p-1}{2}} \equiv 1 \pmod{p} \quad (2)$$

Since p is prime so it can be of the form $4k + 1$ or $4k + 3$.

If p is of the form $4k + 3$ then $\frac{p-1}{2} = 2k + 1$ is odd and therefore from (2) we have

$$-1 \equiv 1 \pmod{p}$$

which is not possible since p is odd.

Therefore p must be of the form $4k + 1$.

So we have a prime p of the form $4k + 1$ and $p \neq p_i$ for $i = 1, 2, \dots, n$. Hence there are infinitely many primes of the form $4k + 1$.

Remark: If p, q are primes of the form $4k + 1$, then pq is also of the same form.

$$\begin{aligned} (4k_1 + 1)(4k_2 + 1) &= 4\{k_1(4k_2 + 1)\} + (4k_2 + 1) \\ &= 4\{k_1(4k_2 + 1)\} + 4k_2 + 1 \\ &= 4\{k_1(4k_2 + 1) + k_2\} + 1 \\ &= 4k_3 + 1 \end{aligned}$$

Theorem 4: There are infinitely many primes of the form $4k+3$ for $k \in \mathbb{N}$.

Proof: Let the primes of the form $4k+3$ are finitely many namely $p_1, p_2, \dots, p_n$. WLOG, we may assume that $p_1 = 3, p_2 = 7, \dots$ and that $p_1 < p_2 < \dots < p_n$. Each $p_2, p_3, \dots, p_n$ is greater than 3.

Let

$$N = 4(p_2 \cdot p_3 \cdots p_n) + 3$$

N is either a prime of the form $4k+3$, which it can not be as $N > p_n$ and p_n is the largest such prime, or a prime of the form $4k+1$ which is not possible. So N is composite and it must contain a prime factor of the form $4k+3$. Otherwise if all prime factors of N are of the form $4k+1$ then N will also be of the form $4k+1 \implies N \equiv 1 \pmod{4}$ which is not true.

So N contains either all prime factors of the form $4k+3$ or both $4k+1$ & $4k+3$ i.e N must contain at least one prime factor of the form $4k+3$.

If no. of factor is 1 then N itself a prime of the form $4k+3$ (Not possible since $N >$ all $4k+3$ primes). So N has more than 1 factor of the form $4k+3$.

Now if $3|N$ then $3|N-3$

$$\implies 3|4(p_2 \cdot p_3 \cdots p_n)$$

$$\implies 3|p_2 \cdot p_3 \cdots p_n$$

$$\implies 3|p_j \text{ some } j = 2, 3, \dots, n \text{ (Not possible)}$$

$$\implies 3 \nmid N$$

Now if $p_j|N$ for some $j > 2$ then $p_j|N-4(p_2 \cdot p_3 \cdots p_n)$

$$\implies p_j|3 \text{ but } p_j > 3$$

$$\implies p_j \nmid 3$$

$$\implies p_j \nmid N \forall j = 2, 3, \dots, n$$

Since N has a factor p of the form $4k+3$ type. But none of $p_1, p_2, \dots, p_n$ divides N this implies that $p \neq p_i \ i = 1, 2, \dots, n$. Which is a contradiction.

So there are infinitely many primes of the form $4k+3$.

6.3 Exercise

1. Let $S = \{A = 0, B = 1, \dots, Z = 25, ! = 26\}$. Encrypt the message "CIPHER" using quadratic cipher.
2. Let $S = \{A = 0, B = 1, \dots, Z = 25\}$. Decrypt the cipher text "DMRXZC" using quadratic cipher.
3. Find all solutions, if there is any, of the congruence $x^2 \equiv 21 \pmod{53}$.
4. Find the Legendre Symbol of $\left(\frac{a}{n}\right)$ if a and n are given as follows:
 - (a) $a = 2, n = 13$
 - (b) $a = 51, n = 97$
 - (c) $a = 1203, n = 1223$
 - (d) $a = 2, n = 391$
5. Show that if $\gcd(b, n) = 1$, then $\left(\frac{ab^2}{p}\right) = \left(\frac{a}{p}\right)$.
6. (a) Prove that for any positive odd n , $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$.
 (b) Prove that $\left(\frac{-2}{p}\right) = 1$ if $p \equiv 1$ or $3 \pmod{8}$, and $\left(\frac{-2}{p}\right) = -1$ if $p \equiv 5$ or $7 \pmod{8}$.

Chapter 7

Rabin Cipher and Some Attacks on RSA

7.1 Rabin Cipher

Let p, q be distinct primes of the form $4k + 3$ and take $n = pq$

Encryption function is defined by

$$\begin{aligned}\mathcal{E}: \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ m &\mapsto c = \mathcal{E}(m) \equiv m^2 \pmod{n}\end{aligned}$$

Decryption function is defined by

$$\begin{aligned}\mathcal{D}: \mathbb{Z}_n &\rightarrow \mathbb{Z}_n \\ c &\mapsto m = \mathcal{D}(c) \equiv \sqrt{c} \pmod{n}\end{aligned}$$

provided c is quadratic residue modulo n .

Example: Let $p = 11, q = 3$ be primes of the form $4k + 3$,

$n = pq = 11 \cdot 3 = 33$

Suppose $\mathcal{A} = \{A = 0, B = 1, \dots, Z = 25, _ = 26, ? = 27, . = 28, @ = 29, ! = 30, \chi = 31, \backslash = 32\}$

To encrypt: "STUDENT"

$$\begin{aligned}
\mathcal{E}(S) &= \mathcal{E}(18) = 18^2 \equiv 324 \pmod{33} \equiv 27 \pmod{33} = ? \\
\mathcal{E}(T) &= \mathcal{E}(19) = 19^2 \equiv 361 \pmod{33} \equiv 31 \pmod{33} = \chi \\
\mathcal{E}(U) &= \mathcal{E}(20) = 20^2 \equiv 400 \pmod{33} \equiv 4 \pmod{33} = \mathbf{E} \\
\mathcal{E}(D) &= \mathcal{E}(3) = 3^2 \equiv 9 \pmod{33} = \mathbf{J} \\
\mathcal{E}(E) &= \mathcal{E}(4) = 4^2 \equiv 16 \pmod{33} = \mathbf{Q} \\
\mathcal{E}(N) &= \mathcal{E}(13) = 13^2 \equiv 169 \pmod{33} \equiv 4 \pmod{33} = \mathbf{E} \\
\mathcal{E}(T) &= \mathcal{E}(19) = 19^2 \equiv 361 \pmod{33} \equiv 31 \pmod{33} = \chi
\end{aligned}$$

So encryption of “STUDENT” is “? χ **EJQE** χ ”

To decrypt: “? χ **EJQE** χ ”

$$\mathcal{D}(?) = \mathcal{D}(27) \equiv \sqrt{27} \pmod{33} \equiv x \pmod{33} \text{ i.e. } x^2 \equiv 27 \pmod{33}$$

To calculate: $x^2 \equiv 27 \pmod{33}$

By Theorem 1 (Previous Chapter): Let p be a prime of the form $p = 4k + 3$ for some $k \in \{0\} \cup \mathbb{N}$. If $a \in \mathbb{Z}$ such that $p \nmid a$ i.e. $\gcd(a, p) = 1$ and if $x^2 \equiv a \pmod{p}$ has a solution $x \in \mathbb{Z}$, then all the solutions are $x = \pm a^{\frac{p+1}{4}}$

Here $a = 27$ for $p = 11 = 4 \times 2 + 3$ and $\gcd(27, 11) = 1$, we have

$$x \equiv \pm 27^{\frac{11+1}{4}} \pmod{11} \equiv \pm 27^3 \pmod{11} \equiv \pm 5^3 \equiv \pm 4 \pmod{11}$$

For $a = 27$ and $p = 3 = 0 \times 4 + 3$

$$a \equiv 27 \pmod{3} \equiv 0 \pmod{3}$$

$$x^2 \equiv a \equiv 0 \pmod{3} \text{ has solution } x \equiv 0 \pmod{3}$$

Now we have

$$x \equiv 4 \pmod{11} \tag{7.1}$$

$$x \equiv -4 \equiv 7 \pmod{11} \tag{7.2}$$

$$x \equiv 0 \pmod{3} \tag{7.3}$$

Recall: Chinese Remainder Theorem

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \end{aligned}$$

m_1 and m_2 are relatively prime then solution of above system is given by

$$x \equiv a_1 y_1 M_1 + a_2 y_2 M_2 \pmod{m_1 m_2}$$

where

$$M_1 = \frac{m_1 \cdot m_2}{m_1} = m_2, \quad M_2 = \frac{m_1 \cdot m_2}{m_2} = m_1$$

$$y_1 \equiv M_1^{-1} \pmod{m_1}, \quad y_2 \equiv M_2^{-1} \pmod{m_2}$$

Now solve (1) and (3) using Chinese Remainder Theorem, we have
 $a_1 = 4, a_2 = 0, m_1 = 11, m_2 = 3$

$$M_1 = \frac{11 \cdot 3}{11} = 3, \quad M_2 = \frac{11 \cdot 3}{3} = 11$$

$$y_1 \equiv 3^{-1} \pmod{11} \equiv 4 \pmod{11}, \quad y_2 \equiv 11^{-1} \pmod{3} \equiv 2 \pmod{3}$$

and

$$x \equiv (4 \cdot 4 \cdot 3) + (0 \cdot 2 \cdot 11) \equiv 48 \equiv 15 \pmod{33}$$

Similarly, solve (2) and (3) using Chinese Remainder Theorem, we get

$$x \equiv 18 \pmod{33}$$

Therefore $\sqrt{27} \equiv 15, 18 \pmod{33}$

$$\mathcal{D}(?) = \mathcal{D}(27) \equiv \sqrt{27} \pmod{33} \equiv 15, 18 \pmod{33} = \mathbf{P \text{ or } S}$$

Now

$$\mathcal{D}(\chi) = \mathcal{D}(31) \equiv \sqrt{31} \pmod{33} \equiv x \pmod{33} \text{ i.e. } x^2 \equiv 31 \pmod{33}$$

To calculate $x^2 \equiv 31 \pmod{33}$

For $a = 31$, $p = 11$ and $\gcd(31, 11) = 1$. By Theorem 1

we have

$$x \equiv \pm 31^{\frac{11+1}{4}} \pmod{11} \equiv \pm 31^3 \pmod{11} \equiv \pm 9^3 \equiv \pm 3 \pmod{11}$$

For $a = 31$, $p = 3$ and $\gcd(31, 3) = 1$. By Theorem 1 we have

$$x \equiv \pm 31^{\frac{3+1}{4}} \pmod{3} \equiv \pm 31^1 \pmod{3} \equiv \pm 1 \pmod{3}$$

Now we have

$$x \equiv 3 \pmod{11} \tag{7.4}$$

$$x \equiv -3 \equiv 8 \pmod{11} \tag{7.5}$$

$$x \equiv 1 \pmod{3} \tag{7.6}$$

$$x \equiv -1 \equiv 2 \pmod{3} \tag{7.7}$$

Solve (4) and (6) using Chinese Remainder Theorem, we have
 $a_1 = 3$, $a_2 = 1$, $m_1 = 11$, $m_2 = 3$

$$M_1 = \frac{11 \cdot 3}{11} = 3, \quad M_2 = \frac{11 \cdot 3}{3} = 11$$

$$y_1 \equiv 3^{-1} \pmod{11} \equiv 4 \pmod{11}, \quad y_2 \equiv 11^{-1} \pmod{3} \equiv 2 \pmod{3}$$

and

$$x \equiv (3 \cdot 4 \cdot 3) + (1 \cdot 2 \cdot 11) \equiv 58 \equiv 25 \pmod{33}$$

Solve (4) and (7) using Chinese Remainder Theorem, we have
 $a_1 = 3$, $a_2 = 2$, $m_1 = 11$, $m_2 = 3$

$$M_1 = \frac{11 \cdot 3}{11} = 3, \quad M_2 = \frac{11 \cdot 3}{3} = 11$$

$$y_1 \equiv 3^{-1} \pmod{11} \equiv 4 \pmod{11}, \quad y_2 \equiv 11^{-1} \pmod{3} \equiv 2 \pmod{3}$$

and

$$x \equiv (3 \cdot 4 \cdot 3) + (2 \cdot 2 \cdot 11) \equiv 80 \equiv 14 \pmod{33}$$

Solve (5) and (6) using Chinese Remainder Theorem, we have
 $a_1 = 8, a_2 = 1, m_1 = 11, m_2 = 3$

$$M_1 = \frac{11 \cdot 3}{11} = 3, \quad M_2 = \frac{11 \cdot 3}{3} = 11$$

$$y_1 \equiv 3^{-1} \pmod{11} \equiv 4 \pmod{11}, \quad y_2 \equiv 11^{-1} \pmod{3} \equiv 2 \pmod{3}$$

and

$$x \equiv (8 \cdot 4 \cdot 3) + (1 \cdot 2 \cdot 11) \equiv 118 \equiv 19 \pmod{33}$$

$$x \equiv 19 \pmod{33}$$

Similarly, Solve (5) and (7) using Chinese Remainder Theorem, we get

$$x \equiv 8 \pmod{33}$$

Hence $\sqrt{31} \equiv 8, 14, 19, 25 \pmod{33}$

$$\mathcal{D}(\chi) = \mathcal{D}(31) \equiv \sqrt{31} \pmod{33} \equiv 8, 14, 19, 25 \pmod{33} = \mathbf{I, O, T \text{ or } Z}$$

In the same way we can do other decryptions, we get finitely many possible plain text. We shall choose meaningful plain texts.

7.2 Attacks on RSA

In RSA cryptosystem, we use two keys one for encryption which is public and another for decryption which is private. Attacks on RSA, we mean that without knowing private key how can someone decrypt the cipher text. There are many attacks on RSA, we will study few of them.

7.2.1 Low exponent attack

In RSA cryptosystem, when small value of exponent is used for encryption then it becomes easy to attack as follows

Suppose Alice wants to send same message m to Bob, Christopher and David with low exponent value $e = 3$ for this Bob, Christopher and David announce

their public key say $(n_B, 3)$, $(n_C, 3)$ and $(n_D, 3)$ respectively
 Alice will send

$$\begin{aligned} c_B &= m^3 \pmod{n_B} \text{ to Bob} \\ c_C &= m^3 \pmod{n_C} \text{ to Christopher} \\ c_D &= m^3 \pmod{n_D} \text{ to David} \end{aligned}$$

Attacker perform the following computations to get message without knowing the private key.
 write

$$\begin{aligned} x &= c_B = m^3 \pmod{n_B} \\ x &= c_C = m^3 \pmod{n_C} \\ x &= c_D = m^3 \pmod{n_D} \end{aligned}$$

Case 1: If n_B, n_C and n_D are relatively prime then by Chinese Remainder Theorem there exist a number x such that

$$x = m^3 \pmod{n_B n_C n_D}$$

Since $m < n_B, n_C$ and n_D implies that $m^3 < n_B n_C n_D$
 Thus message m can be obtained by computing cubic root of x over the integers.

Case 2: If n_B, n_C and n_D are not relatively prime then we have divisors of n_B, n_C and n_D and we can decrypt the cipher easily.

7.2.2 Common modulus attack

In such type of attack it is considered that message is encrypted by taking n common for all receivers but exponents are different and relatively prime i.e. Public key for Bob, Christopher and David will be (n, e_1) , (n, e_2) and (n, e_3) respectively and $\gcd(e_1, e_2, e_3) = 1$.

Alice will send

$$\begin{aligned} c_B &= m^{e_1} \pmod{n} \text{ to Bob} \\ c_C &= m^{e_2} \pmod{n} \text{ to Christopher} \\ c_D &= m^{e_3} \pmod{n} \text{ to David} \end{aligned}$$

Since $\gcd(e_1, e_2, e_3) = 1$ so by Extended Euclidean Algorithm, there exists integers x, y, z such that

$$x \cdot e_1 + y \cdot e_2 + z \cdot e_3 = 1$$

Now the following computation will give the message

$$\begin{aligned} c_B^x \cdot c_C^y \cdot c_D^z &= m^{e_1 \cdot x} m^{e_2 \cdot y} m^{e_3 \cdot z} \pmod{n} \\ &= m^{x \cdot e_1 + y \cdot e_2 + z \cdot e_3} \pmod{n} \\ &= m \pmod{n} \end{aligned}$$

7.2.3 Chosen Ciphertext Attack

Suppose public key is (n, e) , message is ' m ' and cipher is ' c '. Attacker will perform following steps to get message without knowing private key i.e. $d \equiv e^{-1} \pmod{\phi(n)}$

- The attacker will choose a random integer s less than n .
- The attacker will construct a cipher text from the obtained cipher ' c ' (to be decrypted)

$$c' \equiv s^e \cdot c \pmod{n}$$
- The attacker will manage somehow that receiver must decrypt c' with the help of private key d .
- Suppose receiver decrypts c' to m' i.e.

$$m' = (c')^d \pmod{n} \equiv (s^e \cdot c)^d \pmod{n} \equiv s^{e \cdot d} \cdot c^d \pmod{n} \equiv s \cdot m \pmod{n}$$

since $e \cdot d = 1 \pmod{\phi(n)}$ i.e. $e \cdot d = 1 + k\phi(n)$ for some $k \in \mathbb{Z}$.

- Since the attacker has chosen s himself, so attacker knows $s^{-1} \pmod{n}$, therefore message is obtained by

$$m \equiv s^{-1} \cdot m' \pmod{n}$$

7.3 Exercise

1. Alice wants to send a message m to Bob and David using RSA cryptosystem with same modulus 589 but with different exponents 7 and 11 respectively. Ciphertext received by Bob and David is 582 and 331 respectively. Decrypt the message m , assuming factorization of n is hard.
2. Let $n = 893$. Decrypt the cipher "7" using Rabin Cipher.
3. (a) Let $p = 71$ be prime. Find all possible solutions of the quadratic equation $x^2 = 37 \pmod{p}$.
 (b) Let $q = 89$. Find all possible solutions of the quadratic equation $x^2 = 79 \pmod{q}$.
 (c) Let $n = 5293$. Find all possible solutions of the quadratic equation $x^2 = 183 \pmod{n}$.
4. Let $n = 2537$. Find all possible solutions of the quadratic equation $3x^2 + 5x + 2 \equiv 0 \pmod{n}$.

Chapter 8

Strong Primes and Primality Test

8.1 Strong Prime

Let p be a large prime, then p is said to be a strong prime if,

- $p - 1$ has a large prime factor say r
- $r - 1$ has a large prime factor say t
- $p + 1$ has a large prime factor say s

Observe that

Since $p - 1$ has a large prime factor r i.e. $p = kr + 1$ for some k . If k is odd then p will be even which is not possible. Therefore k must be even.

Since $r - 1$ has a large prime factor t i.e. $r = k't + 1$ for some k' . If k' is odd then r will be even which is not possible. Therefore k' must be even.

Also, Since $p + 1$ has a large prime factor s i.e. $p = k''s - 1$ for some k'' . If k'' is odd then p will be even which is not possible. Therefore k'' must be even.

So we can write

- $p = 2jr + 1$
- $p = 2ms - 1$
- $r = 2lt + 1$

for some integres j, l, m where r, s and t are primes.

To construct strong primes, we take large primes r, s and t as above and compute p for different choices for j, l and m . Check whether p is prime or not. If p is prime then it will be strong prime.

Note: Large prime does not imply strong prime.

For e.g. (1.) $p = 3628273133$ is a 10-digits large prime but $p - 1 = 3628273132$ has prime factor $r = 28211$ which is not large prime. So p is not strong prime.

(2.) $p = 10888869450418352160768000001$ is a 29 digits prime no.

Is p a strong prime? "YES" (Verify!)

(3.) $p = 2^{82589933} - 1$ is a large prime (24,862,048- digits) but not a strong prime since $p + 1$ does not have a large prime factor ($p + 1$ has only 2 as prime factor).

Question is given a positive integer n , to decide if it is a prime or not.

8.2 Primality Tests

Recall: Fermat's little theorem: If a is positive integer such that $\gcd(a, p) = 1$ where p is prime then $a^{p-1} \equiv 1 \pmod{p}$.

What about converse of Fermat's little theorem ? i.e. If n and b is a positive integer such that $\gcd(b, n) = 1$ and $b^{n-1} \equiv 1 \pmod{n}$, is then n a prime ?

Converse of the Fermat's little theorem is **NOT** true i.e if n and b is a positive integer such that $\gcd(b, n) = 1$ and $b^{n-1} \equiv 1 \pmod{n}$ then n need not be prime.

For example: Let $b = 2$ and $n = 341$. Observe that n is not a prime since $n = 341 = 11 \times 31$.

But

$$2^{11-1} = 2^{10} \equiv 1 \pmod{11}$$

and

$$2^{31-1} = 2^{30} \equiv 1 \pmod{31}$$

$$2^{340} = (2^{10})^{34} \equiv 1 \pmod{11} \text{ and } 2^{340} = (2^{30})^{11} \cdot 2^{10} \equiv 1 \pmod{31}$$

Now

$$2^{340} \equiv 1 \pmod{11}$$

$$2^{340} \equiv 1 \pmod{31}$$

Then by CRT(Chinese Remainder Theorem), gives that

$$2^{340} \equiv 1 \pmod{11 \times 31} \text{ i.e. } 2^{340} \equiv 1 \pmod{341}.$$

Here $2^{341-1} \equiv 1 \pmod{341}$ but 341 is not a prime.

Some Definition:

- **Prime:** A positive integer $n > 1$ is said to be prime if its positive divisors are only 1 and n itself.
- **Pseudoprime:** A composite number n is said to be pseudoprime to the base b if b is a positive integer such that $\gcd(b, n) = 1$ and $b^{(n-1)} \equiv 1 \pmod{n}$.

From above example, 341 is a pseudoprime to the base 2.

- **Carmichael number:** A composite number n is said to be Carmichael number if n is a pseudoprime to every base b . For eg:- 561, 1105.

For $n = 561 = 3 \cdot 11 \cdot 17$, for any $b \in \mathbb{Z}$

Note that $\gcd(b, 3) = 1$, $\gcd(b, 11) = 1$ and $\gcd(b, 17) = 1$ then by Fermat's Little theorem, we have

$$b^2 \equiv 1 \pmod{3}$$

$$b^{10} \equiv 1 \pmod{11}$$

$$b^{16} \equiv 1 \pmod{17}$$

Also,

$$b^{560} \equiv (b^2)^{280} \equiv 1 \pmod{3}$$

$$b^{560} \equiv (b^{10})^{56} \equiv 1 \pmod{11}$$

$$b^{560} \equiv (b^{16})^{35} \equiv 1 \pmod{17}$$

Now by Chinese Remainder Theorem, we have

$$b^{560} \equiv 1 \pmod{561}$$

So for all base $b \in \mathbb{Z}$, $b^{561-1} \equiv 1 \pmod{561}$.

Hence 561 is Carmichael number. In fact 561 is smallest Carmichael number.

8.2.1 Fermat's Primality Test

Let n be a positive integer. If there exist a positive integer b such that $\gcd(b, n) = 1$ and $b^{n-1} \not\equiv 1 \pmod{n}$ then n is not prime.

Example 1: If $n = 341$ then $b = 3$ is such that $\gcd(3, 341) = 1$

$$3^5 \equiv 243 \pmod{341},$$

$$3^{10} \equiv 56 \pmod{341},$$

$$3^{20} \equiv 67 \pmod{341},$$

$$3^{30} \equiv 1 \pmod{341},$$

$$(3^{30})^{11} = 3^{330} \equiv 1 \pmod{341},$$

$$3^{340} = 3^{330} \cdot 3^{10} \equiv 56 \pmod{341},$$

$$\Rightarrow 3^{341-1} \equiv 56 \not\equiv 1 \pmod{341}.$$

Hence 341 is not prime and we can observe that $341 = 11 \times 31$.

8.2.2 Miller-Rabin test:

Let n be a positive integer and $n - 1 = 2^s t$, $s \geq 0$ and t odd positive integer. We say that n passes the Miller's test for the base b if either $b^t \equiv 1 \pmod{n}$ or $b^{2^k t} \equiv -1 \pmod{n}$ for some $k : 0 \leq k \leq s - 1$.

This test uses the simple fact: If p is prime then $x^2 \equiv 1 \pmod{p} \Rightarrow x \equiv \pm 1 \pmod{p}$.

Let b be a base, then $b^{n-1} \equiv 1 \pmod{n}$. (if not, then by Fermat's primality test, n is composite.)

WLOG, we may assume that $s > 0$ otherwise $n - 1$ will be odd and n will be even i.e n is composite.

So $n - 1$ is an even integer

$$\Rightarrow \frac{n-1}{2} \in \mathbb{N}$$

$$\Rightarrow \left(b^{\frac{n-1}{2}}\right)^2 \equiv 1 \pmod{n}$$

for $x = b^{\frac{n-1}{2}}, x^2 \equiv 1 \pmod{n}$, suppose $y = \frac{n-1}{2}$

1. If $x \not\equiv \pm 1 \pmod{n}$ then n is composite.
2. If $x \equiv -1 \pmod{n}$ then n may be a prime and may not be a prime and also we can not proceed further.
3. If $x \equiv 1 \pmod{n}$ then n may be a prime and may not be a prime and we can proceed further.

(a) If $2|y$ i.e. y is even, assign $y = y/2 \in \mathbb{Z}$ and evaluate

$$x \equiv b^y \pmod{n}$$

consider the three cases earlier.

(b) If $2 \nmid y$ then the last value for x is $x \equiv 1 \pmod{n}$. We can not go further and n passes the test for the base b .

If n passes the test for base b then we choose another base b' and apply Miller-Rabin test for base b' .

Example 1: Let $n = 29, b = 5$

$$5^{28} \equiv 1 \pmod{29}$$

$n - 1 = 28 = 2^2 \cdot 7$ where $t = 7$, and $s = 2$

$$\left(\frac{n-1}{2}\right) = 14$$

$$5^2 \equiv -4 \pmod{29},$$

$$5^4 \equiv 16 \pmod{29},$$

$$5^6 \equiv -6 \pmod{29},$$

$$5^7 \equiv -1 \pmod{29},$$

$$5^{14} \equiv 1 \pmod{29},$$

For $x = 5^{14} \equiv 1 \pmod{29}$

$x \equiv 1 \pmod{29}$ so 29 may be prime we can proceed further.

Since $2|14$ and $5^7 \equiv -1 \pmod{29}$. It may be a prime and may not be a prime and we can not proceed also further i.e 29 passes the test for the base 5.

Example 2:

Let $n = 1387, b = 2$

$$2^{10} \equiv 1024 \pmod{1387},$$

$$2^{12} \equiv -65 \pmod{1387},$$

$$2^{18} \equiv 1 \pmod{1387},$$

$$(2^{18})^{77} = 2^{1386} \equiv 1 \pmod{1387},$$

$$2^{1386} \equiv 1 \pmod{1387}$$

$n - 1 = 1386 = 2 \cdot 693$ where $t = 693$, and $s = 1$
 $\left(\frac{n-1}{2}\right) = 693$

$$(2^{18})^{38} = 2^{684} \equiv 1 \pmod{1387},$$

$$2^{684} \cdot 2^9 = 2^{693} \equiv 512 \not\equiv \pm 1 \pmod{1387}$$

Since $x^2 = (2^{693})^2 = 2^{1386} \equiv 1 \pmod{1387}$ but $x = 2^{693} \equiv 512 \not\equiv \pm 1 \pmod{1387}$.
Hence 1387 fails the test. Therefore 1387 is composite number and $1387 = 19 \cdot 73$.

Remark: Observe that a prime passes the test, but a composite may also. So if n passes the test we can not conclude that n is 'prime'. However if n fails the test then n is 'composite'.

Wilson's theorem:

Statement: Let p be a positive integer then p is a prime iff $(p-1)! \equiv -1 \pmod{p}$.

Proof:

$\Rightarrow$: Let p be a prime. If $p = 2$ then $(2-1)! = 1!$ and $1 \equiv -1 \pmod{2}$. So theorem holds for $p = 2$.

If $p = 3$ then $(3-1)! = 2!$ and $2 \equiv -1 \pmod{3}$. Theorem also true for $p = 3$. Now consider $p > 3$

$$(p-1)! = 1 \cdot 2 \cdot 3 \cdots (p-1)$$

Since $1, 2, \dots, (p-1) \in \mathbb{Z}_p^*$ and $\mathbb{Z}_p^*$ is group under multiplication modulo p . So every integer $1, 2, \dots, (p-1) \in \mathbb{Z}_p^*$ has a unique inverse.

$$2 \cdot 3 \cdots (p-1) = \prod (xx^{-1})$$

writing have the factor x alongwith its inverse $x^{-1} \forall x \in \{2, 3, \dots, (p-1)\}$ being then odd no. there should be some $x \in \{2, 3, \dots, (p-1)\}$ such that $x \equiv x^{-1} \pmod{p}$ i.e. $x^2 \equiv 1 \pmod{p}$.

$$\Rightarrow (p-1)! \equiv x \pmod{p} \text{ for some } x \in \{2, 3, \dots, (p-1)\}$$

such that $x \equiv x^{-1} \pmod{p} \Rightarrow x \equiv \pm 1 \pmod{p}$ since $1 \notin \{2, 3, \dots, (p-1)\}$ we get $x \equiv -1 \pmod{p}$ that is $(p-1)! \equiv -1 \pmod{p}$.

$\Leftarrow$: For if p is not prime, then p has a divisor d with $1 < d < p$. Furthermore, because $d \leq (p-1)$, d occurs as one of the factors in $(p-1)!$, whence $d \mid (p-1)!$. Now we are assuming that $p \mid (p-1)! + 1$, and so $d \mid (p-1)! + 1$, too. The conclusion is that $d \mid 1$, which is nonsense.

Hence If p is composite then $(p-1)! \not\equiv -1 \pmod{p}$ i.e. if $(p-1)! \equiv -1 \pmod{p}$ then p is prime.

Example: For $p = 9$, $(9-1)! = 8! = 40320 \equiv 0 \pmod{9} \not\equiv -1 \pmod{9}$

8.2.3 AKS Primality Test

In 2002 Agrawal, Kayal, and Saxena (IIT Kanpur) introduced the deterministic primality test known as AKS algorithm named after them.

The following steps involve in AKS algorithm to check a given integer $n > 1$ is prime or not:

1. Check $n^{\frac{1}{k}}$ is an integer for $2 \leq k \leq \log_2(n)$. If $n^{\frac{1}{k}}$ is the integer i.e. n is perfect power, then n is composite.
2. Find the smallest integer r such that $\text{ord}_r(n) > (\log_2(n))^2$ where $\text{ord}_r(n)$ is multiplicative order of n modulo r .
3. Calculate $\text{gcd}(a, n)$ for $2 \leq a \leq r$. If $\text{gcd}(a, n) \neq 1, n$ for some a , then n composite.
4. If n has passed step (3) and $n \leq r$, then n is prime (because n is relatively prime to all $a < n$).
5. If n has passed all above steps then check that

$$(X + b)^n \equiv X^n + b \pmod{X^r - 1, n}$$

For all b such that $1 \leq b \leq \sqrt{\phi(r) \log_2(n)}$. If this the case n is prime i.e.

$$(X + b)^n \not\equiv X^n + b \pmod{X^r - 1, n}$$

For some b such that $1 \leq b \leq \sqrt{\phi(r) \log_2(n)}$, then n is composite.

Where $(X + b)^n \equiv X^n + b \pmod{X^r - 1, n}$ means that when we take the difference $(X + b)^n - (X^n + b)$ and divide by $X^r - 1$, the remainder has all of its coefficients divisible by n .

Example: We check whether $n = 37$ is prime or not by AKS Algorithm

$\log_2(n) = \log_2(37) \approx 5.2094$

For $2 \leq k \leq 5.2094$, we calculate $37^{\frac{1}{k}}$ (if $37^{\frac{1}{k}}$ is integer then 37 is composite).

$$37^{\frac{1}{2}} = 6.0827$$

$$37^{\frac{1}{3}} = 3.3322$$

$$37^{\frac{1}{4}} = 2.4663$$

$$37^{\frac{1}{5}} = 2.0589$$

Hence 37 is not perfect power of any integer.

Now we find smallest integer r such that

$$\text{ord}_r(n = 37) \geq (\log_2(n = 37))^2 \approx 27.1384$$

where $\text{ord}_r(n)$ is multiplicative order of n modulo r .

$$\text{ord}_2(37) = 1$$

$$\text{ord}_3(37) = 1$$

$$\text{ord}_4(37) = 1$$

$$\text{ord}_5(37) = 4$$

$$\text{ord}_6(37) = 1$$

and so on, We get

$$\text{ord}_{29}(37) = 28 > 27.1384$$

So we have $r = 29$

and $\gcd(a, 37) = 1$ for $2 \leq a \leq 27$

Since $n \geq r$ So it passes step (4), now we check step(5) i.e.

$$(X + b)^n \equiv X^n + b \pmod{X^r - 1, n}$$

for $n = 37$, $r = 29$ and $1 \leq b \leq \sqrt{\phi(r) \log_2(n)}$. and $\sqrt{\phi(29) \log_2(37)} = \sqrt{28 \log_2(37)} = 27.6$ i.e $1 \leq b \leq 27.6$

For $b = 1$

$$(X + 1)^{37} - (X^{37} + 1) = (X^{29} - 1)q(x) + r(x) \equiv r(x) \pmod{X^{29} - 1}$$

where

$$q(x) = 37X^7 + 666X^6 + 7770X^5 + 66045X^4 + 435897X^3 + 2324784X^2 + 10295472X + 38608020$$

and

$$\begin{aligned} r(x) = & 124403620X^{28} + 348330136X^{27} + 854992152X^{26} + 1852482996X^{25} + 3562467300X^{24} \\ & + 6107086800X^{23} + 9364199760X^{22} + 12875774670X^{21} + 15905368710X^{20} \\ & + 17672631900X^{19} + 17672631900X^{18} + 15905368710X^{17} + 12875774670X^{16} \\ & + 9364199760X^{15} + 6107086800X^{14} + 3562467300X^{13} + 1852482996X^{12} \\ & + 854992152X^{11} + 348330136X^{10} + 124403620X^9 + 38608020X^8 \\ & + 10295435X^7 + 2324118X^6 + 428127X^5 - 428127X^3 \\ & - 2324118X^2 - 10295435X - 38608020 \end{aligned}$$

Observe that each coefficient of $r(x)$ is divisible 37, So we have

$$(X + 1)^{37} \equiv X^{37} + 1 \pmod{X^{29} - 1, 37}$$

Similarly, we can do for all b and we get

$$(X + b)^{37} \equiv X^{37} + b \pmod{X^{29} - 1, 37}$$

for $1 \leq b \leq 27.6$

Therefore 37 is prime.

8.3 Exercise

1. Prove that 91 is pseudoprime to the base 3 but not to the base 2.
2. Prove that 561 is the smallest Carmichael number.
3. Prove that 825265 is a Carmichael number.
4. Is 1601 a Carmichael number? Justify.
5. Prove:
 - (a) A Carmichael number must be odd.
 - (b) A Carmichael number must be the product of distinct primes.
 - (c) A Carmichael number must be the product of atleast three distinct primes.
6. Prove that 187 is composite using Fermat's Primality Test.
7. Apply Rabin-Miller Primality Test to check the primality of 1105.
8. Find the remainder of $120!$ when divided by 127.
9. Apply AKS Primality Test to check the primality of $n = 31$.

Chapter 9

Factorization of Positive Integer

9.1 Fermat's Factorization

Fermat's method for factoring a composite number n tries to write it as the difference of two squares $n = x^2 - y^2$, yielding the factorization $n = (x + y)(x - y)$. Indeed, for a composite odd number $n = ab$ such a representation always exists:

$$\begin{aligned}(a + b)^2 - (a - b)^2 &= a^2 + b^2 + 2ab - (a^2 + b^2 - 2ab) = 4ab \\ \Rightarrow ab &= \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2 \\ \Rightarrow n &= x^2 - y^2 \text{ where } x = \frac{a+b}{2}, y = \frac{a-b}{2}\end{aligned}$$

If $n = x^2 - y^2$ is a positive odd integer then $x^2 - n = y^2$ i.e $x^2 - n$ is a perfect square.

Let m be the positive integer such that $m \geq \sqrt{n}$. Now look for perfect squares in the sequence $m^2 - n, (m + 1)^2 - n, (m + 2)^2 - n, \dots$. The search will end into a perfect square for $m \leq \frac{n+1}{2}$.

Since

$m^2 - n$, for $m = \frac{n+1}{2}$ is

$$m^2 - n = \frac{n^2 + 1 + 2n - 4n}{4} = \left(\frac{n-1}{2}\right)^2 \in \mathbb{Z}$$

Of course the value of m , gives trivial factor.

Remark: To have an initial guess for $m \geq \sqrt{n}, m \in \mathbb{N}$, apply Newton's method for function $f(x) = x^2 - n$, observe that $f'(x) = 2x$. Say r_0 and subsequent guesses by the sequence $r_k = r_{k-1} - \frac{f(r_{k-1})}{f'(r_{k-1})}$ for $k \geq 1$.

Example 1: Let $n = 2183$ and $m \geq \sqrt{2183}$.
 Since $47 > \sqrt{2183} > 46$. Choose $m = 47$.
 Now look for perfect square

$$47^2 - 2183 = 26$$

$$48^2 - 2183 = 121 = 11^2$$

$$\Rightarrow 2183 = 48^2 - 11^2$$

$$\Rightarrow 2183 = (48 - 11)(48 + 11)$$

$$\Rightarrow 2183 = 37 \times 59$$

Factors of 2183 are 37 and 59.

Example 2: Let $n = 3811$ and $m \geq \sqrt{3811}$.
 Since $62 > \sqrt{3811} > 61$. Choose $m = 62$.
 Now look for perfect square

$$62^2 - 3811 = 33$$

$$63^2 - 3811 = 158$$

$$64^2 - 3811 = 285$$

$$65^2 - 3811 = 414$$

$$66^2 - 3811 = 545$$

$$67^2 - 3811 = 678$$

$$68^2 - 3811 = 813$$

$$69^2 - 3811 = 950$$

$$70^2 - 3811 = 1089 = 33^2$$

$$\Rightarrow 3811 = 70^2 - 33^2$$

$$\Rightarrow 3811 = (70 - 33)(70 + 33)$$

$$\Rightarrow 3811 = 37 \times 103$$

Factors of 3811 are 37 and 103.

9.2 Pollard's (p-1) Factorization

Pollard's (p-1) Algorithm is based on :

Suppose a large positive integer n is to be factored, if p is a prime such that $p|n$ and $(p-1)|k!$ for some $k \in \mathbb{Z}^+$

WLOG, assume p is an odd prime i.e. $p > 2$. Let a be a positive integer such that $\gcd(a, n) = 1$ this implies that $\gcd(a, p) = 1$ then by Fermat's Little Theorem $a^{(p-1)} \equiv 1 \pmod{p}$

Since

$$\begin{aligned} (p-1)|k! &\Rightarrow a^{k!} \equiv 1 \pmod{p} \\ &\Rightarrow p|a^{k!} - 1 \\ \text{Let } a^{k!} - 1 &\equiv z \pmod{n} \\ &\Rightarrow a^{k!} - 1 = z + mn \quad \text{for some } m \in \mathbb{Z} \\ &\Rightarrow a^{k!} - 1 - mn = z \end{aligned}$$

Since $p|mn$ and $p|a^{k!} - 1$ this implies that $p|z$ and $p|n$ i.e. $p|\gcd(z, n)$

Note: Since n is odd, we can choose $a = 2$.

Algorithm:

Suppose n is given integer which is to be factored.

Choose a base a such that $\gcd(a, n) = 1$. Let z be the least non-negative residue in modulo n of $a^{k!} - 1$ i.e.

$$a^{k!} - 1 \equiv z \pmod{n} \quad \text{for } k = 1, 2, \dots$$

Case 1: $z \neq 0$

Calculate $\gcd(z, n)$

If $\gcd(z, n) \neq 1$ then we get a non trivial factor on n .

Case 2: If $z = 0$ then test fails and we try it for another a .

Example: $n = 61937$, now take $a = 2$

Note that $\gcd(a, n) = \gcd(2, 61937) = 1$

We calculate $2^{k!} - 1 \equiv z \pmod{n}$ for $k \in \mathbb{Z}^+$ and $\gcd(z, n)$

For $k = 1$, $2^{1!} - 1 \equiv 1 \pmod{61937}$ and $\gcd(1, 61937) = 1$

For $k = 2$, $2^{2!} - 1 \equiv 3 \pmod{61937}$ and

$$61937 = 3 \cdot 20645 + 2$$

$$3 = 2 \cdot 1 + 1$$

$$2 = 1 \cdot 2 + 0$$

Hence $\gcd(3, 61937) = 1$

For $k = 3$, $2^{3!} - 1 \equiv 63 \pmod{61937}$ and

$$61937 = 63 \cdot 983 + 8$$

$$63 = 8 \cdot 7 + 7$$

$$8 = 7 \cdot 1 + 1$$

$$7 = 1 \cdot 7 + 0$$

Hence $\gcd(63, 61937) = 1$

For $k = 4$, $2^{4!} - 1 \equiv 54225 \pmod{61937}$ and

$$61937 = 54225 \cdot 1 + 7712$$

$$54225 = 7712 \cdot 7 + 241$$

$$7712 = 241 \cdot 32 + 0$$

Hence $\gcd(54225, 61937) = 241$

Therefore 241 divides $n = 61937$ and $61937 = 241 \times 257$

9.3 Monte Carlo factorization

Monte Carlo factorization Algorithm (also known as Pollard's Rho Algorithm) is an algorithm for integer factorization. It was invented by John Pollard in 1975.

Setup

1. Let n be a positive composite integer.
2. Let p be a small prime factor of n .
3. Let k be an integer such that k is much larger than $\sqrt{p}$ and much smaller than $\sqrt{n}$, that is

$$\sqrt{p} \ll k \ll \sqrt{n}$$

4. Go to Algorithm.

Algorithm (When a small prime factor p is known)

1. Choose a sequence of positive integers

$$m_0, m_1, m_2, m_3, \dots, m_k$$

such that their least non-negative residues are incongruent modulo n but not all distinct modulo p , that is,

$$m_i \not\equiv m_j \pmod{n} \quad \forall i \neq j, \quad i, j \in \{0, 1, 2, 3, \dots, k\}$$

and

$$m_q \equiv m_r \pmod{p} \quad \text{for some } q \neq r, \quad q, r \in \{0, 1, 2, 3, \dots, k\}$$

2. Observe the sequence

$$m_0, m_1, m_2, m_3, \dots, m_k$$

Note that

p divides n and p divides $(m_q - m_r)$

Now compute

$$\gcd((m_q - m_r), n)$$

If gcd is greater than p , then we have a factor of n other than p .

Algorithm (When a prime factor p is not known)

1. Start with a randomly generated integer m_0 .
2. Choose a sequence of positive integers

$$m_1, m_2, m_3, \dots$$

such that

$$m_i \equiv m_{i-1}^2 + 1 \pmod{n} \quad \forall i \in 1, 2, 3, \dots$$

3. Compute

$$\gcd(m_{2i} - m_i, n)$$
4. If gcd is greater than 1 and less than n for some i , then we get a non trivial factor of n .
5. Proceed in similar way or go to the above algorithm (since we have a non trivial factor now and we can compute a prime factor of n also).

Note

1. Here, the function is choosen as $f(x_i) = x_{i-1}^2 + 1$.
2. We can take any other recursive function also.
3. If $n = pq$ where p and q are primes of the same size (that is, of same length), then Pollard's Rho Algorithm can find a non trivial factor of n in $O(\sqrt[4]{n})$.

Example 2: Let $n = 3293$

Let $m_0 = 3$, and $f(x_i) = x_{i-1}^2 + 1$, then
for each i , $m_i = m_{i-1}^2 + 1$
then,

$$m_1 = 3^2 + 1 = 10$$

$$m_2 = 10^2 + 1 = 101$$

$$m_3 = 101^2 + 1 = 10202 \equiv 323 \pmod{3293}$$

$$m_4 = 323^2 + 1 \equiv 2247 \pmod{3293}$$

and so on.

Compute

$$\gcd(m_{2i} - m_i, n)$$

that is,

$$\gcd(m_2 - m_1, n) = \gcd(91, 3293) = 1$$

$$\gcd(m_4 - m_2, n) = \gcd(2146, 3293) = 37$$

Thus 37 is a factor of $n = 3293$.

Other non trivial factor of n is 89.

Example 2: For $n = 8051$. Let $m_0 = 2$, and $f(x_i) = x_{i-1}^2 + 1$, then
for each i , $m_i = m_{i-1}^2 + 1$
then,

$$m_1 = 2^2 + 1 = 5$$

$$m_2 = 5^2 + 1 = 26$$

$$m_3 = 26^2 + 1 = 677$$

$$m_4 = 677^2 + 1 = 458330 \equiv 7474 \pmod{8051}$$

$$m_5 = 7474^2 + 1 \equiv 2839 \pmod{8051}$$

$$m_6 = 2839^2 + 1 \equiv 871 \pmod{8051}$$

and so on

Now compute

$$\gcd(m_{2i} - m_i, n)$$

that is,

$$\begin{aligned}\gcd(m_2 - m_1, n) &= \gcd(21, 8051) = 1 \\ \gcd(m_4 - m_2, n) &= \gcd(7448, 8051) = 1 \\ \gcd(m_6 - m_3, n) &= \gcd(194, 8051) = 97\end{aligned}$$

Thus 97 is a factor of $n = 8051$.

Other non trivial factor of n is 83.

Example 3: Let $n = 4087$

Let $m_0 = 2$, and $f(x_i) = x_{i-1}^2 + 3$, then

for each i , $m_i = m_{i-1}^2 + 3$

then,

$$\begin{aligned}m_1 &= 2^2 + 3 = 7 \\ m_2 &= 7^2 + 3 = 52 \\ m_3 &= 52^2 + 3 = 2707 \\ m_4 &= 2707^2 + 3 = 3948 \\ m_5 &= 3948^2 + 3 \equiv 2976 \pmod{4087} \\ m_6 &= 2976^2 + 3 \equiv 50 \pmod{4087} \\ m_7 &= 50^2 + 3 \equiv 2503 \pmod{4087} \\ m_8 &= 2503^2 + 3 \equiv 3728 \pmod{4087} \\ m_9 &= 3728^2 + 3 \equiv 2187 \pmod{4087} \\ m_{10} &= 2187^2 + 3 \equiv 1182 \pmod{4087} \\ m_{11} &= 1182^2 + 3 \equiv 3460 \pmod{4087} \\ m_{12} &= 3460^2 + 3 \equiv 780 \pmod{4087} \\ m_{13} &= 780^2 + 3 \equiv 3527 \pmod{4087} \\ m_{14} &= 3527^2 + 3 \equiv 2991 \pmod{4087}\end{aligned}$$

and so on.

Compute

$$\gcd(m_{2i} - m_i, n)$$

that is,

$$\begin{aligned}
\gcd(m_2 - m_1, n) &= \gcd(45, 4087) = 1 \\
\gcd(m_4 - m_2, n) &= \gcd(3896, 4087) = 1 \\
\gcd(m_6 - m_3, n) &= \gcd(-2657, 4087) = \gcd(2657, 4087) = 1 \\
\gcd(m_8 - m_4, n) &= \gcd(-220, 4087) \gcd(220, 4087) = 1 \\
\gcd(m_{10} - m_5, n) &= \gcd(-1794, 4087) \gcd(1794, 4087) = 1 \\
\gcd(m_{12} - m_6, n) &= \gcd(730, 4087) = 1 \\
\gcd(m_{14} - m_7, n) &= \gcd(488, 4087) = 61
\end{aligned}$$

Thus 61 is a factor of $n = 4087$.

Other non trivial factor of n is 67.

9.4 Factor Base Factorization

Factor Base: Let $\mathcal{B} = \{p_1, p_2, \dots, p_h\}$ where $p_1 = -1$ and $p_2, p_3 \dots p_h$ are distinct primes then b^2 is said to be $\mathcal{B}$ -smooth or $\mathcal{B}$ -number modulo n if

$$b^2 = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_h^{\alpha_h} \pmod{n}, \quad \alpha_i \geq 0$$

Example: for $n = 4633$ and $\mathcal{B} = \{-1, 2, 3\}$

$$\begin{aligned}
67^2 &\equiv 4489 \pmod{4633} \\
67^2 &\equiv -144 \pmod{4633} \\
67^2 &\equiv (-1)(2)^4(3)^2 \pmod{4633}
\end{aligned}$$

So $(67)^2$ is $\mathcal{B}$ -smooth in modulo 4633. Similarly we have,

$$\begin{aligned}
(68)^2 &\equiv (-1)(2)^0(3)^2 \pmod{4633} \\
(69)^2 &\equiv (-1)^0(2)^7(3)^0 \pmod{4633}
\end{aligned}$$

- If b^2 is a $\mathcal{B}$ -number for a given positive integer n

$$b^2 \pmod{n} = \prod_{j=1}^h p_j^{\alpha_j}, \quad p_j \in \mathcal{B}$$

Denote $\epsilon_j^b \equiv \alpha_j \pmod{2}$ i.e. powers of primes modulo 2 and $\epsilon_b = \{\epsilon_1^b, \epsilon_2^b, \dots, \epsilon_h^b\}$

From above example we have

$$\epsilon_{67} = \{1, 0, 0\}$$

$$\epsilon_{68} = \{1, 0, 0\}$$

$$\epsilon_{69} = \{0, 1, 0\}$$

Given a positive integer n to be factored, Suppose we have k $\mathcal{B}$ -smooth numbers b_i^2 where $\mathcal{B}$ is a factor base and

$$\epsilon_i = \{\epsilon_{i1}, \epsilon_{i2}, \dots, \epsilon_{ih}\}$$

corresponding to b_i^2 's for $i = 1, 2, \dots, k$ such that $\sum_{i=1}^k \epsilon_i = \{0, 0, \dots, 0\} \pmod{n}$. Let a_i is least absolute residue modulo n for b_i^2 , $i = 1, 2, \dots, k$ then

$$b_i^2 \equiv a_i \equiv \prod_{j=1}^h p_j^{\alpha_j} \pmod{n} \quad p_j \in \mathcal{B}, \quad i = 1, 2, \dots, k$$

So

$$\prod_{i=1}^k a_i = \prod_{i=1}^k \left(\prod_{j=1}^h p_j^{\alpha_{ij}} \right) \pmod{n}$$

$$= \prod_{j=1}^h p_j^{\sum_{i=1}^k \alpha_{ij}} \pmod{n}$$

$$\prod_{i=1}^k b_i^2 = \prod_{i=1}^k a_i = \prod_{j=1}^h p_j^{\sum_{i=1}^k \alpha_{ij}} \pmod{n}$$

Now if $\sum_{i=1}^k \epsilon_i = (0, 0, \dots, 0)$ i.e. sum of powers modulo 2 is 0, i.e. even, then right side become a perfect square,

$$\text{RHS} = \prod_{j=1}^h p_j^{\sum_{i=1}^k \alpha_{ij}} = l^2 \quad (\text{say})$$

and

$$b^2 = \left(\prod_{i=1}^k b_i \right)^2 \quad (\text{say})$$

then we have

$$b^2 \equiv l^2 \pmod{n}$$

i.e. $n \mid b^2 - l^2$ i.e. $n \mid (b - l)(b + l)$

and factors of n is obtained by computing $\gcd(n, b - l)$ and $\gcd(n, b + l)$

Example: From above we have, for $n = 4633$

$\epsilon_{67} = (1, 0, 0)$ and $\epsilon_{68} = (1, 0, 0)$

$\epsilon_{67} + \epsilon_{68} = (0, 0, 0)$

$$\begin{aligned} (67)^2 \cdot (68)^2 &\equiv (-1)^2(2)^4(3)^4 \pmod{n} \\ \implies (67 \cdot 68)^2 &\equiv 2^4 \cdot 3^4 \pmod{n} \\ \implies n \mid (67 \cdot 68)^2 - 2^4 \cdot 3^4 \\ \implies n \mid (67 \cdot 68)^2 - (36)^2 \\ \implies n \mid (67 \cdot 68 - 36)(67 \cdot 68 + 36) \\ \implies n \mid (4520)(4592) \end{aligned}$$

$$\gcd(n, 4520) = \gcd(4633, 4520) = 113$$

$$\gcd(n, 4592) = \gcd(4633, 4592) = 41$$

So factors of $n = 4633$ are 113 and 41.

9.5 Quadratic Sieve Factorization

Suppose n is the given positive integer which is to be factorized.

Let m is the least positive integer such that $m \geq \sqrt{n}$

Take the polynomial $f(x) = (x + m)^2 - n$

Now we calculate $f(j) = (j + m)^2 - n \pmod{n}$ for different values of $j \in \mathbb{Z}$

Choose a factor base $\mathcal{B} = \{p_1, p_2, \dots, p_n\}$ such that we can factor $f(j) \pmod{n}$ using primes in $\mathcal{B}$.

Find some $f(j)$'s such that product of $f(j) \pmod{n}$ is an even power of primes in $\mathcal{B}$ (as in Factor Base Factorization).

Say

$$f(j_1) \cdot f(j_2) \cdots f(j_k) \equiv p_1^{e_1} \cdot p_2^{e_2} \cdots p_n^{e_n} \pmod{n}$$

where e_i 's are even i.e right hand side is perfect square .

Suppose $p_1^{e_1} \cdot p_2^{e_2} \cdots p_n^{e_n} = l^2$ then we have

$$f(j_1) \cdot f(j_2) \cdots f(j_k) \equiv l^2 \pmod{n}$$

i.e.

$$(j_1 + m)^2 \cdot (j_2 + m)^2 \cdots (j_k + m)^2 \equiv l^2 \pmod{n}$$

Suppose $(j_1 + m)^2 \cdot (j_2 + m)^2 \cdots (j_k + m)^2 \equiv r^2 \pmod{n}$

So we have

$$n \mid r^2 - l^2$$

i.e

$$n \mid (r - l)(r + l)$$

A factors of n is given by

$$\gcd(r - l, n)$$

and

$$\gcd(r + l, n)$$

Example: For $n = 7429$, $\sqrt{n} \approx 86.20$, so $m = 87$

take $f(x) = (87 + x)^2 - 7429$

$$f(0) = 87^2 - 7429 \equiv 140 \pmod{7429} \equiv 2^2 \cdot 5 \cdot 7 \pmod{7429}$$

$$f(1) = 88^2 - 7429 \equiv 315 \pmod{7429} \equiv 3^2 \cdot 5 \cdot 7 \pmod{7429}$$

$$f(-2) = 85^2 - 7429 \equiv -204 \pmod{7429} \equiv (-1) \cdot 2^2 \cdot 3 \cdot 17 \pmod{7429}$$

Note that $f(0)f(1) \equiv 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \pmod{7429}$

$$(87^2 - 7429)(88^2 - 7429) \equiv 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \pmod{7429}$$

$$(87 \cdot 88)^2 \equiv (210)^2 \pmod{7429}$$

$$\implies n \mid (87 \cdot 88)^2 - (210)^2$$

$$\implies n \mid (87 \cdot 88 - 210)(87 \cdot 88 + 210)$$

$$\text{i.e. } n \mid (7446)(7866)$$

$$\gcd(n, 7446) = 17$$

$$\gcd(n, 7866) = 1$$

Factors of $n = 7429$ are 17 and 437.

9.6 Continued Fraction Factorization

For a give real number x , we define continued fraction

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \ddots}}}}$$

In compact form $x = [a_0; a_1, a_2, \dots, a_n, \dots]$

where $a_0, a_1, \dots, a_n$ are real numbers and all are positive except $a_0 \geq 0$. Continued Fractions can be finite or infinite.

k^{th} convergent of continued fraction is

$$c_k = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{\ddots + \frac{1}{a_k}}}}}$$

Example:

$$\frac{19}{35} = 0 + \frac{1}{\frac{35}{19}}$$

$$= 0 + \frac{1}{1 + \frac{16}{19}}$$

$$= 0 + \frac{1}{1 + \frac{1}{\frac{19}{16}}}$$

$$= 0 + \frac{1}{1 + \frac{1}{1 + \frac{3}{16}}}$$

$$= 0 + \frac{1}{1 + \frac{1}{1 + \frac{1}{5 + \frac{1}{3}}}}$$

$$\frac{19}{35} = [0; 1, 1, 5, 3]$$

$$c_0 = 0$$

$$c_1 = 0 + \frac{1}{1} = 1$$

$$c_2 = 0 + \frac{1}{1 + \frac{1}{1}} = \frac{1}{2}$$

$$c_3 = 0 + \frac{1}{1 + \frac{1}{1 + \frac{1}{5}}} = \frac{6}{11}$$

$$c_4 = 0 + \frac{1}{1 + \frac{1}{1 + \frac{1}{5 + \frac{1}{3}}}} = \frac{19}{35}$$

Now define A_k and B_k for $k = 0, 1, 2, \dots$

$$A_0 = a_0 \text{ and } B_0 = 1$$

$$A_1 = a_1 a_0 + 1 \text{ and } B_1 = a_1$$

$$A_k = a_k A_{k-1} + A_{k-2} \text{ and } B_k = a_k B_{k-1} + B_{k-2}$$

then we have

$$\begin{aligned} c_0 &= \frac{A_0}{B_0} \\ c_1 &= \frac{A_1}{B_1} \\ c_2 &= \frac{A_2}{B_2} \\ &\vdots \\ c_k &= \frac{A_k}{B_k} \end{aligned}$$

Algorithm for factorization

Suppose n is the given positive integer which is to be factorized.

- Write $\sqrt{n}$ in continued fraction.
 - Let $\frac{A_k}{B_k}$ be the k^{th} convergent of continued fraction.
 - Write $A_k^2 \equiv r_k \pmod{n}$
 - Choose set $\{r_1, r_2, \dots, r_m\}$ such that $\prod_{i=1}^m r_i = X^2 \pmod{n}$ (say) is perfect square.
 - Let $\prod_{i=1}^m A_i \equiv Y \pmod{n}$
- then we have

$$\begin{aligned} X^2 &\equiv \prod_{i=1}^m r_i = \prod_{i=1}^m A_i^2 \equiv Y^2 \pmod{n} \\ \implies X^2 &\equiv Y^2 \pmod{n} \end{aligned}$$

If $X \neq \pm Y$ then factors of n is given by

$$\gcd((X - Y), n)$$

and

$$\gcd((X + Y), n)$$

9.6.1 Algorithm for writing Continued fraction of $\sqrt{n}$

Initialization:

$$\begin{aligned}a_0 &= 2 \lfloor \sqrt{n} \rfloor \\ y_0 &= \lfloor \sqrt{n} \rfloor \\ z_0 &= 1\end{aligned}$$

Iteration process for $k \geq 1$

$$\begin{aligned}y_k &= a_{k-1} \cdot z_{k-1} - y_{k-1} \\ z_k &= \left\lfloor \frac{n - y_k^2}{z_{k-1}} \right\rfloor \\ a_k &= \left\lfloor \frac{\lfloor \sqrt{n} \rfloor + y_k}{z_k} \right\rfloor\end{aligned}$$

Then continued fraction for $\sqrt{n} = [\lfloor \sqrt{n} \rfloor; a_1, a_2, \dots]$

$$\sqrt{n} = \lfloor \sqrt{n} \rfloor + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}$$

Example: For $n = 33$, we write $\sqrt{33}$ in continued fraction
 $\sqrt{n} = \sqrt{33} \approx 5.744$ i.e. $\lfloor \sqrt{33} \rfloor = 5$

$$\begin{aligned}a_0 &= 2 \cdot \lfloor \sqrt{33} \rfloor = 2 \cdot 5 = 10 \\ y_0 &= \lfloor \sqrt{33} \rfloor = 5 \\ z_0 &= 1\end{aligned}$$

$$y_1 = a_0 z_0 - y_0 = 10 \cdot 1 - 5 = 5$$

$$z_1 = \left\lfloor \frac{33 - 25}{1} \right\rfloor = 8$$

$$a_1 = \left\lfloor \frac{[\sqrt{33}] + y_1}{z_1} \right\rfloor = \left\lfloor \frac{5 + 5}{8} \right\rfloor = 1$$

$$y_2 = a_1 z_1 - y_1 = 1 \cdot 8 - 5 = 3$$

$$z_2 = \left\lfloor \frac{33 - 9}{8} \right\rfloor = 3$$

$$a_2 = \left\lfloor \frac{[\sqrt{33}] + y_2}{z_2} \right\rfloor = \left\lfloor \frac{5 + 3}{3} \right\rfloor = 2$$

$$y_3 = a_2 z_2 - y_2 = 2 \cdot 3 - 3 = 3$$

$$z_3 = \left\lfloor \frac{33 - 9}{3} \right\rfloor = 8$$

$$a_3 = \left\lfloor \frac{[\sqrt{33}] + y_3}{z_3} \right\rfloor = \left\lfloor \frac{5 + 3}{8} \right\rfloor = 1$$

$$\sqrt{33} = 5 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}}$$

$$\sqrt{33} = [5; 1, 2, 1 \dots] = [a_0; a_1, a_2, a_3 \dots]$$

k-th convergent $c_k = \frac{A_k}{B_k}$

$$A_0 = a_0$$

$$B_0 = 1$$

$$A_1 = a_1 a_0 + 1$$

$$B_1 = a_1$$

$$A_k = a_k A_{k-1} + A_{k-2}$$

$$B_k = a_k B_{k-1} + B_{k-2}$$

$$A_0 = a_0 = 5$$

$$B_0 = 1$$

$$A_1 = a_1 a_0 + 1 = 5 + 1 = 6$$

$$B_1 = a_1 = 1$$

$$A_2 = a_2 A_1 + A_0 = 17$$

$$B_2 = a_2 B_1 + B_0 = 3$$

$$A_3 = a_3 A_2 + A_1 = 23$$

$$B_3 = a_3 B_2 + B_1 = 4$$

$$c_0 = 5$$

$$c_1 = 6$$

$$c_2 = \frac{17}{3}$$

$$c_3 = \frac{23}{4}$$

So, $A_0 = 5$, $A_1 = 6$, $A_2 = 17$, $A_3 = 23$

$$A_0^2 \equiv 25 \pmod{33} = r_0$$

$$A_1^2 = 36 \equiv 3 \pmod{33} = r_1$$

$$A_2^2 = 17^2 \equiv 25 \pmod{33} = r_2$$

$$A_3^2 = 23^2 \equiv 1 \pmod{33} = r_3$$

Note that $r_2 = 25 = 5^2 = X^2$ and $A_2 = 17 \pmod{33} = Y$

Also

$$X^2 \equiv r_2 = A_2^2 \equiv Y^2 \pmod{33}$$

And $X \neq \pm Y$. Therefore factors of 33 is given by

$$\gcd(Y - X, n) = \gcd(17 - 5, 33) = 3$$

$$\gcd(Y + X, n) = \gcd(17 + 5, 33) = 11$$

Hence $33 = 3 \times 11$

9.7 Exercise

1. Find the factors of the following using Fermat's Factorization:

(a) 6497

- (b) 14129
- 2. Use Pollard's $p - 1$ method to factor each of the following numbers:
 - (a) 1739
 - (b) 220459
- 3. Apply Monte Carlo Factorization Method to find the factors of the following integers:
 - (a) 3741
 - (b) 5251
- 4. Apply Monte Carlo Factorization Method to find the factors of 4087 using $f(x) = x^2 + x + 1$ and $x_0 = 7$.
- 5. If $n = 2701$, find the smallest factor base B such that squares of 52, 53 are B-numbers and then factor n using the factor base B.
- 6. If $n = 6499$, find the smallest factor base B such that squares of 71, 73, 83, 85 are B-numbers and then factor n using the factor base B.
- 7. Apply Quadratic Sieve Factoring Algorithm to find the factors of 8051.
- 8. Apply Quadratic Sieve Factoring Algorithm to find the factors of 1261.
- 9. Find the Continued Fraction representation of the following rational numbers:
 - (a) $\frac{27}{72}$
 - (b) 1.12
- 10. Apply Continued Fraction Factorization Method to find the factors of the following integers:
 - (a) 57
 - (b) 105

Chapter 10

Key Exchange Protocols

10.1 Diffe-Hellman Key Exchange Protocol

Let $G = \langle g | g^n = 1 \rangle$ be a cyclic group generated by g and be public. Alice (sender) and Bob (receiver) exchange the key as follows:

	Alice		Bob
Private	a		$b \in \mathbb{Z}_n$
Public	g^a	$\longrightarrow$	$g^a \in G$
Public	g^b	$\longleftarrow$	$g^b \in G$

Now Alice calculate $(g^b)^a = g^{ab}$ and Bob calculate $(g^a)^b = g^{ab}$. Hence $K = g^{ab}$ which becomes their common knowledge.

Example: Let Alice and Bob be using shift transformation

$$\begin{aligned} \mathcal{E} : \mathcal{A} &\rightarrow \mathcal{A} = \{A = 0, B = 1, \dots, Z = 25\} \\ m &\rightarrow \mathcal{E}(m) = m + k \pmod{26} \end{aligned}$$

To exchange the encryption key k an integer modulo 26 they use Diffe-Hellman key exchange protocol as follows.

Let $G = \mathbb{F}_{53}^* = \mathbb{Z}_{53}^*$

Let $g = 2 \in G$. Note that $\text{order}(g) = |g| = 52$ hence g is the generator of $G = \langle g = 2 | g^{52} = 1 \pmod{53} \rangle$. Alice and Bob make public the group G .

	Alice		Bob
Private	$a = 29$		$b = 19 \in \mathbb{Z}_n$
Public	g^a	$\longrightarrow$	$g^a = 2^{29} \equiv 45 \pmod{53} \in G$
Public	$g^b = 12$	$\longleftarrow$	$g^b = 2^{19} \equiv 12 \pmod{53} \in G$

Now, Alice will calculate

$$\begin{aligned}
 (g^b)^a &= (12)^{29} \pmod{53} \\
 12^2 &= 144 \equiv 38 \pmod{53} \\
 12^4 &= 1444 \equiv 13 \pmod{53} \\
 12^8 &= 169 \equiv 10 \pmod{53} \\
 12^{16} &= 100 \equiv -6 \pmod{53} \\
 12^{24} &= 12^{16} \cdot 12^8 \equiv -60 \equiv -7 \pmod{53} \\
 12^{29} &= 12^{24} \cdot 12^4 \cdot 12^1 \equiv -7 \cdot 13 \cdot 12 = -1092 \equiv 21 \pmod{53}
 \end{aligned}$$

Since $g^b = 12 \implies b = 19$ (as the least non-negative integer x such that $g^x = y$, for $y = 12, g = 2 \in G$)
 $g^a = 45 \quad b = 19$

Now, Bob will calculate

$$\begin{aligned}
 (g^a)^b &= (45)^{19} \equiv (-8)^{19} \pmod{53} \\
 &\equiv (-2^3)^{19} \pmod{53} \\
 &\equiv (-1)^3 2^{57} \pmod{53} \\
 &\equiv (-1)^3 2^{52} \cdot 2^5 \pmod{53} \\
 &\equiv -32 \pmod{53} \\
 &\equiv 21 \pmod{53}
 \end{aligned}$$

Alice choose $a = 29$, Bob choose $b = 19$ and kept as a secret. Made public $g^a = 2^{29} \equiv 45 \pmod{53}$ and $g^b = 2^{19} \equiv 12 \pmod{53}$. Then common shared key $k = 21$.

10.2 Massey-Omura Cryptosystem

Let $G = \langle g | g^n = 1 \rangle$ be a cyclic group generated by g and be public. Suppose a message $M \in G$ is sent by Alice to Bob. They choose integer in $\mathbb{Z}_n$ a

and b respectively and keep secret. Then Alice and Bob exchange the key as follows.

	Alice		Bob
Private	a		$b \in \mathbb{Z}_n$
	M^a	$\longrightarrow$	$M^a \in G$
	$M^{ab} = (M^a)^b$	$\longleftarrow$	$(M^a)^b \in G$
	$(M^{ba})^{a^{-1}}$	$\longrightarrow$	$(M^{ba})^{a^{-1}} = M^b$

Hence Bob receives the message M by computing $(M^b)^{b^{-1}} = M$. Provided a^{-1} and b^{-1} exist in Modulo $\phi(n)$.

Example: Let $\mathcal{A} = \{A = 0, B = 1, \dots, Z = 25, ? = 26, ! = 27, @ = 28\}$, $n = 29$, $\phi(29) = 28$

Alice wants to send message "CARTOON".

Alice: private key $a = 5$

Bob : Private key $b = 3$

To send $M='C'=2$ Alice will send to Bob

$$M^a = 2^5 \equiv 3 \pmod{29}$$

Now Bob will send to Alice

$$(M^a)^b = 3^3 \equiv 27 \pmod{29}$$

Alice will send to Bob

$$(M^{ab})^{a^{-1}} = 27^{5^{-1}} \equiv 27^{17} \equiv 8 \pmod{29}$$

(since $5^{-1} \equiv 17 \pmod{28}$) Bob get the message by computing $(M^b)^{b^{-1}} = 8^{3^{-1}} \equiv 8^{19} \equiv 2 \pmod{29} = 'C'$ (since $3^{-1} \equiv 19 \pmod{28}$).

10.3 ElGamal Cryptosystem

Let $G = \langle g | g^n = 1 \rangle$ be public. All users, say Alice and Bob They choose random integers $a, b, \dots$ respectively and keep it secret and make $g^a, g^b, \dots \in G$ public.

Now, if Bob wants to send message $\mathcal{M}$ to Alice. He chooses a random integer k and creates a mask by doing $(g^a)^k \in G$. Bob then sends to Alice

$(g^k, \mathcal{M}g^{ak}) \in G \times G$.

Alice will receive $(g^k, \mathcal{M}g^{ak}) \in G \times G$, to recover $\mathcal{M}$ from order pair $(g^k, \mathcal{M}g^{ak})$, Alice will recreate the mask by doing $(g^k)^a = g^{ka} = g^{ak} \in G$, as Alice know a (secret) and g^k is first element of order pair $(g^k, \mathcal{M}g^{ak})$ received from Bob. To recover the message, Alice will compute

$$(\mathcal{M}g^{ak})(g^{ak})^{-1} = \mathcal{M} \in G$$

Example: Let

$$G = \mathbb{Z}_{29}^* = \langle 2 \mid 2^{28} = 1 \rangle = \{2^0, 2^1, 2^2, \dots, 2^{27}\}$$

$$= \{1, 2, 4, 8, 16, 3, 6, 12, 24, \dots\},$$

$$|G| = 28$$

Suppose $\mathcal{A} = \{A = 1, B = 2, C = 4, D = 8, E = 16, F = 3, G = 6, H = 12, I = 24, \dots\}$, $|\mathcal{A}| = 28$

Alice: Private key $a = 3$, Public key: $g^a = 2^3 = 8$

Bob wants to send message "HELLO"

To send $\mathcal{M} = H = 12 = 2^7 \in G$,

Bob choose $k = 5$,

create mask:

$$(g^a)^k = (2^3)^5 = 2^{15} \equiv 27 \pmod{29}$$

Bob knows: k , $g^a = 8$, and $\mathcal{M}$,

Bob will send to Alice $(g^k, \mathcal{M}g^{ak})$ i.e.

$$(2^5, 12 \cdot 27) = (3, 5) \in G \times G$$

Alice knows: a and $(g^k, \mathcal{M}g^{ak}) = (3, 5) \in G \times G$

Now, to recover the message Alice recreate the mask

$$(g^k)^a = 3^3 \equiv 27 \pmod{29} \in G$$

and compute

$$\mathcal{M}(g^{ak}) \cdot [g^{ak}]^{-1} = 5 \cdot 27^{-1} \equiv 5 \cdot 14 \equiv 12 \pmod{29}$$

(since $27^{-1} \equiv 14 \pmod{29}$)

So, Alice will get $\mathcal{M} = H$ (as 12 correspond to H).

10.4 Exercise

1. Alice and Bob decide to follow the Diffie-Hellman key exchange protocol to generate a shared key. Alice selects $a = 7$ and Bob selects $b = 9$ as their secret key. Assume that they use the prime $p = 101$. Prove that $g = 2$ is a primitive root of $\mathbb{Z}_{101}^*$. Also describe the steps that Alice and Bob follows to generate a shared key.
2. Alice and Bob decide to follow the Diffie-Hellman key exchange protocol to generate a shared key. Assume that they use the prime $p = 149$. Prove that $g = 3$ is a generator of $\mathbb{Z}_{149}^*$. Alice selects $a = 11$ as her secret key and Bob make 75 public. Find their shared key.
3. Let Alice and Bob use Hill Cipher to encrypt the message m as km for $k \in \mathbb{Z}_{41}^*$. Let $G = \mathbb{Z}_{83}^*$. and 5 be the generator of G . Alice and Bob generate a shared key as $g^{ab} = 52$. Find the decryption of cipher text 9. Also, find the value of a, b, g^a, g^b .
4. Let the message $M = 72 \in \mathbb{Z}_{97}^*$. Let 13 and 17 be the secret keys of Alice and Bob respectively. Show the process of sending the message M to Bob using Massey-Omura Cryptosystem.
5. Alice want to send a message $M \in \mathbb{Z}_{61}^*$ to Bob using Massey-Omura Cryptosystem. She chooses $a = 7$ as her private key and computes $M^a = 30$. Bob chooses $b = 11$ as his private key. Find the value of M .
6. Let $G = \mathbb{Z}_{41}^* = \langle 7 \mid 7^{40} = 1 \rangle$. Suppose private key of Alice is $a = 23$ and Bob chooses $k = 17$ randomly. Bob wants to send the message $M = 7$ to Alice using Elgamal Cryptosystem. Create the mask and the pair (g^k, Mg^{ak}) that Bob will send to Alice. Also, show the recovery process performed by her to get M .
7. Let $G = \mathbb{Z}_{79}^* = \langle 7 \mid 7^{78} = 1 \rangle$. Suppose private key of Alice is $a = 15$. Bob creates the signature $(60, 69)$. Recover the message M .

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Chapter 11

Discrete Logarithm

Definition: Let G be a group and $g, h \in G$, then x is called the **discrete logarithm for h in G to the base g** (denoted as $x = dlog_g h$) if x is the least non negative integer satisfying $g^x = h$.

Note :

1. Discrete Logarithm may or may not exists.
If discrete logarithm for h in G to the base g exists, then we say $dlog_g h$ exists.
If discrete logarithm for h in G to the base g does not exist, then we say $dlog_g h$ does not exist.
2. If G is a finite cyclic group of order n with generator g that is, $G = \langle g \mid g^n = I_G \rangle$, then $\forall h \in G, \exists x$ such that $g^x = h$

Examples:

1. Let $G = \mathbb{Z}_p^*$, where p is prime and $\mathbb{Z}_p^* = \mathbb{Z}_p - \{0\}$, then G is **cyclic**.
Generators of G are all those integers elements of G whose g.c.d. with p is 1. Let $a \in G$ such that $\text{g.c.d.}(a, p) = 1$, then $\forall b \in G, \exists x$ such that $a^x = b$ if $G = \langle a \mid a^{p-1} = 1 \rangle$.
2. Let $G = \mathbb{Z}_2 \oplus \mathbb{Z}_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, then G is **abelian but not cyclic**.
Note that $(0, 1)^2 = (0, 1) + (0, 1) = (0, 0)$, $(1, 0)^2 = (1, 0) + (1, 0) = (0, 0)$, $(1, 1)^2 = (1, 1) + (1, 1) = (0, 0)$.

In this case discrete logarithm for any element of G to any base will not exist except $(0, 0)$.

3. Let $G = S_3 = \{I, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$ be the set of permutations on symbols $\{1, 2, 3\}$ then G is **neither abelian nor cyclic**.

Note that

$$\begin{aligned}(1\ 2)(1\ 2) &= I \\ (1\ 3)(1\ 3) &= I \\ (2\ 3)(2\ 3) &= I \\ (1\ 2\ 3)(1\ 2\ 3) &= (1\ 3\ 2) \\ (1\ 2\ 3)(1\ 2\ 3)(1\ 2\ 3) &= I \\ (1\ 3\ 2)(1\ 3\ 2) &= (1\ 2\ 3) \\ (1\ 3\ 2)(1\ 3\ 2)(1\ 3\ 2) &= I\end{aligned}$$

Observe that $dlog_{(1\ 2\ 3)}(1\ 3\ 2) = 2$

Similarly $dlog_{(1\ 3\ 2)}(1\ 2\ 3) = 2$

But $dlog_{(1\ 2\ 3)}(1\ 2)$ does not exist.

Similarly check for other elements with different bases.

4. Let $G = D_8 = \{1, a, a^2, a^3, b, ab, a^2b, a^3b\} = \langle a, b \mid a^4 = 1, b^2 = 1, ab = ba^3 \rangle$, where D_8 is the dihedral group of order 8 and a be the rotation and b be the reflection, then G is **neither abelian nor cyclic**. Generators of G are a and b .
- (a) Check for each element with base = a .
- (b) Check for each element with base = b .

The computation of discrete logarithm whenever exists is called Discrete Logarithm Problem (DLP).

The discrete logarithm problem is solvable in a cyclic group to the base g , where g is a generator of G . We shall discuss some algorithms to compute the discrete logarithm in a finite cyclic group of order n , say $G = \langle g : g^n = 1 \rangle$:

1. By enumeration : The DLP is solvable. Hence, to compute $x = dlog_g h$ for an element h of G , simply by raising powers of g , till it becomes h . This is the simplest way. It always works. It needs only to store

x, g, g^x . The problem is computation of g powers in a large finite cyclic group, as enumeration requires $x - 1$ multiplications and x number of comparisons in G . The method is not suitable for practical purposes, as x is usually desired to be large. Thus we need to look for other algorithms.

2. Some more algorithms are given as:

- (a) Shank Baby-Step Giant-Step Algorithm (given by Donald Shanks).
- (b) Pollard's Rho Algorithm (given by JH Pollard).
- (c) Pohlig-Hellman Algorithm
- (d) Index Calculus Algorithm (in its most basis setup). This is the fastest known algorithm.

11.1 Shank Baby-Step Giant-Step Algorithm

- This algorithm is given by Daniel Shanks in his paper "Class number, a theory of factorization and genera" in 1971.
- The baby-step giant-step algorithm is a generic algorithm. It works for every finite cyclic group.
- The algorithm is based on a space-time tradeoff. It is a fairly simple modification of trial multiplication, the naive method of finding discrete logarithms.
- The algorithm requires $O(m)$ memory where $m = \sqrt{n}$, and n is equal to the order of group.(that we will see later).
- As the name suggests, this method is divided into two parts. We call them as Baby Steps and Giant Steps.

Set up

Let G be a finite cyclic group.

$$G = \langle g \mid g^n = 1 \rangle$$

Then, $\forall a \in G$, $dlog_g a$ exists.

Let, m be the least positive integer greater than or equal to $\sqrt{n}$.

$$\text{Let } x = dlog_g a \Rightarrow g^x = a$$

$$\begin{aligned}
&\Rightarrow \exists q, r \in \mathbb{Z}, 0 \leq r < m \text{ such that } x = mq + r. \\
&\Rightarrow g^x = g^{mq+r} = a \\
&\Rightarrow g^{mq} = ag^{-r}
\end{aligned}$$

Algorithm :**Baby Steps**

To compute baby steps, we compute a set

$$B = \{(ag^{-r}, r) | 0 \leq r \leq m-1\}$$

that is,

$$B = \{(a, 0), (ag^{-1}, 1), (ag^{-2}, 2), (ag^{-3}, 3), \dots, (ag^{-(m-1)}, m-1)\}$$

If $\exists s, 0 \leq s \leq m-1$ such that $(ag^{-s}, s) = (1, s) \in B$,
then, $ag^{-s} = 1 \Rightarrow a = g^s \Rightarrow s = d\log_g a$

If such s does not exist, we go the Giant Steps.

Giant Steps:

If such s does not exist, that is, there does not any $t, 0 \leq t \leq m-1$ such
that $(ag^{-t}, t) = (1, t) \in B$, we go the Giant Steps.

To compute giant steps, we compute :

$$g^m, (g^m)^2, (g^m)^3, (g^m)^4, \dots$$

that is,

$$(g^m)^i, \quad i = 1, 2, 3, \dots$$

till we get some q such that $(g^m)^q = ag^{-r}$ for some $0 \leq r \leq m-1$.

In that case $(g^m)^q = ag^{-r} \Rightarrow g^{mq+r} = a$

and hence $d\log_g a = mq + r = x$.

Examples :

1. Determine, if exists, $dlog_5 3$ in $\mathbb{Z}_{53}^*$.

Let, m be the least positive integer greater than or equal to $\sqrt{53} = 7.2$ approx.

$$\Rightarrow m = 8$$

$$\text{Let } x = dlog_5 3 \Rightarrow 5^x = 3.$$

$$\Rightarrow \exists q, r \in \mathbb{Z}, 0 \leq r < 8 \text{ such that } x = 8q + r.$$

Baby Steps :

To compute baby steps, we compute a set

$$B = \{(ag^{-r}, r) | 0 \leq r \leq 7\}$$

that is,

$$B = \{(a, 0), (ag^{-1}, 1), (ag^{-2}, 2), (ag^{-3}, 3), (ag^{-4}, 4), (ag^{-5}, 5), (ag^{-6}, 6), (ag^{-7}, 7)\}$$

$$\text{Note that } g^{-1} \equiv 5^{-1} \equiv 32 \pmod{53}$$

$$\Rightarrow B = \{(3, 0), (3 \cdot 32, 1), (3 \cdot (32)^2, 2), (3 \cdot (32)^3, 3), (3 \cdot (32)^4, 4), \\ (3 \cdot (32)^5, 5), (3 \cdot (32)^6, 6), (3 \cdot (32)^7, 7)\}$$

$$\Rightarrow B = \{(3, 0), (43, 1), (51, 2), (42, 3), (19, 4), (25, 5), (5, 6), (1, 7)\}$$

$$\text{Since } \exists s = 7, \text{ such that } (ag^{-s}, s) = (1, s) = (1, 7) \in B,$$

$$\text{then, } 3 \cdot 5^{-7} \equiv 1 \pmod{53} \Rightarrow 3 = 5^7 \Rightarrow 7 = dlog_5 3$$

2. Determine, if exists, $dlog_5 15$ in $\mathbb{Z}_{37}^*$.

Let, m be the least positive integer greater than or equal to $\sqrt{37} = 6.08$ approx.

$$\Rightarrow m = 7$$

$$\text{Let } x = dlog_5 15 \Rightarrow 5^x = 15.$$

$$\Rightarrow \exists q, r \in \mathbb{Z}, 0 \leq r < 7 \text{ such that } x = 7q + r.$$

Baby Steps :

To compute baby steps, we compute a set

$$B = \{(ag^{-r}, r) | 0 \leq r \leq 6\}$$

that is,

$$B = \{(a, 0), (ag^{-1}, 1), (ag^{-2}, 2), (ag^{-3}, 3), (ag^{-4}, 4), (ag^{-5}, 5), (ag^{-6}, 6)\}$$

Note that $g^{-1} \equiv 5^{-1} \equiv 15 \pmod{37}$

$$\Rightarrow B = \{(15, 0), (15 \cdot 15, 1), (15 \cdot (15)^2, 2), (15 \cdot (15)^3, 3), (15 \cdot (15)^4, 4), \\ (15 \cdot (15)^5, 5), (15 \cdot (15)^6, 6), (15 \cdot (15)^7, 7)\}$$

$$\Rightarrow B = \{(15, 0), (3, 1), (8, 2), (9, 3), (24, 4), (27, 5), (35, 6)\}$$

Since there does not exist s , such that $(ag^{-s}, s) = (1, s) \in B$, thus we go to giant step.

Giant Steps :

To compute giant steps, we compute :

$$g^m, (g^m)^2, (g^m)^3, (g^m)^4, \dots$$

that is,

$$(5^7)^i, \quad i = 1, 2, 3, \dots$$

Note that in modulo 37, we have

$$(5^7)^1 = 18$$

$$(5^7)^2 = 28$$

$$(5^7)^3 = 23$$

$$(5^7)^4 = 7$$

$$(5^7)^5 = 15$$

Here we get $q = 5$ such that $(g^m)^q = (5^7)^5 = 15 = ag^{-r}$ for $r = 0$.

$$\text{Hence } (5^7)^5 = 15 \Rightarrow d\log_5 15 = 7 \cdot 5 + 0 = 35.$$

11.2 Pollard's ρ algorithm for discrete logarithm

Let G be a finite multiplicative cyclic group generated by g of order n , i.e.

$$G = \langle g | g^n = 1 \rangle$$

To find : $d \log_g a$ (it exists as G is cyclic)

Let G_1, G_2, G_3 be pairwise disjoint subsets of G such that $G = G_1 \cup G_2 \cup G_3$.

Define a function f

$$f : G \rightarrow G \quad \text{where } G = G_1 \cup G_2 \cup G_3$$

such that for $b \in G$

$$b \rightarrow f(b) = \begin{cases} gb & \text{if } b \in G_1 \\ b^2 & \text{if } b \in G_2 \\ ab & \text{if } b \in G_3 \end{cases}$$

Then we choose a random integer $x_0 \in \{1, 2, 3, \dots, n\}$ and define $b_0 = g^{x_0} \in G$, and a sequence of elements $\{b_0, b_1, \dots\}$ of G by

$$b_{i+1} = f(b_i) \quad \forall i \geq 0$$

for example :

$$b_1 = f(b_0) = \begin{cases} gb_0 & \text{if } b_0 \in G_1 \\ b_0^2 & \text{if } b_0 \in G_2 \\ ab_0 & \text{if } b_0 \in G_3 \end{cases} = \begin{cases} gg^{x_0} & \text{if } b_0 \in G_1 \\ (g^{x_0})^2 & \text{if } b_0 \in G_2 \\ ag^{x_0} & \text{if } b_0 \in G_3 \end{cases} = \begin{cases} g^{x_0+1} & \text{if } b_0 \in G_1 \\ g^{2x_0} & \text{if } b_0 \in G_2 \\ ag^{x_0} & \text{if } b_0 \in G_3 \end{cases}$$

as $b_0 = g^{x_0}$

similarly

$$b_2 = f(b_1) = \begin{cases} gb_1 & \text{if } b_1 \in G_1 \\ b_1^2 & \text{if } b_1 \in G_2 \\ ab_1 & \text{if } b_1 \in G_3 \end{cases}$$

NOTE 1 : $b_i \in G$ and b_i is of the form $b_i = g^{x_i} a^{y_i}$ ($i \geq 0$)

thus $b_{i+1} = f(b_i) = g^{x_{i+1}} a^{y_{i+1}}$

So,

$$b_1 = g^{x_1} a^{y_1} = \begin{cases} g^{x_0+1} & \text{if } b_0 \in G_1 \\ g^{2x_0} & \text{if } b_0 \in G_2 \\ ag^{x_0} & \text{if } b_0 \in G_3 \end{cases} = \begin{cases} g^{x_0+1} a^0 & \text{if } b_0 \in G_1 \\ g^{2x_0} a^0 & \text{if } b_0 \in G_2 \\ g^{x_0} a^1 & \text{if } b_0 \in G_3 \end{cases}$$

thus,

$$x_1 = \begin{cases} x_0 + 1 & \text{if } b_0 \in G_1 \\ 2x_0 & \text{if } b_0 \in G_2 \\ x_0 & \text{if } b_0 \in G_3 \end{cases}$$

and

$$y_1 = \begin{cases} y_0 & \text{if } b_0 \in G_1 \\ 2y_0 & \text{if } b_0 \in G_2 \\ y_0 + 1 & \text{if } b_0 \in G_3 \end{cases}$$

Iterating till i times, we get

$$x_{i+1} = \begin{cases} x_i + 1 \pmod{n} & \text{if } b_i \in G_1 \\ 2x_i \pmod{n} & \text{if } b_i \in G_2 \\ x_i \pmod{n} & \text{if } b_i \in G_3 \end{cases}$$

and

$$y_{i+1} = \begin{cases} y_i \pmod{n} & \text{if } b_i \in G_1 \\ 2y_i \pmod{n} & \text{if } b_i \in G_2 \\ y_i + 1 \pmod{n} & \text{if } b_i \in G_3 \end{cases}$$

NOTE 2 :

- we can prove it by induction
- we won't write $\pmod{n}$ again and again.

The sequence $\{b_m\}_{m=0}$ must be a finite sequence (as G is finite) and thus it terminates at some finite step. Suppose

$$\begin{aligned} b_{i+k} &= b_i \text{ for some } i, k \\ g^{x_{i+k}} a^{y_{i+k}} &= g^{x_i} a^{y_i} \\ g^{x_{i+k} - x_i} &= a^{y_i - y_{i+k}} \text{ in } G = \langle g | g^n = 1 \rangle \end{aligned}$$

If $x = d\log_g a$, then $g^x = a$ Thus,

$$\begin{aligned} x_{i+k} - x_i &\equiv x (y_i - y_{i+k}) \pmod{n} \\ x &\equiv (y_i - y_{i+k})^{-1} (x_{i+k} - x_i) \pmod{n} \end{aligned}$$

- If $(y_i - y_{i+k})^{-1} \pmod{n}$ exists, then $d\log_g a = x$ is uniquely determined.

- If $(y_i - y_{i+k})^{-1} \pmod{n}$ does not exist, the congruence

$$x_{i+k} - x_i \equiv x (y_i - y_{i+k}) \pmod{n}$$

can still be solved and x can be determined from various possible solutions (by putting $g^x = a$ for least possible x).

EXAMPLE 1 : Determine if exists $dlog_5 18$ in $\mathbb{Z}_{23}^* = G$ where $|G| = 22 = n = \phi(23)$.

Assume $G_1 = \{1, 2, \dots, 7\}$, $G_2 = \{8, 9, \dots, 14\}$, $G_3 = \{15, 16, \dots, 22\}$

(we have divided the group in this way but it can be divided in any way by choice)

Let $x_0 = 2$,

$$b_0 = g^{x_0} = 5^2 \equiv 25 \equiv 2 \pmod{23}$$

$$b_0 = g^{x_0} a^{y_0} = (5^2)(18^0) \equiv 2 \pmod{23} \text{ and } 2 \in G_1$$

$$b_1 = g^{x_1} a^{y_1} = g^{x_0+1} a^{y_0} = (5^3)(18^0) \equiv 10 \pmod{23} \text{ and } 10 \in G_2$$

$$b_2 = g^{x_2} a^{y_2} = g^{2x_1} a^{2y_1} = (5^6)(18^0) \equiv 8 \pmod{23} \text{ and } 8 \in G_2$$

$$b_3 = g^{x_3} a^{y_3} = g^{2x_2} a^{2y_2} = (5^{12})(18^0) \equiv 18 \pmod{23} \text{ and } 18 \in G_3$$

$$b_4 = g^{x_4} a^{y_4} = g^{x_3} a^{y_3+1} = (5^{12})(18^1) \equiv 2 \pmod{23} \text{ and } 2 \in G_1$$

Note that

$$\begin{aligned} b_4 &= b_0 \\ \implies g^{x_4} a^{y_4} &= g^{x_0} a^{y_0} \\ \implies g^{x_4-x_0} &= a^{y_0-y_4} \\ \implies x_4 - x_0 &\equiv x(y_0 - y_4) \pmod{22} \\ \implies 12 - 2 &\equiv x(0 - 1) \pmod{22} \\ \implies x &\equiv -10 \equiv 12 \pmod{22} \end{aligned}$$

Thus $dlog_5 18 = 12$

11.3 Index Calculus Algorithm in $\mathbb{Z}_p$:

Let $G = \mathbb{Z}_p^* = \langle g \mid g^{p-1} = 1 \rangle$. and $x = dlog_g a$ where $a \in G$. So $g^x \equiv a \pmod{p}$.

We choose a bound $\mathcal{B}$ and determine factor base $\mathcal{F}(\mathcal{B}) = \{q \in \mathbb{P} | q \leq \mathcal{B}\}$ where $\mathbb{P}$ is set of all prime numbers.

An integer b is said to be $\mathcal{B}$ -smooth number if all prime divisors of b lie in $\mathcal{F}(\mathcal{B})$. That is if p is a prime such that $p|b$ and $p \leq \mathcal{B}$.

Example: If $\mathcal{B} = 15$ then $\mathcal{F}(\mathcal{B}) = \{2, 3, 5, 7, 11, 13\}$. The integer 990 is 15-smooth, since $990 = 2 \times 3^2 \times 5 \times 11$ and $2, 3, 5, 11 \in \mathcal{F}(\mathcal{B})$.

Step-1: Computing the discrete logarithm of the factor base elements.
That is solve for

$$x = x(q) \quad \forall q \in \mathcal{F}(\mathcal{B})$$

such that

$$g^{x(q)} \equiv q \pmod{p}$$

Step-2: Determine an exponent $y \in \{1, 2, \dots, p-1\}$ such that $ag^y \pmod{p}$ is $\mathcal{B}$ -smooth.

That is

$$ag^y \equiv \prod_{q \in \mathcal{F}(\mathcal{B})} q^{e(q)} \pmod{p}$$

where $e(q) \geq 0$ and $q \in \mathcal{F}(\mathcal{B})$.

Now

$$\begin{aligned} ag^y &\equiv \prod_{q \in \mathcal{F}(\mathcal{B})} q^{e(q)} \pmod{p} \\ &\equiv \prod_{q \in \mathcal{F}(\mathcal{B})} [g^{x(q)}]^{e(q)} \pmod{p} \\ &\equiv \prod_{q \in \mathcal{F}(\mathcal{B})} g^{x(q)e(q)} \pmod{p} \\ &\equiv g^{\sum_{q \in \mathcal{F}(\mathcal{B})} x(q)e(q)} \pmod{p} \end{aligned}$$

Hence

$$a \equiv g^{\{(\sum_{q \in \mathcal{F}(\mathcal{B})} x(q)e(q)) - y\}} \pmod{p}$$

This implies

$$\begin{aligned} g^x &\equiv g^{\{(\sum_{q \in \mathcal{F}(\mathcal{B})} x(q)e(q)) - y\}} \pmod{p} \\ \Rightarrow x &\equiv \left\{ \left(\sum_{q \in \mathcal{F}(\mathcal{B})} x(q)e(q) \right) - y \right\} \pmod{p-1} \end{aligned}$$

This gives the discrete log x of a to the base g .

Now for step-1, to determine of $x(q)$ the discrete log of the element $q \in \mathcal{F}(\mathcal{B})$.

For this we choose random integer $z \in \{1, 2, \dots, p-1\}$ and compute $g^z \pmod{p}$, and check if these are $\mathcal{B}$ -smooth. If yes then compute the decomposition:

$$g^z \pmod{p} = \prod_{q \in \mathcal{F}(\mathcal{B})} q^{f(q,z)}$$

Each exponent vector $f(q, z)$ (for $q \in \mathcal{F}(\mathcal{B})$) is called a relation.

Example: Let $p = 2027$, $g = 2$ and determine relations for the factor base $\mathcal{F}(\mathcal{B}) = \mathcal{F}(12) = \{2, 3, 5, 7, 11\}$.

Let $z = 1593$, then $g^z = 2^{1593} \equiv 33 \pmod{2027}$. Since $33 = 3 \times 11$ hence it is $\mathcal{B}$ -smooth.

Similarly $2^{10} = 1024$ and $2^{11} = 2048 \equiv 21 = 3 \times 7 \pmod{2027}$ are $\mathcal{B}$ -smooth number. If we have found as many relations as there are factor base elements, then we try to find the discrete logarithm by solving a system of linear congruences:

$$\begin{aligned} 3 \times 11 &= 33 \equiv 2^{1593} \pmod{2027} \\ 5 \times 7 \times 11 &= 385 \equiv 2^{983} \pmod{2027} \\ 2^7 \times 11 &= 1408 \equiv 2^{1318} \pmod{2027} \\ 3^2 \times 7 &= 63 \equiv 2^{293} \pmod{2027} \\ 2^6 \times 5^2 &= 1600 \equiv 2^{1918} \pmod{2027} \\ 3 \times 7 &= 21 \equiv 2^{11} \pmod{2027} \end{aligned}$$

Thus,

$$\begin{aligned} g^z &\equiv \prod_{q \in \mathcal{F}(\mathcal{B})} q^{f(q,z)} \pmod{p} \\ &\equiv \prod_{q \in \mathcal{F}(\mathcal{B})} [g^{x(q)}]^{f(q,z)} \pmod{p} \\ &\equiv g^{(\sum_{q \in \mathcal{F}(\mathcal{B})} x(q)f(q,z))} \pmod{p} \end{aligned}$$

This implies that

$$g^{(z - \sum_{q \in \mathcal{F}(\mathcal{B})} x(q)f(q, z))} \equiv 1 \pmod{p}$$

Implies that

$$z \equiv \sum_{q \in \mathcal{F}(\mathcal{B})} x(q)f(q, z) \pmod{p-1} \quad \forall z : g^z \pmod{p} \text{ is } \mathcal{B} - \text{smooth}$$

Hence we have a system of linear congruence in variables $x(q)$ for all $q \in \mathcal{F}(\mathcal{B})$

Example: Let $p = 2027$ and $g = 2$

Factor base $\mathcal{F}(\mathcal{B}) = \{2, 3, 5, 7, 11\}$

Step-1 To find $dlog_g q \quad \forall q \in \mathcal{F}(\mathcal{B})$ i.e. $dlog_g 2, dlog_g 3, dlog_g 5, dlog_g 7$, and $dlog_g 11$

For $q = 2 \quad dlog_2 2 = 1$ (since $2^1 \equiv 2 \pmod{p}$)

So we have $x(2) = 1$.

Suppose $g^{x(q)} = q \quad \forall q \in \mathcal{F}(\mathcal{B})$ i.e.

$$g^{x(3)} = 3, g^{x(5)} = 5, g^{x(7)} = 7, g^{x(11)} = 11,$$

$z = 1593 \quad g^z = 2^{1593} \equiv 33 \equiv 11 \times 3 \pmod{2027}$ which is $\mathcal{B} - \text{smooth}$.

i.e.

$$g^z = 2^{1593} \equiv 33 \equiv 11 \times 3 \pmod{2027} \equiv g^{x(3)} \cdot g^{x(11)} \pmod{p}$$

i.e.

$$g^{x(3)+x(11)-z} \equiv 1 \pmod{2027}$$

$$\implies x(3) + x(11) - z \equiv 0 \pmod{2026}$$

$$\implies x(3) + x(11) \equiv 1593 \pmod{2026} \tag{11.1}$$

(since $(p-1) \mid (x(3) + x(11) - z)$ and $z = 1593, p = 2027$)

Next, $z = 983 \implies g^z = 2^{983} \equiv 385 \equiv 5 \times 7 \times 11 \pmod{2027}$ which is $\mathcal{B} - \text{smooth}$.

i.e.

$$g^z = 2^{983} \equiv g^{x(5)} \cdot g^{x(7)} \cdot g^{x(11)} \pmod{p}$$

i.e.

$$g^{x(5)+x(7)+x(11)-z} \equiv 1 \pmod{2027}$$

$$\begin{aligned}
&\implies x(5) + x(7) + x(11) - z \equiv 0 \pmod{2026} \\
&\implies x(5) + x(7) + x(11) \equiv 983 \pmod{2026}
\end{aligned} \tag{11.2}$$

Similarly,

$z = 1318 \implies g^z = 2^{1318} \equiv 1408 \equiv 2^7 \times 11 \pmod{2027}$ which is $\mathcal{B}$ -smooth.
i.e.

$$g^z = 2^{1318} \equiv g^{7x(2)} \cdot g^{x(11)} \pmod{p}$$

i.e.

$$\begin{aligned}
&g^{7x(2)+x(11)-z} \equiv 1 \pmod{2027} \\
&\implies 7x(2) + x(11) - z \equiv 0 \pmod{2026} \\
&\implies 7x(2) + x(11) \equiv 1318 \pmod{2026}
\end{aligned} \tag{11.3}$$

Similarly, $z = 293 \implies g^z = 2^{293} \equiv 3^2 \times 7 \pmod{2027}$ which is $\mathcal{B}$ -smooth.
i.e.

$$g^z = 2^{293} \equiv g^{2x(3)} \cdot g^{x(7)} \pmod{p}$$

i.e.

$$\begin{aligned}
&g^{2x(3)+x(7)-z} \equiv 1 \pmod{2027} \\
&\implies 2x(3) + x(7) - z \equiv 0 \pmod{2026} \\
&\implies 2x(3) + x(7) \equiv 293 \pmod{2026}
\end{aligned} \tag{11.4}$$

Similarly, we have
for $z = 1918$

$$\implies 6x(2) + 2x(5) \equiv 1918 \pmod{2026} \tag{11.5}$$

and for $z = 11$

$$\implies x(3) + x(7) \equiv 11 \pmod{2026} \tag{11.6}$$

Finally we have six equations (1)-(6) to calculate $x(3)$, $x(5)$, $x(7)$, $x(11)$ and we already know $x(2) = 1$.

Note- In fact we need only four equations to calculate four unknowns.

Now we have to solve system of equations (1)-(6) for $x(3)$, $x(5)$, $x(7)$, $x(11)$
 Note that $2026 = 2 \times 1013$. So we solve system of equations Modulo 2 and Modulo 1013.

$$x(3) + x(11) \equiv 1593 \equiv 1 \pmod{2} \quad (11.7)$$

$$x(5) + x(7) + x(11) \equiv 983 \equiv 1 \pmod{2} \quad (11.8)$$

$$7x(2) + x(11) \equiv 1318 \equiv 0 \pmod{2} \quad (11.9)$$

$$2x(3) + x(7) \equiv x(7) \equiv 293 \equiv 1 \pmod{2} \quad (11.10)$$

$$6x(2) + 2x(5) \equiv 1918 \pmod{2} \text{ (This is of form } 0=0 \text{)} \quad (11.11)$$

$$x(3) + x(7) \equiv 11 \equiv 1 \pmod{2} \quad (11.12)$$

we know $x(2) = 1 \pmod{2026} \implies x(2) = 1 \pmod{2}$ also.

On solving above system of equation, we get

$$x(3) \equiv 0 \pmod{2}$$

$$x(5) \equiv 1 \pmod{2}$$

$$x(7) \equiv 1 \pmod{2}$$

$$x(11) \equiv 1 \pmod{2}$$

and system of equation Modulo 1013

$$x(3) + x(11) \equiv 1593 \equiv 580 \pmod{1013} \quad (11.13)$$

$$x(5) + x(7) + x(11) \equiv 983 \pmod{1013} \quad (11.14)$$

$$7x(2) + x(11) \equiv 1318 \equiv 305 \pmod{1013} \quad (11.15)$$

$$2x(3) + x(7) \equiv 293 \pmod{1013} \quad (11.16)$$

$$6x(2) + 2x(5) \equiv 1918 \equiv 905 \pmod{1013} \quad (11.17)$$

$$x(3) + x(7) \equiv 11 \pmod{1013} \quad (11.18)$$

On solving above system , we get

$$\begin{aligned}
x(3) &\equiv 282 \pmod{1013} \\
x(5) &\equiv 956 \pmod{1013} \\
x(7) &\equiv 742 \pmod{1013} \\
x(11) &\equiv 298 \pmod{1013}
\end{aligned}$$

Now we have solutions in Modulo 2 and Modulo 1013, we can easily calculate solution Modulo 2026, using Chinese Remainder Theorem

$$\begin{aligned}
x(3) &\equiv 0 \pmod{2} \\
x(3) &\equiv 282 \pmod{1013} \\
x(3) &\equiv 282 \pmod{2026} \quad (\text{by applying Chinese Remainder Theorem})
\end{aligned}$$

$$\begin{aligned}
x(5) &\equiv 1 \pmod{2} \\
x(5) &\equiv 956 \pmod{1013} \\
x(5) &\equiv 1969 \pmod{2026} \quad (\text{by applying Chinese Remainder Theorem})
\end{aligned}$$

$$\begin{aligned}
x(7) &\equiv 1 \pmod{2} \\
x(7) &\equiv 742 \pmod{1013} \\
x(7) &\equiv 1755 \pmod{2026} \quad (\text{by applying Chinese Remainder Theorem})
\end{aligned}$$

$$\begin{aligned}
x(11) &\equiv 1 \pmod{2} \\
x(11) &\equiv 298 \pmod{1013} \\
x(11) &\equiv 1311 \pmod{2026} \quad (\text{by applying Chinese Remainder Theorem})
\end{aligned}$$

Now to calculate $d\log_g a$ for $g = 2$ and for given a
Factor base $\mathcal{F}(\mathcal{B}) = \{2, 3, 5, 7, 11\}$

(1). For $a = 12$

$a = 12 = 2^2 \cdot 3$ i.e a is $\mathcal{B}$ -smooth and we have $g^{x(2)} = 2^1 = 2$ and $g^{x(3)} = 2^{282} = 3$ Then

$$a = 12 = 2^2 \cdot 3 = (2^1)^2 \cdot 2^{282} = 2^{2+282} = 2^{284}$$

Therefore $dlog_g a = dlog_2 12 = 284$

(2.) For $a = 65$

$a = 65 = 5 \times 13$ is not $\mathcal{B}$ -smooth.

Now we find $y \in \{1, 2, \dots, p-1\}$ such that $ag^y \pmod{p}$ is $\mathcal{B}$ -smooth.

$$\begin{aligned} ag &\equiv 65 \cdot 2 \pmod{2027} \equiv 13 \cdot 5 \cdot 2 \pmod{2027} \\ ag^2 &\equiv 65 \cdot 2^2 \pmod{2027} \equiv 13 \cdot 5 \cdot 2^2 \pmod{2027} \\ ag^3 &\equiv 65 \cdot 2^3 \pmod{2027} \equiv 13 \cdot 5 \cdot 2^3 \pmod{2027} \end{aligned}$$

and so on we get $y = 28$

$$ag^{28} \equiv 65 \cdot 2^{28} \pmod{2027} \equiv 5^3 \pmod{2027}$$

and $5 = g^{x(5)} = 2^{1969}$, we have

$$\begin{aligned} ag^{28} &\equiv 65 \cdot 2^{28} \pmod{2027} \equiv 5^3 \pmod{2027} \equiv (2^{1969})^3 \pmod{2027} \equiv 2^{5907} \\ a &\equiv 2^{5907-28} \equiv 2^{5879} \pmod{2027} \equiv 2^{1827} \pmod{2027} \end{aligned}$$

Therefore, $dlog_g a = dlog_2 65 = 1827$

11.4 Pohlig - Hellman Algorithm for discrete logarithm

Pohlig - Hellman Algorithm for a finite group of order n

About :

1. The Pohlig–Hellman algorithm, sometimes called as the Silver–Pohlig–Hellman algorithm, is a special-purpose algorithm for computing discrete logarithms in a finite abelian group whose order is a smooth integer.

2. The algorithm was introduced by Roland Silver, but first published by Stephen Pohlig and Martin Hellman.
3. The worst-case input for the Pohlig–Hellman algorithm is a group of prime order: In that case, it degrades to the baby-step giant-step algorithm, hence the worst-case time complexity is $\mathcal{O}(\sqrt{n})$. However, it is much more efficient if the order is smooth: Specifically, if $\prod_i p_i^{e_i}$ is the prime factorization of n , then the algorithm's complexity is

$$\mathcal{O}\left(\sum_i e_i(\log n + \sqrt{p_i})\right)$$

Algorithm :

Let G' be a finite abelian group of order k .

We need to compute $d\log_\gamma \alpha$, where $\alpha, \gamma \in G'$.

Assuming that $d\log_\gamma \alpha$ exists, we are considering a subgroup G of G' of order n such that

$$G = \langle \gamma | \gamma^n = 1 \rangle$$

Thus G be a finite cyclic group of order n where $n = \prod_{p|n} p^{e(p)}$.

Then $\gamma \in G$, $O(\gamma) = n$, and, $d\log_\gamma \alpha$ exists.

Let $d\log_\gamma \alpha = x \implies \gamma^x = \alpha$.

For each prime p , $p|n$, denote

$$n_p = \frac{n}{p^{e(p)}}$$

Let, $\gamma_p = \gamma^{n_p} \in G$. Note that, $O(\gamma_p) = p^{e(p)}$

To check $O(\gamma_p) = p^{e(p)}$

Consider $(\gamma_p)^{p^{e(p)}} = (\gamma^{n_p})^{p^{e(p)}} = \gamma^{n_p p^{e(p)}} = \gamma^n = 1$.

$$\implies O(\gamma_p) | p^{e(p)}$$

To show $p^{e(p)} | O(\gamma_p)$, we prove it by contradiction.

Consider,

$$O(\gamma_p) = p^{\epsilon(p)}, \text{ where } \epsilon(p) < e(p)$$

then $(\gamma_p)^{p^{e(p)}} = (\gamma^{n_p})^{p^{e(p)}} = \gamma^{\frac{n}{p^{e(p)} - e(p)}} = 1$
 $\implies O(\gamma) < n$, which is a contradiction.
Hence, $O(\gamma_p) = p^{e(p)}$

Since, $dlog_\gamma \alpha = x$, we assume $\alpha \neq 1$ [$\alpha = 1, \implies x = 0$]

Reduction to prime power order cyclic group :

Consider $\alpha_p = \alpha^{n_p}$, then

$$\begin{aligned} \gamma_p^x &= (\gamma^{n_p})^x = \gamma^{n_p x} = (\gamma^x)^{n_p} = \alpha^{n_p} = \alpha_p \\ \implies \gamma_p^x &= \alpha_p \end{aligned}$$

Let $x(p) = dlog_{\gamma_p} \alpha_p$

Then $\gamma_p^{x(p)} = \alpha_p$ ($x(p)$ is the least non negative integer satisfying this relation.)

Let $H = \langle \gamma_p | \gamma_p^{p^{e(p)}} = 1 \rangle$

Observe that

$$\begin{aligned} (\gamma^{-x} \alpha)^{n_p} &= \gamma_p^{-x} \alpha_p = \alpha_p^{-1} \alpha_p = 1 \\ \implies O(\gamma^{-x} \alpha) &| n_p \quad \forall p|n \\ \implies O(\gamma^{-x} \alpha) &| g.c.d.(n_p) \quad \forall \text{ prime } p|n \end{aligned}$$

and

$$\begin{aligned} g.c.d.(n_p) &= 1 \quad \forall \text{ prime } p|n \\ \implies O(\gamma^{-x} \alpha) &= 1 \\ \implies \gamma^{-x} \alpha &= 1_G \\ \implies x &= dlog_\gamma \alpha \end{aligned}$$

Now,

$$H = \langle \gamma_p | \gamma_p^{p^{e(p)}} = 1 \rangle$$

Then, note that $\alpha_p \in H$, which implies

$$\implies \gamma_p^{x-x(p)} = 1 \in H \quad (\text{since } \gamma_p^{-x} \alpha_p = 1 \implies \alpha_p = \gamma_p^x)$$

Since, $O(\gamma_p) = p^{e(p)}$ implies that

$$(x - x(p)) | p^{e(p)}$$

$$\implies x \equiv x(p) \pmod{p^{e(p)}}$$

Since this relation holds for all prime divisors of n , thus the problem now reduces to the computation of $x(p)$ only. Once, we compute $x(p)$ for every prime divisor p of n , we can apply Chinese remainder theorem to compute x .

Note that the problem now reduces to cyclic group of prime power order and computation of $x(p)$.

Now we further reduce the problem to a cyclic group of prime p order, where $p|n$.

Reduction to prime order group :

Let $H = \langle \gamma_p | \gamma_p^{p^{e(p)}} = 1 \rangle \leq G \quad \forall p|n$

Fix p as a prime divisor of n , then, $O(H) = p^{e(p)}$.

Notations :

Denote $e(p) = e$, $x(p) = x$, $\alpha_p = a$, $\gamma_p = g$.

Since we have to compute $x(p) = d \log_{\gamma_p} \alpha_p$, that is, we have to compute $x = d \log_g a$ in the group $H = \langle g | g^{p^e} = 1 \rangle$

Since $x = d \log_g a \Rightarrow g^x = a$,

where $x = x_0 + x_1 p + x_2 p^2 + \dots + x_{e-1} p^{e-1}$, $0 \leq x_i < p$, $\forall 0 \leq i \leq e-1$
be the p-nary representation of x can be obtained by division algorithm.

For example :

if $x = 9$ and $p = 3$, $e = 3$ then $O(H) = 3^3 = 27$ and $x = 0 + 0 \cdot 3 + 1 \cdot 3^2$,
if $x = 13$ and $p = 3$, $e = 3$ then $O(H) = 3^3 = 27$ and $x = 1 + 1 \cdot 3 + 1 \cdot 3^2$,
if $x = 26$ and $p = 3$, $e = 3$ then $O(H) = 3^3 = 27$ and $x = 2 + 2 \cdot 3 + 2 \cdot 3^2$.

Now,

$$\begin{aligned}
 g^x &= a \\
 \implies (g^x)^{p^{e-1}} &= a^{p^{e-1}} \\
 \implies (g^{p^{e-1}})^x &= a^{p^{e-1}} \\
 \implies g^{p^{e-1}x} &= a^{p^{e-1}} \\
 \implies g^{p^{e-1}(x_0+x_1p+x_2p^2+\dots+x_{e-1}p^{e-1})} &= a^{p^{e-1}} \\
 \implies g^{(x_0p^{e-1}+x_1p^e+x_2p^{e+1}+\dots+x_{e-1}p^{2e-2})} &= a^{p^{e-1}} \\
 \implies g^{x_0p^{e-1}} \cdot g^{x_1p^e} \cdot g^{x_2p^{e+1}} \dots g^{x_{e-1}p^{2e-2}} &= a^{p^{e-1}}
 \end{aligned}$$

Note that $g^{p^e} = 1$, implies

$$\begin{aligned}
 g^{x_0p^{e-1}} &= a^{p^{e-1}} \\
 \implies (g^{p^{e-1}})^{x_0} &= a^{p^{e-1}}
 \end{aligned}$$

Since $O(g) = p^e \implies g^{p^{e-1}} \neq 1 \implies x_0 = d\log_{g^{p^{e-1}}} a^{p^{e-1}}$ [Use same argument that we have used before to prove $x(p) = d\log_{\gamma_p} \alpha_p$]

Also, note that $O(g^{p^{e-1}}) = p$

Let $H_0 = \langle g^{p^{e-1}} | (g^{p^{e-1}})^p = 1 \rangle$ be a subgroup of H of order p ,

then, $x_0 = d\log_{g^{p^{e-1}}} a^{p^{e-1}}$ in H_0 .

Note that the problem now reduces to cyclic group of prime (p) order, namely H_0 and computation of x_0 .

To determine x_i for all i , we first calculate x_0 and then apply induction on x_i .

Suppose $x_0, x_1, x_2, \dots, x_{i-1}$ are determined, we need to determine x_i .

Now consider

$$\begin{aligned}
 g^{(x_ip^i+x_{i+1}p^{i+1}+\dots+x_{e-1}p^{e-1})} &= g^{\left(-x_0-x_1p-\dots-x_{i-1}p^{i-1} + \sum_{j=0}^{e-1} x_jp^j\right)} \\
 &= g^{-x_0-x_1p-\dots-x_{i-1}p^{i-1}} \cdot g^x \\
 &= g^{-x_0-x_1p-\dots-x_{i-1}p^{i-1}} \cdot a \\
 &= a_i \quad (\text{say})
 \end{aligned}$$

Thus, we have

$$g^{(x_i p^i + x_{i+1} p^{i+1} + \dots + x_{e-1} p^{e-1})} = a_i$$

Raising the power p^{e-i-1} , we get

$$\begin{aligned} & \left(g^{(x_i p^i + x_{i+1} p^{i+1} + \dots + x_{e-1} p^{e-1})} \right)^{p^{e-i-1}} = (a_i)^{p^{e-i-1}} \\ \Rightarrow & g^{(p^{e-i-1}(x_i p^i + x_{i+1} p^{i+1} + \dots + x_{e-1} p^{e-1}))} = (a_i)^{p^{e-i-1}} \\ \Rightarrow & g^{(p^{e-1} \cdot p^{-i}(x_i p^i + x_{i+1} p^{i+1} + \dots + x_{e-1} p^{e-1}))} = (a_i)^{p^{e-i-1}} \\ \Rightarrow & g^{(p^{e-1} \cdot (x_i + x_{i+1} p + \dots + x_{e-1} p^{e-i-1}))} = (a_i)^{p^{e-i-1}} \\ \Rightarrow & g^{(p^{e-1} \cdot x_i)} \cdot g^{(p^{e-1} \cdot p(x_{i+1} + \dots + x_{e-1} p^{e-i-2}))} = (a_i)^{p^{e-i-1}} \\ \Rightarrow & g^{(p^{e-1} \cdot x_i)} \cdot g^{p^e \cdot (x_{i+1} + \dots + x_{e-1} p^{e-i-2})} = (a_i)^{p^{e-i-1}} \\ \Rightarrow & g^{(p^{e-1} \cdot x_i)} = (a_i)^{p^{e-i-1}} \quad \text{since } g^{p^e} = 1 \end{aligned}$$

Let $g^{p^{e-1}} = \bar{g}$

Then

$$\begin{aligned} (\bar{g})^{x_i} &= (a_i)^{p^{e-i-1}} \quad \text{in some group } K = \langle \bar{g} \mid \bar{g}^p = 1 \rangle. \\ \Rightarrow x_i &= d\log_{\bar{g}}(a_i)^{p^{e-i-1}} \quad \forall i \end{aligned}$$

Examples :

1. Determine, if exists, $d\log_5 3$ in $\mathbb{Z}_{53}^*$.

Let $x = d\log_5 3$, then $5^x = 3 \pmod{53}$

Note that $G = \langle 5 \mid 5^{52} = 1 \rangle$, thus $\gamma = 5$ and $\alpha = 3$

Also, note that $O(\mathbb{Z}_{53}^*) = 52 = 2^2 \cdot 13$

Let $p_1 = 2$, $p_2 = 13$, $e(2) = 2$, $e(13) = 1$

Then, $n_2 = 13$, $n_{13} = 4$

Let $\gamma_2 = \gamma^{n_2} = 5^{13} = 23 \pmod{53}$

Let $H_1 = \langle \gamma_2 \mid \gamma_2^4 = 1 \rangle = \langle 23 \mid 23^4 = 1 \rangle$

Note that $O(H_1) = 4$

Let $\alpha_2 = \alpha^{n_2} = \alpha^{13} = 3^{13} = 30 \pmod{53}$,

then, $\gamma_2^x = \alpha_2 \Rightarrow 23^x = 30 \pmod{53}$.

We need to compute $x(2) = d\log_{\gamma_2} \alpha_2 = d\log_{23} 30$,

such that $23^{x(2)} = 30$ and $x \equiv x(2) \pmod{4}$.

Since, here, $p_1^{e(2)} = 4$, a small integer, we can either compute $x(2)$ directly or we can reduce the prime power group to a cyclic group of order 2.

One Possible way is we compute $x(2)$ as:

$$23^2 \equiv -1 \pmod{53}$$

$$23^3 \equiv 30 \pmod{53}$$

thus, $x \equiv 3 \pmod{4}$

Similarly, Let $\gamma_{13} = \gamma^4 = 5^4 = 42 \pmod{53}$

Let $H_2 = \langle \gamma_{13} | \gamma_{13}^{13} = 1 \rangle = \langle 42 | 42^{13} = 1 \rangle$

Note that $O(H_2) = 13$, which is prime, thus we don not require to reduce it further.

Let $\alpha_{13} = \alpha^{n_{13}} = \alpha^4 = 3^4 = 28 \pmod{53}$,

then, $\gamma_{13}^x = \alpha_{13} \Rightarrow 42^x = 28 \pmod{53}$.

We need to compute $x(13) = d\log_{\gamma_{13}} \alpha_{13} = d\log_{42} 28$,

such that $42^{x(13)} = 28$ and $x \equiv x(13) \pmod{13}$.

Solving $42^{x(13)} = 28$ in H_2 , we get that $x(13) \equiv 7 \pmod{13}$,

hence $x \equiv 7 \pmod{13}$.

Now we have two equations as :

$$x \equiv 3 \pmod{4}$$

$$x \equiv 7 \pmod{13}$$

Use Chinese remainder theorem to solve these two equation.

On solving, we get that

$$x \equiv 7 \pmod{52}$$

Thus, $d\log_5 3 = 7$ in $\mathbb{Z}_{53}^*$.

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2. Determine, if exists, $dlog_6 7531$ in $\mathbb{Z}_{8101}^*$.

Let $x = dlog_6 7531$, then $6^x = 7531 \pmod{8101}$

Note that $G = \langle 6 \mid 6^{8100} = 1 \rangle$, thus $\gamma = 6$ and $\alpha = 7531$

Also, note that $|\mathbb{Z}_{8101}^*| = 8100 = 2^2 \cdot 3^4 \cdot 5^2$

Let $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, $e(2) = 2$, $e(3) = 4$, $e(5) = 2$

Then, $n_2 = 2025$, $n_3 = 100$, $n_5 = 324$

Let $\gamma_2 = \gamma^{2025} = 6^{2025}$

Let $H_1 = \langle \gamma_2 \mid \gamma_2^4 = 1 \rangle = \langle 6^{2025} \mid (6^{2025})^4 = 1 \rangle$

Note that $O(H_1) = 4$

Let $\alpha_2 = \alpha^{n_2} = \alpha^{2025} = 7531^{2025}$,

then, $\gamma_2^x = \alpha_2 \Rightarrow (6^{2025})^x = 7531^{2025}$.

We need to compute $x(2) = dlog_{\gamma_2} \alpha_2$,

such that $(6^{2025})^{x(2)} = 7531^{2025}$ and $x \equiv x(2) \pmod{4}$.

Note that computation of 6^{2025} and 7531^{2025} is not easy and hence we reduce the problem to prime order group.

Since, $x(2)$ is in modulo 4, thus $x(2) = x_0 + x_1 * p$, where x_0 and x_1 is either 0 or 1.

Then

$$(\gamma_p^{p^{e-1}})^{x_0} = \alpha_p^{p^{e-1}} \in H_1$$

$\Rightarrow$

$$(\gamma_2^2)^{x_0} = \alpha_2^2 \in H_1$$

$\Rightarrow$

$$((6^{2025})^2)^{x_0} = (7531^{2025})^2 \in H_1$$

$\Rightarrow$

$$(6^{4050})^{x_0} = 7531^{4050} \equiv -1 \pmod{8101}$$

$\Rightarrow$

$$(-1)^{x_0} \equiv -1 \pmod{8101}$$

$\Rightarrow$

$$x_0 = 1$$

Thus we need to compute x_1 now.

Note that $x_1 = dlog_{g^{p^e-1}} a_1^{p^{e-1}-1}$, where $a_1 = g^{-x_0} \cdot \alpha_p$ and $g = \gamma_p = 6^{2025}$,

$$\Rightarrow a_1 = g^{-x_0} \cdot \alpha_2 = (6^{2025})^{(-1)} \cdot 7531^{2025}$$

$\Rightarrow$

$$x_1 = dlog_{(6^{2025})^2} a_1^{2^{(2-1-1)}}$$

$\Rightarrow$

$$x_1 = dlog_{(6^{2025})^2} a_1$$

$\Rightarrow$

$$(6^{2025})^{(-1)} \cdot 7531^{2025} = ((6^{2025})^2)^{x_1}$$

$\Rightarrow$

$$(6^{-1})^{(2025)} \cdot 7531^{2025} = (-1)^{x_1}$$

Since, $6^{-1} \equiv 6751 \pmod{8081}$, we have

$$(6751)^{2025} \cdot 7531^{2025} = (-1)^{x_1}$$

$\Rightarrow$

$$(6751 \cdot 7531)^{2025} = (-1)^{x_1}$$

$\Rightarrow$

$$8006^{2025} = (-1)^{x_1}$$

Note that

$$8006^{2025} \equiv 1 \pmod{4} \Rightarrow x_1 = 0$$

Thus

$$x(2) = x_0 + x_1 * 2 = 1$$

and the congruence relation is

$$x \equiv 1 \pmod{4}$$

Similarly, Let $\gamma_3 = \gamma^{100} = 6^{100}$

Let $H_2 = \langle \gamma_3 \mid \gamma_3^{81} = 1 \rangle = \langle 6^{100} \mid (6^{100})^{81} = 1 \rangle$

Note that $O(H_2) = 8$

Let $\alpha_3 = \alpha^{n_3} = \alpha^{100} = 7531^{100}$,

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then, $\gamma_3^x = \alpha_3 \Rightarrow (6^{100})^x = 7531^{100}$.

We need to compute $x(3) = dlog_{\gamma_3} \alpha_3$,

such that $(6^{100})^{x(3)} = 7531^{100}$ and $x \equiv x(3) \pmod{81}$.

Note that computation of 6^{100} and 7531^{100} is not easy and hence we reduce the problem to prime order group.

Since, $x(3)$ is in modulo 81, thus $x(3) = x_0 + x_1.p + x_2.p^2 + x_3.p^3$, where x_0, x_1, x_2 , and x_3 is either 0 or 1 or 2.

Then

$$\begin{aligned} & (\gamma_p^{p^{e-1}})^{x_0} = \alpha_p^{p^{e-1}} \in H_2 \\ \Rightarrow & (\gamma_3^{3^3})^{x_0} = \alpha_3^{3^3} \in H_2 \\ \Rightarrow & ((6^{100})^{27})^{x_0} = (7531^{100})^{27} \in H_2 \\ \Rightarrow & (6^{2700})^{x_0} = 7531^{2700} \equiv 2217 \pmod{8101} \\ \Rightarrow & (5883)^{x_0} \equiv 2217 \pmod{8101} \end{aligned}$$

Since, x_0 can be 0, 1, 2 only, thus after putting x_0 as 0, 1, 2 in the above relation, we get

$$x_0 = 2$$

Thus we need to compute x_1 now.

Note that $x_1 = dlog_{g^{p^{e-1}} a_1} p^{e-1-1}$, where $a_1 = g^{-x_0} . \alpha_p$ and $g = \gamma_p = 6^{100}$,

$$\begin{aligned} \Rightarrow & a_1 = g^{-x_0} . \alpha_3 = (6^{100})^{(-2)} . 7531^{100} \\ \Rightarrow & \\ & x_1 = dlog_{(6^{100})^{27} a_1} 3^{(4-1-1)} \\ \Rightarrow & \\ & x_1 = dlog_{(6^{2700}) . a_1} 9 \\ \Rightarrow & \\ & (6^{2700})^{x_1} = ((6^{100})^{(-2)} . 7531^{100})^9 \end{aligned}$$

$\Rightarrow$

$$(6^{2700})^{x_1} = (7531 \cdot 6^{-2})^{900}$$

Since, $6^{-2} \equiv 7876 \pmod{8081}$, we have

$$(6^{2700})^{x_1} = (7531 \cdot 7876)^{900}$$

$\Rightarrow$

$$(6^{2700})^{x_1} = (6735)^{900}$$

$\Rightarrow$

$$5883^{x_1} = 6735^{900} = 1$$

Note that

$$6735^{900} \equiv 1 \pmod{81} \Rightarrow x_1 = 0$$

Thus

$$x(2) = 2 + 0 \cdot 3 + x_2 \cdot 9 + x_3 \cdot 27$$

Similarly solve for x_2 and x_3 .

On, solving, we get that $x_2 = 2$ and $x_3 = 1$, $\Rightarrow$

$$x(2) = 2 + 0 \cdot 3 + 2 \cdot 9 + 1 \cdot 27 = 47$$

and the congruence relation is

$$x \equiv 47 \pmod{81}$$

Similarly solve for prime $p = 5$, and the congruence relation obtained will be

$$x \equiv 14 \pmod{25}$$

Thus, the problem now reduces to :

$$x \equiv 1 \pmod{4}$$

$$x \equiv 47 \pmod{81}$$

$$x \equiv 14 \pmod{25}$$

Use Chinese remainder theorem to compute the value of x .

11.5 Exercise

1. Determine, if exists, $dlog_2 21$ in $\mathbb{F}_{59}^*$ using Shank baby-step giant-step algorithm for discrete logarithm.
2. Determine, if exists, $dlog_5 15$ in $\mathbb{F}_{83}^*$ using Shank baby-step giant-step algorithm for discrete logarithm.
3. Determine, if exists, $dlog_{21} 500$ in $\mathbb{F}_{86743}^*$ using Shank baby-step giant-step algorithm for discrete logarithm.
4. Determine, if exists, $dlog_5 11$ in $\mathbb{F}_{37}^*$ using Pollard's ρ algorithm for discrete logarithm.
5. Determine, if exists, $dlog_7 5$ in $\mathbb{F}_{97}^*$ using Pollard's ρ algorithm for discrete logarithm.
6. Determine, if exists, $dlog_5 1500$ in $\mathbb{F}_{19417}^*$ using Pollard's ρ algorithm for discrete logarithm.
7. Determine, if exists, $dlog_6 13$ in $\mathbb{F}_{229}^*$ with factor base $\beta = \{2, 3, 5, 7, 11\}$ using Index Calculus algorithm for discrete logarithm.
8. Determine, if exists, $dlog_5 5872$ in $\mathbb{F}_{14087}^*$ with factor base $\beta = \{2, 3, 5, 7, 11, 13\}$ using Index Calculus algorithm for discrete logarithm.
9. Determine, if exists, $dlog_5 15$ in $\mathbb{F}_{37}^*$ using Pohlig-Hellman algorithm for discrete logarithm.
10. Determine, if exists, $dlog_7 5$ in $\mathbb{F}_{23}^*$ using Pohlig-Hellman algorithm for discrete logarithm.
11. Determine, if exists, $dlog_7 300$ in $\mathbb{F}_{32401}^*$ using Pohlig-Hellman algorithm for discrete logarithm.

R.K.Sharma

Chapter 12

Introduction to Finite Field

12.1 Some Definitions and Results

Group : A non empty set G is said to be a group if $\exists \ * : G \times G \rightarrow G$ defined as $(x, y) \rightarrow x * y \ \forall \ x, y \in G$ such that :

1. Closure : $\forall \ x, y \in G, x * y \in G$
2. Associative : $(x * y) * z = x * (y * z) \ \forall \ x, y, z \in G$
3. Existence of identity : $\exists \ e \in G$ such that $e * x = x * e = x$ (the element $e \in G$ is unique)
4. Existence of inverse : for $x \in G, \exists \ y \in G$ such that $x * y = y * x = e$ (the element $y \in G$ is unique)

*It is denoted by $(G, *)$.*

Commutative or abelian Group : A group $(G, *)$ is said to be commutative or abelian if $*$ is commutative over G , i.e. $x * y = y * x \ \forall \ x, y \in G$.

Cyclic Group : A group $(G, *)$ is said to be a cyclic group if there exist an element $x \in G$ such that every element $y \in G$ can be written as power of x .

Note : Every cyclic group is abelian but converse need not be true.

Proof : Let G be a cyclic group with a generator $g \in G$ i.e. $G = \langle g \rangle$ (every element in G is some power of g .) Let a and b be arbitrary elements in G .

Then there exists $n, m \in \mathbb{Z}$ such that $a = g^n$ and $b = g^m$. It follows that $ab = g^n g^m = g^{n+m} = g^m g^n = ba$. Hence we obtain $ab = ba$ for arbitrary $a, b \in G$. Thus G is an abelian group.

Converse need not be true.

Let $G = \{e, a, b, ab\}$ such that order of $e = 1$, , order of $a =$ order of $b =$ order of $ab = 2$ and $ab = ba$, thus G is abelian. But G is not a cyclic.

Ring : A non empty set $(R, +, \cdot)$ is said to be a ring if :

1. $(R, +)$ is an abelian group.
2. R is closed w.r.t. multiplication, i.e. $\forall r, s \in R, r \cdot s \in R$
3. The multiplication distributes over addition, i.e. $x(y + z) = xy + xz$ and $(x + y)z = xz + yz \quad \forall x, y, z \in R$

Note : We write $x \cdot y$ as xy .

Associative ring : A ring R is said to be an associative ring if multiplication is associative over R , i.e. $(xy)z = x(yz) \in R \quad \forall x, y, z \in R$

Commutative ring : A ring R is s.t.b. a commutative or abelian ring if it is abelian w.r.t. multiplication over R , i.e. $x \cdot y = y \cdot x \quad \forall x, y \in R$

Ring with unity : A ring R is said to be a ring with unity if $\exists 1 \in R$ such that $x \cdot 1 = 1 \cdot x = x \quad \forall x \in R$. 1 is called multiplicative unity or (simply) unity of R .

Unit of ring : An element $x \in R$ is said to be a unit or an invertible element if there exists $y \in R$ such that $xy = yx = 1$ in a ring with unity. Such an element y is called multiplicative inverse or (simply) inverse of x and denoted by x^{-1} .

Ideal: For an arbitrary ring $(R, +, \cdot)$, a subset I is called a left ideal of R if:

1. $(I, +)$ is a subgroup of $(R, +)$
2. $\forall r \in R, \forall x \in I$, the product $rx \in I$

Similarly, we define right ideal. A two-sided ideal is a left ideal that is also a right ideal, and is sometimes simply called an ideal.

Maximal Ideal : A maximal ideal is an ideal that is maximal (with respect to set inclusion) amongst all proper ideals. In other words, I is a maximal ideal of a ring R if there are no other ideals contained between I and R , i.e. For any ideal J with $I \subseteq J \subseteq R$, either $J = I$ or $J = R$.

Principal Ideal : A principal ideal is an ideal I in a ring R that is generated by a single element a of R through multiplication by every element of R , denoted by $I = \langle a \rangle$ for some $a \in R$.

Field : A non empty set $(\mathbb{F}, +, \cdot)$ is said to be a field if $(\mathbb{F}, +)$ is an abelian group and $(\mathbb{F} \setminus \{0\}, \cdot)$ is also an abelian group and multiplication distributes over addition.

Division ring or Skew field : $(D, +, \cdot)$ is said to be a division ring or a skew field if $(D, +)$ is an abelian group and $(D \setminus \{0\}, \cdot)$ form a group and multiplication distributed over addition.
($(D \setminus \{0\}, \cdot)$ need not be commutative.)

Zero Divisor : In a ring R , elements, if they exists, $a, b \in R$ such that $a \neq 0, b \neq 0$ but $ab = 0$, then a, b are called zero divisors in R .

Example : 2,3,4 are zero divisors of ring $(\mathbb{Z}_6, +_6, \cdot_6)$, since $3 \cdot_6 4 = 12 = 0 \in \mathbb{Z}_6$.

Integral Domain (ID) : A commutative ring without zero divisors is called an integral domain.

Principal Ideal Domain : A principal ideal domain (PID) is an integral domain in which every ideal is principal, i.e., can be generated by a single element.

Results :

1. **A field has no zero divisors.**

Proof: Let $x, y \in \mathbb{F}$ and $xy = 0$. We will show either $x = 0$ or $y = 0$.

If $x \neq 0$ then x^{-1} exists and $x^{-1}xy = x^{-1} \cdot 0 \implies 1 \cdot y = 0 \implies y = 0$.

If $y \neq 0$ then y^{-1} exists and $xyy^{-1} = 0 \cdot y^{-1} \implies x \cdot 1 = 0 \implies x = 0$.

2. **A finite Integral Domain is a field.**

Proof: We shall show that every non zero element in finite integral domain R is unit. Let r be a non zero element in R . To show that r is

a unit. Define a map $f : R \rightarrow R$ as $x \rightarrow rx$.

Claim: f is injective map.

Suppose $f(x) = f(y) \implies rx = ry \implies rx - ry = 0 \implies r(x - y) = 0$
 since R is integral domain $r \neq 0 \implies (x - y) = 0 \implies x = y$

Hence f is injective. Since R finite therefore f is surjective. $1 \in R$ and f is surjective, so there exist $s \in R$ such that $f(s) = 1$ i.e. $rs = 1$ implies that r is unit in R .

3. If R is an infinite ID, then R can be embedded into a field.

4. Every finite Division ring is a field.

Proof of above results : try yourself

Theorem: Let R be a commutative ring with unity. Then, an ideal M of R is maximal if and only if quotient ring R/M is a field.

Characteristics of an Integral Domain : Let R be an Integral domain. The Characteristic of R denoted $\text{char}(R)$ or $\text{ch}(R)$ is the least non negative integer n such that $n.1 = 0$. If no such n exists then we define the $\text{char}(R) = 0$.

Example : Consider the fields $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. It is easy to see that the characteristic of each of these fields is 0, for clearly $n.1 = n = 0$ if and only if $n = 0$.

Theorem: Let $\mathbb{F}$ be a field. Then $\text{char}(\mathbb{F}) = p$ for some prime p or $\text{char}(\mathbb{F}) = 0$.

proof: If $\text{char}(\mathbb{F}) = 0$, then we are done.

Otherwise, suppose that $\text{char}(\mathbb{F}) = n$ for some $n \in \mathbb{N}$, then $n.1 = 0$. Suppose that n is a composite number, say $n = rs$, then

$$rs.1 = 0$$

$$(r.1)(s.1) = 0$$

Since $\mathbb{F}$ is a field, $\mathbb{F}$ has no zero divisors. So $r.1 = 0$ or $s.1 = 0$. But this contradicts n being the least such non negative integer with this property. So $n = p$ for some prime p .

Euclidean Ring : An Integral Domain R is called an Euclidean Domain if $\forall 0 \neq a \in R, \exists$ a function $d : R \setminus \{0\} \rightarrow \mathbb{N} \cup \{0\}$ such that

1. $d(a) \leq d(ab) \quad \forall a, b \in R \quad (b \neq 0)$
2. $a = qb + r \quad \forall a, b \in R \quad (b \neq 0)$ and for some $q \in R$ and $r \in R$ such that $r = 0$ or $d(r) < d(b)$

Moreover, for $\forall a, b \in R, \exists \gcd(a, b) = d \in R$ such that $d = \alpha a + \beta b$ for some $\alpha, \beta \in R$ and d is unique upto units.

Polynomial Ring : It is denoted by $\mathbb{F}[x]$ and defined as

$$\mathbb{F}[x] = \{f(x) = a_0 + a_1x + \dots + a_nx^n \mid a_0, a_1, \dots, a_n \in \mathbb{F}\}$$

Note : $a_0, a_1, \dots, a_n \in \mathbb{F}$ are called coefficients of polynomial $f(x) = a_0 + a_1x + \dots + a_nx^n$

Result : If $\mathbb{F}$ is a field, then $\mathbb{F}[x]$ is a Euclidean Ring.

Content of a polynomial : The content of a polynomial with integer coefficients is the greatest common divisor of its coefficients.

The primitive part of a polynomial with integer coefficient is the quotient of the polynomial by its content. Thus a polynomial is the product of its primitive part and its content.

12.2 Factorization of polynomials

Irreducible polynomial: Let D be an integral domain. A polynomial $f(x)$ from $D[x]$ that is neither the zero polynomial nor a unit in $D[x]$ is said to be irreducible over D if, whenever $f(x)$ is expressed, as a product $f(x) = g(x)h(x)$ with $g(x)$ and $h(x)$ from $D[x]$, then $g(x)$ or $h(x)$ is a unit in $D[x]$.

Reducible Polynomial: A nonzero, nonunit element of $D[x]$ that is not irreducible over D is called reducible over D .

Note: In the case that an integral domain is a field $\mathbb{F}$, it is equivalent and more convenient to define a nonconstant $f(x) \in \mathbb{F}[x]$ to be irreducible if $f(x)$ cannot be expressed as a product of two polynomials of lower degree.

Examples

1. The polynomial $f(x) = 2x^2 + 4$ is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{Z}$, since $2x^2 + 4 = 2(x^2 + 2)$ and neither 2 nor $x^2 + 2$ is a unit in $\mathbb{Z}[x]$.
2. The polynomial $f(x) = 2x^2 + 4$ is irreducible over $\mathbb{R}$ but reducible over $\mathbb{C}$.
3. The polynomial $f(x) = x^2 - 2$ is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{R}$.
4. The polynomial $f(x) = x^2 + 1$ is irreducible over $\mathbb{Z}_3$ but reducible over $\mathbb{Z}_5$.

Reducibility Test for Degree 2 and 3

Let $\mathbb{F}$ be a field. If $f(x) \in \mathbb{F}[x]$ and $\deg f(x)$ is 2 or 3, then $f(x)$ is reducible over $\mathbb{F}$ if and only if $f(x)$ has a zero in $\mathbb{F}$.

Theorem: Reducibility over $\mathbb{Q}$ Implies Reducibility Over $\mathbb{Z}$

Let $f(x) \in \mathbb{Z}[x]$. If $f(x)$ is reducible over $\mathbb{Q}$, then it is reducible over $\mathbb{Z}$.

Note :

- Converse of above theorem need not true .(See example 1).
- Condition $f(x) \in \mathbb{Z}[x]$ in above theorem is important.

12.3 Irreducibility Tests

Theorem: Mod p Irreducibility Test

Let p be a prime and suppose that $f(x) \in \mathbb{Z}[x]$ with $\deg f(x) \geq 1$. Let $\overline{f(x)}$ be the polynomial in $\mathbb{Z}_p[x]$ obtained from $f(x)$ by reducing all the coefficients of $f(x)$ modulo p . If $\overline{f(x)}$ is irreducible over $\mathbb{Z}_p$ and $\deg \overline{f(x)} = \deg f(x)$, then $f(x)$ is irreducible over $\mathbb{Q}$.

Example : Let $f(x) = 21x^3 - 3x^2 + 2x + 9$. Then, over $\mathbb{Z}_2$, we have $\overline{f(x)} = x^3 + x^2 + 1$ and, since $\overline{f(0)} = 1$ and $\overline{f(1)} = 1$, we see that $\overline{f(x)}$ is irreducible over $\mathbb{Z}_2$. Thus $f(x)$ is irreducible over $\mathbb{Q}$. Notice that over $\mathbb{Z}_3$, $\overline{f(x)} = 2x$ is irreducible, but we can not apply above theorem to conclude that $f(x)$ is irreducible over $\mathbb{Q}$ since $\deg \overline{f(x)} \neq \deg f(x)$.

Note : Be careful not to use the converse of Theorem (Mod p Irreducibility Test). If $f(x) \in \mathbb{Z}[x]$ and $\overline{f(x)}$ is reducible over $\mathbb{Z}_p$ for some p , $f(x)$ need not be irreducible over $\mathbb{Q}$.

For example, consider $f(x) = 21x^3 - 3x^2 + 2x + 8$. Then, over $\mathbb{Z}_2$, $\overline{f(x)} = x^3 + x^2 = x^2(x + 1)$. But over $\mathbb{Z}_5$, has no zeros and therefore is irreducible over $\mathbb{Z}_5$. So, $f(x)$ is irreducible over $\mathbb{Q}$.

Note that this example shows that the Mod p Irreducibility Test may fail for some p and work for others. To conclude that a particular $f(x) \in \mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$, all we need to do is find a single p for which the corresponding polynomial in $\overline{f(x)} \in \mathbb{Z}_p$ is irreducible. However, this is not always possible, since $f(x) = x^4 + 1$ is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{Z}_p$ for every prime p .

The Mod p Irreducibility Test can also be helpful in checking for irreducibility of polynomials of degree greater than 3 and polynomials with rational coefficients.

Theorem: Eisenstein's Irreducibility Criterion :

Let $f(x) = a_n x^n + \dots + a_1 x + a_0$ be a polynomial with integer coefficients. Suppose that there exists a prime p , such that

- $p \nmid a_n$
- $p \mid a_{n-1}, a_{n-2}, \dots, a_1, a_0$
- $p^2 \nmid a_0$

Then $f(x)$ is irreducible over the integers.

Note : If $f(x) \in \mathbb{Z}[x]$, and it satisfies all 3 conditions, then $f(x)$ is irreducible over the $\mathbb{Q}$ and hence over $\mathbb{Z}$. But in general $f(x)$ need not be in $\mathbb{Z}[x]$.

Corollary: (Irreducibility of p th Cyclotomic Polynomial)

For any prime p , the p th cyclotomic polynomial

$$\phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + x + 1$$

is irreducible over $\mathbb{Q}$.

Theorem Let $\mathbb{F}$ be a field and let $f(x) \in \mathbb{F}[x]$. Then $\langle f(x) \rangle$ is maximal in $\mathbb{F}[x]$ if and only if $f(x)$ is irreducible over $\mathbb{F}$.

Theorem: Let $\mathbb{F}$ be a field and let $f(x)$ be an irreducible polynomial over $\mathbb{F}$. Then $\frac{\mathbb{F}[x]}{\langle f(x) \rangle}$ is a field.

12.4 Finite field

Theorem: If $\mathbb{F}$ is a finite field, then $|\mathbb{F}| = q = p^n$ for some prime p and some positive integer n .

Theorem: For every prime p and every positive integer n , there exist a field having p^n elements.

Theorem: If $\mathbb{F}$ is a finite field, then $\mathbb{F}^* = \mathbb{F} \setminus \{0\}$ is a cyclic group with respect to multiplication.

Splitting field : Let K be a field and let $f(x) \in K[x]$ be a polynomial. The splitting field of f is the smallest field extension L of K so that f splits into linear factors in $L[x]$.

Result : Let K be a field and let $f(x) \in K[x]$ be a polynomial. The splitting field of f exists and is unique up to isomorphism.

Example :

- The splitting field of $f(x) = x + 1 \in \mathbb{F}_2[x]$ is $\mathbb{F}_2$ itself since f is linear.
- The splitting field of $g(x) = x^2 + x + 1$ is $\mathbb{F}_4$ – by constructing the class of x in $L = \mathbb{F}_2[x]/g\mathbb{F}_2[x]$ is a root of g . To see this consider

$g(y) = y^2 + y + 1$ as polynomial in $L[y]$ and note that we compute modulo $x^2 + x + 1$ in L

$$(y + x)(y + x + 1) = y^2 + (x + x + 1)y + x^2 + x = y^2 + y + 1 = g(y)$$

- Put $h(x) = (x^3 + x + 1) \in \mathbb{F}_2[x]$. This polynomial is irreducible over $\mathbb{F}_2$ and it thus allows to define a field with 8 elements as $\mathbb{F}_8$ is isomorphic to $\mathbb{F}_2[x]/h\mathbb{F}_2[x]$. By the same considerations as above the splitting field of h is $\mathbb{F}_8$.

Result : Let $f(x) = (x^p)^n - x \in \mathbb{F}_p[x]$ for some integer n . The splitting field of f is a finite field K with $|K| = p^n$ elements and f splits as

$$(x^p)^n - x = \prod_{a \in K} (x - a)$$

Theorem: Existence and uniqueness of finite fields

For any prime p and any natural number n there exists a finite field with p^n elements. Every field with p^n elements is isomorphic to the splitting field of $f(x) = (x^p)^n - x$ over $\mathbb{F}_p$.

12.4.1 Construction of a finite field

Points to remember :

1. We write $\mathbb{Z}/\langle p \rangle$ and $\mathbb{F}_p$ interchangeably for the field of size p .
2. Every finite field has prime power order.
3. For every prime power, there is a finite field of that order.
4. For a prime p and positive integer n , there is an irreducible $f(x)$ of degree n in $\mathbb{F}_p[x]$, and $\frac{\mathbb{F}_p[x]}{\langle f(x) \rangle}$ is a field of order p^n .
5. Any two finite fields of the same size are isomorphic.

Theorem: For a prime p and a monic irreducible $f(x) \in \mathbb{F}_p[x]$ of degree n , the ring $\mathbb{F}_p[x]/\langle f(x) \rangle$ is a field of order p^n .

Construction:

We will explain the construction by an example.

$\mathbb{Z}_2[x]$ is a commutative ring with identity 1. Since for every field $\mathbb{F}$, the polynomial ring $\mathbb{F}[x]$ is a PID (also UFD -unique factorization domain) and every ideal is of the form $\langle f(x) \rangle$ and $\langle f(x) \rangle$ is maximal in $\mathbb{F}[x]$ if and only if $\langle f(x) \rangle$ is irreducible over $\mathbb{F}$.

Therefore $\mathbb{Z}_2[x]/\langle f(x) \rangle$ is a field, if $f(x)$ is irreducible over $\mathbb{Z}_2$.

If degree of $f(x)$ is n then $\mathbb{Z}_2[x]/\langle f(x) \rangle$ has finitely many elements and

$$\left| \frac{\mathbb{Z}_2[x]}{\langle f(x) \rangle} \right| = p^n = 2^n$$

If

$$f(x) = a_0 + a_1x + a_{n-1}x^{n-1} + a_nx^n, \quad a_n \neq 0$$

then

$$\frac{\mathbb{Z}_p[x]}{\langle f(x) \rangle} = \{g(x) + \langle f(x) \rangle : g(x) \in \mathbb{Z}_p[x]\}$$

implies

$$\frac{\mathbb{Z}_p[x]}{\langle f(x) \rangle} = \{f(x)q(x) + r(x) + \langle f(x) \rangle : q(x), r(x) \in \mathbb{Z}_p[x], r(x) = 0 \text{ or } \deg r(x) < \deg f(x)\}$$

(by Euclidean Algorithm)

implies

$$\frac{\mathbb{Z}_p[x]}{\langle f(x) \rangle} = \{r(x) + \langle f(x) \rangle : r(x) \in \mathbb{Z}_p[x], r(x) = 0 \text{ or } \deg r(x) < \deg f(x) = n\}$$

implies

$$\frac{\mathbb{Z}_p[x]}{\langle f(x) \rangle} = \{\alpha_0 + \alpha_1x + \dots + \alpha_{n-1}x^{n-1} + \langle f(x) \rangle : \alpha_i \in \mathbb{Z}_p\}$$

$$\sim \{(\alpha_0, \alpha_1, \dots, \alpha_{n-1}) : \alpha_i \in \mathbb{Z}_p\}$$

Example 1 : $p = 2, n = 2$

The quadratic polynomial of degree 2 in $\mathbb{Z}_2[x]$ are :

$$\begin{array}{ll}
 x^2 = x \cdot x & \text{reducible} \\
 x^2 + 1 = (x + 1)(x + 1) & \text{reducible} \\
 x^2 + x + 1 & \text{irreducible} \quad \text{verify!} \\
 x^2 + x = x(x + 1) & \text{reducible}
 \end{array}$$

Only one of these namely $f(x) = x^2 + x + 1$ is irreducible over $\mathbb{Z}_2$

So,

$$\begin{aligned}
 \frac{\mathbb{Z}_2[x]}{\langle f(x) \rangle} &= \{ \alpha_0 + \alpha_1 x + \langle x^2 + x + 1 \rangle : \alpha_0, \alpha_1 \in \mathbb{Z}_2 \} \\
 &= \{ 0 + \langle x^2 + x + 1 \rangle, 1 + \langle x^2 + x + 1 \rangle, x + 1 + \langle x^2 + x + 1 \rangle, x + \langle x^2 + x + 1 \rangle \} \\
 &\sim \{ 0, 1, x, x + 1 \}
 \end{aligned}$$

Cayley's Table:

+	0	1	x	$x + 1$
0	0	1	x	$x + 1$
1	1	0	$x + 1$	x
x	x	$x + 1$	0	1
$x + 1$	$x + 1$	x	1	0

.	0	1	x	$x + 1$
0	0	0	0	0
1	0	1	x	$x + 1$
x	0	x	$x + 1$	1
$x + 1$	0	$x + 1$	1	x

Note that $x^2 + x + 1 = 0 \Rightarrow x^2 = -x - 1 = x + 1 \pmod{2}$

Thus, it is closed with respect to addition and multiplication.

Verify that $\{0, 1, x, x + 1\}$ is a field of order 4 with above defined addition and multiplication.

Example 2: Polynomials of degree 4 in $\mathbb{Z}_2$:

x^4	$x^4 + 1$	$x^4 + x$	$x^4 + x + 1$
$x^4 + x^3$	$x^4 + x^3 + x^2$	$x^4 + x^3 + x$	$x^4 + x^3 + 1$
$x^4 + x^2 + x + 1$	$x^4 + x^3 + x^2 + x$	$x^4 + x^3 + x^2 + x + 1$	$x^4 + x^2 + 1$
$x^4 + x^2 + x$	$x^4 + x^2$	$x^4 + x^3 + x + 1$	$x^4 + x^3 + x + 1$

1. x^4 reducible since x is a factor
2. $x^4 + 1$ reducible since $x + 1$ is a factor
3. $x^4 + x$ reducible since x is a factor
4. $x^4 + x + 1$ -
5. $x^4 + x^3$ reducible since x is a factor
6. $x^4 + x^3 + x^2$ reducible since x is a factor
7. $x^4 + x^3 + x$ reducible since x is a factor
8. $x^4 + x^3 + 1$ -
9. $x^4 + x^2 + x + 1$ reducible since $x + 1$ is a factor
10. $x^4 + x^3 + x^2 + x$ reducible since x is a factor
11. $x^4 + x^3 + x^2 + x + 1$ -
12. $x^4 + x^2 + 1$ reducible since $x^2 + x + 1$ is a factor
13. $x^4 + x^2 + x$ reducible since x is a factor
14. $x^4 + x^2$ reducible since x is a factor
15. $x^4 + x^3 + x + 1$ reducible since $x + 1$ is a factor
16. $x^4 + x^3 + x + 1$ reducible since $x + 1$ is a factor

Now we check the reducibility of remaining polynomial

$$f(x) = x^4 + x + 1, \quad g(x) = x^4 + x^3 + 1, \quad h(x) = x^4 + x^3 + x^2 + x + 1$$

None of them is divisible by a polynomial of degree 1. So they can be reducible if and only if divisible by an irreducible polynomial of degree 2 in $\mathbb{Z}_2[x]$ i.e. x^2 , $x^2 + x$, $x^2 + 1$, $x^2 + x + 1$.

There is only one irreducible polynomial of degree 2 over $\mathbb{Z}_2$, namely $k(x) = x^2 + x + 1$.

By division, Observe that $k(x) \nmid f(x)$, $k(x) \nmid g(x)$ and $k(x) \nmid h(x)$.

Therefore $f(x)$, $g(x)$ and $h(x)$ are the only irreducible polynomial of degree 4 over $\mathbb{Z}_2$ and quotient over $\mathbb{Z}_2$ will become field of order $2^4 = 16$.

$$\frac{\mathbb{Z}_2[x]}{\langle f(x) \rangle} \cong \frac{\mathbb{Z}_2[x]}{\langle g(x) \rangle} \cong \frac{\mathbb{Z}_2[x]}{\langle h(x) \rangle}$$

Example 3: Two finite fields of order 8 are $\frac{\mathbb{F}_2[x]}{\langle x^3+x+1 \rangle}$ and $\frac{\mathbb{F}_2[x]}{\langle x^3+x^2+1 \rangle}$.

Example 4: Two finite fields of order 9 are $\frac{\mathbb{F}_3[x]}{\langle x^2+1 \rangle}$ and $\frac{\mathbb{F}_3[x]}{\langle x^2+x+2 \rangle}$.

Example 5: The polynomial $x^3 - 2$ is irreducible in $\mathbb{F}_7[x]$, so $\frac{\mathbb{F}_7[x]}{\langle x^3-2 \rangle}$ is a field of order $7^3 = 343$.

12.4.2 Primitive Polynomial

If $\mathbb{F}$ is a finite field, then $\mathbb{F}^* = \mathbb{F} \setminus \{0\}$ is a cyclic group with respect to multiplication. Let $q = p^n$, then $\mathbb{F}_q^*$ is cyclic and thus, $\exists \alpha \in \mathbb{F}_q^*$ such that

$$\mathbb{F}_q^* = \langle \alpha \rangle = \langle \alpha^i : 0 \leq i \leq q-1 \rangle$$

and $|\alpha| = q-1$ with respect to multiplication.

Thus α is a primitive root of unity in $\mathbb{F}_q^*$.

Corollary : Every finite field contains at least one primitive element. More precisely there are exactly $\phi(q-1)$ primitive elements.

Primitive Polynomial : A polynomial $f(x)$ such that α is a root of $f(x)$ and α is a primitive element of $\mathbb{F}_q$ is called a primitive polynomial.

Example : From above example we have $\frac{\mathbb{Z}_2[x]}{\langle x^4+x^3+x^2+x+1 \rangle}$ is a field of order 2^4 as $f(x) = x^4 + x^3 + x^2 + x + 1$ is irreducible over $\mathbb{Z}_2$.

If α is a root of $f(x)$, then $f(\alpha) = 0$, implies,

$$\alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$$

multiplying both sides by $\alpha - 1$, we have

$$\alpha^5 - 1 = 0$$

implies

$$\alpha^5 = 1$$

Hence $|\alpha| = 5 \neq 15 = |\mathbb{F}_{2^4}^*|$

So α cannot be primitive element and so $f(x)$ is not a primitive polynomial though $f(x)$ is irreducible over $\mathbb{Z}_2$.

Result : If $\frac{\mathbb{Z}_p[x]}{\langle f(x) \rangle}$ and $\frac{\mathbb{Z}_p[x]}{\langle g(x) \rangle}$ are fields of order p^n , then they are isomorphic. But if $g(x)$ is a primitive polynomial, then $f(x)$ need not be primitive polynomial.

In above example $f(x)$ is not a primitive polynomial but $g(x) = x^4 + x + 1$ is a primitive polynomial.

$g(x)$ is irreducible over $\mathbb{Z}_2$.

If α is a root of $g(x)$, then $g(\alpha) = 0$, implies,

$$\alpha^4 + \alpha + 1 = 0 \implies \alpha^4 = \alpha + 1 \text{ (as } 1 \equiv -1 \pmod{2} \text{)}$$

The set

$$\begin{aligned} & \{\alpha^i : 0 \leq i \leq q-1\} \\ &= \{\alpha, \alpha^2, \alpha^3, \alpha^4 = \alpha + 1, \alpha^5 = \alpha^2 + \alpha, \alpha^6 = \alpha^3 + \alpha^2, \alpha^7 = \alpha^3 + \alpha + 1, \\ & \alpha^8 = \alpha^2 + 1, \alpha^9 = \alpha^3 + \alpha^1, \alpha^{10} = \alpha^2 + \alpha + 1, \alpha^{11} = \alpha^3 + \alpha^2 + \alpha, \\ & \alpha^{12} = \alpha^3 + \alpha^2 + \alpha + 1, \alpha^{13} = \alpha^3 + \alpha^2 + 1, \alpha^{14} = \alpha^3 + 1, \alpha^{15} = 1\} \end{aligned}$$

Hence α is a primitive root of $g(x)$.

Result : A polynomial $g(x) \in \mathbb{Z}_p[x]$ of degree k is said to be a primitive polynomial of degree k if $g(x) \mid x^m - 1$, $m = p^k - 1$ and for no smaller values of m .

Verify : The polynomial $g(x) = x^4 + x^2 + 1$ is a primitive polynomial of

degree 4 in $\mathbb{Z}_2[x]$ using above result.

Example of Application of Primitive Polynomial

Whenever we need a cyclic group with a generator, we can take multiplicative group of a finite field with primitive element. For example:

Recall the El-Gamal Cryptosystem, all users agree on a finite cyclic group, say $\mathbb{F}_q^* = \langle \alpha : \alpha^{q-1} = 1 \rangle$ where $\mathbb{F}_q = \mathbb{F}_q^* \cup \{0\}$ is a finite field of order $q = p^n$ for some prime p and some positive integer n .

Every user, say Alice, Bob,..., will choose and keep secret positive integers $a, b, \dots$ and make public $\alpha^a, \alpha^b, \dots$ respectively.

To send a message $M \in \mathbb{F}_q^*$, to Alice, we create a mask by choosing a random integer k and doing $(\alpha^a)^k$ and hide the message $M(\alpha^a)^k \in \mathbb{F}_q^*$.

Now send Alice the ordered pair : $(\alpha^k, M(\alpha^a)^k) \in \mathbb{F}_q^* \times \mathbb{F}_q^*$

So, She retrieves the message by doing $M(\alpha^a)^k \cdot [(\alpha^k)^a]^{-1} = M \in \mathbb{F}_q^*$

Exercise : Find a primitive polynomial of degree 6 and 7 over $\mathbb{Z}_2$.

12.5 Cryptography Over Finite Field

Let $\mathbb{F}_q$ be a finite field. Then order $|\mathbb{F}_q| = q = p^n$ for some prime p and some positive integer n .

We know that $G = \mathbb{F}_q^* = \mathbb{F}_q \setminus \{0\}$ is cyclic group with respect to multiplication of order $q - 1$. So

$$\mathbb{F}_q^* = \langle \alpha : \alpha^{q-1} \equiv 1 \pmod{p} \rangle$$

and $\mathbb{F}_q = \mathbb{F}_q^* \cup \{0\}$

Hence all the cryptographic applications done on finite cyclic groups are applicable on $G = \mathbb{F}_q^*$. For example

1. Diffie-Hellman Key Exchange Protocol
2. Massy-Omura Cryptosystem

3. El-Gamal Cryptosystem

4. Discrete Logarithm Problem and Algorithm to calculate the Discrete Logarithm

Question: Decrypt the following El-Gamal cipher text

(H,K) (X,P) (K,N) (R,H) (F,T) (Y,V) (H,E) (A,F) (W,T) (D,J)
(J,U)

Take field $\mathbb{F}_{p^3} = \frac{\mathbb{Z}_3[x]}{\langle x^3+2x^2+1 \rangle}$

A	1	N	$x^2 + x + 1$
B	2	O	$x^2 + 2x$
C	x	P	$x^2 + 2x + 1$
D	$x + 1$	Q	$x^2 + 2x + 2$
E	$x + 2$	R	$2x^2$
F	$2x$	S	$2x^2 + 1$
G	$2x + 1$	T	$2x^2 + 2$
H	$2x + 2$	U	$2x^2 + x$
I	x^2	V	$2x^2 + x + 1$
J	$x^2 + 1$	W	$2x^2 + x + 2$
K	$x^2 + 2$	X	$2x^2 + 2x$
L	$x^2 + x$	Y	$2x^2 + 2x + 1$
M	$x^2 + x + 1$	Z	$2x^2 + 2x + 2$

The correspondence of the normal English alphabet letters with field elements

A	x^{26}	1
B	x^{13}	2
C	x	x
D	x^{18}	$x + 1$
E	x^{11}	$x + 2$
F	x^{14}	$2x$
G	x^{24}	$2x + 1$
H	x^5	$2x + 2$
I	x^2	x^2
J	x^7	$x^2 + 1$
K	x^3	$x^2 + 2$
L	x^{19}	$x^2 + x$
M	x^{22}	$x^2 + x + 1$
N	x^8	$x^2 + x + 1$
O	x^{12}	$x^2 + 2x$
P	x^{10}	$x^2 + 2x + 1$
Q	x^4	$x^2 + 2x + 2$
R	x^{15}	$2x^2$
S	x^{16}	$2x^2 + 1$
T	x^{20}	$2x^2 + 2$
U	x^{25}	$2x^2 + x$
V	x^{17}	$2x^2 + x + 1$
W	x^{23}	$2x^2 + x + 2$
X	x^6	$2x^2 + 2x$
Y	x^{21}	$2x^2 + 2x + 1$
Z	x^9	$2x^2 + 2x + 2$

Correspondence of the field elements to the normal text letters

C	x	F	x^{14}
I	x^2	R	x^{15}
K	x^3	S	x^{16}
Q	x^4	V	x^{17}
H	x^5	D	x^{18}
X	x^6	L	x^{19}
J	x^7	T	x^{20}
N	x^8	Y	x^{21}
Z	x^9	M	x^{22}
P	x^{10}	W	x^{23}
E	x^{11}	G	x^{24}
O	x^{12}	U	x^{25}
B	x^{13}	A	x^{26}

El-Gamal Cipher text $(H,K) = (x^5, x^3)$

$(X,P) = (x^6, x^{10})$

$(K,N) = (x^3, x^8)$

$(R,H) = (x^{15}, x^5)$

$(F,T) = (x^{14}, x^{20})$

$(Y,V) = (x^{21}, x^{17})$

$(H,E) = (x^5, x^{11})$

$(A,F) = (x^{26}, x^{14})$

$(W,T) = (x^{23}, x^3)$

$(D,J) = (x^{18}, x^3)$

$(J,U) = (x^7, x^{25})$

Recall: El-Gamal Cryptosystem Let $G = \langle g | g^n = 1 \rangle$ be public. All users, say Alice and Bob $\dots$. They choose random integers $a, b, \dots$ respectively and keep it secret and make $g^a, g^b, \dots \in G$ public.

Now, if Bob wants to send message $\mathcal{M}$ to Alice. He choose a random integer k and create a mask by doing $(g^a)^k \in G$. Bob then sends to Alice $(g^k, \mathcal{M}g^{ak}) \in G \times G$.

Alice will receive $(g^k, \mathcal{M}g^{ak}) \in G \times G$, to recover $\mathcal{M}$ from order pair $(g^k, \mathcal{M}g^{ak})$, Alice will recreate the mask by doing $(g^k)^a = g^{ka} = g^{ak} \in G$, as Alice know a (secret) and g^k is first element of order pair $(g^k, \mathcal{M}g^{ak})$ received from Bob. To recover the message, Alice will compute

$$(\mathcal{M}g^{ak})(g^{ak})^{-1} = \mathcal{M} \in G$$

Decryption: Suppose secret $a = 11$, public encryption key $g^a = g^{11} = x^{11} = x + 2 = E$

The first cipher message unit is (H,K) or (x^5, x^3)

The message lies masked in the first coordinate i.e. H or x^5

Recreate the mask from the second coordinate i.e. K or x^3 by doing $(x^3)^{11} = x^{33} = x^7$. Remove the mask from the first coordinate

$$x^5(x^7)^{-1} = x^5(x^{19}) = x^{24} \text{ Corresponds English alphabet G}$$

So the plain text corresponding to the first cipher (H,K) is G.
Similarly for others.

12.6 Exercise

1. Describe the elements of the fields K

(a) $K = \mathbb{Q}(\sqrt{3})$

(b) $K = \mathbb{Q}(\sqrt[5]{7})$

(c) $K = \mathbb{Q}(\omega)$ where ω a primitive cube root of unity.

In each of the above cases, write $K = \frac{\mathbb{Q}[x]}{\langle f(x) \rangle}$ for an appropriate polynomial f . In each of the above cases, what is the dimension of K over $\mathbb{Q}$ when K is viewed as a $\mathbb{Q}$ vector space.

2. For a prime $p \geq 3$, show that there are exactly $(p+1)/2$ elements $x \in F_p$ with $x = y^2$ for some $y \in F_p$. We call such x a square. Conclude that there is some $x \in F_p$ which is not a square whenever $p > 2$.
3. Construct a field of order 8.
4. Construct a field of order 125.
5. Check the irreducibility of the polynomial $f(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$ over $\mathbb{Q}$.
6. Construct a field of order 27 as a ring of polynomials and find the inverse of $x^2 + 2x + 1$ in it.

7. Check the irreducibility of the polynomial $x^5 + 9x^4 + 12x^2 + 6$ over $\mathbb{Z}_3$.
8. Let $f(x) = x^4 + x^2 + x + 1$. Show that $f(x)$ is irreducible over $\mathbb{Z}_5$.
9. Factor the following polynomials into a product of irreducible polynomials in the given fields:
 - (a) $x^3 + x + 1$ over $\mathbb{Z}_3$.
 - (b) $x^4 + x + 1$ over $\mathbb{Z}_5$.
 - (c) $x^4 + 10x^3 + 13x^2 - 32x + 12$ over $\mathbb{Q}$.
10. Show that $f(x) = x^2 + 2x - 1 \in \mathbb{Z}_3[x]$ is irreducible over $\mathbb{Z}_3$. Using this fact construct a finite field $\mathbb{F}_9$ of 9 elements. If α is a root of $f(x)$, then find the elements of this field. Also find the discrete logarithms of $(\alpha - 1)$ to the base α and of α to the base $(\alpha - 1)$.
11. Let $f(x) = x^4 + x^2 + x + 1$. Show that $f(x)$ is irreducible over $\mathbb{Z}_5$. Using this fact construct a finite field $\mathbb{F}_{125}$ of 125 elements. Also check if $f(x)$ is a primitive polynomial over $\mathbb{Z}_5$. If yes, then assuming α to be a root of $f(x)$, find the elements of this field.
12. Find all possible irreducible polynomials of degree 6 and 7 over $\mathbb{Z}_2$ and check which of them is a primitive polynomial over $\mathbb{Z}_2$.
13. Prove the following results and theorems:
 - (a) Prove : Let R be a commutative ring with unity. Then, an ideal M of R is maximal if and only if R/M is a field.
 - (b) Prove : Let $\mathbb{F}$ be a field. Then $\text{char}(\mathbb{F}) = p$ for some prime p or $\text{char}(\mathbb{F}) = 0$.
 - (c) Let K be a field and Let $f \in K[x]$ be a monic irreducible polynomial. Then prove that $\frac{K[x]}{\langle f \rangle}$ is a field.
 - (d) Prove : If $\mathbb{F}$ is a finite field, then $|\mathbb{F}| = p^n$ for some prime p and some positive integer n .
 - (e) Prove : For every prime p and every positive integer n , there exist a field having p^n elements.

Chapter 13

Elliptic Curves

13.1 Elliptic Equation

The most general elliptic curve equation is the **Weierstrass Equation**:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6 \quad (13.1)$$

where the coefficient come from a ring or a field. Just for the time being consider the real Weierstrass equation, i.e. the coefficients a_i 's are real numbers.

13.1.1 Conversion of Weierstrass Equation to a Normal Form:

With a suitable change of variables, Weierstrass equation i.e. (1) can be converted to a simpler form known as its normal form:

$$y^2 = x^3 + ax + b \quad (13.2)$$

for some real numbers a, b .

There are several methods to do this . We take the simplest way.

In equation(1), put

$$y = \frac{v - a_1x - a_3}{2}.$$

The LHS of equation becomes:

$$\begin{aligned}
 y^2 + a_1xy + a_3y &= y(y + a_1x + a_3) \\
 &= \left(\frac{v - a_1x - a_3}{2} \right) \left(\frac{v - a_1x - a_3}{2} + a_1x + a_3 \right) \\
 &= \left(\frac{v - a_1x - a_3}{2} \right) \left(\frac{v + a_1x + a_3}{2} \right) \\
 &= \frac{v^2 - (a_1x + a_3)^2}{4} \\
 &= \frac{v^2 - (a_1^2x^2 + 2a_1a_3x + a_3^2)}{4}
 \end{aligned}$$

Hence equation(1) becomes:

$$\begin{aligned}
 \frac{v^2 - (a_1^2x^2 + 2a_1a_3x + a_3^2)}{4} &= x^3 + a_2x^2 + a_4x + a_6 \\
 v^2 &= 4x^3 + 4a_2x^2 + 4a_4x + 4a_6 + a_1^2x^2 + 2a_1a_3x + a_3^2 \\
 v^2 &= 4x^3 + (a_1^2 + 4a_2)x^2 + 2(a_1a_3 + 2a_4)x + (a_3^2 + 4a_6) \quad (13.3)
 \end{aligned}$$

In equation(3) put $x = \frac{X - 3(a_1^2 + 4a_2)}{36}$, then RHS of equation (3) becomes:

$$\begin{aligned}
 4 \left(\frac{X - 3(a_1^2 + 4a_2)}{36} \right)^3 &+ (a_1^2 + 4a_2) \left(\frac{X - 3(a_1^2 + 4a_2)}{36} \right)^2 \\
 &+ 2(a_1a_3 + 2a_4) \left(\frac{X - 3(a_1^2 + 4a_2)}{36} \right) + (a_3^2 + 4a_6)
 \end{aligned}$$

And equation(3) becomes:

$$\begin{aligned}
 v^2 &= \frac{4}{36^3} \{X - 3(a_1^2 + 4a_2)\}^3 + \frac{(a_1^2 + 4a_2)}{36^2} \{X - 3(a_1^2 + 4a_2)\}^2 \\
 &+ \frac{2}{36} (a_1a_3 + 2a_4) \{X - 3(a_1^2 + 4a_2)\} \\
 &+ (a_3^2 + 4a_6)
 \end{aligned}$$

Now put $v = \frac{Y}{108}$, in this equation we get:

$$\begin{aligned} \left(\frac{Y}{108}\right)^2 &= \frac{4}{36^3} \{X - 3(a_1^2 + 4a_2)\}^3 + \frac{(a_1^2 + 4a_2)}{36^2} \{X - 3(a_1^2 + 4a_2)\}^2 \\ &\quad + \frac{2}{36} (a_1 a_3 + 2a_4) \{X - 3(a_1^2 + 4a_2)\} \\ &\quad + (a_3^2 + 4a_6) \end{aligned}$$

Multiplying both sides by $(108)^2$, and $108 = 3 \times 36$; and

$$\begin{aligned} (108)^2 \cdot \frac{4}{(36)^3} &= 4 \frac{3^2 \cdot 36^2}{36^3} \\ &= \frac{3^2 \cdot 4}{36} \\ &= 1 \end{aligned}$$

$$\frac{108^2}{36^2} = \frac{3^2 \cdot 36^2}{36} = 9$$

$$108^2 \cdot \frac{2}{36} = \frac{3^2 \cdot 36^2 \cdot 2}{36} = 3^2 \cdot 36 \cdot 2 = 3 \cdot 3 \cdot 2 \cdot 36 = 6 \cdot 108$$

We get

$$\begin{aligned} Y^2 &= \{X - 3(a_1^2 + 4a_2)\}^3 + 9(a_1^2 + 4a_2) \{X - 3(a_1^2 + 4a_2)\}^2 \\ &\quad + 6 \cdot 108(a_1 a_3 + 2a_4) \{X - 3(a_1^2 + 4a_2)\} + 108^2(a_3^2 + 4a_6) \end{aligned}$$

On expanding each bracket term we get

$$\begin{aligned} Y^2 &= [\{X^3 + 3X^2\{-3(a_1^2 + 4a_2)\} + 3X\{-3(a_1^2 + 4a_2)\}^2 + \{-3(a_1^2 + 4a_2)\}^3] \\ &\quad + [9(a_1^2 + 4a_2)\{X^2 - 6(a_1^2 + 4a_2)X + 9(a_1^2 + 4a_2)^2\}] \\ &\quad + [6 \cdot 108(a_1 a_3 + 2a_4)X - 18 \cdot 108(a_1 \cdot a_3 + 2a_4)(a_1^2 + 4a_2)] \\ &\quad + [(108)^2(a_3^2 + 4a_6)] \end{aligned}$$

Writing the RHS as polynomial in X , the equation becomes:

$$\begin{aligned} Y^2 = & X^3 + \{-9(a_1^2 + 4a_2) + 9(a_1^2 + 4a_2)\}X^2 \\ & + \{27(a_1^2 + 4a_2)^2 - 54(a_1^2 + 4a_2)^2 + 6 \cdot 108(a_1a_3 + 2a_4)\}X \\ & + \{-27(a_1^2 + 4a_2)^3 + 81(a_1^2 + 4a_2)^3 - 18 \cdot 108(a_1a_3 + 2a_4)(a_1^2 + 4a_2) \\ & + 108^2(a_3^2 + 4a_6)\} \end{aligned}$$

Or:

$$\begin{aligned} Y^2 = & X^3 + \{-27(a_1^2 + 4a_2)^2 + 6 \cdot 108(a_1a_3 + 2a_4)\}X \\ & + \{54(a_1^2 + 4a_2)^3 - 18 \cdot 108(a_1a_3 + 2a_4)(a_1^2 + 4a_2) + 108^2(a_3^2 + 4a_6)\} \end{aligned}$$

Or

$$Y^2 = X^3 + AX + B$$

$$\text{Where } A = \{-27(a_1^2 + 4a_2)^2 + 6 \cdot 108(a_1a_3 + 2a_4)\}$$

$$B = \{54(a_1^2 + 4a_2)^3 - 18 \cdot 108(a_1a_3 + 2a_4)(a_1^2 + 4a_2) + 108^2(a_3^2 + 4a_6)\}$$

Note: In the process of transforming an elliptic curve to normal form we, have used division by 2 and 3. Hence if the coefficients come from a field K with, $\text{Char}(K) \neq 2, 3$, then the above equation is the normal form of Weierstrass equation over K . That is $y^2 = x^3 + ax + b$; $a, b \in K$, where $\text{Char}(K) \neq 2, 3$.

13.1.2 Normal Forms Of Weierstrass Equation Over Fields

Let K be a field of characteristic p . Then $p = 0$ or p is a prime. Recall the Weierstrass equation in general form:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

Also recall that if $p \neq 2, 3$ then the equation can be transformed into simpler equation called normal form

$$Y^2 = X^3 + AX + B; \quad A, B \in K$$

Similarly, the normal form in case $p = 2$,
either

$$Y^2 + XY = X^3 + AX^2 + B$$

or

$$Y^2 + CY = X^3 + AX + B; \quad A, B, C \in K.$$

And in case $p = 3$, the normal form is

$$Y^2 = X^3 + AX^2 + BX + C \quad A, B, C \in K.$$

This can be split into following two equivalent forms, $Y^2 = X^3 + CX^2 + D$
or $Y^2 = X^3 + AX + B$, $A, B, C, D \in K$.

13.1.3 Real Elliptic Curves

The normal form of a real elliptic curve is

$$E = E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + ax + b, a, b \in \mathbb{R}\} \cup \{\infty\}.$$

Or

$$E = \{(x, y) \in \mathbb{R} \times \mathbb{R} : F(x, y) = 0 \text{ where } F(x, y) = y^2 - x^3 - ax - b, a, b \in \mathbb{R}\} \cup \{\infty\}.$$

The representation of E_K for a field of $\text{Char}(K) \neq 2, 3$, is same as above. However, considering $K = \mathbb{R}$ have the advantage of knowledge of calculus and plane geometry. We shall move to other fields as and when convenient. For the moment, let K be an algebraically closed field. Best is the field $\mathbb{C}$ of complex numbers.

13.2 Elliptic Curves

Let K be a field, and

$$E = E_K = \{(x, y) \in K \times K | y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6\} \cup \{\infty\},$$

or

$$E_K = \{(x, y) \in K \times K | F(x, y) = 0\} \cup \{\infty\},$$

where $F(x, y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6$

Then E_K is said to be an elliptic curve over K . That is E consists of all points $(x, y) \in K \times K$, satisfying the Weierstrass equation (W.E.), together with the point at infinity denoted by ∞ . We shall talk about the ‘point of infinity’ later on, where develop some basics of algebraic geometry. Till that let it be there. We shall set up some properties on E .

13.2.1 Singular and Non-Singular Curves

Let $E = E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} | y^2 = x^3 + ax + b; a, b \in \mathbb{R}\} \cup \{\infty\}$
 $= \{(x, y) \in \mathbb{R} \times \mathbb{R} | 0 = F(x, y) = y^2 - x^3 - ax - b; a, b \in \mathbb{R}\} \cup \{\infty\}$

Then E can be traced by mapping a point $P(x, y) \in E$, with (x, y) as coordinates of P in $\mathbb{R}^2$ and ∞ as the point on the end of every ordinate.

Observe that E is symmetric, the graph $\{(x, y) \in \mathbb{R} \times \mathbb{R} | y^2 = x^3 + ax + b; a, b \in \mathbb{R}\}$ about x -axis. Many more properties can be seen easily. First of all:

Singular Point: A point $P(x, y)$ on a curve $F(x, y) = 0$, is said to be a singular point if

$$\left(\frac{\partial f}{\partial x}\right)_P = 0 \text{ and } \left(\frac{\partial f}{\partial y}\right)_P = 0.$$

Recall an ordinary point or a smooth point on the curve $f(x, y) = 0$, where the curve has a definite tangent.

Let $f(x, y) = 0$ be the curve. If both $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist, then we get

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 0$$

and so the slope of the tangent at $P(x, y)$:

$$\frac{dy}{dx} = -\frac{\left(\frac{\partial f}{\partial x}\right)_P}{\left(\frac{\partial f}{\partial y}\right)_P}$$

Now, if at least one of the partial derivatives does not vanish, one of $\frac{dy}{dx}$ or $\frac{dx}{dy}$, can be determined (at P). The point then has surely a tangent and the point P is called an ordinary point. However, if both the partial derivative

$\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at P vanish, the point P is called a singular point.

Further, a curve given by $f(x, y) = 0$ is said to be singular, if it has at least one singular point.

We shall discuss general real curves first, and then Elliptic curves in particular.

Example 1 Let the curve be given by $C : f(x, y) = 0$, where $f(x, y) = x^2 + y^2 - 1$, i.e. unit circle with origin as its centre. Then

$$\frac{\partial f}{\partial x} = 2x = 0 \implies x = 0$$

$$\frac{\partial f}{\partial y} = 2y = 0 \implies y = 0$$

Both $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ is zero only at $(0, 0)$ i.e. origin. But this does not lie on C . Hence, the curve C has no singular points.

Example 2 Let the curve be given by $C : f(x, y) = (x-1)^2 - (y-1)(y-2)^2$, then

$$\frac{\partial f}{\partial x} = 0 \implies x = 1$$

$$\frac{\partial f}{\partial y} = 0 \implies -(y-1) \cdot 2(y-2) - (y-2)^2 = 0$$

$$\implies (y-2)\{2y-2+(y-2)\} = 0$$

$$\implies (y-2)(3y-4) = 0$$

$$\implies y = 2, \frac{4}{3}$$

Possible singular points are $(1, 2)$ and $(1, \frac{4}{3})$.

However $(1, \frac{4}{3})$ does not lie on C . The point $(1, 2)$ is the only singular point of C .

Double, Triple, Multiple Point: A point $P(x, y)$ on a curve C given by $f(x, y) = 0$ is said to be a multiple point of C , if more than one branch of C pass through P . In particular, if two branches pass through $P \in C$, P is called a double point of C . Like wise, if three branches pass through $P \in C$, P is called a triple point of C . A multiple point $P \in C$ is said to be of order m if m branches of C pass through P .

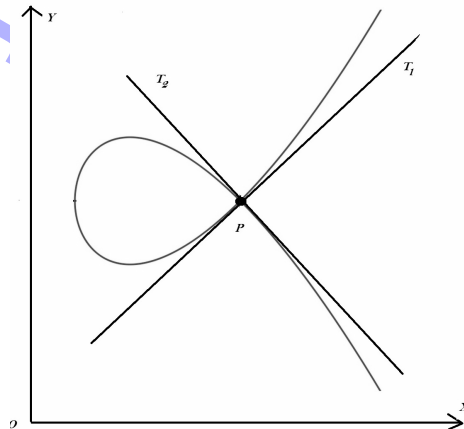
We shall discuss double point in more detail.

Types of Double Point: Let $P(x, y)$ be a double point of a curve C given by $f(x, y) = 0$. So there are two branches pass through P , and thus two tangent at P to C . These may be real and distinct, real and coincident, or imaginary. Accordingly, P is said to be

1. Node
2. Cusp
3. Conjugate or isolated point of C

More details on double points

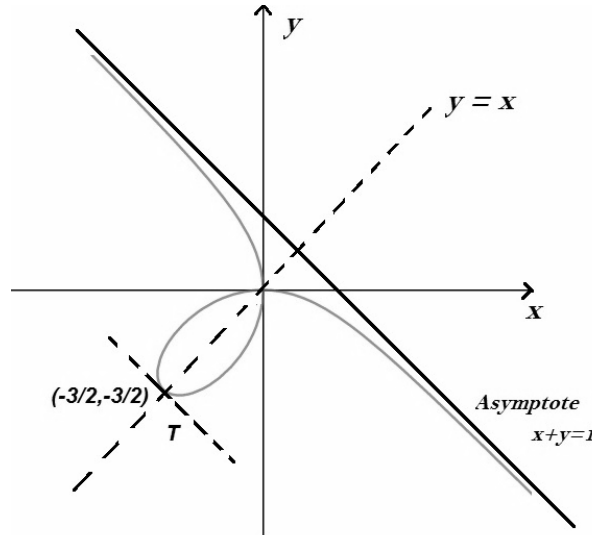
Node: If $P(x, y)$ is a node of the curve $C: f(x, y) = 0$, then there exist two real, distinct tangent. The shape will be of the type



The curve crosses itself.

Example 3 Let C be given by : $f(x, y) = 0$, where $f(x, y) = x^3 + y^3 + 3xy$ or $x^3 + y^3 = -3xy$.

Let us trace the curve:



Observe that

1. The curve is symmetric about the line $y = x$. The point of intersection of line $y = x$ with C are given by substituting $y = x$ in the equation of C . That is:

$$2x^3 + 3x^2 = 0 \implies x^2(2x + 3) = 0 \implies x = 0, -\frac{3}{2}$$

Points of intersection are $(0, 0), (-\frac{3}{2}, -\frac{3}{2})$.

You may verify that:

2. The curve passes through origin but does not meet any of the coordinate axis at any other point.
3. There is no curve in the first quadrant.

4. Slope of the tangent at $(-\frac{3}{2}, -\frac{3}{2})$ is

$$\left(\frac{dy}{dx}\right)_{(-\frac{3}{2}, -\frac{3}{2})} = -\frac{\left(\frac{\partial f}{\partial x}\right)_{(-\frac{3}{2}, -\frac{3}{2})}}{\left(\frac{\partial f}{\partial y}\right)_{(-\frac{3}{2}, -\frac{3}{2})}} = -\left(\frac{3x^2 + 3y}{3y^2 + 3x}\right)_{(-\frac{3}{2}, -\frac{3}{2})} = -1$$

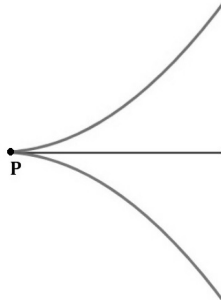
Hence the equation of the tangent T at $(-\frac{3}{2}, -\frac{3}{2})$:

$$y + \frac{3}{2} = -\left(x + \frac{3}{2}\right) \implies T : x + y + 3 = 0$$

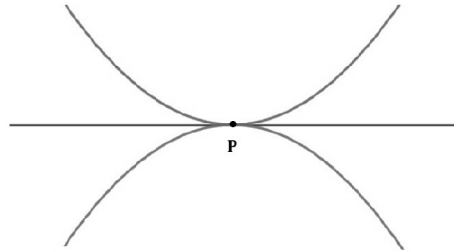
5. The line $A : x + y = 1$ is the only asymptote to the curve.
6. The curve lies between the tangent $T : x + y + 3 = 0$ and the asymptote $A : x + y = 1$.
7. All points of C except origin are ordinary.
8. Origin is a point, where two coordinate axis namely $y = 0$, and $x = 0$ are two distinct and real tangents.
9. Origin is a node for the curve $C : f(x, y) = 0$, where $f(x, y) = x^3 + y^3 + 3xy$

Cusp: If the two tangent at a double point P on a curve C are real but coincident, P is called its cusp.

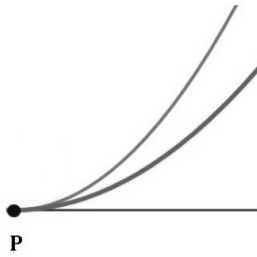
Depending of the two branches of the curve and the tangent at P , the curve at P may look like any of the following:



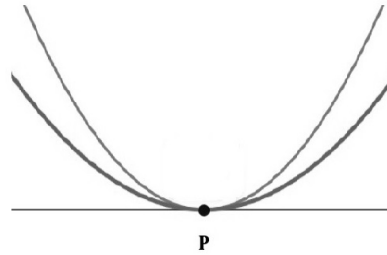
(a)



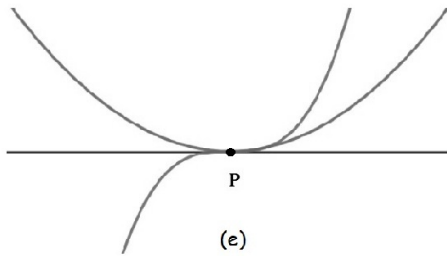
(b)



(c)



(d)



(e)

Accordingly, P is called.

Single Cusp: Figure (a) and (c), when the branches do not extend on both sides of the normal at P .

Double Cusp: Figures (d) and (e), when the branches extend on the both sides of the normal at P .

First Species: Figures (a) and (b), when two branches are on the opposite sides of the tangent at P .

Second Species: Figures (c) and (d), when the two branches of the curve are on the same side of the tangent at P .

Point of Oscul-inflexion: A double cusp if it is a mix of the two species e.g. Figure (e), then it is called a point of oscul-inflexion.

Example 5: Let C be given by $f(x, y) = 0$, where $f(x, y) = x^3 - y^3 + x^2$. Observe that

$$\frac{\partial f}{\partial x} = 3x^2 + 2x = 0 \implies x = 0, -\frac{2}{3}$$

and

$$\frac{\partial f}{\partial y} = -3y^2 = 0 \implies y = 0.$$

Hence possible singular points are $(0, 0)$ and $(-\frac{2}{3}, 0)$. Since $(0, 0)$ lies on C , and $(-\frac{2}{3}, 0)$ does not lie on C . We get that $(0, 0)$ is the only singular point on C .

Further, to know more about the nature of the point $(0, 0)$, note that the tangent(s) at origin are given by $x^2 = 0$. The two tangents are real but coincident. In fact, it is a cusp of first species. (Prove it).

Example 6: Let C be given by $f(x, y) = x^2 + y^2 - x^3 = 0$

For $P(x, y) \in C$ to be singular point of C , $(\frac{\partial f}{\partial x})_P = 0$ and $(\frac{\partial f}{\partial y})_P = 0$
i.e.

$$\begin{aligned} 2x - 3x^2 = 0 &\implies x(2 - 3x) = 0 \implies x = 0, \frac{2}{3}, \\ 2y = 0 &\implies y = 0 \end{aligned}$$

Possible singular points are : $(0, 0)$ and $(\frac{2}{3}, 0)$. The point $(0, 0)$ lies on C , but $(\frac{2}{3}, 0)$ does not lie on C . Hence $(0, 0)$ is the only singular point of C . Note that at $(0, 0)$, the tangents are given by $x^2 + y^2 = 0$, which means the tangents are imaginary. Thus $(0, 0)$ is a double point, in fact only one. It is a conjugate point.

Necessary Condition For Double Points A necessary condition for the existence of a singular point P on a curve C is that at P , the values of $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$. And, therefore a necessary condition for a double point or a multiple point is same i.e. $(\frac{\partial f}{\partial x})_P = 0$, $(\frac{\partial f}{\partial y})_P = 0$.

Further, a double point P on a curve $C : f(x, y) = 0$ can be classified as:

1. **Node:** if $\left(\frac{\partial^2 f}{\partial x \partial y}\right)_P^2 - \left(\frac{\partial^2 f}{\partial x^2}\right)_P \left(\frac{\partial^2 f}{\partial y^2}\right)_P > 0$
2. **Cusp:** if $\left(\frac{\partial^2 f}{\partial x \partial y}\right)_P^2 - \left(\frac{\partial^2 f}{\partial x^2}\right)_P \left(\frac{\partial^2 f}{\partial y^2}\right)_P = 0$
3. **Conjugate Point:** if $\left(\frac{\partial^2 f}{\partial x \partial y}\right)_P^2 - \left(\frac{\partial^2 f}{\partial x^2}\right)_P \left(\frac{\partial^2 f}{\partial y^2}\right)_P < 0$

4. If $\left(\frac{\partial^2 f}{\partial x^2}\right)_P = 0$, $\left(\frac{\partial^2 f}{\partial y^2}\right)_P = 0$ and $\left(\frac{\partial^2 f}{\partial x \partial y}\right)_P = 0$

then the singular point $P \in C$ is called triple point.

Exercise: Prove above statements.

Example 7: Let the curve C be given by : $f(x, y) = 0$, where

$$f(x, y) = x^2(x - y) + y^2 = x^3 - x^2y + y^2$$

For singular points :

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0$$

$$\frac{\partial f}{\partial x} = 3x^2 - 2xy = 0 \implies x(3x - 2y) = 0 \implies x = 0, y = \frac{3}{2}x$$

$$\frac{\partial f}{\partial y} = -x^2 + 2y = 0 \implies -x^2 + (3x) = 0 \implies x(-x + 3) = 0 \implies x = 0, 3$$

Possible singular points : $(0, 0)$, $(3, \frac{9}{2})$.

Out of these $(3, \frac{9}{2})$ does not lie on C as

$$3^2(3 - \frac{9}{2}) + \left(\frac{9}{2}\right)^2 = 3^2\left(-\frac{3}{2}\right) + \frac{9^2}{4} = \frac{81}{4} - \frac{27}{2} \neq 0$$

However, $(0, 0)$ lies on C . So $(0, 0)$ is the only singular point. We discuss further, the nature of the singular point. We shall use the second derivative test.

$$\frac{\partial f}{\partial x} = 3x^2 - 2xy \implies \frac{\partial^2 f}{\partial x^2} = 6x - 2y$$

$$\frac{\partial f}{\partial y} = -x^2 + 2y \implies \frac{\partial^2 f}{\partial y^2} = 2$$

$$\frac{\partial^2 f}{\partial x \partial y} = -2x$$

$$\left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 - \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) = (-2x^2) - (6x - 2y)(2) = 4x^2 - 12x + 4y$$

At $(0, 0)$, this value is 0. Hence $(0, 0)$ is a cusp.

Discuss the nature of the cusp at $(0, 0)$ as follows:

1. Prove that $y = 0$ i.e. x -axis is a tangent at origin.
2. Note that $f(0, y) = y^2$ implies that in the neighbourhood of origin, the curve is symmetric about x -axis i.e. the common tangent. Hence, origin is a cusp of first species.

13.3 Singular and Non singular Elliptic Curve

Let $E = E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} | y^2 = x^3 + Ax + B, A, B \in \mathbb{R}\} \cup \{\infty\}$ be a real Elliptic Curve. The curve is singular if $\left(\frac{\partial f}{\partial x}\right)_P = 0$ and $\left(\frac{\partial f}{\partial y}\right)_P = 0$, where $f = f(x, y) = y^2 - x^3 - Ax - B$, at some point $P(x, y) \in E$, then such a point is called a singular point.

For example: If $E = E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} | f(x, y) = y^2 - x^3 + x^2 = 0\} \cup \{\infty\}$, then $(0, 0) \in E$ is such that

$$\left(\frac{\partial f}{\partial x}\right)_{(0,0)} = (-3x^2 + 2x)_{(0,0)} = 0 \text{ and } \left(\frac{\partial f}{\partial y}\right)_{(0,0)} = (2y)_{(0,0)} = 0$$

Hence $(0, 0)$ is a singular point of E , and E is a singular curve.

The point $P(x, y) \in E_{\mathbb{R}}$ is a non singular, if at least one of $\left(\frac{\partial f}{\partial x}\right)$ and $\left(\frac{\partial f}{\partial y}\right)$ is non zero at P . The curve E is non singular curve if every point of E is non singular point.

Theorem: Let $E = E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} | y^2 = x^3 + Ax + B, A, B \in \mathbb{R}\} \cup \{\infty\}$. The following statement are equivalent.

1. E is a non singular curve.
2. The polynomial $f(x) = x^3 + Ax + B = 0$ has no repeated root.
3. $4A^3 + 27B^2 \neq 0$

Proof: (1) and (2) are equivalent is obvious.

For (2) and (3) are equivalent, we prove that $f(x) = x^3 + Ax + B = 0$ has repeated root if and only if $4A^3 + 27B^2 = 0$

Suppose, $f(x) = 0$ has a repeated root α . Two cases arises.

(a) α is repeated thrice

(b) α is repeated twice

In case (a): $f(x) = x^3 + Ax + B = (x - \alpha)^3 = x^3 - 3\alpha x^2 + 3\alpha x - \alpha^3$

On comparing the coefficient we get

$$-3\alpha = 0 \implies \alpha = 0$$

$$A = 3\alpha = 0; \quad B = -\alpha^3 = 0 \implies A = 0, B = 0$$

$$\implies 4A^3 + 27B^2 = 0$$

In case (b): $f(x) = x^3 + Ax + B = (x - \alpha)^2(x - \beta)$, for some $\beta \neq \alpha$.
Again, this implies that

$$x^3 + Ax + B = x^3 - (\beta + 2\alpha)x^2 + (\alpha^2 + 2\alpha\beta)x - \alpha^2\beta$$

On comparing the coefficient we get

$$\beta + 2\alpha = 0; \quad \alpha^2 + 2\alpha\beta = A; \quad -\alpha^2\beta = B$$

$$\implies \beta = -2\alpha; \quad \alpha^2 + 2\alpha(-2\alpha) = A; \quad -\alpha^2(-2\alpha) = B$$

$$\implies \beta = -2\alpha; \quad -3\alpha^2 = A; \quad 2\alpha^3 = B$$

$$\alpha = \left(-\frac{A}{3}\right)^{\frac{1}{2}}$$

and

$$\alpha = \left(\frac{B}{2}\right)^{\frac{1}{3}} \implies \left(-\frac{A}{3}\right)^{\frac{1}{2}} = \left(\frac{B}{2}\right)^{\frac{1}{3}}$$

Raising 6-th power on both side, we have

$$\begin{aligned} \left(-\frac{A}{3}\right)^3 &= \left(\frac{B}{2}\right)^2 \\ \implies -\frac{A^3}{27} &= \frac{B^2}{4} \\ \implies 4A^3 + 27B^2 &= 0 \end{aligned}$$

Conversely, suppose $4A^3 + 27B^2 = 0$

This gives that

$$\begin{aligned} -\frac{A^3}{27} = \frac{B^2}{4} &\implies \left(-\frac{A}{3}\right)^3 = \left(-\frac{B}{2}\right)^2 \\ &\implies \left(-\frac{A}{3}\right)^{\frac{1}{2}} = \left(-\frac{B}{2}\right)^{\frac{1}{3}} = \alpha \text{ (say)} \\ &\implies \alpha^2 = -\frac{A}{3} \text{ and } \alpha^3 = \frac{B}{2} \end{aligned}$$

Now

$$(x - \alpha)^2(x + 2\alpha) = x^3 - 3\alpha^2x + 2\alpha^3 = x^3 + Ax + B$$

This proves the equivalence of (2) and (3).

13.4 More On Roots Of A Polynomial

Recall, an element $b \in F$ said to be a root or a zero of a polynomial $f(x) \in F[x]$, if $f(b) = 0$. Obviously, $b \in F$ is a root of $f(x)$ if and only if $(x - b)$ divides $f(x)$. Further, $b \in F$ is said to be a root of $f(x)$ of multiplicity k , if $(x - b)^k$ divides $f(x)$, but $(x - b)^{k+1}$ does not divide $f(x)$.

In case $k = 1$, the root is called a simple root, if $k \geq 2$, then it is called a multiple root or repeated root. It is easy to see that $b \in F$ is a multiple root of $f(x)$ if and only if $f(x)$ and its derivative $f'(x)$ has b as a common root.

A polynomial $f(x) \in K[x]$ is said to split in F , (an extension F of K) if

$$f(x) = a(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$$

where a is the coefficient of the highest power of x in $f(x)$. The field F is called the splitting field of $f(x)$, if F is the smallest such extension of K . In that case $F = K(\alpha_1, \alpha_2, \dots, \alpha_n)$.

Splitting fields can be useful in knowing if a polynomial has a multiple root. For this we need:

13.4.1 Discriminant $D(f)$ of a polynomial $f(x) \in K[x]$

Let $f(x) \in K[x]$ be a polynomial of degree n splits into linear factors in $F[x]$, where F is the splitting field of K , as :

$$f(x) = a(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$$

then the discriminant $D(f)$ of $f(x)$ is defined as

$$D(f) = a^{2n-2} \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2; \quad \alpha_i \in F \quad \forall i = 1, 2, \dots, n$$

Obviously $f(x)$ has a multiple root if and only if its discriminant $D(f) = 0$. Although α_i 's $\in F$, but $D(f) \in K$ (Prove)

For example, if $f(x) = ax^2 + bx + c = a(x - \alpha_1)(x - \alpha_2)$, then

$$\begin{aligned} D(f) &= a^2(\alpha_1 - \alpha_2)^2 \\ &= a^2\{(\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2\} \\ &= a^2 \left\{ \left(-\frac{b}{a}\right)^2 - 4 \cdot \frac{c}{a} \right\} \\ &= b^2 - 4ac \end{aligned}$$

Thus $D(ax^2 + bx + c) = b^2 - 4ac$

And the two roots are equal if and only if $b^2 = 4ac$.

Exercise: Prove that:

$$D(ax^3 + bx^2 + cx + d) = b^2c^2 - 4b^3d - 4ac^3 - 27a^2d^2 + 18abcd$$

Corollary 1 $D(x^3 + cx + d) = -(4c^3 + 27d^2)$, hence $D(x^3 + ax + b) = -(4a^3 + 27b^2)$.

Corollary 2 $x^3 + ax + b$ has a repeated root if and only if $4a^3 + 27b^2 = 0$.

For cryptographic applications, we mainly consider non-singular Elliptic Curves. Henceforth, take K to be a field of $\text{Char}(K) = p$

1. When $p \neq 2, 3$

$$E_K = \{(x, y) \in K \times K \mid y^2 = x^3 + ax + b; \quad a, b \in K\} \cup \{\infty\}$$

where $x^3 + ax + b = 0$ has no repeated root

$$E_K = \{(x, y) \in K \times K \mid y^2 = x^3 + ax + b; \quad 4a^3 + 27b^2 \neq 0\} \cup \{\infty\}$$

2. When $p = 2$

$$(a) \quad E_K = \{(x, y) \in K \times K \mid y^2 + cy = x^3 + ax + b; \ a, b, c \in K\} \cup \{\infty\}$$

such that $x^3 + ax + b = 0$ has no repeated root.

$$(b) \quad E_K = \{(x, y) \in K \times K \mid y^2 + xy = x^3 + ax^2 + b; \ a, b \in K\} \cup \{\infty\}$$

such that $x^3 + ax^2 + b = 0$ has no repeated root.

3. When $p = 3$

$$(a) \quad E_K = \{(x, y) \in K \times K \mid y^2 = x^3 + ax + b; \ a, b \in K\} \cup \{\infty\}$$

such that $x^3 + ax + b = 0$ has no repeated root.

$$(b) \quad E_K = \{(x, y) \in K \times K \mid y^2 = x^3 + ax^2 + b; \ a, b \in K\} \cup \{\infty\}$$

such that $x^3 + ax^2 + b = 0$ has no repeated root.

Or combining the two cases:

$$E_K = \{(x, y) \in K \times K \mid y^2 = x^3 + ax^2 + bx + c; \ a, b, c \in K\} \cup \{\infty\}$$

such that $x^3 + ax^2 + bx + c = 0$ has no repeated root.

13.5 Graph of an Elliptic Curve over $\mathbb{R}$

Let $E = E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + ax + b; \ a, b \in \mathbb{R} \text{ be such that } x^3 + ax + b = 0 \text{ has no repeated roots}\} \cup \{\infty\}$.

We had seen this is equivalent to $4a^3 + 27b^2 \neq 0$ or $D(x^3 + ax + b) = -(4a^3 + 27b^2)$ the discriminant of $x^3 + ax + b$ is non zero. In the context of elliptic curves, the discriminant is denoted by

$$\Delta(E) = -16(4a^3 + 27b^2).$$

Thus $4a^3 + 27b^2 \neq 0$ iff $\Delta(E) \neq 0$.

Further, if $x^3 + ax + b = 0$ has no repeated root, then it has either one real

root, or all the three distinct real roots. This happens (Prove that) depending on $\Delta(E)$ is negative or positive.

Consider, the following Examples:

Example 1: $E_1 = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + 2x - 3\} \cup \{\infty\}$

Example 2: $E_2 = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 - 7x + 6\} \cup \{\infty\}$.

Observe that $\Delta(E_1) = -16(4 \cdot 2^3 + 27(-3)^2) < 0$. So, $x^3 + 2x - 3$ has exactly one real root. This is evident as

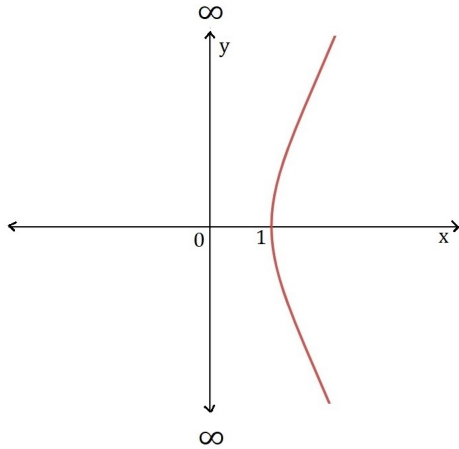
$$x^3 + 2x - 3 = (x - 1)(x^2 + x + 3)$$

has only one real root namely $x = 1$, as $x^2 + x + 3$ has no real root. Similarly,

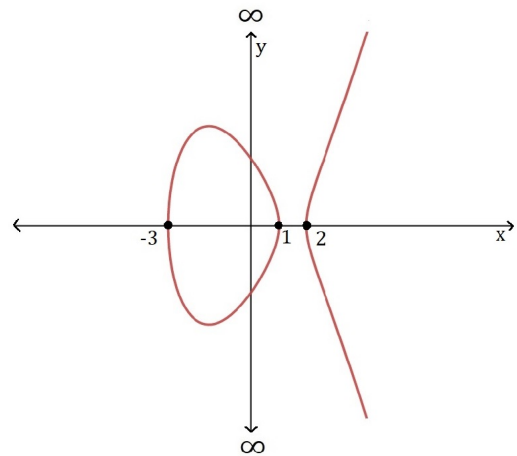
$$\begin{aligned} \Delta(E_2) &= -16(4 \cdot (-7)^3 + 27 \cdot 6^2) \\ &= -64\{(-7)^3 + 27 \cdot 9\} \\ &= -64(-343 + 243) = 6400 > 0 \end{aligned}$$

Hence $x^3 - 7x + 6 = 0$ has all the three roots to be real. This too is clear that the roots are $x = 1, 2, -3$ as $x^3 - 7x + 6 = (x - 1)(x - 2)(x + 3)$.

We can now draw these graphs.



$$E_1 = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + 2x - 3\} \cup \{\infty\}$$

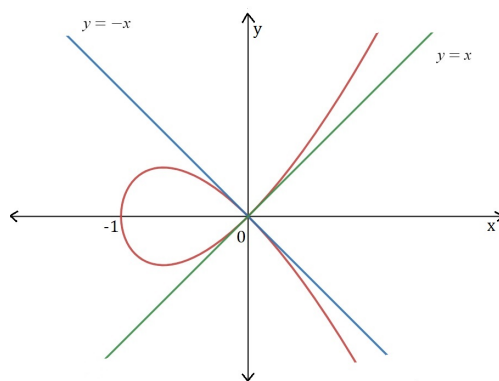


$$E_2 = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 - 7x + 6\} \cup \{\infty\}$$

13.5.1 Examples of singular real Elliptic Curve:

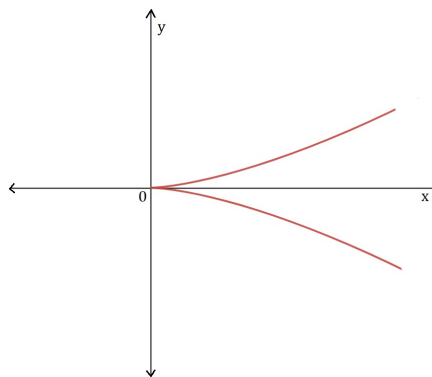
Example 3: $E_1 = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + x^2\} \cup \{\infty\}$.

Note that the $x^3 + x^2 = x^2(x + 1)$. The only singular point of E is $(0, 0)$. This is a node as $y = \pm x$ are two real tangents to the curve at this point. The graph is like:



$$E_1 = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + x^2\} \cup \{\infty\}$$

Example 4: $E = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3\} \cup \{\infty\}$. Note that the curve has only one singular point, namely $(0, 0)$, as $x^3 = 0$, has root $x = 0$, repeated thrice. The graph of the curve:



$$E = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3\} \cup \{\infty\}$$

Analyzing the nature of the double point $(0, 0)$ of E , we find that $(0, 0)$ is a cusp of the first species as $y = 0$ is the common tangent, and the branches of the curve lie on both sides of the tangent.

13.6 Examples Of An Elliptic Curve Over A Finite Field

Example 5: Let the curve E be given by

$$E = \{(x, y) \in \mathbb{Z}_p \times \mathbb{Z}_p : y^2 = x^3 + ax + b; a, b \in \mathbb{Z}_p\} \cup \{\infty\}$$

For $p = 11$, $a = 3$, $b = 1$, this is

$$E = \{(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11} : y^2 = x^3 + 3x + 1; a, b \in \mathbb{Z}_p\} \cup \{\infty\}$$

The discriminant

$$\begin{aligned} \Delta(E) &= -16(4a^3 + 27b^2) = -16(4 \cdot 3^3 + 27 \cdot 1^2) \\ &= -16 \cdot 5 \cdot 27 = -2160 \\ &= 7 - 2167 = 7 - (197 \cdot 11) \equiv 7 \not\equiv 0 \pmod{11} \end{aligned}$$

Hence the curve is non-singular.

As x, y both come from $\mathbb{Z}_p = \mathbb{Z}_{11}$, the curve has finitely many points. The number of points on E , except the point at infinity, is at most $11^2 = 121$. That is total number of points is at most 122. This number, however can be brought down to 20, using Hasse's Theorem. The points in E may be further less.

We calculate the points as follows:

x	0	1	2	3	4	5	6	7	8	9	10
y^2	1	5	15	37	77	141	235	365	537	757	100031
$y^2 \pmod{11}$	1	5	4	4	0	9	4	2	9	9	10

y	0	1	2	3	4	5	6	7	8	9	10
$y^2 \pmod{11}$	0	1	4	9	5	3	3	5	9	4	1

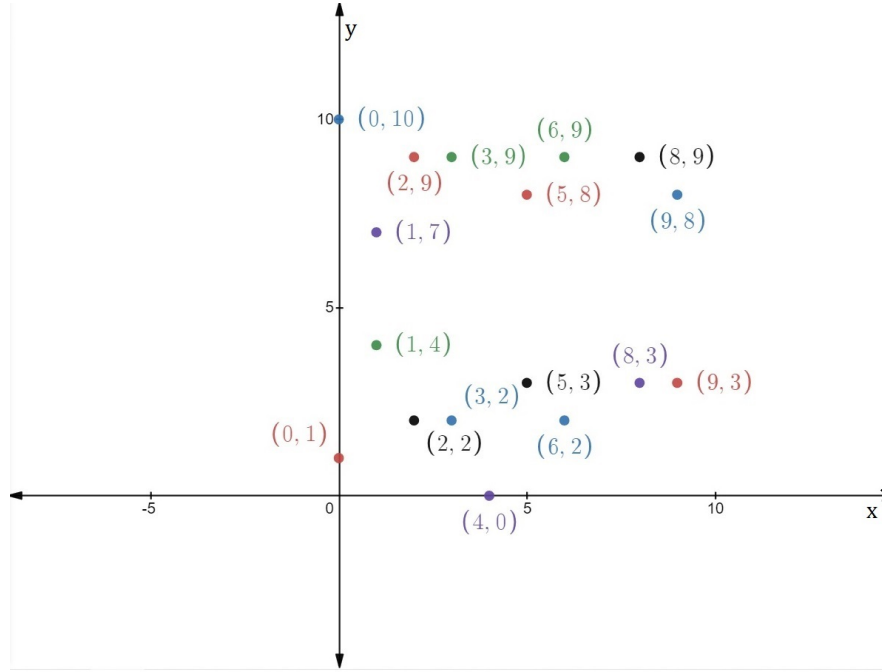
$y^2 \pmod{11}$	0	1	3	4	5	9
$y \pmod{11}$	0	1,10	5,6	2,9	4,7	3,8

this gives the correspondence :

x	0	0	1	1	2	2	3	3	4	5	5	6	6	8	8	9	9
y	1	10	4	7	2	9	2	9	0	3	8	2	3	3	9	3	8

Hence the curve E is:

$$E := \{(0, 1), (0, 10), (1, 4), (1, 7), (2, 2), (2, 9), (3, 2), (3, 9), (4, 0), (5, 3), (5, 8), (6, 2), (6, 9), (8, 3), (8, 9), (9, 3), (9, 8)\} \cup \{\infty\}$$



$$E = \{(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11} : y^2 = x^3 + x + 1; a, b \in \mathbb{Z}_{11}\} \cup \{\infty\}$$

Example 6: Let the curve be defined by

$$E = E(\mathbb{Z}_{11}) = \{(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11} : y^2 = x^3 + 4x + 5\} \cup \{\infty\}$$

Its discriminant

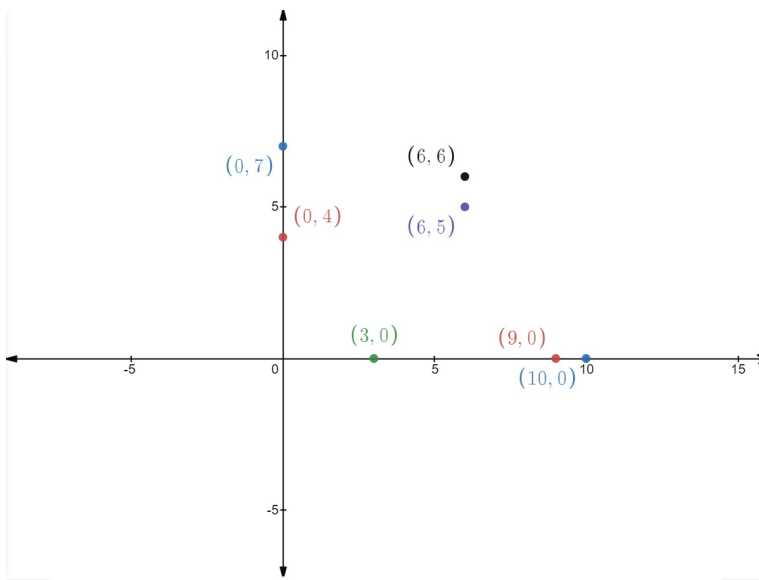
$$\begin{aligned} \Delta(E) &= -16(4a^3 + 27b^2) \\ &\equiv 6(4a^3 + 5b^2) \pmod{11} \\ &\equiv 6(4 \cdot 4^3 + 5 \cdot 5^2) \pmod{11} \\ &\equiv 6(4 \cdot 64 + 5 \cdot 25) \pmod{11} \\ &\equiv 6(4(-2) + 5 \cdot 3) \pmod{11} \\ &\equiv 6(15 - 8) \equiv 6 \cdot 7 \pmod{11} \\ &\equiv 42 \pmod{11} \\ &\equiv -4 \equiv 7 \pmod{11} \not\equiv 0 \pmod{11} \end{aligned}$$

Hence the curve is non-singular. The number of points on E , using Hasse's Theorem, can not exceed 20. In fact, they are much less. Since $\mathbb{Z}_{11}$ is a small field, you can find all points of E by taking every $(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11}$ and checking if $P(x, y) \in E$, that is by observing $y^2 = x^3 + 4x + 5$ or not. And list all:

Points of E : $(0, 4)$, $(0, 7)$, $(3, 0)$, $(6, 5)$, $(6, 6)$, $(9, 0)$, $(10, 0)$, ∞

Hence $E = \{(0, 4), (0, 7), (3, 0), (6, 5), (6, 6), (9, 0), (10, 0)\} \cup \{\infty\}$

Note that E has just eight points, including the point at infinity.



$$E = E(\mathbb{Z}_{11}) = \{(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11} : y^2 = x^3 + 4x + 5\} \cup \{\infty\}$$

13.7 Examples of Curves Over Field of Characteristic 2

Example 7: Let $E = \{(x, y) \in \mathbb{Z}_2 \times \mathbb{Z}_2 | y^2 + y = x^3 + x\} \cup \{\infty\}$.

Recall that the Elliptic Curve over finite fields K of characteristic 2 are of the form

1.

$$E_1 = \{(x, y) \in K \times K | y^2 + cy = x^3 + ax + b; \ a, b, c \in K\} \cup \{\infty\}$$

such that $x^3 + ax + b = 0$ has no repeated root.

2.

$$E_2 = \{(x, y) \in K \times K | y^2 + xy = x^3 + ax^2 + b; \ a, b \in K\} \cup \{\infty\}$$

such that $x^3 + ax^2 + b = 0$ has no repeated root.

The above example is an example of the form of E_1 . We calculate point of E as follows:

x	$y^2 + y = x^3 + x$	y
0	0	0,1
1	0	0,1

Points: $(x, y) = (0, 0), (0, 1), (1, 0), (1, 1)$

Hence $E = \{(0, 0), (0, 1), (1, 0), (1, 1)\} \cup \{\infty\}$.

Number of points $\#(E) = 5$

Example 8: Let $E = \{(x, y) \in \mathbb{Z}_2 \times \mathbb{Z}_2 | y^2 + xy = x^3 + 1\} \cup \{\infty\}$.

This is an example of the form of E_2 . We calculate points as follows:

x	$y^2 + xy = x^3 + 1$	y
0	$y^2 = 1$	1
1	$y^2 + y = 0$	0,1

Hence $E = \{(0, 1), (1, 0), (1, 1)\} \cup \{\infty\}$

The number of points $\#(E) = 4$

Example 9: Let $E = \{(x, y) \in F_4 \times F_4 | y^2 + y = x^3 + x\} \cup \{\infty\}$, where F_4 is a finite field of order 4.

Observe that $f(x) = x^2 + x + 1 \in \mathbb{Z}_2[x]$ is an irreducible quadratic polynomial over $\mathbb{Z}_2$. Let ω be a root of $f(x) = 0$. Then $\omega \neq 1, \omega^2 + \omega + 1 = 0$ and $\omega^3 = 1$.

We can take $F_4 = \{0, 1, \omega, \omega^2 | \omega^2 + \omega + 1 = 0\}$

We calculate points of E as follows:

x	$y^2 + y = x^3 + x$	y
0	0	0,1
1	0	0,1
ω	$1 + \omega = \omega^2$	-
$\omega^2 = 1 + \omega$	$1 + \omega^2 = \omega = \omega^4$	-

Hence the curve $E = \{(0, 0), (0, 1), (1, 0), (1, 1)\} \cup \{\infty\}$.

Number of points of E is $\#(E) = 5$.

Example 10: $E = \{(x, y) \in F_4 \times F_4 \mid y^2 + xy = x^3 + 1\} \cup \{\infty\}$, where F_4 is a finite field of order 4.

Recall

$$F_4 \cong \frac{\mathbb{Z}_2[x]}{\langle f(x) = x^2 + x + 1 \rangle}$$

If ω is the primitive root of $f(x)$, then $\omega \neq 1, \omega^2 + \omega + 1 = 0$.

We can take $F_4 = \{0, 1, \omega, \omega^2 \mid \omega^2 + \omega + 1 = 0\}$

We calculate points of E as follows:

x	$y^2 + xy = x^3 + 1$	y
0	$y^2 = 1$	1
1	$y^2 + y = 0$	0, 1
ω	$y^2 + \omega y = 0$	0, ω
ω^2	$y^2 + \omega^2 y = 0$	0, ω^2

Hence $E = \{(0, 1), (1, 0), (1, 1), (\omega, 0), (\omega, \omega), (\omega^2, 0), (\omega^2, \omega^2)\} \cup \{\infty\}$

Number of points of E is $\#(E) = 8$.

13.8 Example of Elliptic Curves Over Fields of Characteristic 3

Example 11: Let $E = \{(x, y) \in \mathbb{Z}_3 \times \mathbb{Z}_3 \mid y^2 = x^3 - x + 2\} \cup \{\infty\}$

Recall the Elliptic Curves defined over a finite field of characteristic 3 are of the form:

$$E_K = \{(x, y) \in K \times K \mid y^2 = x^3 + ax^2 + bx + c; \ a, b, c \in K\} \cup \{\infty\}$$

such that $x^3 + ax^2 + bx + c = 0$ has no repeated root.

These can be further divided into the following two types:

1.

$$E_1 = \{(x, y) \in K \times K \mid y^2 = x^3 + ax + b; \ a, b \in K\} \cup \{\infty\}$$

such that $x^3 + ax + b = 0$ has no repeated root.

2.

$$E_2 = \{(x, y) \in K \times K | y^2 = x^3 + ax^2 + b; \ a, b \in K\} \cup \{\infty\}$$

such that $x^3 + ax^2 + b = 0$ has no repeated root.

The given example curve is of the form $E_1 = E_1(\mathbb{Z}_3) = \{(x, y) \in \mathbb{Z}_3 \times \mathbb{Z}_3 | y^2 = x^3 - x + 2\} \cup \{\infty\}$.

We calculate points of E as follows:

x	$y^2 = x^3 - x + 2$	y
0	2	-
1	2	-
2	2	-

Recall the Legendre Symbol $\left(\frac{2}{3}\right) = -1$

That is 2 is a quadratic non residue mod 3 i.e. $y^2 \equiv 2 \pmod{3}$ has no solution.

Hence $E = \{\infty\}$ and $\#(E) = 1$

Further, the curve is non singular as $\frac{\partial f}{\partial x} = 3x^2 - 1 = 2 \in \mathbb{Z}_3[x]$ is non zero for all $P(x, y) \in E$

Example 12: Let

$$\begin{aligned} E &= \{(x, y) \in \mathbb{Z}_3 \times \mathbb{Z}_3 : y^2 = x^3 - x^2 + 2\} \cup \{\infty\} \\ &= \{(x, y) \in \mathbb{Z}_3 \times \mathbb{Z}_3 : f(x, y) = 0\} \cup \{\infty\} \end{aligned}$$

where $f(x, y) = y^2 - x^3 + x^2 - 2$

$$\frac{\partial f}{\partial x} = -3x^2 + 2x \equiv 2x \text{ over } \mathbb{Z}_3$$

Also

$$\frac{\partial f}{\partial x} = 0 \implies x = 0$$

$$\frac{\partial f}{\partial y} = 2y = 0 \implies y = 0$$

Hence, possible singular point is $(0, 0)$. But $(0, 0)$ does not lie on E . So the curve is non singular. We calculate points of E as follows:

13.8. EXAMPLE OF ELLIPTIC CURVES OVER FIELDS OF CHARACTERISTIC 3239

x	0	1	2
$y^2 = x^3 - x^2 + 2$	2	2	0
y	-	-	0

Points of E : $(2, 0), \infty$; Hence $E = \{(2, 0), \infty\}$, $\#(E) = 2 \cong C_2$

Example 13: Let $E = \{(x, y) \in \mathbb{F}_9 \times \mathbb{F}_9 : y^2 = x^3 - x + 2\} \cup \{\infty\}$,
and

$$\mathbb{F}_9 \cong \frac{\mathbb{Z}_3[x]}{\langle g(x) = x^2 + x + 2 \rangle}$$

If ξ is a root of $x^2 + x + 2 = 0$, then ξ is a primitive 8th root of unity over $\mathbb{Z}_3$, and we can take

$$\begin{aligned} \mathbb{F}_9 &= \{0, \xi^i : 1 \leq i \leq 8; \xi^2 + \xi + 2 = 0\} \\ &= \{0, \xi, \xi^2 = 1 + 2\xi, \xi^3 = 2 + 2\xi, \xi^4 = -1 = 2, \xi^5 = 2\xi, \\ &\quad \xi^6 = 1 + \xi, \xi^7 = 1 + \xi, \xi^8 = 1\} \end{aligned}$$

We compute points of E as follows:

x	0	ξ	ξ^2	ξ^3	$\xi^4 = -1 = 2$
$y^2 = x^3 - x + 2$	2	$\xi^3 - \xi + 2$	$\xi^6 - \xi^2 + 2$	$\xi^9 - \xi^3 + 2$	$\xi^{12} - \xi^4 + 2$
y^2	2	$1 + \xi$	2ξ	2ξ	2
y^2	ξ^4	ξ^7	ξ^5	ξ^5	ξ^4
y	$\xi^2, 2\xi^2$	-	-	-	$\xi^2, 2\xi^2$

x	ξ^5	ξ^6	ξ^7	$\xi^8 = 1$
$y^2 = x^3 - x + 2$	$\xi^{15} - \xi^5 + 2$	$\xi^{18} - \xi^6 + 2$	$\xi^{21} - \xi^7 + 2$	$\xi^{24} - \xi^8 + 2$
y^2	2ξ	$1 + \xi$	$1 + \xi$	2
y^2	ξ^5	ξ^7	ξ^7	ξ^4
y	-	-	-	$\xi^2, 2\xi^2$

Point of E are :

$$(0, \xi^2), (0, 2\xi^2), (\xi^4, \xi^2), (\xi^4, 2\xi^2), (\xi^8, \xi^2), (\xi^8, 2\xi^2)$$

or

$$(0, 1 + 2\xi), (0, 2 + \xi), (2, 1 + 2\xi), (2, 2 + \xi), (1, 1 + 2\xi), (1, 2 + \xi)$$

and the point at infinity ' ∞ '

Hence,

$$E := \{(0, 1 + 2\xi), (0, 2 + \xi), (2, 1 + 2\xi), (2, 2 + \xi), (1, 1 + 2\xi), (1, 2 + \xi), \infty\}.$$

and $\#(E) = 7$

Example 14: Let $E = \{(x, y) \in \mathbb{F}_9 : y^2 = x^3 - x^2 + 2\} \cup \{\infty\}$

This is an example of type (ii) case curve, when $\text{char}(\mathbb{F}) = 3$.

Let

$$\mathbb{F}_9 \cong \frac{\mathbb{Z}_3[x]}{\langle g(x) = x^2 + x + 2 \rangle}$$

If ξ is a root of $x^2 + x + 2 = 0$, then ξ is a primitive 8th root of unity over $\mathbb{Z}_3$, and we can take

$$\begin{aligned} \mathbb{F}_9 &= \{0, \xi^i : 1 \leq i \leq 8; \xi^2 + \xi + 2 = 0\} \\ &= \{0, \xi, \xi^2 = 1 + 2\xi, \xi^3 = 2 + 2\xi, \xi^4 = -1 = 2, \xi^5 = 2\xi, \\ &\quad \xi^6 = 1 + \xi, \xi^7 = 1 + \xi, \xi^8 = 1\} \end{aligned}$$

We compute points of E as follows:

x	0	ξ	$\xi^2 = 1 + 2\xi$
$y^2 = x^3 - x^2 + 2$	2	$\xi^3 - \xi^2 + 2$	$\xi^6 - \xi^4 + 2$
y^2	ξ^4	$(2 + 2\xi) - (1 + 2\xi) + 2$	$(2 + \xi) - 2 + 2$
y^2	ξ^4	0	$2 + \xi = \xi^6$
y	$\xi^2, 2\xi^2$	0	$\xi^3, 2\xi^3$
y	$1 + 2\xi, 2 + \xi$	0	$2 + 2\xi, 1 + \xi$

x	$\xi^3 = 2 + 2\xi$	$\xi^4 = 2 = -1$	$\xi^5 = 2\xi$
$y^2 = x^3 - x^2 + 2$	$\xi^9 - \xi^6 + 2$	$\xi^{12} - \xi^8 + 2$	$\xi^{15} - \xi^{10} + 2$
y^2	$\xi - (2 + \xi) + 2$	$\xi^4 - 1 - 1$	$\xi^7 - \xi^2 + 2$
y^2	0	0	$(1 + \xi) - (1 + 2\xi) + 2 = 2 - \xi$ $= 2 + 2\xi = \xi^3$
y	0	0	-
y	0	0	-

x	$\xi^6 = 2 + \xi$	$\xi^7 = 1 + \xi$	$\xi^8 = 1$
$y^2 = x^3 - x^2 + 2$	$\xi^{18} - \xi^{12} + 2$	$\xi^{21} - \xi^{14} + 2$	$\xi^{24} - \xi^{16} + 2$
y^2	$\xi^2 - \xi^4 + 2$	$\xi^5 - \xi^6 + 2$	$1 - 1 + 2$
y^2	$(1 + 2\xi) - 2 + 2$	$(2\xi) - (2 + \xi) + 2$	2
y^2	$(1 + 2\xi) = \xi^2$	ξ	ξ^4
y	$\xi, 2\xi$	-	$\xi^2, 2\xi^2$
y	$\xi, 2\xi$	-	$1 + 2\xi, 2 + \xi$

This gives us the curve

$$E = \{(0, 1 + 2\xi), (0, 2 + \xi), (\xi, 0), (1 + 2\xi, 2 + 2\xi), (1 + 2\xi, 1 + \xi), (2 + 2\xi, 0), (2, 0), (2 + \xi, \xi), (2 + \xi, 2\xi), (1, 1 + 2\xi), (1, 2 + \xi)\} \cup \{\infty\}$$

Number of points of E : $\#(E) = 12$

13.9 Algebraic Structure on Elliptic Curve E

Let $\mathbb{F}$ be a field of characteristic 0 or a prime p , $p \neq 2$ and $p \neq 3$. Define the set

$$E_{\mathbb{F}} = \{(x, y) \in \mathbb{F} \times \mathbb{F} : y^2 = x^3 + ax + b\} \cup \{\infty\} \quad \text{where } a, b \in \mathbb{F}.$$

Define an **Operation** on $E_{\mathbb{F}}$:

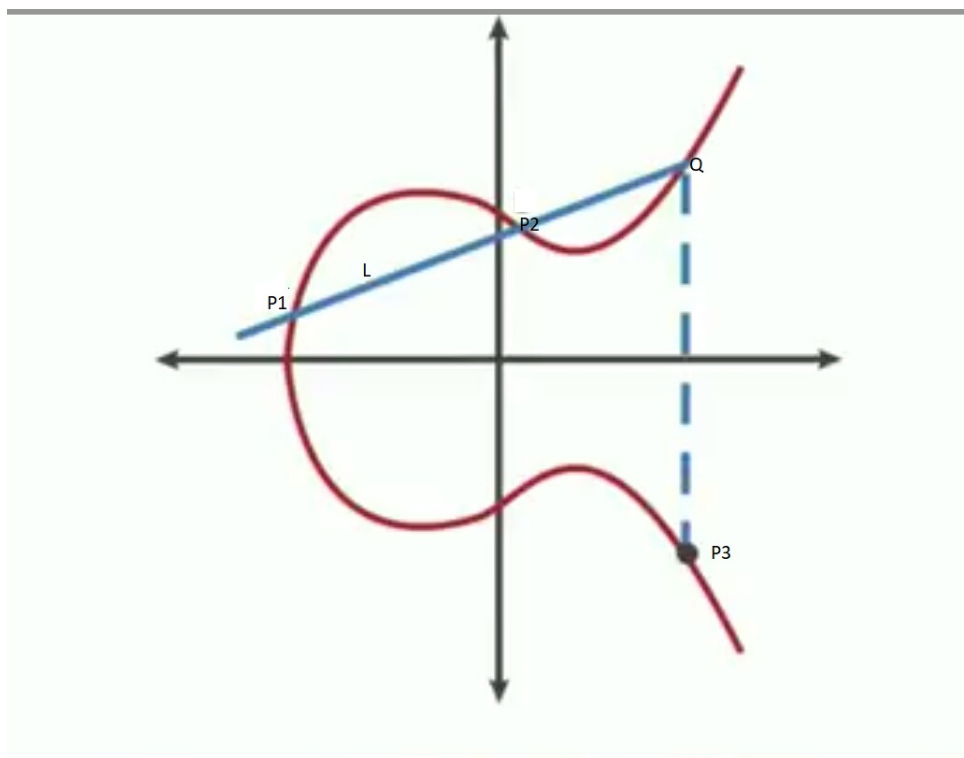
If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be two points in $E_{\mathbb{F}}$ ($P(x_1, y_1)$ is the point (x_1, y_1) represented by P_1), then define

$$P_3(x_3, y_3) = P_1(x_1, y_1) + P_2(x_2, y_2)$$

as follows :

Take L to be the line P_1P_2 joining the points P_1 and P_2 . Extend the same.

Let the line L meet the curve at Q . Define P_3 to be the reflection about x -axis of Q on the curve $E_{\mathbb{F}}$. For example, see the curve in the following figure :



13.9.1 Properties desired by the relation (addition of points of Elliptic Curves $E_{\mathbb{F}}$)

We desire the following Rules for addition:

1. *Closure property:* For every $P_1, P_2 \in E_{\mathbb{F}}$ their sum $P_1 + P_2 \in E_{\mathbb{F}}$
2. *Existence of (additive) identity property:* $P(x, y) + \infty = \infty + P(x, y) = P$ including: $\infty + \infty = \infty$ (We shall see the meaning of ∞ by example and figure given below)
3. *Existence of (additive) inverse:* If $P(x, y) = P \in E_{\mathbb{F}}$, then $P(x, -y) = -P(x, y) = -P \in E_{\mathbb{F}}$. That is $P(x, y) + P(x, -y) = P + (-P) = \infty$
4. *Commutative property:* $P_1(x_1, y_1) + P_2(x_2, y_2) = P_2(x_2, y_2) + P_1(x_1, y_1)$ for all $P_1, P_2 \in E_{\mathbb{F}}$

5. *Associative property:* $(P_1(x_1, y_1) + P_2(x_2, y_2)) + P_3(x_3, y_3) = P_1(x_1, y_1) + (P_2(x_2, y_2) + P_3(x_3, y_3)) \forall P_1, P_2, P_3 \in E_{\mathbb{F}}$

Note: Assuming above rules and ' ∞ ' as additive identity, the set $E_{\mathbb{F}}$ becomes an (additive) Abelian group.

Verify for $E_{\mathbb{F}}$: The properties listed above, using the geometry for addition of points. You will see that The Associative Property (5th property) is the most difficult to prove, although it appears to be trivial.

13.9.2 Meaning of point at infinity ∞ and geometry of point addition on an elliptic curve

Although, things can be best explained using algebraic geometry (projective space P_K^2 over a field K), we define the point at infinity ' ∞ ' to be a point and a member of $E_{\mathbb{F}}$ defined to be geometrically sitting at the end point of every ordinate. We defer the algebraic geometric treatment of the concept for the time being.

Recall : If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are two distinct points (none of them point at infinity ' ∞ ') on $E_{\mathbb{F}}$, then the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ can be represented as the points $(x_1, y_1), (x_2, y_2)$ by P_1 and P_2 respectively, in the figure. Recall, how their addition is defined.

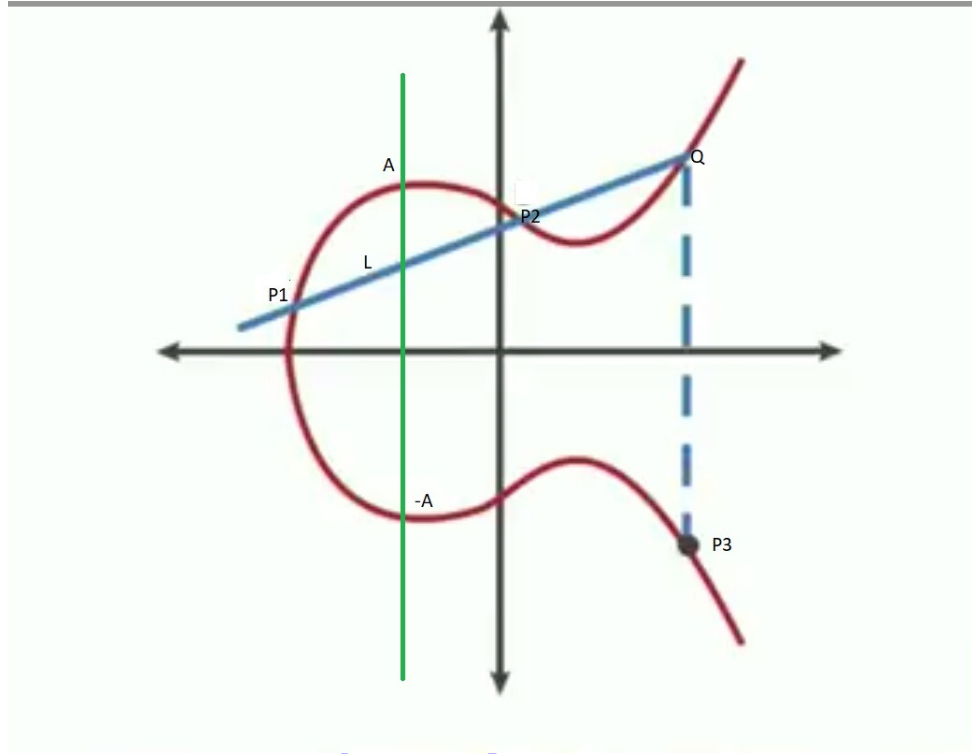
$$P_3(x_3, y_3) = P_1(x_1, y_1) + P_2(x_2, y_2)$$

This was done in case $x_1 \neq x_2$ by : Take the line $L = P_1P_2$ joining P_1 and P_2 and extend the same. If the line L meet the curve at Q , then define P_3 to be the reflection of Q on $E_{\mathbb{F}}$.

Similarly, if we join (i.e. add) A and $-A$, case of two distinct points (none of them point at infinity ' ∞ ') on $E_{\mathbb{F}}$ having same x -coordinate (as shown in the figure by green line), the line we get is a straight line parallel to y -axis, and thus it will not intersect the curve further (except at A and $-A$).

Since the line joining A and $-A$ goes to ∞ without intersecting the curve at any other point except A and $-A$, thus we assume

$$A + (-A) = \text{Image of } \infty = \infty.$$



If the points P_1, P_2 are identical (except the point at infinity, ' ∞ '), the line $L = P_1P_2$ becomes tangent to the curve at P_1 , Q is the point on the curve, where ' L ' meets the curve, and the sum $P_1 + P_2 = P_3$ is the reflection on the curve about x -axis of the point Q . Similarly, other rules too can be explained.

We need to develop algebraic formula for the coordinates of the point addition.

13.9.3 Algebraic Formula For Point Addition

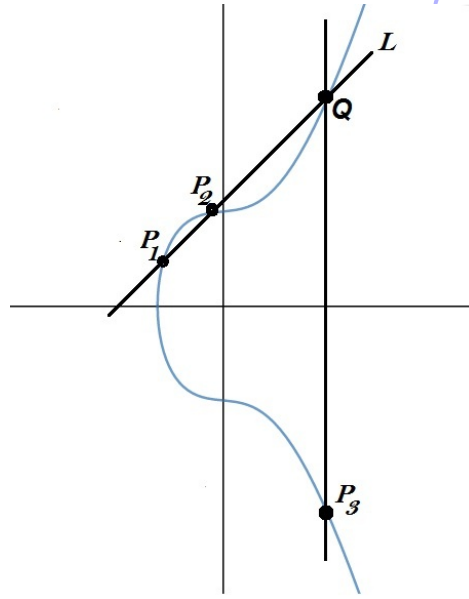
Recall from geometry, discussed as earlier, the following rules hold for adding points $P_1(x_1, y_1), P_2(x_2, y_2)$, and let $P_3(x_3, y_3) = P_1(x_1, y_1) + P_2(x_2, y_2)$ on an Elliptic Curve

$$E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} | y^2 = x^3 + ax + b; \quad a, b \in \mathbb{R}\} \cup \{\infty\}$$

1. If $P_2 = \infty$, the point at infinity, then $P_1 + \infty = P_1 \quad \forall P_1 \in E$

2. If $x_1 = x_2$ but $y_1 \neq y_2$, then $P_1 + P_2 = \infty$.
3. If $P_1 = P_2$, and $y_1 = 0$, then $P_1 + P_2 = 2P_1 = \infty$.
4. (a) $P_1 \neq P_2$ with $x_1 \neq x_2$, and
 (b) $P_1 = P_2$ with $y_1 \neq 0$

Case(a)



Let L be the line joining P_1 to P_2 . Suppose on extending it meets at Q , the curve E , and suppose P_3 is the reflection of Q , about x -axis on E . Then $P_1 + P_2 = P_3$

To determine coordinates (x_3, y_3) of P_3 in terms of the coordinates (x_1, y_1) of P_1 and (x_2, y_2) of P_2 . Since $P_3(x_3, y_3) \in E$ is the point reflected by Q , we get the coordinates of Q as $(x_3, -y_3)$.

Now equation of $L : y = mx + c$, where $m = \frac{y_2 - y_1}{x_2 - x_1}$, since it passes through P_1 and P_2 , we get :

$$y_2 = mx_2 + c, \quad y_1 = mx_1 + c.$$

$$\implies y_2 - y_1 = m(x_2 - x_1)$$

since $x_2 \neq x_1 \implies m = \frac{y_2 - y_1}{x_2 - x_1}$

Further,

$$y_1 = mx_1 + c \implies c = y_1 - mx_1$$

Hence equation of L: $y = mx + y_1 - mx_1$ or L: $y - y_1 = m(x - x_1)$, where $m = \frac{y_2 - y_1}{x_2 - x_1}$ or L: $y = m(x - x_1) + y_1$

Since L meets E at Q , we get:

$$\begin{aligned} y^2 &= x^3 + ax + b \\ \implies \{m(x - x_1) + y_1\}^2 &= x^3 + ax + b \\ \implies m^2(x - x_1)^2 + y_1^2 + 2m(x - x_1)y_1 &= x^3 + ax + b \\ \implies m^2(x^2 - 2xx_1 + x_1^2) + y_1^2 + 2m(xy_1 - x_1y_1) &= x^3 + ax + b \\ \implies x^3 - m^2x^2 + (a + 2m^2x_1 - 2my_1)x + (b - m^2x_1^2 - y_1^2 + 2mx_1y_1) &= 0 \end{aligned}$$

This is a cubic equation with roots x_1, x_2, x_3 as $P_1(x_1, y_1), P_2(x_2, y_2)$ and $Q(x_3, -y_3) \in E$

We observe that $x_1 + x_2 + x_3 = m^2$

$$\implies x_3 = m^2 - x_1 - x_2$$

Also L: $y = m(x - x_1) + y_1$, as $Q(x_3, -y_3)$ lies on this line L, we get:

$$\begin{aligned} -y_3 &= m(x_3 - x_1) + y_1 \\ \text{or } -y_3 &= -m(x_1 - x_3) + y_1 \\ \implies y_3 &= m(x_1 - x_3) - y_1 \end{aligned}$$

Hence $P_3(x_3, y_3)$ can be given by:

$$\begin{aligned} x_3 &= m^2 - x_1 - x_2 \\ y_3 &= m(x_1 - x_3) - y_1, \end{aligned}$$

where $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Case(b) $P_1 = P_2$, and $y_1 \neq 0$. The line L becomes tangent to the curve E at $P_1(x, y)$. The slope of L, can be obtained as $\left(\frac{dy}{dx}\right)_{P_1(x_1, y_1)}$ on differentiation of $y^2 = x^3 + ax + b$ i.e.

$$2y \frac{dy}{dx} = 3x^2 + a \implies m = \left(\frac{dy}{dx}\right)_{P_1} = \frac{3x_1^2 + a}{2y_1}$$

This gives $P_1 + P_2 = 2P_1 = P_3(x_3, y_3)$,
where

$$\begin{aligned}x_3 &= m^2 - 2x_1 \\y_3 &= m(x_1 - x_3) - y_1 \\m &= \frac{3x_1^2 + a}{2y_1}\end{aligned}$$

We now summarize the definition of point addition on an Elliptic Curve over $\mathbb{R}$ given by:

$$E_{\mathbb{F}} := \{(x, y) \in \mathbb{F} \times \mathbb{F} : y^2 = x^3 + ax + b; a, b \in \mathbb{F}\} \cup \{\infty\}$$

where $\Delta(E) \neq 0$,

as follows : for $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$, and $P_3 = (x_3, y_3)$

1. If $P_1, P_2 \neq \infty$ then $P_1 + P_2 = P_3$

- (a) When $x_1 \neq x_2$,

$$x_3 = m^2 - x_1 - x_2, \quad y_3 = m(x_1 - x_3) - y_1$$

$$\text{where } m = \frac{y_2 - y_1}{x_2 - x_1}$$

- (b) When $x_1 = x_2$ but $y_1 \neq y_2$, $P_3 = \infty$

- (c) When $P_1 = P_2$ but $y_1 \neq 0$ then

$$x_3 = m^2 - 2x_1, \quad y_3 = m(x_1 - x_3) - y_1$$

$$\text{where } m = \frac{3x_1^2 + a}{2y_1}$$

- (d) When $P_1 = P_2$ and $y_1 = 0$ then $P_1 + P_2 = \infty$

2. $P + \infty = P \quad \forall P \in E$

(So $\infty + \infty = \infty$ is included)

These rules are defined for a field $\mathbb{F}$ of characteristic $\text{char}(\mathbb{F}) > 3$. In case $\text{char}(\mathbb{F}) = 2$ or 3 , the rules are different.

It is easy to see that $P_1 + P_2 \in E \quad \forall P_1, P_2 \in E$ and that:

1. The point addition is commutative.

That is $P_1 + P_2 = P_2 + P_1 \quad \forall P_1, P_2 \in E$

2. The point at infinity ∞ serves as the identity of the this addition, since $P + \infty = P = \infty + P \quad \forall P \in E$
3. If $P(x, y) \in E$, then there is a point Q on E , such that

$$P + Q = \infty = Q + P$$

This point Q is the additive inverse of P , and is usually denoted by $-P$. So for $P(x, y) \in E$, $-P(x, y) = (x, -y)$

4. The point addition is associative. That is

$$(P_1 + P_2) + P_3 = P_1 + (P_2 + P_3) \quad \forall P_1, P_2, P_3 \in E.$$

Although, it appears to be trivial. But this is the most difficult of all to prove as it requires to consider all cases for each P_1, P_2 and P_3 .

Once all the properties of point addition are verified, we get that E becomes an Abelian group with respect to addition (of points on E).

In case $\mathbb{F}$ is a finite field, $E_{\mathbb{F}}$ or $E(\mathbb{F})$ becomes a finite Abelian group. By the fundamental theorem for finite Abelian groups, E is isomorphic onto a finite direct sum of finite cyclic groups. In fact more can be said. We shall discuss that later on.

First, we shall discuss various examples of Elliptic curves over the field $\mathbb{F} = \mathbb{R}$, finite fields $\mathbb{F}$ of characteristic $p > 3$, of characteristic $p = 2$, as well as over characteristic $p = 3$, fields.

13.10 Computations of integer multiples of a point on a curve E and the group E

We compute kP , for an integer k , using the laws of addition and duplication of points on E .

Example 1: Let $E := \{(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11} : y^2 = x^3 + 3x + 1\} \cup \{\infty\}$

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If $P(x_1, y_1) \in E$, and if $P \neq \infty$, then $2P(x_3, y_3)$, where

$$\begin{aligned}x_3 &= m^2 - 2x_1 \\y_3 &= m(x_1 - x_3) - y_1 \\m &= \frac{3x_1^2 + a}{2y_1}\end{aligned}$$

For $x_1 = 4, y = 0$ i.e. for $P(4, 0)$,
we calculate $m = \frac{3 \cdot 4^2 + 3}{0} = \infty$, $2P = \infty$ (point at infinity). Order of P is 2.

For $P(5, 8)$ i.e. $x_1 = 5, y_1 = 8$,

$$\begin{aligned}m &= \frac{3 \cdot 5^2 + 3}{2 \cdot 8} = \frac{78}{16} \\&= \frac{1}{5} \equiv 5^{-1} \pmod{11} \\&\equiv 9 \pmod{11}\end{aligned}$$

$$x_3 = 9^2 - 2 \cdot 5 = 81 - 10 = 71 \equiv 5 \pmod{11};$$

$$y_3 = 9 \cdot (5 - 5) - 8 \equiv -8 \equiv 3 \pmod{11}$$

Hence $2P = (5, 3)$. It is easy to see that $3P = P + 2P = (5, 8) + (5, 3) = \infty$.

This gives order of $(5, 8)$ to be 3.

If $P(0, 1)$, then it can be seen that

$$2P = (5, 8), \quad 3P = (4, 0), \quad 4P = (5, 3), \quad 5P = (0, 10), \quad 6P = \infty$$

Finally, for $P(1, 4) \in E$ verify that

$$\begin{array}{llll}2P = (2, 9) & 3P = (0, 1) & 4P = (8, 8) & 5P = (3, 9) \\6P = (5, 8) & 7P = (6, 2) & 8P = (9, 8) & 9P = (4, 0) \\10P = (9, 3) & 11P = (6, 9) & 12P = (5, 3) & 13P = (3, 2) \\14P = (8, 3) & 15P = (0, 10) & 16P = (2, 2) & 17P = (1, 7) \\& 18P = \infty & & \end{array}$$

Computation of order of a point $P(x, y)$ on an Elliptic Curve E .

Let $P(x, y)$ be point of the Elliptic Curve E , then the least non-negative

integer k is said to be the order of P in E if $kP = \infty$ (point at infinity). Recall the curve

$$E := \{(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11} : y^2 = x^3 + 3x + 1\} \cup \{\infty\}$$

We have calculated all the points of E . They are 18 in all. To be specific:

$$E := \{(0, 1), (0, 10), (1, 4), (1, 7), (2, 2), (2, 9), (3, 2), (3, 9), (4, 0), (5, 3), (5, 8), (6, 2), (6, 9), (8, 3), (8, 9), (9, 3), (9, 8)\} \cup \{\infty\}$$

Compute and observe that if $P = (4, 0)$, then $2P = \infty$.

Hence order of P is 2.

If $P = (5, 8)$, then $3P = \infty$. Hence order of $P(5, 8)$ is 3.

If $P = (0, 1)$, then $6P = \infty$. Hence order of $P(0, 1)$ is 6.

If $P = (2, 2)$, then $9P = \infty$. Hence order of $P(2, 2)$ is 9.

If $P = (1, 4)$, then $18P = \infty$. Hence order of $P(1, 4)$ is 18.

This also proves that E is a cyclic group C_{18} of order 18. The point $P(1, 4)$ is a generator of this group (with respect to point addition). Find all other generators. We can write E as:

$$E := \langle P(1, 4) : 18P = \infty \rangle = \{kP(1, 4) : 1 \leq k \leq 18\} \cong C_{18}$$

Recall the Elliptic Curve

Example 2:

$$\begin{aligned} E &:= E(\mathbb{Z}_{11}) = \{(x, y) \in \mathbb{Z}_{11} \times \mathbb{Z}_{11} : y^2 = x^3 + 4x + 5\} \cup \{\infty\} \\ &= \{(0, 4), (0, 7), (3, 0), (6, 5), (6, 6), (9, 0), (10, 0)\} \cup \{\infty\} \end{aligned}$$

Note that : $\#(E) = 8$

Every element has order 2 or 4, as follows:

Points of order 2 are: $(3, 0), (9, 0), (10, 0)$

Points of order 4 are: $(0, 4), (0, 7), (6, 5), (6, 6)$

For $P(0, 4) = (x_1, y_1)$, $x_1 = 0, y_1 = 4, 2P = (x_3, y_3)$, where

$$x_3 = m^2 - x_1 - x_2$$

$$y_3 = m(x_1 - x_3) - y_1,$$

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and

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \text{ if } x_1 \neq x_2 \\ &= \frac{3x_1^2 + a}{2y_1} \text{ if } P_1 = P_2 \text{ \& } y_1 \neq 0 \end{aligned}$$

Here

$$\begin{aligned} m &= \frac{3 \cdot 0^2 + 4}{2 \cdot 4} = \frac{4}{8} \quad \text{as } a = 4, b = 5, \\ &\equiv \frac{1}{2} \\ &\equiv 6 \quad \because 2 \times 6 \equiv 12 \equiv 1 \pmod{11} \end{aligned}$$

$$\implies x_3 \equiv 6^2 - 0 - 0 = 36 \equiv 3 \pmod{11} \text{ and}$$

$$\implies y_3 = 6(0 - 3) - 4 = -18 - 4 = -22 \equiv 0 \pmod{11}$$

Hence $2P(x_3, y_3) = (3, 0)$.

Clearly $4P(0, 4) = 2P + 2P = (3, 0) + (3, 0) = \infty$. It is easy to see that $3P = P + 2P = (0, 4) + (3, 0) = (0, 7)$, using the addition of points formula. Since, there is no element of order 8, and the Elliptic curve E has 8 points in all, the group of points E is an Abelian group which can not be cyclic. Therefore, by the fundamental theorem of finite abelian group, E is isomorphic to $C_2 \times C_2 \times C_2$ or $C_2 \times C_4$. It can not be isomorphic to $C_2 \times C_2 \times C_2$ as it has points of order 4, e.g. if $P(0, 4)$, then $2P = (3, 0)$ and $4P = \infty$. This leaves, $E \cong C_2 \times C_4$. In fact, this is the case.

Observe that, subgroups of E of order 2: are the subgroups generated by $\langle(3, 0)\rangle, \langle(9, 0)\rangle, \langle(10, 0)\rangle$. These are $\{(3, 0), \infty\}, \{(9, 0), \infty\}, \{(10, 0), \infty\}$.

Similarly, subgroup of order 4:

$$\begin{aligned} \langle(0, 4) : 4(0, 4) = \infty\rangle &= \{(0, 4), 2(0, 4) = (3, 0), 3(0, 4) = (0, 7), 4(0, 4) = \infty\} \\ \langle(6, 5) : 4(6, 5) = \infty\rangle &= \{(6, 5), 2(6, 5) = (3, 0), 3(6, 5) = (6, 6), 4(6, 5) = \infty\} \end{aligned}$$

You can prove that $E = \langle(9, 0)\rangle \oplus \langle(0, 4)\rangle$

For, $\langle(9, 0)\rangle \oplus \langle(0, 4)\rangle \subseteq E$ as $\langle(9, 0)\rangle$ and $\langle(0, 4)\rangle$ are subgroups of E .

Conversely

$$\begin{aligned}(0, 4) &= \infty + (0, 4), & (0, 7) &= \infty + 3(0, 4), \\ (3, 0) &= \infty + 2(0, 4), & (6, 5) &= (9, 0) + (0, 7) = (9, 0) + 3(0, 4), \\ (6, 6) &= (9, 0) + (0, 4), & (9, 0) &= (9, 0) + \infty, \\ (10, 0) &= (9, 0) + (3, 0) = (9, 0) + 2(0, 4), & \infty &= \infty + \infty.\end{aligned}$$

This gives that $E \subseteq \langle (9, 0) \rangle \oplus \langle (0, 4) \rangle$ or that $E = \langle (9, 0) \rangle \oplus \langle (0, 4) \rangle \cong C_2 \times C_4$.

Example 3: Consider $E = \{(x, y) \in \mathbb{Z}_{29} \times \mathbb{Z}_{29} : y^2 = x^3 + 4x + 7\} \cup \{\infty\}$. Observe that $(6, 0) \notin E$, since

$$6^3 + 4 \cdot 6 + 7 = 36 \cdot 6 + 31 \equiv 7 \cdot 6 + 2 \equiv 42 + 2 = 44 \equiv 15 \not\equiv 0 \pmod{29}$$

However,

$$(0, 6) \in E, \text{ since } x^3 + 4x + 7 = 0 + 7 = 7 \equiv 29 + 7 \equiv 36 \equiv 6^2 \pmod{29}$$

$\implies (0, -6)$ i.e. $(0, 23)$ is also a point on E .

Since 29 is not a big number, all points on E can be calculated simply taking $(x, y) \in \mathbb{Z}_{29} \times \mathbb{Z}_{29}$ and verifying if $P(x, y) \in E$.

The order of E , $\#(E)$ can not exceed $29^2 = 841$. The order, however can not exceed, by Hasse's Theorem:

$$p + 1 + 2\sqrt{p} \text{ i.e. } 29 + 1 + 2\sqrt{29} = 30 + 11 = 41.$$

The order infact is much less. It is $\#(E) = 32$ You may verify that:

$$\begin{aligned}E = \{ & (0, 6), (0, 23), (2, 9), (2, 20), (4, 0), (5, 6), (5, 23), (7, 1), (7, 28), \\ & (8, 0), (13, 9), (13, 20), (14, 9), (14, 20), (15, 7), (15, 22), (16, 7), \\ & (16, 22), (17, 0), (18, 13), (18, 16), (20, 5), (20, 24), (22, 10), (22, 19), \\ & (23, 12), (23, 17), (24, 6), (24, 23), (27, 7), (27, 22) \} \cup \{\infty\}\end{aligned}$$

Next to determine the structure of the group of point E with respect to point addition, observe the following:

If P is $(0, 6)$, the $P(0, 6) \in E$, and multiples of P as:

$$\begin{aligned}P &= (0, 6), \quad 2P = (13, 9), \quad 3P = (20, 5), \quad 4P = (4, 0), \\ 5P &= (20, 24), \quad 6P = (13, 20), \quad 7P = (0, 23), \quad 8P = \infty.\end{aligned}$$

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Hence $H = \langle P \rangle = \{kP : 1 \leq k \leq 8\}$ is a subgroup of E of order 8.

And if, $Q = (22, 10)$, then $Q(22, 10) \in E$, as

$$\begin{aligned} 22^3 + 4 \cdot 22 + 7 &\equiv (-7)^3 + 4(-7) + 7 \pmod{29} \\ &\equiv -7^3 - 4 \cdot 7 + 7 \pmod{29} \\ &\equiv -7^3 - 3 \cdot 7 \equiv -7(7^2 + 3) \pmod{29} \\ &\equiv -7 \cdot 52 \equiv -7(-6) \pmod{29} \\ &\equiv 42 \equiv 13 \equiv 13 + 3 \cdot 29 \pmod{29} \\ &\equiv 13 + 87 = 100 = (\pm 10)^2 \pmod{29} \end{aligned}$$

and $-Q(22, 10) = (22, -10) = (22, 19) \in E$.

Also, observe that

$$Q = (22, 10), \quad 2Q = (8, 0), \quad 3Q = (22, 19), \quad 4Q = \infty$$

So, if $K = \langle Q \rangle = \{jQ : 1 \leq j \leq 4\}$, then K is a subgroup of order 4.

Further $K \cap H = \{\infty\}$. $H \oplus K \leq E$ and $E \leq H \oplus K$, as

$$(2, 20) = P + Q, \quad (18, 13) = 2P + Q$$

and so on (verify for remaining elements).

Hence $E = H + K$ and $H \cap K = \{\infty\}$.

This gives that $E = H \oplus K \implies E \cong C_8 \times C_4 \cong C_4 \times C_8$

13.10.1 Point Addition In Elliptic Curves Over Fields Of Characteristic 3

Recall the Elliptic curve $E(F)$ over a field F of Characteristics 3:

$$E_K = \{(x, y) \in F \times F \mid y^2 = x^3 + ax^2 + bx + c; \quad a, b, c \in F\} \cup \{\infty\}$$

Let $P(x, y), P_1(x_1, y_1), P_2(x_2, y_2), P_3(x_3, y_3)$ be points on E . We define the sum of two points $P_1 + P_2 = P_3$ as follows (if $P_1, P_2 \neq \infty$):

1. $P + \infty = P \quad \forall P \in E$ (including $\infty + \infty = \infty$).

2. If $P_1 \neq P_2$ with $x_1 \neq x_2$, then $P_1 + P_2 = P_3(x_3, y_3)$, where

$$\begin{aligned} x_3 &= m^2 - a - x_1 - x_2 \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \end{aligned}$$

3. $P_1 \neq P_2$ with $x_1 = x_2$ but $y_1 \neq y_2$, then $P_1 + P_2 = \infty$

4. If $P_1 = P_2$ with $y_1 \neq 0$, then $2P_1 = P_1 + P_2 = P_3(x_3, y_3)$, where

$$\begin{aligned} x_3 &= m^2 - a - 2x_1, \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{3x_1^2 + 2ax_1 + b}{2y_1} \\ &= \frac{2ax_1 + b}{2y_1} \text{ since } \text{Char}(F) = 3 \end{aligned}$$

5. If $P_1 = P_2$ and $y_1 = 0$, then $2P_1 = P_1 + P_2 = \infty$

Note that the curve E can be represented as:

$$E = \{(x, y) \in F \times F \mid g(x, y) = 0, \text{ where } g(x, y) = y^2 - x^3 - ax^2 - bx - c; a, b, c \in F\} \cup \{\infty\}$$

And at a non singular point P , the slope of the line (tangent in case $P_1 = P_2$) is m which can be computed using:

$$\begin{aligned} m &= \frac{\left(\frac{\partial g}{\partial x}\right)_{P=P_1=P_2}}{\left(\frac{\partial g}{\partial y}\right)_{P=P_1=P_2}} \\ m &= - \left(\frac{-3x^2 - 2ax - b}{2y} \right)_{P(x_1, y_1)} \\ m &= \left(\frac{3x_1^2 + 2ax_1 + b}{2y_1} \right) \\ m^2 &= \left(\frac{3x_1^2 + 2ax_1 + b}{2y_1} \right)^2 \end{aligned}$$

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Equation of tangent at $P_1(x_1, y_1)$:

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ \implies T : y &= m(x - x_1) + y_1 \end{aligned}$$

$Q(x_3, -y_3)$ lies on T : this implies that $-y_3 = m(x_3 - x_1) + y_1$
Since Q is reflection of $P_3(x_3, y_3)$ about x -axis on E , hence

$$\begin{aligned} y_3 &= -m(x_3 - x_1) - y_1 \\ \text{Or } y_3 &= m(x_1 - x_3) - y_1 \end{aligned}$$

Particular Case (1): If the curve $E = E(F)$, where $\text{Char}(F) = 3$ is given by:

$$E_K = \{(x, y) \in F \times F \mid y^2 = x^3 + ax + b; \ a, b \in F\} \cup \{\infty\}$$

We define the sum of two points $P_1 + P_2 = P_3$ as follows :

1. $P + \infty = P \ \forall P \in E$ (including $\infty + \infty = \infty$).
2. If $P_1 \neq P_2$ with $x_1 \neq x_2$, then $P_1 + P_2 = P_3(x_3, y_3)$, where

$$\begin{aligned} x_3 &= m^2 - x_1 - x_2, \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \end{aligned}$$

3. $P_1 \neq P_2$ with $x_1 = x_2$ but $y_1 \neq y_2$, then $P_1 + P_2 = \infty$
4. If $P_1 = P_2$ with $y_1 \neq 0$, then $2P_1 = P_1 + P_2 = P_3(x_3, y_3)$, where

$$\begin{aligned} x_3 &= m^2 - 2x_1, \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{3x_1^2 + a}{2y_1} \\ &= \frac{a}{2y_1} \text{ since } \text{Char}(F) = 3 \end{aligned}$$

5. If $P_1 = P_2$ and $y_1 = 0$, then $2P_1 = P_1 + P_2 = \infty$

Particular Case (2): If the curve $E = E(F)$, where $\text{Char}(F) = 3$ is given by:

$$E_K = \{(x, y) \in F \times F \mid y^2 = x^3 + ax^2 + b; \ a, b \in F\} \cup \{\infty\}$$

then for $P_1(x_1, y_1), P_2(x_2, y_2) \neq \infty$, and $P(x, y) \in E$, we define the sum of two points $P_1 + P_2 = P_3$ as follows:

1. $P + \infty = P \ \forall P \in E$ (including $\infty + \infty = \infty$).
2. If $P_1 \neq P_2$ with $x_1 \neq x_2$, then $P_1 + P_2 = P_3(x_3, y_3)$, where

$$\begin{aligned} x_3 &= m^2 - a - x_1 - x_2 \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \end{aligned}$$

3. $P_1 \neq P_2$ with $x_1 = x_2$ but $y_1 \neq y_2$, then $P_1 + P_2 = \infty$
4. If $P_1 = P_2$ with $y_1 \neq 0$, then $2P_1 = P_1 + P_2 = P_3(x_3, y_3)$, where

$$\begin{aligned} x_3 &= m^2 - a - 2x_1 \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{3x_1^2 + 2ax_1}{2y_1} \\ &= \frac{2ax_1}{2y_1} \quad \text{since } \text{Char}(F) = 3 \end{aligned}$$

5. If $P_1 = P_2$ and $y_1 = 0$, then $2P_1 = P_1 + P_2 = \infty$

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Example 4.: Let $E = \{(x, y) \in \mathbb{Z}_3 \times \mathbb{Z}_3 \mid y^2 = x^3 - x + 2\} \cup \{\infty\}$

We have seen that E has only one point, namely the point at infinity ∞ . Hence $E = \{\infty\}$, and therefore $E \cong C_1$

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Example 5: Let $E = \{(x, y) \in \mathbb{Z}_3 \times \mathbb{Z}_3 | y^2 = x^3 - x^2 + 2\} \cup \{\infty\}$

We have computed all points of E , and found that $E \setminus \{(2, 0), \infty\}$. Also, note that if $P_1(x_1, y_1) = (2, 0)$, then $2P_1 = \infty$. Hence $E \cong C_2$.

Example 6: Let $E = \{(x, y) \in \mathbb{F}_9 \times \mathbb{F}_9 : y^2 = x^3 - x + 2\} \cup \{\infty\}$, where

$$\begin{aligned} \mathbb{F}_9 &= \{0, \xi^i : 1 \leq i \leq 8; \xi^2 + \xi + 2 = 0\} \\ &= \{0, \xi, \xi^2 = 1 + 2\xi, \xi^3 = 2 + 2\xi, \xi^4 = -1 = 2, \xi^5 = 2\xi, \\ &\quad \xi^6 = 2 + \xi, \xi^7 = 1 + \xi, \xi^8 = 1\} \end{aligned}$$

We have computed all the points of the curve E , we have seen that:

$$E := \{(0, 1 + 2\xi), (0, 2 + \xi), (2, 1 + 2\xi), (2, 2 + \xi), (1, 1 + 2\xi), (1, 2 + \xi), \infty\}.$$

Let $P(0, 1 + 2\xi)$ and $x_1 = 0, y_1 = 1 + 2\xi$ then $2P(x_1, y_1) = P_3(x_3, y_3)$, where

$$\begin{aligned} x_3 &= m^2 - 2x_1, \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{3x_1^2 + a}{2y_1} = \frac{a}{2y_1} \end{aligned}$$

Here $a = -1 = 2 \in F$, Also $a = \xi^4$, as ξ is a primitive 8-th root of unity over $\mathbb{F}_9 \cong \mathbb{Z}_3$. Now

$$\begin{aligned} m &= \frac{2}{2 \cdot (1 + 2\xi)} = \frac{1}{1 + 2\xi} = \frac{1}{\xi^2} = (\xi^2)^{-1} = \xi^6 \\ \Rightarrow m &= 2 + \xi \\ x_3 &= m^2 - 2x_1 = (2 + \xi)^2 - 2 \cdot 0 = (2 + \xi)^2 \\ &= (\xi - 1)^2 = \xi^2 - 2\xi + 1 = \xi^2 + \xi + 1 \\ &= (1 + 2\xi) + \xi + 1 = 2 + 3\xi = 2 = -1 = \xi^4 \\ y_3 &= m(x_1 - x_3) - y_1 \\ &= (2 + \xi)(0 - 2) - (1 + 2\xi) \\ &= (2 + \xi) \cdot 1 - 1 - 2\xi \\ &= 2 + \xi - 1 + \xi = 1 + 2\xi = 1 - \xi = \xi^2 \end{aligned}$$

Hence $2P_1 = 2(0, 1 + 2\xi) = (2, 1 + 2\xi)$

Next, we calculate $3P_1$ as $P_1 + 2P_1$.

Let $P_2(x_2, y_2) = 2P_1$, have the coordinates $x_2 = 2, y_2 = 1 + 2\xi$. Also recall $x_1 = 0, y_1 = 1 + 2\xi$.

Suppose $P_3(x'_3, y'_3) = P_1 + P_2$, then

$$\begin{aligned}x'_3 &= m^2 - x_1 - x_2, \\y'_3 &= m(x_1 - x'_3) - y_1 \\m &= \frac{y_2 - y_1}{x_2 - x_1}\end{aligned}$$

$$\begin{aligned}m &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{(1 + 2\xi) - (1 + 2\xi)}{2 - 0} = 0 \\x'_3 &= m^2 - x_1 - x_2 = 0^2 - 0 - 2 = -2 = 1 \\y'_3 &= m(x_1 - x'_3) - y_1 = -y_1 = -(1 + 2\xi) = (2 + \xi)\end{aligned}$$

Hence $3P_1 = (x'_3, y'_3) = (1, 2 + \xi)$

We now calculate $4P_1$ as $(2P_1) + (2P_1)$.

Recall $2P_1 = (2, 1 + 2\xi) = (x''_1, y''_1)$ (say) then $4P_1 = (x''_3, y''_3)$, where

$$\begin{aligned}x''_3 &= m^2 - 2x''_1, \\y''_3 &= m(x''_1 - x''_3) - y''_1 \\m &= \frac{3(x''_1)^2 + a}{2y''_1} = \frac{a}{2y''_1}\end{aligned}$$

Thus

$$m = \frac{-1}{2 \cdot (1 + 2\xi)} = 2 + \xi$$

and

$$m^2 = -1 = 2$$

Now

$$x''_3 = m^2 - 2x''_1 = 2 - 2 \cdot 2 = 2 - 4 = -2 = 1$$

$$\begin{aligned}y''_3 &= m(x''_1 - x''_3) - y''_1 \\&= (2 + \xi)(2 - 1) - (1 + 2\xi) \\&= (2 + \xi) - (1 + 2\xi) \\&= 1 - \xi = 1 + 2\xi\end{aligned}$$

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Hence $4P_1 = (1, 1 + 2\xi)$

Similarly calculate $5P_1, 6P_1, 7P_1$. You may also note that

$$7P_1 = 3P_1 + 4P_1 = (1, 2 + \xi) + (1, 1 + 2\xi) = \infty$$

Since $\#(E) = 7$. Therefore

$$E = \langle P_1 | 7P_1 = \infty \rangle \cong C_7$$

Example 7: Let $E = \{(x, y) \in \mathbb{F}_9 \times \mathbb{F}_9 : y^2 = x^3 - x^2 + 2\} \cup \{\infty\}$

$$\begin{aligned} \mathbb{F}_9 &= \{0, \xi^i : 1 \leq i \leq 8; \xi^2 + \xi + 2 = 0\} \\ &= \{0, \xi, \xi^2 = 1 + 2\xi, \xi^3 = 2 + 2\xi, \xi^4 = -1 = 2, \xi^5 = 2\xi, \\ &\quad \xi^6 = 1 + \xi, \xi^7 = 1 + \xi, \xi^8 = 1\} \end{aligned}$$

We compute points of E as follows:

$$E = \{(0, 1 + 2\xi), (0, 2 + \xi), (\xi, 0), (1 + 2\xi, 2 + 2\xi), (1 + 2\xi, 1 + \xi), (2 + 2\xi, 0), (2, 0), \\ (2 + \xi, \xi), (2 + \xi, 2\xi), (1, 1 + 2\xi), (1, 2 + \xi)\} \cup \{\infty\}$$

Now calculate integer multiples of the point $P_1(0, 1 + 2\xi)$ as follows:

Let $x_1 = 0, y_1 = 1 + 2\xi$ then $2P_1 = P_3(x_3, y_3)$

where

$$\begin{aligned} x_3 &= m^2 - a - 2x_1, \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{3x_1^2 + 2ax_1}{2y_1} = \frac{2ax_1}{2y_1} \end{aligned}$$

Observing $a = -1 = 2 \pmod{3}$, $x_1 = 0, y_1 = 1 + 2\xi$, we get

$$m = \frac{2 \cdot 2 \cdot 0}{2 \cdot (1 + 2\xi)} = 0$$

$$x_3 = -a - 2x_1 = 1 + x_1 = 1$$

$$y_3 = m(x_1 - x_3) - y_1 = -y_1 = -(1 + 2\xi) = 2 + \xi$$

Hence $(2P_1)(x_1, y_3) = (1, 2 + \xi)$

next, we compute $3P_1$ by doing $P_1 + 2P_1$

Let $P_1 = (x_1, y_1)$, $2P_1 = (x_2, y_2)$ where $x_1 = 0, y_1 = 1 + 2\xi, x_2 = 1, y_2 = 2 + \xi$ then

$$\begin{aligned} x'_3 &= m^2 - a - x_1 - x_2, \\ y'_3 &= m'(x_1 - x'_3) - y_1 \\ m' &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{(2 + \xi) - (1 + 2\xi)}{1 - 0} = 1 - \xi = 1 + 2\xi = \xi^2 \\ (m')^2 &= \xi^4 = -1 = 2 \end{aligned}$$

This gives

$$\begin{aligned} x'_3 &= 2 - 2 - 0 - 1 = 2 \\ y'_3 &= m'(x_1 - x_3) - y_1 \\ &= (1 + 2\xi)(0 - 2) - (1 + 2\xi) \\ &= (1 + 2\xi) - (1 + 2\xi) = 0 \end{aligned}$$

Hence we get $3P_1 = (2, 0)$ this implies that $6P_1 = \infty$

Therefore $K = \langle P_1 | 6P_1 = \infty \rangle \leq E$ of order 6.

Compute $4P_1$ and $5P_1$ also and write all elements of K .

Let $(\xi, 0) = Q_1$. Then $H = \langle Q_1 | 2Q_1 = \infty \rangle \leq E$ is a subgroup of order 2.

Is $E \cong H \oplus K$?

Is $E = \langle (2 + 2\xi, 0) \rangle \oplus \langle (0, 2 + \xi) \rangle$?

Is $E \cong C_2 \times C_6$?

Write all possible combinations?

Example 8: Let

$$\begin{aligned} E &= \{(x, y) \in \mathbb{F}_9 \times \mathbb{F}_9 | y^2 = x^3 + x^2 + x + 1\} \cup \{\infty\} \\ &= \{(x, y) \in \mathbb{F}_9 \times \mathbb{F}_9 | g(x, y) = 0 \text{ where } g(x, y) = y^2 - x^3 - x^2 - x - 2\} \cup \{\infty\} \end{aligned}$$

$$\begin{aligned} \mathbb{F}_9 &= \{0, \xi^i : 1 \leq i \leq 8; \xi^2 + \xi + 2 = 0\} \\ &= \{0, \xi, \xi^2 = 1 + 2\xi, \xi^3 = 2 + 2\xi, \xi^4 = -1 = 2, \xi^5 = 2\xi, \\ &\quad \xi^6 = 1 + \xi, \xi^7 = 1 + \xi, \xi^8 = 1\} \end{aligned}$$

$$\begin{aligned} \frac{\partial g}{\partial x} &= 3x^2 + 2x + 1 = 0 \implies x = 1 \\ \frac{\partial g}{\partial y} &= 2y = 0 \implies y = 0 \end{aligned}$$

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So, possible singular point is $(1, 0)$. But $(1, 0)$ is not a point on the curve E .

Hence, the curve is non-singular. Now we compute points on E .

Some points are easily noticeable on E .

For example

$(0, 1) \in E$. Hence $(0, -1) = (0, 2) \in E$

$(1, 1) \in E$. Hence $(1, 2) \in E$. Also $(2, 0) \in E$

It is better to calculate all the points of E by taking $x, y \in F_9$ and verifying g if $(x, y) \in E$.

x	$y^2 = x^3 + x^2 + x + 1$	y^2	y^2	y
0	1	1	1	1,2
ξ	$\xi^3 + \xi^2 + \xi + 1$	$\xi^3 - 1 = 1 + 2\xi$	ξ^2	$\xi, 2\xi$
$\xi^2 = 1 + 2\xi$	$\xi^6 + \xi^4 + \xi^2 + 1$	$\xi^6 + \xi^2$	0	0
$\xi^3 = 2 + 2\xi$	$\xi^9 + \xi^6 + \xi^3 + 1$	$2 + \xi$	ξ^6	$2 + 2\xi, 1 + \xi$
$\xi^4 = -1 = 2$	$(-1)^3 + (-1)^2 + (-1) + 1$	0	0	0
$\xi^5 = 2\xi$	$(2\xi)^3 + (2\xi)^2 + (2\xi) + 1$	(2ξ)	ξ^5	-
$\xi^6 = 2 + \xi$	$\xi^{18} + \xi^{12} + \xi^6 + 1$	$\xi^2 + \xi^4 + \xi^6 + 1$	0	0
$\xi^7 = 1 + \xi$	$\xi^{21} + \xi^{14} + \xi^7 + 1$	$\xi^5 + \xi^6 + \xi^7 + 1$	$1 + \xi = \xi^7$	-
$\xi^8 = 1$	1	1	1	1,2

We can see that E has 12 points in all.

Points of E are

$$E = \{(0, 1), (0, 2), (\xi, \xi), (\xi, 2\xi), (1 + 2\xi, 0), (2 + 2\xi, 2 + 2\xi), (2 + 2\xi, 1 + \xi), (2, 0), (2 + \xi, 0), (1, 1), (1, 2), \infty\}$$

Points of order 2 are $(1+2\xi, 0), (2, 0), (2+\xi, 0)$. We now compute the subgroup generated by $P_1 = (0, 1)$.

Observe that $2P_1 = (0, 2)$

For let $P_1 = (0, 1), x_1 = 0, y_1 = 1$ and $2P_1 = (x_3, y_3)$ then

$$\begin{aligned}x_3 &= m^2 - a - 2x_1, \\y_3 &= m(x_1 - x_3) - y_1 \\m &= \frac{3x_1^2 + 2ax_1 + b}{2y_1} = \frac{2ax_1 + b}{2y_1}\end{aligned}$$

using $a = b = 1, x_1 = 0, y_1 = 1$, we get

$$m = \frac{2 \cdot 1 \cdot 0 + 1}{2 \cdot 1} = \frac{1}{2} = 2 \implies m^2 = 1$$

$$x_3 = 1 - 1 - 2 \cdot 0 = 0$$

$$y_3 = 1 \cdot (0 - 0) - 1 = -1 = 2$$

Hence $2P_1 = (x_3, y_3) = (0, 2) = (0, -1)$.

Next we compute $3P_1$ as follows, by doing $P_1 + 2P_1$.

Let $3P_1 = (x'_3, y'_3)$. Suppose $2P_1 = (x_2, y_2) = (0, 2)$ and $P_1 = (x_1, y_1) = (0, 1)$

Since $x_1 = x_2 = 0$ this implies that $3P_1 = \infty$.

Hence $(0, 1)$ is a point of order 3 on E .

Let $Q_1 = (1, 1) = (x_1, y_1)$, and $2Q_1 = (x_3, y_3)$, then

$$\begin{aligned}x_3 &= m^2 - a - 2x_1, \\y_3 &= m(x_1 - x_3) - y_1 \\m &= \frac{3x_1^2 + 2ax_1 + b}{2y_1} = \frac{2ax_1 + b}{2y_1}\end{aligned}$$

Using $a = 1, x_1 = 1, y_1 = 1$, we get

$$m = \frac{2 \cdot 1 \cdot 1 + 1}{2 \cdot 1} = \frac{3}{2} = 0$$

$$x_3 = 0^2 - 1 - 2 \cdot 1 = -3 = 0$$

$$y_3 = 0 \cdot (1 - 0) - 1 = 0 - 1 = 2$$

This implies that $2Q_1 = (x_3, y_3) = (0, 2)$

Note that $2P_1 = 2Q_1 = (0, 2)$

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We compute $3Q_1 = (x_3'', y_3'') = Q_1 + 2Q_1 = (1, 1) + (0, 2)$. Then

$$\begin{aligned}x_3'' &= m^2 - a - x_1 - x_2, \\y_3'' &= m(x_1 - x_3'') - y_1 \\m &= \frac{y_2 - y_1}{x_2 - x_1}\end{aligned}$$

Using $a = 1, x_1 = 1, y_1 = 1, x_2 = 0, y_2 = 2$, we get

$$m = \frac{2 - 1}{0 - 1} = \frac{1}{-1} = -1 \implies m^2 = 1$$

$$x_3'' = 1 - 1 - 1 - 0 = -1 = 2$$

$$y_3'' = 1 \cdot (1 - 0) - 1 = 0$$

Hence $3Q_1 = (2, 0)$ this implies that $6Q_1 = \infty$.

Therefore $K = \langle Q_1 | 6Q_1 = \infty \rangle$.

Compute all elements of K .

If $H = \langle (1 + 2\xi, 0) \rangle$, then $H \leq E$ is a subgroup of order 2.

Is $E = H \oplus K$?

Write E as a direct sum of all possible subgroups of E . Is $E \cong C_2 \times C_6$.

13.10.2 Groups Of Elliptic Curves over fields of characteristic 2

Rules of point addition and duplication

There are two normal forms of Elliptic curves over fields of characteristic 2 as follows :

1. $E_1 = \{(x, y) \in \mathbb{F} \times \mathbb{F} | y^2 + cy = x^3 + ax + b; \ a, b, c \in \mathbb{F}\} \cup \{\infty\}$
2. $E_2 = \{(x, y) \in \mathbb{F} \times \mathbb{F} | y^2 + xy = x^3 + ax^2 + b; \ a, b \in \mathbb{F}\} \cup \{\infty\}$

We now state rules of addition of points on these curves

First for E_1 :

1. $P + \infty = \infty + P = P \quad \forall P \in E_1$ (including $\infty + \infty = \infty$).

2. If $P_1(x_1, y_1), P_2(x_2, y_2) \neq \infty$ and $P_1 \neq P_2$ with $x_1 \neq x_2$, then $P_1 + P_2 = P_3(x_3, y_3)$ Where

$$x_3 = m^2 - x_1 - x_2 = m^2 + x_1 + x_2,$$

$$y_3 = m(x_1 - x_3) - y_1 - c = m(x_1 + x_3) + y_1 + c$$

where $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_2 + y_1}{x_2 + x_1}$ since $\text{char}(\mathbb{F}) = 2$

3. If $P_1(x_1, y_1), P_2(x_2, y_2) \neq \infty$ and $P_1 \neq P_2$ with $x_1 = x_2$, but $y_1 \neq y_2$ then $P_1 + P_2 = \infty$
4. If $P_1(x_1, y_1), P_2(x_2, y_2) \neq \infty$ and $P_1 = P_2$ then $2P_1 = P_1 + P_2 = P_3(x_3, y_3)$ Where

$$x_3 = m^2 - 2x_1 = m^2, \quad \because \text{char}(\mathbb{F}) = 2$$

$$y_3 = m(x_1 - x_3) - y_1 - c = m(x_1 + x_3) + y_1 + c$$

as $\text{char}(\mathbb{F}) = 2$ where $m = \frac{x^2 + a}{c}$ [If $c = 0$ then $2P_1 = \infty$]

Theorem: E_1 is an additive abelian group with respect to (point) addition (defined as above).

Rules of point addition on the curve $E_2(\mathbb{F}); \text{Char}(\mathbb{F}) = 2$:

$$E_2 = \{(x, y) \in \mathbb{F} \times \mathbb{F} | y^2 + xy = x^3 + ax^2 + b; \quad a, b \in \mathbb{F}\} \cup \{\infty\}$$

1. $P + \infty = \infty + P = P \quad \forall P \in E_2$ (including $\infty + \infty = \infty$).
2. If $P_1(x_1, y_1), P_2(x_2, y_2) \neq \infty$ and $P_1 \neq P_2$ with $x_1 \neq x_2$, then $P_1 + P_2 = P_3(x_3, y_3)$ Where

$$x_3 = m^2 + m + a + x_1 + x_2,$$

$$y_3 = m(x_1 + x_3) + y_1 + x_3$$

where $m = \frac{y_2 + y_1}{x_2 + x_1}$ [$\because \text{char}(\mathbb{F}) = 2$]

3. If $P_1(x_1, y_1), P_2(x_2, y_2) \neq \infty$ and $P_1 \neq P_2$ with $x_1 = x_2$, but $y_1 \neq y_2$ then $P_1 + P_2 = \infty$

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4. If $P_1(x_1, y_1), P_2(x_2, y_2) \neq \infty$ and $P_1 = P_2$ with $x_1 \neq 0$ then
 $2P_1 = P_1 + P_2 = P_3(x_3, y_3)$ Where

$$x_3 = x_1^2 + \frac{b}{x_1^2}, \quad y_3 = \frac{x_1^2 + y_1}{x_1} x_3 + x_1^2 + x_3$$

[If $x_1 = 0$ then $2P_1 = \infty$]

Theorem: The points on the curve E_2 form an additive Abelian group.

Example 9: Let $E = \{(x, y) \in \mathbb{Z}_2 \times \mathbb{Z}_2 : y^2 + y = x^3 + x\} \cup \{\infty\}$.
 We have seen that :

$$E = \{(0, 0), (0, 1), (1, 0), (1, 1), \infty\}$$

$$\#(E) = 5$$

Let $P = (0, 0) = (x_1, y_1) \implies x_1 = 0, y_1 = 0$, suppose $2P = (x_3, y_3)$, where

$$x_3 = m^2, \quad y_3 = m(x_1 + x_3) + y_1 + c \text{ where } m = \frac{x_1^2 + a}{c}$$

i.e.

$$x_3 = \left(\frac{x_1^2 + a}{c} \right)^2, \quad y_3 = \frac{x_1^2 + a}{c} (x_1 + x_3) + y_1 + c$$

For this curve $y^2 + y = x^3 + x$, $c = 1, a = 1, b = 0$ and $x_1 = 0, y_1 = 0$ we get:

$$x_3 = \left(\frac{0^2 + 1}{1} \right)^2 = 1, \quad y_3 = \frac{0^2 + 1}{1} (0 + 1) + 0 + 1 = 1 \cdot 1 + 1 = 2 = 0$$

$\implies (x_3, y_3) = (1, 0)$. Hence $2P = (1, 0)$

We next, compute $3P = P + 2P = (0, 0) + (1, 0)$

Let $x_1 = 0, y_1 = 0, x_2 = 1, y_2 = 0$ then $3P = (x'_3, y'_3)$ where

$$x'_3 = m^2 + x_1 + x_2, \quad y'_3 = m(x_1 + x_3) + y_1 + c \text{ where } m = \frac{y_1 + y_2}{x_1 + x_2}$$

Here

$$m = \frac{0 + 0}{0 + 1} = 0 \implies x'_3 = x_1 + x_2 = 0 + 1 = 1$$

and

$$y'_3 = y_1 + c = 0 + c = c = 1$$

Hence $3P = (x'_3, y'_3) = (1, 1)$.

We now compute $4P$ as $2P + 2P$ i.e. $4P = (x_2, y_2) + (x_2, y_2)$, $x_2 = 1, y_2 = 0$

Let $4P = (x_2, y_2) + (x_2, y_2) = (x''_3, y''_3)$,

then

$$x''_3 = \left(\frac{x_2^2 + a}{c} \right)^2 = \left(\frac{1^2 + 1}{1} \right)^2 = \frac{2^2}{1} = 0$$

$$y''_3 = \frac{x_2^2 + a}{c}(x_2 + x_3) + y_2 + c = 0(1 + 0) + 0 + c = c = 1$$

$$\Rightarrow 4P = (x''_3, y''_3) = (0, 1)$$

Finally, we compute $6P$ as $6P = 3P + 3P$.

Let $3P = (1, 1) = (u, v)$ with $u = 1, v = 1$

If $6P = (x_5, y_5)$ then

$$x_5 = \left(\frac{u^2 + a}{c} \right)^2 = \left(\frac{1^2 + 1}{1} \right)^2 = \left(\frac{2}{1} \right)^2 = 0$$

$$y_5 = \frac{u^2 + a}{c}(u + x_5) + v + c = 0(u + x_5) + v + c = v + c = 1 + 1 = 0$$

$$\Rightarrow 6P = (x_5, y_5) = (0, 0) = P \Rightarrow 5P = \infty$$

This can be obtained by observing that

$$2P + 3P = (1, 0) + (1, 1) = \infty$$

Hence

$$\begin{aligned} E &= \langle P : 5P = \infty \rangle \\ &= \{P = (0, 0), 2P = (1, 0), 3P = (1, 1), 4P = (0, 1), 5P = \infty\} \\ &\cong C_5 \end{aligned}$$

Example 10: Let $E = \{(x, y) \in \mathbb{Z}_2 \times \mathbb{Z}_2 : y^2 + xy = x^3 + 1\} \cup \{\infty\}$.

We have computed all the points on E , and observe that

$$E = \{(0, 1), (1, 0), (1, 1), \infty\}$$

$$\#(E) = 4$$

Now let $P = (0, 1)$, then $2P = \infty$.

Next if $Q = (1, 0)$, then $2Q = (x_3, y_3)$, where

$$x_3 = x_1^2 + \frac{b}{x_1^2}, \quad y_3 = \frac{x_1^2 + y_1}{x_1}x_3 + x_1^2 + x_3$$

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where $x_1 = 1, y_1 = 0, a = 0, b = 1$ implies that

$$x_3 = 1^2 + \frac{1}{1} = 1 + 1 = 0,$$

$$y_3 = \frac{1^2 + 0}{1} \cdot 0 + 1^2 + 0 = 1$$

$$\implies 2Q = (0, 1) = P \implies 4Q = 2Q + 2Q = P + P = 2P = \infty.$$

Hence $E = \langle Q : 4Q = \infty \rangle = \langle (1, 0) : 4(1, 0) = \infty \rangle \cong C_4$

Example 11: Let $E = \{(x, y) \in \mathbb{F}_4 \times \mathbb{F}_4 \mid y^2 + y = x^3 + 3\} \cup \{\infty\}$, where

$$\begin{aligned} \mathbb{F}_4 &= \{0, \omega^i \mid 1 \leq i \leq 3; \omega^3 = 1, \omega \neq 1\} \\ &= \{0, 1, \omega, \omega^2 \mid \omega^2 + \omega + 1 = 0\} \\ &= \{0, 1, \omega, 1 + \omega\} \end{aligned}$$

We have computed all the points of E , and observe that:

$$E = \{(0, 0), (0, 1), (1, 0), (1, 1)\} \cup \{\infty\}$$

$\#(E) = 5$. We determine the subgroup generated by the point $P(0, 0) \in E$, as follows.

Let $x_1 = 0, y_1 = 0 \implies P(0, 0) = (x_1, y_1)$

$2P = (x_2, y_2)$, then

$$x_2 = \left(\frac{x_1^2 + a}{c} \right)^2, \quad y_2 = \frac{x_1^2 + a}{c} (x_1 + x_2) + y_1 + c$$

Here $a = 1, c = 1$.

We get $x_2 = \left(\frac{0^2 + 1}{1} \right)^2 = 1^2 = 1, \quad y_2 = 1(0 + 1) + 0 + 1 = 2 + 0 = 0$

$\implies (x_2, y_2) = 2P = (1, 0)$

Let $3P = (x_3, y_3) = P + 2P = (x_1, y_1) + (x_2, y_2)$ where

$x_1 = 0, y_1 = 0, x_2 = 1, y_2 = 0$, We have

$$x_3 = m^2 + x_1 + x_2, \quad y_3 = m(x_1 + x_3) + y_1 + c$$

where

$$m = \frac{y_1 + y_2}{x_1 + x_2} = \frac{0 + 0}{0 + 1} = \frac{0}{1} = 0$$

This gives $x_3 = 0^2 + 0 + 1 = 1$ and $y_3 = 0 \cdot (0 + 1) + 0 + 1 = 1$ [$\because c = 1$]
Hence $3P = (x_3, y_3) = (1, 1)$.

Next, let $4P = 2P + 2P = (1, 0) + (1, 0) = (x_2, y_2) + (x_2, y_2)$ have coordinates $4P = (x_4, y_4)$. Then

$$x_4 = \left(\frac{x_2^2 + a}{c} \right)^2 = m^2 = \left(\frac{1^2 + 1}{1} \right)^2 = \frac{2^2}{1} = 0 \quad [\text{as } a = 1, c = 1]$$

$$y_4 = m \cdot (x_2 + x_4) + y_2 + c = 0(1 + 0) + 0 + 1 = 1$$

Hence $4P = (x_4, y_4) = (0, 1)$.

We now calculate $6P = 3P + 3P = (1, 1) + (1, 1)$

Let $6P = (x_6, y_6) = (x_3, y_3) + (x_3, y_3)$ where $x_3 = 1, y_3 = 1$

$$x_6 = \left(\frac{x_3^2 + a}{c} \right)^2 = m^2 = \left(\frac{1^2 + 1}{1} \right)^2 = \frac{2^2}{1} = 0 \text{ i.e. } m = 0$$

$$y_6 = m \cdot (x_3 + x_6) + y_3 + c = 0(1 + 0) + 1 + 1 = 0$$

$$\Rightarrow 6P = (0, 0) = P \Rightarrow 5P = \infty$$

Hence

$$\begin{aligned} E &= \langle P \rangle = \langle P : 5P = \infty \rangle \\ &= \{P = (0, 0), 2P = (1, 0), 3P = (1, 1), 4P = (0, 1), 5P = \infty\} \\ &\cong C_5 \end{aligned}$$

Example 12: Let $E = \{(x, y) \in \mathbb{F}_4 \times \mathbb{F}_4 \mid y^2 + xy = x^3 + 1\} \cup \{\infty\}$, where

$$\begin{aligned} \mathbb{F}_4 &= \{0, \omega^i \mid 1 \leq i \leq 3; \omega^3 = 1, \omega \neq 1\} \\ &= \{0, 1, \omega, \omega^2 \mid \omega^2 + \omega + 1 = 0\} \\ &= \{0, 1, \omega, 1 + \omega\} \end{aligned}$$

We have computed all the points of E . We found that E has eight points in all.

$$E = \{(0, 1), (1, 0), (1, 1), (\omega, 0), (\omega, \omega), (\omega^2, 0), (\omega^2, \omega^2), \infty\}$$

or

$$E = \{(0, 1), (1, 0), (1, 1), (\omega, 0), (\omega, \omega), (1 + \omega, 0), (1 + \omega, 1 + \omega), \infty\}$$

Prove that $Q = (0, 1)$ is a point in E of order 2.

We compute multiples of $P = (\omega, 0) = (x, y)$ as follows:

Let $2P = (x_2, y_2)$, then

$$x_2 = x_1^2 + \frac{c}{x_1^2} = \omega^2 + \frac{1}{\omega^2} = \omega^2 + \omega = 1$$

$$y_2 = \frac{x_1^2 + y_1}{x_1} x_2 + x_1^2 + x_2 = \frac{\omega^2 + 0}{\omega} \cdot 1 + \omega^2 + 1$$

$$\implies y_2 = \omega + \omega^2 + 1 = 0$$

$$\implies 2P = (x_2, y_2) = (1, 0)$$

We next compute $4P = 2P + 2P = (x_2, y_2) + (x_2, y_2) = (1, 0) + (1, 0)$.

Let $4P = (x_4, y_4)$. Then

$$x_4 = x_2^2 + \frac{a}{x_2^2} = 1^2 + \frac{1}{1^2} = 1 + 1 = 0$$

$$y_4 = \frac{x_2^2 + y_2}{x_2} x_4 + x_2^2 + x_4 = \frac{1^2 + 0}{1} \cdot 0 + 1^2 + 0 = 1$$

$$\implies (x_4, y_4) = (0, 1) \text{ or } 4P = (0, 1) = Q$$

But Q is of order 2. Hence $8P = 2Q = \infty$

Hence $E = \langle P \rangle = \langle P : 8P = \infty \rangle \cong C_8$

Write all multiples kP ($k = 1, 2, 3, 4, 5, 6, 7, 8$). In fact, you have to compute $3P, 5P$ and $7P$ only.

13.11 Exercise

1. Given that the point $P_1(1, 2)$ and $P_2(3, 4)$ lies on the elliptic curve $y^2 = x^3 - 7x + 10$, find the sum $P_1 + P_2$ and $(P_1 + P_2) + P_2$.
2. Verify that the elliptic curve $y^2 = x^3 + 4x + 4$ is non-singular mod p and then find all the points on the curve mod p , for $p = 5, 7, 11, 13$.
3. Show that the elliptic curve $E = y^2 = x^3 + 4x + 4$ in modulo 13 has points of order 15.

4. Examine what happens when you add the point $P_1 = (1, 3)$ and $P_2 = (15, 4)$ on the Elliptic curve $E : y^2 = x^3 + 4x + 4$ in modulo 21, and when you double the point P_1 .
5. The Curve $E : y^2 = x^3 - 5x = 8 \pmod{37}$ contains the point $P = (6, 3) \in E(\mathbf{F}_{37})$ and $Q = (9, 10) \in E(\mathbf{F}_{37})$. Using the addition formula, Compute $E(\mathbf{F}_{37})$ and identify the structure of the group.
6. The points $P = (2, 9)$, $Q = (3, 10)$ and $R = (-4, -3)$ lie on the elliptic curve $E : y^2 = x^3 + 73$ over any field $\mathbb{F}$, then
 - (a) compute $P + Q$ and $(P + Q) + R$.
 - (b) Also compute $Q + R$ and $P + (Q + R)$. Your answer for $(P + Q) + R$ should agree with the result of part (1).
7. Let (x, y) be a point on the elliptic curve E given by $y^2 = x^3 + Ax + B$. Show that if $y = 0$ then $3x^2 + A \neq 0$.
8. Show that three points on an elliptic curve add to ∞ if and only if they are collinear.
9. The point $(3, 5)$ lies on the elliptic curve $E : y^2 = x^3 - 2$ defined over $\mathbf{Q}$. Find a point other than ∞ with rational, non-integral coordinates in $\mathbf{Q}$.

Chapter 14

Applications Of Elliptic Curves

14.1 Factorization Of Integers

We have studied several algorithm for integer factorization, earlier. We now see how Elliptic Curves can be used for this purpose. We begin with an example.

Example 1. Let $n = 589$, and

$$E := \{(x, y) \in \mathbb{Z}_n \times \mathbb{Z}_n | y^2 = x^3 + 4x + 9\} \cup \{\infty\}$$

If $P_1 = (x_1, y_1) = (2, 5)$, then $2^3 + 4 \times 2 + 9 = 25 = (\pm 5)^2$ implies that $P_1 \in E$ (and also $(2, -5)$ i.e. $(2, 584) \in E$.)

We compute integer multiples of P_1 as follows:

Let $2P_1 = P_2(x_2, y_2)$. Then

$$\begin{aligned}x_2 &= m^2 - 2x_1 \\y_2 &= m(x_1 - x_2) - y_1 \\m &= \frac{3x_1^2 + a}{2y_1}\end{aligned}$$

Since $x_1 = 2$, $y_1 = 5$ and $a = 4$, we get

$$m = \frac{3 \cdot 2^2 + 4}{2 \cdot 5} = \frac{16}{10} \implies m = 16 \times 10^{-1} \pmod{589}$$

As 10 and 589 are relatively prime, 10^{-1} can be computed, using division algorithm. we find that

$$10^{-1} \equiv 59 \implies m \equiv 16 \times 59 \equiv 944 \equiv 355 \pmod{589}.$$

Hence

$$x_2 = m^2 - 2x_1 = (355)^2 - 2 \cdot 2 = 126025 - 4 = 126021 \equiv 564 \pmod{589}$$

$$y_2 = m(x_1 - x_2) - y_1 = 355(2 - 564) - 5 = -199515 \equiv 156 \pmod{589}$$

This gives that

$$2P_1 = P_2(x_2, y_2) = (564, 156)$$

Let $3P_1 = P_3(x_3, y_3)$ then

$$\begin{aligned} x_3 &= m^2 - x_1 - x_2 \\ y_3 &= m(x_1 - x_3) - y_1 \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \end{aligned}$$

Since $x_1 = 2, y_1 = 5, x_2 = 564, y_2 = 156$, we get

$$m = \frac{156 - 5}{564 - 2} = \frac{151}{562} = 151 \times 562^{-1} \equiv 151 \times 349 \equiv 52699 \equiv 278 \pmod{589}$$

$$x_3 = m^2 - x_1 - x_2 = 278^2 - 2 - 564 = 76718 = 148 \pmod{589}$$

$$y_3 = m(x_1 - x_3) - y_1 = 278(2 - 148) - 5 = -40593 \equiv 48 \pmod{589}$$

This gives $P_3 = 3P_1(x_3, y_3) = (148, 48)$

Next, let $P_4(x_4, y_4) = P_2 + P_2 = (564, 156) + (564, 156)$ then

$$\begin{aligned} x_4 &= m^2 - 2x_2 \\ y_4 &= m(x_2 - x_4) - y_2 \\ m &= \frac{3x_2^2 + a}{2y_2} \end{aligned}$$

Since $x_2 = 564, y_2 = 156, a = 4$, we get

$$m = \frac{3 \cdot 564^2 + 4}{2 \cdot 156} = \frac{954292}{312}$$

$$\implies m = \frac{112}{312} \equiv 112 \times 312^{-1} \equiv 112 \times 202 \equiv 22624 \equiv 242 \pmod{589}$$

$$x_4 = m^2 - 2x_2 = 242^2 - 2 \times 564 = 57436 = 303 \pmod{589}$$

$$y_4 = m(x_2 - x_4) - y_2 = 242(564 - 303) - 156 = 63006 \equiv 572 \pmod{589}$$

Hence $P_4(x_4, y_4) = (303, 572)$.

We now compute $7P$.

Let $P_7 = (x_7, y_7) = 7P = 3P + 4P = (148, 48) + (303, 572)$

$$\begin{aligned}x_7 &= m^2 - x_3 - x_4 \\y_7 &= m(x_3 - x_7) - y_3 \\m &= \frac{y_4 - y_3}{x_4 - x_3}\end{aligned}$$

Since $x_3 = 148, y_3 = 48, x_4 = 303, y_4 = 572$, we get

$$m = \frac{572 - 48}{303 - 148} = \frac{524}{155} = 524 \times 155^{-1} \pmod{589}$$

We compute the gcd, the greatest common divisor of 155 and 589. Using Euclidean Algorithm as follows:

$$\begin{aligned}589 &= 3 \times 155 + 124 \\155 &= 1 \times 124 + 31 \\124 &= 4 \times 31 + 0\end{aligned}$$

We get $\gcd(155, 589) = 31$ (as the least non zero divisor).

Since $d = \gcd(155, 589) > 1$, this implies that $155^{-1} \pmod{589}$, does not exist.

In fact $d \mid 589$. We get 31 as a factor of 589. The other factor can be easily seen to be 19. Hence $589 = 31 \times 19$.

The Algorithm by Hendrik Lenstra using Elliptic Curves for integer factorization is based on the J. M. Pollard's $p - 1$, factorization algorithm. We recall the $(p - 1)$ - algorithm.

First, the idea of Pollard's $(p - 1)$ - algorithm:

Let m be a positive odd integer greater than 1 (to be factored). Define

$$d_n = \gcd(2^{n!} - 1, m),$$

the gcd of $2^{n!} - 1$ and m for every positive integer n .

Since

$$d_{n+1} = \gcd(2^{(n+1)!} - 1, m),$$

and

$$2^{(n+1)!} - 1 = (2^{n!})^{n+1} - 1 = k^{n+1} - 1 = (k - 1)(k^n + k^{n-1} + \cdots + k + 1),$$

where $k = 2^{n!}$, gives that $2^{n!} - 1 = k - 1$ is a factor of $2^{(n+1)!} - 1$.
Hence

$$d_n | (2^{n!} - 1) \implies d_n | (2^{(n+1)!} - 1) \implies d_n | d_{n+1}, \text{ as } d_n | m.$$

Thus $d_n | d_{n+1} \quad \forall n = 1, 2, \dots$

Further, observe that if p is a prime which divides m , and is such that $p - 1$ divides $n!$, then $d_n > 1 \quad \forall n = 1, 2, \dots$. In this process, we get a proper divisor of m .

We apply this idea to $m = 403$. We compute the sequence d_n as follows:

$$\begin{aligned} d_1 &= \gcd(2^{1!} - 1, 403) = \gcd(1, 403) = 1, \\ d_2 &= \gcd(2^{2!} - 1, 403) = \gcd(3, 403) = 1, \\ d_3 &= \gcd(2^{3!} - 1, 403) = \gcd(2^6 - 1, 403) = \gcd(63, 403) = 1, \\ d_4 &= \gcd(2^{4!} - 1, 403) = \gcd(2^{24} - 1, 403) = \gcd(16777215, 403) = 13 > 1 \end{aligned}$$

We get 13 as a proper divisor of 403. The other divisor of 403 is 31. We also observe that $n = 4$ is the least positive integer such that d_n gives a proper factor of 403.

We also compute/conclude that $n = 5$ is the least n such that $d_n = d_5 = 403$.
Next, we apply this idea on $m = 589$. We compute the sequence d_n as follows:

$$\begin{aligned} d_1 &= \gcd(2^{1!} - 1, 589) = \gcd(1, 589) = 1, \\ d_2 &= \gcd(2^{2!} - 1, 589) = \gcd(3, 589) = 1, \\ d_3 &= \gcd(2^{3!} - 1, 589) = \gcd(2^6 - 1, 589) = \gcd(63, 589) = 1, \\ d_4 &= \gcd(2^{4!} - 1, 589) = \gcd(2^{24} - 1, 589) = \gcd(16777215, 589) = 1 \end{aligned}$$

What is d_5 ?

What is the least n for which 589 has a proper divisor?

What is the least n such that $d_n = 589$?

Observe that one of the major difficulty in the idea discussed is the growing size of $d_n = \gcd(2^{n!} - 1, m)$, as n increase. This can be overcome by observing that $\gcd(a, b) = \gcd(a + bx, b) \quad \forall a, b, x \in \mathbb{Z}$. So if $2^{n!} - 1 \equiv b \pmod{m}$ i.e. $2^{n!} - 1 = qm + b$ ($0 \leq b < m$), then $\gcd(2^{n!} - 1, m) = \gcd(b + mq, m) = \gcd(b, m) \implies d_m = \gcd(b, m)$. Therefore, we need not calculate exact value

of $2^{n!} - 1$ but need to know $b \equiv (2^{n!} - 1) \pmod{m}$. Since $0 \leq b < m$, it reduces computations a lot. For example d_5 for $m = 403$, can be obtained as:

$$d_4 = \gcd(2^{4!} - 1, 403) = \gcd(16677215, 403) = \gcd(325, 403) = 13$$

Next observe that

$$2^{5!} - 1 = (2^{4!})^5 - 1 = (2^{4!} - 1)\{1 + 2^{4!} + (2^{4!})^2 + (2^{4!})^3 + (2^{4!})^4\}$$

$$\begin{aligned} k &= 2^{4!} = 2^{24} = 16677216 \equiv 326 \pmod{403} \\ k^2 &= 326^2 \equiv 106276 \equiv 287 \pmod{403} \\ k^3 &= 287 \times 326 \equiv 93562 \equiv 66 \pmod{403} \\ k^4 &= 66 \times 326 = 21516 \equiv 157 \pmod{403} \end{aligned}$$

Hence

$$2^{5!} - 1 \equiv (k - 1)(k^4 + k^3 + k^2 + k + 1) \pmod{403}$$

where $k = 2^{4!} \equiv 326 \pmod{403}$, We get

$$2^{5!} - 1 \equiv (326 - 1)(157 + 66 + 287 + 326 + 1) \equiv 0 \pmod{403}$$

This implies that $d_5 = \gcd(2^{5!} - 1, 403) = \gcd(0, 403) = 403$

Remark: If $a = 3$, then $a^{3!} = 3^6 = 729 = 326 \pmod{403}$. Hence in this case $d_3 = \gcd(a^{3!} - 1, 403) = \gcd(326, 403) = 13$ and $d_4 = 403$.

We generalize the idea little further. In d_n , instead of taking integer $2^{n!} - 1$, we take $a^{n!} - 1$, where a is an arbitrary positive integer greater than 1. Define the sequence $\{g_n\}$ as follows:

$$g_n = \gcd(a_n - 1, m)$$

where m is the positive integer asked to be factored, and the a'_n s are define as

$$a_n = a_{n-1}^{r_{n-1}}, \text{ with } a_1 = a \quad \forall n \geq 2,$$

where $\{r_k\}$ is a sequence of positive integer greater than 1. Thus

$$\begin{aligned} g_1 &= \gcd(a_1 - 1, m) = \gcd(a - 1, m), \\ g_2 &= \gcd(a_2 - 1, m) = \gcd(a_1^{r_1} - 1, m) = \gcd(a^{r_1} - 1, m), \\ g_3 &= \gcd(a_3 - 1, m) = \gcd(a_2^{r_2} - 1, m) = \gcd((a_1^{r_1})^{r_2} - 1, m) = \gcd(a^{r_1 r_2} - 1, m), \\ &\vdots \\ g_n &= \gcd(a_n - 1, m) = \gcd(a^{r_1 r_2 \cdots r_{n-1}} - 1, m) \quad \forall n \geq 2 \\ g_{n+1} &= \gcd(a_{n+1} - 1, m) = \gcd(a_n^{r_n} - 1, m) \end{aligned}$$

Now

$$\begin{aligned} g_n = \gcd(a_n - 1, m) &\implies g_n \mid a_n - 1 \text{ and} \\ g_n \mid m &\implies g_n \mid (a_n^{r_n} - 1) \quad \because (a_n - 1) \mid (a_n^{r_n} - 1) \\ &\implies g_n \mid g_{n+1} \quad \forall n \geq 1 \\ \text{That is } &g_1 \mid g_2, g_2 \mid g_3, \dots \end{aligned}$$

Thus, we wish to determine one n (least positive integer in fact) such that $1 \leq g_n \leq m$, as this makes g_n to be a proper factor of m .

For $r_n = n \quad \forall n \in \mathbb{N}$ and $a = 2$,

$$\begin{aligned} g_n &= \gcd(a^{r_1 r_2 \cdots r_{n-1}} - 1, m) \\ &= \gcd(2^{1 \cdot 2 \cdot 3 \cdots (n-1)} - 1, m) \\ &= \gcd(2^{(n-1)!} - 1, m) = d_{n-1} \end{aligned}$$

in our earlier discussion.

Next, suppose p is a prime such that $p \mid m$, and $(p-1) \mid r_1 r_2 \cdots r_{n-1}$ then

$$a_n = a^{r_1 r_2 \cdots r_{n-1}} = [a^{(p-1)}]^t \equiv 1 \pmod{p},$$

where $(p-1)t = r_1 r_2 \cdots r_{n-1} \implies p \mid a_n - 1$. Since $p \mid m$, we get $p \mid g_n$ and we get a proper factor of m .

Can there be a better choice for r_n other than n ?

Let $\{q_n\}$ be a sequence of all the positive powers of all the primes arranged in the increasing manner. Suppose $\{r_n\}$ is the sequence of positive integers such that $r_n \mid q_n$ and r_n is a prime $\forall n \in \mathbb{N}$.

For example: Consider positive integers 1 to 100. The terms of the sequence $\{q_n\}$ are:

$$q_1 = 2^1, q_2 = 3^1, q_3 = 2^2, q_4 = 5^1, q_5 = 7^1, q_6 = 2^3, q_7 = 3^2, q_8 = 11^1, q_9 = 13^1, q_{10} = 17$$

The terms of the sequence r_n are:

$$r_1 = 2, r_2 = 3, r_3 = 2, r_4 = 5, r_5 = 7, r_6 = 2, r_7 = 3, r_8 = 11, r_9 = 13, r_{10} = 17$$

Observe that: lcm , the least common multiple of the numbers 1 to 100, i.e

$$[1, 2, 3, \dots, 10] = lcm[q_1, q_2, \dots, q_7] = r_1, r_2, \dots, r_7.$$

Prove in general that: $lcm[1, 2, 3, \dots, q_n] = lcm[q_1, q_2, \dots, q_n] = r_1, r_2, \dots, r_n \quad \forall n \in \mathbb{N}$.

Since $r_1 r_2 \dots r_n \leq n!$, we expect less number of calculations. Obviously, Pollard's $(p-1)$ -algorithm is most efficient, when $p-1$ has a large number of small prime factors.

We summarize the algorithm as follows:

Input: The integer m (to be factored)

1. Set a positive integer greater than 1. It can be 2 itself.
2. Loop $j = 2, 3, 4, \dots$, up to specific bound $\mathcal{B}$
3. Set $a = a^j \pmod{m}$
4. Compute $d = gcd(a-1, m)$
5. If $1 \leq d \leq m$, then return d as a proper factor of m .
6. If not increment j and loop again at step 2.

Note 1. Hopping can be done to increase efficiency by computing (in step 4) gcd only for every k -th position, for some suitable k .

Thus

$$\begin{aligned} 2^{51} - 1 &= 325(157 + 66 + 287 + 326 + 1) \\ &= 325 \times 837 = 272025 \\ &\equiv 0 \pmod{403} \quad \text{as } 272025 = 675 \times 403 \end{aligned}$$

$$\implies d_5 = \gcd(2^{5!-1}, 403) = \gcd(0, 403) = 403$$

Of course, you could also have argued as follows:

Since $d_4 < d_5$, $d_4 \mid d_5$, $d_5 \leq 403$, and $d_4 = 13$, we get $d_5 = 403$.

Do this for $m = 589$.

Compute d_{120} for $m = 3870573781$.

Remark 1: In calculating $2^{n!} - 1 \pmod{m}$, we can use that

$$a \equiv b \pmod{m} \implies a + c \equiv b + c \pmod{m}.$$

Thus if $2^{n!} \equiv r \pmod{m}$, then $2^{n!} - 1 \equiv r - 1 \pmod{m}$. Apply this in proving that

$$2^{4!} \equiv 326 \pmod{403} \implies 2^{4!} - 1 \equiv 326 - 1 \equiv 325 \pmod{403}.$$

To do this use the method of repeated squaring and multiplying if required. observe that

$$\begin{aligned} 2^{4!} &= (2^{3!})^4 = (2^6)^4 \\ \text{and that } 2^6 &= 64 \implies (2^6)^2 = 64^2 = 4096 \equiv 66 \pmod{403} \\ &\implies (2^6)^4 \equiv 66^2 \equiv 4356 \equiv 326 \pmod{403} \end{aligned}$$

$$\text{Similarly } 2^{5!} = (2^{4!})^5 = ((2^{4!})^2)^2 (2^{4!})$$

14.2 Lenstra's Elliptic Curve Factorization Algorithm

Recall the Pollard's $(p-1)$ factorization algorithm of a positive integer $m > 1$. The crux of algorithm was a prime factor p of m such that $p-1$, has small prime factors and $(p-1) \mid (a^{r_1 r_2 \cdots r_{n-1}})$ where r_i 's are primes not exceeding a bound, at most the primes occurring in $n! \forall n \in \mathbb{N}$. The condition $(p-1) \mid (a^{r_1 r_2 \cdots r_{n-1}})$ implies $a^{p-1} \equiv 1 \pmod{p}$. This happens in the multiplicative group $\mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\}$, that $\alpha^{p-1} \equiv 1 \pmod{p}$, for every $0 \neq \alpha \in \mathbb{F}_p$. Lenstra replaced the multiplicative group by the additive Abelian group of an Elliptic Curve modulo n , where n is the composite number to be factored.

Thus an Elliptic Curve $E := \{(x, y) \in \mathbb{Z}_n \times \mathbb{Z}_n \mid y^2 = x^3 + Ax + B; A, B \in$

$\mathbb{Z}_n\} \cup \{\infty\}$ is searched and a point $P(a, b)$ on E is found. Multiples of P are computed till the computation fails. In that case, a proper factor is found. otherwise, start with a new curve E and a new point P on E .

How to choose a curve E ?

Start with a point $P(a, b)$, compute $B \equiv b^2 - a^3 - Aa \pmod{n}$ for randomly chosen A, a, b so that $B + Aa + a^3$ is a quadratic residue $\pmod{n}$.

The Algorithm (Lenstra):

Input: Integer n to be factored.

1. Choose random values A, a and $b \pmod{n}$
2. Set $P = (a, b)$ and $B \equiv b^2 - a^3 - Aa \pmod{n}$.
3. Loop $j = 2, 3, \dots$ up to a specified bound
4. Compute $Q = jP \pmod{n}$ and set $P = Q$.
5. If computation in step 4 fails, there exist a proper factor d of n .
6. If $d < n$, then success, return d .
7. If $d = n$, go to step 1 and choose a new curve and a new point P on E .
8. Increment j and loop again at step 2.

Example: Factorize $n = 589$ using Lenstra's Elliptic Curve Algorithm.

Let $P = (2, 5)$ i.e. $a = 2, b = 5$ and $A = 4$ then

$$B = b^2 - a^3 - Aa \pmod{589} \equiv 5^2 - 2^3 - 4 \cdot 2 = 25 - 8 - 8 = 9 = 3^2 \pmod{589}.$$

Hence choose the curve

$$E := \{(x, y) \in \mathbb{Z}_{589} \times \mathbb{Z}_{589} \mid y^2 = x^3 + Ax + B; A, B \in \mathbb{Z}_{589}\} \cup \{\infty\}; A = 4, B = 9$$

Then clearly $P(2, 5) \in E$. let $P = (x_1, y_1)$, $x_1 = 2, y_1 = 5$

Recall:

$$2P = (x_2, y_2) = (564, 156), \quad \text{so}$$

$$(2!)P = P_2 = (x_2, y_2) = (564, 156)$$

$$3(2P) = (3!)P = 6P = P_6(x_6, y_6) = 2P + 4P = (564, 156) + (303, 572)$$

We have calculated $4P = (x_4, y_4) = 2P_2 = (303, 572)$

$$\begin{aligned}
 x_6 &= m_6^2 - x_2 - x_4 \\
 y_6 &= m_6(x_2 - x_6) - y_2 \text{ where} \\
 m_6 &= \frac{y_4 - y_2}{x_4 - x_2} \\
 &= \frac{572 - 156}{303 - 564} = \frac{416}{-261} \\
 \Rightarrow m_6 &= \frac{416}{328} \equiv 416 \times 328^{-1} \pmod{589} \\
 &\equiv 416 \times 422 \equiv 30 \pmod{589} \\
 \Rightarrow m_6^2 &\equiv 900 \equiv 311 \pmod{589}. \\
 \Rightarrow x_6 &= 900 - 564 - 303 \equiv 311 - 564 - 303 \\
 &\equiv -556 \equiv 33 \pmod{589} \\
 y_6 &= 311(564 - 33) - 572 = 311 \times 531 - 572 \\
 &= 165141 - 572 = 164569 \equiv 238 \pmod{589}
 \end{aligned}$$

Hence $(3!)P = 6P = P_6 = (x_6, y_6) = (33, 238)$.

Now $(4!)P = 4(3!)P = 4Q$ where $Q = P_6 = (33, 238)$

Now $Q_1 = (u_1, v_1) = (33, 238)$.

Let $2Q = Q_2 = (u_2, v_2)$. Then

$$u_2 = m_2^2 - u_1 - u_2, \quad v_2 = m_2(u_1 - u_2) - v_1$$

where

$$\begin{aligned}
 m_2 &= \frac{3u_1^2 + A}{2u_1} = \frac{3 \cdot 33^2 + 4}{2 \cdot 238} \\
 &= \frac{3271}{476} \equiv 3271 \times 476^{-1} \pmod{589} \\
 \Rightarrow m_2 &= 3271 \times 172 \equiv 562612 \equiv 117 \pmod{589} \\
 \Rightarrow m_2^2 &\equiv 117^2 \equiv 13689 \equiv 142 \pmod{589} \\
 \Rightarrow u_2 &\equiv 142 - 33 - 33 \equiv 142 - 66 \pmod{589} \equiv 76 \pmod{589} \\
 v_2 &\equiv 117(33 - 76) - 238 \equiv 117 \times (-43) - 238 \pmod{589} \\
 &\equiv -5269 \equiv 32 \pmod{589} \\
 \Rightarrow 2Q &= Q_2 = (u_2, v_2) = (76, 32)
 \end{aligned}$$

$$\begin{aligned}
&\implies 4Q = Q_4 = 2Q + 2Q = Q_2 + Q_2 = (76, 32) + (76, 32) \\
&\implies u_4 = m_4^2 - 2u_2, \quad v_4 = m_4(u_2 - u_4) - v_2, \\
&m_4 = \frac{3 \cdot u_2^2 + 4}{2 \cdot v_2} = \frac{3 \cdot 76^2 + 4}{2 \cdot 32} \\
&m_4 = \frac{17332}{64} = \frac{251}{64} \equiv 251 \times 64^{-1} \pmod{589} \\
&\quad \equiv 251 \times 543 \equiv 234 \pmod{589} \\
&m_4^2 \equiv 234^2 \equiv 568 \pmod{589} \\
&\implies u_4 = m_4^2 - 2u_2 = 568 - 2 \times 76 = 568 - 152 \equiv 396 \pmod{589} \\
&\quad v_4 \equiv 568(76 - 396) - 32 = -181792 \equiv 209 \pmod{589}
\end{aligned}$$

Hence $(4!)P = 4Q = (396, 209)$

So far, we have been able to compute $P, (2!)P, (3!)P, (4!)P$. Continue as per the Lenstra's Algorithm and determine non trivial factor of 589, if there is any.

14.3 Elliptic Curve Primality Test

Most of the Elliptic Curve methods are based on classical methods. For example, Lenstra's Elliptic Curve in factorization is based on the Pollard's(p-1) factorization method. Likewise, the Elliptic Curve primality test (Pocklington or Pocklington- Lehmar Primality Test is the basis due to Goldwasser, Kilian and Alkin). We shall therefore first discuss the Pocklington-Lehmar Primality test. Usually, it is lot more easier to prove that a given positive integer is not a prime (also called a composite number) rather than prime. For example $a(1 < a \in \mathbb{N})$ is a composite number then there exist positive integer b, c such that $1 < b < a, 1 < c < a$ and $a = bc$. Let $b \leq c$. Then $1 < b \leq \sqrt{a}$, and by the fundamental theorem of arithmetic, that every positive integer greater than 1 is a product of primes, there exists a prime p such that $p|b$. Thus p is a prime such that $1 < p \leq b \leq \sqrt{a}; p|b, b|a \implies p|a$.

Conclusion: If $1 < a \in \mathbb{N}$ is composite, then a has a prime factor not exceeding $\sqrt{a}$.

Hence if $a \in \mathbb{N}$ does not have a prime factor $p (\leq \sqrt{a},)$ then a is a prime.

For example $a = 5179$ is a prime, since none of the primes up to $\sqrt{a}$, that is 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 71 (as

$\sqrt{a} \leq 72$), divide a . Of course, the availability of primes up to $\sqrt{a}$, when a is large could be a problem.

Proposition 1: Let $n > 1$ be a positive integer such that there exist a prime $q > \sqrt{n} - 1$ and $q|(n-1)$. If there exists an integer a such that

1. $a^{n-1} \equiv 1 \pmod{n}$
2. $\gcd(a^{\frac{n-1}{q}} - 1, n) = 1$

then n is prime.

Proof: Suppose contrary, that n is composite, then as discussed above there exists a prime $p \leq \sqrt{n}$ such that $p|n$. Now $q > \sqrt{n} - 1 \geq p - 1 \implies q > p - 1 \implies \gcd(q, p-1) = 1 \implies u, v \in \mathbb{Z}$ such that $uq + v(p-1) = 1$ this implies that

$$\begin{aligned} a^{\frac{n-1}{q}} &= a^{1 \cdot \frac{n-1}{q}} = a^{[uq+v(p-1)] \frac{n-1}{q}} \\ &= a^{uq \cdot \frac{n-1}{q}} \cdot a^{v(p-1) \frac{n-1}{q}} \\ &\equiv a^{uq \cdot \frac{n-1}{q}} \pmod{p} \end{aligned}$$

since $a^{v(p-1) \frac{n-1}{q}} = (a^{p-1})^{v \cdot \frac{n-1}{q}} \equiv 1 \pmod{p}$ (using Fermat's little Theorem , also $\frac{n-1}{q} \in \mathbb{N}$ as $q|(n-1)$).

$$\begin{aligned} \implies a^{\frac{n-1}{q}} &\equiv a^{uq \cdot \frac{n-1}{q}} \pmod{p} \\ &\equiv a^{u(n-1)} \pmod{p} \\ &\equiv (a^{n-1})^u \pmod{p} \\ &\equiv 1 \pmod{p} \quad \text{Since } p|n \text{ and } a^{n-1} \equiv 1 \pmod{n} \\ \implies p &| a^{\frac{n-1}{q}} - 1 \\ \implies \gcd(a^{\frac{n-1}{q}} - 1, n) &\geq p > 1 \text{ as } p|n \end{aligned}$$

This is a contradiction to the second condition, Hence, n is a prime.

The Pocklington-Lehmar Test, stated above, is certain to return n as a prime, unlike the probabilistic primality test such as Miller-Rabin and many others.

The Elliptic Curve Primality Test due to Golawasser and Kilian is based

on the following proposition, which is analogous to the Pocklington-Lehmar primality test.

Proposition 2: Let n be a positive integer, and let

$$E := \{(x, y) \in \mathbb{Z}_n \times \mathbb{Z}_n \mid y^2 = x^3 + Ax + B; A, B \in \mathbb{Z}_n\} \cup \{\infty\}$$

Also let $m \in \mathbb{Z}$ be an integer. Further, let there exist a prime q such that $q \mid m$ and $q > (n^{\frac{1}{4}} + 1)^2$. If there exists a point $P \in E$ such that

1. $mP = \infty$
2. $\left(\frac{m}{q}\right)P \neq \infty$

is defined, then n is a prime.

Proof: Again assume n is composite, then there exists a prime p such that $p \mid n$ and $p \leq \sqrt{n}$. Let E_p be the Elliptic Curve E , defined mod p . That is:

$$E_p := \{(x, y) \in \mathbb{Z}_p \times \mathbb{Z}_p \mid y^2 = x^3 + A_px + B_p; A_p, B_p \in \mathbb{Z}_p\} \cup \{\infty\}$$

Same A, B but now mod p . That is

$$\begin{aligned} A &\equiv A_p \pmod{p} \\ B &\equiv B_p \pmod{p} \end{aligned}$$

Let $\#(E_p) = m_p$ (order of the group E_p). By Hasse's Theorem:

$$\begin{aligned} |p + 1 - m_p| &\leq 2\sqrt{p} \\ \implies m_p &\leq p + 1 + 2\sqrt{p} = (\sqrt{p} + 1)^2 \\ \implies m_p &\leq (\sqrt{p} + 1)^2 \leq \left(n^{\frac{1}{4}} + 1\right)^2 < q \text{ since } p \leq \sqrt{n} \\ \implies 1 &\leq m_p < q \\ \implies \gcd(q, m_p) &= 1 \\ \implies 1 &= uq + vm_p \text{ for some integer } u, v \in \mathbb{Z} \\ \implies uq &\equiv 1 \pmod{m_p}, \text{ for some } u \in \mathbb{Z}_p \end{aligned}$$

Since $P = P(x, y) \in E$, we get $Q(x', y') \in E_p$, where $x, y \in \mathbb{Z}_n, x', y' \in \mathbb{Z}_p$ are such that $x \equiv x' \pmod{p}, y \equiv y' \pmod{p}$.

Now

$$q \mid m \implies \frac{m}{q} \in \mathbb{Z} \implies \left(\frac{m}{q}\right)Q = uq \left(\frac{m}{q}Q\right) = (um)Q = u(mQ) = \infty$$

Since $mP = \infty$ on E and Q is the same point considered mod p , instead mod n . Thus $\left(\frac{m}{q}\right) Q = \infty$ on E_p .

But this is a contradiction on to the condition 2. As this will lead to $\left(\frac{m}{q}\right) P = \infty$. Because, if $\left(\frac{m}{q}\right) P$ is defined and not ∞ on E , then as earlier $\left(\frac{m}{q}\right) Q \neq \infty$. The proposition is proved.

Remark: Even without knowing n (prime or composite), an Elliptic Curve E and a point P on E can be found, point addition (that includes doubling of a point) may lead to the factorization of n , and thus confirming n to be a composite.

14.4 Discrete Logarithm Problem on an Elliptic Curve

Let $E_{\mathbb{F}}$ be an elliptic curve on a field $\mathbb{F}$. If P, Q are two points on E such that $Q = xP = P + P + \cdots + P$ (x -copies) for some non-negative integer x , and x is such least value, then it is called the discrete logarithm of Q to the base (point) P , and written as $x = dlog_P Q$ (in E).

If no such x - exists, then it is said that $dlog_P Q$ in E , does not exist. The discrete logarithm problem is said to be solvable in an elliptic curve E with respect to the base point P on E , if for every $Q \in E$, $dlog_P Q$ exists. This of course is the case if E is an additive (or multiplicative) cyclic group and P is a generator with respect to addition (or multiplication). But this may not be the case if the group is not cyclic, even if it is Abelian.

Example: Let $E := \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$.

If $P(1, 4)$, then $P \in E$ as

$$1^3 + 3 \times 1 + 1 = 5 \equiv 16 \equiv (\pm 4)^2$$

$$\implies (1, 4) \text{ and } (1, -4) = (1, 7) \in E.$$

And $-(1, 4) = (1, -4)$. Also $2(1, 4) = (2, 9) \in E$ as can be seen by the addition of point formula.

So $dlog_P Q = 2$ where $Q = (2, 9), P = (1, 4)$. This can be verified that the DLP in E with respect to $P(1, 4)$ is solvable, as $dlog_P Q$ in E exists for every

$Q \in E$, since E is an additive cyclic group and P is a generator of this group. Determine $dlog$ of $(6, 9)$ in E with respect to the base $(1, 4)$. Does $dlog_{(2,9)}(6, 2)$ in E exist? If yes, then what is it?

Example: Let $E := \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 4x + 5\} \cup \{\infty\}$.

If $Q = (0, 7), R = (3, 0)$, then it can be verified that $Q, R \in E$

Further, if $P = (0, 4)$, then $P \in E$.

Verify that $3P = (0, 7) = Q$. hence $dlog_P Q = 3$, also $2P = (3, 0) = R \implies dlog_P R = 2$. Also, verify that if $B = (9, 0)$, then also $B \in E$, and $2B = \infty$. Hence $dlog_B Q$ as well as $dlog_B R$ does not exist.

Write all pairs P, Q of points on E , for which $dlog_P Q$ exists, as well as the pairs for which it does not exist in above two examples.

Algorithms to Compute Discrete Logarithm in Elliptic Curves

All the algorithms (except the index calculus method) to compute discrete logarithm in a multiplicative cyclic group can be applied to the additive Abelian group E (of elliptic curve) by suitably changing the multiplicative operation to the additive operation. We shall therefore, first recall the multiplicative cyclic group $G = \langle g \mid g^n = 1 \rangle$ and algorithms to compute $dlog_g h$ in G and then restate its Elliptic Curve group additive analogue, if required.

1. **Enumeration:** Let $E_{\mathbb{F}}$ be an elliptic curve $P, Q \in E_{\mathbb{F}}$. To compute $dlog_P Q$ in E , simply compute kP , $k = 1, 2, 3, \dots$ if there exists such a k and is least with the property $kP = Q$, then $x = k = dlog_P Q$ in E . If no such k exists, then $dlog_P Q$ in E does not exist. This method always works, and is quite satisfactory of small order elliptic curves. For example, $dlog_{(1,4)}(2, 9) = 2$ in

$$E := \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 3x + 3\} \cup \{\infty\}.$$

The problem is quite time consuming if the order of the group E is large. Compute the order of the group (of points)

$$E := \{(x, y) \in \mathbb{F}_{97} \times \mathbb{F}_{97} \mid y^2 = x^3 + 14x + 19\} \cup \{\infty\}$$

Let $P = (57, 58)$ and $Q = (56, 48)$

- (a) Is $P \in E$?
- (b) Is $Q \in E$?
- (c) Does $dlog_P Q$ exist? If yes, then compute it.
- (d) Does $dlog_Q P$ exist? If yes, then compute it.
- (e) What is the group structure E ?

14.4.1 Shanks Baby Step Giant Step Algorithm for Elliptic Curves

First, recall the algorithm in a multiplicative cyclic group $G = \langle g \mid g^n = 1 \rangle$ to compute $x = dlog_g h$ in G . Set m to be the least positive integer greater or equal to $\sqrt{n}$.

By division algorithm/Euclidean algorithm there exists integers q and r such that

$$\begin{aligned} x &= qm + r \quad (0 \leq r < m) \\ \implies h &= g^x = g^{qm+r} = g^{qm} \times g^r = (g^m)^q \times g^r \\ \implies (g^m)^q &= hg^{-r} \end{aligned}$$

Compute the Baby steps $B = \{(hg^{-r}, r) \mid 0 \leq r < m\}$ i.e.

$$B = \{(h, 0), (hg^{-1}, 1), (hg^{-2}, 2), \dots, (hg^{-(m-1)}, m-1)\}$$

There are m -Baby steps. If $(1_G, s) \in B$ for some $0 \leq s < m$, then

$$hg^{-s} = 1_G \implies h = g^s \implies s = dlog_g h$$

and we are done.

If $(1_G, s) \notin B$ for any $s = 0, 1, \dots, m-1$ then compute the Giant steps,

$$g^m, (g^m)^2 = g^{2m}, (g^m)^3, \dots$$

If $(g^m)^q = g^{mq}$, is the first component in some Baby step with second component r , then

$$(g^{mq}, r) \in B \implies g^{mq} = hg^{-r} \implies g^{mq+r} = h \implies x = dlog_g h$$

Algorithm:

We can now state the analogue of Shanks Baby step-Giant step algorithm for elliptic curves.

Let E be an elliptic curve of order $n = \#(E)$. Suppose $P, Q \in E$ are such that $Q = kP$ and k is the least non negative integer with this property. Then $dlog_P Q = k$.

So, we set $m =$ the least integer greater or equal to $\frac{n}{2}$. Then $k = mq + r$, for some integer q, r such that $0 \leq r < m$

We get

$$Q = kP = (mq + r)P = (mq)P + rP = q(mP) + rP \implies Q - rP = q(mP)$$

Baby steps $B := \{(Q - rP, r) \mid 0 \leq r < m\}$ that is

$$B = \{(Q, 0), (Q - P, 1), (Q - 2P, 2), \dots, (Q - (m-1)P, m-1)\} \subseteq E \times \mathbb{Z}$$

m -Baby steps.

Now, if there is a Baby step $(\infty, j) \in B$ for some j ($0 \leq j \leq m-1$), then

$$Q - jP = \infty \implies Q = jP + \infty = jP \text{ and } j = dlog_P Q.$$

If there is no such Baby step then compute Giant steps:

$$mP, 2(mP), 3(mP), \dots$$

The moment the Giant step $q(mP)$ is a point in the first component of a Baby step $(q(mP), r) \in B$ then

$$\begin{aligned} q(mP) = Q - rP &\implies Q = q(mP) + rP = (qm + r)P \\ &\implies k = qm + r = dlog_P Q \in E. \end{aligned}$$

Since $dlog_P Q$ exists, therefore, such integers q and r exist.

Example: Let $E := \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$

Suppose $P(1, 4)$ and $Q = (3, 2)$. Then $P, Q \in E$.

Recall $n =$ order of E is 18 and $E \cong C_{18} = \langle P = (1, 4) \rangle$. Then $dlog_P Q$ exists.

Let $k = dlog_P Q$. Then $Q = kP$.

We compute k using the Shanks Baby step Giant step algorithm as follows:

Set $m = \frac{18}{2} = 9$.

Now $k = qm + r$ for some integers q, r , $k = 9q + r$ ($0 \leq r < m$) Baby step

$$B := \{(Q, 0), (Q - P, 1), (Q - 2P, 2), \dots, (Q - 8P, 8)\}$$

Observe that $P = (1, 4) \implies -P = (1, -4) = (1, 7) = R$ (say)

$$B := \{(Q, 0), (Q + R, 1), (Q + 2R, 2), \dots, (Q + 8R, 8)\}$$

where $Q = (3, 2)$ and $R = (1, 7)$

The Baby steps:

$$\begin{aligned} B &:= \{((3, 2), 0), ((3, 2) + (1, 7), 1), ((3, 2) + 2(1, 7), 2), ((3, 2) + 3(1, 7), 3), ((3, 2) + 4(1, 7), 4), \\ &\quad ((3, 2) + 5(1, 7), 5), ((3, 2) + 6(1, 7), 6), ((3, 2) + 7(1, 7), 7), ((3, 2) + 8(1, 7), 8)\} \\ &= \{((3, 2), 0), ((3, 2) + (1, 7), 1), ((3, 2) + (2, 2), 2), ((3, 2) + (0, 10), 3), ((3, 2) + (8, 3), 4), \\ &\quad ((3, 2) + (3, 2), 5), ((3, 2) + (5, 3), 6), ((3, 2) + (6, 9), 7), ((3, 2) + (9, 3), 8), ((3, 2), 0), \\ &\quad ((5, 3), 1), ((6, 9), 2), ((9, 3), 3), ((4, 0), 4), ((9, 8), 5), ((6, 2), 6), ((5, 8), 7), ((3, 9), 8)\} \end{aligned}$$

We observe that the point at infinity ∞ is not appearing as the first coordinate in any of the Baby step.

Hence, we need to compute the Giant steps as follows:

$$mP, 2(mP), 3(mP), \dots \text{ i.e. } 9P, 2(9P), 3(9P), \dots$$

$$P = (1, 4) \implies 9P = (4, 0)$$

So the Giant-steps are:

$$(4, 0), 2(4, 0), 3(4, 0), \dots \text{ i.e. } (4, 0), \infty, (4, 0), \dots$$

since order of $(4, 0)$ is 2.

We also observe that $((4, 0), 4)$ is a Baby-step.

Hence $q = 1, r = 4, m = 9$

$$\implies k = qm + r = 1 \times 9 + 4 = 13 \implies d\log_P Q = 13$$

This can be directly too verified as

$$P = (1, 4), 2P = (2, 9), 4P = (8, 8), 8P = (9, 8).$$

We calculate using doubling repeatedly and adding if required.

$$\text{Hence } 13P = 8P + 4P + P = (9, 8) + (8, 8) + (1, 4) = (5, 3) + (1, 4) = (3, 2) = Q$$

14.4.2 Pollard's Rho Algorithm For Discrete Logarithm

Recall the Pollard's ρ algorithm for computing discrete logarithm in a finite cyclic group of an element w.r.t its generator. We give here its analogue for the group E (the additive group of an elliptic curve E) as follows:

Let E be an Elliptic Curve, P, Q be points on E . Then $dlog_P Q$ exists if $Q \in \langle P \rangle \leq E$ where $G = \langle P \rangle$, denotes the subgroup of E (w.r.t. point addition in E).

So suppose, $|G| = n$, and $Q \in G = \langle P | nP = \infty \rangle$. Then $dlog_P Q$ exists. That is there is exist an integer $k \geq 0$, such that $kP = Q$ ($0 \leq k < n$), and k is least with this property. That means $k = dlog_P Q$ in G and thus in E . Now divide G into three pairwise disjoint subsets, G_1, G_2, G_3 such that G is their union. That is

$$G = G_1 \cup G_2 \cup G_3 \quad (G_i \cap G_j = \phi \text{ for } i \neq j)$$

Define a function

$$f : G = G_1 \cup G_2 \cup G_3 \rightarrow G$$

$$R \rightarrow f(R) = \begin{cases} P + R, & \text{if } R \in G_1 \\ 2R, & \text{if } R \in G_2 \\ Q + R, & \text{if } R \in G_3 \end{cases}$$

Choose a random integer $m_0 \in \{1, 2, \dots, n\}$. Let $R_0 = m_0P$, $R_1 = f(R_0)$, $R_{i+1} = f(R_i) \quad \forall i \geq 0$. This suggest that $R_i = m_iP + n_iQ \quad i \geq 0, n_0 = 0$

$$\Rightarrow R_1 = f(R_0) = \begin{cases} P + m_0P = (m_0 + 1)P & \text{if } R_0 \in G_1 \\ 2m_0P & \text{if } R_0 \in G_2 \\ m_0P + Q & \text{if } R_0 \in G_3 \end{cases}$$

By induction, prove that (assuming $R_i = m_iP + n_iQ$)

$$R_{i+1} = f(R_i) = f(m_iP + n_iQ) = m_{i+1}P + n_{i+1}Q \quad \forall i \geq 0, n_0 = 0$$

$$R_{i+1} = \begin{cases} m_{i+1}P + n_iQ & \text{if } R_i \in G_1 \\ 2m_{i+1}P + 2n_{i+1}Q & \text{if } R_i \in G_2 \\ m_iP + n_{i+1}Q & \text{if } R_i \in G_3 \end{cases}$$

Thus,

$$R_{i+1} = m_{i+1}P + n_{i+1}Q,$$

where

$$m_{i+1} = \begin{cases} m_i + 1, & \text{if } R_i \in G_1 \\ 2m_i, & \text{if } R_i \in G_2 \\ m_i, & \text{if } R_i \in G_3 \end{cases}$$

and

$$n_{i+1} = \begin{cases} n_i, & \text{if } R_i \in G_1 \\ 2n_i, & \text{if } R_i \in G_2 \\ n_i + 1, & \text{if } R_i \in G_3 \end{cases}$$

Since, $P, Q \in G$ and each $R_i = m_i P + n_i Q$ is an integer linear combination of P and Q , we get each $R_i \in G$.

Now $\#(E)$ is finite this implies $\#(G)$ is also finite, this gives that the points $R_0, R_1, R_2, \dots$ thus constructed in G can not be distinct. So for some integer $j \geq 0$, $R_i = R_{i+j}$. But then

$$\begin{aligned} m_i P + n_i Q &= m_{i+j} P + n_{i+j} Q \\ \implies (m_i - m_{i+j})P &= (n_{i+j} - n_i)Q \end{aligned}$$

If $k = d \log_P Q$ in G , then $Q = kP$, and we get

$$\begin{aligned} (m_i - m_{i+j})P &= (n_{i+j} - n_i)kP \\ \implies k &= (n_{i+j} - n_i)^{-1}(m_i - m_{i+j}) \text{ (if } \gcd(n_{i+j} - n_i) \text{ and } n \text{ is } 1). \end{aligned}$$

If $\gcd((n_{i+j} - n_i), n) = d > 1$, then $\gcd((n_{i+j} - n_i), \frac{n}{d}) = 1$, and $(n_{i+j} - n_i)^{-1}$ exists $(\text{mod } \frac{n}{d})$. In that case

$$k \equiv (n_{i+j} - n_i)^{-1}(m_i - m_{i+j}) \pmod{\frac{n}{d}}$$

will give d number of choices for $k = d \log_P Q$, namely

$$k, k + \frac{n}{d}, k + \frac{2n}{d}, \dots, k + \frac{(d-1)n}{d},$$

i.e. $k + \frac{sn}{d}$, $0 \leq s \leq d-1$

Example. Compute if it exists $d \log_P Q$ in the curve

$$E = \{(x, y) \in F_{11} \times F_{11} | y^2 = x^3 + 3x + 1\} \cup \{\infty\}$$

where $P = (1, 4)$, and $Q = (3, 2)$. It can be easily verified that $P, Q \in E$ (and $d \log_P Q = k$ exists, as we know, but otherwise, we can assume that it

exists and proceed). Since $\#(E)$ is small, in fact we have already computed all points on E , and saw that $\#(G)=\#(E) = 18$, where $G = \langle P \rangle$. The group G can be written as the disjoint union.

$$\begin{aligned} E = G &= \{(0, 1), (0, 10), (1, 4), (1, 7), (2, 2), (2, 9)\} \\ &\cup \{(3, 2), (3, 9), (4, 0), (5, 3), (5, 8), (6, 2)\} \\ &\cup \{(6, 9), (8, 3), (8, 8), (9, 3), (9, 8), \infty\} \\ &= G_1 \cup G_2 \cup G_3 \end{aligned}$$

where

$$\begin{aligned} G_1 &= \{(0, 1), (0, 10), (1, 4), (1, 7), (2, 2), (2, 9)\} \\ G_2 &= \{(3, 2), (3, 9), (4, 0), (5, 3), (5, 8), (6, 2)\} \\ G_3 &= \{(6, 9), (8, 3), (8, 8), (9, 3), (9, 8), \infty\} \end{aligned}$$

Let $m_0 = 2, \quad n_0 = 0, \quad R_0 = m_0P + n_0Q \implies R_0 = 2P = 2(1, 4) = (2, 9)$
Define

$$\begin{aligned} f : G &\rightarrow G \\ R \mapsto f(R) &= \begin{cases} P + R, & \text{if } R \in G_1 \\ 2R, & \text{if } R \in G_2 \\ Q + R, & \text{if } R \in G_3 \end{cases} \end{aligned}$$

Then $R_{i+1} = f(R_i) \quad \forall i \geq 0 \implies R_1 = f(R_0) = P + R_0$ as $R_0 \in G_1$ This gives $R_1 = (1, 4) + (2, 9) = (0, 1)$
Since $R_1 \in G_1$, we get

$$R_2 = f(R_1) = P + R_1 \quad \text{i.e.} \quad R_2 = (1, 4) + (0, 1) = (8, 8)$$

As $R_2 = (8, 8) \in G_3$, we get

$$R_3 = f(R_2) = Q + R_2 = (3, 2) + (8, 8) = (1, 7)$$

Since $R_3 = (1, 7) \in G_1$, we get

$$R_4 = f(R_3) = P + R_3 = (1, 4) + (1, 7) \implies R_4 = \infty$$

Observe that $R_4 = \infty \in G_3$, we get

$$R_5 = f(R_4) = Q + R_4 = (3, 2) + \infty \implies R_5 = (3, 2)$$

Since $R_5 = (3, 2) \in G_2$, we get

$$R_6 = f(R_5) = 2R_5 = 2(3, 2) \implies R_6 = (9, 8)$$

We see that $R_6 = (9, 8) \in G_3$ this implies

$$R_7 = f(R_6) = Q + R_6 = (3, 2) + (9, 8) \implies R_7 = (0, 1)$$

We note that $R_1 = R_7 = (0, 1)$

We also know that

$$R_i = m_i P + n_i Q \quad \forall i \geq 0 \quad \text{with } m_0 = 2, n_0 = 0$$

This implies that

$$\begin{aligned} R_1 &= m_1 P + n_1 Q = R_7 = m_7 P + n_7 Q \\ \implies (m_1 - m_7)P &= (n_7 - n_1)Q \\ \implies Q &= (n_7 - n_1)^{-1}(m_1 - m_7)P \\ \implies k &= (n_7 - n_1)^{-1}(m_1 - m_7) = d \log_P Q. \quad \text{Provided } (n_7 - n_1)^{-1} \pmod{\#(G)} \text{ exist.} \end{aligned}$$

The integer m'_i s and n'_i s are calculated using the recurrence relations:

$$m_{i+1} = \begin{cases} m_i + 1, & \text{if } R_i \in G_1 \\ 2m_i, & \text{if } R_i \in G_2 \\ m_i, & \text{if } R_i \in G_3 \end{cases} \quad \text{and } n_{i+1} = \begin{cases} n_i, & \text{if } R_i \in G_1 \\ 2n_i, & \text{if } R_i \in G_2 \\ n_i + 1, & \text{if } R_i \in G_3 \end{cases} \quad \forall i \geq 0 \quad (m_0 = 2, n_0 = 0)$$

This gives :

$$\begin{aligned} m_1 &= m_0 + 1 = 2 + 1 = 3 \\ n_1 &= n_0 = 0 \quad \text{since } R_0 \in G_1 \end{aligned}$$

$$\begin{aligned} \implies m_2 &= m_1 + 1 = 3 + 1 = 4 \\ n_2 &= n_1 = 0 \quad \text{since } R_1 \in G_1 \end{aligned}$$

$$\begin{aligned} \implies m_3 &= m_2 = 4 \\ n_3 &= n_2 + 1 = 0 + 1 = 1 \quad \text{since } R_2 \in G_3 \end{aligned}$$

$$\begin{aligned} \implies m_4 &= m_3 + 1 = 4 + 1 = 5 \\ n_4 &= n_3 = 1 \quad \text{since } R_3 \in G_1 \end{aligned}$$

$$\begin{aligned} \implies m_5 &= m_4 = 5 \\ n_5 &= n_4 + 1 = 1 + 1 = 2 \quad \text{since } R_4 \in G_3 \end{aligned}$$

$$\begin{aligned} \implies m_6 &= 2m_5 = 2 \times 5 = 10 \\ n_6 &= 2n_5 = 2 \times 2 = 4 \quad \text{since } R_5 \in G_2 \end{aligned}$$

$$\begin{aligned} \implies m_7 &= m_6 = 10 \\ n_7 &= n_6 + 1 = 4 + 1 = 5 \quad \text{since } R_6 \in G_3 \end{aligned}$$

This gives $m_1 = 3, n_1 = 0, m_7 = 10, n_7 = 5$ and therefore

$$\begin{aligned} (m_1 - m_7)P &= (n_7 - n_1)Q \\ (3 - 10)P &= (5 - 0)Q \\ \text{or } -7P &= 5Q \\ \text{or } Q &= (-5^{-1} \times 7 \pmod{18})P \end{aligned}$$

Since $(5, 18) = 1$, we get using the Euclidean /Division Algorithm, $5^{-1} \pmod{18}$ to be 11 as $11 \times 5 = 55 \equiv 1 \pmod{18}$

Hence $Q = (-11 \times 7)P = -77P = -5P = 13P \implies Q = 13P$

Hence $k = d\log_P Q = 13$

This can be directly verified by repeatedly doubling and adding that $13P = Q$

Silver-Pohlig-Hellman Algorithm For Computing Logarithm In Elliptic Curves

Recall the idea of the Silver-Pohlig Hellman Algorithm for computing discrete logarithm in a cyclic group $G = \langle g | g^n = 1 \rangle$ of order n for an element h to the base g the generator of G . The idea was first to reduce the DLP to cyclic groups of prime power order p^e , where $p^e | n$ but $p^{e+1} \nmid n$, and p is a prime, and then reduce the problem of prime order cyclic groups, evaluating g discrete logarithm in these groups and then computing discrete logarithm g , by applying Chinese Remainder Theorem.

So, we shall reiterate the algorithm but with changes according to the group $G = \langle P | nP = \infty \rangle \leq E$, where E is an elliptic curve and $Q, P \in G$ to compute $d \log_P Q$ in G and thus in E .

Writing prime factorization of n as :

$$n = \prod_{(p|n)} p^{e(p)} \implies nP = \infty \text{ as } O(P) = n$$

For each prime $p|n$, let $n_p = \frac{n}{p^{e(p)}}$, and

$$P_p = n_p P \in G$$

Then

$$p^{e(p)}[P_p] = p^{e(p)}[n_p P] = (p^{e(p)} \cdot n_p)P = nP = \infty \implies O(P_p) | p^{e(p)}$$

$P_p \neq \infty$ as this will imply that order of G is less than n .

Further, if $O(P_p) = p^{\epsilon(p)}$, then $\epsilon(p) < n$

$$\begin{aligned} &\implies p^{\epsilon(p)}(P_p) = \infty \\ &\implies p^{\epsilon(p)}(n_p P) = \infty \\ &\implies (p^{\epsilon(p)} \cdot n_p)P = \infty \\ &\implies n | (p^{\epsilon(p)} \cdot n_p) \end{aligned}$$

which is not possible as $n > (p^{\epsilon(p)} \cdot n_p)$. Hence $O(P_p) = p^{e(p)} \quad \forall p|n$

Reduction of DLP to prime power order additive cyclic groups

in Elliptic Curves

Let $G = \langle P | nP = \infty \rangle$, and $n = \prod_{p|n} p^{e(p)}$ be the prime factorization of n . If $n_p = \frac{n}{p^{e(p)}}$, then $p \nmid n_p$, and if $P_p = n_p P$ then $P_p \in G \leq E$ (the group of the Elliptic Curve), we have observed that order $O(P_p) = p^{e(p)}$. Let

$$H = \langle P_p | p^{e(p)} P_p = \infty \rangle \leq G \leq E$$

Suppose $k = d\log_P Q \in G$. Then

$$\begin{aligned} kP &= Q \\ \implies k(P_p) &= k(n_p P) = (kn_p)P = (n_p k)P = n_p(kP) = n_p Q = Q_p \quad (\text{say}) \\ \implies Q_p &\in H = \langle P_p | p^{e(p)} P_p = \infty \rangle \end{aligned}$$

Let $k(p) = d\log_{P_p} Q_p$. Then $k(p)[P_p] = Q_p \quad \forall p|n$.

Observe that

$$\begin{aligned} n_p(-kP + Q) &= -k(n_p P) + n_p Q \\ &= -k(P_p) + Q_p = -Q_p + Q_p = \infty \\ \implies O(-kp + Q) &| n_p \quad \forall p|n \\ \implies O(-kp + Q) &| \gcd(n_p) \quad \forall p|n \\ \implies O(-kp + Q) &| 1 \\ \implies O(-kp + Q) &= 1 \\ \implies -kp + Q &= \infty \\ \implies kP &= Q \\ \implies k &= d\log_P Q \in E \end{aligned}$$

Recall, $Q_p = n_p Q \in H = \langle P_p | p^{e(p)} P_p = \infty \rangle$ Also,

$$k(p) = d\log_{P_p} Q_p \implies k(p)P_p = Q_p \text{ and } k(P_p) = Q_p$$

Hence $k(P_p) = k(p)P_p$ in H , $|H| = p^{e(p)}$

$$\implies (k - k(p))P_p = \infty \text{ in } H$$

$$\implies k \equiv k(p) \pmod{p^{e(p)}}$$

Thus k can be computed using Chinese Remainder Theorem.

Reduction of DLP to prime order cyclic group of an Elliptic Curve

Let $H = \langle P \mid p^{e(p)}P = \infty \rangle \leq E$ for an Elliptic Curve E .

Let $R \in H$, then $k = d\log_P R$ exists. If $A = \infty$, then $k = 0$.

Let $A \neq \infty$. Then $0 < k < m = p^e$.

Write

$$k = k_0 + k_1p + k_2p^2 + \cdots + k_{e-1}p^{e-1}$$

[This can be done (prove) that every positive integer a can be written an integer polynomial in the powers of an another integer b .]

Now $kP = R \quad \because k = d\log_P R$ in H

$$\begin{aligned} &\implies p^{e-1}(kP) = p^{e-1}R \\ &\implies (p^{e-1}k)P = p^{e-1}R \\ &\implies p^{e-1}(k_0 + k_1p + k_2p^2 + \cdots + k_{e-1}p^{e-1})P = p^{e-1}R \\ &\implies (k_0p^{e-1})P + (k_1p^e + k_2p^{e+1} + \cdots + k_{e-1}p^{2e-2})P = p^{e-1}R \\ &\implies (k_0p^{e-1})P + (k_1 + k_2p + \cdots + k_{e-1}p^{e-2})p^eP = p^{e-1}R \\ &\implies (k_0p^{e-1})P = p^{e-1}R \end{aligned}$$

$\because (k_1 + k_2p + \cdots + k_{e-1}p^{e-2})p^eP = \infty$ as $p^eP = \infty$ in H .

Thus

$$\begin{aligned} &(k_0p^{e-1})P = p^{e-1}R \quad \text{or} \quad k_0(p^{e-1}P) = p^{e-1}R \in H \\ &\implies k_0 = d\log_{(p^{e-1}P)}(p^{e-1}R) \in H \quad \text{or} \quad k_0 = d\log_{\bar{P}}\bar{R} \end{aligned}$$

where

$$\bar{P} = p^{e-1}P \quad \text{and} \quad \bar{R} = p^{e-1}R \in H$$

Further

$$p(\bar{P}) = p(p^{e-1}P) = p^eP = \infty$$

Let $K = \langle \bar{P} \mid p\bar{P} = \infty \rangle$, then $\bar{R} \in K$, and the problem of computing k_0 is to compute logarithm in K , a cyclic prime order group in E .

Computing other coefficients $k_1, k_2, \dots, k_{e-1}$

The other coefficient $k_1, k_2, \dots, k_{e-1}$, are computed recursively as follows:

Suppose $k_1, k_2, \dots, k_{i-1}$ have been computed then

$$\begin{aligned}
 (k_i p^i + k_{i+1} p^{i+1} + \dots + k_{e-1} p^{e-1})P &= \left\{ (-k_0 - k_1 p - k_2 p^2 - \dots - k_{i-1} p^{i-1}) + \sum_{i=0}^{e-1} k_i p^i \right\} P \\
 &= (-k_0 - k_1 p - k_2 p^2 - \dots - k_{i-1} p^{i-1} + k)P \\
 &= (-k_0 - k_1 p - k_2 p^2 - \dots - k_{i-1} p^{i-1})P + kP \\
 &= (-k_0 - k_1 p - k_2 p^2 - \dots - k_{i-1} p^{i-1})P + R \\
 &= R_i \quad (\text{say}), \text{ as } kP = R
 \end{aligned}$$

Thus

$$\begin{aligned}
 (k_i p^i + k_{i+1} p^{i+1} + \dots + k_{e-1} p^{e-1})P &= R_i \\
 &= (-k_0 - k_1 p - k_2 p^2 - \dots - k_{i-1} p^{i-1})P + R, \\
 &\quad \forall i \geq 1 \quad \text{and} \quad R_0 = R
 \end{aligned}$$

$$(k_i p^i + k_{i+1} p^{i+1} + \dots + k_{e-1} p^{e-1})P = R_i$$

Adding both sides p^{e-i-1} times. we get:

$$\begin{aligned}
 \{p^{e-i-1}(k_i p^i + k_{i+1} p^{i+1} + \dots + k_{e-1} p^{e-1})\}P &= p^{e-i-1} R_i \\
 \implies p^{e-1} p^{-i} (k_i p^i + k_{i+1} p^{i+1} + \dots + k_{e-1} p^{e-1})P &= p^{e-i-1} R_i \\
 \implies p^{e-1} (k_i + k_{i+1} p + \dots + k_{e-1} p^{e-i-1})P &= p^{e-i-1} R_i \\
 \implies \{p^{e-1} k_i + p^{e-1} p (k_{i+1} + k_{i+2} p + \dots + k_{e-1} p^{e-i-2})\}P &= p^{e-i-1} R_i \\
 \implies (p^{e-1} k_i)P + p^e (k_{i+1} + k_{i+2} p + \dots + k_{e-1} p^{e-i-2})P &= p^{e-i-1} R_i \\
 \implies (p^{e-1} k_i)P = p^{e-i-1} R_i \quad \forall i \geq 1
 \end{aligned}$$

$$\begin{aligned}
 \text{Since } p^e (k_{i+1} + k_{i+2} p + \dots + k_{e-1} p^{e-i-2})P & \\
 = (k_{i+1} + k_{i+2} p + \dots + k_{e-1} p^{e-i-2})[p^e P] & \\
 = \infty &
 \end{aligned}$$

as $p^e P = \infty$, since $P \in H$, and $|H| = p^e$.

Hence

$$\implies (p^{e-1} k_i)P = p^{e-i-1} R_i \quad \forall i \geq 1$$

$$\text{Or } k_i (p^{e-1} P) = p^{e-i-1} R_i$$

$$k_i \bar{P} = p^{e-i-1} R_i \implies p^{e-i-1} R_i \in K = \langle \bar{P} | p\bar{P} = \infty \rangle$$

Hence solutions k_i 's are the discrete logs of some element in K . in particular, we get

$$k_i = d\log_{(p^{e-1}P)}(p^{e-i-1}R_i) \quad \forall i$$

Or

$$k_i = d\log_{\bar{P}}(p^{e-i-1}R_i) \quad \text{where } \bar{P} = p^{e-1}P$$

Since $\bar{P}$ is an element of $K \leq E$, so $\bar{P}$ is a point of order p , the prime $p|n$. Thus the problem is now completely reduced to finding discrete logarithm in a cyclic group of order a prime p .

Recall

$$\begin{aligned} R_i &= (-k_0 - k_1p - k_2p^2 - \cdots - k_{i-1}p^{i-1})P + R \\ &= (k_ip^i + k_{i+1}p^{i+1} + \cdots + k_{e-1}p^{e-1})P \end{aligned}$$

In particular,

$$R_0 = (k_0 + k_1p + \cdots + k_{e-1}p^{e-1})P = kP = R \quad \text{Or} \quad R_0 = R$$

Hence, $k_0 = d\log_{\bar{P}}(p^{e-1}R_0) = d\log_{(p^{e-1}P)}(p^{e-1}R)$ as $R_0 = R$

$$\implies k_0 = d\log_{(p^{e-1}P)}(p^{e-1}R)$$

Next,

$$\begin{aligned} R_1 &= (k_1p + k_2p^2 + \cdots + k_{e-1}p^{e-1})P \\ &= (k_0 + k_1p + \cdots + k_{e-1}p^{e-1} - k_0)P \\ &= (k - k_0)P \\ &= kP - k_0P \end{aligned}$$

$$\text{Or } R_1 = R - k_0P$$

$$\text{and so, } k_1 = d\log_{\bar{P}}(p^{e-2}R_1)$$

Since k_0 has already been determined, R_i is determined and hence k_1 can be determined,

Similarly,

$$\begin{aligned} R_2 &= (k_2p^2 + k_3p^3 + \cdots + k_{e-1}p^{e-1})P \\ &= (k_0 + k_1p + \cdots + k_{e-1}p^{e-1} - k_0 - k_1p)P \\ &= (k - k_0 - k_1p)P \\ &= kP + (-k_0 - k_1p)P \\ \implies R_2 &= kP - k_0P - (k_1p)P \\ k_2 &= d\log_{\bar{P}}(p^{e-3}R_2) \quad \text{where } \bar{P} = pP \end{aligned}$$

can be determined as k_0, k_1 have been determined, then k_i (and hence k) can be determined.

Example: Let $E = \{(x, y) \in F_{11} \times F_{11} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$ using Silver-Pohlig-Hellman Algorithm determined $dlog_{(1,4)}(3, 2)$ in E , if it exists. Suppose $k = dlog_P Q$ exists.

Let $P = (1, 4)$, $Q = (3, 2)$. Then it can be verified that P has order 18 in E . Also note that $Q \in G = \langle P = (1, 4) \mid 18P = \infty \rangle \leq E$,

Recall, the notations, explained in the Algorithm order of G is

$$n = 18 = 2 \times 3^2 = \prod_{p \mid n} p^{e(p)} \implies e(p) = 1 \text{ for } p = 2, e(p) = 2 \text{ for } p = 3.$$

Since

$$\begin{aligned} n_p &= \frac{n}{p^{e(p)}} \\ \implies n_2 &= \frac{18}{2} = 9, \text{ So } n_2 = 9 \text{ for } p = 2 \\ \implies n_3 &= \frac{18}{3^2} = 2, \text{ So } n_3 = 2 \text{ for } p = 3 \\ P_{p=2} &= n_p P = n_2 P = 9P = 9(1, 4) = (4, 0) \\ P_{p=3} &= n_p P = n_3 P = 2P = 2(1, 4) = (2, 9) \end{aligned}$$

Since $Q_p = n_p Q$ this implies that

$$\begin{aligned} Q_{p=2} &= n_2 Q = 9(3, 2) = (4, 0) \\ Q_{p=3} &= n_3 Q = 2(3, 2) = (9, 8) \end{aligned}$$

Corresponding subgroup H of G

$$\begin{aligned} H_{p=2} &= \langle P_{(p=2)} = (4, 0) \mid 2P_{p=2} = \infty \rangle \\ H_{p=3} &= \langle P_{(p=3)} = (2, 9) \mid 9P_{p=3} = \infty \rangle \\ \implies |H_{p=2}| &= p^{e(p)} = 2^1 = 2 \text{ and } |H_{p=3}| = p^{e(p)} = 3^2 = 9 \end{aligned}$$

We know that $Q_{p=2} \in H_{p=2}$ and $Q_{p=3} \in H_{p=3}$ and the problem is now reduced to determine discrete logarithm in these subgroups of order $p^{e(p)}$ for $p = 2, 3$ respectively i.e.

$$\begin{aligned} k(2) &= dlog_{(P_{p=2})} Q_{p=2} = dlog_{(4,0)}(4, 0) \in H_{p=2} \\ k(3) &= dlog_{(P_{p=3})} Q_{p=3} = dlog_{(2,9)}(9, 8) \in H_{p=3} \end{aligned}$$

Obviously $k(2) = 1$. We determine $k(3)$ by reducing the DLP further to the prime order cyclic group for $p = 3$ i.e. C_3 . Since $e(p) = 2$ for $p = 3$, we can assume for some integers $k_0(3)$ and $k_1(3)$, that

$$k(3) = k_0(3) + 3k_1(3)$$

The integer $k_0(3)$ and $k_1(3)$ are determined as follows: $R_{p=3} = (9, 8)$ [follows the notations]. Recall

$$\bar{P}_{p=3} = p^{e(p)-1}P_{p=3} = 3^{2-1}(2, 9) = 3(2, 9) = (5, 8)$$

and

$$\bar{R}_{p=3} = p^{e(p)-1}R_{p=3} \implies \bar{R}_{p=3} = 3^{2-1}(9, 8) = 3(9, 8) = (5, 8)$$

Hence $k_0(3) = d\log_{(\bar{P}_{p=3})}\bar{R}_{p=3} = d\log_{(5,8)}(5, 8) = 1$. We next determine $k_1(3) = d\log_{(\bar{P}_{p=3})}\bar{R}_{1(p=3)}$ where

$$\begin{aligned}\bar{R}_{1(p=3)} &= p^{e(p)-2}R_1 \text{ for } p = 3 \\ &= 3^{2-2}R_{1(p=3)} = R_{1(p=3)}\end{aligned}$$

$$\begin{aligned}\text{where } R_{1(p=3)} &= R - k_0(p)P_{p=3} = R_{p=3} - k_0(3)P_{p=3} \\ &= (9, 8) - 1 \cdot (2, 9) = (9, 8) - (2, 9) \\ &= (9, 8) + (2, -9) = (9, 8) + (2, 2) \\ &= (5, 8)\end{aligned}$$

This gives $\bar{R}_{1(p=3)} = \bar{R}_{(p=3)} = (5, 8)$. And therefore $k_1(3) = d\log_{(\bar{P}_{p=3})}\bar{R}_{1(p=3)} = d\log_{(5,8)}(5, 8) = 1$. Hence

$$k(3) = k_0(3) + 3k_1(3) = 1 + 3 \cdot 1 = 1 + 3 = 4$$

These values, $k(2) = 1$ and $k(3) = 4$, give value $k = d\log_P Q$, since

$$\begin{aligned}k &\equiv k(2) \pmod{2} \\ k &\equiv k(3) \pmod{9} \\ \implies k &\equiv 1 \pmod{2} \\ k &\equiv 4 \pmod{9}\end{aligned}$$

Apply now the Chinese Remainder Theorem to get $k \equiv 13 \pmod{18}$. Hence

$$d\log_{(1,4)}(3, 2) = 13 \in E$$

This can be verified directly by computing $13P$ for $P = (1, 4)$ using the method of repeatedly doubling and adding if required.

14.5 Elliptic Curve Cryptography

We shall see how elliptic curves can be used in communication. Both the plain text space as well as the cipher text space is the group $E = E(\mathbb{F})$ of points of an elliptic curve E over a field $\mathbb{F}$. The group E is an additive Abelian group.

We present here analogues of various cryptosystem already discussed on cyclic groups. We begin with:

14.5.1 Diffie-Hellman Key Exchange Protocol

Let $\mathbb{F}_q$ be a finite field of order q (here q is prime or prime power). Then $\mathbb{F}_q^* = \mathbb{F}_q \setminus \{0\}$ is a multiplicative cyclic group of order $q - 1$. And we know how to set up Diffie-Hellman Key Exchange Protocol on $\mathbb{F}_q^*$. The corresponding analogue for $E = E(\mathbb{F}_q)$ is as follows:

Let $E = E(\mathbb{F}_q)$ be an elliptic curve over a finite field $\mathbb{F}_q$. Suppose Alice and Bob wants to have a common knowledge (a point on E). For this, Alice chooses a positive integer ' a ' and keeps it secret. Similarly, Bob chooses a positive integer ' b ' and keeps it secret.

Suppose E and a point P on E are known to them. Alice computes the point aP (by adding P repeatedly a -times) and sends it to Bob. Bob computes the point bP and sends it to Alice.

Alice, who received bP from Bob, computes $a(bP)$, adding bP (repeatedly a -times) and get $(ab)P$. Similarly, Bob has received aP from Alice. He computes $b(aP)$. Now Alice has the point $a(bP)$ while Bob has the point $b(aP)$ of E . Both of them in fact have the same point namely $(ab)P$ as $a(bP) = (ab)P = (ba)P = b(aP)$ in E . This point becomes their common knowledge.

Example: Let $E := \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$ be public. Also let the point $P(3, 2)$ be also public.

Suppose Alice chooses integer $a = 3$ and Bob chooses the integer $b = 5$. They keep these integers secret. Alice computes $aP = 3P = (0, 1)$ and sends it to Bob. Bob Computes $bP = 5P = (6, 9)$. Alice computes

$$a(bP) = 3(6, 9) = (0, 10).$$

Similarly Bob computes

$$b(aP) = 5(0, 1) = (0, 10).$$

Now Alice and Bob both have their common knowledge the point $(0, 10) \in E$. They can use it in various applications of Diffie-Hellman key exchange protocol.

Verify all the calculations.

The Diffie-Hellman Problem and The Diffie-Hellman Assumption

Let $E = E(\mathbb{F}_q)$ be an elliptic curve P, aP, bP are points on E , where a, b are some integers. Suppose E, P, aP, bP are public. Then the Diffie-Hellman Problem (DHP) is to compute (from this knowledge) the point $(ab)P$ (of E). The Diffie-Hellman Assumption (DHA) is that it is difficult to do that. Of course the Diffie-Hellman Problem is solvable if the discrete logarithm can be computed of the point(s) aP, bP to the base point P in E . There may be some other ways as well to do the same. There is another related problem:

Decision Diffie-Hellman Problem and The Diffie-Hellman Assumption

If $P, aP, bP \in E(\mathbb{F}_q)$ are known is then possible to decide for a given point $Q \in E(\mathbb{F}_q)$, that $Q = (ab)P$ or not. Accordingly, the DDH Assumption is that it is not easy to do that.

14.5.2 Massey-Omura Cryptosystem

Alice and Bob setup the Massey-Omura cryptosystem for communication as follows:

1. They both agree on a Elliptic Curve $E = E(\mathbb{F}_q)$.
2. They also agree that the plain text space as well as cipher text space are the points of E i.e. the group E .

Now suppose Alice wants to send a message (a point $M \in E$) to Bob.

- (a) She chooses and keeps it secret a positive integer a , computes the point $aM \in E$ and sends it to Bob
- (b) Bob also chooses and keeps secret a positive integer b . He received aM from Alice. He computes $b(aM) = baM = abM \in E$ and sends it to Alice.

- (c) Alice received $(ab)M$ from Bob, she computes $a^{-1}(ab)M = bM$ and sends it to Bob.
- (d) Bob received bM from Alice, he computes $b^{-1}(bM) = M$ and thus gets the message M from Alice.

Note: Alice computes a^{-1} from a and Bob computes b^{-1} from b . This, therefore, requires $\gcd(a, N) = 1$ as well as $\gcd(b, N) = 1$, where N is the order of the group. Hence, while choosing integers a, b , they have to be careful. The $a^{-1} \pmod{N}$ and $b^{-1} \pmod{N}$ can be computed using Division Algorithm/Euclidean Algorithm.

Example: Let Alice and Bob want to communicate using Massey-Omura cryptosystem.

1. They agree on the Elliptic curve:

$$E = E(\mathbb{F}_{11}) = \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} | y^2 = x^3 + 3x + 1\} \cup \{\infty\}$$

Verify N =order of E is 18.

2. Alice chooses and keeps it secret the integer $a = 7$. She computes $7^{-1} \pmod{18}$. This is 13 (as $13 \times 7 = 91 = 1 + 5 \times 18 \equiv 1 \pmod{18}$).
3. Bob chooses and keeps it secret the integer $b = 11$. He computes $11^{-1} \pmod{18}$. This is 5 since $11 \times 5 = 55 = 1 + 3 \times 18 \equiv 1 \pmod{18}$.

Now suppose Alice wants to send the message the point $M(6, 2) \in E$ to Bob.

- (a) She computes $aM = 7M = 7(6, 2) = (3, 2)$ and sends to Bob.
- (b) Bob received $aM = (3, 2)$ from Alice.
He computes $b(aM) = 11(3, 2) = (1, 7)$ and sends to her.
- (c) Alice received $b(aM) = (ba)M = (ab)M = a(bM) = (1, 7)$ from Bob.
She computes $a^{-1}(ab)M = 7^{-1}(1, 7) = 13(1, 7) = (3, 9)$.
She sends $(3, 9)$ to Bob.
- (d) Bob received the point $(3, 9)$ from Alice. this is $a^{-1}(ab)M = bM$.
He retrieves the message M by doing $b^{-1}(bM)$ i.e.
 $11^{-1}(3, 9) = 5(3, 9) = (6, 2)$.

Verify all the computations.

14.5.3 Elliptic Curve ElGamal Cryptosystem

1. All user, Alice, Bob, ... agree upon an Elliptic Curve $E = E(\mathbb{F}_q)$, and a point P (called a Base point) on E .
2. To send a message $M \in E$ to a user say Alice, use her public (encrypting) key aP (a is a secretly chosen positive integer by her) as follows:
 - (a) Choose a positive integer k . Create a mask by computing the point $k(aP) \in E$.
 - (b) Hide the message as the point $Q_2 = M + k(aP) \in E$.
 - (c) Send to Alice the ordered pair of points $(Q_1, Q_2) \in E \times E$ where $Q_1 = kP$.
 - (d) Alice receives the pair of points (Q_1, Q_2) . She knows that the message is hidden in the second coordinate point Q_2 . She recreates the mask by doing

$$aQ_1 = a(kP) = (ak)P = (ka)P = k(aP) \in E$$

- (e) Alice retrieves the message from Q_2 by doing

$$\begin{aligned} Q_2 - aQ_1 &= \{M + k(aP)\} - a(kP) \\ &= M + \{k(aP) - a(kP)\} \\ &= M \in E \end{aligned}$$

Example:

1. Let the publicly agreed elliptic curve be

$$E = E(\mathbb{F}_{11}) = \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$$

and the public base point $P(1, 4) \in E$.

2. Let Alice choose the integer $a = 11$, keep it secret and publish her public (encrypting) key $aP = 11(1, 4) = (6, 9)$.
 - (a) Choose a positive integer $k = 13$. Create a mask

$$k(aP) = 13(6, 9) = (1, 7).$$

- (b) Suppose the message to be sent to Alice is the point $M = (3, 2) \in E$. Hide the message in

$$Q_2 = M + k(aP) = (3, 2) + (1, 7) = (5, 3).$$

- (c) Send Alice the ordered pair of points (Q_1, Q_2) . Here

$$Q_1 = kP = 13P = 13(1, 4) = (3, 2)$$

and $Q_2 = (5, 3)$. That is the pair $((3, 2), (5, 3)) \in E \times E$ of points $(3, 2)$ and $(5, 3)$ is sent to Alice.

- (d) Alice received $(Q_1, Q_2) = ((3, 2), (5, 3))$. She knows the message M is hidden in the (second coordinate) point $Q_2 = (5, 3)$. She recreates the mask. She computes $aQ_1 = 11(3, 2) = (1, 7)$. This point $aQ_1 = a(kP)$.
- (e) Alice retrieves the message by doing

$$Q_2 - aQ_1 = (5, 3) - (1, 7) = (5, 3) + (1, -7) = (5, 3) + (1, 4) = (3, 2)$$

Remark 1: In the Elliptic Curve ElGamal Cryptosystem the creation of mask by Bob $k(aP)$, as well as the recreation of mask by Alice $a(kP)$ uses the Diffie-Hellman Key Exchange.

Remark 2: In the Elliptic Curve Diffie-hellman Key Exchange Protocol (ECDHKEP) on an elliptic curve $(\text{mod } p)$ for $p > 3$, say

$$E := \{(x, y) \in \mathbb{F}_p \times \mathbb{F}_p \mid y^2 = x^3 + Ax + B; A, B \in \mathbb{F}_p\} \cup \{\infty\}$$

the common knowledge is a point say $K = a(bP) = b(aP) = abP$, where a, b are positive integers, secretly chosen by Alice and Bob. So, if $K = (x, y) \in E$ is the common knowledge, then $x, y \in E$ such that $y^2 = x^3 + Ax + B; A, B \in \mathbb{F}_p$. Because x, y satisfy an (elliptic) equation, they are not independent. If an eavesdropper Eve knows the x -coordinate of K , then she can compute $y^2 = x_K^3 + Ax_K + B$, since the curve E is a public knowledge. As $\text{mod } p$, half are quadratic residues and half are quadratic non residues. In case of a quadratic residue only two values of y (other $-y$) congruent $\text{mod } p$ are possible and not very difficult to calculate. Thus Eve gets $K = (x_K, y_K)$ or in the worst case $(x_K, -y_K) = -K$. But it hardly makes big difference as K or $-K$ are only two cases of the common knowledge, Eve posses. Therefore, in practical situation, it is enough to send their common knowledge, the x -

coordinate only. In that case, they get either (in case of Alice) $\pm a(bP)$ or (in case of Bob) $\pm b(aP)$ i.e. $\pm K$. But in both cases x -coordinate (of K) is same.

Example: recall the curve

$$E = E(\mathbb{F}_{11}) = \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$$

and the point $P(3, 2)$. Suppose Alice and Bob have agreed on these publicly. Let Alice choose and keep it secret the integer $a = 3$, and Bob $b = 5$.

Alice computes $aP = 3P = (0, 1)$, and Bob computes $bP = 5P = (6, 9)$. Suppose Alice sends to Bob instead the point $(0, 1)$, just its x -coordinate i.e. 0, and like wise Bob sends to Alice only the x -coordinate of bP i.e. 6. Alice received from Bob only the x -coordinate say $x_B = 6$. The corresponding y_B satisfies

$$\begin{aligned} y_B^2 &= x_B^3 + 3x_B + 1 \pmod{11} \implies y_B^2 = 6^3 + 18 + 1 \\ &\equiv 235 \equiv 4 \pmod{11} \\ &\implies y_B = \pm 2 \pmod{11} \\ &\implies y_B = 2 \text{ or } -2 \end{aligned}$$

Hence the point, she may assume sent by Bob to be

$$B(6, 2) \text{ or } B(6, -2) = (6, 9)$$

Similarly, Bob may assume the point sent by Alice $A(0, 1)$ or $(0, 10)$ This happened in their first exchange i.e. the points (bP) and (aP) as sent by Bob to Alice and by Alice to Bob respectively.

In the next and final round, Alice computes aB (or $-aB$ as the other possibility) i.e. $3(6, 2)$ or $-3(6, 2)$ [and $5(0, 1)$ or $5(0, 10)$ by Bob].

Observe that

$$3(6, 2) = (0, 1); \quad -3(6, 2) = 3(6, -2) = 3(6, 9) = (0, 10)$$

Hence common knowledge by Alice calculation is $(0, 1)$ or $(0, 10)$.

Similarly, the common knowledge by Bob calculation is

$$5(0, 1) = (0, 10) \text{ or } -5(0, 1) = 5(0, 10) = (0, 1).$$

So their common point is $(0, 1)$ or $(0, 10)$. They are additive inverse of each other, make no difference as long as their common knowledge which is x -coordinate is concerned.

14.6 Message mapping on an Elliptic Curve

Since the plain text space as well as the cipher text space is the group E of the points of an elliptic curve, it requires a method by which normal message units are mapped onto a point in E , and vice versa while decryption. This can be done in several ways.

(1). Pre-arranged table between message units and points on E . For example consider the elliptic curve

$$E = E(\mathbb{F}_{11}) = \{(x, y) \in \mathbb{F}_{11} \times \mathbb{F}_{11} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$$

We know that $\#(E) = 18$. So we can map 18-message units to 18 points of E and vice versa.

For example, the English Language letters 'A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R' can be put into one-one correspondence with points of E .

The problem is that the one-one correspondence putting message units with points on an elliptic curve has to be shared by the users.

Exercises 1 Set up Diffie-Hellman Key Exchange Protocol on the elliptic curve E of discussed above and use to communicate letters 'A' to 'R' only.

2 Explain Massey-Omura cryptosystem by an example using letters, 'A' to 'R' only.

3 Discuss ElGamal cryptosystem to encrypt and decrypt a message using letters 'A' to 'R' by mapping them on the elliptic curve E , of the above discussion.

(2). Koblitz Method

In Koblitz method message units are already represented by an integer $m \pmod{p}$, where $p > 3$ is a prime. To map the integer m on to a point P on the elliptic curve $E := \{(x, y) \in \mathbb{F}_p \times \mathbb{F}_p \mid y^2 = x^3 + Ax + B; A, B \in \mathbb{F}_p\} \cup \{\infty\}$, look for a point $P(x, y) \in E$ if there is any such that $x = m$. If there is no such point, look for the points $(m + 1, y), (m + 2, y), \dots$ (a value of y) such that $(m + j, y) \in E$. The integer m is represented by that corresponding point on E .

If the probability of failure acceptable is $\frac{1}{2^k}$, then for the message (represented by the integers) $m : 0 \leq m < \frac{p}{k}$, let $x_j = j = j + mk$ ($0 \leq j < k$). For these values of $j = 0, 1, 2, \dots, k - 1$ compute

$$\beta_j = x_j^3 + Ax_j + B.$$

If $\beta_j^{\frac{p-1}{2}} \equiv 1 \pmod{p}$, then β_j is a square $\pmod{p}$. In that we have got a corresponding point on E . Recall how the square roots modulo a prime are calculated. For example, in case $p \equiv 3 \pmod{4}$, a square root $\pmod{p}$ is given by $y_j = \beta_j^{\frac{p+1}{4}}$. The point $P_m(x_j, y_j)$ lies on E , and thus can represent m on E . The process is reversible. To retrieve m from $P_m(x_j, y_j)$ is to compute $\left[\frac{x_j}{k}\right]$, the greatest integer less or equal to $\frac{x_j}{k}$. This is m since

$$\begin{aligned} \frac{x_j}{k} &= \frac{mk + j}{k} = m + \frac{j}{k} \quad (0 \leq j < k) \\ \implies \left[\frac{x_j}{k}\right] &= m \text{ since } 0 \leq \frac{j}{k} < 1 \end{aligned}$$

Exercise: Let $E = E(\mathbb{F}_{271}) = \{(x, y) \in \mathbb{F}_{271} \times \mathbb{F}_{271} \mid y^2 = x^3 + 3x + 1\} \cup \{\infty\}$. If it is acceptable that the probability of failure in representing an integer modulo 271 to a point on E can be $\frac{1}{2^{10}}$, then represent English Language letters A to Z, first by representing them by integers $1, 2, \dots, 26$ and then by points on the elliptic curve E .

14.7 The Group E of an Elliptic Curve: Order and Structure of E (Some Basic Theorems)

Let $E(\mathbb{F})$ be an elliptic curve defined over a field $\mathbb{F}$. Then an operation, called point addition can be defined with respect to which E becomes an (additive) Abelian group. Further, if $\mathbb{F} = \mathbb{F}_q$ is a finite field, then $E(\mathbb{F}_q)$ is a finite Abelian group, and therefore by the Fundamental Theorem for Finite Abelian groups, E is isomorphic to a direct product of finitely many cyclic groups of finite order [Prove the theorem].

Can there be more information on the structure of E , since E is a particular finite Abelian group?

Yes, indeed. The answer is the following.

Theorem: Let $\mathbb{F}_q$ be a finite field and $E = E(\mathbb{F}_q)$ be the group of points on E . Then either $E \cong C_n$ or $E \cong C_{n_1} \times C_{n_2}$ for some integer $n \geq 1$ or for some integers $n_1, n_2 \geq 1$ and $n_1 \mid n_2$.

since E is an additive Abelian group some authors prefer to write above results as either $E \cong \mathbb{Z}_n$ or $E \cong \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2}$, $n, n_1, n_2 \in \mathbb{N}$ and $n_1 \mid n_2$.

We have discussed examples of Elliptic curves of each type.

The next question is about the integers n, n_1, n_2 . Can they be precisely determined? or estimated? At least a bound on the order of the E ?

Let $\mathbb{F}_q$ be a finite field and $E = E(\mathbb{F}_q)$ be the group of points on elliptic curve E over the finite field $\mathbb{F}_q$ (So q is a prime or a power of prime). Then the order $N = (\text{also denoted by } \#(E))$ of E , satisfies :

$$\#(E) = q + 1 - t_q \text{ where } |t_q| \leq 2\sqrt{q}.$$

Hence $q + 1 - 2\sqrt{q} \leq \#(E) \leq q + 1 + 2\sqrt{q}$. We have used this theorem in some examples to compute the order $\#(E)$.

Can the order be computed precisely? Is there any algorithm? Schoof's algorithm is the answer to count points on E .

14.8 Exercise

1. Let $G = E(\mathbf{F}_{41})$, where E is the elliptic curve given by $y^2 = x^3 + 2x + 1$. Let $P = (0, 1)$ and $Q = (30, 40)$. Use Hasse's theorem to find the bound on order of group G . Also Find k such that $Q = kP$ using Baby step-Giant step Algorithm for discrete logarithm.
2. Let $G = E(\mathbf{F}_{599})$, where E is the elliptic curve given by $y^2 = x^3 + 1$. Let $P = (60, 19)$ and $Q = (277, 239)$. Given P has order 600, Find k such that $Q = kP$ using Pohlig-Hellman Method.
3. Let E is the elliptic curve given by $y^2 = x^3 + 1$ over $\mathbf{F}_q$ where $q \equiv 2 \pmod{3}$. Let $w \in \mathbf{F}_{q^2}$ be a primitive third root of unity. Define a map

$$\begin{aligned} \beta : E(\mathbf{F}_q) &\longrightarrow E(\mathbf{F}_q) \\ (x, y) &\mapsto (wx, y), \quad \beta(\infty) = \infty \end{aligned}$$

using formulas for addition prove that β is an isomorphism.

4. Use Pocklington-Lehmar primality test to check the primality of $n = 153533$.
5. Given $P = (819, 784)$ lies on the elliptic curve given by $E : y^2 = x^3 + 10x - 2$. Check the primality of $n = 907$.

6. Use Hasse's theorem in the following:-

- (a) Let E be the elliptic curve given by $y^2 = x^3 + 7x + 1$ over $\mathbf{F}_{101}$. If 29 divides the order of the point $(0, 1)$ lying on E . Find the order of E .
- (b) Let E be the elliptic curve given by $y^2 = x^3 - 10x + 21$ over $\mathbf{F}_{557}$. Point $(2, 3)$ lying on E has order 189. Find the order of E .
- (c) Let E be the elliptic curve given by $y^2 = x^3 + 7x + 12$ over $\mathbf{F}_{103}$. The points $(-1, 2)$ and $(19, 0)$ has order 13 and 2 respectively. Find the order of E .

7. Use Elliptic-curve Diffie-Hellman to construct a secret shared key using $E : y^2 = x^3 + 7x + 1$, $p = 44927$ and $P = (27844, 29401)$, where Alice's secret number is $a = 40006$ and Bob's secret number is $b = 18846$.

Chapter 15

Elliptic Curves: Algebraic Geometry Approach

Definition 15.0.1. Let k be field. Affine n -space over k is the set of n -tuples of elements of k and denoted as $\mathbb{A}^n(k)$ or simply $\mathbb{A}^n$. In particular when $n = 2$, $\mathbb{A}^2(k)$ is called affine plane.

Definition 15.0.2. The set of zeros of a non constant polynomial $f(x, y) \in k[x, y]$ in $\mathbb{A}^2(k)$ is called as affine plane curve.

Definition 15.0.3. Projective n -space over k is defined to be the set of all lines through $(0, 0, \dots, 0)$ in $\mathbb{A}^{n+1}(k)$ and it is denoted as $\mathbb{P}^n(k)$ or simply $\mathbb{P}^n$. Any point $(x) = (x_1, x_1, \dots, x_{n+1}) \neq (0, 0, \dots, 0)$ in $\mathbb{A}^{n+1}$ determine the unique line $\{(\lambda x_1, \lambda x_2, \dots, \lambda x_{n+1}) : \lambda \in k\}$. Two points $(x) = (x_1, x_1, \dots, x_{n+1})$ and $(y) = (y_1, y_2, \dots, y_{n+1})$ in $\mathbb{P}^n$ determines same line if and only if there exist a nonzero scalar $\lambda \in k$ such that $y_i = \lambda x_i$ for $i = 1, 2, \dots, n+1$. That is we can identify $\mathbb{P}^n$ with the set of equivalence classes of points in $\mathbb{A}^{n+1} \setminus \{(0, 0, \dots, 0)\}$, where the equivalence relation $\sim$ is defined by $(x) = (x_1, x_2, \dots, x_{n+1}) \sim (y) = (y_1, y_2, \dots, y_{n+1})$ if $y_i = \lambda x_i$ for $i = 1, 2, \dots, n+1$ and for some non zero λ belongs to k . We denote a point in $\mathbb{P}^n$ by $(x_1 : x_2 : \dots : x_{n+1})$.

Definition 15.0.4. Projective plane over k written $\mathbb{P}^2(k)$ is defined to be the set of equivalence classes of points in $\mathbb{A}^3(k) \setminus \{(0, 0, 0)\}$ where the equivalence relation is as defined above.

If $F(x, y, z)$ be an arbitrary polynomial in x, y, z then we cannot say that a point p in $\mathbb{P}^2$ is a zero of $F(x, y, z)$, since this depends on the representatives

that we choose from the equivalence class.

For example consider the polynomial

$$F(x, y, z) = x^3 + 5y - 6z^2,$$

then $F(1, 1, 1) = 0$, but in $\mathbb{P}^2$ $(1, 1, 1) = (2, 2, 2)$ by the equivalence relation we defined. In fact they are just two different representatives of the same class $(1 : 1 : 1)$ but

$$F(2, 2, 2) = -6 \neq 0.$$

To avoid such problems in projective space, we need to consider homogeneous polynomials instead.

If $f(x, y)$ is any polynomial then we can convert it to a homogeneous polynomial by inserting one more variable say z . For example if we take

$$f(x, y) = x^3 + 2y^2 + 7x - 1$$

then we can homogenize this polynomial by introducing a new variable z and we obtained the corresponding homogeneous polynomial

$$F(x, y, z) = x^3 + 2y^2z + 7xz^2 - z^3.$$

In general if $f(x, y)$ is any polynomial in variable x, y of degree n , then we can define the corresponding homogeneous polynomial $F(x, y, z)$ by

$$F(x, y, z) = z^n f\left(\frac{x}{z}, \frac{y}{z}\right).$$

Similarly if $F(x, y, z)$ is any homogeneous polynomial then we can form the corresponding non-homogeneous polynomial by de-homogenizing with respect to z (also we can do with respect to x and y) as follows:

$$f(x, y) = F(x, y, 1)$$

For example if $F(x, y, z) = x^3 + 2y^2z - z^3$, then the corresponding non-homogeneous polynomial is $f(x, y) = x^3 + 2y^2 - 1$.

Definition 15.0.5. A projective plane curve is the zero set of a homogeneous polynomial $F(x, y, z) \in k[x, y, z]$ in $\mathbb{P}^2$.

Definition 15.0.6. A projective plane curve of degree three is called a cubic, that is the set of zeros of a homogeneous polynomial in $f(x, y, z)$ of degree three in $k[x, y, z]$.

Definition 15.0.7. A projective plane curve $F(x, y, z)$ is called non singular or smooth if and only if for every point p on F , $F(p), F_x(p), F_y(p), F_z(p)$ does not vanishes simultaneously, where $F_x(p), F_y(p), F_z(p)$ denote partial derivatives of F with respect to x, y and z respectively.

15.0.1 Elliptic Curves

An Elliptic curve over a field k is a smooth projective curve of degree three. By projective change of coordinates we can transform elliptic curve to some standard useful form called Weierstrass general form, given by

$$y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3,$$

with $a_i \in k$ for $i = 1, 2, 3, 4$ or 6 .

If $(x : y : z)$ is a point in $\mathbb{P}^2$ with $z \neq 0$, then $(x : y : z) = (\frac{x}{z} : \frac{y}{z} : 1)$. Points of this form that is $(x : y : z) \in \mathbb{P}^2$ such that $z \neq 0$ is called the finite points of $\mathbb{P}^2$. However if $z = \infty$ then we dividing by z should be thought of as giving ∞ in either x or y coordinate, therefor the points of the form $(x : y : 0)$ are called points at infinity in $\mathbb{P}^2$.

The affine plane can be identified with the finite points of of $\mathbb{P}^2$ by the inclusion map

$$\begin{aligned} \mathbb{A}^2 &\rightarrow \mathbb{P}^2 \\ (x, y) &\rightarrow (x : y : 1) \end{aligned}$$

Hence $\mathbb{P}^2 = \{[x : y : 1] | (x, y) \in \mathbb{A}^2\} \cup \{[x : y : 0] | [x : y] \in \mathbb{P}^1(k)\}$.

That is $\mathbb{P}^2 = \mathbb{A}^2 \cup \{\text{Points at infity of } \mathbb{P}^2\}$.

Let $(x : y : z)$ be any point on the elliptic curve E given by Weierstrass general form defined above. If $z \neq 0$ then we can write $(x : y : z) = (\frac{x}{z} : \frac{y}{z} : 1)$

in $\mathbb{P}^2$, that is if $z \neq 0$ then $(x : y : z)$ is of the form $(x' : y' : 1)$. Check that $(x' : y' : 1) \in E(x, y, z) \Leftrightarrow (x, y) \in E(x, y, 1)$. Or (x, y) is a point on the corresponding affine curve

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Now if $(x : y : z) \in E(x, y, z)$ with $z = 0$ then this gives $x^3 = 0 \Rightarrow x = 0$ and y can be any non zero number. That is for $z = 0$ we get the point $(0 : y : 0) = (0 : 1 : 0)$ as only point at infinity on E .

Intersection of two parallel lines: Recall that in coordinate geometry, we say two parallel lines intersect at infinity, we can now see what it meant exactly. Let

$$y = mx + b_1, \quad y = mx + b_2$$

be two distinct non-vertical parallel lines that is $b_1 \neq b_2$ and $m \neq \infty$. Then they have homogeneous form as follows:

$$y = mx + b_1z, \quad y = mx + b_2z.$$

As these lines are parallel hence they do not have any intersection point in $\mathbb{A}^2$, that is for $z \neq 0$ there is no solution. For $z = 0$ we have $y = mx$, hence we get the point of intersection as $(x : mx : 0) = (1 : m : 0)$ ($x \neq 0$ since for $x = 0$ this corresponds to the point $(0,0,0)$ which is not a projective point).

If we take two distinct vertical lines $x = c_1$ and $x = c_2$ then this corresponds the solution of homogeneous equation $x = c_1z$ and $x = c_2z$ in $\mathbb{P}^2$. For $z = 0$ this gives $x = 0$, hence they intersect at the $(0 : 1 : 0)$ in projective plane $\mathbb{P}^2$. For $z \neq 0$ there is no solution since this corresponds to $c_1 = c_2$, but we have taken distinct lines. Hence two distinct vertical lines does not intersect at any finite point (that is points of $\mathbb{A}^2$) of $\mathbb{P}^2$.

From the above argument it is clear that any vertical line passes through the point $(0 : 1 : 0)$ in $\mathbb{P}^2$. This point $(0 : 1 : 0)$ is also lie on the elliptic curve $E(x, y, z)$, hence any vertical line intersects the elliptic curve at the point $(0 : 1 : 0)$ (Point at infinity). Moreover since $(0 : 1 : 0) = (0 : -1 : 0)$, therefore each vertical line at both end contains the same point at infinity

$(0 : 1 : 0)$ or in simple we can say point at infinity is situated at both end of every vertical line.

As we have seen that

$$\mathbb{P}^2 = \{[x : y : 1] | (x, y) \in \mathbb{A}^2\} \cup \{[x : y : 0] | [x : y] \in \mathbb{P}^1(k)\}$$

$$\text{Or } \mathbb{P}^2 = \{[x : y : 1] | (x, y) \in \mathbb{A}^2\} \cup \{\text{points at infinity}\}.$$

That is projective plane($\mathbb{P}^2$) is the union of affine plane($\mathbb{A}^2$) and the points at infinity which corresponds to $z = 0$.

Recall that we said an elliptic curve in Weierstrass form is the set of zero of the polynomial

$$y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3,$$

with $a_i \in k$ for $i = 1, 2, 3, 4, 6$. in $\mathbb{P}^2$. Hence we can transform this elliptic curve(In projective plane) to the affine curve which we get by letting $z = 1$ in the equation of elliptic curve $E(x, y, z) = 0$ in addition with point at infinity which we get by letting $z = 0$. That is elliptic curve in general form is the set of zeros of the equation:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

with the point at infinity $(0 : 1 : 0)$. We simply denote this point at infinity by the symbol ∞ .

Definition 15.0.8. If k is any field we define elliptic curve over k by

$$E_k = \{(x, y) \in k \times k : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6\} \cup \{\infty\}. \quad (15.1)$$

We mostly use this definition of elliptic in general form, as there is one variable less hence easier for computational point of view.

15.1 Weierstrass Normal Form

With a suitable change of variables Weierstrass equation that is equation (1) can be converted to a simpler form, known as its normal form

$$y^2 = x^3 + ax + b \quad (15.2)$$

for a and $b \in k$. There are several methods to do this, we choose the simplest way.

In equation(1), put

$$y = \frac{v - a_1x - a_3}{2}.$$

The RHS of equation becomes:

$$\begin{aligned} & y^2 + a_1xy + a_3y \\ &= y(y + a_1x + a_3) \\ &= \left(\frac{v - a_1x - a_3}{2} \right) \left(\frac{v - a_1x - a_3}{2} + a_1x + a_3 \right) \\ &= \left(\frac{v - a_1x - a_3}{2} \right) \left(\frac{v + a_1x + a_3}{2} \right) \\ &= \frac{v^2 - (a_1x + a_3)^2}{4} \\ &= \frac{v^2 - (a_1^2x^2 + 2a_1a_3x + a_3^2)}{4} \end{aligned}$$

Hence equation(1) becomes:

$$\frac{v^2 - (a_1^2x^2 + 2a_1a_3x + a_3^2)}{4} = x^3 + a_2x^2 + a_4x + a_6$$

$$\text{Or } v^2 = 4x^3 + 4a_2x^2 + 4a_4x + 4a_6 + a_1^2x^2 + 2a_1a_3x + a_3^2$$

$$v^2 = 4x^3 + (a_1^2 + 4a_2)x^2 + 2(a_1a_3 + 2a_4)x + (a_3^2 + 4a_6) \quad (15.3)$$

In equation(3) put $x = \frac{X - (3a_1^2 + 4a_2)}{36}$, then RHS of equation (3) becomes:

$$\begin{aligned} & 4 \left(\frac{X - (3a_1^2 + 4a_2)}{36} \right)^3 + (a_1^2 + 4a_2) \left(\frac{X - (3a_1^2 + 4a_2)}{36} \right)^2 \\ & \quad + 2(a_1a_3 + 2a_4) \left(\frac{X - (3a_1^2 + 4a_2)}{36} \right) + (a_3^2 + 4a_6) \end{aligned}$$

And equation(3) becomes:

$$\begin{aligned} v^2 = \frac{4}{36^3} \{X - 3(a_1^2 + 4a_2)\}^3 &+ \frac{(a_1^2 + 4a_2)}{36^2} \{X - 3(a_1^2 + 4a_2)\}^2 \\ &+ \frac{2}{36} (a_1 a_2 + 2a_4) \{X - 3(a_1^2 + 4a_2)\} \\ &+ (a_3^2 + 4a_6) \end{aligned}$$

Now put $v = \frac{Y}{108}$, in this equation we get:

$$\begin{aligned} \left(\frac{Y}{108}\right)^2 = \frac{4}{36^3} \{X - 3(a_1^2 + 4a_2)\}^3 &+ \frac{(a_1^2 + 4a_2)}{36^2} \{X - 3(a_1^2 + 4a_2)\}^2 \\ &+ \frac{2}{36} (a_1 a_2 + 2a_4) \{X - 3(a_1^2 + 4a_2)\} \\ &+ (a_3^2 + 4a_6) \end{aligned}$$

Multiplying both sides by $(108)^2$, and $108 = 3 \times 36$; and

$$\begin{aligned} (108)^2 \cdot \frac{4}{(36)^3} &= 4 \frac{3^2 \cdot 36^2}{36^3} \\ &= \frac{3^2 \cdot 4}{36} \\ &= 1 \end{aligned}$$

$$\frac{108^2}{36^2} = \frac{3^2 \cdot 36^2}{36} = 9$$

$$108^2 \cdot \frac{2}{36} = \frac{3^2 \cdot 36^2 \cdot 2}{36} = 3^2 \cdot 36 \cdot 2 = 3 \cdot 3 \cdot 2 \cdot 36 = 6 \cdot 108$$

We get

$$\begin{aligned} Y^2 = X - 3(a_1^2 + 4a_2)\}^3 &+ 9(a_1^2 + 4a_2)\{X - 3(a_1^2 + 4a_2)\}^2 \\ &+ 6 \cdot 108(a_1 a_2 + 2a_4)\{X - 3(a_1^2 + 4a_2)\} + 108^2(a_3^2 + 4a_6) \end{aligned}$$

On expanding each bracket term we get

$$\begin{aligned} Y^2 = & [\{X^3 + 3X^2\{-3(a_1^2 + 4a_2)\} + 3X\{-3(a_1^2 + 4a_2)\}^2 + \{-3(a_1^2 + 4a_2)\}^3\} \\ & + [9(a_1^2 + 4a_2)\{X^2 - 6(a_1^2 + 4a_2)X + 9(a_1^2 + 4a_2)^2\}] \\ & + [6 \cdot 108(a_1a_3 + 2a_4)X - 18 \cdot 108(a_1a_3 + 2a_4)(a_1^2 + 4a_2)] \\ & + [(108)^2(a_3^2 + 4a_6)]] \end{aligned}$$

Writing the RHS as polynomial in X , the equation becomes:

$$\begin{aligned} Y^2 = & X^3 + \{-9(a_1^2 + 4a_2) + 9(a_1^2 + 4a_2)\}X^2 \\ & + \{27(a_1^2 + 4a_2)^2 - 54(a_1^2 + 4a_2)^2 + 6 \cdot 108(a_1a_3 + 2a_4)\}X \\ & + \{-27(a_1^2 + 4a_2)^3 + 81(a_1^2 + 4a_2)^3 - 18 \cdot 108(a_1a_3 + 2a_4)(a_1^2 + 4a_2) + 108^2(a_3^2 + 4a_6)\} \end{aligned}$$

Or:

$$\begin{aligned} Y^2 = & X^3 + \{-27(a_1^2 + 4a_2)^2 + 6 \cdot 108(a_1a_3 + 2a_4)\}X \\ & + \{54(a_1^2 + 4a_2)^3 - 18 \cdot 108(a_1a_3 + 2a_4)(a_1^2 + 4a_6) + 108^2(a_3^2 + 4a_6)\} \end{aligned}$$

Or

$$Y^2 = X^3 + AX + B$$

$$\text{Where } A = \{-27(a_1^2 + 4a_2)^2 + 6 \cdot 108(a_1a_3 + 2a_4)\}$$

$$B = \{54(a_1^2 + 4a_2)^3 - 18 \cdot 108(a_1a_3 + 2a_4)(a_1^2 + 4a_6) + 108^2(a_3^2 + 4a_6)\}$$

Note: In the process of transforming an elliptic curve to normal form we, have used division by 2 and 3. Hence if the coefficients are come from a field with, $\text{Char}(k) \neq 2, 3$, then the above equation is the normal form of Weierstrass equation over k . That is $y^2 = x^3 + ax + b$; $a, b \in k$, where $\text{Char}(F) \neq 2, 3$.

15.2 Normal Forms Of Weierstrass Equation Over Fields

Let k be a field of characteristic p . Then $p = 0$ or p is a prime.

Recall the Weierstrass equation in general form:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

Also recall that if $p \neq 2, 3$ then the equation can be transformed into simpler equation called normal form

$$y^2 = x^3 + Ax + B; \quad A, B \in k$$

Similarly, the normal form in case $p = 2$,

either $y^2 + xy = x^3 + Ax^2 + B$,

or $y^2 + Cy = x^3 + Ax + B; \quad A, B, C \in k$.

And in case $p = 3$, the normal form is $y^2 = x^3 + Ax^2 + Bx + C, \quad A, B, C \in k$.

This can be split into following two equivalent forms, $y^2 = x^3 + Cx^2 + D$ or $y^2 = x^3 + Ax + B, \quad A, B, C, D \in k$.

15.3 Real Elliptic Curves

The normal form of a real elliptic curve is

$$E = E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + ax + b, a, b \in \mathbb{R}\} \cup \{\infty\}.$$

Or

$$E = \{(x, y) \in \mathbb{R} \times \mathbb{R} : F(x, y) = 0 \text{ where } F(x, y) = y^2 - x^3 - ax - b, a, b \in \mathbb{R}\} \cup \{\infty\}.$$

The representation of E_k for a field of $\text{Char}(k) \neq 2, 3$, is same as above. However considering $k = \mathbb{R}$ have the advantage of knowledge of calculus and plane geometry. We shall move to other field as and when convenient. For the moment let k be an algebraically closed field. Best is the field $\mathbb{C}$ of complex numbers.

Let E be a elliptic curve over the field $\mathbb{R}$. Then we can define E by the Weierstrass normal form over $\mathbb{R}$;

$$E_{\mathbb{R}} = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y^2 = x^3 + ax + b, a \text{ and } b \in \mathbb{R}\} \cup \{\infty\}.$$

Recall that $E_{\mathbb{R}}$ form a abelian group under the operation $+$ with additive identity ∞ , defined by

$$\begin{aligned} + : E_{\mathbb{R}} \times E_{\mathbb{R}} &\rightarrow E_{\mathbb{R}} \\ (p_1, p_2) &\rightarrow p_3 \end{aligned}$$

Where p_3 denote the reflection through x -axis, of the third point of intersection of the line through p_1 and p_3 with $E_{\mathbb{R}}$.

In any group we can talk about the order of its elements.

Definition 15.3.1. Let E be the elliptic curve over the field $\mathbb{R}$. A point $P \in E$ is said to be a point of order m if

$$mP = \underbrace{P + P + \dots + P}_{m \text{ summand}} = \infty,$$

but $m'P \neq \infty$ for any integer m' satisfying $2 \leq m' < m$. If such an integer m exists, then p has finite order otherwise it has infinite order.

Points of finite orders are called as torsion points, in particular a point of order m is called a m -torsion point.

Theorem 15.3.1 (Points of Order Two). Let $E : y^2 = f(x) = x^3 + ax + b$ be an elliptic curve over $\mathbb{R}$. A point $P = (x, y) \neq \infty$ on E has order two if and only if $y = 0$.

Proof. $P = (x, y)$ is a point of order two is same as

$$\begin{aligned} 2P &= P + P = \infty \\ \Leftrightarrow P &= -P \end{aligned}$$

Since $-P = -(x, y) = (x, -y)$

(Recall that $-P$ denote the intersection of the vertical line through P and ∞ , hence $-P$ is just the reflection of the points P through x -axis).

Therefore $P = (x, y)$ is a point of order two $\Leftrightarrow (x, y) = (x, -y)$

$$\Leftrightarrow y = -y$$

$$\Leftrightarrow 2y = 0$$

$$\Leftrightarrow y = 0.$$

□

15.4 Discriminant of an Elliptic Curve

If E is a elliptic curve given by Weierstrass normal form;

$$y^2 = f(x) = x^3 + ax + b,$$

then discriminant of the cubic polynomial $f(x)$ is the quantity

$$D = -4a^3 - 27b^2.$$

If we factor $f(x)$ over the complex numbers, and α_1 , α_2 and α_3 are roots of $f(x)$, then it can be checked that

$$D = (\alpha_1 - \alpha_2)^2(\alpha_2 - \alpha_3)^2(\alpha_1 - \alpha_3)^2.$$

Hence D is nonzero is same as $f(x)$ has distinct roots and conversely. Recall the definition of elliptic curve (It is non singular), it follows that elliptic curve are curves with discriminant nonzero.

If $f(x)$ is any monic polynomial in $\mathbb{Z}[x]$ then the discriminant of $f(x)$ belongs to the ideal of $\mathbb{Z}[x]$ generated by the $f(x)$ and $f'(x)$. This follows from general theory of discriminant but here it is a particular case that we can check directly. For $f(x) = x^3 + ax + b$, we have $f'(x) = 3x^2 + a$ and

$$\begin{aligned} & (18ax - 27b)f(x) + (-6ax^2 + 9bx - 4a^2)f'(x) \\ &= (18ax - 27b)(x^3 + ax + b) + (-6ax^2 + 9bx - 4a^2)(3x^2 + a) \\ &= 18ax^4 - 27bx^3 + 18a^2x^2 - 27abx + 18abx - 27b^2 - 18ax^4 + 27bx^3 - 12a^2x^2 - 6a^2x^2 + 9abx - 4a^3 \\ &= -4a^3 - 27b^2 \\ &= D \end{aligned}$$

Hence $D \in \langle f(x), f'(x) \rangle$ in $\mathbb{Z}[x]$. That is D can be written as

$$D = r(x)f(x) + s(x)f'(x),$$

with $r(x)$ and $s(x)$ belongs to $\mathbb{Z}[x]$.

15.5 Elliptic Curve Over $\mathbb{Q}$

If E is a elliptic curve over $\mathbb{Q}$, then we can always write its equation in normal form

$$y^2 = x^3 + ax + b,$$

where a, b are rational numbers.

By applying change of variables $X = d^2x$ and $Y = d^3y$, the equation becomes

$$Y^2 = X^3 + d^2aX^2 + d^4bX + d^6c.$$

By choosing a large integer d , we can clear the denominator in a, b . Hence any elliptic curve over $\mathbb{Q}$, can be transferred into an elliptic curve over $\mathbb{Z}$. Therefore if E is an elliptic curve over $\mathbb{Q}$ then the discriminant $D = -4a^3 - 27b^2$ of E is an integer.

Lemma 15.5.1. *Let $P = (x, y)$ be a point on the elliptic curve E over $\mathbb{Q}$ such that both P and $2P$ have integer coordinates. Then either $y = 0$ or $y|D$.*

Proof. If $y = 0$ then there is nothing to prove. If $y \neq 0$, and we will show that $y|D$. Now assume that $y \neq 0$, and this implies $p = (x, y)$ is not a point of order 2, that is $2p \neq \infty$. Hence we can write $2p = (X, Y)$. We can get X and Y by using algebraic formula of point addition, as follows;

$$X = m^2 - 2x, \quad \text{where } m = \frac{3x^2 + a}{2y},$$

and $Y = m(x - X) - y$.

Since we have assumed that x, y, X and Y are integers, hence

$$m^2 = X + 2x,$$

is an integer. As $m = \frac{3x^2 + a}{2y}$, hence m is rational since both $3x^2 + a$ and $2y$ are integers. Therefore m is an integer.

$$\begin{aligned} \text{Now } m \cdot 2y &= 3x^2 + a = f'(x) \\ \Rightarrow 2y &| f'(x) \\ \Rightarrow y &| f'(x) \end{aligned}$$

But $y^2 = f(x) \Rightarrow y \mid f(x)$.

Now using the facts

$$D = r(x)f(x) + s(x)f'(x)$$

where $r(x)$ and $s(x)$ are polynomials with integers coefficients. Since y divides $f(x)$ and $f(x)'$ therefore $y \mid D$. $\square$

Definition 15.5.1. A point $P = (x, y)$ on the elliptic curve E is said to be a rational point if x and y are both rational numbers.

Lemma 15.5.2. Let $\mathfrak{p}$ be a prime, R be the ring of rational numbers with denominator prime to $\mathfrak{p}$, and let $E(\mathfrak{p}^\nu)$ be the set of rational points on the elliptic curve E for which x has a denominator divisible by $\mathfrak{p}^{2\nu}$, plus the point ∞ on the curve.

(a) $E(\mathfrak{p})$ consists of all rational points (x, y) for which the denominator of either x or y divisible by $\mathfrak{p}$.

(b) For every $\nu \geq 0$, the set $E(\mathfrak{p}^\nu)$ is a subgroup of the group of rational points on E .

(c) The map

$$\begin{aligned} \frac{E(\mathfrak{p}^\nu)}{E(\mathfrak{p}^{3\nu})} &\longrightarrow \frac{\mathfrak{p}^\nu R}{\mathfrak{p}^{3\nu} R}, \text{ defined by} \\ \text{if } P \neq \infty \quad P = (x, y) &\longrightarrow t(P) = \frac{x}{y} \\ \text{and } \infty &\longrightarrow 0 \end{aligned}$$

is one-to-one homomorphism.

Theorem 15.5.3. (a) For every prime $\mathfrak{p}$, the subgroup $E(\mathfrak{p})$ contains no points of finite order (other than ∞).

(b) Let $P = (x, y) \neq \infty$ be a rational point of finite order, then x and y are integers.

Proof. (a) Let $P = (x, y)$ be a point of order m , where $m \neq 1$. We want to show that $P \notin E(\mathfrak{p})$, for any prime $\mathfrak{p}$. Suppose $P \in E(\mathfrak{p})$, for some prime $\mathfrak{p}$. Recall from the lemma above that $P \in E(\mathfrak{p}^\nu)$ if the denominator of x is divisible by $\mathfrak{p}^{2\nu}$. Note that P cannot belong to all of $E(\mathfrak{p}^\nu)$, as the denominator of x cannot not be divisible by arbitrary high powers of $\mathfrak{p}$. So we can get

a $\nu > 0$ such that $P \in E(\mathfrak{p}^\nu)$, but $P \notin E(\mathfrak{p}^{\nu+1})$.

We can separate the proof in the following two cases.

Case-1 If $\mathfrak{p} \nmid m$, as the mapping t given in part(c) of the above lemma is a homomorphism, we have;

$$t(P_1 + P_2) = t(P_1) + t(P_2) \pmod{\mathfrak{p}^{3\nu}R}.$$

Hence for finite m ,

$$t(mP) = t(\underbrace{P + P + \dots + P}_{m \text{ times}}) = m t(P) \pmod{\mathfrak{p}^{3\nu}R}.$$

Since $mP = \infty$, we get

$$t(mP) = t(\infty) = 0.$$

Now $\mathfrak{p} \nmid m$ that implies $\frac{1}{m} \in R$. Hence m is a unit in R . That is,

$$\begin{aligned} 0 &= m t(P) \pmod{\mathfrak{p}^{3\nu}R} \\ \Rightarrow t(P) &= 0 \pmod{\mathfrak{p}^{3\nu}R} \\ \Rightarrow P &\in E(\mathfrak{p}^{3\nu}). \end{aligned}$$

But $E(\mathfrak{p}^{3\nu}) \subset E(\mathfrak{p}^{\nu+1})$ since if denominator of x is divisible by $\mathfrak{p}^{6\nu}$ then it is divisible by $\mathfrak{p}^{2\nu+2}$, as $6\nu \geq 2\nu + 2$. This contradicts the fact that $P \notin E(\mathfrak{p}^{\nu+1})$.

$\therefore P$ does not belongs to $E(\mathfrak{p})$ for any prime $\mathfrak{p}$.

case-2 If $\mathfrak{p} \mid m$, we can write $m = \mathfrak{p}.n$, for some integer n . Let $P' = nP$. Since P has order m , that is $mP = \infty$, hence $\mathfrak{p}P' = \mathfrak{p}(nP) = mP = \infty$. Conversely for any $k < \mathfrak{p}$ then

$$kP' < \mathfrak{p}P' = \mathfrak{p}(nP) = (\mathfrak{p}n)P = mP = \infty.$$

Hence order of P' is $\mathfrak{p}$.

Now $P \in E(\mathfrak{p})$ and $E(\mathfrak{p})$ is a subgroup, so $nP = P' \in E(\mathfrak{p})$. By following same argument as in part-(a) we can find some $\nu > 0$ such that,

$$P' \in E(\mathfrak{p}^\nu) \text{ but } P' \notin E(\mathfrak{p}^{\nu+1}).$$

Observe that

$$\begin{aligned} t(\mathfrak{p}P') &= \mathfrak{p} t(P') \pmod{\mathfrak{p}^{3\nu}R} \\ \Rightarrow t(\infty) &= \mathfrak{p} t(P') \pmod{\mathfrak{p}^{3\nu}R} \\ \Rightarrow \mathfrak{p} t(P') &\pmod{\mathfrak{p}^{3\nu}R} = 0 \\ \Rightarrow t(P') &= 0 \pmod{\mathfrak{p}^{3\nu-1}R} \end{aligned}$$

This implies $P' \in E(\mathfrak{p}^{3\nu-1})$, but $3\nu - 1 \geq \nu + 1$, this contradicts the fact that $P' \notin E(\mathfrak{p}^{\nu+1})$. This completes the proof of part (a).

(b) Given $P = (x, y)$ is a point of finite order, then by part(a), we have P does not belongs to $E(\mathfrak{p})$ for any prime $\mathfrak{p}$. So by definition of $E(\mathfrak{p})$, we have denominators of x and y are not divisible by any prime. This implies, denominator of x, y are 1, that is x and y are integers. $\square$

Theorem 15.5.4. (Nagell-Lutz) Let

$$y^2 = f(x) = x^3 + ax + b$$

be an elliptic curve with integer coefficients a, b and D be the discriminant of the cubic polynomial $f(x)$,

$$D = -4a^3 - 27b^2.$$

If $p = (x, y)$ be a rational point of finite order, then x and y are integers and either $y = 0$ or $y|D$.

Proof. In part(b) of theorem-2.3, we have shown that points of finite order have integer coordinates. Hence if $P = (x, y)$ is a point of finite order then, we have x and y are integers. Now if $y = 0$ then this is equivalent to P is a point of order two. If $y \neq 0$, then this is same as $2P \neq \infty$. But $2P$ is also a point of finite order, hence $2P$ has integers coordinates. From lemma-2.1 we get $y|D$ as we have assumed $y \neq 0$. $\square$

15.6 Reduction modulo a prime

An elliptic curve $E : y^2 = x^3 + ax + b$ is said to be defined over a finite field $\mathbb{F}_p$, if the coefficients a and b belongs to $\mathbb{F}_p$. Given an Elliptic E over $\mathbb{Z}$, we

are interested in the points of finite order, so let us collect all points of finite order in E and denote it by Φ . That is

$$\Phi = \{P = (x, y) \in E : P \text{ has finite order}\}.$$

Note that Φ is subgroup of E because if p_1 and p_2 are points of finite order, say m_1 and m_2 respectively, that is

$$m_1 P_1 = \infty \quad \text{and} \quad m_2 P_2 = \infty.$$

Then $(m_1 m_2)(P_1 \pm P_2) = \infty$.

Hence both $P_1 + P_2$ and $P_1 - P_2$ are elements of finite orders.

Recall that Φ consists of points with integer coordinates together with ∞ . For any prime $\mathfrak{p}$, we can define reduction modulo $\mathfrak{p}$ map

$$\begin{aligned} \pi : \Phi &\rightarrow E(\mathbb{F}_{\mathfrak{p}}) \\ P &\rightarrow \overline{P} \end{aligned}$$

$$\text{where} \quad \overline{P} = \begin{cases} (\overline{x}, \overline{y}), & \text{if } P = (x, y); \\ \infty, & \text{if } P = \infty. \end{cases}$$

Theorem 15.6.1. *Let $E : y^2 = x^3 + ax + b$ be an elliptic curve with integer coefficients, D denote discriminant of E given by*

$$D = -4a^3 - 27b^2.$$

$\Phi \subseteq E$ denote the subgroup consisting of all points of finite order. For any prime $\mathfrak{p}$, let

$$\begin{aligned} \pi : \Phi &\rightarrow E(\mathbb{F}_{\mathfrak{p}}) \\ P &\rightarrow \overline{P} \end{aligned}$$

denote the reduction modulo $\mathfrak{p}$ map. Then π is a homomorphism. In particular if $\mathfrak{p} \nmid 2D$, then π is an isomorphism of Φ onto a subgroup of $E(\mathbb{F}_{\mathfrak{p}})$.

15.7 Finding Torsion Points on Elliptic Curves

Nagell-Lutz theorem and reduction modulo $\mathfrak{p}$ theorem can be used to find torsion points on elliptic curves with integer coefficients. Lets see some examples how it works.

Example 2. Let $E : y^2 = x^3 + 3$ be an elliptic curve.

1. By Reduction modulo $\mathfrak{p}$ theorem;

The discriminant of E is

$$D = -27(3^2) = -243 = -3^5,$$

$$\text{hence } 2D = -2 \cdot 3^5.$$

As the only primes dividing $2D$ are 2 and 3, hence by reduction modulo $\mathfrak{p}$ theorem, for all primes other than 2 and 3, the reduction modulo $\mathfrak{p}$ map is a one-to-one homomorphism from Φ to $E(\mathbb{F}_{\mathfrak{p}})$. That is for all prime $\mathfrak{p} \geq 5$, there is a one to one homomorphism from Φ to $E(\mathbb{F}_{\mathfrak{p}})$.

For $\mathfrak{p} = 5$, $E(\mathbb{F}_5)$ denote the curve $y^2 = x^3 + 3$ with considering coefficients in $\mathbb{F}_5$. As there are finitely many elements in $\mathbb{F}_5$, we can find $E(\mathbb{F}_{\mathfrak{p}})$ easily by substituting all five possible values of x in the equation of elliptic curve and can cheack for which values of x this gives solutions for y in $\mathbb{F}_5$.

For $x = 0 \Rightarrow y^2 = 3$, there is no solution for y in $\mathbb{F}_5$.

For $x = 1 \Rightarrow y^2 = 4, \Rightarrow y = \pm 2 \in \mathbb{F}_5$.

For $x = 2 \Rightarrow y^2 = 11 = 1, \Rightarrow y = \pm 1 \in \mathbb{F}_5$.

For $x = 3 \Rightarrow y^2 = 30 = 0, \Rightarrow y = 0 \in \mathbb{F}_5$.

For $x = 4 \Rightarrow y^2 = 2$, there is no solution for y in $\mathbb{F}_5$

Hence $E(\mathbb{F}_5) = \{(1, 2), (1, -2), (2, 1), (2, -1), (3, 0), \infty\}$.
 $\#E(\mathbb{F}_5) = 6$.

As π is a homomorphism hence $\#\Phi$ divides $\#E(\mathbb{F}_p) = 6$.

For $p = 7$, similarly we can compute the all points on $E(\mathbb{F}_7)$, as follows;

For $x = 0 \Rightarrow y^2 = 3$, there is no solution for y in $\mathbb{F}_7$.

For $x = 1 \Rightarrow y^2 = 4, \Rightarrow y = \pm 2 \in \mathbb{F}_7$.

For $x = 2 \Rightarrow y^2 = 4, \Rightarrow y = \pm 2 \in \mathbb{F}_7$.

For $x = 3 \Rightarrow y^2 = 2, \Rightarrow y = \pm 3 \in \mathbb{F}_7$.

For $x = 4 \Rightarrow y^2 = 4, \Rightarrow y = \pm 2 \in \mathbb{F}_7$.

For $x = 5 \Rightarrow y^2 = 2, \Rightarrow y = \pm 3 \in \mathbb{F}_7$.

For $x = 6 \Rightarrow y^2 = 2, \Rightarrow y = \pm 3 \in \mathbb{F}_7$.

Here $E(\mathbb{F}_7) = \{(1, \pm 2), (2, \pm 2), (3, \pm 3), (4, \pm 2), (5, \pm 3), (6, \pm 3), \infty\}$.
 $\#E(\mathbb{F}_7) = 13$. Similarly $\#\Phi$ divides $\#E(\mathbb{F}_7) = 13$. Hence we get $\#\Phi$ must divides $\gcd(6, 13) = 1$.

So $\#\Phi = 1$. That is our elliptic curve $E: y^2 = x^3 + 3$ over integers does not contain any point of finite order than ∞ .

2. By Nagell-Lutz Theorem; We can also determine Φ by Nagell-Lutz theorem as follows. Let $P = (x, y)$ be a point of finite order. Recall that $P = (x, y)$ is a point of finite order if $y = 0$ or $y|D = -3^5 = -243$. If $y = 0$ this gives $x^3 + 3 = 0$ but there is no integer whose cube is -3 . Hence E contains no points of order 2. If $y \neq 0$ then y must belongs to the set

$$\{\pm 1, \pm 3, \pm 9, \pm 27, \pm 81, \pm 243\}.$$

For $y = \pm 1 \Rightarrow y^2 = 1$ this gives $x^3 + 3 = 1$ or $x^3 = -2$

there is no such integer x , hence there is no point on E with y -coordinate ± 1 .

If $y \neq \pm 1$, then y is divisible by 3, and y satisfies $y^2 = x^3 + 3$, hence x^3 is divisible by 3 that implies x is also divisible by 3. Look at the equation

$$3 = x^3 - y^2,$$

the RHS is divisible by $3^2 = 9$, this implies LHS is divisible by 9, that is 3 is divisible by 9 which is meaningless. Hence there is no points on E with y

coordinates in the set;

$$\{\pm 1, \pm 3, \pm 9, \pm 27, \pm 81, \pm 243\}.$$

$\therefore$ There is no point of finite order on E other than ∞ , that is $\#\Phi = 1$.

Example 3. Let E be the elliptic curve $y^2 = x^3 + x$.

Here discriminant of E is $D = -4$. By Nagell-lutz theorem we have $P = (x, y)$ is point of finite order if $y = 0$ or y belongs to the set $\{\pm 1, \pm 2, \pm 4\}$.

$$\text{If } y = 0 \Rightarrow x^3 + x = 0 \Rightarrow x = 0$$

$$\text{If } y = \pm 1 \Rightarrow x^3 + x = 1, \text{ there is no such integer } x.$$

$$\text{If } y = \pm 2 \Rightarrow x^3 + x = 4, \text{ there is no such integer } x.$$

$$\text{If } y = \pm 4 \Rightarrow x^3 + x = 16, \text{ there is no such integer } x.$$

$$\text{Hence } \Phi = \{(0, 0), \infty\}.$$

By reduction modulo $\mathfrak{p}$ theorem; We can use the reduction modulo $\mathfrak{p}$ theorem to find Φ as follows. Here $D = -4$, hence $2D = -8$. Hence by reduction modulo $\mathfrak{p}$ theorem we have, for all prime $\mathfrak{p} \geq 3$, we have one-to-one map

$$\pi : \Phi \rightarrow E(\mathbb{F}_{\mathfrak{p}}).$$

For $\mathfrak{p} = 3$, we can compute all points on $E(\mathbb{F}_{\mathfrak{p}})$

$$\text{For } x = 0 \Rightarrow y^2 = 0, \rightarrow y = 0.$$

$$\text{For } x = 1 \Rightarrow y^2 = 2, \text{ there is no solution for } y \text{ in } \mathbb{F}_3.$$

$$\text{For } x = 2 \Rightarrow y^2 = 1, \Rightarrow y = \pm 1.$$

$$\text{Hence } E(\mathbb{F}_{\mathfrak{p}}) = \{(0, 0), (2, 1), (2, -1), \infty\}.$$

Similarly for $\mathfrak{p} = 5$, we can compute $E(\mathbb{F}_5)$ which is given by

$$E(\mathbb{F}_5) = \{(0, 0), (2, 0), (3, 0), \infty\}.$$

Note that in $E(\mathbb{F}_3)$, $(2, 1) \in E(\mathbb{F}_3)$ has order 4 (As its order divides the order of the group $E(\mathbb{F}_3) = 4$ and $y \neq 0$, hence it is not of order 2). Hence $E(\mathbb{F}_3)$ is isomorphic to the cyclic group of order 4, that is $E(\mathbb{F}_3) \simeq \frac{\mathbb{Z}}{4\mathbb{Z}}$. Similarly it can be noted that in $E(\mathbb{F}_5)$ every nontrivial elements have order 2, hence

$$E(\mathbb{F}_5) \simeq \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \frac{\mathbb{Z}}{2\mathbb{Z}}.$$

As Φ subgroup of both of these groups, so the only possibilities for Φ is that either Φ is trivial or isomorphic to $\frac{\mathbb{Z}}{2\mathbb{Z}}$. As $(0, 0) \in \Phi$ is a point of order 2 hence $\Phi = \{(0, 0), \infty\}$.

Let E be an elliptic curve defined over a field $\mathbb{K}$ and n be a positive integer. Define

$$E[n] = \{P \in E(\overline{\mathbb{K}}) \mid nP = \infty\}$$

where $\overline{\mathbb{K}}$ = algebraic closure of $\mathbb{K}$.

Note that $E[n]$ contains points with coordinates in $\overline{\mathbb{K}}$, not necessarily in $\mathbb{K}$.

Proposition 15.7.1. *Let E be an elliptic curve over a field $\mathbb{K}$. If the characteristic of $\mathbb{K} \neq 2$, then*

$$E[2] \simeq \mathbb{Z}_2 \oplus \mathbb{Z}_2,$$

if the characteristic of $\mathbb{K}$ is 2, then

$$E[2] \simeq 0 \text{ or } \mathbb{Z}_2.$$

Proof. For char($\mathbb{K} \neq 2$).

Recall that when the characteristic of $\mathbb{K} = 3$, we can write any elliptic curve E in the normal form $y^2 = x^3 + ax^2 + bx + c$, and if the characteristic not equals to 2 and 3 we can write E as $y^2 = x^3 + ax^2 + b$. Hence in general if the characteristic of $\mathbb{K} \neq 2$, we can write E of the form $y^2 = x^3 + ax^2 + bx + c$. Lets determine $E[2]$ in this case;

$$\begin{aligned} y^2 &= x^3 + ax^2 + bx + c, \\ y^2 &= (x - \alpha_1)(x - \alpha_2)(x - \alpha_3). \end{aligned}$$

where $\alpha_1, \alpha_2, \alpha_3 \in \overline{\mathbb{K}}$.

Recall that $P = (x, y)$ is a point of order 2 that is $2P = \infty$ if and only if $y = 0$. Letting $y = 0$ in the equation of elliptic curve, we get the equation, we get

$$(x - \alpha_1)(x - \alpha_2)(x - \alpha_3) = 0.$$

This equation has three roots in $\overline{\mathbb{K}}$, namely $\alpha_1, \alpha_2, \alpha_3$. Hence

$$E[n] = \{\infty, (\alpha_1, 0), (\alpha_2, 0), (\alpha_3, 0)\}.$$

As $E[n]$ has three elements of order two hence it is non cyclic group of order four. Therefore it is isomorphic to $\mathbb{Z}_2 \oplus \mathbb{Z}_2$.

For $\text{char}(\mathbb{K} = 2)$

Recall that if E is an elliptic curve over a field $\mathbb{K}$ of $\text{char}(\mathbb{K}) = 2$, then E has one of the following two forms;

$$E_1 : y^2 + cy = x^3 + ax + b \quad \text{or} \quad E_2 : y^2 + xy = x^3 + ax^2 + b$$

Now if $p = (x, y)$ is point of order two on E_1 that is $2P = \infty$. Using the algebraic formula for point addition, we get $2P = (x_3, y_3)$,

$$\begin{aligned} \text{where } x_3 &= m^2 - 2x = m^2, & [\text{as } \text{char}(\mathbb{F}) = 2] \\ \text{and } y_3 &= m(x + x_3) + y + c, \\ \text{here } m &= \frac{3x^2 + a}{2y + c} = \frac{x^2 + a}{c} & [\text{as } \text{char}(\mathbb{F}) = 2]. \end{aligned}$$

It can be noted that P is a point of order two if and only if $m = \infty$, that is $c = 0$. But if $c = 0$, then partial derivative of E_1 with respect to y is $2y = 0$ (as $\text{char}(\mathbb{K}) = 2$). Hence this implies that E_1 is singular, it is not possible as elliptic curve is a non singular curve. Therefore there is no point of order two on E_1 . Hence $E_1 = \{\infty\}$.

Similarly we can check that in E_2 , $P = (x, y)$ is a point of order two if and only if $m = \frac{3x^2 + 2ax + y}{2y + x} = \frac{x^2 + y}{x} = \infty$, that implies $x = 0$. Now plugging $x = 0$ in equation of E_2 we get,

$$\begin{aligned}
y^2 &= b \\
\Rightarrow y^2 + 2y\sqrt{b} + \sqrt{b}^2 &= 0 && [\text{square root is unique in characteristic two.}] \\
\Rightarrow (y + \sqrt{b})^2 &= 0 \\
\Rightarrow y &= -\sqrt{b}
\end{aligned}$$

So $E_2 = \{\infty, (0, \sqrt{b})\}$.

Hence from both of the cases we conclude that $E[2] \simeq 0$ or $\mathbb{Z}_2$. $\square$

Now the general situation is given by the following theorem. We will simply state it and omit the proof as our main goal is to study Weil pairing.

Theorem 15.7.1. *Let E be an elliptic curve over a field $\mathbb{K}$ and let n be a positive integer. If the characteristic of $\mathbb{K}$ neither divide n , nor 0, then*

$$E[n] \simeq \mathbb{Z}_n \oplus \mathbb{Z}_n.$$

If the characteristic of $\mathbb{K}$ is $p > 0$ and $p|n$, then we can write $n = p^r n'$ with $p \nmid n'$. Then

$$E[n] \simeq \mathbb{Z}_{n'} \oplus \mathbb{Z}_{n'} \quad \text{or} \quad E[n] \simeq \mathbb{Z}_n \oplus \mathbb{Z}_{n'}.$$

Let n be a positive integer not divisible by characteristic of $\mathbb{K}$, then by the above theorem $E[n] \simeq \mathbb{Z}_n \oplus \mathbb{Z}_n$. We say $\beta_1, \beta_2 \in E[n]$ is called a basis of $E[n]$ if every element of $E[n]$ can be expressed as $m_1\beta_1 + m_2\beta_2$ for some m_1, m_2 belongs to $\mathbb{Z}$ note that m_1, m_2 is uniquely determined modulo n .

15.8 Endomorphism of Elliptic Curves

Recall that $E[n]$ (collection of all elements of order dividing n) form a abelian group, hence we can talk about various maps from $E[n]$ to $E[n]$. Let $\alpha : E(\overline{\mathbb{K}}) \rightarrow E(\overline{\mathbb{K}})$ be a homomorphism (homomorphism of additive group) that is $\alpha(P_1 + P_2) = \alpha(P_1) + \alpha(P_2)$. Note that if α is a homomorphism from $E(\overline{\mathbb{K}})$ to $E(\overline{\mathbb{K}})$ then order of $\alpha(P)$ divides order of P for any point $P \in E(\overline{\mathbb{K}})$ of finite order. Now if $P \in E[n]$ that is $|P| \mid n$ then $|\alpha(P)| \mid |P|$, that implies $|\alpha(P)| \mid n$. Therefore $\alpha(P)$ belongs to $E[n]$ whenever P belongs to $E[n]$ that is α restrict a map from $E[n]$ to $E[n]$, lets denote it by α_n

Now let $\{\beta_1, \beta_2\}$ is a basis for $E[n]$, hence there exists $a, b, c, d \in \mathbb{Z}$ such that

$$\alpha(\beta_1) = a\beta_1 + c\beta_2,$$

$$\alpha(\beta_2) = b\beta_1 + d\beta_2.$$

Therefore each homomorphism $\alpha : E(\overline{\mathbb{K}}) \rightarrow E(\overline{\mathbb{K}})$ is represented by a 2×2 matrix

$$\alpha_n = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Example 4. let $E : y^2 = x^3 - 2$ be an elliptic curve over $\mathbb{R}$. It can be checked that

$$E[2] = \{\infty, (2^{\frac{1}{3}}, 0), (\omega 2^{\frac{1}{3}}, 0), (\omega^2 2^{\frac{1}{3}}, 0)\}$$

where ω denote primitive cube root of unity.

Let $\beta_1 = (2^{\frac{1}{3}}, 0)$ and $\beta_2 = (\omega 2^{\frac{1}{3}}, 0)$.

As $(\omega^2 2^{\frac{1}{3}}, 0) = (2^{\frac{1}{3}}, 0) + (\omega 2^{\frac{1}{3}}, 0) = \beta_1 + \beta_2$, hence $\{\beta_1, \beta_2\}$ forms a basis for $E[2]$.

Let $\alpha : E(\mathbb{C}) \rightarrow E(\mathbb{C})$ defined by $\alpha(x, y) = (\bar{x}, \bar{y})$, where $\bar{x}$ denote the complex conjugation of x . Then α is a homomorphism, it is easy to check; Let $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2) \in E(\mathbb{C})$ and $P_3 = (x_3, y_3) = (x_1, y_1) + (x_2, y_2)$. Using algebraic formula of point addition:

$$\begin{aligned} \alpha(P_1 + P_2) &= \overline{(P_1 + P_2)} \\ &= \overline{(x_1 + y_1) + (x_2 + y_2)} \\ &= \overline{(x_3, y_3)} \\ &= (\bar{x}_3, \bar{y}_3) \\ &= \left(\overline{m^2 - x_1 - x_2}, \overline{m(x_1 - x_3) + y_1} \right) \quad \left[\text{Here } m = \frac{3x_1^2}{2y_1} \right] \\ &= \left(\left(\frac{3\bar{x}_1^2}{2\bar{y}_1} \right)^2 - \bar{x}_1 - \bar{y}_1, \frac{3\bar{x}_1^2}{2\bar{y}_1}(\bar{x}_1 - \bar{x}_3) + \bar{y}_1 \right) \\ &= (\bar{x}_1, \bar{y}_1) + (\bar{x}_2, \bar{y}_2) \\ &= \bar{P}_1 + \bar{P}_2 \\ &= \alpha(P_1) + \alpha(P_2) \end{aligned}$$

We can get the matrix of α as follows;

$$\alpha(\beta_1) = \alpha(2^{\frac{1}{3}}, 0) = (\overline{2^{\frac{1}{3}}}, \overline{0}) = (2^{\frac{1}{3}}, 0) = \beta_1 = 1.\beta_1 + 0.\beta_2,$$

$$\alpha(\beta_2) = \alpha(\omega 2^{\frac{1}{3}}, 0) = (\overline{\omega 2^{\frac{1}{3}}}, \overline{0}) = (\omega^2 2^{\frac{1}{3}}, 0) = \beta_3 = 1.\beta_1 + 1.\beta_2.$$

Hence we obtain the matrix $\alpha_n = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

Definition 15.8.1. Let E be an elliptic curve over $\mathbb{K}$. We define elliptic curve endomorphism α as a homomorphism $\alpha : E(\overline{\mathbb{K}}) \rightarrow E(\overline{\mathbb{K}})$ given by rational functions that it $\alpha(x, y) = (F_1(x, y), F_2(x, y))$ where F_1, F_2 are rational functions in variables x and y .

Example 5. Let $E : y^2 = x^3 + ax + b$ be an elliptic curve given in normal form. Define α on points of E as $\alpha(P) = 2P$.

$$\alpha(x, y) = (x \text{ coordinate } 2P, y \text{ coordinate of } 2P)$$

$$\alpha(x, y) = \left(\left(\frac{3x^2 + a}{2y} \right)^2 - 2x, \left(\frac{3x^2 + a}{2y} \right) \left(3x - \left(\frac{3x^2 + a}{2y} \right)^2 \right) - y \right)$$

Using algebraic formula for point addition it can be verified easily that

$$\alpha(P_1 + P_2) = \alpha(P_1) + \alpha(P_2).$$

Hence α is an endomorphism of E .

15.8.1 Transformation of Rational Functions

Let $R(x, y)$ be any rational function in variable x and y . Suppose $R(x, y)$ is a rational function on points of $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, then we can replace y^2 with $(x^3 + a_2x^2 + a_4x + a_6) + y(-a_1x - a_3)$. Then proceeding this way, for any positive integer k we can write

$$y^k = f(x) + yg(x) \quad \text{for some } f(x), g(x) \in K[x].$$

Hence more precisely, we can write

$$\begin{aligned}
 R(x, y) &= \frac{P_1(x) + P_2(x)y}{P_3(x) + P_4(x)y} \\
 &= \frac{P_1(x) + P_2(x)y}{P_3(x) + P_4(x)y} \times \frac{P_3(x) - P_4(x)y}{P_3(x) - P_4(x)y} \\
 &= \frac{(P_1(x)P_3(x) - P_2(x)P_4(x)y^2) + (P_2(x)P_3(x) - P_1(x)P_4(x))y}{P_3(x)^2 - P_4(x)^2y^2}
 \end{aligned}$$

Now replacing $y^2 = (x^3 + a_2x^2 + a_4x + a_6) + y(-a_1x - a_3)$ in the above equation and simplifying, we can write

$$R(x, y) = \frac{q_1(x) + q_2(x)y}{q_3(x)}, \text{ where } q_1(x), q_2(x), q_3(x) \in K[x].$$

$$R(x, y) = \frac{q_1(x)}{q_3(x)} + \frac{q_2(x)}{q_3(x)}y$$

So we can write $R(x, y) = (f_1(x), f_2(x)y)$, where $f_1(x), f_2(x)$ are rational functions in variable x .

Definition 15.8.2. Let $\alpha(x, y) = (f_1(x), f_2(x)y)$, where $f(x) = \frac{p(x)}{q(x)}$ with $P(x), q(x) \in K[x]$. Define degree of endomorphism α :

$$\deg(\alpha) = \max\{\deg(p(x)), \deg(q(x))\}.$$

Definition 15.8.3. If $\alpha = (f_1(x), f_2(x)y)$ is a non zero endomorphism, then α is called separable endomorphism:

If $f_1'(x) \neq 0 \Leftrightarrow$ at least one of the $p'(x)$ and $q'(x)$ is not zero.

Example 6. Let $E : y^2 = x^3 + ax + b$ be the elliptic curve. $\alpha(P) = 2P$ defines an endomorphism.

$$\text{Here } \alpha(x, y) = (F_1(x, y), F_2(x, y))$$

Where,

$$F_1(x, y) = \left(\frac{3x^2 + a}{2y}\right)^2 - 2x \text{ and } F_2(x, y) = \left(\frac{3x^2 + a}{2y}\right) \left(3x - \left(\frac{3x^2 + a}{2y}\right)^2\right) - y$$

Here

$$\begin{aligned}
 F_1(x, y) &= \frac{(3x^2 + a)^2 - 2x \cdot 4y^2}{(2y)^2} \\
 &= \frac{(3x^2 + a)^2 - 8xy^2}{4y^2} \\
 &= \frac{9x^4 + 6ax^2 + a^2 - 8x(x^3 + ax + b)}{4(x^3 + ax + b)} \\
 &= \frac{9x^4 + 6ax^2 + a^2 - 8x^4 - 8ax^2 - 8bx}{4(x^3 + ax + b)} \\
 &= \frac{x^4 - 2ax^2 - 8bx + a^2}{4(x^3 + ax + b)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence } \deg(\alpha) &= \max\{\deg(x^4 - 2ax^2 - 8bx + a^2), \deg(4x^3 + 4ax + 4b)\} \\
 \deg(\alpha) &= \max\{4, 3\}
 \end{aligned}$$

$$\therefore \deg(\alpha) = 4.$$

α is separable as $q'(x) \neq 0$.

Because if

$$\begin{aligned}
 q'(x) &= 0 \\
 \Rightarrow 4(3x^2 + a) &= 0 \\
 \Rightarrow (3x^2 + a) &= 0
 \end{aligned}$$

Which imply that the polynomial $x^3 + ax + b$ has multiple zero hence the discriminant of $x^3 + ax + b$ is zero. Therefore E can not be non-singular.

$\therefore \alpha$ is separable.

15.9 The Weil Pairing

Let E be an elliptic curve over a field $\mathbb{K}$ and let n be an integer not divisible by the characteristic of $\mathbb{K}$, then $E[n] \simeq \mathbb{Z}_n \oplus \mathbb{Z}_n$. Let μ_n be the group of n th roots of unity in $\overline{\mathbb{K}}$, that is;

$$\mu_n = \{x \in \overline{\mathbb{K}} : x^n = 1\}.$$

Since the characteristic of $\mathbb{K}$ does not divide n , hence the polynomial $f(x) = x^n - 1$ and $f'(x) = nx^{n-1}$ has no common factor. Therefore the equation $x^n = 1$ has no multiple roots, hence has n distinct roots in $\overline{\mathbb{K}}$. μ_n is a cyclic group of order n .

Any generator ζ of μ_n is called primitive n th root of unity or equivalently $\zeta^k = 1$ if and only if $n|k$.

Theorem 15.9.1. *Let E be an elliptic curve defined over a field $\mathbb{K}$ and let n be a positive integer. Assume that the characteristic of $\mathbb{K}$ does not divide n . Then there is a pairing*

$$e_n : E[n] \times E[n] \rightarrow \mu_n,$$

called the Weil pairing, that satisfies the following properties:

(1) e_n is bilinear in each variable. That is

$$e_n(S_1 + S_2, T) = e_n(S_1, T) \cdot e_n(S_2, T)$$

and

$$e_n(S, T_1 + T_2) = e_n(S, T_1) \cdot e_n(S, T_2)$$

for all $S, S_1, S_2, T, T_1, T_2 \in E[n]$.

(2) e_n is non-degenerate in each variable, that means if $e_n(S, T) = 1$ for all $T \in E[n]$ then $S = \infty$ and also if $e_n(S, T) = 1$ for all $S \in E[n]$ then $T = \infty$.

(3) $e_n(T, T) = 1$ for all $T \in E[n]$.

(4) $e_n(T, S) = e_n(S, T)^{-1}$ for all $S, T \in E[n]$.

(5) $e_n(\sigma(S), \sigma(T)) = \sigma(e_n(S, T))$ for all automorphism σ of $\mathbb{K}$ such that σ is the identity map on the coefficients of E . For example if E is in Weierstrass normal form $y^2 = x^3 + ax + b$, then $\sigma(a) = a$ and $\sigma(b) = b$.

(6) $e_n(\alpha(S), \alpha(T)) = e_n(S, T)^{\deg(\alpha)}$ for any endomorphism α of E , where $\deg(\alpha)$ denote the degree the endomorphism α .

Corollary 15.9.1.1. *Let T_1, T_2 be a basis of $E[n]$. Then $e_n(T_1, T_2)$ is a primitive n th root of unity.*

Proof. As e_n is a map from $E[n] \times E[n]$ to μ_n , hence $e_n(T_1, T_2)$ is a n th root of unity. let $e_n(T_1, T_2) = \zeta$ with $\zeta^n = 1$. To show ζ is primitive n th root of unity, for that we need to show $\zeta^d = 1$ if and only if $n|d$. Let $\zeta^d = 1$.

$$\text{Now } e_n(T_1, dT_2) = e_n(T_1, T_2)^d = \zeta^d.$$

Let $S \in E[n]$, as T_1, T_2 is a basis fro $E[n]$, hence we can write S as;

$$S = aT_1 + bT_2, \quad \text{for some integer a and b.}$$

$$\begin{aligned} e_n(S, dT_2) &= e_n(aT_1 + bT_2, dT_2) \\ &= e_n(T_1, dT_2)^a \cdot e_n(T_2, dT_2)^b && \text{[Using property(1)]} \\ &= 1 \cdot e_n(T_2, T_2)^{bd} && \text{[Using property(3)]} \\ &= 1 \end{aligned}$$

Sine this hold for all $S \in E[n]$, hence by property(2) we get $dT_2 = \infty$. Since T_1, T_2 is a basis for $e[n]$, so T_2 is an element of order n , that is $dT_2 = \infty$ if and only if $n|d$.

Therefore we get $\zeta^d = 1 \Leftrightarrow n|d$.

$\therefore \zeta = e_n(T_1, T_2)$ is primitive n th root of unity. $\square$

Corollary 15.9.1.2. *If $E[n] \subseteq E(\mathbb{K})$, then $\mu_n \subset \mathbb{K}$.*

Proof. Let σ be an automorphism of $\overline{\mathbb{K}}$ such that σ fixes elements of $\mathbb{K}$. Let T_1, T_2 be a basis of $E[n]$, since T_1, T_2 are assumed to have coordinates in $\mathbb{K}$, we have $\sigma(T_1) = T_1$ and $\sigma(T_2) = T_2$.

$$\begin{aligned} \zeta &= e_n(T_1, T_2) \\ &= e_n(\sigma(T_1), \sigma(T_2)) \\ &= \sigma(e_n(T_1, T_2)) \\ &= \sigma(\zeta). \end{aligned}$$

The fundamental theorem of Galois theory says that if an element $a \in \overline{\mathbb{K}}$ is fixed by all such automorphisms σ , then $a \in \mathbb{K}$. Therefore, $\zeta \in \mathbb{K}$. Since ζ is a primitive n th root of unity by previous corollary, it follows that $\mu_n \subset \mathbb{K}$. $\square$

Corollary 15.9.1.3. *Let E be an elliptic curve defined over $\mathbb{Q}$. Then $E[n] \not\subseteq E(\mathbb{Q})$ for $n \geq 3$.*

Proof. Suppose $E[n] \subseteq E(\mathbb{Q})$, then $\mu_n \subset \mathbb{Q}$. But for $n \geq 3$, $\mathbb{Q}$ does not contain any n th root of unity other than 1 and -1. Hence $E[n] \not\subseteq E(\mathbb{Q})$. $\square$

Remark: When $n = 2$, it may happen that $E[2] \subseteq E(\mathbb{Q})$. For example, let E be the elliptic curve given by $y^2 = x(x-1)(x+1)$. Let's find $E[2]$. Recall that $P = (x, y)$ is a point of order two, then $y = 0$. Hence solving the above equation for $y = 0$, we get the corresponding values of x as 0, 1, -1. Therefore $E[2] = \{\infty, (0, 0), (1, 0), (-1, 0)\} \subseteq E(\mathbb{Q})$.

Weil pairing can be used to prove the following two proposition that will be used in the proof of Hasse's theorem.

Proposition 15.9.1. *Let α be an endomorphism of an elliptic curve defined over a finite field $\mathbb{K}$. Let n be a positive integer not divisible by the characteristic of $\mathbb{K}$. Then $\det(\alpha_n) \equiv \deg(\alpha) \pmod{n}$.*

Proof. let $\{T_1, T_2\}$ be a basis for $E[n]$. Let $\zeta = e_n(T_1, T_2)$. Corollary-2.1.1 imply that ζ is a primitive n th root of unity.

Recall that if α is an endomorphism of elliptic curve E , then get the matrix of α on $E[n]$ that is α_n , by describing its action on a basis $\{T_1, T_2\}$ of $E[n]$.

$$\alpha_n = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ with } a, b, c, d \in \mathbb{Z}_n.$$

$$\begin{aligned} \zeta^{\deg(\alpha)} &= e_n(\alpha(T_1), \alpha(T_2)) \quad [\text{By property (6) of Weil pairing.}] \\ &= e_n(aT_1 + cT_2, bT_1 + dT_2) \\ &= e_n(T_1, T_1)^{ab} e_n(T_1, T_2)^{ad} e_n(T_2, T_1)^{cb} e_n(T_2, T_2)^{cd} \quad [\text{Using property (1).}] \\ &= e_n(T_1, T_2)^{ad} e_n(T_2, T_1)^{cb} \quad [\text{Since } e_n(T, T) = 1, \text{ for all } T \in E[n].] \\ &= e_n(T_1, T_2)^{ad} e_n(T_1, T_2)^{-cb} \quad [\text{By property (6) of Weil pairing.}] \\ &= e_n(T_1, T_2)^{ad-bc} \\ &= \zeta^{ad-bc} \end{aligned}$$

Since ζ is a primitive n th root of unity, hence order of ζ is n . Therefore $\zeta^{\deg(\alpha)} = \zeta^{ad-bc}$ if and only if $\deg(\alpha) \equiv ad - bc \pmod{n}$.
 $\therefore \deg(\alpha) \equiv \det(\alpha_n) \pmod{n}$. $\square$

Proposition 15.9.2. *Let α and β be endomorphisms of E and let a, b be integers. The endomorphism $a\alpha + b\beta$ is defined by $(a\alpha + b\beta)(P) = a\alpha(P) + b\beta(P)$. Then $\deg(a\alpha + b\beta) = a^2\deg(\alpha) + b^2\deg(\beta) + ab\deg(\alpha + \beta) - \deg(\alpha) - \deg(\beta)$.*

Proof. Let n be any integer not divisible by characteristic of $\mathbb{K}$. Let represent α and β on $E[n]$ by the matrices $\alpha_n = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ and $\beta_n = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$ with respect to some basis of $E[n]$. Then

$$\begin{aligned} \det(a\alpha + b\beta) &= \det(a\alpha_n + b\beta_n) \\ &= \det \begin{pmatrix} aa_1 + ba_2 & ab_1 + bb_2 \\ ac_1 + bc_2 & ad_1 + bd_2 \end{pmatrix} \\ &= (aa_1 + ba_2)(ad_1 + bd_2) - (ab_1 + bb_2)(ac_1 + bc_2) \\ &= a^2a_1d_1 + aba_1d_2 + aba_2d_1 + b^2a_2b_2 - a^2b_1c_1 - abb_1c_2 - abb_2c_1 - b^2b_2c_2 \\ &= a^2(a_1d_1 - b_1c_1) + b^2(a_2b_2 - b_2c_2) + ab(a_1d_2 + a_2d_1 - b_1c_2 - b_2c_1) \end{aligned}$$

$$\det(a\alpha + b\beta) = a^2\det(\alpha_n) + b^2\det(\beta_n) + ab((a_1d_1 + a_2d_1 + a_1d_2 + a_2d_2 - c_1b_1 - c_1b_2 - c_2b_1c_2b_2) - (a_1d_1 - c_1b_1) - (a_2d_2 - b_2c_2))$$

$$\text{Or } \det(a\alpha + b\beta) = a^2\det(\alpha_n) + b^2\det(\beta_n) + ab((a_1d_1 + a_1d_2 + a_2d_1 + a_2d_2) - (b_1c_1 + c_1b_2 + c_2b_1 + c_2b_2) - (a_1d_1 - c_1b_1) - (a_2d_2 - b_2c_2))$$

$$\det(a\alpha + b\beta) = a^2\det(\alpha_n) + b^2\det(\beta_n) + ab((a_1 + a_2)(d_1 + d_2) - (c_1 + c_2)(b_1 + b_2) - (a_1d_1 - c_1b_1) - (a_2d_2 - b_2c_2))$$

$$\det(a\alpha + b\beta) = a^2\det(\alpha_n) + b^2\det(\beta_n) + ab(\det(\alpha_n + \beta_n) - \det(\alpha_n) - \det(\beta_n)).$$

By previous proposition, we have

$$\det(\alpha_n) \equiv \deg(\alpha) \pmod{n}.$$

Applying this here, we get

$$\deg(a\alpha + b\beta) \equiv \det(a\alpha + b\beta) \pmod{n}$$

$$\therefore \deg(a\alpha + b\beta) \equiv a^2 \deg(\alpha) + b^2 \deg(\beta) + ab(\deg(\alpha + \beta) - \deg(\alpha) - \deg(\beta)) \pmod{n}.$$

□

15.10 Hasse's Theorem

Let E denote the elliptic curve given by Weierstrass equation,

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

with a_1, a_2, a_3, a_4, a_6 belongs the finite field $\mathbb{F}_q$, where $q = p^k$, for some $k \in \mathbb{Z}^+$. Then we denote the set of all points with coordinates in $\mathbb{F}_q$ by

$$E(\mathbb{F}_q) = \{(x, y) : x, y \in \mathbb{F}_q \text{ and } E(x, y) = 0\} \cup \{\infty\}.$$

Before doing more general theory lets look at an example.

Example 7. Consider the elliptic curve $E : y^2 = x^3 + x + 1$ over the field $\mathbb{F}_5$. Find $\#E(\mathbb{F}_5)$.

Since every (x, y) belongs to $E(\mathbb{F}_5)$, we have x and y are in $\mathbb{F}_5$. Hence we can take each of the five possible values of x and plug them into the equation of elliptic curve and see for which value of x , y^2 is square of elements of $\mathbb{F}_5$.

Following this method and calculating each x in $\mathbb{F}_5$, we get:

- For $x = 0$, we get $y^2 = 1$, this implies $y = \pm 1$.
- For $x = 1$, we get $y^2 = 3$, but 3 is quadratic non residue modulo 5 that is no such y exist in $\mathbb{F}_5$ whose square is 3. Hence no point (x, y) belongs to $E(\mathbb{F}_5)$ with $x = 1$.
- For $x = 2$, we get $y^2 = 1$, that implies $y = \pm 1$.
- For $x = 3$, we get $y^2 = 1$, that implies $y = \pm 1$.
- For $x = 4$, we get $y^2 = 4$, that implies $y = \pm 2$.

So we get eight points with coordinates in $E(\mathbb{F}_5)$ with one additional point ∞ . Hence $E(\mathbb{F}_5) = \{(0, \pm 1), (2, \pm 1), (3, \pm 1), (4, \pm 2), \infty\}$ and $\#E(\mathbb{F}_5) = 9$.

It is natural that $E(\mathbb{F}_q)$ is finite as there are finitely many choices for x and y . Next interesting question is that how big is it? Can we bound its size? It turns out that our guess is true and Hasse's theorem for elliptic curve provides an estimate on the number of points on an elliptic curve over a finite field which is both bounded above and below. Notably the bounds depend only on the finite field but not on the elliptic curve. Which states as follows:

Theorem 15.10.1. (Hasse) *Let E be an elliptic curve over finite field $\mathbb{F}_q$. Then*

$$|\#E(\mathbb{F}_q) - q - 1| \leq 2\sqrt{q}.$$

Before proving Hasse's theorem let us see how it works. Let us revisit example 3.1. As we have seen that $\#E(\mathbb{F}_5) = 9$, now we will check this with the bounds given by Hasse. Let us apply Hasse's theorem to the elliptic curve E over $\mathbb{F}_5$.

$$2\sqrt{q} \leq |\#E(\mathbb{F}_q) - q - 1| \leq 2\sqrt{q},$$

here $q = 5$, hence $2\sqrt{5} \leq |\#E(\mathbb{F}_5) - 5 - 1| \leq 2\sqrt{5}$, and $\sqrt{5} \sim 2.23606$. Hence we have $2 \times -2.2360 + 6 \leq \#E(\mathbb{F}_5) \leq 2 \times 2.2360 + 6$ that implies $1.5272 \leq \#E(\mathbb{F}_5) \leq 10.4721$ or more precisely we can say that $\#E(\mathbb{F}_5)$ is greater than or equal to 2 and less than or equal to 10. Here $2 \leq \#E(\mathbb{F}_5) = 9 \leq 10$.

Definition 15.10.1. *The Frobenius Endomorphism for an elliptic curve E over a finite field $\mathbb{F}_q$ is defined as the map $\phi_q : E(\overline{\mathbb{F}_q}) \rightarrow E(\overline{\mathbb{F}_q})$ such that $\phi_q(x, y) = (x^q, y^q)$.*

Proposition 15.10.1. *Let ϕ_q be the Frobenius endomorphism of elliptic curve E defined over finite field $\mathbb{F}_q$. Then $\text{Ker}(\phi_q - 1) = E(\mathbb{F}_q)$.*

Proof. Consider E be an elliptic curve given in Weierstrass general form

$$y^2 + a_1xy + a_3 = x^3 + a_2x^2 + a_4x + a_6$$

Claim: $a \in \mathbb{F}_q \Leftrightarrow a^q = a$.

Recall that the subset $\mathbb{F}_q^\times$ of nonzero elements of $\mathbb{F}_q$ forms a group under multiplication of order $q - 1$. That is for all $0 \neq a \in \mathbb{F}_q$ we have $a^{q-1} = 1$. Therefore $a^q = a$ for all $a \in \mathbb{F}_q$.

Conversely polynomial $x^q - x$ over $\mathbb{F}_q$ has degree q , hence it has exactly q roots over $\overline{\mathbb{F}_q}$ counting multiplicity. As the polynomial $x^q - x$ and $qx^{q-1} - 1$ does not have any common factor, hence the polynomial $x^q - x$ has q distinct roots.

Hence the roots of $x^q - x = 0$ are precisely the elements of $\mathbb{F}_q$. therefore $a \in \mathbb{F}_q \Leftrightarrow a^q = a$.

$$\begin{aligned} \text{Let } (x, y) \in E(\mathbb{F}_q) &\Leftrightarrow x \in E(\mathbb{F}_q) \text{ and } y \in E(\mathbb{F}_q). \\ &\Leftrightarrow x^q = x \text{ and } y^q = y \\ &\Leftrightarrow \phi_q(x, y) = (x, y) \end{aligned}$$

Therefore fixed points of ϕ_q are those which precisely belongs to $E(\mathbb{F}_q)$. It follows that the points $(x, y) \in E(\mathbb{F}_q)$ are exactly those for which

$$\begin{aligned} (\phi_q - 1)(x, y) &= \phi_q(x, y) - (x, y) = (x, y) - (x, y) = 0 \\ \therefore \text{Ker}(\phi_q - 1) &= E(\mathbb{F}_q). \end{aligned}$$

□

Lemma 15.10.2. *Let q be a prime, and let r, s be integers such that $\gcd(s, q) = 1$.*

Let $a = q + 1 - \deg(\phi_q - 1)$, where ϕ_q is the frobenious endomorphism on some elliptic curve E over the finite field F_q . Then $\deg(r\phi_q - s) = r^2q + s^2 - rsa$.

Proof. Here we use the result on degree function of endomorphisms of elliptic curve which we proved in last section.

If α and β are endomorphism of E and a, b are integers. Then

$$\deg(a\alpha + b\beta) = a^2\deg(\alpha) + b^2\deg(\beta) + ab(\deg(\alpha + \beta) - \deg(\alpha) - \deg(\beta))$$

Take $\alpha = \phi_q, \beta = -1, a = r$ and $b = s$.

Since ϕ_q is the frobenious endomorphism sending (x, y) to (x^q, y^q) , therefore $\deg(\phi_q) = \max\{\deg(x^q), 1\} = q$. $\deg(-1) = 1$, from previous lemma, we get $\deg(\phi_q - 1) = \text{Ker}(\phi_q - 1)$. Using the above fact we get,

$$\begin{aligned} \deg(r\phi_q - s) &= r^2\deg(\phi_q) + s^2\deg(-1) + rs(\deg(\phi_q - 1) - \deg(\phi_q) - \deg(-1)) \\ &= r^2q + s^2 + rs(\deg(\phi_q - 1) - q - 1) \\ &= r^2q + s^2 - rsa, \quad \text{where } a = q + 1 - \deg(\phi_q - 1). \end{aligned}$$

□

Proposition 15.10.2. *Let $\alpha \neq 0$ be a separable endomorphism of an elliptic curve E defined over a field $\mathbb{K}$. Then, $\deg(\alpha) = |\ker(\alpha)|$.*

Proof. Let $\alpha(x, y) = (f_1(x), yf_2(x))$ where f_1, f_2 are rational functions. We can write $f_1(x) = \frac{p(x)}{q(x)}$, $p(x), q(x) \in \mathbb{K}[x]$. As given that α is separable hence

$$\begin{aligned} f_1(x)' &\neq 0 \\ \Rightarrow \frac{p'(x)q(x) - q'(x)p(x)}{q^2(x)} &\neq 0 \\ \Rightarrow p'(x)q(x) - q'(x)p(x) &\neq 0 \end{aligned}$$

Let $S = \{x \in \overline{\mathbb{K}} : (p'(x)q(x) - q'(x)p(x))q(x) = 0\}$.

Since $(p'(x)q(x) - q'(x)p(x)) \neq 0$ and $q(x) \neq 0$, hence S contains roots of the polynomial $(p'(x)q(x) - q'(x)p(x))q(x) \in \mathbb{K}[x]$. So S is a finite set.

Let $(a, b) \in E(\mathbb{K})$, satisfying

- (1) $a \neq 0, b \neq 0, (a, b) \neq \infty$
- (2) $\deg(p(x) - aq(x)) = \max\{\deg(p), \deg(q)\} = \deg(\alpha)$
- (3) $a \notin f_1(S)$
- (4) $(a, b) \in \alpha(E(\overline{\mathbb{K}}))$

Let $R = \{(x, y) \in E(\overline{\mathbb{K}}) : \alpha(x, y) = (a, b)\}$

Claim: $\#R = \deg(\alpha)$. Let $(x_1, y_1) \in R \Rightarrow \alpha(x_1, y_1) = (a, b)$.

That is

$$f_1(x_1) = \frac{p(x_1)}{q(x_1)} = a,$$

$$y_1 f_2(x_1) = b.$$

Recall that denominator of both of the rational function $f_1(x_1)$ and $f_2(x_1)$ are same and is equals to $q(x_1)$. Since $(a, b) \neq \infty$ it follows that $q(x_1) \neq 0$, hence $f_2(x_1)$ is well defined.

$$\begin{aligned} y_1 f_2(x_1) &= b \\ \Rightarrow y_1 &= \frac{b}{f_2(x_1)} \end{aligned}$$

Hence y_1 depends on x_1 . Therefore

$$|R| = \#\{x : \text{there exists } y \in \overline{\mathbb{K}} \text{ such that } (x, y) \in R\}$$

As we have choose (a, b) in such way that degree of the polynomial $p(x) - aq(x)$ is equals to $\deg(\alpha)$. Fundamental theorem of arithmetic guarantee that the polynomial $p(x) - aq(x) = 0$ has $\deg(\alpha)$ numbers of roots in $\overline{\mathbb{K}}$ and we want to show that these roots are all distinct. For contradiction, suppose there exist a root x_0 of multiplicity ≥ 2 .

Then $p(x_0) - aq(x_0) = 0$ and $p'(x_0) - aq'(x_0) = 0$.

Then rearranging these equations and multiplying we get,

$$\begin{aligned} ap(x_0)q'(x_0) &= ap'(x_0)q(x_0) \\ \Rightarrow ap(x_0)q'(x_0) - ap'(x_0)q(x_0) &= 0 \end{aligned}$$

As $a \neq 0$, $\Rightarrow x_0$ is a root of $p(x)q'(x) - p'(x)q(x)$. Hence $x_0 \in S$.

But $a = f_1(x_0)$ that implies $a \in f_1(S)$, that contradicts the way we have choose a . Therefore the polynomial $p(x) - aq(x)$ has exactly $\deg(\alpha)$ number of distinct roots. That is there are $\deg(\alpha)$ number of points (x_1, y_1) such that $\alpha(x_1, y_1) = (a, b)$ so $|R| = \ker(\alpha)$. As α is a homomorphism this result hold for any point in $E(\overline{\mathbb{K}})$ including the point ∞ .

$\therefore |Ker(\alpha)| = \deg(\alpha)$. □

Now we are ready to prove Hasse's theorem. Lets restate it.

Theorem 15.10.3. (Hasse) Let E be an elliptic curve over a finite field $\mathbb{F}_q$. Then

$$-2\sqrt{q} + q + 1 \leq \#E(\mathbb{F}_q) \leq 2\sqrt{q} + q + 1$$

Proof. Consider the endomorphism $(r\phi_q - s)$ from lemma-3.2. By definition degree of any endomorphism is a non negative integer. Hence $\deg(r\alpha - s) \geq 0$, from the lemma we have $\deg(r\alpha - s) = r^2q + s^2 - rsa$. That implies

$$\begin{aligned} r^2q + s^2 - rsa &\geq 0 \\ \Rightarrow q\left(\frac{r}{s}\right)^2 - a\left(\frac{r}{s}\right) + 1 &\geq 0 \end{aligned}$$

for all r, s with $\gcd(s, q) = 1$.

Claim: The set of $M = \{\frac{r}{s} : r, s \in \mathbb{Z} \text{ and } \gcd(s, q) = 1\}$ is dense in $\mathbb{R}$.

proof of Claim: For proving this claim we need some concept from analysis. As $q = p^k$, where p is a prime and k is a positive integer. We can take s as power 2 or 3 depending on $p \neq 2$ or $p \neq 3$ respectively, then $\gcd(q, s) = \gcd(p^k, s) = 1$. In particular

If $p \neq 2$, take $A = \{\frac{r}{s} : r \in \mathbb{Z}, s = 2^n, n \in \mathbb{N}\}$,

If $p \neq 3$, take $A = \{\frac{r}{s} : r \in \mathbb{Z}, s = 3^n, n \in \mathbb{N}\}$.

Then $A \subset M$ and we know that this set A is dense in $\mathbb{R}$. (Since for any real number a in $\mathbb{R}$ is we can construct a sequence a_n in A that converges to a .) Therefore for any real number x belongs to $\mathbb{R}$, we have

$$qx^2 - ax + 1 \geq 0$$

That means $qx^2 - ax + 1$ can have at most one distinct real root (If $qx^2 - ax + 1$ have two distinct real root then for infinitely many values of $x \in \mathbb{R}$, $qx^2 - ax + 1 \leq 0$). As we know for any polynomial $f(x)$ has no real root if and only if discriminant of $f(x) < 0$ and $f(x)$ has two repeated real root if and only if discriminant of $f(x) = 0$. So we conclude that discriminant of $qx^2 - ax + 1 \leq 0$, that is

$$a^2 - 4q \leq 0$$

$$\Rightarrow a^2 \leq 4q$$

$$\Rightarrow |a| \leq 2\sqrt{q}$$

$$\Rightarrow -2\sqrt{q} \leq a \leq 2\sqrt{q}$$

$$\Rightarrow -2\sqrt{q} \leq \#E(\mathbb{F}_q) - q - 1 \leq 2\sqrt{q} \quad [\because a = \#E(\mathbb{F}_q) - q - 1]$$

$$\therefore -2\sqrt{q} + q + 1 \leq \#E(\mathbb{F}_q) \leq 2\sqrt{q} + q + 1$$

□

15.11 Attacks With Weil Pairing

Recall the discrete logarithm problem on elliptic curve over a field $\mathbb{K}$. It is a computationally infeasible problem. Here we will discuss a way to attack a discrete logarithm problem on elliptic curve over finite field. One strategy for attacking a discrete logarithm problem is to reduce it to an easier discrete logarithm problem. This can often be done with Weil pairing, it reduces the discrete logarithm problem on elliptic curve to the multiplicative group of finite field.

15.11.1 The MOV Attack

The MOV attack, uses the Weil pairing to convert a discrete logarithm problem in a finite field $E(\mathbb{F}_q)$ to one in $\mathbb{F}_{q^m}^\times$. It is named after mathematicians Menezes, Okamoto, and Vanstone. Since discrete log problems in finite fields can be attacked by index calculus methods, they can be solved faster than elliptic curve discrete log problems, as long as the field $\mathbb{F}_{q^m}$ is not much larger than $\mathbb{F}_q$.

Recall that for an elliptic curve E defined over $\mathbb{F}_q$.

$$E[n] = \{P \in \overline{\mathbb{F}_q} : nP = \infty\}$$

If $\gcd(q, n) = 1$ and $S, T \in E[n]$, then the Weil pairing $e_n(S, T)$ is an n th root of unity and this can be computed easily.

We want to find discrete logarithm of Q to the base P , that is to find k such that $kP = Q$.

Lemma 15.11.1. *Let E be an elliptic curve over $\mathbb{F}_q$ and $P, Q \in E(\mathbb{F}_q)$. Let N be the order of P . Assume that $\gcd(N, q) = 1$. There exists k such that $Q = kP$ if and only if $NQ = \infty$ and the Weil pairing $e_N(P, Q) = 1$.*

Proof. Suppose there exist k such that $Q = kP$. Then $NQ = NkP = k(NP) = \infty$.

$e_N(P, Q) = e_N(P, kP) = e_N(P, P)^k = 1$ (Using property (3) of Weil pairing).

Conversely suppose that $NQ = \infty$ and $e_N(P, Q) = 1$.

Now $NQ = \infty$ imply that $Q \in E[N]$. Since $\gcd(N, q) = 1$, from section-1, we have

$$E[N] = \mathbb{Z}_N \oplus \mathbb{Z}_N.$$

Now choose a point $R \in E[N]$ such that $\{P, R\}$ forms a basis for $E[N]$. Hence we can write $Q \in E[N]$ as

$$Q = kP + tR, \quad \text{for some integer } k, t.$$

Recall that if $\{P, R\}$ is a basis for $E[N]$ then $e_N(P, R) = \zeta$, a primitive N th root of unity. As we have assumed that $1 = e_N(P, Q)$, hence

$$1 = e_N(P, Q) = e_N(P, kP + tR) = e_N(P, P)^k e_N(P, R)^t = \zeta^t.$$

As ζ is primitive N th root of unity, therefore $t \equiv 0 \pmod{N}$. That is $N|t$, as $R \in E[N] \Rightarrow tR = \infty$.

$$\therefore Q = kP + tR = kP + \infty = kP. \quad \square$$

The idea used in the proof of the above lemma gives the MOV attack on discrete logarithms for elliptic curves.

Choose m such that

$$E[N] \subseteq E(\mathbb{F}_{q^m}).$$

Since points of $E[N]$ have coordinates in $\overline{\mathbb{F}_q}$, such an m exists.

Recall that the group μ_N of N th roots of unity is contained in $\mathbb{F}_{q^m}$, whenever $E[N] \subseteq E(\mathbb{F}_{q^m})$.

All of our calculations will be done in $\mathbb{F}_{q^m}$.

Given that E is an elliptic curve over $\mathbb{F}_q$ and $P, Q \in E(\mathbb{F}_q)$. Let N be the order of P . We want to solve the discrete log problem $kP = Q$ in $E(\mathbb{F}_q)$

The algorithm is as follows;

1. Choose a random point $T \in E(\mathbb{F}_{q^m})$.
2. Compute the order M of T .
3. Let $d = \gcd(M, N)$, and let $T_1 = (\frac{M}{d})T$. Then T_1 has order d , which divides N , so $T_1 \in E[N]$.

4. Compute $\zeta_1 = e_N(P, T_1)$ and $\zeta_2 = e_N(Q, T_1)$. Then both ζ_1 and ζ_2 are in $\mu_d \subseteq \mathbb{F}_{q^m}^\times$.
5. Solve the discrete logarithm problem $\zeta_2 = \zeta_1^k$ in $\mathbb{F}_{q^m}^\times$. This will give $k \pmod{d}$.
6. Repeat with random points T until the least common multiple of the various d 's obtained is N . This determines $k \pmod{N}$.

15.12 Supersingular Elliptic Curve

Definition 15.12.1. An elliptic curve E over a field of characteristic p is called supersingular if $E[p] = \{\infty\}$.

In other words, we can say there are no points of order p , even with coordinates in an algebraically closed field. Supersingular curves have many interesting properties, we will discuss some of them. These curves are very much useful in Weil pairing cryptography. An attractive feature of supersingular curves is that computations involving an integer times a point can be sometimes much faster.

The following results give a useful way to check whether or not an elliptic curve over a finite field is supersingular.

Proposition 15.12.1. Let E be an elliptic curve over $\mathbb{F}_q$, where q is a power of the prime p . Let $a = \#E(\mathbb{F}_q) - q - 1$. Then E is supersingular if and only if $a \equiv 0 \pmod{p} \Leftrightarrow \#E(\mathbb{F}_q) \equiv 1 \pmod{p}$.

Corollary 15.12.0.1. Suppose $p \geq 5$ is a prime and E is defined over $\mathbb{F}_p$. Then E is supersingular if and only if $a = 0$, which is the case if and only if $\#E(\mathbb{F}_p) = p + 1$.

Proof. If $a = 0$, then $a \equiv 0 \pmod{p}$, by the above theorem we have E is supersingular. When $a \neq 0$, this implies $\#E(\mathbb{F}_p) = p + 1$.

Now suppose that E is supersingular but $a \neq 0$. Then by the above theorem $a \equiv 0 \pmod{p}$ implies $|a| \geq p$. By Hasse's theorem, $|a| \leq 2\sqrt{p}$

$$\begin{aligned}
 p &\leq |a| \leq 2\sqrt{p} \\
 \Rightarrow p &\leq 2\sqrt{p} \\
 \Rightarrow \sqrt{p} &\leq 2 \\
 \Rightarrow p &\leq 4.
 \end{aligned}$$

This is a contradiction as we have assumed that $p \geq 5$, hence $a = 0$ and this imply $\#E(\mathbb{F}_p) = p + 1$ $\square$

The integer m could be large, in this case the discrete logarithm problem in $E(\mathbb{F}_q)$ is transformed in the group $\mathbb{F}_{q^m}^\times$, and order of $\mathbb{F}_{q^m}^\times = q^m - 1$. Then the problem is even hard as the original discrete log problem in the smaller group $E(\mathbb{F}_q)$, which has order approximately $q + 1$, by Hasse's theorem. For this purpose we will choose certain elliptic curves.

Proposition 15.12.2. *Let E be an elliptic curve over $\mathbb{F}_q$ and suppose $a = q + 1 - \#E(\mathbb{F}_q) = 0$. If there exists a point $P \in E(\mathbb{F}_q)$ of order N , then $E[N] \subseteq E(\mathbb{F}_{q^2})$, where N is a positive integer.*

Proof. Here we will use one result on Frobenius endomorphism ϕ_q , that is ϕ_q satisfies; $\phi_q^2 - a\phi_q + q = 0$. Since $a = 0$, we get;

$$\begin{aligned}\phi_q^2 + q &= 0 \\ \phi_q^2 &= -q\end{aligned}$$

let $S \in E[N]$, we will show that $S \in E(\mathbb{F}_{q^2})$.
As $a = 0 \Rightarrow \#E(\mathbb{F}_q) = q + 1$ and $E(\mathbb{F}_q)$ contains a points of order N , hence $N \mid \#E(\mathbb{F}_q) = q + 1$.

$$\begin{aligned}\text{That is } q + 1 &\equiv 0 \pmod{N}. \\ &\Rightarrow -q \equiv 1 \pmod{N}.\end{aligned}$$

Therefore for any $S = (x, y) \in E[N]$, we have

$$\phi_q^2(S) = -q(S) = 1.S = S$$

Observe that $\phi_q^2(x, y) = \phi_q(\phi_q(x, y)) = \phi_q(x^q, y^q) = (x^{q^2}, y^{q^2}) = \phi_{q^2}(x, y)$.

Now $\phi_{q^2}(x, y) = (x, y)$ if and only if $(x, y) \in E(\mathbb{F}_{q^2})$.

Since for $S = (x, y) \in E[N]$, $\phi_{q^2}(S) = S$. [Using property of frobenious endomorphism]

$\therefore S \in E(\mathbb{F}_{q^2})$ or $E[N] \subseteq E(\mathbb{F}_{q^2})$. $\square$

Therefore, discrete logarithm problems over $\mathbb{F}_q$ for supersingular curves with $a = 0$ can be reduced to discrete logarithm problems in $\mathbb{F}_{q^2}^\times$. These are much easier to solve.

Here we will see some special supersingular curve that will mostly used in Weil pairing cryptosystem.

Proposition 15.12.3. *Suppose q is a odd positive integer and $q \equiv 2 \pmod{3}$. Let $B \in \mathbb{F}_q^\times$, then the elliptic curve $E : y^2 = x^3 + B$ is supersingular.*

Proof. let $\psi : \mathbb{F}_q^\times \rightarrow \mathbb{F}_q^\times$ be the homomorphism defined by $\psi(x) = x^3$. Since $3 \nmid q - 1$, $\mathbb{F}_q^\times$ does not contain any element of order 3. Hence $\text{Ker}(\psi) = \{1\}$, that imply ψ is one-one. Then ψ is onto also as it is a homomorphism from a finite group to itself. Every element in $\mathbb{F}_q^\times$ has unique pre-image in $\mathbb{F}_q^\times$ or equivalently every element of $\mathbb{F}_q^\times$ has unique cube root in $\mathbb{F}_q^\times$. Since $0 \in \mathbb{F}_q$ has unique cube root that is 0 in $\mathbb{F}_q$. Hence we can say every element of $\mathbb{F}_q$ has unique cube root.

Now $x^3 = y^2 - B$. Hence for each $y \in \mathbb{F}_q$, there exist a unique cube root of $y^2 - B$ that is x such that $x^3 = y^2 - B$. For each q choices of y , we get q values of x , that correspond q points on E with coordinates in $\mathbb{F}_q$. Including the point ∞ , we have

$$\#E(\mathbb{F}_q) = q + 1.$$

Here $a = \#E(\mathbb{F}_q) - q + 1 = 0$, hence E is super singular. □

15.13 Diffie-Hellman Problem

In symmetric encryption scheme, same key is used for both encryption and decryption. Alice and Bob want to agree on common key that they can use for exchanging data. Suppose there is no prior contact between Alice and Bob, only communication channels between them are public. So this gives a challenging problem to the symmetric encryption system. One of the efficient way for key exchange given by Diffie and Hellman. This is described as below.

1. Alice and Bob agree on an elliptic curve E over a finite field $\mathbb{F}_q$ such that the discrete logarithm problem is hard in $E(\mathbb{F}_q)$. They also agree on a point $P \in E(\mathbb{F}_q)$ such that the subgroup generated by P has large order.

2. Alice chooses a secret integer a , computes $P_a = aP$, and sends P_a to Bob.
3. Bob chooses a secret integer b , computes $P_b = bP$, and sends P_b to Alice.
4. Alice computes $a P_b = a(bP) = abP$.
5. Bob computes $bP_a = b(aP) = abP$.

Hence the common key that is abP is now shared between Alice and Bob.

Lets see the cryptanalysis part. If Eve is any unauthentic body, she has only the knowledge of the curve E , the finite field $\mathbb{F}_q$ and the points P , aP , bP . Therefore she need to solve the following problem;

Diffie Hellman Problem Given P, aP , and bP in $E(\mathbb{F}_q)$, to compute abP . If Eve can solve discrete logarithm in $E(\mathbb{F}_q)$, then she can use P and aP to find a . Then she can compute $a(bP)$ to get abP .

Decision Diffie Hellman Problem: Given P, aP , and bP in $E(\mathbb{F}_q)$, and given a point $Q \in E(\mathbb{F}_q)$ determine whether or not $Q = abP$.

Let E be the curve $y^2 = x^3 + 1$ over $\mathbb{F}_q$, where $q \equiv 2 \pmod{3}$. Then E is supersingular. Let $\omega \in \mathbb{F}_{q^2}$ be a primitive third root of unity. Note that $\omega \notin \mathbb{F}_q$, as the order of $\mathbb{F}_q^\times$ is $q - 1$ and $3 \nmid q - 1$.

Now define the map β from $\overline{\mathbb{F}}$ to $\overline{\mathbb{F}}$ as follows:

$$\begin{aligned} \beta : \overline{\mathbb{F}}_q &\longrightarrow \overline{\mathbb{F}}_q \\ (x, y) &\longrightarrow (\omega x, y), \\ \infty &\longrightarrow \infty. \end{aligned}$$

It can be easily checked by using the point addition formula on elliptic curve $\overline{\mathbb{F}}$ that β is an isomorphism. Hence if $P \in \overline{\mathbb{F}}$ has order n then $\beta(P)$ also has order n .

Define the modified Weil pairing;

$$\bar{e}_n(P_1, P_2) = e_n(P_1, \beta(P_2))$$

where $e_n(P_1, P_2)$ is the usual Weil paring and $P_1, P_2 \in E[n]$.

Lemma 15.13.1. *Let E be the curve $y^2 = x^3 + 1$ over $\mathbb{F}_q$, where $q \equiv 2 \pmod{3}$. Assume $3 \nmid n$. If $P \in E(\mathbb{F}_q)$ has order exactly n , then $\bar{e}_n(P, P)$ is a primitive n th root of unity.*

Proof. To show P and $\beta(P)$ are linearly independent in $E[n]$.

On contrary suppose that there is a nontrivial relation between P and $\beta(P)$ of the form

$$uP = v\beta(P), \text{ for some integers } u, v.$$

$$\text{Then } \beta(vP) = v\beta(P) = uP \in E(\mathbb{F}_q)$$

If $vP = \infty$ then $\beta(vP) = \beta(\infty) = \infty = uP$. As order of P is n and $uP = \infty$ this imply $u \equiv 0 \pmod{n}$.

If $vP \neq \infty$, we can write $vP = (x, y) \in \mathbb{F}_q$.

$$\text{Now } \beta(vP) = \beta(x, y) = (\omega x, y) \in E(\mathbb{F}_q)$$

Recall that we have just discussed that $\omega \notin \mathbb{F}_q$ as $3 \nmid q - 1$. Hence $x = 0$, we get the values of y by substituting $x = 0$ in the equation of the curve. This gives $y^2 = 1$ which imply that $y = \pm 1$. Therefore $vP = (x, y) = (0, \pm 1)$.

Using point addition formula for elliptic curve, you can check easily that $(0, \pm 1)$ is point of order 3. It is impossible as we have assumed that $3 \nmid n$ and order of P is n .

Hence, it follows that the only relation of the form $uP = v\beta(P)$ has $u, v \equiv 0 \pmod{n}$, so P and $\beta(P)$ independent on each other, so they form a basis of $E[n]$. Therefore $\bar{e}_n(P, P) = e_n(P, \beta(P))$ is a primitive n th root of unity. □

15.13.1 Solving Decision Diffie Hellman Problem:

In this method we will use modified Weil pairing to solve Decision Diffie Hellman Problem. This was given by Joux and Nguyen. Suppose P, aP, bP, Q

are known and we want to decide whether or not $Q = abP$. First, use the usual Weil pairing to decide whether or not Q is a multiple of P . Recall that

$$e_n(P, tP) = e_n(P, P)^t = 1 \quad [\text{for any integer } t]$$

. Hence Q is a multiple of P if and only if $e_n(P, Q) = 1$. If this is the case, that is $Q = tP$ for some t . We have

$$\bar{e}_n(aP, bP) = \bar{e}_n(P, P)^{ab} = \bar{e}_n(P, abP),$$

$$\text{and } \bar{e}_n(Q, P) = \bar{e}_n(tP, P) = \bar{e}_n(P, P)^t.$$

Assume that $3 \nmid n$, then $\bar{e}_n(P, P)$ is a primitive n th root of unity. Therefore

$$Q = abP \Leftrightarrow t \equiv ab \pmod{n} \Leftrightarrow \bar{e}_n(aP, bP) = \bar{e}_n(Q, P).$$

This solves the decision Diffie Hellman Problem in this case.

Note that here it is needed to compute the Weil pairing, not required to compute any discrete logs, even in finite fields.

15.13.2 Solving tripartite Diffie-Hellman key exchange

Joux has given another application of modified Weil pairing.

Suppose Alice, Bob, and Chris want to establish a common key. The standard Diffie-Hellman procedure requires two rounds of interaction. The modified Weil pairing allows this to be cut to one round.

Let E be the elliptic curve $y^2 = x^3 + 1$ over $\mathbb{F}_q$, where $q \equiv 2 \pmod{3}$. Let P be a point of order n . Usually, n should be chosen to be a large prime. Alice, Bob, and Chris do the following.

1. Alice, Bob, and Chris choose secret integers $a, b, c \pmod{n}$, respectively.
2. Alice broadcasts aP , Bob broadcasts bP , and Chris broadcasts cP .
3. Alice computes $\bar{e}_n(bP, cP)^a$, Bob computes $\bar{e}_n(aP, cP)^b$, and Chris computes $\bar{e}_n(aP, bP)^c$.
4. Since each of the three users has computed the same number, they use this number to produce a key, using some publicly prearranged method.

15.14 A Cryptosystem Based on Weil Pairing

Let E be defined by $y^2 = x^3 + 1$ over $\mathbb{F}_p$, where $p \equiv 2 \pmod{3}$. Let $\omega \in \mathbb{F}_{p^2}$ be a primitive third root of unity. Since E is supersingular this is equivalent to $a = \#E(\mathbb{F}_p) - p - 1 = 0$, therefore $\#E(\mathbb{F}_p) = p + 1$.

Add the further assumption that $p = 6l - 1$ for some prime l . Then $6P$ has order l or 1 for each $P \in E(\mathbb{F}_p)$.

In most public key systems, when Alice wants to send a message to Bob, she looks up Bob's public key. However, she needs some way of verifying that this key actually belongs to Bob, rather than someone such as Eve who is masquerading as Bob.

We will describe a cryptosystem where each user has a public key based on her or his identity, such as an email address. A central trusted authority assigns a corresponding private key to each user.

To set up the system the trusted authority does the following:

1. Chooses a large prime $p = 6l - 1$.
2. Chooses a point P of order l in $E(\mathbb{F}_p)$.
3. Chooses hash functions H_1 and H_2 . The function H_1 takes a string of bits of arbitrary length and outputs a point of order l on E . The function H_2 inputs an element of order l in $\mathbb{F}_{p^2}^\times$ and outputs a binary string of length n , where n is the length of the messages that will be sent.
4. Chooses a secret random $s \in \mathbb{F}^\times$ and computes $P_{pub} = sP$.
5. Makes $p, H_1, H_2, n, P, P_{pub}$ public, while keeps s secret.

If a user with identity ID wants a private key, the trusted authority does the following:

1. Computes $Q_{ID} = H_1(ID)$, this is a point on E .
2. Lets $D_{ID} = sQ_{ID}$.

3. After verifying that ID is the identification for the user with whom he is communicating, sends D_{ID} to this user.

Encryption

If Alice wants to send a message M to Bob, she does the following:

1. Looks up Bob's identity, for example, $ID = bob@computer.com$ and computes $Q_{ID} = H_1(ID)$.
2. Choose a random $r \in \mathbb{F}_l^\times$.
3. Computes $g_{ID} = \bar{e}_n(Q_{ID}, P_{pub})$.
4. let $c = (rP, M \oplus H_2(g_{ID}^r))$, where $\oplus$ denotes XOR (bitwise addition mod 2).

Decryption

Bob decrypts a ciphertext (u, v) as follows:

1. Uses his private key D_{ID} to compute $h_{ID} = \bar{e}_l(D_{ID}, u)$.
2. Computes $m = v \oplus H_2(h_{ID})$.

The decryption works because

$$\bar{e}_l(D_{ID}, u) = \bar{e}_l(sQ_{ID}, rP) = \bar{e}_l(Q_{ID}, P)^{sr} = \bar{e}_l(Q_{ID}, sP)^r = \bar{e}_l(Q_{ID}, P_{pub})^r = g_{ID}^r.$$

Therefore,

$$m = v \oplus H_2(\bar{e}_l(D_{ID}, u)) = (M \oplus H_2(g_{ID}^r)) \oplus H_2(g_{ID}^r) = M \oplus H_2(g_{ID}^r) = M.$$

Chapter 16

Stream Ciphers

16.1 Encryption and Decryption

Stream Cipher is a symmetric key cryptosystem. Symmetric key cryptosystem means that the encryption and the decryption keys are same. In a stream cipher, we write plain text in binary form, and encryption is done bitwise (i.e. one symbol at a time).

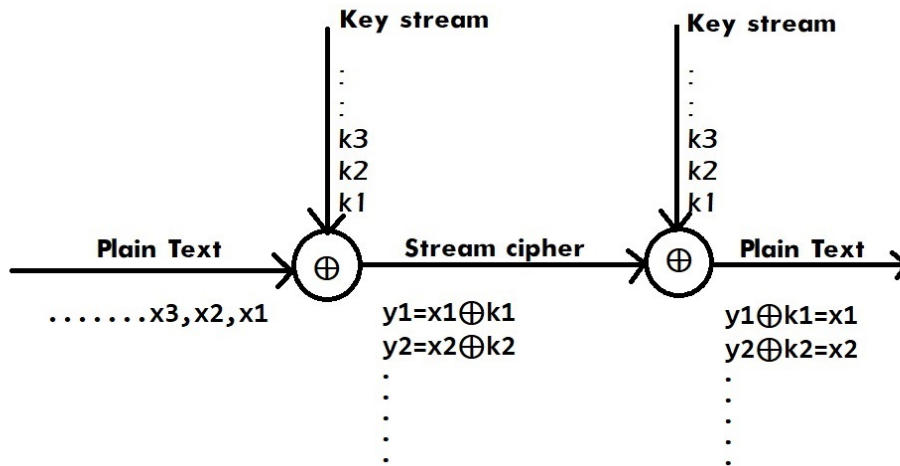


Figure 16.1: Block Diagram of Stream Cipher

- **Encryption:** Suppose plain text is $x = \dots x_3, x_2, x_1$ and key is $k =$

$\dots k_3, k_2, k_1$ in binary form then stream cipher is obtained by

$$y = \dots, y_3, y_2, y_1 \text{ where } y_i = x_i \oplus k_i$$

- **Decryption:** Decryption of $y = \dots, y_3, y_2, y_1$ is obtained by

$$x = y \oplus k = \dots, x_3, x_2, x_1 \text{ where } y_i \oplus k_i = x_i \oplus k_i \oplus k_i = x_i$$

Since in stream cipher encryption and decryption is done bitwise. So we must have a binary representation (equivalent) of plain text space and cipher text space.

ASCII (American Standard Code for Information Interchange) codes of English Alphabets and their binary representation in 8-bit:

Letter	ASCII Code	Binary	Letter	ASCII Code	Binary
a	097	01100001	A	065	01000001
b	098	01100010	B	066	01000010
c	099	01100011	C	067	01000011
d	100	01100100	D	068	01000100
e	101	01100101	E	069	01000101
f	102	01100110	F	070	01000110
g	103	01100111	G	071	01000111
h	104	01101000	H	072	01001000
i	105	01101001	I	073	01001001
j	106	01101010	J	074	01001010
k	107	01101011	K	075	01001011
l	108	01101100	L	076	01001100
m	109	01101101	M	077	01001101
n	110	01101110	N	078	01001110
o	111	01101111	O	079	01001111
p	112	01110000	P	080	01010000
q	113	01110001	Q	081	01010001
r	114	01110010	R	082	01010010
s	115	01110011	S	083	01010011
t	116	01110100	T	084	01010100
u	117	01110101	U	085	01010101
v	118	01110110	V	086	01010110
w	119	01110111	W	087	01010111
x	120	01111000	X	088	01011000
y	121	01111001	Y	089	01011001
z	122	01111010	Z	090	01011010

Example: Encryption “PEN” with key $k = 011010$

P=01010000

E=01000101

N=01001110

Plain Text (m)=01010000 01000101 01001110

m	0	1	0	1	0	0	0	0	0	1	0	0	0	1	0	1	0	1	0	0	1	1	1	0
$k = 011010$	0	1	1	0	1	0	0	1	1	0	1	0	0	1	1	0	1	0	0	1	1	0	1	0
$c = m \oplus k$	0	0	1	1	1	0	0	1	1	1	1	0	0	0	1	1	1	1	0	1	0	1	0	0

Cipher Text=00111001 11100011 11010100

Decryption: Cipher Text=00111001 11100011 11010100

c	0	0	1	1	1	0	0	1	1	1	1	0	0	0	1	1	1	1	0	1	0	1	0	0
$k = 011010$	0	1	1	0	1	0	0	1	1	0	1	0	0	1	1	0	1	0	0	1	1	0	1	0
$m = c \oplus k$	0	1	0	1	0	0	0	0	0	1	0	0	0	1	0	1	0	1	0	0	1	1	1	0

Plain Text (m)=01010000 01000101 01001110

The security of a stream cipher depends on the choice of the key. Key is a sequence of 0's and 1's. In practice, it is assumed that the size of the key must be equal to the size of plain text to be secure. The key's best choice is a truly random sequence of 0's and 1's.

Truly Random sequence can be generated by tossing a coin and note 1 if tails come and 0 if head comes.

But in practice, we do not use truly random sequences. We use pseudo-random sequences as the encryption key in stream ciphers.

• A **Pseudo-random sequence** must have two properties:

1. **Randomness:** It means that pseudo-random sequence must look like random sequence where it is difficult to find the rules how the sequence is generated by looking at sequence itself.
2. **Reproduction:** It means that for given small amount of data the pseudo-random sequence can be reproduced.

16.2 Pseudo-Random Number Generators(PRNGs)

Pseudo-Random Sequences Generators means mathematical algorithms that generate pseudo random sequence/number. There are many algorithms to generate pseudo random numbers. We shall discuss few of them.

16.2.1 Linear Congruential Generator

Linear Congruential Generator is a classical Pseudo Random Generator. It is based on linear congruences. The generator is defined by the following recurrence relation

$$x_{n+1} \equiv ax_n + b \pmod{m} \quad \text{for } n \geq 1$$

where x_0 is seed or initial value, $a, b \in \mathbb{Z}_m$. Observe that sequence terms lies in $\mathbb{Z}_m$

Example 1: Suppose the recurrence relation is defined by

$$x_{n+1} \equiv 3x_n + 2 \pmod{5}; \quad x_0 = 1 \quad \text{for } n \geq 1$$

then the pseudo random sequence is $x_0, x_1, x_2, x_3, x_4, x_5, x_6 \dots$ where

$$\begin{aligned} x_0 &\equiv 1 \pmod{5} \\ x_1 &\equiv 3x_0 + 2 \pmod{5} \equiv 3 \cdot 1 + 2 \equiv 3 + 2 = 5 \equiv 0 \pmod{5} \\ x_2 &\equiv 3x_1 + 2 \pmod{5} \equiv 3 \cdot 0 + 2 \equiv 0 + 2 = 2 \equiv 2 \pmod{5} \\ x_3 &\equiv 3x_2 + 2 \pmod{5} \equiv 3 \cdot 2 + 2 \equiv 6 + 2 = 8 \equiv 3 \pmod{5} \\ x_4 &\equiv 3x_3 + 2 \pmod{5} \equiv 3 \cdot 3 + 2 \equiv 9 + 2 = 11 \equiv 1 \pmod{5} \\ x_5 &\equiv 3x_4 + 2 \pmod{5} \equiv 3 \cdot 1 + 2 \equiv 3 + 2 = 5 \equiv 0 \pmod{5} \\ x_6 &\equiv 3x_5 + 2 \pmod{5} \equiv 3 \cdot 0 + 2 \equiv 0 + 2 = 2 \equiv 2 \pmod{5} \\ &\vdots \end{aligned}$$

i.e. sequence is $1, 0, 2, 3, 1, 0, 2, \dots$. Observe that the period of the sequence is 4 i.e. $x_{i+4} = x_i \quad \forall i \geq 0$

Linear Congruential Generator is fast but not secure, security depends on the choice of a, b and initial value x_0 . It is not suitable for cryptographic purposes.

16.2.2 Lagged Fibonacci Generator

Fibonacci sequence is defined by following recurrence relation

$$s_n = s_{n-1} + s_{n-2} \quad \forall n \geq 2$$

where s_0 and s_1 are the initial values of recurrence relation. The new term in the sequence is the sum of the last two terms in the sequence. Pseudo Random can be generated by same Fibonacci relation modulo a positive integer m i.e. $s_n = s_{n-1} + s_{n-2} \pmod{m}$.

Example: Let initial value $s_0 = 1, s_1 = 1$ then sequence modulo 7 is given

by

$$\begin{aligned}
 s_2 &= s_1 + s_0 = 1 + 1 = 2 \equiv 2 \pmod{7} \\
 s_3 &= s_2 + s_1 = 2 + 1 = 3 \equiv 3 \pmod{7} \\
 s_4 &= s_3 + s_2 = 3 + 2 = 5 \equiv 5 \pmod{7} \\
 s_5 &= s_4 + s_3 = 5 + 3 = 8 \equiv 1 \pmod{7} \\
 s_6 &= s_5 + s_4 = 1 + 5 = 6 \equiv 6 \pmod{7} \\
 &\vdots
 \end{aligned}$$

Sequence $s_0, s_1, s_2, s_3 \dots = 1, 1, 2, 3, 5, 1, 6 \dots$

16.3 Linear Feedback Shift Register (LFSR)

Definition: A sequence $s_0, s_1, \dots$ of elements from $\mathbb{F}_2$ satisfying the relation

$$s_{n+k} = a_{k-1}s_{n+k-1} \oplus a_{k-2}s_{n+k-2} \oplus \dots \oplus a_0s_n \oplus a \quad \text{for } n = 0, 1, 2, \dots$$

is called a k^{th} -order linear recurring sequence in $\mathbb{F}_2$.

• The terms $s_0, s_1, \dots, s_{k-1}$ are called initial value which determines the rest of the sequence uniquely.

Definition: If $a = 0$ then the linear recurring sequence is called a homogeneous linear recurring sequence.

A linear feedback shift register is a special kind of electronic circuit handling information in the form of element of $\mathbb{F}_2$. Four types of devices are used:

1. **Adder:** It has two input and one output, the output is the sum in $\mathbb{F}_2$ of two inputs.
2. **Constant Multiplier:** It has one input and yields as output the product of input with a constant element of $\mathbb{F}_2$.
3. **Constant Adder:** It adds a constant element of $\mathbb{F}_2$ to the input.
4. **Delay Element:** It has one input and one output and it is regulated by an external synchronous clock so that its input at a time appear as its output one unit of time later.

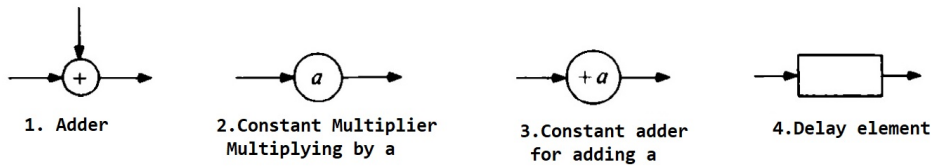


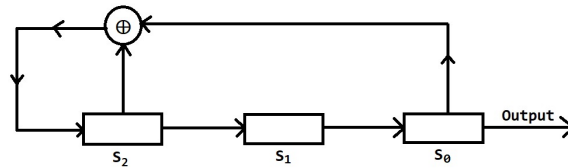
Figure 16.2: Building Blocks of LFSR

Definition: Let $s_0, s_1, \dots$ be a sequence of elements of $\mathbb{F}_2$. If there exist integers $r > 0$ and $n_0 \geq 0$ such that $s_{n+r} = s_n$ for all $n \geq n_0$, then the sequence is called ultimately periodic and r is called its period. The smallest number among all possible periods of ultimately periodic sequence is called the least period of the sequence.

Example: Consider 3^{rd} order homogeneous linear recurrence relation in $\mathbb{F}_2$

$$s_{n+3} = s_{n+2} \oplus s_n \text{ for } n = 0, 1, \dots$$

The LFSR is For initial value $(s_0, s_1, s_2) = (1, 0, 1)$

Figure 16.3: LFSR for $s_{n+3} = s_{n+2} \oplus s_n$

s_2	s_1	s_0	Output
1	0	1	1
0	1	0	0
0	0	1	1
1	0	0	0
1	1	0	0
1	1	1	1
0	1	1	1
1	0	1	

The sequence generated by LFSR is ... 1100101 and the least period is 7.

Theorem: Every k^{th} -order linear recurring sequence in $\mathbb{F}_2$ with least period r satisfies $r \leq 2^k$ and $r \geq 2^{k-1}$ if the sequence is homogeneous.

For cryptography, a sequence (key) is considered to be good if it has highest possible period.

Definition: An ultimately periodic sequence $s_0, s_1, \dots$ with least period r is called periodic if $s_{n+r} = s_n$ for all $n \geq 0$.

Definition: Let $s_0, s_1, \dots$ be homogeneous linear recurring sequence in $\mathbb{F}_2$ satisfying

$$s_{n+k} = a_{k-1}s_{n+k-1} \oplus a_{k-2}s_{n+k-2} \oplus \dots \oplus a_0s_n \text{ for } n = 0, 1, 2, \dots$$

where $a_i \in \mathbb{F}_2$ for $0 \leq i \leq k-1$. The polynomial

$$f(x) = x^k + a_{k-1}x^{k-1} + a_{k-2}x^{k-2} + \dots + a_0 \in \mathbb{F}_2[x]$$

is called the characteristic polynomial of linear recurring sequence (Characteristic polynomial of LFSR).

For example Consider 3^{rd} order homogeneous linear recurrence relation in $\mathbb{F}_2$

$$s_{n+3} = s_{n+2} \oplus s_n \text{ for } n = 0, 1, \dots$$

Characteristic polynomial is $f(x) = x^3 + x^2 + 1$.

Definition: A homogeneous linear recurring sequence in $\mathbb{F}_2$ whose characteristic polynomial is a primitive polynomial over $\mathbb{F}_2$ and which has non-zero initial value is called maximal period sequence in $\mathbb{F}_2$. Corresponding LFSR is called primitive LFSR.

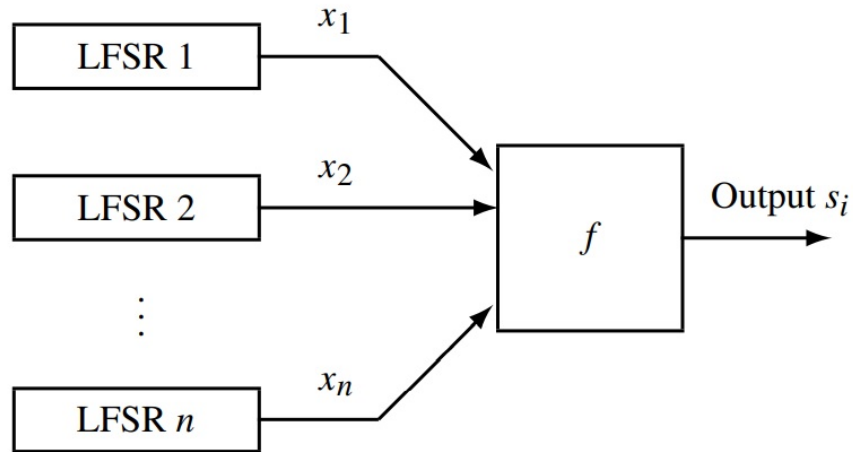
- A k^{th} -order homogeneous linear recurring sequence whose characteristic polynomial is primitive polynomial over $\mathbb{F}_2$ has highest possible period i.e. $2^k - 1$. When the Characteristic polynomial of an LFSR is primitive, for any nonzero initial value, the LFSR will produce a sequence with very good pseudo-random properties.

It is not easy to generate a sequence with high linear complexity unless the order of LFSR (for k^{th} -order linear recurring sequence, order of LFSR is k) is very high, which is in practice not possible. Nonlinear feedback shift registers can be used to generate a sequence with high linear complexity.

16.3.1 Non Linear Combiner

The efficient way to generate a sequence with high linear complexity is to use one or more LFSR's with a nonlinear Boolean function which acts as a combining function.

Nonlinear combiner is composed of few LFSRs, and output of all LFSRs forms the input of the nonlinear combining function $f(x)$, the output $f(x)$ is the final output of the generator. Suppose the output of n LFSRs are $x_1, x_2, \dots, x_n$ then the output of the non linear combiner function $f(x)$ is $f(x_1, x_2, \dots, x_n)$.



16.3.2 Non Linear Filtering Generators

A nonlinear filtering generator is composed of an LFSR and a nonlinear function $f(x)$, where the content of the cells in the LFSR is taken as the input of $f(x)$, and the output of $f(x)$ is the final output of the generator.

Example: Consider 4^{th} order homogeneous linear recurrence relation in $\mathbb{F}_2$

$$s_{n+4} = s_{n+3} \oplus s_{n+1} \oplus s_n \text{ for } n = 0, 1, \dots$$

For initial value $(s_0, s_1, s_2, s_3) = (1, 0, 1, 1)$

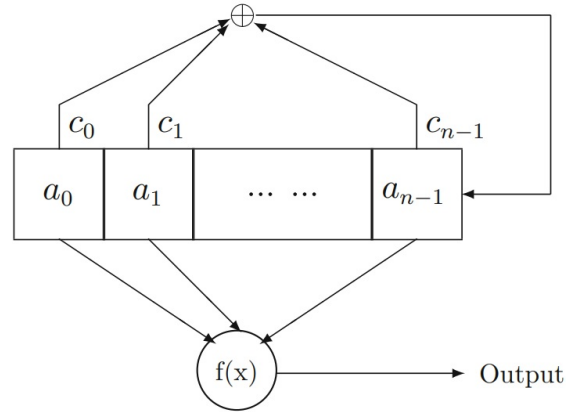
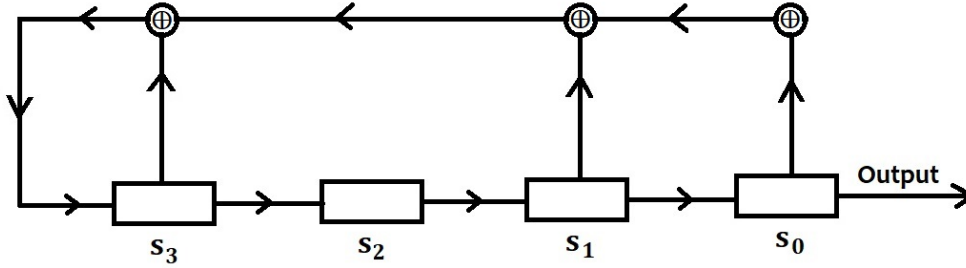


Figure 16.4: Block Diagram of Non Linear Filtering Generator

Content of cells after Shifts	s_3	s_2	s_1	s_0	Output
Shift-1	1	0	1	1	1
Shift-2	1	1	0	1	1
Shift-3	0	1	1	0	0
Shift-4	1	0	1	1	1
Shift-5	1	1	0	1	1
Shift-6	0	1	1	0	0
Shift-7	1	0	1	1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Let $f(x) : \mathbb{F}_2^4 \rightarrow \mathbb{F}_2$ be non linear filtering function defined by

$$f(x_1, x_2, x_3, x_4) = x_1x_2 + x_3x_4$$

Figure 16.5: Block Diagram $s_{n+4} = s_{n+3} \oplus s_{n+1} \oplus s_n$

$$f(1, 0, 1, 1) = 1 \cdot 0 + 1 \cdot 1 = 1$$

$$f(1, 1, 0, 1) = 1 \cdot 1 + 0 \cdot 1 = 1$$

$$f(0, 1, 1, 0) = 0 \cdot 1 + 1 \cdot 0 = 0$$

$$f(1, 0, 1, 1) = 1 \cdot 0 + 1 \cdot 1 = 1$$

$$f(1, 1, 0, 1) = 1 \cdot 1 + 0 \cdot 1 = 1$$

$$f(0, 1, 1, 0) = 0 \cdot 1 + 1 \cdot 0 = 0$$

$$f(1, 0, 1, 1) = 1 \cdot 0 + 1 \cdot 1 = 1$$

Sequence generated by taking content of cells as input of non linear filtering function is ...1011011.

16.4 Some Examples of Stream Cipher

16.4.1 Vigenère Cipher

Vigenère Cipher is a stream cipher (and block cipher also).

Recall: Vigenère Cipher

The process of encryption and decryption, suppose keyword

$$K = k_1 k_2 k_3 \dots k_l, \quad k_i \in \mathbb{Z}_n$$

message block is therefore

$$x = x_1x_2 \dots x_l, \quad x_i \in \mathbb{Z}_n$$

then encryption function is given by

$$\mathcal{E}(x) = y = y_1y_2 \dots y_l \quad \text{where } y_i \equiv x_i + k_i \pmod{n}$$

decryption function is given by

$$\mathcal{D}(y) = x = x_1x_2 \dots x_l \quad \text{such that } x_i \equiv y_i - k_i \pmod{n}$$

where $n = |\mathcal{A}|$ =number of symbols written in their numerical equivalent.

We can treat

$$x = (x_1, x_2, \dots, x_l) \in \underbrace{\mathbb{Z}_n \times \mathbb{Z}_n \times \dots \times \mathbb{Z}_n}_{l \text{ times}} = \mathbb{Z}_n^l$$

Now consider plaintext space $\mathcal{P} = \mathbb{Z}_n^l$ and cipher text space $\mathcal{C} = \mathbb{Z}_n^l$ Then encryption is defined as

$$\mathcal{E} : \mathcal{P} \rightarrow \mathcal{C}$$

$$\mathcal{E}(x) = y = (y_1, y_2, \dots, y_l) \quad \text{where each } y_i \in \mathbb{Z}_n$$

Observe that only one symbol is encrypted and decrypted at a time.

Example: Let $\mathcal{A} = \{A=0, B=1, \dots, Y=24, Z=25, ?=26\}$ i.e $|\mathcal{A}| = 27$
Encrypt the following plain text with keyword “EGYPT”

“HELLO HOW ARE YOU?”

Solution:

Keyword: EGYPT, Keylength=5

Encrypt: HELLO HOW ARE YOU?

First divide the message into blocks of length 5 i.e.

$$b_1 = \text{HELLO} \quad b_2 = \text{HOWAR} \quad b_3 = \text{EYOU?}$$

Encryption of block $b_1 = \text{HELLO} = x = x_1x_2x_3x_4x_5 = [7 \ 4 \ 11 \ 11 \ 14]$

$$K=\text{EGYPT}=k_1k_2k_3k_4k_5 = [4 \ 6 \ 24 \ 15 \ 19]$$

$$\mathcal{E}(x) = y = y_1y_2y_3y_4y_5 \text{ where } y_i \equiv x_i + k_i \pmod{27}$$

$$y_1 = x_1 + k_1 = 7 + 4 \equiv 11 \pmod{27}$$

$$y_2 = x_2 + k_2 = 4 + 6 \equiv 10 \pmod{27}$$

$$y_3 = x_3 + k_3 = 11 + 24 = 35 \equiv 8 \pmod{27}$$

$$y_4 = x_4 + k_4 = 11 + 15 \equiv 26 \pmod{27}$$

$$y_5 = x_5 + k_5 = 14 + 19 = 33 \equiv 6 \pmod{27}$$

$$\mathcal{E}(x) = y = y_1y_2y_3y_4y_5 = [11 \ 10 \ 8 \ 26 \ 6] = \text{LKI?G}$$

So encryption of 'HELLO' is LKI?G

$$\text{Similarly for block } b_2 = \text{HOWAR} = [7 \ 14 \ 22 \ 0 \ 17] = x = x_1x_2x_3x_4x_5$$

$$\mathcal{E}(x) = y = y_1y_2y_3y_4y_5 \text{ where } y_i \equiv x_i + k_i \pmod{27}$$

$$y_1 = x_1 + k_1 = 7 + 4 \equiv 11 \pmod{27}$$

$$y_2 = x_2 + k_2 = 14 + 6 \equiv 20 \pmod{27}$$

$$y_3 = x_3 + k_3 = 22 + 24 = 46 \equiv 19 \pmod{27}$$

$$y_4 = x_4 + k_4 = 0 + 15 \equiv 15 \pmod{27}$$

$$y_5 = x_5 + k_5 = 17 + 19 = 33 \equiv 9 \pmod{27}$$

$$\mathcal{E}(x) = y = y_1y_2y_3y_4y_5 = [11 \ 20 \ 19 \ 15 \ 9] = \text{LUTPJ}$$

So encryption of 'HOWAR' is LUTPJ.

Similarly do for block block b_3 .

$$\textbf{Decryption: } y = \text{LKI?G} = y_1y_2y_3y_4y_5 = [11 \ 10 \ 8 \ 26 \ 6]$$

$$\mathcal{D}(y) = x = x_1x_2x_3x_4x_5 \text{ such that } x_i \equiv y_i - k_i \pmod{27}$$

$$x_1 = y_1 - k_1 = 11 - 4 \equiv 7 \pmod{27}$$

$$x_2 = y_2 - k_2 = 10 - 6 \equiv 4 \pmod{27}$$

$$x_3 = y_3 - k_3 = 8 - 24 = -16 \equiv 11 \pmod{27}$$

$$x_4 = y_4 - k_4 = 26 - 15 \equiv 11 \pmod{27}$$

$$x_5 = y_5 - k_5 = 6 - 19 = -13 \equiv 14 \pmod{27}$$

$\mathcal{D}(y)=x=[7\ 4\ 11\ 11\ 14]=\text{HELLO}$

So decryption of 'LKI?G' is 'HELLO'.

16.4.2 A5/1 Cipher

A5/1 is a symmetric key encryption algorithm which is used for confidentiality in GSM mobile phones.

Set Up: It uses three Linear Feedback Shift Registers (LFSRs) to generate key stream say X, Y and Z .

The three LFSRs X, Y and Z have lengths 19, 22 and 23 respectively. LFSR X represented as $(X_0, X_1, \dots, X_{18})$

LFSR Y represented as $(Y_0, Y_1, \dots, Y_{21})$

LFSR Z represented as $(Z_0, Z_1, \dots, Z_{22})$

Output of LFSR X is given by $X_{13} \oplus X_{16} \oplus X_{17} \oplus X_{18}$.

Output of LFSR Y is given by $Y_{20} \oplus Y_{21}$.

Output of LFSR Z is given by $Z_7 \oplus Z_{20} \oplus Z_{21} \oplus Z_{22}$

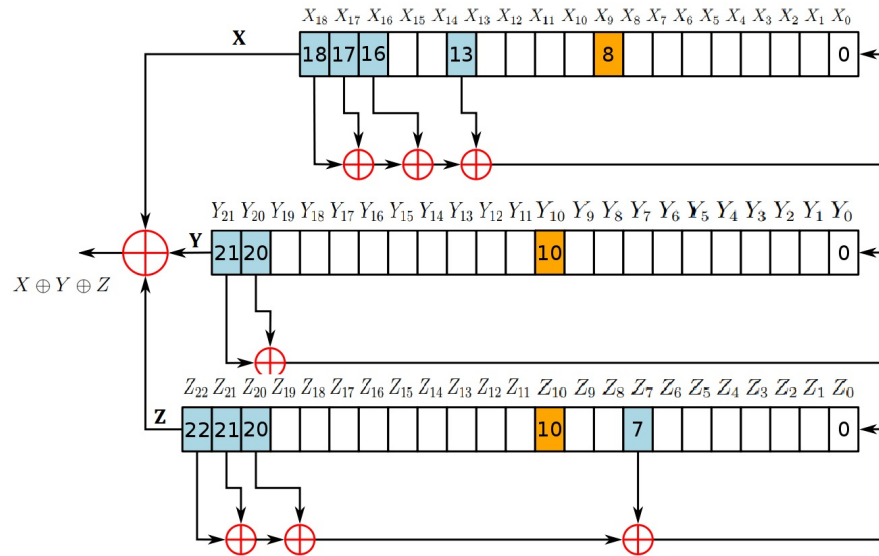


Figure 16.6: A5/1 Stream Cipher

Key Stream Generation Algorithm

To generate key stream, not all three LFSRs X, Y and Z shift. The shifting of registers is based on a function $maj(X_8, Y_{10}, Z_{10})$ which is defined as:

If the majority of bits X_8, Y_{10} and Z_{10} are 0, then function returns 0; If the majority of bits X_8, Y_{10} and Z_{10} are 1, then function returns 1.

To generate one key stream bit, first we calculate

$$m = maj(X_8, Y_{10}, Z_{10})$$

Then the registers X, Y and Z shift as follows:

- If $X_8 = m$ then register X shifts, i.e. if the 9th bit of register X equal to m , the shift operation applied to register X .
- If $Y_{10} = m$ then register Y shifts, i.e. if the 11th bit of register Y equal to m , the shift operation applied to register Y .
- If $Z_{10} = m$ then register Z shifts, i.e. if the 11th bit of register Z equal to m , the shift operation applied to register Z .

Example:

X_8	Y_{10}	Z_{10}	m	Registers to Shift
0	0	0	0	X, Y, Z
0	0	1	0	X, Y
0	1	0	0	X, Z
0	1	1	1	Y, Z
1	0	0	0	Y, Z
1	0	1	1	X, Z
1	1	0	1	X, Y
1	1	1	1	X, Y, Z

In the last, one key stream bit k is generated by addition modulo 2 (XOR) of outputs of LFSRS X, Y and Z .

$$k = X \oplus Y \oplus Z$$

This generates only one bit of key stream K . To generate key stream of length (length of plain text) n , we have to apply above steps n times.

R.K.Sharma

Chapter 17

Symmetric Key Ciphers

17.1 Data Encryption Standard(DES)

Data Encryption Standard is a symmetric key algorithm known as DES. It was chosen as a standard for encryption in 1977 by National Institute of Standard and Technology(NIST).

17.1.1 Description of DES

The design of this algorithm is based on 16 round Feistel structure. Here, the data is divided into chunks of 64 bit and each 64 bit is called a block. Each block of 64 bit is encrypted by this algorithm using a 64 bit of key. Out of 64 bit key actually it uses only 56 bits and rest 8 bits are redundant. The output of this algorithm is 64 bits. Decryption is done by reversing the encryption process by using the same key. This process is illustrate in fig. 1.

17.1.2 Encryption Process

The encryption process of DES algorithm is discussed in the subsequent sections and broad level of architecture is shown in figure 2 and each function used in this figure is explained in subsections.

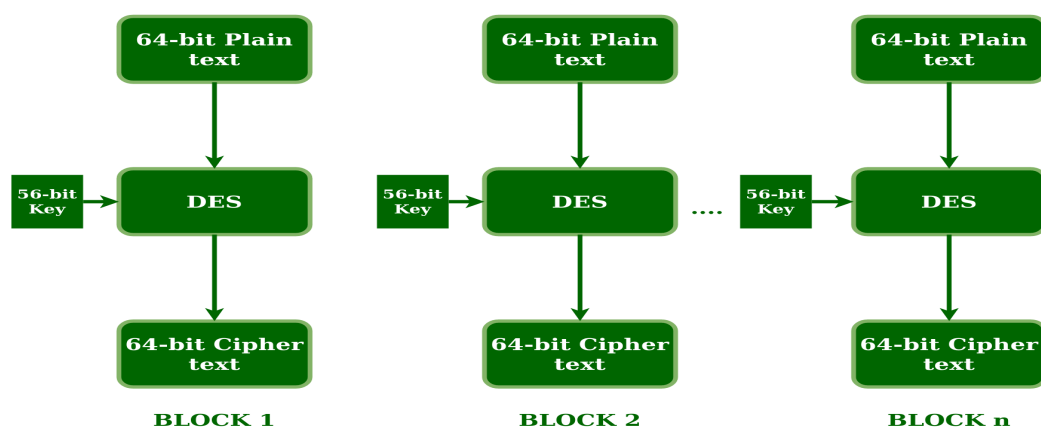


Figure 17.1: Description of DES

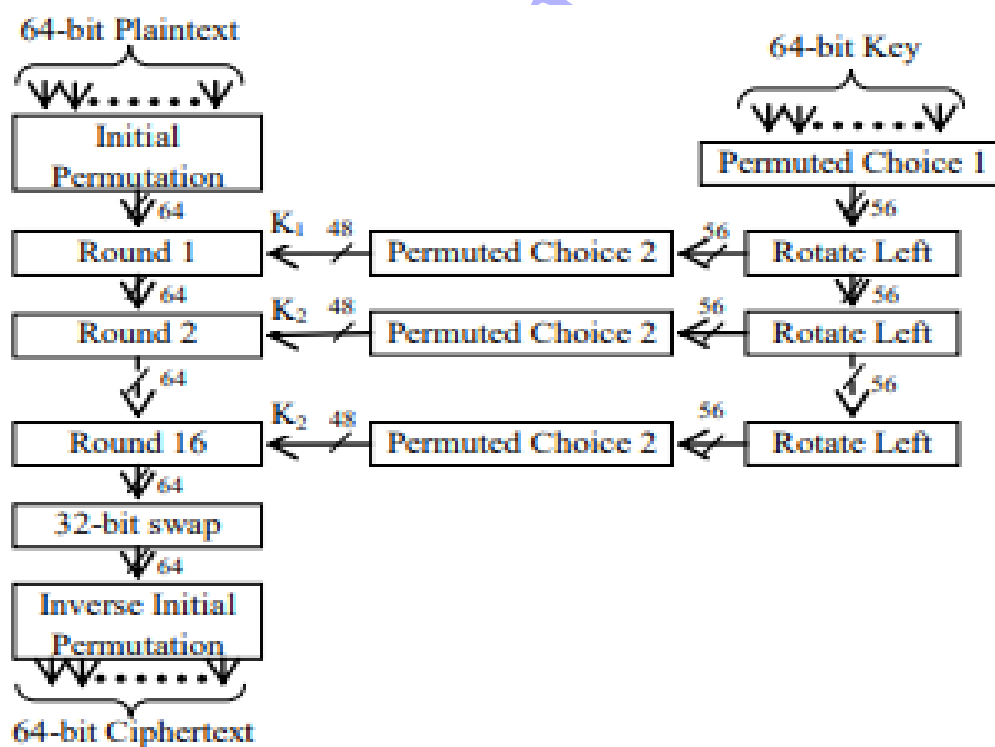


Figure 17.2: Encryption of DES

Initial Permutation

In first step, a 64 bit of plain text passes through the initial permutation IP, which is given in equation 1. As per this permutation, the bit at 58 position will go at the 1 position and so on.

$$IP = \begin{matrix} & 58 & 50 & 42 & 34 & 26 & 18 & 10 & 2 \\ & 60 & 52 & 44 & 36 & 28 & 20 & 12 & 4 \\ & 62 & 54 & 46 & 38 & 30 & 22 & 14 & 6 \\ & 64 & 56 & 48 & 40 & 32 & 24 & 16 & 8 \\ & 57 & 49 & 41 & 33 & 25 & 17 & 9 & 1 \\ & 59 & 51 & 43 & 35 & 27 & 19 & 11 & 3 \\ & 61 & 53 & 45 & 37 & 29 & 21 & 13 & 5 \\ & 63 & 55 & 47 & 39 & 31 & 23 & 15 & 7 \end{matrix} \quad (17.1)$$

Round Function

After initial permutation, the permuted bits passes through the round function, which is based on feistel structure. The process is shown in figure 3. During this process, the 64 bit value is divided in two equal parts say L_0 and R_0 i.e. left and right 32 bits. The following operations are applied on these values.

$$\begin{aligned} L_i &= R_{i-1} \\ R_i &= L_{i-1} \oplus F(R_{i-1}, K_i) \end{aligned} \quad (17.2)$$

where, $1 \leq i \leq 16$ i.e. number of rounds and K_i is the 48 bit i^{th} round key, which is generated from 64 bit initial key value by key generation algorithm. Key generation algorithm will be discussed separately. F is a Substitution function, take 32 bit R_{i-1} and 48 bit K_i as input values. This 32 bit R_{i-1} value is converted in 48 bit by following Expansion function:-

$$E = \begin{matrix} & 32 & 1 & 2 & 3 & 4 & 5 \\ & 4 & 5 & 6 & 7 & 8 & 9 \\ & 8 & 9 & 10 & 11 & 12 & 13 \\ & 12 & 13 & 14 & 15 & 16 & 17 \\ & 16 & 17 & 18 & 19 & 20 & 21 \\ & 20 & 21 & 22 & 23 & 24 & 25 \\ & 24 & 25 & 26 & 27 & 28 & 29 \\ & 28 & 29 & 30 & 31 & 32 & 1 \end{matrix} \quad (17.3)$$

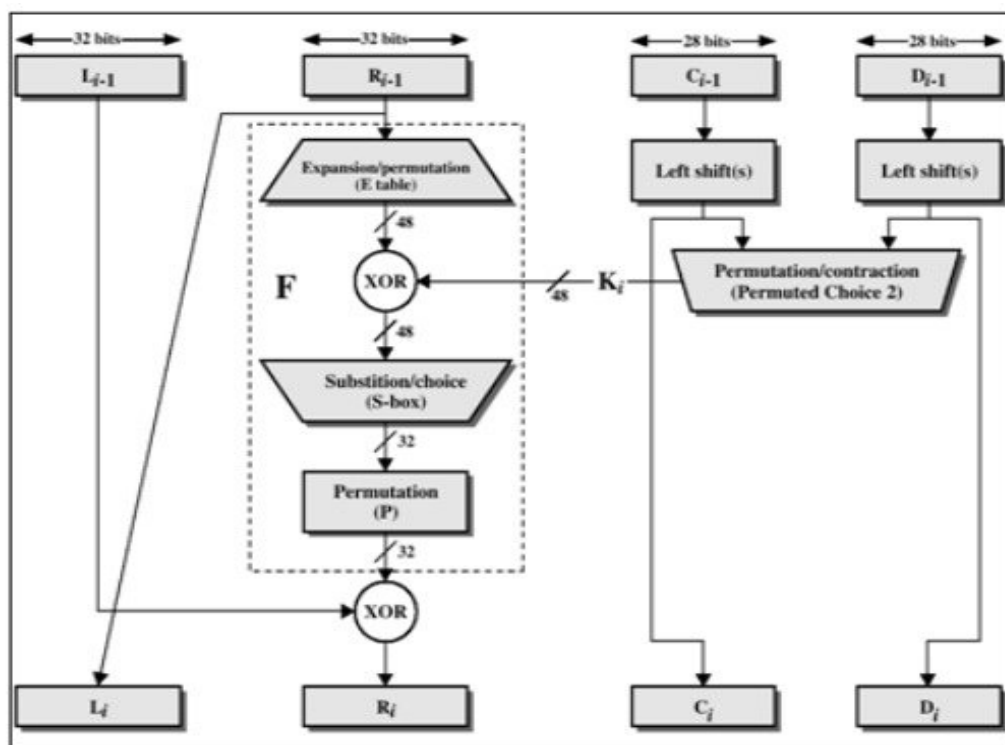


Figure 17.3: Round Function of DES

After expansion these bits are XORed with the 48 bit value of K_i . After XORing these 48 bits are input to the substitution functions F , which convert them into 32 bits. For this, It divides these 48 bits into 8 chunks of 6 bits each and each 6 bits go as input to the different S-box which convert them into 4 bits. 8 different S-boxes are defined as follows:-

$$S_1 = \begin{matrix} 14 & 4 & 13 & 1 & 2 & 15 & 11 & 8 & 3 & 10 & 6 & 12 & 5 & 9 & 0 & 7 \\ 0 & 15 & 7 & 4 & 14 & 2 & 13 & 1 & 10 & 6 & 12 & 11 & 9 & 5 & 3 & 8 \\ 4 & 1 & 14 & 8 & 13 & 6 & 2 & 11 & 15 & 12 & 9 & 7 & 3 & 10 & 5 & 0 \\ 15 & 12 & 8 & 2 & 4 & 9 & 1 & 7 & 5 & 11 & 3 & 14 & 10 & 0 & 6 & 13 \end{matrix} \quad (17.4)$$

$$S_2 = \begin{matrix} 15 & 1 & 8 & 14 & 6 & 11 & 3 & 4 & 9 & 7 & 2 & 13 & 12 & 0 & 5 & 10 \\ 3 & 13 & 4 & 7 & 15 & 2 & 8 & 14 & 12 & 0 & 1 & 10 & 6 & 9 & 11 & 5 \\ 0 & 14 & 7 & 11 & 10 & 4 & 13 & 1 & 5 & 8 & 12 & 6 & 9 & 3 & 2 & 15 \\ 13 & 8 & 10 & 1 & 3 & 15 & 4 & 2 & 11 & 6 & 7 & 12 & 0 & 5 & 14 & 9 \end{matrix} \quad (17.5)$$

$$S_3 = \begin{matrix} 10 & 0 & 9 & 14 & 6 & 3 & 15 & 5 & 1 & 13 & 12 & 7 & 11 & 4 & 2 & 8 \\ 13 & 7 & 0 & 9 & 3 & 4 & 6 & 10 & 2 & 8 & 5 & 14 & 12 & 11 & 15 & 1 \\ 13 & 6 & 4 & 9 & 8 & 15 & 3 & 0 & 11 & 1 & 2 & 12 & 5 & 10 & 14 & 7 \\ 1 & 10 & 13 & 0 & 6 & 9 & 8 & 7 & 4 & 15 & 14 & 3 & 11 & 5 & 2 & 12 \end{matrix} \quad (17.6)$$

$$S_4 = \begin{matrix} 7 & 13 & 14 & 3 & 0 & 6 & 9 & 10 & 1 & 2 & 8 & 5 & 11 & 12 & 4 & 15 \\ 13 & 8 & 11 & 5 & 6 & 15 & 0 & 3 & 4 & 7 & 2 & 12 & 1 & 10 & 14 & 9 \\ 10 & 6 & 9 & 0 & 12 & 11 & 7 & 13 & 15 & 1 & 3 & 14 & 5 & 2 & 8 & 4 \\ 3 & 15 & 0 & 6 & 10 & 1 & 13 & 8 & 9 & 4 & 5 & 11 & 12 & 7 & 2 & 14 \end{matrix} \quad (17.7)$$

$$S_5 = \begin{matrix} 2 & 12 & 4 & 1 & 7 & 10 & 11 & 6 & 8 & 5 & 3 & 15 & 13 & 0 & 14 & 9 \\ 14 & 11 & 2 & 12 & 4 & 7 & 13 & 1 & 5 & 0 & 15 & 10 & 3 & 9 & 8 & 6 \\ 4 & 2 & 1 & 11 & 10 & 13 & 7 & 8 & 15 & 9 & 12 & 5 & 6 & 3 & 0 & 14 \\ 11 & 8 & 12 & 7 & 1 & 14 & 2 & 13 & 6 & 15 & 0 & 9 & 10 & 4 & 5 & 3 \end{matrix} \quad (17.8)$$

$$S_6 = \begin{matrix} 12 & 1 & 10 & 15 & 9 & 2 & 6 & 8 & 0 & 13 & 3 & 4 & 14 & 7 & 5 & 11 \\ 10 & 15 & 4 & 2 & 7 & 12 & 9 & 5 & 6 & 1 & 13 & 14 & 0 & 11 & 3 & 8 \\ 9 & 14 & 15 & 5 & 2 & 8 & 12 & 3 & 7 & 0 & 4 & 10 & 1 & 13 & 11 & 6 \\ 4 & 3 & 2 & 12 & 9 & 5 & 15 & 10 & 11 & 14 & 1 & 7 & 6 & 0 & 8 & 13 \end{matrix} \quad (17.9)$$

$$S_7 = \begin{matrix} 4 & 11 & 2 & 14 & 15 & 0 & 8 & 13 & 3 & 12 & 9 & 7 & 5 & 10 & 6 & 1 \\ 13 & 0 & 11 & 7 & 4 & 9 & 1 & 10 & 14 & 3 & 5 & 12 & 2 & 15 & 8 & 6 \\ 1 & 4 & 11 & 13 & 12 & 3 & 7 & 14 & 10 & 15 & 6 & 8 & 0 & 5 & 9 & 2 \\ 6 & 11 & 13 & 8 & 1 & 4 & 10 & 7 & 9 & 5 & 0 & 15 & 14 & 2 & 3 & 12 \end{matrix} \quad (17.10)$$

$$S_8 = \begin{matrix} 13 & 2 & 8 & 4 & 6 & 15 & 11 & 1 & 10 & 9 & 3 & 14 & 5 & 0 & 12 & 7 \\ 1 & 15 & 13 & 8 & 10 & 3 & 7 & 4 & 12 & 5 & 6 & 11 & 0 & 14 & 9 & 2 \\ 7 & 11 & 4 & 1 & 9 & 12 & 14 & 2 & 0 & 6 & 10 & 13 & 15 & 3 & 5 & 8 \\ 2 & 1 & 14 & 7 & 4 & 10 & 8 & 13 & 15 & 12 & 9 & 0 & 3 & 5 & 6 & 11 \end{matrix} \quad (17.11)$$

Now the rule for S-box transformation says that first and last bit of 6 bit sequence defines the row number of the S-box and middle 4 bits defines the column number of S-box. The binary value corresponding to this row and column in that particular S-box is the output. Let 100101 is the input to first S- box then 11 i.e 3 in decimal is the row number and 0010 i.e 2 is the column number. The value corresponding to 4 row and 3 column in S-box one is 8. so output is 1000. The output of each S-box (four bits) is concatenated and 32 bits are obtained. These 32 bits are permuted by the permutation P,

which is defined as follows:-

$$P = \begin{matrix} & 16 & 7 & 20 & 21 & 29 & 12 & 28 & 17 \\ 1 & 1 & 15 & 23 & 26 & 5 & 18 & 31 & 10 \\ 2 & 2 & 8 & 24 & 14 & 32 & 27 & 3 & 9 \\ 19 & 19 & 13 & 30 & 6 & 22 & 11 & 4 & 25 \end{matrix} \quad (17.12)$$

This permuted value is the output of function F. The description of round function F is shown in figure 4. This output value is XORed with the value of L_{i-1} and this output called as R_i . This process of round function is repeated

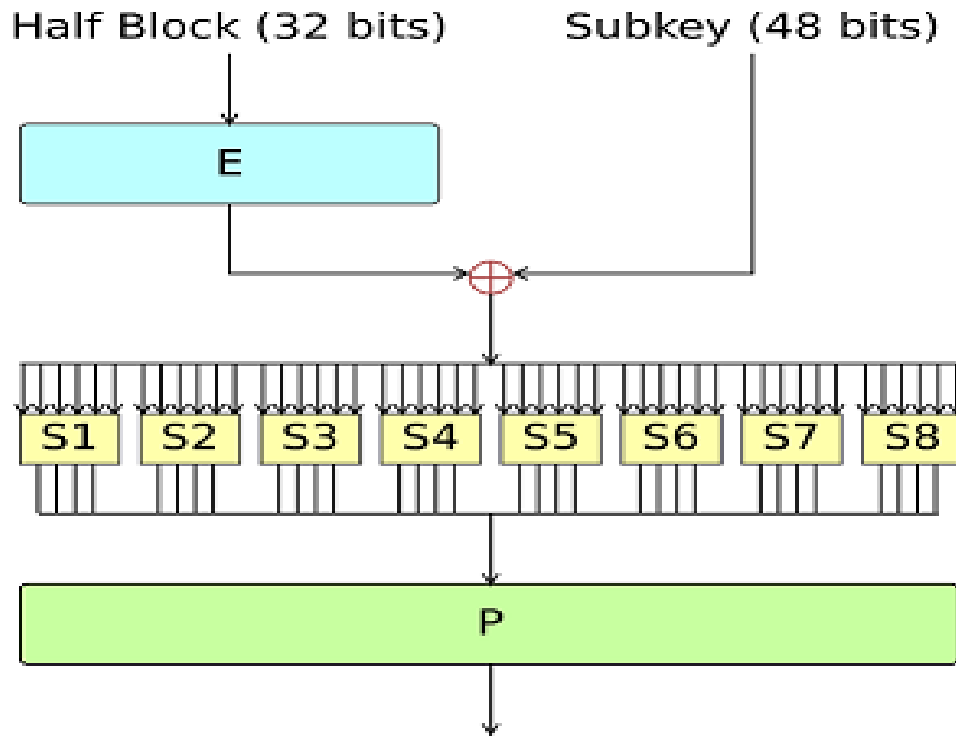


Figure 17.4: Calculation of F

for sixteen times, i.e. for $1 \leq i \leq 16$.

After 16 rounds the out values L_{16} and R_{16} are swapped with each other. Finally after swapping these two 32 bits values are concatenated and passes through the permutation which is basically inverse of initial permutation and

defines as follows (refer figure 2):-

$$IP^{-1} = \begin{matrix} 40 & 8 & 48 & 16 & 56 & 24 & 64 & 32 \\ 39 & 7 & 47 & 15 & 55 & 23 & 63 & 31 \\ 38 & 6 & 46 & 14 & 54 & 22 & 62 & 30 \\ 37 & 5 & 45 & 13 & 53 & 21 & 61 & 29 \\ 36 & 4 & 44 & 12 & 52 & 20 & 60 & 28 \\ 35 & 3 & 43 & 11 & 51 & 19 & 59 & 27 \\ 34 & 2 & 42 & 10 & 50 & 18 & 58 & 26 \\ 33 & 1 & 41 & 9 & 49 & 17 & 57 & 25 \end{matrix} \quad (17.13)$$

These 64 permuted bits are the cipher output.

17.1.3 Key Generation

The 64 bit secret key is used to generate the keys for all 16 rounds. The process of generation is defined in figure 5.

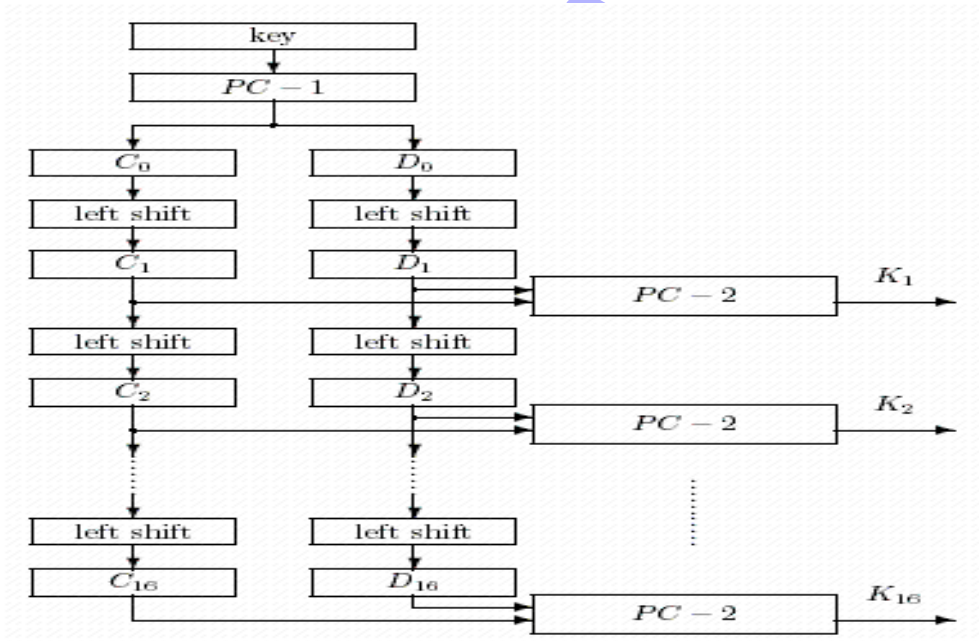


Figure 17.5: Round Key generation

First the bits of secret key are numbered from 1 to 64 and every eighth bit

is ignored. The remaining 56 bits numbered from 1 to 7, 9 to 15, 17 to 23, 25 to 31, 33 to 39, 41 to 47, 49 to 55, and 57 to 63 are permuted by the permutation PC-1, which is defined as follows:-

$$PC - 1 = \begin{array}{cccccc} 57 & 49 & 41 & 33 & 25 & 17 & 9 \\ 1 & 58 & 50 & 42 & 34 & 26 & 18 \\ 10 & 2 & 59 & 51 & 43 & 35 & 27 \\ 19 & 11 & 3 & 60 & 52 & 44 & 36 \\ 63 & 55 & 47 & 39 & 31 & 23 & 15 \\ 7 & 62 & 54 & 46 & 38 & 30 & 22 \\ 14 & 6 & 61 & 53 & 45 & 37 & 29 \\ 21 & 13 & 5 & 28 & 20 & 12 & 4 \end{array} \quad (17.14)$$

These permuted bits are divided into two equal parts C_0 and D_0 , each 28 bits. Both of these 28 bits are rotated left either one or two bits decided as per the following table:-

$$\begin{array}{rcccccccccccccccccccc} \text{No. of round -} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 \\ \text{Bits rotation -} & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 1 \end{array} \quad (17.15)$$

These rotated bits (C_1, D_1) are the input for generation of next round key and also used as key for current round after applying the permutation PC-2, which is defined as follows:-

$$PC - 2 = \begin{array}{cccccc} 14 & 17 & 11 & 24 & 1 & 5 & 3 & 28 \\ 15 & 6 & 21 & 10 & 23 & 19 & 12 & 4 \\ 26 & 8 & 16 & 7 & 27 & 20 & 13 & 2 \\ 41 & 52 & 31 & 37 & 47 & 55 & 30 & 40 \\ 51 & 45 & 33 & 48 & 44 & 49 & 39 & 56 \\ 34 & 53 & 46 & 42 & 50 & 36 & 29 & 32 \end{array} \quad (17.16)$$

In the same manner round keys are generated for all 16 rounds.

17.1.4 Decryption

Decryption is the reverse process of encryption and in Feistel cipher, decryption uses the same algorithm as encryption, except that the application of round keys are reversed. The decryption is shown in figure 6.

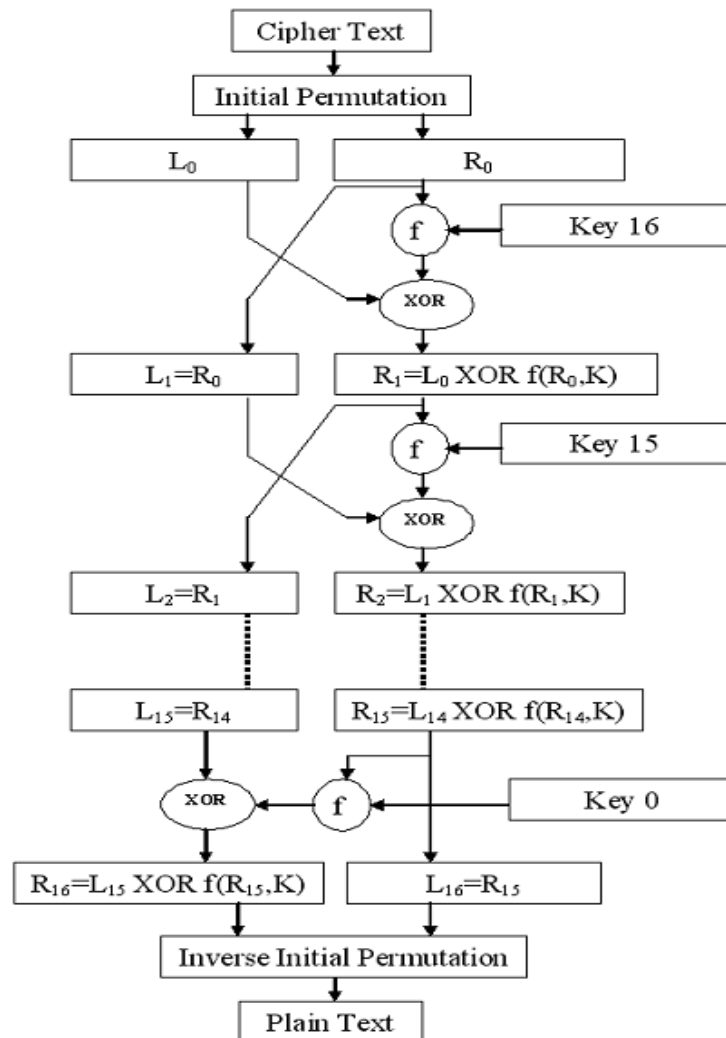


Figure 17.6: DES Decryption

17.2 Triple DES

The speed of exhaustive key searches against DES after 1990 began to cause discomfort amongst users of DES. However, users did not want to replace DES as it takes an enormous amount of time and money to change encryption algorithms that are widely adopted and embedded in large security architectures. The pragmatic approach was not to abandon the DES completely, but to change the manner in which DES is used. This led to the modified schemes of Triple DES (sometimes known as 3DES).

Incidentally, there are two variants of Triple DES known as 3-key Triple DES (3TDES) and 2-key Triple DES (2TDES).

17.2.1 3-Key Triple DES (3TDES)

Encryption

Before using 3TDES, user first generate and distribute a 3TDES key K , which consists of three different DES keys K_1 , K_2 and K_3 . This means that the actual 3TDES key has length $3 \times 56 = 168$ bits.

1. Encrypt the plaintext blocks using single DES with key K_1 .
2. Now decrypt the output of step 1 using single DES with key K_2 .
3. Finally, encrypt the output of step 2 using single DES with key K_3 .
4. The output of step 3 is the cipher text.

Decryption

Decryption of a ciphertext is a reverse process. User first decrypt using K_3 , then encrypt with K_2 , and finally decrypt with K_1 . The encryption and decryption of this scheme is illustrated in figure 7.

Due to this design of Triple DES as an encrypt–decrypt–encrypt process, it is possible to use a 3TDES (hardware) implementation for single DES by setting K_1 , K_2 , and K_3 to be the same value. This provides backwards compatibility with DES.

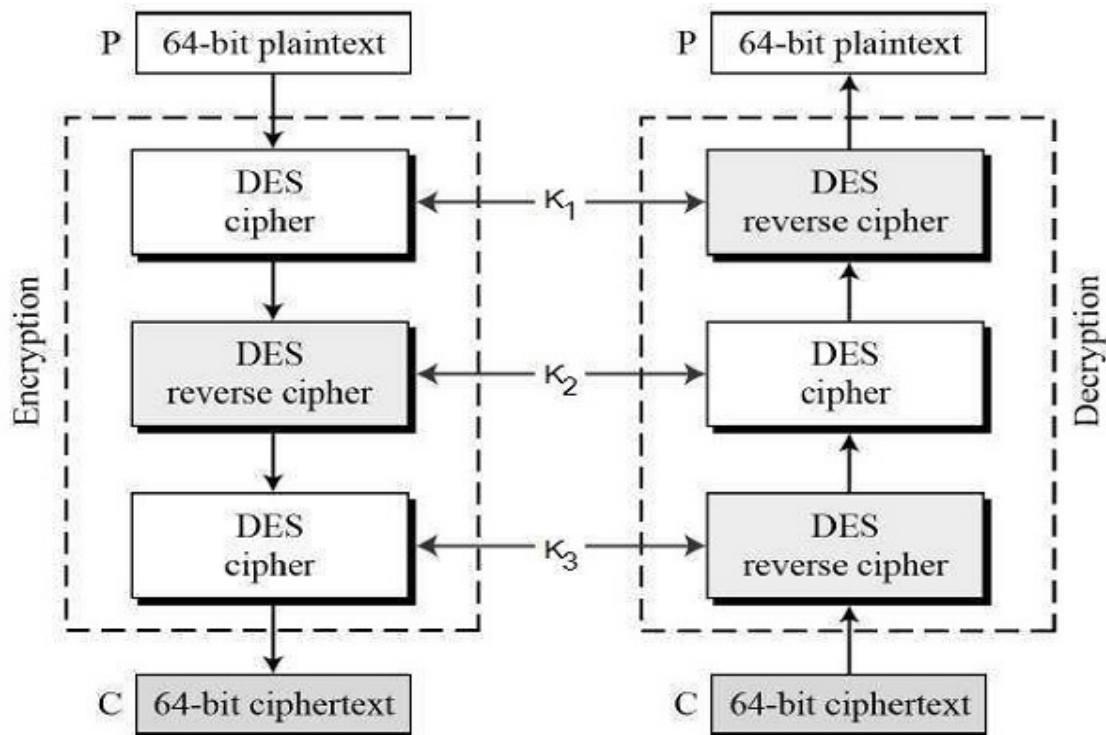


Figure 17.7: Triple DES

17.2.2 2-Key Triple DES (2TDES)

Second variant of Triple DES (2TDES) is identical to 3TDES except that K_3 is replaced by K_1 . In other words, user encrypt plaintext blocks with key K_1 , then decrypt with key K_2 , and finally encrypt with K_1 again. Therefore, 2TDES has a key length of $2 \times 56 = 112$ bits. In this case the resultant key space is about 2^{112} .

17.3 Advance Encryption Standard (AES)

Nowadays, Advanced Encryption Standard (AES) is the more popular and widely adopted symmetric encryption algorithm. It is at least six time faster than triple DES. A replacement for DES was needed as its key size was too small. With increasing computing power, it was considered vulnerable against exhaustive key search (Brute force) attack. Triple DES was designed to overcome this drawback but it was found slow. The features of AES are as follows:

1. Block Size = 128-bit data
2. Variable key size = 128/192/256 -bit keys
3. Stronger and faster than Triple-DES

17.3.1 Operation of AES

AES is an iterative cipher based on ‘substitution–permutation network (SPN)’. It has two operations substitution which replaces inputs by specific outputs and permutation which shuffles bits. AES performs all its computations on bytes rather than bits. Hence, AES treats the 128 bits of a plaintext block as 16 bytes. These 16 bytes are arranged in four columns and four rows for processing as a matrix. It’s stacked in a 4×4 matrix of 128 bits (16 bytes) per block in an array that’s known as a “state.” A state looks something like this: The number of rounds in AES is variable and depends on the length

$$\begin{bmatrix} A_0 & A_4 & A_8 & A_{12} \\ A_1 & A_5 & A_9 & A_{13} \\ A_2 & A_6 & A_{10} & A_{14} \\ A_3 & A_7 & A_{11} & A_{15} \end{bmatrix}$$

Figure 17.8:

of the key. AES uses 10 rounds for 128-bit keys, 12 rounds for 192-bit keys and 14 rounds for 256-bit keys. Each of these rounds uses a different 128-bit round key, which is calculated from the original AES key. The structure of AES is shown in figure 9:

17.3.2 Encryption Process

Each round of AES consists of four operations AddRoundKey, SubBytes, ShiftRows, and MixColumns. The description of one round is shown in figure 10 and these four operations are explained in subsections:

AddRoundKey

The AddRoundKey operation takes the current state of the data and executes the XOR Boolean operation against the current round subkey. XOR means “Exclusively Or,” which will yield a result of true if the inputs differ (e.g. one input must be 1 and the other input must be 0 to be true). There will be a unique subkey per round, plus one more. Suppose $a_{2,2} = 10001011$ and $k_{2,2} = 00111010$, then $b_{2,2} = 10001011 \oplus 00111010 = 10110001$

Byte Substitution (SubBytes)

The SubBytes operation, which stands for substitute bytes, will take the 16-byte block and run it through an S-Box (substitution box) to produce an alternate value. Simply put, the operation will take a value and then replace it by spitting out another value. The S box of AES is shown in figure 13: Suppose the value of input to sbox is 10001011 i.e. let $a_{2,2} = 10001011 \implies a_{2,2} = 8B$. As per above table the output of sbox will be 3D i.e. $S(10001011) = S(8B) = 3D = 00111101 = b_{2,2}$. Similarly all the 16 bytes are substituted with the help of this sbox.

ShiftRows

The ShiftRows operation is a little more straightforward and is easier to understand. Based off the arrangement of the data, the idea of ShiftRows is to move the positions of the data in their respective rows with wrapping. Remember, the data is arranged in a stacked arrangement and not left to right like most of us are used to reading. The image provided helps to visualize this operation. The first row goes unchanged. The second row shifts the bytes to the left by one position with row wrap around. The third row shifts the bytes one position beyond that, moving the byte to the left by a total of two positions with row wrap around. Likewise, this means that the fourth row shifts the bytes to the left by a total of three positions with row wrap around.

MixColumn

The MixColumns operation is a linear transformation of the columns of the dataset. It uses matrix multiplication and bitwise XOR addition to output the results. The column data, which can be represented as a 4×1 matrix, will be multiplied against a 4×4 matrix in a format called the Gallois field, and set as an inverse of input and output. That will look something like the

following: Here $c(X) = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$

To understand this multiplication let us take an example, say $\begin{bmatrix} a_{0,1} \\ a_{1,1} \\ a_{2,1} \\ a_{3,1} \end{bmatrix} = \begin{bmatrix} d4 \\ bf \\ 5d \\ 30 \end{bmatrix}$.

Now, let $\begin{bmatrix} r_0 \\ r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \otimes \begin{bmatrix} d4 \\ bf \\ 5d \\ 30 \end{bmatrix}$

To get the r_0 value, the formula goes like this: $r_0 = 02 \otimes d4 \oplus 03 \otimes bf \oplus 01 \otimes 5d \oplus 01 \otimes 30$. All the operations in AES are done in the finite field $GF(2^8)$ with irreducible polynomial $m(x) = x^8 + x^4 + x^3 + x + 1$. Now, for $02 \otimes d4$, convert 02 and d4 in binary, i.e. 02 = 0000 0010, d4 = 1101 0100 and write in polynomial form say $p(x)$ and $q(x)$. $p(x) = x$ and $q(x) = x^7 + x^6 + x^4 + x^2$. Now $p(x) \cdot q(x) \bmod(m(x)) = x \cdot (x^7 + x^6 + x^4 + x^2) \bmod(x^8 + x^4 + x^3 + x + 1) = x^7 + x^5 + x^4 + x + 1$. Now write coefficients in binary form i.e 10110011 i.e. equal to b3.

Similarly $03 \cdot bf = 11011010$

$01 \cdot 5d = 01011101$

$01 \cdot 30 = 00110000$

$r_0 = 02 \otimes d4 \oplus 03 \otimes bf \oplus 01 \otimes 5d \oplus 01 \otimes 30 = 10110011 \oplus 11011010 \oplus 01011101 \oplus 00110000 = 00000100 = 04$ (in Hex).

The value of r_1, r_2 and r_3 can be calculated in same way.

It may also be noted that Mix column operation will not be applied in last round i.e in 10^{th} round.

17.3.3 Decryption Process

The process of decryption of an AES ciphertext is similar to the encryption process in the reverse order and shown in figure 9. Each round consists of the four processes conducted in the reverse order:

1. Add round key
2. Inverse Mix columns
3. Inverse Shift rows
4. Inverse Byte substitution

The above four process are explained in following subsections.

Add Round Key

This operation is same as used in encryption. Only difference is the round keys are used in reverse order.

Inverse Mix columns

The Inverse matrix of Mix column matrix $C = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$ is

$$C^{-1} = \begin{bmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{bmatrix}$$

It can be verified that $C \times C^{-1} = I$

Inverse Shift Rows

The inverse shift rows transformation performs the circular shifts in opposite direction for each of the last three rows, with a one byte circular right shift for the second row, and so on.

Inverse Byte Substitution

For inverse byte substitution the following table will be used:

17.3.4 Key Generation

Let K is the secret key and the size of K can be 128/192/256. N is the length of the key in 32-bit words i.e. 4 words for AES-128, 6 words for AES-192, and 8 words for AES-256. $K_0, K_1, \dots, K_{N-1}$ are the 32-bit words of the original key. R is the number of round keys needed: 11 round keys for AES-128, 13 keys for AES-192, and 15 keys for AES-256. $W_0, W_1, \dots, W_{4R-1}$ are the 32-bit words of the expanded key. Define a function RotWord as a one-byte left circular shift:

$$\text{RotWord}([b_0 \ b_1 \ b_2 \ b_3]) = [b_1 \ b_2 \ b_3 \ b_0]$$

SubWord as an application of the AES S-box to each of the four bytes of the word: $\text{SubWord}([b_0 \ b_1 \ b_2 \ b_3]) = [S(b_0) \ S(b_1) \ S(b_2) \ S(b_3)]$

The round keys for different rounds will be generated by following function:

$$W_j = \begin{cases} K_j & \text{if } j < N \\ W_{j-N} \oplus \text{SubWord}(\text{RotWord}(W_{j-1})) \oplus rcon_{j/N} & \text{if } j \geq N \text{ and } j \equiv 0 \pmod{N} \\ W_{j-N} \oplus \text{SubWord}(W_{j-1}) & \text{if } j \geq N, N > 6, \text{ and } j \equiv 4 \pmod{N} \\ W_{j-N} \oplus W_{j-1} & \text{otherwise.} \end{cases}$$

where $0 \leq j \leq (4R-1)$ and the value of $rcon_j$ for $1 \leq j \leq 10$ is given in the following table: In case of 128 bit key $R=11$, $N=4$ and $0 \leq j \leq 43$

$$W_0 = K_0$$

$$W_1 = K_1$$

$$W_2 = K_2$$

$$W_3 = K_3$$

$$W_4 = W_0 \oplus \text{SubWord}(\text{RotWord}(W_3)) \oplus rcon_1$$

$$W_5 = W_1 \oplus W_4$$

$$W_6 = W_2 \oplus W_5$$

$$W_7 = W_3 \oplus W_6$$

$$W_8 = W_4 \oplus \text{SubWord}(\text{RotWord}(W_7)) \oplus rcon_2$$

$$W_9 = W_5 \oplus W_8$$

$$W_{10} = W_6 \oplus W_9$$

$$W_{11} = W_7 \oplus W_{10}$$

·
·
·

$$W_{40} = W_{36} \oplus \text{SubWord}(\text{RotWord}(W_{39})) \oplus \text{rcon}_{10}$$

$$W_{41} = W_{37} \oplus W_{40}$$

$$W_{42} = W_{38} \oplus W_{41}$$

$$W_{43} = W_{39} \oplus W_{42}$$

R.K.Sharma

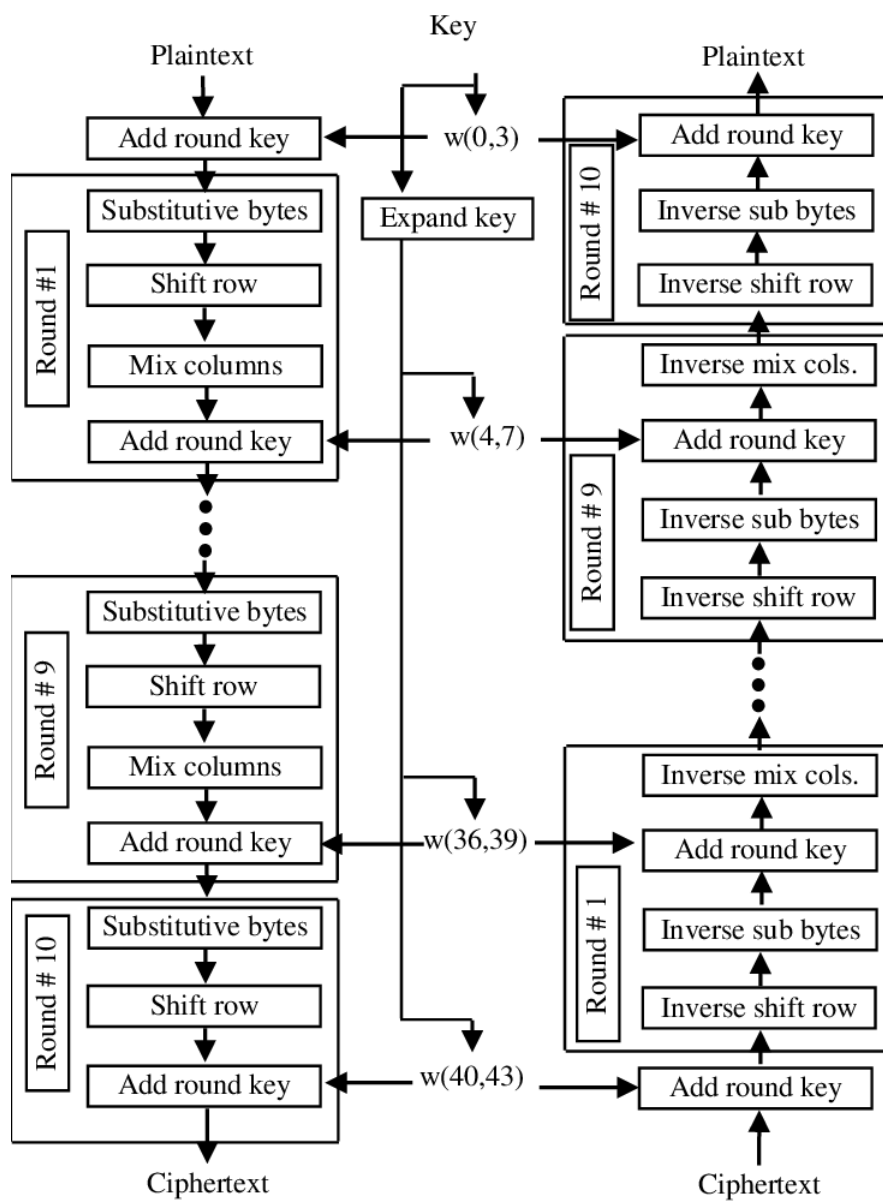


Figure 17.9: AES

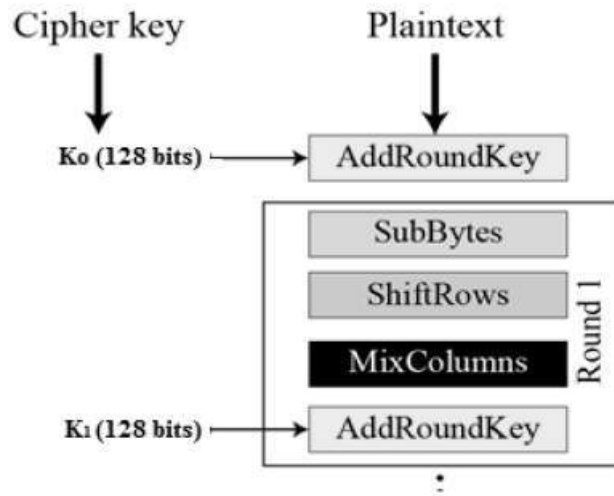


Figure 17.10:

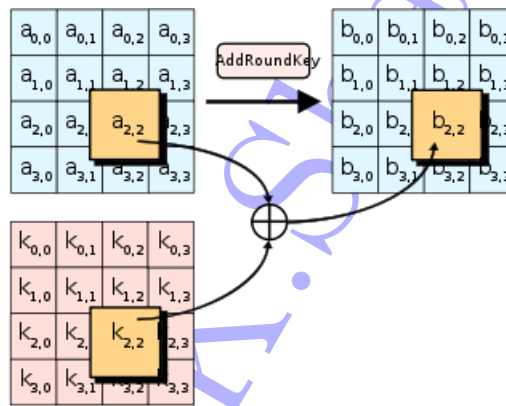


Figure 17.11:

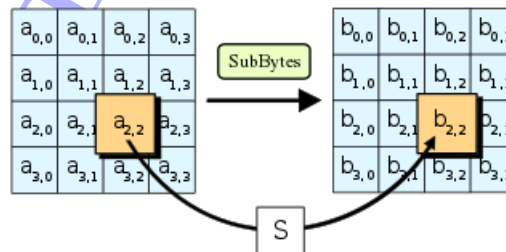


Figure 17.12:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

Figure 17.13: AES SBox

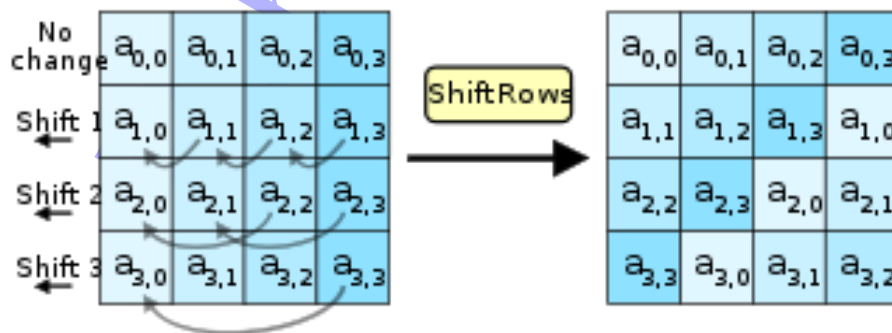


Figure 17.14:

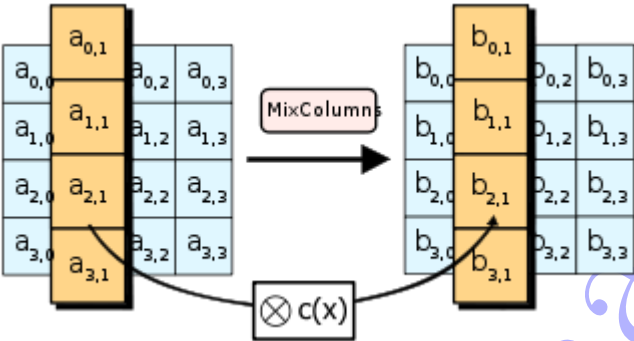


Figure 17.15:

		y															
		0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
x	0	52	09	6a	d5	30	36	a5	38	bf	40	a3	9e	81	f3	d7	fb
	1	7c	e3	39	82	9b	2f	ff	87	34	8e	43	44	c4	de	e9	cb
	2	54	7b	94	32	a6	c2	23	3d	ee	4c	95	0b	42	fa	c3	4e
	3	08	2e	a1	66	28	d9	24	b2	76	5b	a2	49	6d	8b	d1	25
	4	72	f8	f6	64	86	68	98	16	d4	a4	5c	cc	5d	65	b6	92
	5	6c	70	48	50	fd	ed	b9	da	5e	15	46	57	a7	8d	9d	84
	6	90	d8	ab	00	8c	bc	d3	0a	f7	e4	58	05	b8	b3	45	06
	7	d0	2c	1e	8f	ca	3f	0f	02	c1	af	bd	03	01	13	8a	6b
	8	3a	91	11	41	4f	67	dc	ea	97	f2	cf	ce	f0	b4	e6	73
	9	96	ac	74	22	e7	ad	35	85	e2	f9	37	e8	1c	75	df	6e
	a	47	f1	1a	71	1d	29	c5	89	6f	b7	62	0e	aa	18	be	1b
	b	fc	56	3e	4b	c6	d2	79	20	9a	db	c0	fe	78	cd	5a	f4
	c	1f	dd	a8	33	88	07	c7	31	b1	12	10	59	27	80	ec	5f
	d	60	51	7f	a9	19	b5	4a	0d	2d	e5	7a	9f	93	c9	9c	ef
	e	a0	e0	3b	4d	ae	2a	f5	b0	c8	eb	bb	3c	83	53	99	61
	f	17	2b	04	7e	ba	77	d6	26	e1	69	14	63	55	21	0c	7d

Figure 17.16: Inverse S-Box

j	1	2	3	4	5	6	7	8	9	10
RC[j]	01	02	04	08	10	20	40	80	1B	36

Figure 17.17:

17.4 RC5

RC5 is a block cipher based on symmetric key which was developed by Ron Rivest in 1994. RC stands for "Rivest Cipher", or alternatively, "Ron's Code". The main feature of this is, it is quite fast as it uses only primitive computer operations. It allows a variable number of rounds and variable bit size key to add flexibility. Another advantage of using RC5 is that it requires less memory for execution. This feature enables RC 5 to be used for various purposes like desktop operation, smart cards, etc.

17.4.1 Description of RC5

The RC5 encryption algorithm is a Feistel network based block cipher that converts plain text data blocks of 16, 32, and 64 bits into cipher text blocks of the same length. It uses a key of selectable length b (0, 1, 2, ..., 255) byte. The algorithm is organized as a set of iterations called rounds r that takes values in the range (0, 1, 2, ..., 255) as illustrated in fig. 18. The RC5 is basically denoted as RC5- $w/r/b$ where w =word size in bits, r =number of rounds, b =number of 8-bit byte in the key. RC5 is a fully parameterized family of encryption algorithms.

The original suggested choice of parameters are a block size of 64 bits(two words of size 32 each), a 128-bit key and 12 rounds. The operations performed on the data blocks include bitwise exclusive-OR of words, data-dependent rotations by means of circular left and right rotations and Two's complement addition/subtraction of words, which is modulo- 2^w addition/subtraction. For all variants, RC5- $w/r/b$ operates on units of two w -bit words and the operations used in RC5 are defined as follows:

1. $A+B$ integer addition modulo 2^w
 let $A=10, B=12$ and $w=4$.
 $(A+B) \bmod 2^w = (10 + 12) \bmod 2^4 = 22 \bmod 16 = 6$
2. $A-B$ integer subtraction modulo 2^w
 let $A=10, B=3$ and $w=4$.
 $(A-B) \bmod 2^w = (10 - 3) \bmod 2^4 = 7 \bmod 16 = 7$
3. $A \oplus B$ bitwise exclusive-or of w -bit words
 let $A=10 = (1010)_2, B=3 = (0011)_2$ and $w=4$.

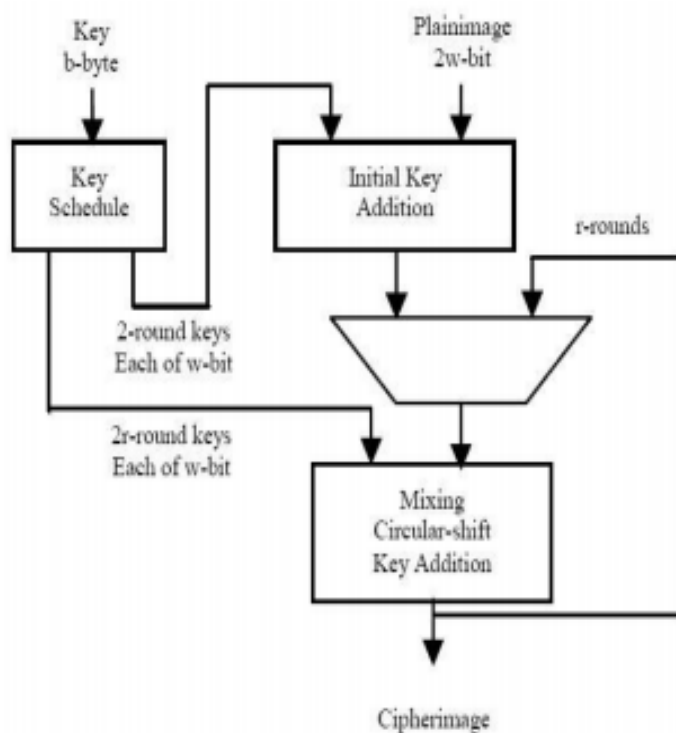


Figure 17.18: RC5 Encryption

$$10 \oplus 3 = 1010 \oplus 0011 = 1001 = (9)_{10}$$

4. $A \lll B$ rotation of the w -bit word A to the left by the amount given by the least significant $\lg w$ bits of B

let $A=10 = (1010)_2, B=3 = (0011)_2$ and $w=4$.

$$A \lll B = 10 \lll 3 = (1010)_2 \lll 3 = (0101)_2 = (5)_{10}$$

5. $A \ggg B$ rotation of the w -bit word A to the right by the amount given by the least significant $\lg w$ bits of B . let $A=10 = (1010)_2, B=3 = (0011)_2$ and $w=4$.

$$A \ggg B = 10 \ggg 3 = (1010)_2 \ggg 3 = (0101)_2 = (5)_{10}$$

There are three routines in RC5: key expansion, encryption, and decryption. We will discuss each of them in next subsections. KeyExpansion algorithm is used to generate the round sub keys that will be use in both encryption and decryption algorithms. RC5 has different algorithms for encryption and de-

ryption, in encryption it uses integer addition modulo 2^w but in decryption it uses integer subtraction modulo 2^w . RC5 is a symmetric key encryption so encryption and decryption algorithms uses same key.

17.4.2 Key Expansion Algorithm

Key Expansion algorithm can be understand in three steps:-

1. **Initialization of constants P and Q:** RC5 makes use of 2 magic constants P and Q whose value is defined by the word size w .

For word size w , P and Q can be determined as:

$$\begin{aligned} P &= \text{Odd}((e - 2)2^w) \\ Q &= \text{Odd}((v - 1)2^w) \end{aligned} \quad (17.17)$$

Where e is the base of natural logarithms ($e = 2.718281828459$) and v is the golden ratio ($v = 1.618033988749$).
Odd (x) is the odd integer nearest to x .

Take an example, let $w=16$,

$$P = \text{Odd}((2.718281828459 - 2)2^{16}) = 47073 = (b7e1)_{16}$$

$$Q = \text{Odd}((1.618033988749 - 1)2^{16}) = 40503 = (9e37)_{16}$$

The value of P and Q is tabulated in following table for different word size w .

Word Size in bits	P(Hexa Decimal)	Q(Hexa Decimal)
16	b7e1	9e37
32	b7e15163	9e3779b9
64	b7e151628aed2a6b	9e3779b97f4a7c15

2. **Converting secret key K from bytes to words:** Secret key K of size b bytes is used to initialize an array L consisting of c words, where $c = \lceil b/u \rceil$ words, $u = w/8$ no of bytes per word and w = word size used for that particular instance of RC5. For example, if we choose $w=32$ bits and Key K is of size 96 bytes then, $u=32/8=4$, $c=b/u=96/4=24$.

L is pre initialized to 0 value before adding secret key K to it.

for $i=b-1$ to 0

$L[i/u] = (L[i/u] \lll 8) + K[i]$

3. Initializing sub-key S:

Sub-key S of size $t=2(r+1)$ is initialized using magic constants P and Q.

$S[0] = P$

for $i = 1$ to $2(r+1)-1$

$S[i] = S[i-1] + Q$

4. Sub-key mixing:

The RC5 encryption algorithm uses Sub key S. L is merely, a temporary array formed on the basis of user entered secret key.

Mix in user's secret key with S and L.

$i = j = 0$

$X = Y = 0$

do $3 * \max(t, c)$ times:

$X = S[i] = (S[i] + X + Y) \lll 3$

$Y = L[j] = (L[j] + X + Y) \lll (X + Y)$

$i = (i + 1) \% t$

$j = (j + 1) \% c$

17.4.3 Encryption Process

We divide the input plain text block into two registers A and B each of size w bits. After undergoing the encryption process the result of A and B together forms the cipher text block. RC5 Encryption Algorithm is explained in the following steps:

1. One time initialization of plain text blocks A and B by adding $S[0]$ and $S[1]$ to A and B respectively. These operations are over $\text{mod}2^w$.
2. XOR A and B, i.e. $A=A \oplus B$
3. Cyclic left shift new value of A by B bits.
4. Add $S[2*i]$ to the output of previous step. This is the new value of A.

5. XOR B with new value of A and store in B.
6. Cyclic left shift new value of B by A bits.
7. Add $S[2*i+1]$ to the output of previous step. This is the new value of B.
8. Repeat entire procedure (except one time initialization) r times.

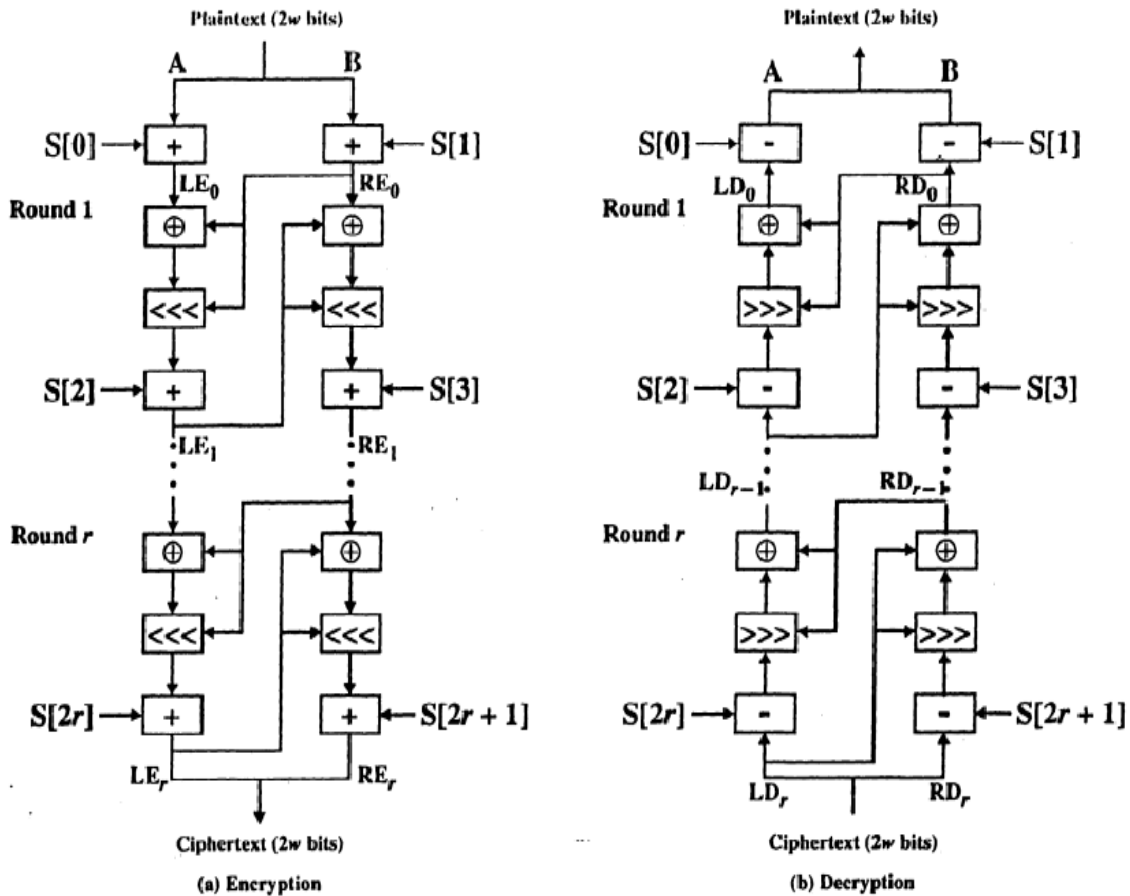


Figure 17.19: RC5 Encryption and Decryption

17.4.4 Decryption process

For decryption of cipher text first divide the cipher text into two words w bit words say A and B then apply these following operations to convert cipher text into plain text.

for $i = r$ down to 1

do

{

$B = ((B - S[2i + 1]) \ggg A) \oplus A$

$A = ((A - S[2i]) \ggg B) \oplus B$

}

$B = B - S[1]$

$A = A - S[0]$ The encryption and decryption procedure is illustrate in figure 17.19.

Chapter 18

Digital Signature Standards (DSS)

18.1 Introduction

Physical signature is a way of authenticating the physical paper coming from a trusted individual. Similarly, digital signature is a way of authenticating a digital data coming from a trusted source.

Digital Signature Standard (DSS) is a Federal Information Processing Standard (FIPS) which defines algorithms that are used to generate digital signatures with the help of Secure Hash Algorithm (SHA) for the authentication of electronic documents. The process of signing at sender end and verification of document at receiver end is explained in following sections and shown in figure 18.1.

Sender Side:

In DSS Approach, a hash code is generated out of the message and following inputs are given to the signature function

1. The hash code.
2. The random number 'k' generated for that particular signature.
3. The private key of the sender i.e., $PR(a)$.

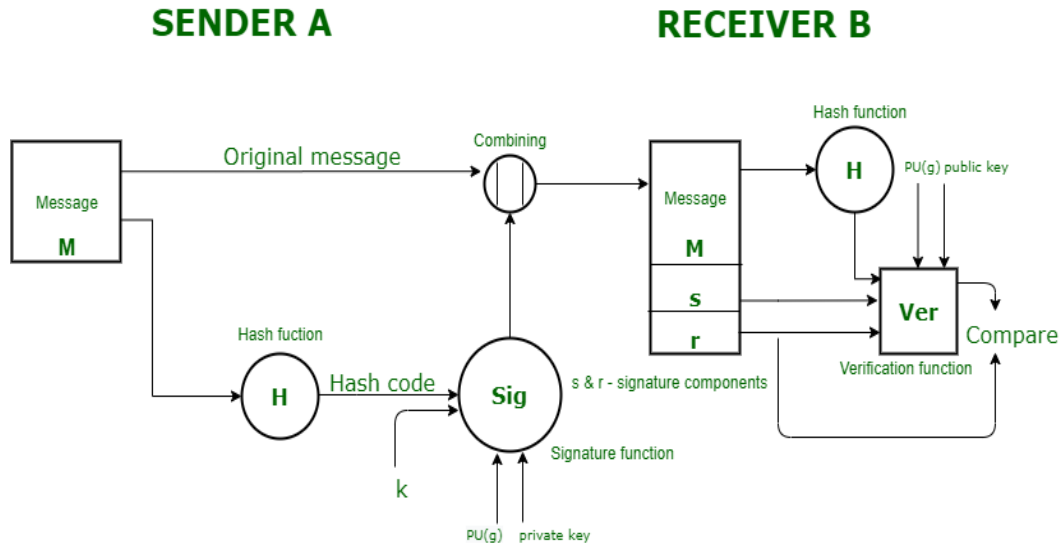


Figure 18.1: DSS

4. A global public key (which is a set of parameters for the communicating principles) i.e., $PU(g)$.

These input to the function will provide the output signature containing two components – 's' and 'r'. Therefore, the original message concatenated with the signature is sent to the receiver.

Receiver Side:

At the receiver end, verification of the sender is done. The hash code of the sent message is generated. There is a verification function which takes the following inputs:

1. The hash code generated by the receiver.
2. Signature components 's' and 'r'.
3. Public key of the sender.
4. Global public key.

The output of the verification function is compared with the signature component 'r'. Both the values will match if the sent signature is valid because only the sender with the help of its private key can generate a valid signature.

18.2 Digital Signature Algorithm(DSA)

The Digital Signature Algorithm (DSA) is a Federal Information Processing Standard (FIPS) for digital signatures, based on the mathematical concept of modular exponentiation and the discrete logarithm problem. The National Institute of Standards and Technology (NIST) proposed DSA for use in their Digital Signature Standard (DSS) in 1991, and adopted it as FIPS 186 in 1994. Four revisions to the initial specification have been released. The newest specification is FIPS 186-4 from July 2013.

The algorithm uses a key pair consisting of a public key and a private key. The private key is used to generate a digital signature for a message, and such a signature can be verified by using the signer's corresponding public key. The digital signature provides message authentication (the receiver can verify the origin of the message), integrity (the receiver can verify that the message has not been modified since it was signed) and non-repudiation (the sender cannot falsely claim that they have not signed the message).

18.2.1 DSA Operations

DAS has four operations, which are explained in following subsections:

Key Generation

It has two phases. The first phase is a choice of algorithm parameters which may be shared between different users of the system, while the second phase computes a single key pair for one user.

Parameters generation:

1. Choose cryptographic hash function H with output length $|H|$ bits. In the original DSS, SHA-1 was used as hash function and SHA-2 is being used in current DSS. If $|H|$ is greater than the modulus length N , only the leftmost N bits of the hash output are used.
2. Choose a key length L . The original DSS constrained L to be a multiple of 64 bit between 512 and 1024 inclusive. NIST recommends lengths of 2048 (or 3072).

3. Choose the modulus length N such that $N < L$ and $N \leq |H|$. FIPS specifies L and N to have one of the values: (1024, 160), (2048, 224), (2048, 256), or (3072, 256).
4. Choose an N -bit prime q .
5. Choose an L -bit prime p such that $(p - 1)$ is a multiple of q .
6. Choose an integer h randomly from $\{2 \dots p - 2\}$.
7. Compute $g = h^{(p-1)/q} \bmod p$. If $g = 1$ try again with a different h .
8. The algorithm parameters are (p, q, g) . These may be shared between different users of the system.

User Key Generation:

For the given set of parameters, the user's keys can be generated as follows:

1. Choose an integer x randomly from $\{1 \dots q - 1\}$.
2. Compute $y = g^x \bmod p$.
3. x is the private key and y is the public key.

Key Distribution

The signer publish public key y . That is, they should send the key to the receiver via a reliable, but not necessarily secret, mechanism. The signer should keep the private key x secret.

Signing

1. Take a message m to be signed.
2. Choose an integer k randomly from $\{1 \dots q - 1\}$
3. Compute $r = (g^k \bmod p) \bmod q$. if $r = 0$, take a different random k .
4. Compute $s = (k^{-1} (H(m) + xr)) \bmod q$. if $s = 0$, take a different k .
5. (r, s) is the signature

Verification of Signature

Signature (r, s) for a message m can be verified as follows:

1. Verify that $0 < r < q$ and $0 < s < q$.
2. Compute $w := s^{-1} \bmod q$.
3. Compute $u_1 = H(m) \cdot w \bmod q$.
4. Compute $u_2 = r \cdot w \bmod q$.
5. Compute $v = (g^{u_1} y^{u_2} \bmod p) \bmod q$.
6. The signature is valid if and only if $v = r$.

18.2.2 Correctness of Algorithm

Correctness of algorithm means that verifier always accept the correct signatures. It is shown as follows:-

Since $g = h^{(p-1)/q} \bmod p$

$\Rightarrow g^q \equiv h^{p-1} \equiv 1 \bmod p$ by Fermat's little theorem.

Now, the signer computes

$$s = k^{-1}(H(m) + xr) \bmod q$$

$$k \equiv H(m)s^{-1} + xrs^{-1}$$

$$\equiv H(m)w + xrw \pmod{q}$$

$$g^k \equiv g^{H(m)w} g^{xrw}$$

$$\equiv g^{H(m)w} y^{rw}, \text{ since } g^x = y$$

$$\equiv g^{u_1} y^{u_2} \pmod{p}$$

Finally,

$$r = (g^k \bmod p) \bmod q$$

$$= (g^{u_1} y^{u_2} \bmod p) \bmod q$$

$$= v$$

18.3 DSA illustration

Parameters generation

Let us take $q = 3$, since $(p - 1)$ is a multiple of q , take $p = 2 \times q + 1 = 2 \times 3 + 1 = 7$.

Now Choose $g = h^{(p-1)/q} \bmod p$, where h randomly from $\{2 \dots p - 2\}$. If $h = 2$ then $g = 2^2 \bmod (7) = 4$

The algorithm parameters are $(p, q, g) = (7, 3, 4)$.

User key generation

Let take an integer $x = 2$ randomly from $\{1 \dots q - 1\}$ i.e $\{1 \dots 2\}$.

Compute $y := g^x \bmod p = 4^2 \bmod 7 = 2$. private key $x = 2$ and public key $y = 2$.

Signing

Suppose the hash value of message m is 3, i.e $H(m) = 3$.

Choose an integer $k = 2$ randomly from $\{1 \dots q - 1\}$

Compute $r = (g^k \bmod p) \bmod q = (4^2 \bmod 7)(\bmod 3) = 2$.

Compute $s = (k^{-1} (H(m) + xr)) \bmod q = (2^{-1} \times (3 + 2 \times 2)) \bmod 3 = (2 \times 7) \bmod 3 = 2$.

$(r, s) = (2, 2)$ is the signature

Verification

Here, $0 < r = 2 < q$ and $0 < s = 2 < q$ verifies.

Compute $w := s^{-1} \bmod q = 2^{-1} \bmod 3 = 2$.

Compute $u_1 = H(m) \cdot w \bmod q = (3 \times 2) \bmod 3 = 0$.

Compute $u_2 = r \cdot w \bmod q = (2 \times 2) \bmod 3 = 1$.

Compute $v = (g^{u_1} g^{u_2} \bmod p) \bmod q = (4^0 \times 2^1) \bmod 7 = 2$.

Since $v = r$, the signature is valid.

Chapter 19

Modes of Encryption

Encryption is the process of encoding information. This process converts the original message, known as plaintext, into a coded message known as ciphertext.

The two main kinds of encryption are symmetric encryption and asymmetric encryption. Asymmetric encryption is also known as public key encryption.

In symmetric encryption, there is only one key, and all communicating parties use the same (secret) key for both encryption and decryption. In asymmetric, or public key, encryption, there are two keys: one key is used for encryption, and a different key is used for decryption. The decryption key is kept private (hence the "private key" name), while the encryption key is shared publicly, for anyone to use.

Why there is need of data encryption ?

Privacy: Encryption ensures that no one can read data except the intended recipient or the rightful data owner.

Security: Encryption helps in preventing data breaches. If a device is lost or stolen and its hard drive is properly encrypted, the data on that device will still be secured.

Data integrity: Encryption also helps in preventing malicious behaviour such as on-path attacks.

Authentication: It is the process of determining whether someone or something is, in fact, who or what it declares itself to be. During authentication, credentials provided by the user are compared to those on file in a database

of authorized users either on the local operating system or through an authentication server.

Regulations: For all these reasons, many industry and government regulations require companies that handle user data to keep that data encrypted. Examples of regulatory and compliance standards that require encryption include HIPAA, PCI-DSS, and the GDPR.

Commonly used symmetric encryption algorithms include:

- Advanced Encryption Standard (AES)
- Triple Data Encryption Algorithm (3DES or TDES or Triple DES)
- SNOW

Commonly used asymmetric encryption algorithms include:

- RSA
- Elliptic curve cryptography

Advanced Encryption Standard (AES) and **Triple Data Encryption Algorithm** are symmetric block ciphers whereas **SNOW** is a stream cipher.

Modes of Encryption :

1. Electronic Code Book (ECB)
2. Cipher Block Chaining (CBC)
3. Cipher Feedback (CFB)
4. Output Feedback (OFB)
5. Counter (CTR).

Electronic Code Book mode

1. About

- This mode is a most straightforward way of processing a series of sequentially listed message blocks.
- It is considered as the simplest mode where each block of data is simply encrypted with the same key.
- There is a one-to-one mapping between the plaintext block and corresponding ciphertext block for any particular key.
- The ECB mode is deterministic, that is, if plaintext block $P_1, P_2, \dots, P_m$ are encrypted twice under the same key, the output ciphertext blocks will be the same.

2. Process

- Encryption :
 - Input : Let K be k -bit Key and $P = P_1 P_2 \dots P_m$ be n -bit plaintext blocks.
 - Output :

$$C_i = E_K(P_i) \quad \forall 1 \leq i \leq m$$
 and $C = C_1 C_2 \dots C_m$ is n -bit ciphertext blocks.
- Decryption :
 - Input : Let K be k -bit Key and $C = C_1 C_2 \dots C_m$ be n -bit ciphertext blocks.
 - Output :

$$P_i = D_K(C_i) \quad \forall 1 \leq i \leq m$$
 and $P = P_1 P_2 \dots P_m$ is n -bit plaintext blocks.

3. Analysis

- Very weak if used for general-purpose encryption
- least secure to use it for a file or a message
- Any application data usually have partial information which can be guessed

- Attacker can build up codebook
- A ciphertext from ECB can allow an attacker to guess the plaintext by trial-and-error also if the plaintext message is within predictable

4. Examples

- If a ciphertext from the ECB mode is known to encrypt a salary figure, then a small number of trials will allow an attacker to recover the figure.
- Shift cipher, Playfair cipher, Permutation cipher, Hill cipher, affine cipher, etc are examples of ECB mode.

Cipher Block Chaining mode

1. About

- CBC tries to improve on ECB by making the encryption of each block dependent not just on the key but also on ciphertext of the previous block and thus makes the system non-deterministic.
- Each block of ciphertext depends on all the plaintext blocks processed up to that point, which prevents parallelization of the encryption process.
- Also, any error can propagate to the subsequent block.
- The security of the cryptosystem is based on the secrecy of the key, not on the secrecy of initialization vector.
- Initialization vectors : To randomize the ciphertext of the first block (and thus make each ciphertext unique, even if the plaintext message is repeated), an “Initialization Vector” (IV) is used. The IV is a random number known to both the encrypting and decrypting systems and should only be used once.

2. Process

- Encryption :
 - Input : Let K be k -bit Key, C_0 be n -bit initialization vector and $P = P_1 P_2 \dots P_m$ be n -bit plaintext blocks.

– Output :

$$C_i = E_K(C_{i-1} \oplus P_i) \quad \forall 1 \leq i \leq m$$

and $C = C_1 C_2 \dots C_m$ is n -bit ciphertext blocks.

• Decryption :

– Input : Let K be k -bit Key and $C = C_1 C_2 \dots C_m$ be n -bit ciphertext blocks.

– Output :

$$P_i = C_{i-1} \oplus D_K(C_i) \quad \forall 1 \leq i \leq m$$

and $P = P_1 P_2 \dots P_m$ is n -bit plaintext blocks.

3. Analysis

- In CBC mode, the current plaintext block is added to the previous ciphertext block, and then the result is encrypted with the key. Decryption is thus the reverse process, which involves decrypting the current ciphertext and then adding the previous ciphertext block to the result.
- The greatest advantage CBC has over ECB is that, with CBC mode, identical blocks do not have the same cipher. This is because the initialization vector adds a random factor to each block; hence, the same blocks in different positions will have different ciphers.
- CBC is vulnerable to “bit flipping” attack.
- Also, its encryption is not tolerant of block losses. This is because blocks depend on their previous blocks for encryption.

4. Examples

- Let $S = \{ "A", "B", "C", \dots, "Z", ".", "?", "!", ",", "-", " " \}$ be the set of alphabets and few symbols of order 32

Let $m = HELLO$ be the message and plaintext corresponding to the message m is $P = 00111 00100 01011 01011 01110$.

Let $K = 11001$, $C_0 = 01101$ and $E_K(*) = * \oplus K$.

Then,

$$C_1 = E_K(C_0 \oplus P_1) = E_K(01101 \oplus 00111) = E_K(01010) = 01010 \oplus 11001 = 10011$$

$$C_2 = E_K(C_1 \oplus P_2) = E_K(10011 \oplus 00100) = E_K(10111) = 10111 \oplus 11001 = 01110$$

$$C_3 = E_K(C_2 \oplus P_3) = E_K(01110 \oplus 01011) = E_K(00101) = 00101 \oplus 11001 = 11100$$

$$C_4 = E_K(C_3 \oplus P_4) = E_K(11100 \oplus 01011) = E_K(10111) = 10111 \oplus 11001 = 01110$$

$$C_5 = E_K(C_4 \oplus P_5) = E_K(01110 \oplus 01110) = E_K(00000) = 00000 \oplus 11001 = 11001$$

Thus, ciphertext is $C = 10011 \ 01110 \ 11100 \ 01110 \ 11001$

Similarly, we can decrypt the message by taking $D_K(*) = * \oplus K$.

Cipher Feedback mode

1. About

- The CFB (Cipher FeedBack) mode of operation allows the block encryptor to be used as a stream cipher. It also needs an IV.
- It allows one to process blocks of size $r < n$ at a time. The typical value for r is 1, while n may be of size 64, using DES.
- CFB is similar to CBC but has the advantage of being self-synchronizing – if one or more blocks are lost, it doesn't affect the decryption of the remaining blocks.
- Underlying block cipher used only in encryption mode, that is, the encryption and decryption functions are identical, and it doesn't require the plaintext data to be padded.

2. Process

- Encryption :
 - Input : Let K be k -bit Key, I_1 be n -bit initialization vector and $P = P_1 \ P_2 \ \dots \ P_m$ be r -bit plaintext blocks.
 - Output : $\forall 1 \leq i \leq m$, we have

$$C_i = P_i \oplus L_r(E_K(I_i))$$

and

$$I_{i+1} = R_{n-r}(I_i) || C_i$$

where L_r and R_{n-r} are the operators which take the left most r -bits and the right most $n - r$ -bits respectively and $||$ is the concatenation operators.

- Decryption :

- Input : Let K be k -bit Key, I_1 be n -bit initialization vector and $C = C_1 C_2 \dots C_m$ be r -bit ciphertext blocks.
- Output : Compute $I_1, \dots, I_m$ as in the enciphering algorithm, which can be generated independently of the deciphered message text, and then compute

$$P_i = C_i \oplus L_r(E_K(I_i)) \quad \forall 1 \leq i \leq m$$

Note that CFB deciphering requires only the block cipher E_K , not the inverse block deciphering map D_K .

3. Analysis

- In CFB mode, user decrypts the ciphertext using only the encryption process of the block cipher. The decryption algorithm of the underlying block cipher is never used.
- Apparently, CFB mode is converting a block cipher into a type of stream cipher. The encryption algorithm is used as a key-stream generator to produce key-stream that is placed in the bottom register. This key stream is then XORed with the plaintext as in case of stream cipher.
- By converting a block cipher into a stream cipher, CFB mode provides some of the advantageous properties of a stream cipher while retaining the advantageous properties of a block cipher.
- On the flip side, the error of transmission gets propagated due to the change of blocks.

4. Examples

- Let $m = HELLO$ be the message and plaintext corresponding to the message m is $P = 00111\ 00100\ 01011\ 01011\ 01110$. Let $K = 111001$, $I_1 = 101101$ and $E_K(*) = * \oplus K$. Then,

$$\begin{aligned} C_1 &= P_1 \oplus L_r(E_K(I_1)) = 00111 \oplus L_r(101101 \oplus 111001) \\ &= 00111 \oplus L_r(010100) = 01101 \end{aligned}$$

$$I_2 = R_1(I_1) || C_1 = R_1(101101) || 01101 = 101101$$

$$\begin{aligned} C_2 &= P_2 \oplus L_r(E_K(I_2)) = 00100 \oplus L_r(101101 \oplus 111001) \\ &= 00100 \oplus L_r(010100) = 01110 \end{aligned}$$

$$I_3 = R_1(I_2) || C_2 = R_1(101101) || 01110 = 101110$$

$$\begin{aligned} C_3 &= P_3 \oplus L_r(E_K(I_3)) = 01011 \oplus L_r(101110 \oplus 111001) \\ &= 01011 \oplus L_r(010111) = 00000 \end{aligned}$$

$$I_4 = R_1(I_3) || C_3 = R_1(101110) || 00000 = 100000$$

$$\begin{aligned} C_4 &= P_4 \oplus L_r(E_K(I_4)) = 01011 \oplus L_r(100000 \oplus 111001) \\ &= 01011 \oplus L_r(011001) = 00111 \end{aligned}$$

$$I_5 = R_1(I_4) || C_4 = R_1(100000) || 00111 = 100111$$

$$\begin{aligned} C_5 &= P_5 \oplus L_r(E_K(I_5)) = 01110 \oplus L_r(100111 \oplus 111001) \\ &= 01110 \oplus L_r(011110) = 00001 \end{aligned}$$

Thus, ciphertext is $C = 01101 \ 01110 \ 00000 \ 00111 \ 00001$

Output Feedback mode

1. About

- Output feedback mode has a similar use as cipher feedback mode, but is relevant to applications for which error propagation must be avoided.
- Output feedback mode is an example of a synchronous stream cipher (constructed from a block cipher), in which the keystream is created independently of the plaintext stream.
- In OFB, active attacker can make controlled changes to plaintext.

2. Process

- Encryption :
 - Input : Let K be k -bit Key, I_0 be n -bit initialization vector and $P = P_1 P_2 \dots P_m$ be r -bit plaintext blocks.
 - Output : $\forall 1 \leq i \leq m$, we have

$$I_i = E_K(I_{i-1})$$

and

$$C_i = P_i \oplus L_r(I_i)$$

- Decryption :
 - Input : Let K be k -bit Key, I_0 be n -bit initialization vector and $C = C_1 C_2 \dots C_m$ be r -bit ciphertext blocks.
 - Output : Compute $I_i, \dots, I_m$ as in the enciphering algorithm.

$$P_i = C_i \oplus L_r(I_i) \quad \forall 1 \leq i \leq m$$

3. Examples

- Let $m = HELLO$ be the message and plaintext corresponding to the message m is $P = 00111 00100 01011 01011 01110$.
Let $K = 111001$, $I_0 = 101101$ and $E_K(*) = * \oplus K$.
Then,

$$I_1 = E_K(I_0) = 101101 \oplus 111001 = 010100$$

$$C_1 = P_1 \oplus L_r(I_1) = 00111 \oplus L_r(010100) = 01101$$

$$I_2 = E_K(I_1) = 010100 \oplus 111001 = 101101$$

$$C_2 = P_2 \oplus L_r(I_2) = 00100 \oplus L_r(101101) = 10010$$

$$I_3 = E_K(I_2) = 101101 \oplus 111001 = 010100$$

$$C_3 = P_3 \oplus L_r(I_3) = 01011 \oplus L_r(010100) = 00001$$

$$I_4 = E_K(I_3) = 010100 \oplus 111001 = 101101$$

$$C_4 = P_4 \oplus L_r(I_4) = 01011 \oplus L_r(101101) = 11101$$

$$I_5 = E_K(I_4) = 101101 \oplus 111001 = 010100$$

$$C_5 = P_5 \oplus L_r(I_5) = 01110 \oplus L_r(010100) = 00100$$

Thus, ciphertext is $C = 01101\ 10010\ 00001\ 11101\ 00100$

Counter mode

1. About

- The Counter Mode is a simple counter based block cipher implementation in cryptography.
- Each or every time a counter initiated value is encrypted and given as input to XOR with plaintext or original text which results in ciphertext block.
- The CTR mode is independent of feedback use and thus can be implemented in parallel in this mode.
- It generates the next keystream block by encrypting successive values as named as "counter".
- This counter can be any purpose or function which generates a sequence that is guaranteed not to call for a long time, although an actual increment-by-one counter is the simplest or easiest and most popular or famous.
- CTR can be considered as a counter-based version of CFB mode without the feedback in this cryptography.
- In this mode, both the sender and receiver need to get access to a reliable counter, which finds out a new shared value each time a ciphertext block is exchanged or changes the value which is shared.
- This shared counter is not necessarily a secret value, but the challenge is that both sides must keep the counter synchronized at a time when both are active.

2. Process

- Encryption :
 - Input : Let K be k -bit Key, $P = P_1 P_2 \dots P_m$ be n -bit plaintext blocks and $Counter(1)$ be initial counter value known to both sender and receiver in this mode and

$$Counter(i) = Counter(1) + (i - 1) \quad \forall 1 \leq i \leq m$$

- Output :

$$C_i = E_K(Counter(i)) \oplus P_i \quad \forall 1 \leq i \leq m$$

- Decryption :

- Input : Let K be k -bit Key, $C = C_1 C_2 \dots C_m$ be n -bit ciphertext blocks and $Counter(1)$ be initial counter value known to both sender and receiver in this mode and

$$Counter(i) = Counter(1) + (i - 1) \quad \forall 1 \leq i \leq m$$

- Output :

$$P_i = E_K(Counter(i)) \oplus C_i \quad \forall 1 \leq i \leq m$$

Note : In general $Counter(i)$ is a function of $Counter(1)$ and i .

3. Analysis

- It does not have message dependency and hence a ciphertext block does not depend on the previous plaintext or original text blocks.
- Similar to CFB mode, CTR mode does not involve the decryption process of the block cipher in that way. This is because the CTR mode is using the block cipher to generate a key-stream, which is encrypted using the XOR function in cryptography. In other words, CTR mode also converts a block cipher to a stream cipher in cryptography.
- CTR mode has mostly all benefits of CFB mode.
- Also, it does not propagate the error of transmission at all in this mode.
- The main disadvantage of CTR mode is that it requires a synchronous counter at the sender and receiver in this mode. Decreases of synchronization lead to the incorrect recovery of plaintext or original text.

4. Examples

- Let $m = HELLO$ be the message and plaintext corresponding to the message m is $P = 00111\ 00100\ 01011\ 01011\ 01110$.
Let $K = 11001$, $Counter(1) = 01101$, $E_K(*) = * \oplus K$, and

$Counter(i+1) = Counter(i) \oplus B.R.(i)$, where

$B.R.(i) = L_5(\text{binary representation of } i)$.

Then,

$$Counter(2) = Counter(1) \oplus B.R.(1) = 01101 \oplus 00001 = 01100$$

$$Counter(3) = Counter(2) \oplus B.R.(2) = 01100 \oplus 00010 = 01110$$

$$Counter(4) = Counter(3) \oplus B.R.(3) = 01110 \oplus 00011 = 01101$$

$$Counter(5) = Counter(4) \oplus B.R.(4) = 01101 \oplus 00100 = 01001$$

$$C_1 = P_1 \oplus E_K(Counter(1)) = 00111 \oplus E_K(01101) = 10011$$

$$C_2 = P_2 \oplus E_K(Counter(2)) = 00100 \oplus E_K(01100) = 10001$$

$$C_3 = P_3 \oplus E_K(Counter(3)) = 01011 \oplus E_K(01110) = 11100$$

$$C_4 = P_4 \oplus E_K(Counter(4)) = 01011 \oplus E_K(01101) = 11111$$

$$C_5 = P_5 \oplus E_K(Counter(5)) = 01110 \oplus E_K(01001) = 11110$$

Thus, ciphertext is $C = 10011 \ 10001 \ 11100 \ 11111 \ 11110$

R.K.Sharma

Chapter 20

Quantum Cryptography

20.1 Introduction

In 1941, German engineer Konard Zeise created the first computer based on the ideas of Charles Babbage and world got a technological advancement in the name of development. The size of a computer has been decreasing exponentially since 1950 that is physical system needed to represent a bit of memory is getting smaller and smaller. The whole process involved a sequence of changes from one type of physical realization to another from gears to relays to valves to transistors to integrated circuits to chip and so on. However the task of the computer is the same: to manipulate and interpret an encoding of binary bits into a useful computational result. We might be within the reach of atomic scale computations with in a decade.

Atoms follows the rules of quantum mechanics, which are quite different from the classical rules on which classical computers are based. Classical computers are built by electrical circuit containing logical gates on which classical theory of physics is applied. So if computers are to become smaller in future, new, quantum technology must replace or supplement what we have now, classical technology. With the size of components in classical computers shrinking to where the behavior of the components, is practically dominated by quantum theory than classical theory.

20.2 Limitation of Classical Computer and Birth of Quantum Computer

The idea of quantum computing emerged when scientists were pondering on the fundamental limits of computation.

In 1970's and early 1980's, many physicists and computer scientists such as Charles H. Bennet, Paul A. Beniof and David Deutsch proposed the idea of computational device based on quantum mechanics.

In 1982 Feynman suggested a new kind of computers which could be devised based on the principles of quantum physics. He constructed an abstract model to show how a quantum system could be used to do computations and also explain how such a machine would be able to act as a simulator for physical problems pertaining to quantum physics.

Later, in 1985, Deutsch showed that any physical process, in principle could be modelled perfectly by a quantum computer. Peter Shor in 1994 set out a method for using quantum computers to crack an important problem in number theory which was namely factorization.

20.2.1 Quantum factoring

Many classical cryptographic algorithms are based on mathematical concepts. The security of the algorithms depend on computational hardness of the assumptions. For example security of RSA cryptosystem totally depends on factoring the large positive integer. Peter Shor showed how an ensemble of mathematical operations, designed specifically for a quantum computer could be organized to make such a machine to factor huge numbers extremely rapidly, much faster than is possible on conventional computers.

20.2.2 Quantum searching

Searching of an item with desired property from a collection of N items is another problem in the classical computer that require $N/2$ operation for getting the right items. However Grover invented quantum logic based algorithm that admits tremendous speed up and accomplishes the same task in an average of $\sqrt{N}$ number of operations.

20.2.3 Quantum simulation

Richard P. Feynman, proposed that a quantum physical system of N particles with its quantum probabilities can not be simulated by the usual computer without an exponential slowdown in the efficiency of simulation. However, a quantum computer can simulate a quantum mechanical system.

20.3 Quantum Cryptography

Quantum cryptography is the science of exploiting the quantum mechanical properties to perform the cryptographic task.

There are some counter intuitive points that are very useful in quantum cryptography:

- It is not possible to take measurement without perturbing the quantum system.
- One cannot determine simultaneously the position and the momentum of a particle with arbitrarily high accuracy.
- It is not possible to make a duplicate of unknown quantum state.

20.4 Quantum Bits

An ordinary computer chip uses bits. These are information carrier which is used in communication. Bits are like tiny switches that can either be in off and on i.e. 0 or 1 state. But in nature things are not just on or off. There is uncertainty. Therefore instead of bits, quantum computers use quantum bits, or *qubit* for short.

A qubit (or quantum-bit) is a unit vector in $\mathbb{C}^2$. We fix orthonormal basis of column vectors denoted as $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ corresponding to classical bits 0 and 1.

Note: Notation like $|\psi\rangle$ is the Dirac notation. The entire object $|\psi\rangle$ is called a ‘ket ψ ’. For example $|0\rangle$ is called as ‘ket 0’.

If there are two pair of classical bits, there would be four possible states 00, 01, 10, 11. In analogues way we can define two qubit system that has four *computational basis states* denoted as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

More generally, we may consider n qubit system. The computational basis states are $|x_1 x_2 x_3 \dots x_n\rangle$ where x_i 's either 0 or 1.

20.5 Quantum Superposition

A qubit can exist not only in a state corresponding to the logical state 0 or 1 as in a classical state bit but also in states corresponding to a blend of superposition of those classical states. In other words a qubit can exist as a zero, a one, or simultaneously as both 0 and 1, with numerical coefficient representing the probability for each state.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

where $|0\rangle, |1\rangle$ are known as *computational basis state* can be represented as $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in two dimensional vector space, and $\alpha, \beta \in \mathbb{C}$ satisfying $|\alpha|^2 + |\beta|^2 = 1$

For example, a single qubit can be in the state:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

and

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

$|+\rangle$ and $|-\rangle$ known as Hadamard basis.

State vector describing the two qubits is

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

such that $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$

For Example, two qubit can be in the state

$$|\phi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

which is the important two qubit state known as Bell state.

20.6 Quantum Gates

A classical computer is built from an electrical circuit containing wires and logic gates, similarly a Quantum computer is composed of a quantum circuit

that contains wires and quantum gates. Quantum gates perform the manipulation of quantum information by converting one qubit state to another qubit state. A Quantum gate can be described as a unitary operator. Amazingly, any unitary matrix specifies a valid quantum gate. An important feature of quantum circuits is that, between initializing the qubits and measuring them, the operations (gates) are always reversible.

20.6.1 Single Qubit Gate

Quantum gates on a single qubit can be described by 2×2 matrix. There are many single qubit gates, we will discuss some of them.

Pauli X-gate can be represented by Pauli X- matrix

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Pauli X-gate is known as Not gate or Bit Flip gate. This gate flips the bit i.e. it change $|0\rangle$ to $|1\rangle$ and vice versa.

$$X |0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X |1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Pauli Y-gate can be represented by Pauli Y- matrix

$$Y = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

$$Y |0\rangle = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i |1\rangle$$

$$Y |1\rangle = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i |0\rangle$$

Pauli Z-gate can be represented by Pauli Z- matrix

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Z |0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

Hadamard gate can be represented as

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

This gate change $|0\rangle$ to $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ and $|1\rangle$ to $\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$.
Phase gate can be represented as

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$\frac{\pi}{8}$ gate be represented as

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

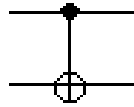
20.6.2 Multiple Qubit Gate

Multiple qubit gate takes multiple qubit as an input and we get multiple qubit as an output.

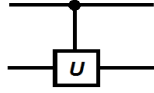
Controlled-Not gate is two qubit gate which takes two qubit, known as control qubit and target qubit, as an input and gives two qubit as an output. In terms of computational basis the action of the CNOT gate is given by $|c\rangle|t\rangle \rightarrow |c\rangle|t \oplus c\rangle$. Thus matrix representation of CNOT gate is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Circuit representation for the controlled -NOT gate



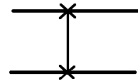
More generally, suppose U is an arbitrary single qubit unitary operation. A controlled- U gate can be represented as



Swap gate is also two qubit gate which swaps the qubit. The matrix representation of the swap gate is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

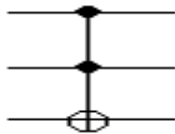
Circuit representation for the swap gate



Toffoli gate is the three qubit gate, in which two qubits are control qubit and one qubit is target qubit. The matrix representation of the toffoli gate is

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Circuit representation for the toffoli gate



20.7 Quantum Entanglement

Quantum entanglement plays a very important role in quantum cryptography mainly in quantum key distribution (QKD). Entanglement can be defined as a correlation between the particles that cannot be separated even though they are separated by physical distance.

Two qubits are considered to be entangled when the measurement of one qubit has affected the state of the other qubit.

Consider two arbitrary quantum systems A and B, with respective Hilbert spaces H_A and H_B . The Hilbert space of the composite system is the tensor product

$$H_A \otimes H_B$$

If the first system is in state $|\psi_A\rangle$ and the second in $|\psi_B\rangle$, the state of the composite system is

$$|\psi_A\rangle \otimes |\psi_B\rangle$$

States of the composite system that can be represented in this form are called separable states, or product states.

Note that all states of composite system are not necessary in the product form. If qubits are not prepared independently, they are allowed to interact, then it may not possible to write the state in product form these qubits are called Entangled state.

The simplest entangle state is Bell states:

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}};$$

$$|\beta_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}};$$

$$|\beta_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}};$$

$$|\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

In QKD, entangle state can be used, since any examination on entangled bits by an eavesdropper leads to a disturbance.

20.8 Quantum Measurement

We can not examine a qubit in its actual quantum state. When we perform a measurement on a qubit in its quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then it collapse one of the basis state, we get either 0 with probability $|\alpha|^2$, or 1 with probability $|\beta|^2$. We can not determine the values of α and β exactly.

Quantum measurements are described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result m occurs is given by

$$p(m) = \langle\psi| M_m^\dagger M_m |\psi\rangle$$

and the state of the system after the measurement is

$$\frac{M_m |\psi\rangle}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$$

The measurement operators satisfy the completeness equation,

$$\sum_m M_m^\dagger M_m = I$$

The completeness equation expresses the fact that probabilities sum to one:

$$1 = \sum_m p(m) = \sum_m \langle\psi| M_m^\dagger M_m |\psi\rangle$$

Example: (Measurement of a qubit in the computational basis) Let the state being measured is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Measurement operators are

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|$$

Observe that $M_0^\dagger = M_0$, $M_1^\dagger = M_1$ and $M_0^2 = M_0$, $M_1^2 = M_1$. Then the probability of obtaining measurement outcome 0 is

$$p(0) = \langle\psi| M_0^\dagger M_0 |\psi\rangle = \langle\psi| M_0 |\psi\rangle = |\alpha|^2$$

The state of the system after the measurement is

$$\frac{M_0 |\psi\rangle}{|\alpha|} = \frac{\alpha}{|\alpha|} |0\rangle$$

Then the probability of obtaining measurement outcome 1 is

$$p(1) = \langle \psi | M_1^\dagger M_1 | \psi \rangle = \langle \psi | M_1 | \psi \rangle = |\beta|^2$$

The state of the system after the measurement is

$$\frac{M_1 |\psi\rangle}{|\beta|} = \frac{\beta}{|\beta|} |1\rangle$$

20.9 No Cloning Theorem

It is impossible to build a machine that can clone any given quantum state. This is in stark contrast to the classical case, where we copy information all the time. It is the tip of the iceberg for the differences between quantum and classical computing.

Proof. Suppose we have a machine with two slots: A for the quantum state $|\psi\rangle$ to be cloned, and B in some fixed initial state $|s\rangle$, and the machine makes a copy of the quantum state A. By the rules of quantum mechanics, the evolution U is unitary, so we have

$$|\psi\rangle \otimes |\psi\rangle \xrightarrow{U} |\psi\rangle \otimes |\psi\rangle$$

Now suppose we have two states we wish to clone, $|\psi\rangle$ and $|\phi\rangle$, giving

$$U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$$U(|\phi\rangle \otimes |s\rangle) = |\phi\rangle \otimes |\phi\rangle$$

Taking the inner product of these two equations, and using $U^\dagger U = I$

$$\begin{aligned} (\langle \phi | \otimes \langle s |) U^\dagger U (|\psi\rangle \otimes |s\rangle) &= (\langle \phi | \otimes \langle \phi |) (|\psi\rangle \otimes |\psi\rangle) \\ \langle \phi | \psi \rangle \langle s | s \rangle &= \langle \phi | \psi \rangle \langle \phi | \psi \rangle \\ \langle \phi | \psi \rangle &= (\langle \phi | \psi \rangle)^2 \end{aligned}$$

This has solutions if and only if $\langle \phi | \psi \rangle$ is 0 or 1, so cloning cannot be done for general states.

20.10 Quantum Parallelism

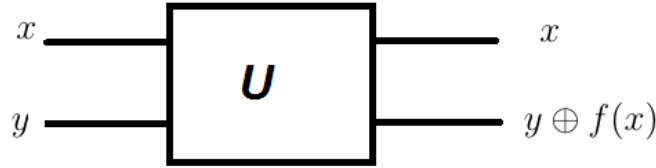
Many quantum algorithm depends on a fundamental property known as quantum parallelism. Quantum parallelism arises from the ability of a quantum memory register to exist in a superposition of base states. A quantum memory register can exist in a superposition of states, each component of this superposition may be thought of as a single argument to a function.

In simple words, quantum parallelism allows a quantum computer to compute a function $f(x)$ in one time for different values of x . Suppose we have a function f two bit domain to one bit co-domain i.e.

$$f(x) : \{0, 1\}^2 \rightarrow \{0, 1\}$$

To evaluate f on all four possible two bits 00, 01, 10, 11, we will need to call f four times: $f(0, 0)$, $f(0, 1)$, $f(1, 0)$, $f(1, 1)$. Exploiting quantum parallelism allows us to evaluate all four inputs in a single call to f . Note, however, that our function f is not reversible, and all operations on qubits must be reversible. So we first define a reversible version of f as follows:

$$U |x\rangle |y\rangle \rightarrow |x\rangle |y \oplus f(x)\rangle$$



The quantum oracle U takes two inputs x and y and outputs two values. The first is x itself. The second is the value: $y \oplus f(x)$. It is easy to see that when $y = 0$, the second output equals $f(x)$. Now we take input state

$$|\phi\rangle = \frac{1}{2} |00\rangle + |01\rangle + |10\rangle + |11\rangle$$

Next, we apply our quantum function U on this state by setting $y = 0$. This

gives us:

$$\begin{aligned}
 U |\phi\rangle |0\rangle &= \frac{1}{2} U(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \otimes |0\rangle \\
 &= \frac{1}{2} U(|00, 0\rangle + |01, 0\rangle + |10, 0\rangle + |11, 0\rangle) \\
 &= \frac{1}{2} (|00, f(00)\rangle + |01, f(01)\rangle + |10, f(10)\rangle + |11, f(11)\rangle)
 \end{aligned}$$

If we separate out the two qubits, we see that the second output qubit contains the superposition of all four results we are interested in. Thus, we evaluated four inputs using a single application of the target function.

20.11 Quantum Key distribution (QKD)

Quantum Key distribution protocol provides a provably secure way to exchange the keys over a public channel between two parties. The key can be used to implement a classical symmetric key cryptosystem. The security of the QKD protocol is guaranteed by the fundamental laws of quantum physics that is Eves can not clone Alice's qubit, by no-cloning theorem and any information gain by Eve disturbs the transmitted qubit states.

Proposition 20.11.1. *(Information gain implies disturbance) In any attempt to distinguish between two non-orthogonal quantum states, information gain is only possible at the expense of introducing disturbance to the signal.*

20.11.1 BB84 Protocol

In 1984 Bennett and Brassard describe how two bases $\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$ can be used in quantum key distribution protocol. This protocol is based on two fundamental: Measurement causes perturbation and no-cloning theorem. Thus, measuring the qubit on the wrong basis destroys the information.

1. Alice choose n random data bits
2. Now Alice encode the each data bit either in the basis $A = \{|0\rangle, |1\rangle\}$ or $B = \{|+\rangle, |-\rangle\}$ randomly.
3. Alice sends encoded state to Bob.

4. Now Bob chooses a random sequence of basis A and B and gets a sequence of n bits.
5. Alice and Bob announce their basis sequence and compare their measurement basis, discard unmatched. Finally, they have a common key.

For eg:

Alice choose:	0	1	0	1	1	0	0	0	1	0	1	0	1	1
Alice's bases:	B	A	A	B	A	A	B	A	A	A	A	B	B	B
States sent:	$ +\rangle$	$ 1\rangle$	$ 0\rangle$	$ -\rangle$	$ 1\rangle$	$ 0\rangle$	$ +\rangle$	$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$	$ -\rangle$
Bob's bases:	A	B	A	A	B	A	A	A	B	A	B	B	B	B
Bob's results:	1	+	0	0	+	0	-	0	+	0	+	+	-	-

So Bob and Alice are left with 7 useable bits out of 14 bits

0000011

20.11.2 TWO-STATE PROTOCOL, B92

Besides BB84, other protocols were designed. In 1992, C. H. Bennett showed (Bennett [1992b]) that two nonorthogonal states are already sufficient to implement secure QKD. Let Alice choose two nonorthogonal states and send them to Bob in random order. When Bob performs projections onto subspaces orthogonal to the signal states, he sometimes learns Alice's bit with certainty and sometimes he obtains an inconclusive outcome. After the transmission, Bob tells Alice when he detected a bit. In this case, he does not announce the used basis, because a basis in which he detected a photon, uniquely identifies the bit Alice had sent. This protocol is usually called B92.

However, such a scheme is secure only in lossless systems or if the losses are very low. In the case of higher losses, an eavesdropper could sit in the middle and make measurements on the quantum states. If she has obtained an inconclusive result, she blocks the signal, while if she has detected the sent state, she re-sends a correct copy to Bob, because she knows the state with certainty. To compensate for the blocked photons, she can send a pulse of higher intensity so that Bob cannot observe any decrease in the expected transmission rate.

20.11.3 SIX-STATE PROTOCOL

In the six-state protocol, three non-orthogonal bases are used (Bruss [1998], BechmannPasquinucci and Gisin [1999]) that Alice and Bob randomly alternate. If we denote the two conjugate bases employed in the BB84 protocol as $\{|0\rangle, |1\rangle\}$ and $\{|\bar{0}\rangle, |\bar{1}\rangle\}$, where

$$|\bar{0}\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

and

$$|\bar{1}\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

then the third basis is $\{|\bar{\bar{0}}\rangle, |\bar{\bar{1}}\rangle\}$ with

$$|\bar{\bar{0}}\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}i|1\rangle$$

and

$$|\bar{\bar{1}}\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}i|1\rangle$$

The probability that Alice and Bob choose the same basis is $1/3$ now.⁷ But this disadvantage against BB84 is outweighed by the fact that eavesdropping causes higher error rate. For example, a continuous intercept-resend attack induces in average 33% maximal mutual information between Eve and Alice is smaller than in the BB84 scenario. Besides, the symmetry of the signal states simplifies the security analysis.

20.12 Quantum Factoring Algorithm (Shor's Algorithm)

For a given positive integer N , it is very difficult to find out its non trivial factor, that is factoring is considered as a hard problem in classical computer. But using quantum computer it can be easily simplified. There is quantum factoring algorithm which is given by Shor in 1994 [?].

The reduction of factoring to order-finding proceeds in two basic steps. The first step is to show that we can compute a factor of N if we can find a non-trivial solution $x \not\equiv \pm 1 \pmod{N}$ to the equation $x^2 \equiv 1 \pmod{N}$. The

second step is to show that a randomly chosen y co-prime to N is quite likely to have an order r which is even, and such that $y^{r/2} \not\equiv \pm 1 \pmod{N}$, and thus $x \equiv y^{r/2} \pmod{N}$ is a non-trivial solution to $x^2 = 1 \pmod{N}$.

Theorem 20.12.1. *Suppose N is an L bit composite number, and x is a non-trivial solution to the equation $x^2 = 1 \pmod{N}$ in the range $1 \leq x \leq N$, that is, neither $x = 1 \pmod{N}$ nor $x = N - 1 = -1 \pmod{N}$. Then at least one of $\gcd(x - 1, N)$ and $\gcd(x + 1, N)$ is a non-trivial factor of N that can be computed using $O(L^3)$ operations.*

Theorem 20.12.2. *Suppose $N = p_1^{\alpha_1} \cdots p_m^{\alpha_m}$ is the prime factorization of an odd composite positive integer. Let x be an integer chosen uniformly at random, subject to the requirements that $1 \leq x \leq N - 1$ and x is co-prime to N . Let r be the order of x modulo N . Then*

$$p(r \text{ is even and } x^{r/2} \not\equiv -1 \pmod{N}) \geq 1 - \frac{1}{2^m}$$

Shor's Algorithm:

- **Input** A composite number N
- **Output** A non trivial factor of N
- **Procedure**
 1. If N is even, return the factor 2
 2. Determine whether $N = a^b$ for integer $a \geq 1$ and $b \geq 2$, and if so return the factor a .
 3. Randomly choose x in the range 1 to $N - 1$. If $\gcd(x, N) > 1$ then return the factor $\gcd(x, N)$.
 4. Use the order-finding subroutine to find the order r of x modulo N .
 5. If r is even and $x^{r/2} \not\equiv -1 \pmod{N}$ then compute $\gcd(x^{r/2} - 1, N)$ and $\gcd(x^{r/2} + 1, N)$, and test to see if one of these is a non-trivial factor, returning that factor if so. If r is not even then again choose random x and go to step 3.

Note: In above algorithm we can observe that only 4th step of the algorithm can not be done by classical computer in polynomial time. To do this we need quantum computer which can find order of an element in polynomial time. So factoring of a large integer can be done by quantum computer in polynomial time.

20.13 Quantum Simulator

A universal quantum simulator is a quantum computer proposed by Yuri Manin in 1980 and Richard Feynman in 1982. Feynman showed that a classical Turing machine would not be able to simulate a quantum effect.

IBM offers a high-performance simulation framework called Qiskit Aer for the open source Qiskit software stack. Qiskit is the most modular and extensible tool for leveraging today's quantum processors in research, education, and business. Explore and optimize solutions that aren't yet possible on today's hardware

Allow you to explore circuits for future hardware, rapidly prototype circuits you are designing for real hardware, and characterize noise response and sensitivity of a circuit.